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Research article

Electrical breakdown in liquid-phase processing on an enhancement of 7-hydroxymitragynine conversion from mitragynine in *Mitragyna speciose* (Kratom)

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ABSTRACT

This study investigates the impact of the Electrical Breakdown in Liquid-phase (EBL) process on alkaloid transformation in *Mitragyna speciose* (Kratom) leaves, focusing on the conversion of mitragynine (MG) to 7-hydroxy mitragynine (7-OH-MG) by using advanced oxidation processes (AOPs). A novel reactor has been developed to enhance plasma exposure to Kratom leaf powdered solutions during the EBL process. Two distinct electrical voltage characteristics, half-positive and negative half-waves, have been utilized for the EBL, with an output voltage of 4.57 kV_{peak} at a no-load condition and a frequency of 50 Hz. The experimental findings demonstrate a time-dependent enhancement in the transformation process. The highest yield of 7-OH-MG, reaching 2,485 ± 134 μg/g of dried Kratom leaves weight, has been attained with the EBL processing generated by positive half-wave voltage after 20 min of EBL exposure. Notably, the EBL processing generated by positive half-wave voltage has outperformed the one generated by negative half-wave voltage by a significant factor of 2.01.

Nomenclatures

Abbreviations	Meaning
7-OH-MG	7-hydroxymitragynine
ANOVA	Analysis of variance
AOPs	Advanced oxidation processes
CCD	Compact charge-coupled device
I_D	Discharge current
V_D	Discharge voltage
pKa	Dissociation constant
EBL	Electrical Breakdown in Liquid-phase
NV-EBL	EBL processing generated by the negative half-wave supplied voltage

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PV-EBL	EBL processing generated by the positive half-wave supplied voltage
EC	Electrical conductivity
HWVD	Half-wave voltage doubler
HPLC	High-performance liquid chromatography
LOD	Limit of detection
LOQ	Limit of quantification
MG	Mitragynine
NV	Negative half-wave supplied voltage
OES	Optical emission spectra
PV	Positive half-wave supplied voltage
RNSs	Reactive nitrogen species
RONSs	Reactive oxygen and nitrogen species
ROSs	Reactive oxygen species
RSD	Relative standard deviation
SPN	Second positive system

1. Introduction

Kratom (*Mitragyna speciosa*), native of the tropical regions of Thailand, Myanmar, Malaysia, Indonesia, and other Southeast Asian countries, has gained significant recognition for its analgesic properties. This acknowledgment predominantly stems from a distinct alkaloid present in the plant known as mitragynine (MG), which is exclusive to Kratom. Consequently, the medicinal utilization of Kratom in Southeast Asia boasts a lengthy historical background. However, in recent times, Kratom has surged in popularity in Western nations, with individuals incorporating it into their routines for various health reasons, in order to address chronic pain and mood disorders. Although some nations have recently legalized the cultivation and sale of Kratom, its efficacy for these purposes remains a contentious and politically charged subject, which necessitates scientific scrutiny [1,2].

The conversion of MG into 7-hydroxymitragynine (7-OH MG) through a liver enzymatic process offers the potential for more powerful pain relief with reduced addictive properties [3–5]. This conversion, catalyzed by enzymes, represents a unique biological pathway within the human body [6]. Consequently, the efficacy of pain relief through the consumption of boiled Kratom water may be limited, as this transformation is not a common biological occurrence. Typically, extracting alkaloids from Kratom requires acidic solutions to dissolve them in water, posing challenges for subsequent applications. Following the extraction, conditions should be adjusted to an alkaline state in order to facilitate alkaloid extraction by using organic solvents. Unfortunately, this complex extraction process often results in the loss of essential compounds, leading to increased extraction costs and environmental concerns [4]. Therefore, developing a cost-effective, non-acidic organic extraction method for MG conversion holds significant promise for potential pharmaceutical applications [3–5].

Recent advancements in atmospheric-pressure plasma technology, an innovative and environmentally sustainable approach, have found applications in various fields, including food technology [7], agriculture [8–10], environmental management [11], medical treatments [12], and pharmacology [13]. The Electrical Breakdown in Liquid-phase (EBL) process is an emerging plasma technique that harnesses advanced oxidation processes (AOPs) driven by free radicals, reactive oxygen species (ROSs), reactive nitrogen species (RNSs), and reactive oxygen and nitrogen species (RONSs) in a liquid-phase environment. This results in chemically active oxidation during plasma generation [7,11,14–16]. These species play critical roles as precursors and potent oxidants with exceptionally high oxidative activity. Valuable byproducts, such as hydrogen ions (H^+), hydroxyl radicals (OH•), singlet oxygen (O•), ozone (O_3), and hydrogen peroxide (H_2O_2), are produced in this process. Furthermore, ultraviolet radiation, photons, electromagnetic waves, and x-rays, which are present during plasma generation, contribute to the unique properties of plasma [11,17–20]. Given these merits and the suitability for MG and 7-OH MG conversion, plasma presents a promising and eco-friendly solution for 7-OH MG extraction from Kratom solutions.

This research aims to address the research gap identified in previous studies concerning the complex and environmentally taxing extraction process of alkaloids from Kratom leaves. In pursuit of this objective, the study introduces the novel Electrical Breakdown in Liquid-phase (EBL) process as a potential alternative [3–6]. The primary objective of this research is to evaluate the potential of the EBL process in converting MG into 7-OH MG through advanced plasma-driven oxidation. A custom-designed plasma reactor, meticulously developed by the author, is employed to enhance plasma contact with Kratom powder during EBL. By evolving from its predecessor, the EBL reactor in Ref. [21], this innovative system has been meticulously refined in order to optimize the conversion of mitragynine into 7-hydroxymitragynine. In previous iterations, the necessity of transferring the solution outside for cooling has hindered effective plasma interaction. In order to overcome this limitation, internal cooling via a dedicated solution pipe now facilitates seamless circulation within the reactor. Furthermore, the strategic inclusion of a connecting pipe aims to maximize the interaction between the solution, Kratom powder, and plasma, thereby augmenting conversion efficiency. The primary objective of this experiment is to discern the optimal conditions for efficiently extracting 7-OH MG from Kratom leaves. Two distinct electrically driven methods (half-positive and negative half-wave voltages, potentially inducing differing corona discharge characteristics) have been employed for EBL generation, with varying processing times, in order to analyze their time-dependent effects. Subsequent to the extraction, purification procedures will be executed, and the compound structures will be confirmed through column chromatography. The quantity of extracted material will be quantified by using high-performance liquid chromatography (HPLC) techniques.

2. Experimental setup and procedure

2.1. Plasma model and experimental setup

Fig. 1 illustrates a schematic representation of the Electrical Breakdown in Liquid (EBL) process along with the experimental setup. The plasma reactor, designed in the shape of a wine Pyrex glass structure with a volume of 150 mL, features a recirculation tube on the side in order to ensure continuous plasma contact. At the midpoint, where plasma is generated, there are two tungsten pin electrodes with diameters of 0.1 mm, serving as the high-voltage and ground electrodes, strategically spaced 1 mm apart.

The Kratom solution enters the reactor through the upper solution port. At the lowest section of reactor, a gas inlet pipe aligned with its central core introduces air from the HAILEA ACO-318 air pump at a rate of 1.20 LPM, crucial for initiating plasma generation between the electrodes. In order to maintain EBL process stability, the upper reactor section controls the solution temperature within 50-55 °C. Equipped with a Kratom filter at its terminus, the upper reactor pipe conveys the solution at a constant rate of 42 mL per minute to the 5 °C cooling system via a Peristaltic Pump DP-385. Then the cooled solution is reintroduced into the reactor at its bottom for further treatment, with only 10 percent of the total solution volume extracted for cooling purposes.

The initiation of the EBL process involves connecting one end of the high-voltage electrode to a high-voltage source through a positive or negative half-wave voltage doubler (HWVD) circuit, derived from an 800 W microwave transformer. The counter electrode connects to a 50-Ω monitoring resistor in series, subsequently grounded. The maximum output voltage in this setup reaches 4.57 kV_{peak}, achieved by utilizing a 150 V_{rms} input voltage from 50 Hz variable AC power supplies [21,22].

The monitoring of the discharge voltage (V_D) and the current (I_D) electrical characteristics is conducted by using an oscilloscope (LECROY, WaveRunner 8208HD, 2 GHz, 10 GS/s). A high-voltage probe (Pintek HVP-28 H F, 75 MHz bandwidth) is employed across the plasma model, while a general-purpose passive probe (GW Instek, 100 MHz bandwidth) is connected across the monitored resistor. In order to analyze optical emission spectroscopy (OES) during plasma formation, a compact charge-coupled device (CCD) spectrometer (Newport 71SI00087) is utilized, with the optical fiber detector tip positioned 1 cm perpendicular to the electrode plane.

2.2. Kratom solution preparation and EBL treatment condition

The preparation of the Kratom leaf solution begins by selecting mature Kratom leaves from plants aged one year or older. Next, these leaves are dried at 60 °C for 24 h. Once dried, the leaves are grinded to a particle size smaller than 125 μm. Finally, 1 g of the resulting Kratom powder is combined with 100 mL of distilled water. The prepared Kratom solution will undergo treatment via the EBL process, by employing an input voltage of 150 Vrms for processing times of 5, 10, 15, 20, 40, 60, and 80 min.

2.3. Characterization of plasma-treated Kratom solution

1) Physical and chemical analysis of Kratom solution

The physical and the chemical properties of both the Kratom solution and the distilled water (serving as a reference) have been meticulously examined, by encompassing parameters such as pH, temperature, and electrical conductivity (EC). A Mettler Toledo SevenEasyTMpH meter and a Hanna HI4321 conductivity meter have been utilized to measure pH levels and EC, respectively. The solution temperature has been monitored by using a mercury thermometer submerged within the treatment solution. In order to confirm the presence of hydrogen peroxide (H_2O_2) in the solution, Quantofix peroxide 100 test strips (1–100 mg/L H_2O_2) have been employed. Additionally, the concentration of dissolved ozone in the solution has been determined by using an ozone colorimeter test kit (MColortestTM, 10 mg/L O_3).

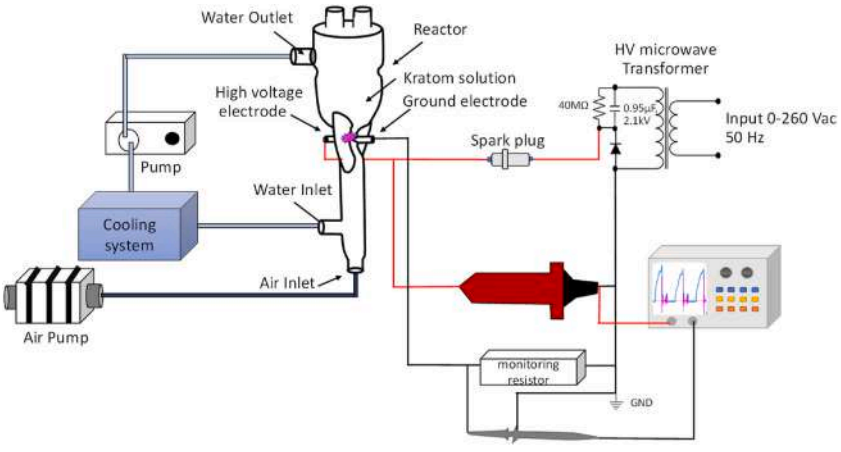


Fig. 1. A schematic illustration of the Electrical Breakdown in Liquid (EBL) process and the corresponding experimental setup.

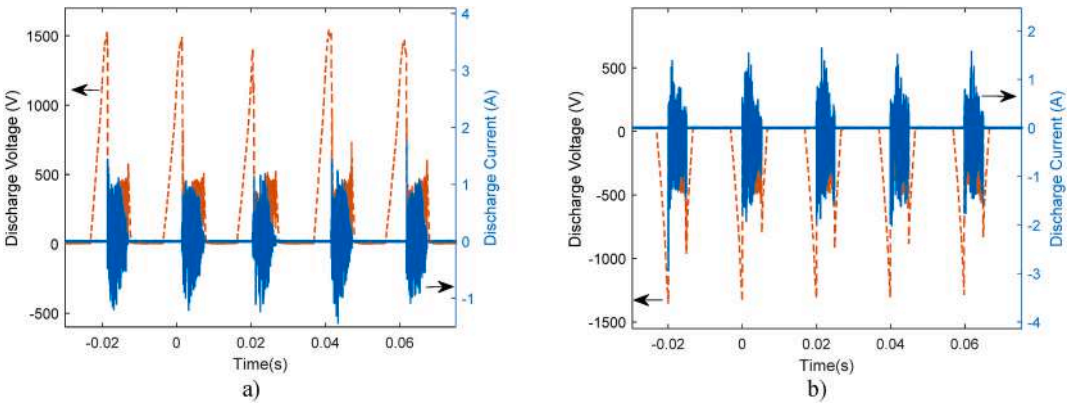


Fig. 2. Discharge voltage and current waveforms during the EBL process driven by a) positive half-wave voltage and b) negative half-wave voltage.

2) HPLC-DAD quantification of 7-hydroxymitragynine and mitragynine in Kratom solution

The HPLC system (Agilent, Santa Clara, CA, USA) has comprised the 1260 Infinity II quaternary pump, 1260 Infinity II autosampler, 1260 Infinity II multi-column thermostat, and 1260 Infinity II photodiode array detector. The separation process has occurred within an ACE 5C18-AR column (4.6 × 250 mm, 5 μm) paired with a C18 guard column. The mobile phase, consisting of acetonitrile and ammonium acetate buffer with a pH of 9.5 in a fixed ratio of 45:55, has maintained a steady flow rate of 1 mL/min. Prior to use, the mobile phase has undergone degassing and has been freshly prepared for each analysis. The detection wavelength has been consistently set at 246 nm, with an injection volume of 10 μl. All the analyses have been performed at an operating temperature of 40 °C. Under these specified chromatographic conditions, the average retention times for 7-OH-MG and MG reference standards have been determined to be 8.7 and 61.6 min, respectively. The method has undergone validation in a Kratom solution, following the guidelines established by the International Conference on Harmonization (ICH), in order to assess linearity, selectivity, accuracy, precision, limit of detection (LOD), and limit of quantification (LOQ) [23].

2.4. Statistical analysis

All the experimental conditions and measurements have undergone rigorous validation through triplicate testing to derive the mean result value. Statistical analyses have been conducted by using *t*-test analysis and one-way analysis of variance (ANOVA) in order to identify significant differences among samples at a significance level of $p < 0.05$.

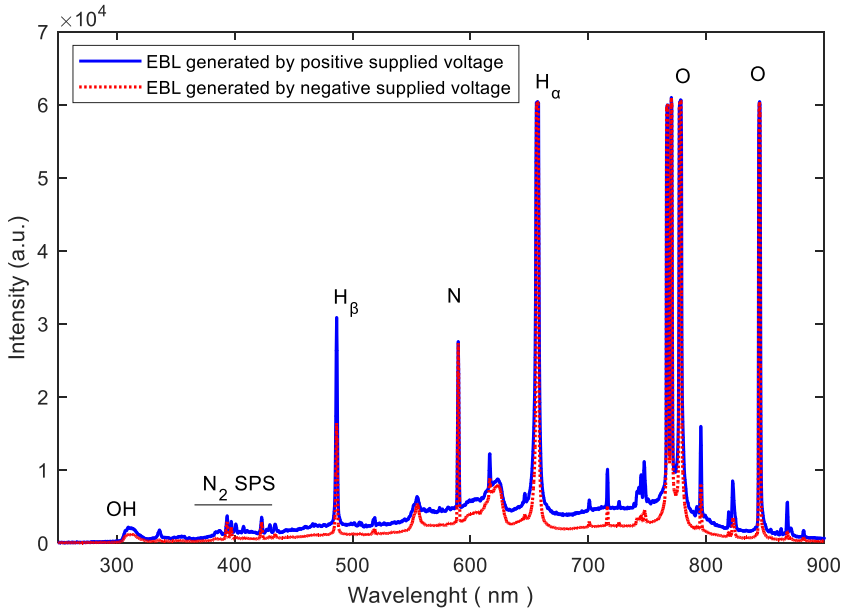


Fig. 3. Optical emission spectra (OES) in the Kratom solution during the EBL process generated by positive and negative half-wave supplied voltages.

Table 1
Quantities of MG and 7-OH-MG in Kratom solution at various EBL processing durations*.

Time (min.)	Mean concentration ± SD (μg/mL), n = 3			Productivity of 7-OH-MG (%)	7-OH-MG content (μg/g of dried weight of Kratom leaves)
	7-OH-MG	7-OH-MG production	MG		
EBL processing generated by POSITIVE half-wave supplied voltage (PV-EBL)					
0	9.03 ± 0.43	0.00	36.19 ± 2.24	0.00	9.03 ± 0.43
5	13.69 ± 1.96	4.66 ± 0.67	33.89 ± 0.70	12.88 ± 1.84	13.69 ± 1.96
10	17.16 ± 1.12	8.13 ± 0.53	30.89 ± 0.94	22.46 ± 1.46	17.16 ± 1.12
15	21.07 ± 1.37	12.04 ± 0.78	29.36 ± 2.87	33.27 ± 2.16	21.07 ± 1.37
20	24.85 ± 1.34	15.81 ± 0.85	23.31 ± 0.98	43.70 ± 2.36	24.85 ± 1.34
40	15.86 ± 1.67	6.82 ± 0.72	0.00	18.86 ± 1.99	15.86 ± 1.67
60	7.35 ± 0.52	−1.68 ± 0.12	0.00	−4.66 ± 0.33	7.35 ± 0.52
80	3.68 ± 0.58	−5.36 ± 0.84	0.00	−14.80 ± 2.32	3.68 ± 0.58
EBL processing generated by NEGATIVE half-wave supplied voltage (NV-EBL)					
0	9.03 ± 0.43	0.00	36.19 ± 2.24	0.00	9.03 ± 0.43
5	9.65 ± 0.66	0.62 ± 0.04	32.33 ± 1.99	1.71 ± 0.12	9.65 ± 0.66
10	10.50 ± 0.44	1.47 ± 0.06	29.25 ± 1.68	4.05 ± 0.17	10.50 ± 0.44
15	11.03 ± 0.25	1.99 ± 0.05	25.90 ± 2.43	5.52 ± 0.12	11.03 ± 0.25
20	11.88 ± 0.57	2.85 ± 0.14	22.56 ± 2.97	7.88 ± 0.38	11.88 ± 0.57
40	6.42 ± 0.56	−2.62 ± 0.23	0.00	−7.22 ± 0.63	6.42 ± 0.56
60	2.41 ± 0.43	−6.62 ± 1.18	0.00	−18.30 ± 3.26	2.41 ± 0.43
80	0.00	−9.03 ± 0.00	0.00	−24.96 ± 0.00	0.00

* Regarding the *t*-test analysis and one-way ANOVA conducted at a significance level of $p < 0.05$, all the data show significant differences across all the dimensions, including treatment times and supplied voltage polarities.

3. Experimental results and discussion

3.1. Electrical characteristics of the EBL processing

The electrical characteristics of discharge voltage and current waveforms under different electrically driven methods during the EBL process are illustrated in Fig. 2. In this work, the positive and the negative half-wave supplied voltage (PV and NV) have been utilized for EBL process generation. Under no-load conditions, the electrodes have been supplied at the input voltage of 150 V_{rms}, resulting in an absolute output voltage across electrodes of approximately 4.48 kV_{peak}.

Regarding Fig. 2, it could be noticed that the positive discharge case (Fig. 2a)) has slightly higher pulse amplitudes, while the negative discharge case (Fig. 2b)) has slightly lower amplitudes of inception voltage with higher repetition rates of discharge current, called Trichel pulses [24–27]. The absolute peak voltages at the breakdown points of PV case are in the range of 1.40-1.52 kV slightly higher than the ones of the NV case, which are 1.32-1.37 kV. The rms of discharge voltage and currents are 477.10 V and 41.30 mA, and 331.1 V and 49.6 mA for PV and NV cases, respectively. The detailed electrical characteristics of EBL process can be found in Ref. [21].

3.2. Characteristics of the optical emission spectrum during EBL processing

During the EBL process, optical emission spectra (OES) have been examined to gain insights into the chemical reactions occurring. Fig. 3 illustrates the comparative OES plots during the EBL process of the Kratom solution under positive and negative half-wave supplied voltages (PV and NV). Both conditions exhibit similar OES patterns but with varying magnitudes. Consistent observations from both conditions include the presence of highly reactive OH• species at 309.6 nm. Additionally, atomic oxygen emissions are detected at wavelengths of 777.4 nm and 844 nm. The N_2 second positive system (SPN) is evident in the range from 310 nm to 400 nm. Notably, spectral lines for H_β, atomic nitrogen (N), and Hα appear at 486.13, 589.72, and 656.28 nm, respectively. In the realm of Advanced Oxidation Processes (AOPs), these radical species play a pivotal role as potent agents that initiate oxidation. Their involvement extends beyond mere reactivity, actively contributing to the creation of beneficial synergistic reactive oxygen and nitrogen species (RONSs) [28–34]. The chemical reactions observed during the EBL process, as related to the observed peaks, are listed in Table 2 [35–37]. Regarding Fig. 3, it can be noticed that the magnitude of the broad spectrum is more significant in the PV condition. This broad spectrum is attributed to black-body radiation resulting from joule heating during spark discharge, potentially originating from molecular bands such as N_2 bands commonly observed in air discharges [38–41].

3.3. Characteristics of treated Kratom solution and 7-hydroxymitragynine conversion after EBL processing

In this study, the physical-chemical characteristics of both treated DI water and Kratom solution have been examined during the 0–20 min duration of the EBL process, as depicted in Fig. 4. Notably, as the treatment time of the EBL process has increased, the electrical conductivity (EC), hydrogen peroxide (H_2O_2), and dissolved ozone (O_3) concentrations have increased, while the pH (Fig. 4a)) has exhibited the opposite trend for both EBL processes generated by positive and negative half-wave supplied voltages (PV-

Table 2
Chemical reactions during EBL process [35–37].

Reactive species	Chemical reactions
Hydrogen ion (H^+), Hydrogen radical (H•), Hydroxyl radical (OH•), and Hydroxide (OH^-)	$H_2O + e^- \rightarrow OH\bullet + H\bullet + e^-$
	$H_2O + e^- \rightarrow H+ + OH\bullet + 2e^-$
	$H_2O + e^- \rightarrow H\bullet + O\bullet + H\bullet + e^-$
	$H_2O + M^* \rightarrow OH\bullet + H\bullet + M$
	$H_2O + UV \rightarrow H_2O^*$
	$H_2O^* + UV \rightarrow H^+ + OH^-$
	$OH^- \rightarrow OH\bullet + e^-$
	$H_2O_2 + h\nu \rightarrow OH\bullet + OH\bullet$
Atomic oxygen (O), Oxygen ion (O^+), Singlet oxygen ($^1O_2$), and Superoxide ($O_2{}^{*-}$)	$O_2 + e^- \rightarrow O^+ + O + 2e^-$
	$O_2 + e^- \rightarrow O^- + O$
	$O_2 + M^* \rightarrow 2O + M$
	$O_2 + H\bullet \rightarrow OH\bullet + O$
	$O_2 + e^- \rightarrow {}^1O_2$
	$O + O \rightarrow {}^1O_2$
	$O_2 + e^- \rightarrow O_2{}^{*-}$
	$OH\bullet + HO_2 \rightarrow O_2{}^{*-} + H_2O$
Hydrogen peroxide (H_2O_2)	$OH\bullet + OH\bullet \rightarrow H_2O_2$
Ozone (O_3)	$O + O_2 \rightarrow O_3$
Excited Nitrogen (N)	$N_2 + e^- \rightarrow N_2^* + e^-$
	$N_2 + e^- \rightarrow N + N$
	$N_2 + O_2{}^{*-} \rightarrow O_2 + N + N$
Nitric oxide (NO)	$N + O_2 \rightarrow NO + O$
	$N_2 + O \rightarrow NO + N$
	$NO_2 + O \rightarrow NO + O_2$
	$NO_2 + h\nu \rightarrow NO + O\bullet$
	$NO_3 + h\nu \rightarrow NO + O_2$
Nitrate (NO_2) and Nitrous acid (HNO_2)	$NO + O_3 \rightarrow NO_2 + O_2$
	$2NO + O_2 \rightarrow 2NO_2$
	$NO + OH\bullet \rightarrow HNO_2$
	$HNO_2 + OH\bullet \rightarrow NO_2 + H_2O$
Nitrite (NO_3), Nitric acid (HNO_3), Peroxynitrate ($ONOO^-$), and Peroxynitrous acid (ONOOH)	$NO_2 + O\bullet \rightarrow NO_3$
	$NO_2 + OH\bullet \rightarrow HNO_3$
	$2NO_2 + H_2O \rightarrow NO_2^- + NO_3^- + 2H^+$
	$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$
	$NO_2^- + O_3 \rightarrow NO_3^- + O_2$
	$NO_2^- + H_2O_2 + H^+ \rightarrow ONOOH + H_2O$
	$NO_2 + OH\bullet \rightarrow ONOOH \rightarrow ONOO^- + H^+$
	$ONOOH \rightarrow NO_3^- + H^+$
	$O_2{}^{*-} + NO \rightarrow ONOO^-$

Where M and M* are neutral gas molecule and its excited state.

and NV- EBL).

The significant increase in EC of the plasma-treated Kratom solution illustrated in Fig. 4b) can be attributed to the abundant active ions generated during the EBL process [35,42–46]. Initially, the EC values of DI water and Kratom solution have been approximately 0.81 ± 0.02 and 716.47 ± 7.05 μS/cm, respectively. It is observed that the EC values of the PV-EBL case for both DI water and Kratom solution are slightly higher than those of the NV-EBL case. After 20 min of the EBL process, the EC values of the PV-EBL case for DI water and Kratom solution have increased by 342.18 and 1.40 times their initial values, respectively, and have been 1.17 and 1.13 times higher compared to the ones of the NV-EBL case after the same duration.

During and after the EBL process, various compounds, including H_2O_2, nitric acid (HNO_3), peroxynitrous acid (ONOOH), and others, have been formed, resulting in a decrease in pH levels. While the initial pH of DI water has been around 6.86 ± 0.07, when mixed with Kratom powder, it has decreased to 4.80 ± 0.24, indicating the slight acidity of the Kratom powder due to phenolic acids [47]. Subsequently, as shown in Fig. 2b), the pH of all the cases has decreased to around 3.18–4.25. The lowest pH has been observed in the Kratom solution subjected to PV-EBL, at 3.18 ± 0.19.

Concerning the concentration of dissolved H_2O_2 in the plasma-treated solution, it has been observed that the H_2O_2 concentration has increased over time, nearly reaching 100 mg/L, the upper limit of the strip, in the case of Kratom solution subjected to both PV- and NV-EBL processing, as illustrated in Fig. 4c). The H_2O_2 concentration in plasma-treated Kratom solutions has been higher than that in plasma-treated DI water. Notably, in the PV-EBL cases, the H_2O_2 concentration has been higher than that in the NV-EBL cases.

Fig. 4d) illustrates the concentration of dissolved O_3 in the solution. It can be noticed that the O_3 concentration in both cases of DI water has increased rapidly, exceeding 10 mg/L (the limit of the testing kit) within the first 10 min after the EBL process. Conversely, for the Kratom solution subjected to PV- and NV-EBL processing, the O_3 concentration has gradually increased to around 7 and 2.7 mg/L, respectively.

It is important to note that the assessment of H_2O_2 and O_3 concentrations has relied on a color-based testing method, with human observers providing their judgments. When values have not aligned precisely with the scale indicated on the test kit label, they have

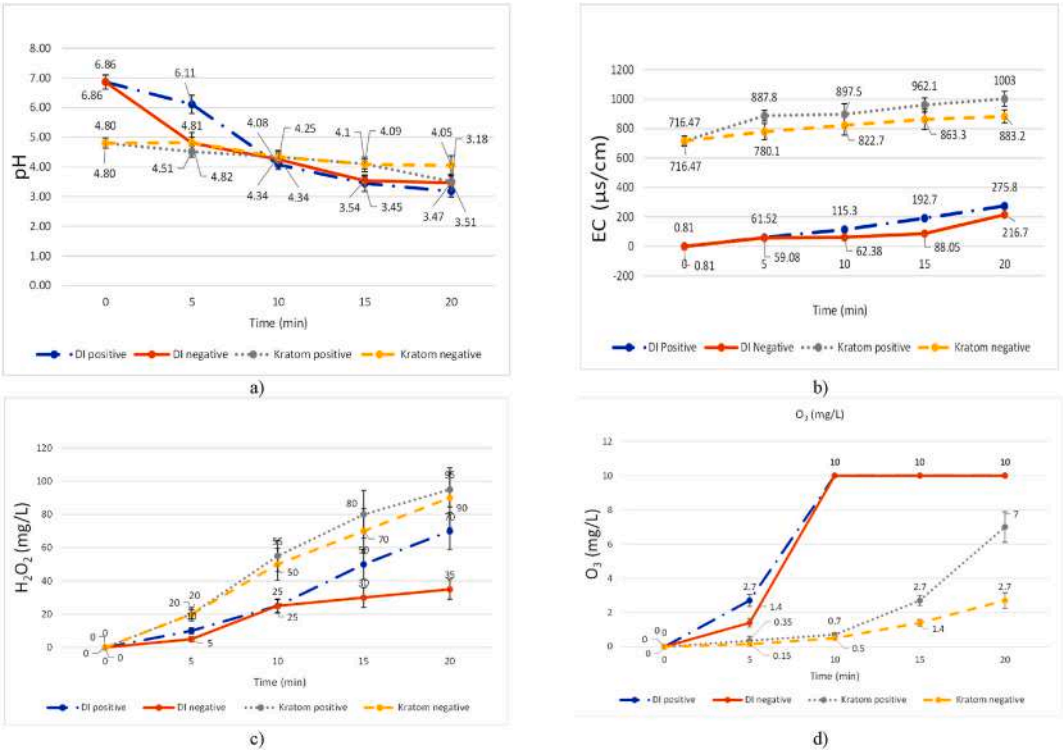


Fig. 4. Kratom solution characteristics under different supplied voltage polarities (positive and negative half-wave voltage), depicted as: a) Electrical Conductivity (EC), b) pH, c) Hydrogen Peroxide (H_2O_2), and d) Ozone (O_3).

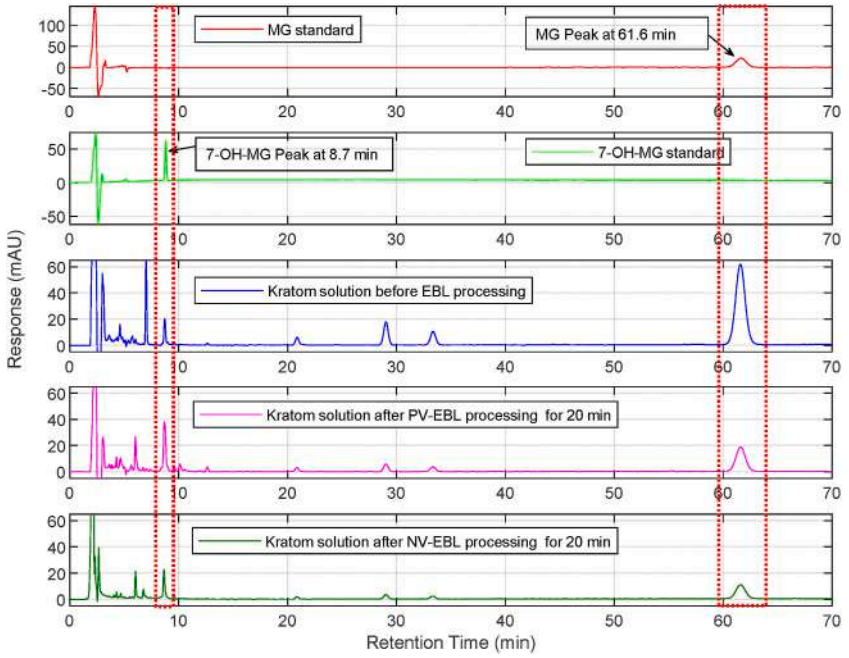


Fig. 5. Chromatograms of MG standard, 7-OH-MG standard, Kratom solution before EBL processing, and Kratom solution after PV- and NV-EBL processing for 20 min, obtained by HPLC-DAD at 246 nm.

been considered indicative of concentrations approximated by comparing the observed color with the reference color range, employing a color interpolation approach.

3.4. Validation of HPLC method for the determination of 7-hydroxymitragynine and mitragynine in Kratom solutions

This study has focused on validating a high-performance liquid chromatography (HPLC) methodology for quantifying 7-Hydroxymitragynine (7-OH-MG) and Mitragynine (MG) in Kratom solutions. The developed HPLC system has demonstrated exceptional specificity in detecting 7-OH-MG and MG, with mean retention times of 8.7 and 61.6 min, respectively, as depicted in Fig. 5. All the other analytical parameters have met the established standards.

The analytical range for 7-OH-MG has ranged from 5 to 30 μg/mL, exhibiting high linearity (r^2 = 0.9998), a limit of detection (LOD) of 0.1 μg/mL, and a limit of quantification (LOQ) of 0.5 μg/mL. Similarly, for MG, the analytical range has been 10–60 μg/mL, with notable linearity (r^2 = 0.9999), an LOD of 0.2 μg/mL, and an LOQ of 0.5 μg/mL. Method precision has been rigorously evaluated, with both repeatability (intra-day) and intermediate precision (inter-day) showing relative standard deviation (RSD) values consistently below 2 % (ranging from 0.96 % to 1.82 %), indicating exceptional precision.

Accuracy has been assessed by analyzing the recovery of spiked 7-OH-MG and MG in Kratom solutions and comparing the measured values with the actual concentrations. The method has exhibited robust recovery rates ranging from 97.24 % to 102.86 %. Thus, the HPLC method has demonstrated reliability in quantifying 7-OH-MG and MG in Kratom solutions, making it suitable for analytical applications in this context.

3.5. Mitragynine conversion and 7-hydroxymitragynine production after EBL process

In the context of validating the method for analyzing Kratom solutions, the validation process has involved examining the contents of both 7-OH-MG and MG before and after the EBL process, as illustrated in Fig. 5. As indicated in Table 1, the initial concentrations of MG and 7-OH-MG have been quantified at 36.19 ± 2.24 μg/mL and 9.03 ± 0.43 μg/mL, respectively.

In order to evaluate the production of 7-OH-MG, the post-plasma treatment concentration has been subtracted from the initial concentration. Then, productivity has been determined by calculating the ratio of 7-OH-MG production to the initial MG concentration, which has been multiplied by 100. Notably, the highest 7-OH-MG productivity, reaching 43.70 ± 2.36 %, has been achieved after subjecting the Kratom solution to a 20-min plasma treatment. Additionally, the highest concentration of 7-OH-MG quantified has been 2,485 ± 134 μg/g of dried Kratom leaves weight. Further experimental results, including the relationship between MG and 7-OH-MG concentrations at different EBL processing times, are presented in Table 1 and Fig. 6, respectively. It is important to note that comprehensive statistical analyses, including *t*-test and one-way ANOVA with a significance level set at $p < 0.05$, have revealed significant differences across all dimensions, including treatment times and supplied voltage polarities, indicating a comprehensive differentiation in the data.

As depicted in Fig. 6, the MG concentration in both Kratom solutions subjected to EBL processing generated by positive half-wave and negative half-wave supplied voltage (PV- and NV- EBL) cases exhibits a decreasing trend with increasing concentrations of 7-OH-MG. However, after 40 min of EBL processing, the MG concentration has vanished in both cases, while the concentration of 7-OH-MG has decreased at a slower rate compared to its conversion to MG. It is important to note that due to operational limitations associated with the power supply duration, the EBL process has been originally constrained to a maximum duration of 20 min. Consequently, efforts have

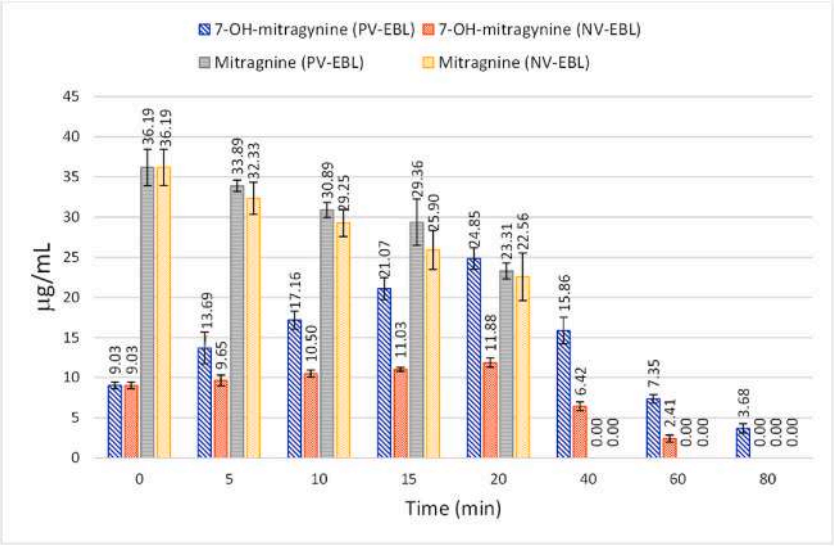


Fig. 6. Concentrations of MG and 7-OH-MG at different EBL processing times under "PV-EBL" and "NV-EBL" conditions, representing positive and negative half-wave supplied voltages, respectively.

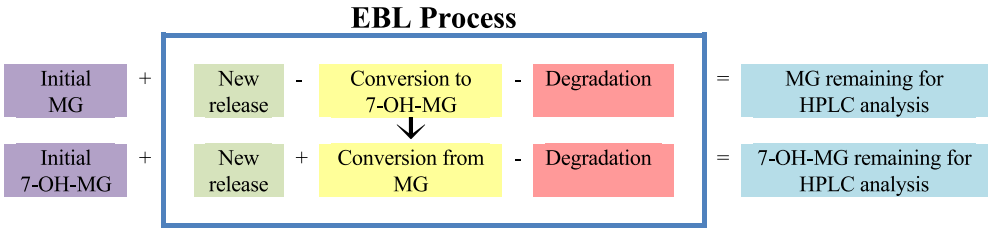


Fig. 7. Impact of the EBL process on MG and 7-OH-MG contents in the sample solution.

been made to extend the operational time limit of the power supply in subsequent experiments, encompassing treatment durations of 5, 10, 15, and 20 min. As a result, the examination of MG and 7-OH-MG contents at prolonged operational intervals of 40, 60, and 80 min has been undertaken, by using the 20-min EBL-treated solution for the second, the third, and the fourth iterations of the EBL process, respectively. However, it can be noticed that the conversion of 7-OH-MG has been significantly suppressed and ultimately ceased, as demonstrated in Fig. 6. The underlying causes of this phenomenon remain a topic of ongoing research. Nonetheless, it is important to note that this specific observation holds potential implications for future studies in this field.

Regarding the presence of both 7-OH-MG and MG contents in the Kratom solution before and after the EBL process, as well as the data provided in Table 1, Fig. 7 offers insight into this phenomenon. In the sample solution, the MG content comprises the fraction initially dissolved in water before entering the EBL process, the amount subsequently released during the EBL process, and a portion transformed into 7-OH-MG and related compounds. Similarly, the 7-OH-MG content in the sample solution includes the 7-OH-MG initially dissolved in water before the EBL process, the quantity liberated during the EBL process, the portion that increases due to the transformation of MG, and a specific amount lost as it converts into other substances.

3.6. Influence of EBL process on mitragynine to 7-hydroxymitragynine conversion

In this research, the Electrical Breakdown in Liquid-phase (EBL) process has been employed to convert mitragynine (MG) into 7-hydroxymitragynine (7-OH MG). The breakdown mechanism within liquid electrical discharges is a complex process involving various phases and the generation of multiple reactive species. The intricacy is attributed to the unique characteristics of liquid media and it is further influenced by factors like electrode geometry, voltage properties, and liquid purity. This complexity characterizes the breakdown in liquids and is exemplified by the gas bubble theory and in-bubble contacted discharge in the EBL process. The cycle of discharge in the EBL system can be briefly elucidated as follows. The pre-breakdown phase is initiated by the corona streamer discharge, which generates small gas bubbles through the heating of highly stressed electrode tips in the liquid. Over time, the streamer evolves into a plasma column, triggering a spark streamer discharge when an air bubble, released from the gas inlet at the lowest section of the reactor, approaches closely. The discharge within the air bubbles expands due to the non-uniform distribution of the electric field. The cycle concludes as the air bubble undergoes a change in shape and traverses the electrode gap. This sequence initiates a new cycle in the process [21,42,48–51].

Generally, the conversion of MG to 7-OH MG likely involves several key factors. Firstly, it necessitates the provision of acidic conditions. Furthermore, the transformation of MG into 7-OH MG involves an oxidation reaction catalyzed by an enzyme. This catalysis process [4,52] is closely associated with the generation of plasma in the presence of air, giving rise to protons, electrons, ions, and plenty of free radicals, most notably oxygen free radicals, nitrogen free radicals, as well as reactive oxygen species (ROSs), reactive nitrogen species (RNSs), and reactive oxygen and nitrogen species (RONS). These components play a pivotal role, functioning as precursors and potent oxidizing agents.

This intricate chemical interplay leads to the creation of valuable byproducts integral to the Advanced Oxidation Processes (AOPs). These byproducts include hydrogen ions (H^+), hydroxyl radicals (·OH), singlet oxygen (·O), ozone (O_3), and hydrogen peroxide (H_2O_2) [17–19]. Some of the essential chemical reactions during the EBL process can be seen in Table 2. Hence, the efficiency of plasma under atmospheric conditions for alkaloid extraction and the conversion of MG to 7-OH MG in Kratom leaves is rooted in a conceptual framework. The concept posits that plasma stimulation of Kratom leaves facilitates the dispersion of protons and free radicals generated during plasma formation throughout the tissue surface and into the fiber of the Kratom powder. These protons make the solution acidic, causing the ionization of alkaloids and rendering them soluble in water because the dissociation constant (pKa) of MG conjugate acid is approximately 8.11 [53]. Simultaneously, the generated free radicals interact with MG components, triggering oxidation reactions that result in the conversion of MG derivatives into 7-OH MG [4,6,17–19,54–56].

By analyzing the experimental results depicted in Fig. 6, a compelling trend has emerged within the EBL processing generated by a positive half-wave supplied voltage case. It can be noticed that as the experimental duration extended, there has been a notable decline in the MG substance, coinciding with a concurrent increase in the abundance of 7-OH-MG. Notably, extending the EBL processing time to 20 min has facilitated the most enhanced modifications in MG. A comparative evaluation of 7-OH-MG production from the EBL processing generated by the positive half-wave and the negative half-wave supplied voltage (PV- and NV-EBL) cases unveiled a distinct contrast in their effectiveness in augmenting substances. Remarkably, in the PV-EBL case, the enhancement has been 2.09 times greater than the NV-EBL case after a 20-min EBL process. These findings underscore the consistent superiority of the PV-EBL case over its negative counterpart. The disparity in performance is likely attributed to the reduced production of ozone-free radicals in the

NV-EBL case as shown in Fig. 4d), signifying a less pronounced impact of plasma in this context.

Concerning the experimental findings depicted in Fig. 4d), it can be noticed that the O_3 concentration in plasma-treated Kratom solutions is notably lower when compared to plasma-treated DI water, in both PV- and NV-EBL cases. This observation aligns consistently with the conversion rate of MG to 7-OH MG in the PV case, as opposed to the NV-EBL case. Consequently, it is reasonable to infer that O_3 plays a significant role in the conversion of MG to 7-OH MG. Additionally, ultraviolet rays, shock waves, and photons produced during plasma generation may further contribute to the synergistic stimulation of chemical reactions within Kratom leaves [16,28–34,57].

The hypothetical mechanism of converting mitragynine (MG) into 7-hydroxymitragynine (7-OH-MG) through the EBL process in this study can be elucidated as depicted in Fig. 8. The environment created by the EBL process, including the presence of reactive oxygen species, acidity, and heat, plays a pivotal role in this transformation mechanism. This process consists of three essential steps, as follows. (1) Formation of epoxide: MG, an intrinsic indole alkaloid, exhibits the oxidation potential when exposed to reactive oxygen species. The theoretical framework for this indole oxidation mechanism begins with the electrophilic addition of reactive oxygen species to the carbon-carbon double bond, giving rise to an intermediate epoxide of considerable instability [58,59]. (2) Formation of alcohol from epoxide: in an acidic environment, the epoxide accepts a proton, making the carbon at position 2 more electrophilic. Furthermore, the carbon at position 2 is also surrounded by highly electronegative atoms, such as oxygen and nitrogen, which further increase its electrophilic nature. This electron deficiency renders it receptive to nucleophilic substitution reactions, primarily initiated by the presence of the hydroxyl group (OH). The leaving group is the oxygen atom of the epoxide in the form of the alkoxide, which is converted to alcohol during an acidic workup. As a consequence of these reactions, the cascade leads to the formation of 2,7-dihydroxymitragynine [60,61]. (3) In an acidic condition, and with heat generated by plasma stimulation, water loss occurs at the tertiary alcohol at position 2 of the 2,7-dihydroxymitragynine molecule, resulting in the final product, 7-hydroxymitragynine [62,63].

The research findings presented herein not only contribute to a deeper scientific understanding of alkaloid conversion within Kratom leaves but also hold promise for practical applications in healthcare, pharmaceuticals, and sustainable technologies. The EBL process emerges as a novel technique for 7-OH-MG conversion, demonstrating considerable potential for further exploration. However, it is important to acknowledge areas for improvement. Initially, the EBL process has been limited to a maximum duration of 20 min due to constraints related to power supply duration. Attempts to extend this duration have resulted in a notable suppression of 7-OH-MG conversion, warranting ongoing investigation into the underlying causes. Nevertheless, this observation underscores the need for careful consideration in future research endeavors within this domain. This study represents a significant step forward in unlocking the potential of natural resources to advance medical science while promoting environmental sustainability.

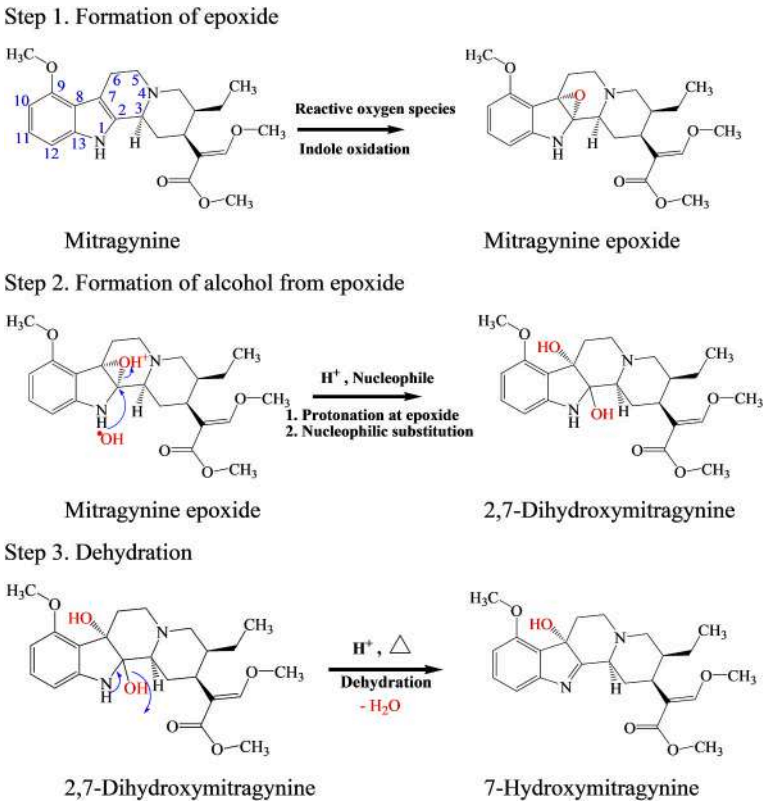


Fig. 8. Proposed mechanism for the conversion of mitragynine to 7-hydroxymitragynine via the EBL process.

4. Conclusion

This research explores the impact of the Electrical Breakdown in Liquid-phase (EBL) process on the conversion of mitragynine (MG) in Kratom leaves to 7-hydroxy mitragynine (7-OH-MG) through advanced oxidation processes (AOPs). In order to enhance plasma exposure to Kratom leaf powdered solutions during EBL, a novel reactor is utilized, incorporating both positive and negative half-wave supplied voltages (PV and NV). The findings indicate that the PV-EBL case outperforms the NV-EBL case in transforming MG into 7-OH-MG, with the highest yield achieved after 20 min of EBL due to its greater production of ozone and free radicals. Ozone, ROSs, RNSs, RONSs, and other plasma-induced factors significantly influence this conversion. These results lay the groundwork for the advancement of methodologies for the production of 7-OH-MG from Kratom leaves, offering potential applications in pharmacology and medical research. The proposed mechanism for this conversion involves epoxide formation, succeeded by the creation of alcohol from the epoxide, ultimately resulting in the production of 7-OH-MG, underscoring the role of plasma-induced factors in this process.

Data availability

Data included in article/supp. material/referenced in article.

CRediT authorship contribution statement

Weerasak Samee: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Khanit Matra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Nattawut Lakkham:** Investigation. **Beelawan Dongkaew:** Investigation. **Passakorn Sumkhum:** Investigation. **Wutthichok Sangwang:** Visualization, Investigation. **Wasin Nupangtha:** Validation, Investigation. **Jiraporn Promping:** Validation, Investigation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT [chat.openai.com] in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Supporting Information

Studies on the Synthesis and Opioid Agonistic Activities of Mitragynine-Related Indole Alkaloids: Discovery of Opioid Agonists Structurally Different from Other Opioid Ligands

Hiromitsu Takayama,*[1] Hayato Ishikawa,[1] Mika Kurihara,[1] Mariko Kitajima,[1] Norio Aimi,[1] Dhavadee Ponglux,[2] Fumi Koyama,[3] Kenjiro Matsumoto,[3] Tomoyuki Moriyama,[3] Leonard T. Yamamoto,[3] Kazuo Watanabe,[3] Toshihiko Murayama,[3] Syunji Horie*[3]

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Experimental

Preparation of 9-ethoxycorynantheidine (6). To a stirred solution of **5** (32 mg, 0.083 mmol) in benzene (3 mL) was successively added aqueous 15% NaOH solution (73 μL, 0.249 mmol), tetrabutylammonium hydrogensulfate (8.4 mg, 0.025 mmol), and then EtI (20 μL, 0.166 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 1 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (*n*-hexane/AcOEt 7:3) to give **6** (27 mg, 80%) as an amorphous powder. UV (MeOH) λ_{max} : 293, 227 nm. IR ν_{max} ($CHCl_3$) : 1702, 1628 cm^{-1}. ^{1}H-NMR ($CDCl_3$) δ : 7.69 (1H, br.s, *Na*-H), 7.43 (1H, s, H-17), 6.96 (1H, dd, *J*=8.0, 8.0, H-11), 6.86 (1H, d, *J*=7.3, H-12), 6.43 (1H, d, *J*=7.5, H-10), 4.10 (1H, m, 9-O$\underline{CH_2}CH_3$), 3.75 and 3.70 (each 3H, each s, 17-OCH_3 and 22-OCH_3), 3.15 (1H, br.d, *J*=11.2, H-3), 3.14 (1H, m, H-6), 3.05 (1H, m, H-15), 3.00 (1H, m, H-21), 2.95 (1H, m, H-6), 2.93

(1H, dd, *J*=9.3, 3.7, H-5), 2.55 (2H, each m, H-5 and H-14), 2.45 (1H, m, H-21), 1.80 (1H, m, H-14), 1.75 (1H, m, H-19), 1.63 (1H, m, H-20), 1.41 (1H, m, H-19), 1.40 (3H, dd, *J*=7.1, 7.1, 9-OCH$_2$<u>CH$_3$</u>), 0.86 (3H, dd, *J*=7.4, 7.4, H-18). ^{13}C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5(C-17), 153.0 (C-9), 137.0 (C-13), 133.4 (C-2), 121.2 (C-11), 117.1 (C-7), 111.5 (C-16), 107.9 (C-8), 107.2 (C-12), 99.7 (C-10), 63.4 (9-O<u>CH$_2$</u>CH$_3$), 61.5 (17-O<u>CH$_3$</u>), 61.2 (C-3), 57.7 (C-21), 53.7 (C-5), 51.4 (22-OCH$_3$), 40.5 (C-20), 40.0 (C-15), 29.9 (C-14), 23.9 (C-6), 19.4 (C-19), 14.9 (9-OCH$_2$<u>CH$_3$</u>), 12.8 (C-18). EI-MS *m/z* (%) : 412 (M^+, 99), 397 (35), 381 (10), 283 (27), 228 (100). HR-FABMS : calcd. for $C_{24}H_{33}O_4N_2$ [MH^+] : 413.2440, found: 413.2432.

Preparation of 9-isopropoxycorynantheidine (7). To a stirred solution of **5** (30 mg, 0.078 mmol) in benzene (3 mL) was successively added aqueous 15% NaOH solution (70 μL, 0.234 mmol), tetrabutylammonium hydrogensulfate (8.4 mg, 0.025 mmol), and then *i*-PrI (20 μL, 0.234 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 8 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (*n*-hexane/AcOEt 7:3) to give **7** (22 mg, 67%) as an amorphous powder. UV (MeOH) λ_{max} : 293, 225 nm. ^{1}H-NMR ($CDCl_3$) δ : 7.64 (1H, br.s, *Na*-H), 7.35 (1H, s, H-17), 6.88 (1H, dd, *J*=8.0, 8.0, H-11), 6.80 (1H, d, *J*=8.1, H-12), 6.38 (1H, d, *J*=7.8, H-10), 4.54 (1H, m, 9-O<u>CH$_2$</u>(CH$_3$)$_2$), 3.65 and 3.63 (each 3H, each s, 17-OCH$_3$ and 22-OCH$_3$), 3.08 (1H, br.d, *J*=11.0, H-3), 3.05 (1H, m, H-15), 3.02 (1H, m, H-6), 2.95 (1H, m, H-6), 2.92 (1H, m, H-21), 2.87 (1H, dd, *J*=11.2, 5.4, H-5), 2.50 (1H, m, H-14), 2.46 (2H, each m, H-21 and H-5), 1.70 (1H, m, H-14), 1.68 (1H, m, H-19), 1.55 (1H, br.d, *J*=13.2, H-20), 1.12 (1H, m, H-19), 1.30 and 1.28 (each 3H, each d, *J*=6.1 and 5.9, 9-OCH$_2$(<u>CH$_3$</u>)$_2$), 0.79 (3H, dd, *J*=7.3, 7.3, H-18). ^{13}C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5 (C-17), 152.4 (C-9), 137.5 (C-13), 133.6 (C-2), 121.8 (C-11), 118.0 (C-7), 111.4 (C-16) , 108.0 (C-8), 103.7 (C-12), 102.3 (C-10), 69.8 (9-O<u>CH$_2$</u>(CH$_3$)$_2$), 61.5 (17-OCH$_3$), 61.2 (C-3), 57.7 (C-21), 53.7 (C-5), 51.3 (22-OCH$_3$), 40.1 (C-20), 39.9(C-15), 29.9 (C-14), 23.9 (C-6), 22.3 and 22.4 (9-OCH$_2$(<u>CH$_3$</u>)$_2$), 19.1 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 426 (M^+, 100), 411 (39), 383 (11), 297 (17), 242 (78). HR-FABMS : calcd. for $C_{25}H_{35}O_4N_2$ [MH^+] : 427.2564, found: 427.2597.

Preparation of (8). To a stirred solution of **5** (53 mg, 0.138 mmol) in CH_2Cl_2 (0.5 mL) was successively added aqueous 15% NaOH solution (110 μL, 0.414 mmol), Adogen 464 [methyltrialkyl(C_8-C_{10})ammonium chloride] (0.01 mL, 0.028 mmol), and

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then methoxymethylchloride (56 μL, 0.552 mmol) at 0 °C under argon atmosphere and the mixture was stirred at the room temperature for 1 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (3% MeOH-$CHCl_3$) to give **8** (42 mg, 71%) as an amorphous powder. UV (MeOH) λ_{max}: 291, 223 nm. ^{1}H-NMR ($CDCl_3$) δ : 7.73 (1H, br.s, *Na*-H), 7.43 (1H, s, H-17), 6.99~6.93 (2H, m, H-11 and H-12), 6.65 (1H, dd, *J*=7.1, 1.2, H-10), 5.48 (1H, d, *J*=6.3, 9-O$\underline{CH_2}$OCH$_3$), 5.27 (1H, d, *J*=6.3, 9-O$\underline{CH_2}$OCH$_3$), 3.78 and 3.72 (each 3H, each s, 17-OCH$_3$ and 22-OCH$_3$), 3.51 (3H, s, 9-OCH$_2$O$\underline{CH_3}$), 3.17 (1H, m, H-3), 3.15 (1H, m, H-6), 3.05 (1H, m, H-21), 3.03 (2H, m, H-6 and H-15), 2.93 (1H, dd, *J*=10.7, 5.1, H-5), 2.57 (1H, m, H-14), 2.53 (1H, m, H-5), 2.45 (1H, dd, *J*=10.5, 2.4, H-21), 1.79 (1H, br.d, *J*=12.0, H-14), 1.73 (1H, m, H-19), 1.63 (1H, m, H-20), 1.21 (1H, m, H-19), 0.86 (3H, dd, *J*=7.3, 7.3, H-18). ^{13}C-NMR ($CDCl_3$) δ : 169.2 (C-22), 160.5 (C-17), 151.5 (C-9), 137.6 (C-13), 134.2 (C-2), 121.8 (C-11), 118.3 (C-7), 111.6 (C-16), 107.6 (C-8), 105.1 (C-12), 103.4 (C-10), 94.5 (9-O$\underline{CH_2}$OCH$_3$), 61.5 (17-OCH$_3$), 61.3 (C-3), 57.8 (C-21), 56.0 (9-OCH$_2$O$\underline{CH_3}$), 53.8 (C-5), 51.3 (22-OCH$_3$), 40.7 (C-20), 39.9 (C-15), 29.9 (C-14), 23.9 (C-6), 19.1 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 428 (M^+, 100), 413 (36), 383 (16), 299 (18), 244 (88). HR-FABMS : calcd. for $C_{24}H_{33}O_5N_2$ [MH^+] : 429.2389, found: 429.2350.

Preparation of 7-methoxy-7*H*-mitragynine (13). A solution of idosobenzene diacetate (24 mg, 0.078 mmol) in dry MeOH (1 mL) was added dropwise at 0 °C to a solution of mitragynine (**1**) (30 mg, 0.078 mmol) in dry MeOH (3 mL) under argon atmosphere. After stirring at room temperature for 1 h, the mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (*n*-hexane/AcOEt 7:3) to give **14** (31 mg, 96%) as an amorphous powder. UV (MeOH) λ_{max}: 320 (sh), 246 (sh), 221 nm. IR ν_{max} ($CHCl_3$) : 2930, 1698, 1600 cm^{-1}. ^{1}H-NMR ($CDCl_3$) δ : 7.44 (1H, s, H-17), 7.32 (1H, dd, *J*=7.9, 7.9, H-11), 7.25 (1H, d, *J*=7.8, H-12), 6.70 (1H, d, *J*=8.3, H-10), 3.87 (3H, s, 9-OCH$_3$), 3.87 (3H, s, 17-OCH$_3$) , 3.70 (3H, s, 22-OCH$_3$), 3.05 (1H, m, H-3), 3.03 (1H, m, H-21), 3.00 (1H, m, H-15), 2.86 (3H, s, 7-OCH$_3$), 2.83 (1H, m, H-5), 2.80 (1H, m, H-14), 2.70 (1H, m, H-6), 2.60 (1H, m, H-5), 2.48 (1H, br-d, *J*=8.8, H-21), 1.87 (1H, m, H-14), 1.70 (1H, m, H-19), 1.60 (2H, m, H-6 and H-20), 1.25 (1H, m, H-19), 0.82 (3H, dd, *J*=7.4, 7.4, H-18). ^{13}C-NMR ($CDCl_3$) δ : 184.1 (C-2), 169.3 (C-22), 160.7 (C-17), 156.6 (C-9), 155.8 (C-13), 131.0 (C-11), 122.7 (C-8), 114.4 (C-12), 111.3 (C-16), 108.8 (C-

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10), 86.7 (C-7), 61.9 (17-OCH_3), 61.8 (C-3), 58.1 (C-21), 55.6 (9-OCH_3), 51.9 (7-OCH_3), 51.3 (22-OCH_3), 50.2 (C-5), 40.6 (C-20), 39.3 (C-15), 35.1 (C-6), 26.0 (C-14), 19.0 (C-19), 12.8 (C-18). EI-MS *m/z* (%) : 428 (M^+, 3), 397 (100), 256 (10). HR-FABMS : calcd. for $C_{24}H_{33}O_5N_2$ [MH^+] : 429.2389, found: 429.2364.

Preparation of 7-ethoxy-7*H*-mitragynine (14). A solution of idosobenzene diacetate (24 mg, 0.078 mmol) in dry EtOH (1 mL) was added dropwise at 0 °C to a solution of mitragynine (**1**) (30 mg, 0.078 mmol) in dry MeOH (0.5 mL) under argon atmosphere. After stirring at room temperature for 1.5 h, the mixture was poured onto the chilled water and was extracted with $CHCl_3$ three times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (*n*-hexane/AcOEt 1:1) to give **15** (18 mg, 53%) as an amorphous powder. UV (MeOH) λ_{max}: 320 (sh), 246 (sh), 221 nm. IR ν_{max} ($CHCl_3$) : 2930, 1698, 1600 cm^{-1}. ^{1}H-NMR ($CDCl_3$) δ : 7.45 (1H, s, H-17), 7.33 (1H, dd, *J*=7.9, 7.9, H-11), 7.25 (1H, d, *J*=7.6, H-12), 6.70 (1H, d, *J*=8.3, H-10), 3.86 (3H, s, 9-OCH_3), 3.81 (3H, s, 17-OCH_3) , 3.70 (3H, s, 22-OCH_3), 3.1~2.8 (2H, m, 7-O$\underline{CH_2}CH_3$), 3.05 (2H, m, H-3 and H-5), 3.00 (1H, m, H-15), 2.98 (1H, m, H-21), 2.79 (1H, m, H-14), 2.70 (1H, br.d, *J*=13.9, H-6), 2.57 (1H, m, H-5), 2.47 (1H, dd, *J*=11.5, 2.9, H-21), 1.87 (1H, br.d, *J*=13.4, H-14), 1.72 (1H, m, H-19), 1.60 (1H, m, H-20), 1.55 (1H, m, H-6), 1.25 (1H, m, H-19), 1.10 (3H, dd, *J*=7.0, 7.0, 7-$OCH_2\underline{CH_3}$), 0.82 (3H, dd, *J*=7.3, 7.3, H-18). ^{13}C-NMR ($CDCl_3$) δ [ppm] : 184.6 (C-2), 169.4 (C-22), 160.7 (C-17), 156.5 (C-9), 155.5 (C-13), 130.7 (C-11), 123.6 (C-8), 114.3 (C-12), 111.3 (C-16), 108.8 (C-10), 86.3 (C-7), 62.1 (17-OCH_3), 61.8 (C-3), 59.9 (7-O$\underline{CH_2}OCH_3$), 58.2 (C-21), 55.5 (9-OCH_3), 51.3 (22-OCH_3), 50.2 (C-5), 40.6 (C-20), 39.4 (C-15), 35.4 (C-6), 26.0 (C-14), 19.0 (C-19), 15.5 (7-$OCH_2O\underline{CH_3}$), 12.8 (C-18). EI-MS *m/z* (%) : 442 (M^+, 18), 397 (100), 365 (8). HR-FABMS : calcd. for $C_{25}H_{35}O_5N_2$ [MH^+] : 443.2552, found: 443.2546.

Preparation of 4-quinolone derivative (15). A mixture of **1** (30 mg, 0.075 mmol) and NaH (50% suspension, 15 mg, 0.38 mmol) in dry DMF (0.5 mL) was stirred at room temperature under air for 15 h. The reaction mixture was poured onto the chilled water and was extracted with $CHCl_3$ five times. The combined organic layer was washed with brine, dried over $MgSO_4$ and evaporated. The residue was separated by SiO_2 column chromatography (5% MeOH-$CHCl_3$) to give **16** (15 mg, 49%) as an amorphous powder. UV (MeOH) λ_{max}: 334, 323, 233, 201 nm. ^{1}H-NMR ($CDCl_3$) δ : 9.69 (1H, s, *Na*-H), 7.67 (1H, d, *J*=8.6, H-12), 7.47 (1H, dd, *J*=8.2, 8.2, H-11), 7.44 (1H, s, H-17), 6.80 (1H, d, *J*=7.9, H-10), 4.26 (1H, d, *J*=12.2, H-5), 4.06 (3H, s, 9-OCH_3),

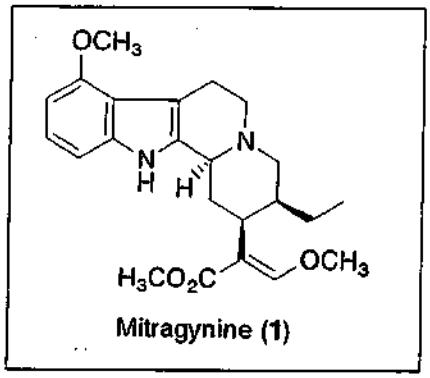
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3.74 (3H, s, 17-OCH_3) , 3.71 (3H, s, 22-OCH_3), 3.54 (1H, d, *J*=12.2, H-5), 3.37 (1H, d, *J*=11.3, H-3), 3.24 (1H, d, *J*=11.0, H-21), 3.02 (1H, d, *J*=12.7, H-15), 2.69 (1H, d, *J*=11.3, H-21), 2.54 (1H, m, H-14), 2.25 (1H, d, *J*=12.5, H-14), 1.73 (1H, m, H-19), 1.72 (1H, m, H-20), 1.25 (1H, m, H-19), 0.88 (3H, dd, *J*=7.3, 7.0, H-18). ^{13}C-NMR ($CDCl_3$) δ : 168.3 (C-7), 169.3 (C-22), 160.7 (C-17), 156.2 (C-9), 155.5 (C-2), 151.3 (C-13), 128.1 (C-11), 122.9 (C-12), 115.3 (C-6), 111.0 (C-8), 103.9 (C-10), 69.4 (C-3), 61.6 (17-OCH_3), 56.4 (9-OCH_3), 53.5 (C-21), 52.9 (C-5), 51.3 (22-OCH_3), 40.5 (C-20), 39.8 (C-15), 27.6 (C-14), 19.5 (C-19), 12.9 (C-18). EI-MS *m/z* (%) : 412 (M^+, 100), 397 (28), 283 (21), 228 (50). HR-FABMS : calcd. for $C_{23}H_{29}O_5N_2$ [MH^+] : 413.2076, found: 413.2089.

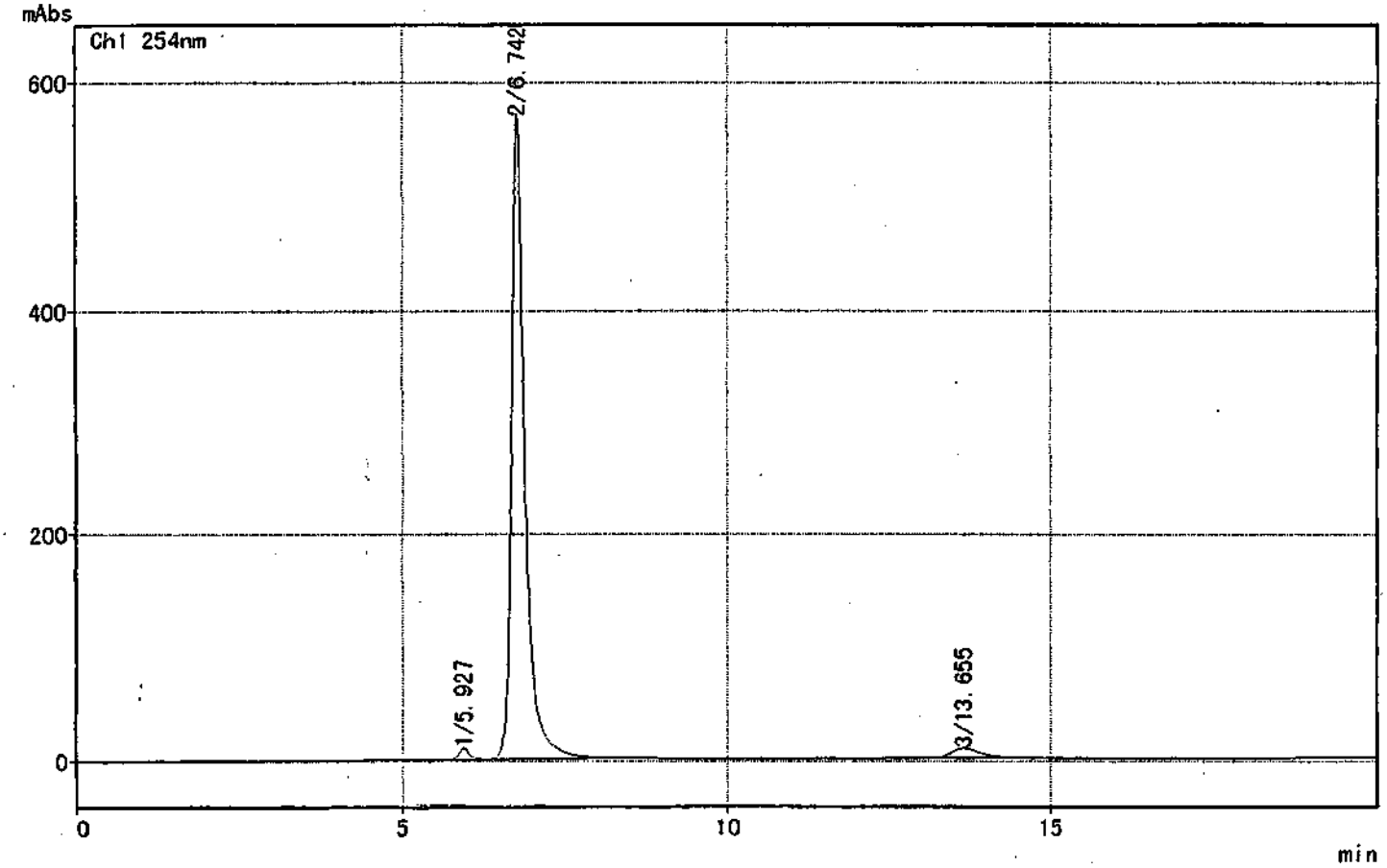
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Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine



Mitragynine (1)



*** ピークレポート ***

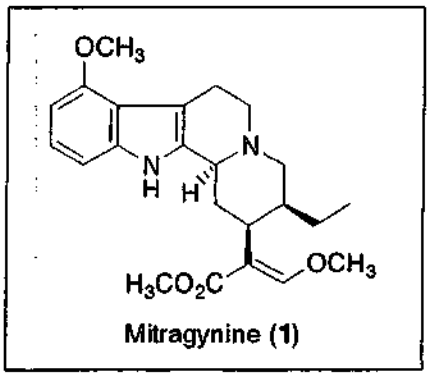
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2	1	8072496	6.742
3	1	252350	13.655
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Purity : 95.7 %

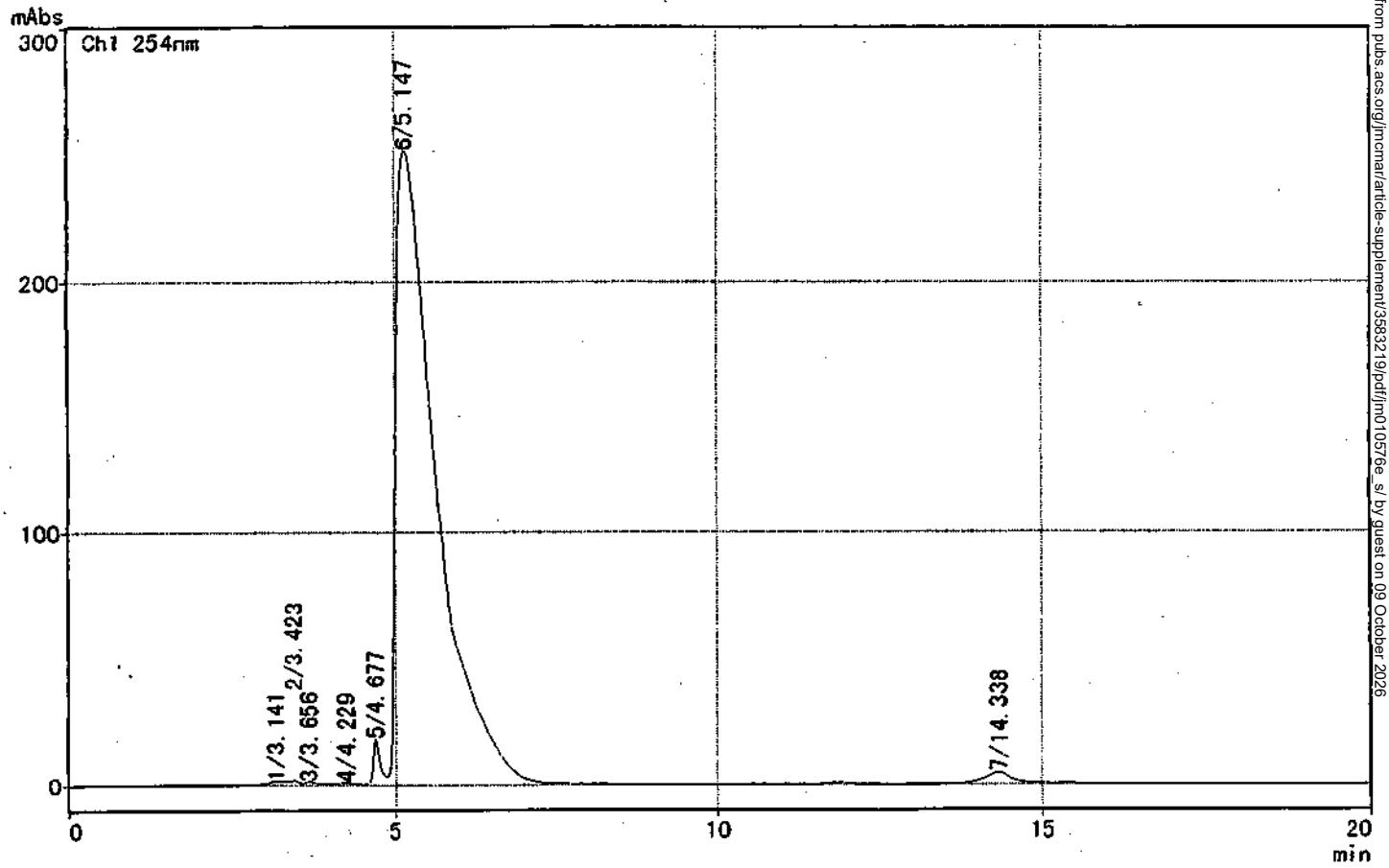
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Condition

Column : YMC-Pack A-012 SIL
(YMC) S-5μm
Column size : 6.0mm x 150mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine



Mitragynine (1)



*** ピークレポート ***

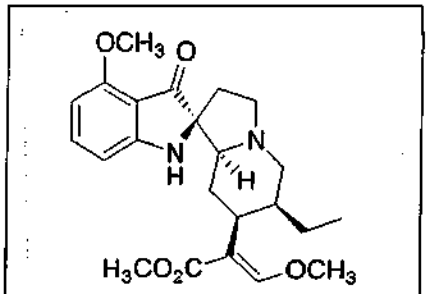
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2	1	15317	3.423
3	1	5572	3.656
4	1	2474	4.229
5	1	135005	4.677
6	1	10916515	5.147
7	1	108241	14.338
		11200057	

Purity : 97.5 %

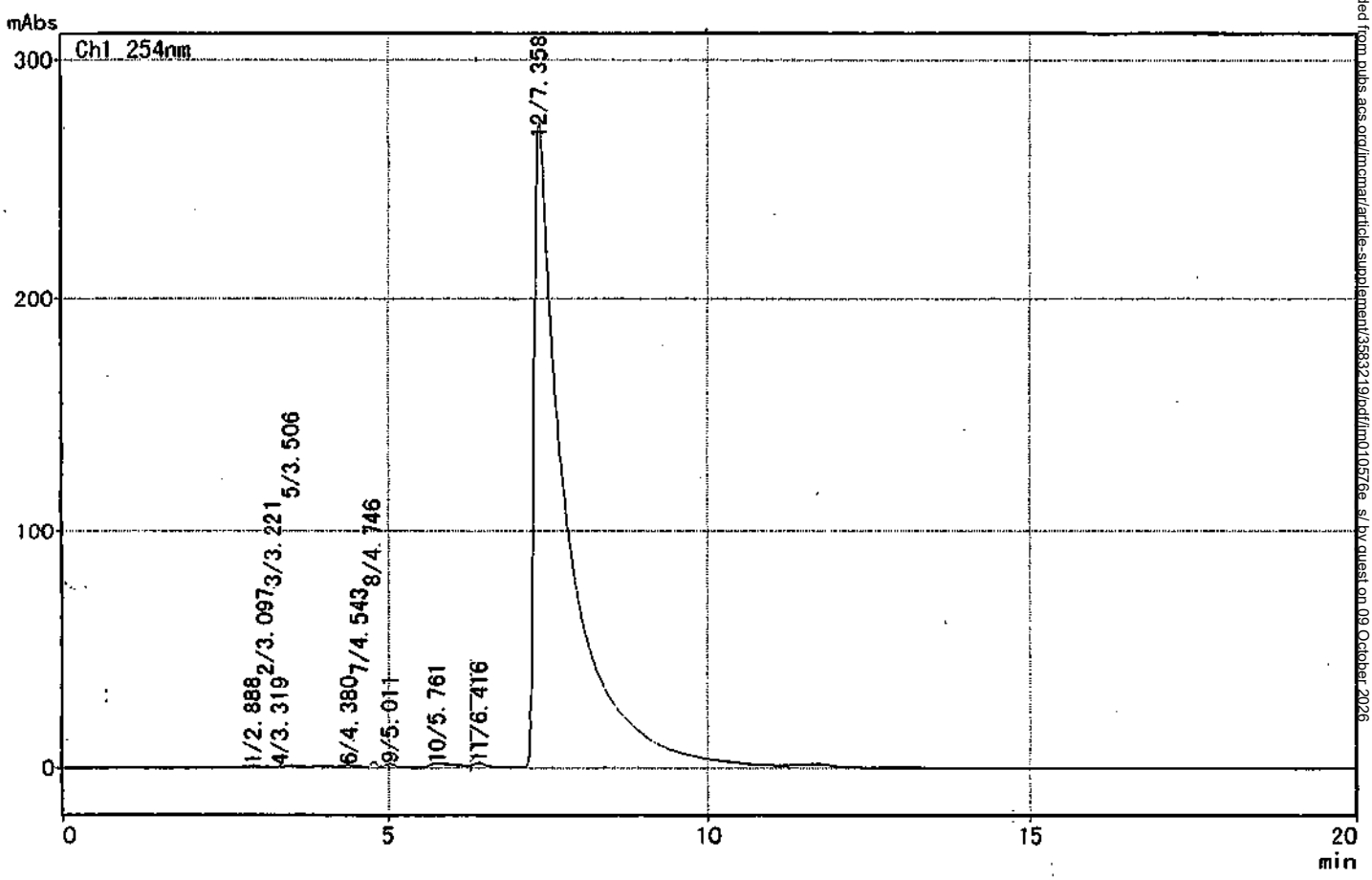
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Condition

Column : PEGASIL Silica 60-5
(Senshu Pak)
Column size : 4.6mm x 250mm
Flow Rate : 1.0mL/min
Eluent : 3% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine pseudoindoxyl



Mitragynine pseudoindoxyl (**2**)



*** ピークレポート ***

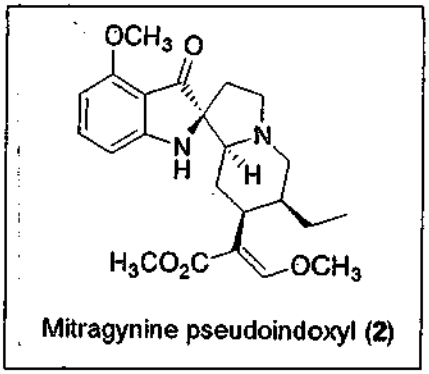
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2	1	2440	3.097
3	1	1309	3.221
4	1	2608	3.319
5	1	6443	3.506
6	1	3654	4.380
7	1	1490	4.543
8	1	13828	4.746
9	1	11755	5.011
10	1	40782	5.761
11	1	22444	6.416
12	1	9530087	7.358
		9641805	

Purity : 98.8 %

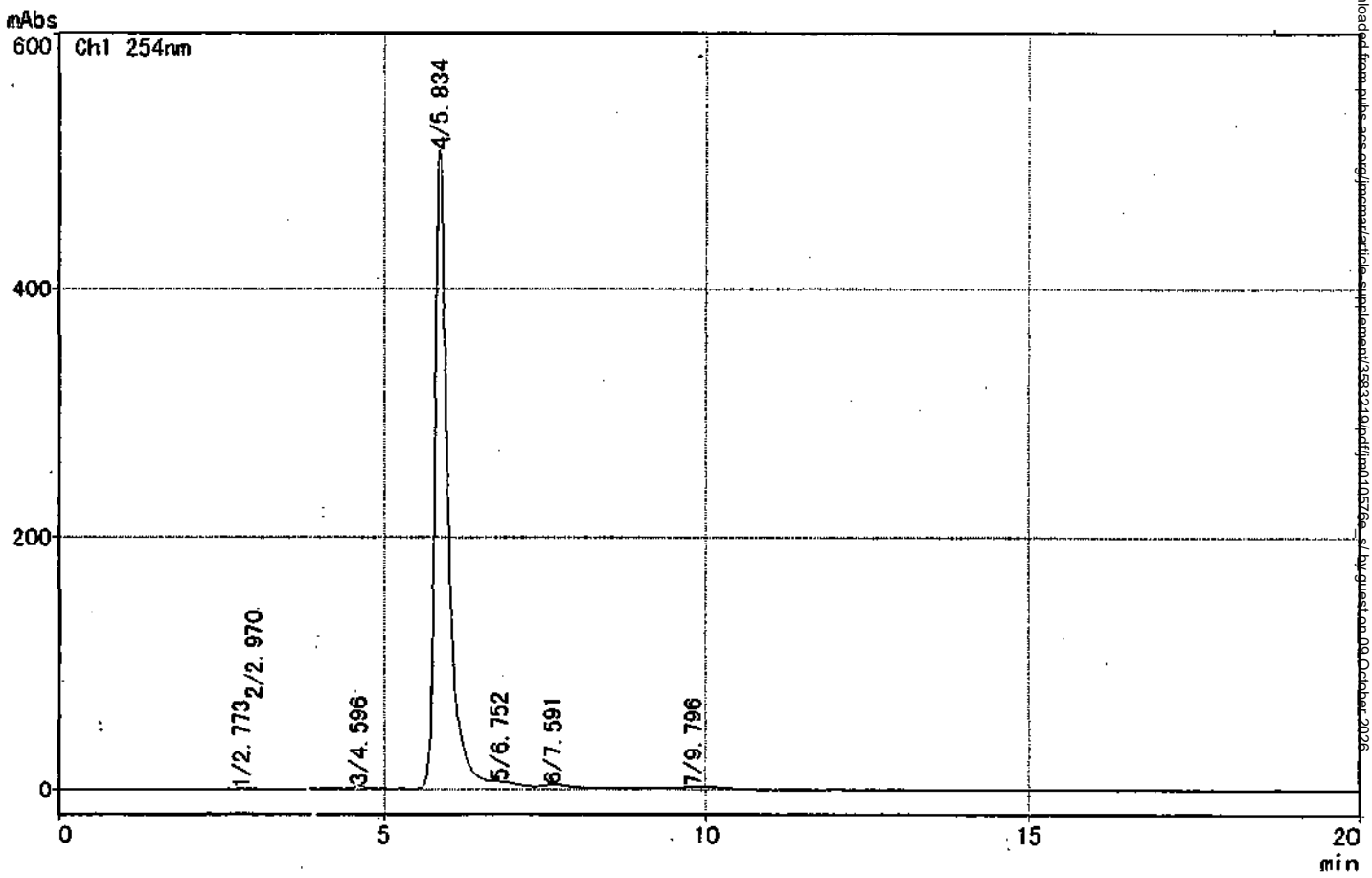
Downloaded from pubs.acs.org/jmcmar/article-supplement/3583219/pdf/jm010576e_s/ by guest on 09 October 2026

Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 60%Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Mitragynine pseudoindoxyl



Mitragynine pseudoindoxyl (**2**)



*** ピークレポート ***

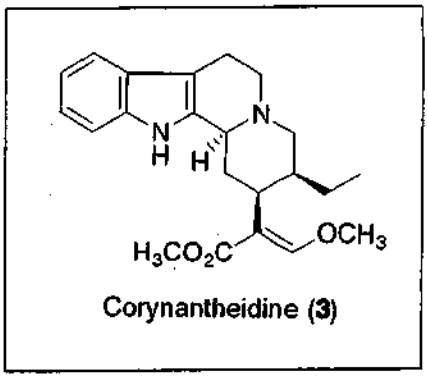
*KNO	ChNO	AREA	TIME
1	1	1308	2.773
2	1	2538	2.970
3	1	28437	4.596
4	1	7168618	5.834
5	1	12388	6.752
6	1	50334	7.591
7	1	56022	9.796
		7319645	

Purity : 97.9 %

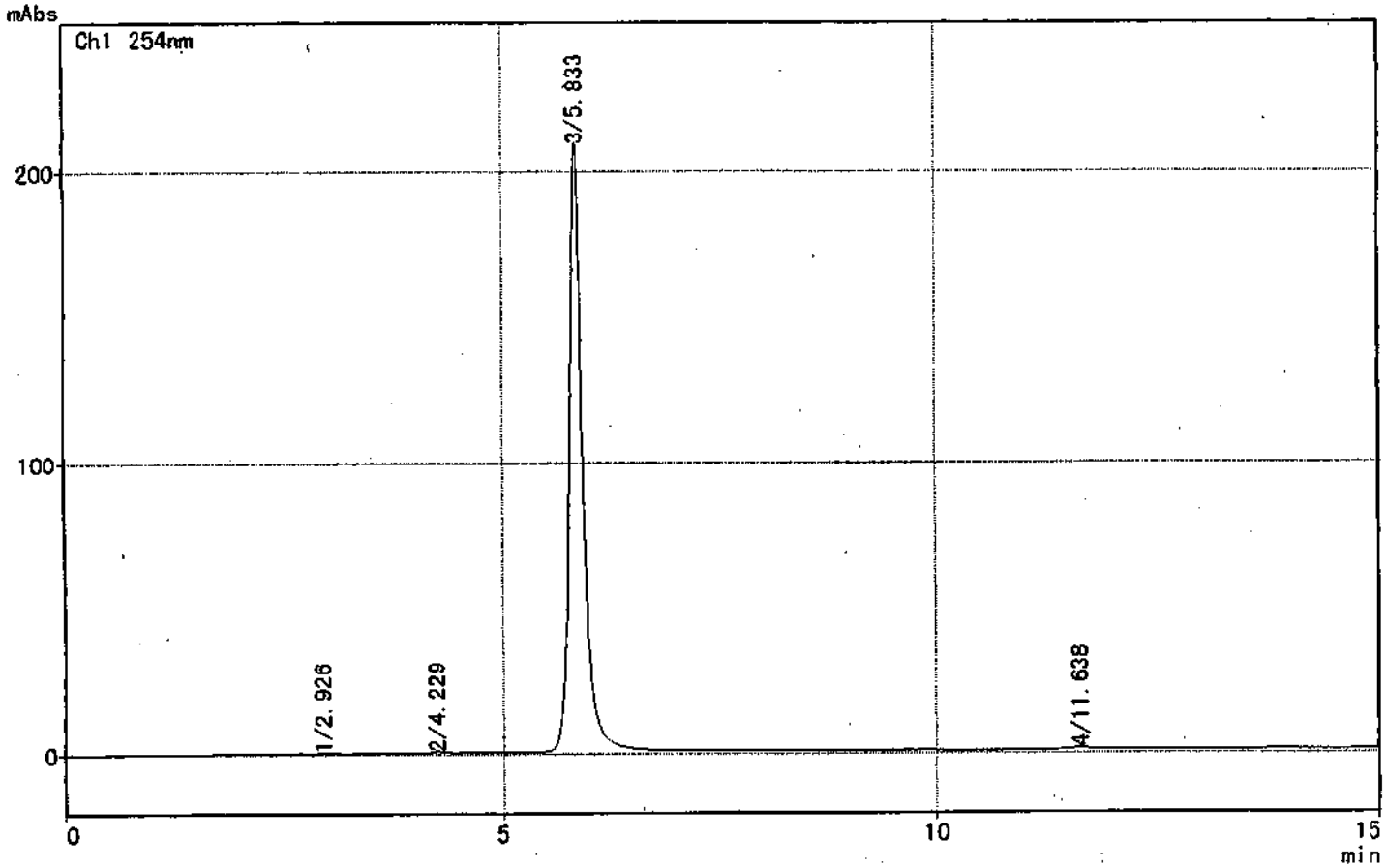
Downloaded from pubs.acs.org/jmcmar/article_supplement/358321g/pdf/jm010576e_s/by guest on 09 October 2026

Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : Corynantheidine



Corynantheidine (**3**)



*** ピークレポート ***

PKNO	ChNO	AREA	TIME
1	1	2801	2.926
2	1	2870	4.229
3	1	2288486	5.833
4	1	12663	11.638
		2306820	

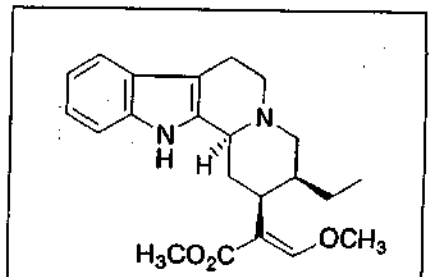
Purity : 99.2 %

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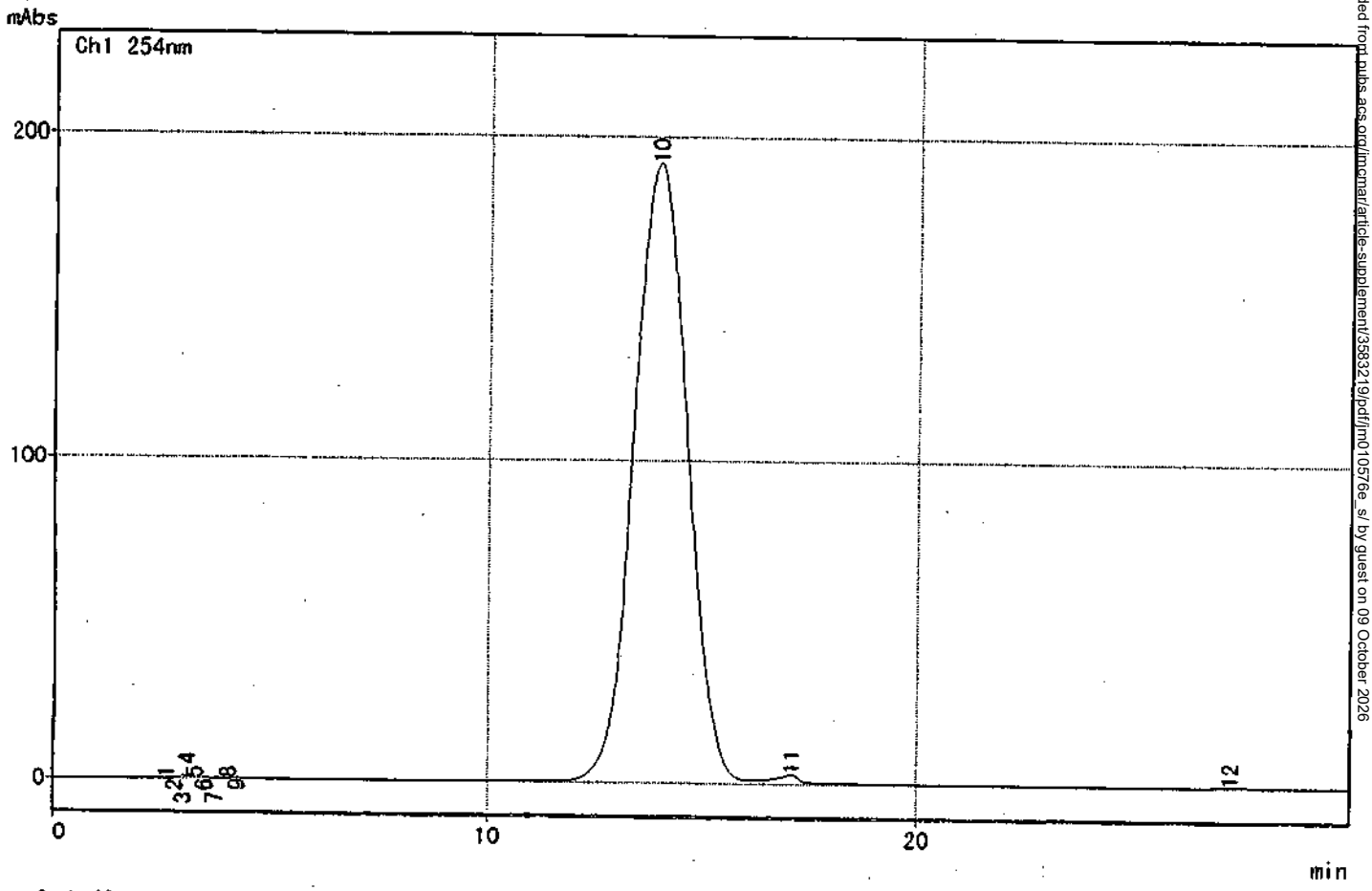
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Condition

Column : PEGASIL Silica 60-5
(Senshu Pak) S-5μm
Column size : 4.6mm x 250mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : Corynantheidine



Corynantheidine (3)



*** ピークレポート ***

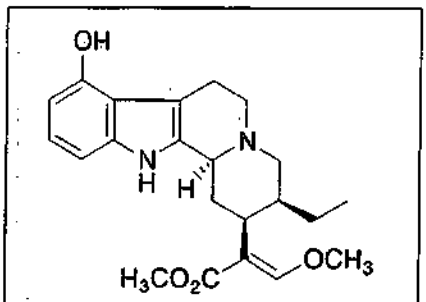
PKNO	ChNO	AREA	TIME
1	1	318	2.549
2	1	487	2.720
3	1	6073	2.930
4	1	2560	3.028
5	1	3154	3.219
6	1	4063	3.431
7	1	2599	3.654
8	1	664	3.957
9	1	1664	4.159
10	1	16709486	13.943
11	1	73537	17.086
12	1	9188	27.265
		16813793	

Purity : 99.4 %

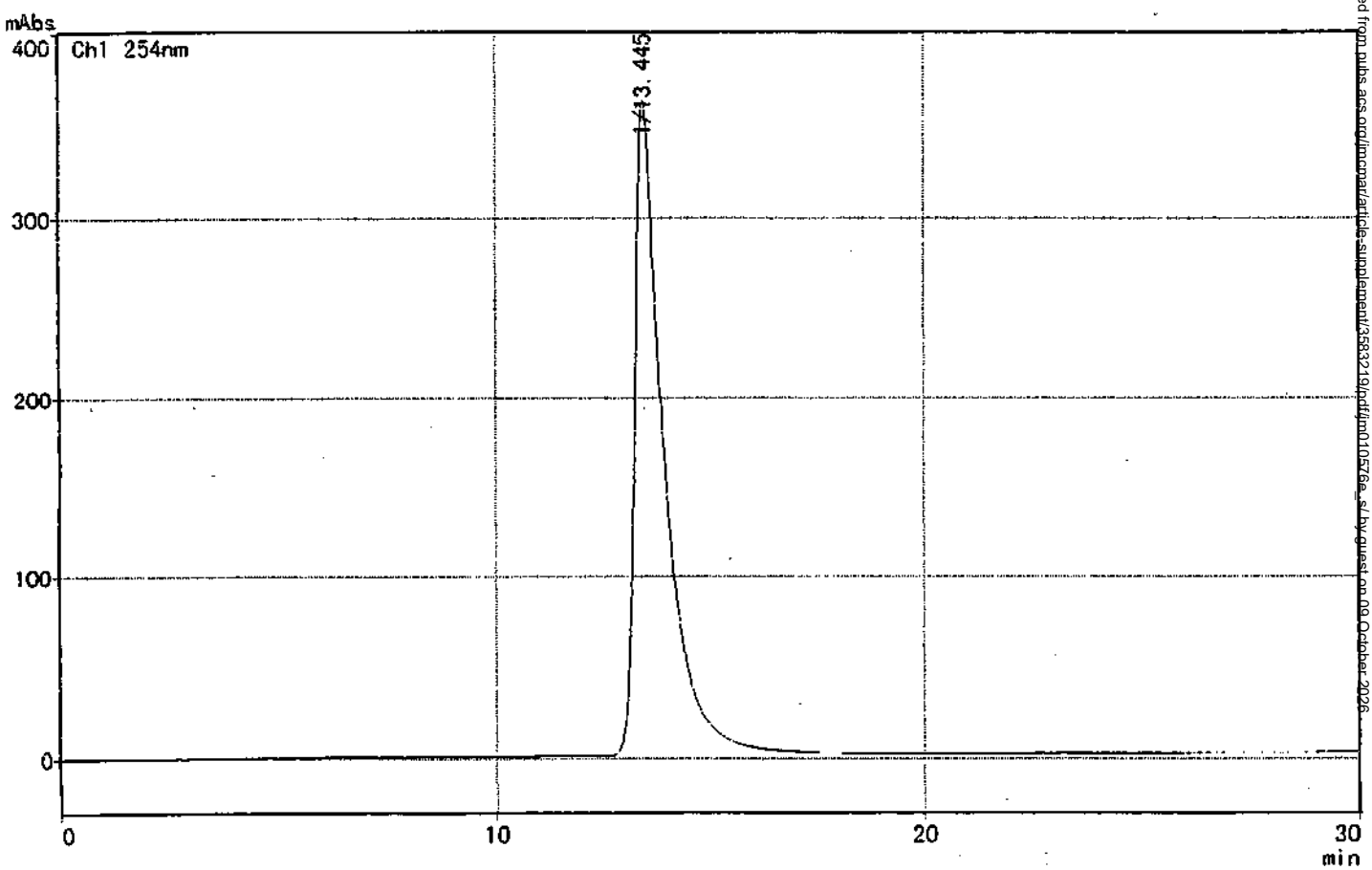
Downloaded from pubs.acs.org/jmcmar/article-supplement/3583219/pdf/jm010576e_s/ by guest on 09 October 2026

Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 80%Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : 9-Hydroxycorynantheidine



9--Hydroxycorynantheidine (**5**)



*** ピークレポート ***

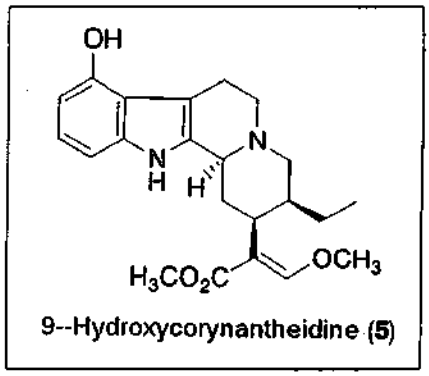
PKNO	ChNO	AREA	TIME
1	1	16766512	13.445
		16766512	

Purity : 100 %

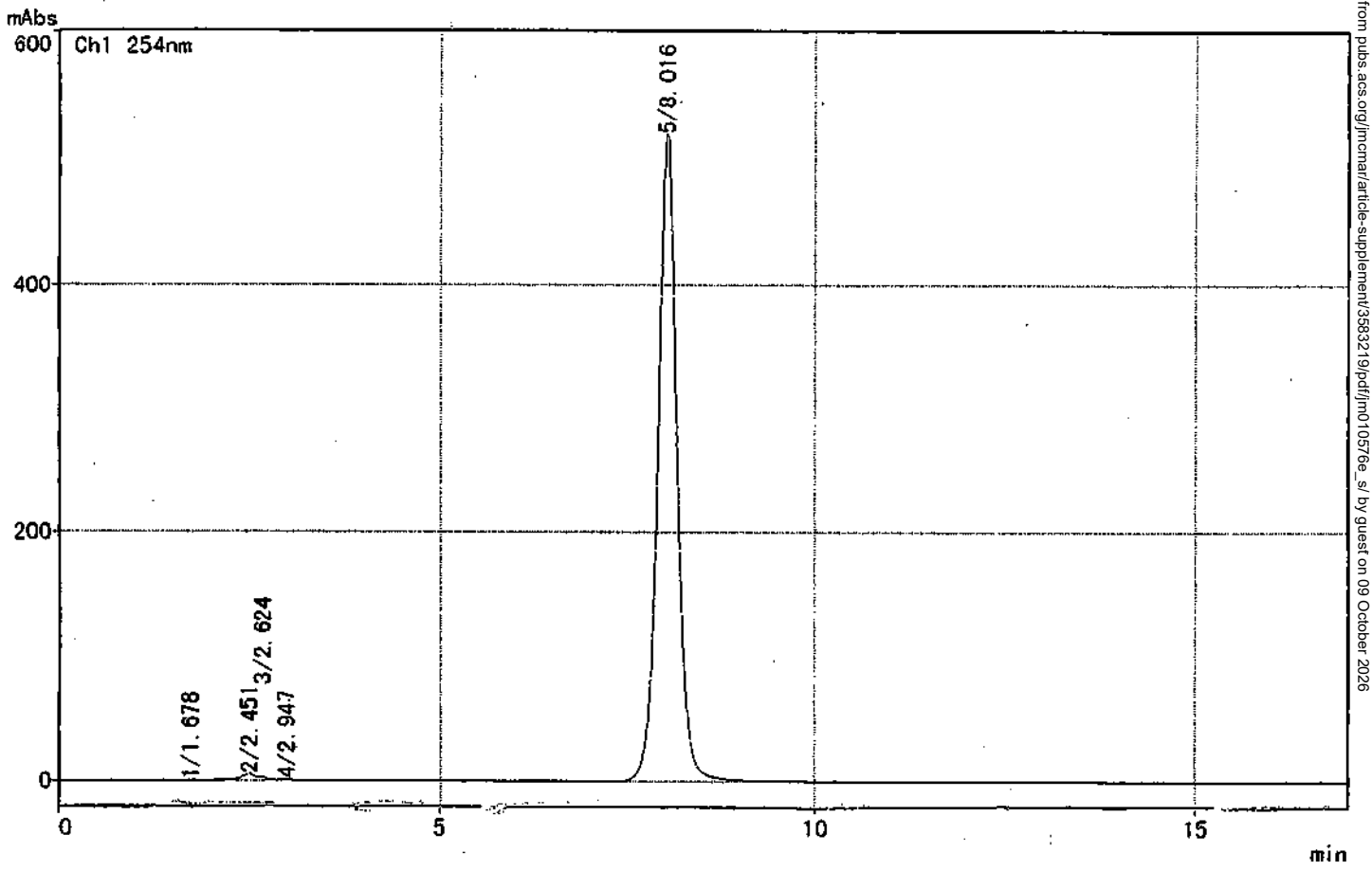
Downloaded from pubs.acs.org/jmcmar/article-supplement/358321/pdf/jm010576e_s/by guest on 09 October 2026

Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 1%MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : 9-Hydroxycorynantheidine



9--Hydroxycorynantheidine (**5**)



*** ピークレポート ***

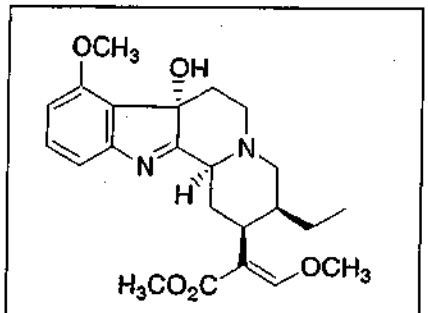
PKNO	ChNO	AREA	TIME
1	1	1015	1.678
2	1	62202	2.451
3	1	22885	2.624
4	1	8151	2.947
5	1	9187358	8.016
		9281611	

Purity : 99.0 %

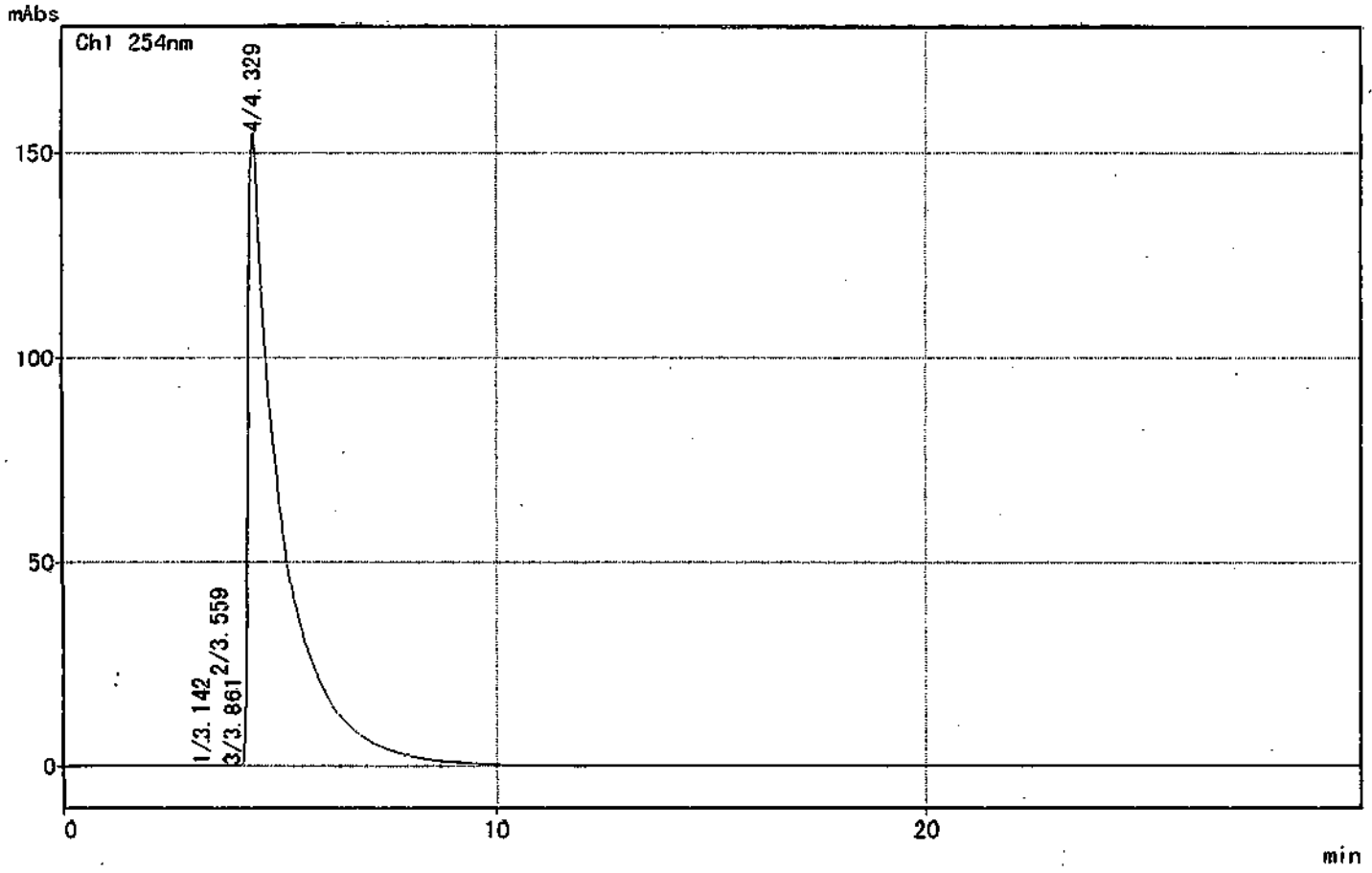
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Condition

Column : YMC-Pack A-012 SIL
(YMC) S-5μm
Column size : 6.0mm x 150mm
Flow Rate : 1.0mL/min
Eluent : 5% MeOH / $CHCl_3$
Detector : UV254nm
Temperature : 30°C
Sample : 7-Hydroxymitragynine



7-Hydroxymitragynine **(12)**



*** ピークレポート ***

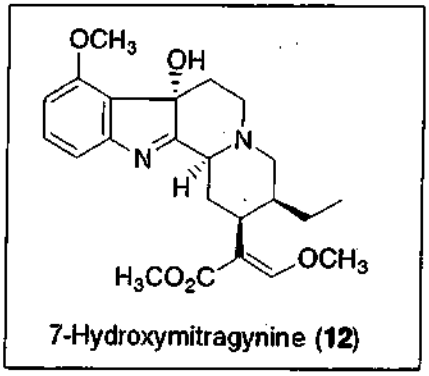
PKNO	ChNO	AREA	TIME
1	1	3085	3.142
2	1	2716	3.559
3	1	108	3.861
4	1	8119304	4.329
		8125213	

Purity : 99.9 %

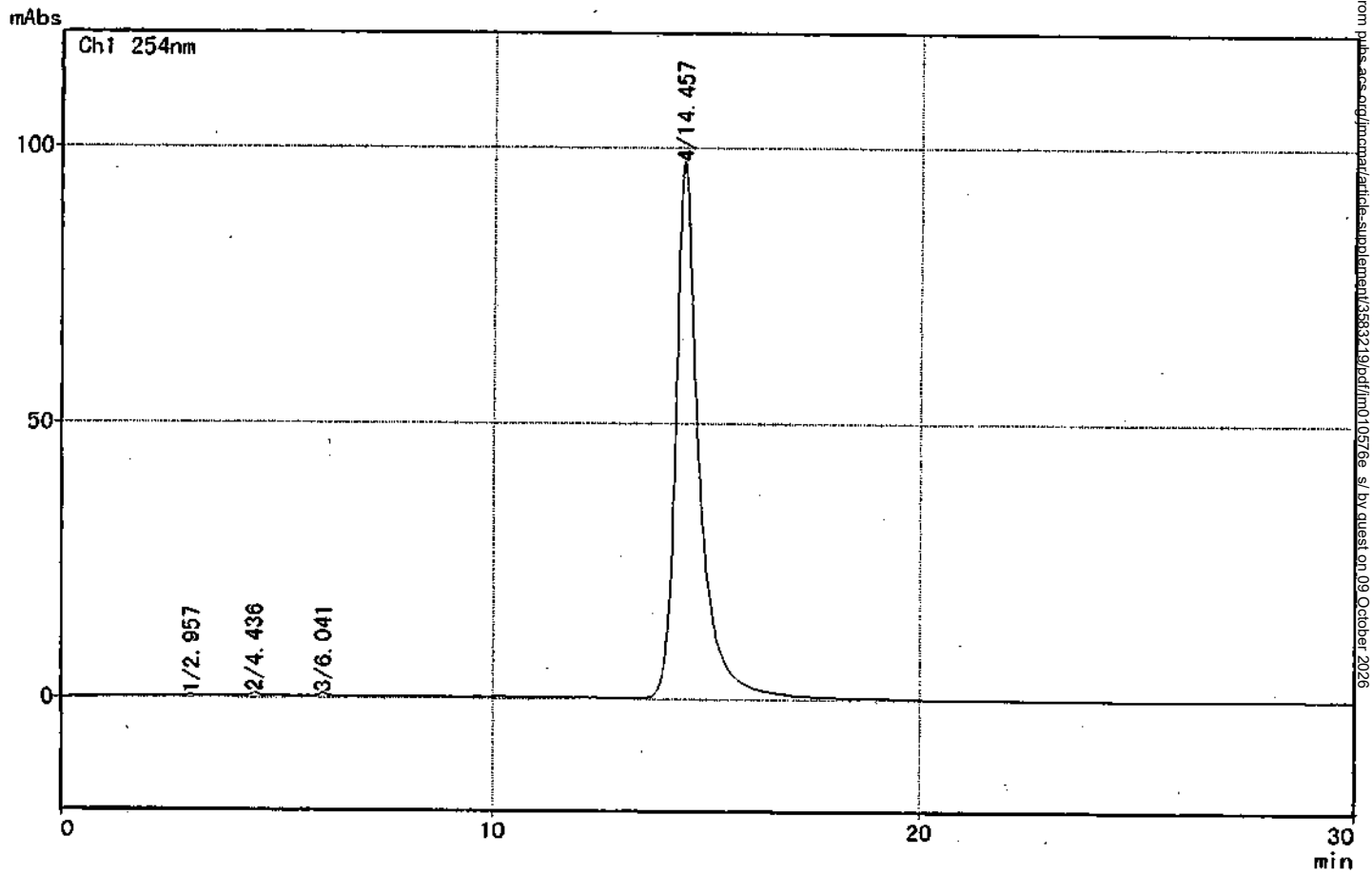
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Condition

Column : CAPCELL PAK NH2 UG80
(SHISEIDO FINE CHEMICAL) S-5μm
Column size : 4.6mmI.D. x 250mm
Flow Rate : 1.0mL/min
Eluent : 50% Ethyl Acetate / *n*-Hexane
Detector : UV254nm
Temperature : 30°C
Sample : 7-Hydroxymitragynine



7-Hydroxymitragynine (**12**)



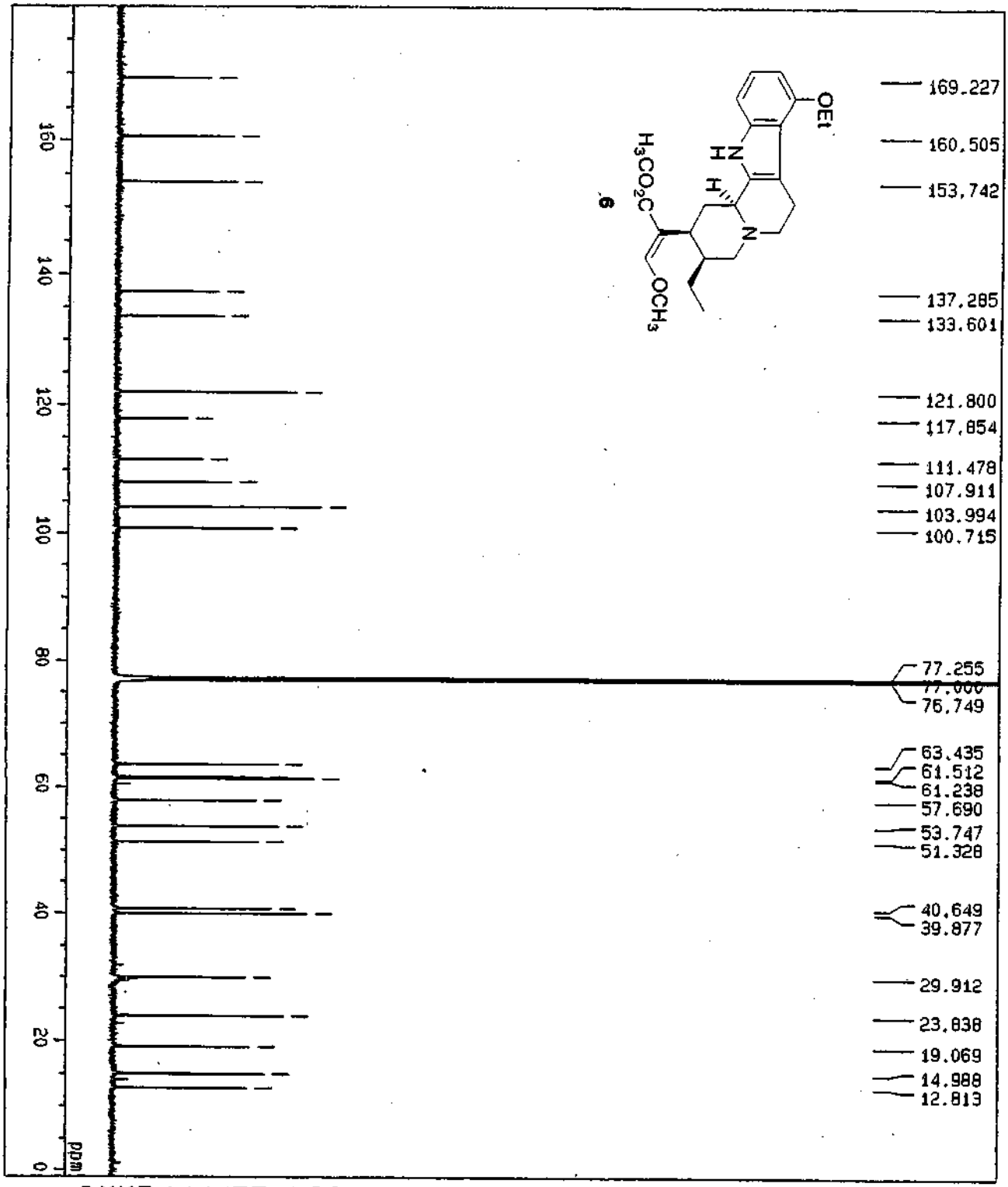
*** ピークレポート ***

PKNO	ChNO	AREA	TIME
1	1	1980	2.957
2	1	5371	4.436
3	1	5748	6.041
4	1	3606465	14.457
		3619563	

Purity : 99.6 %

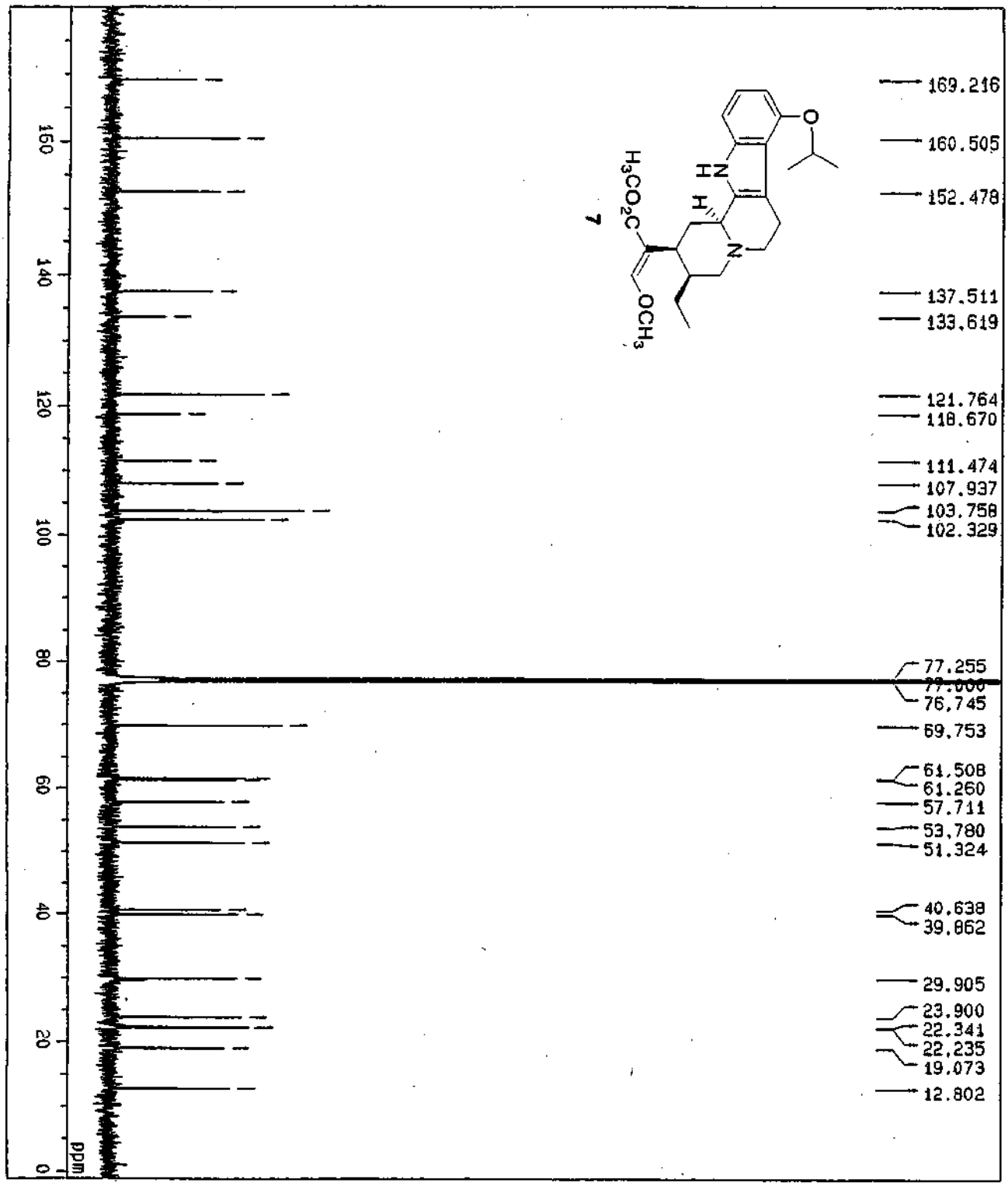
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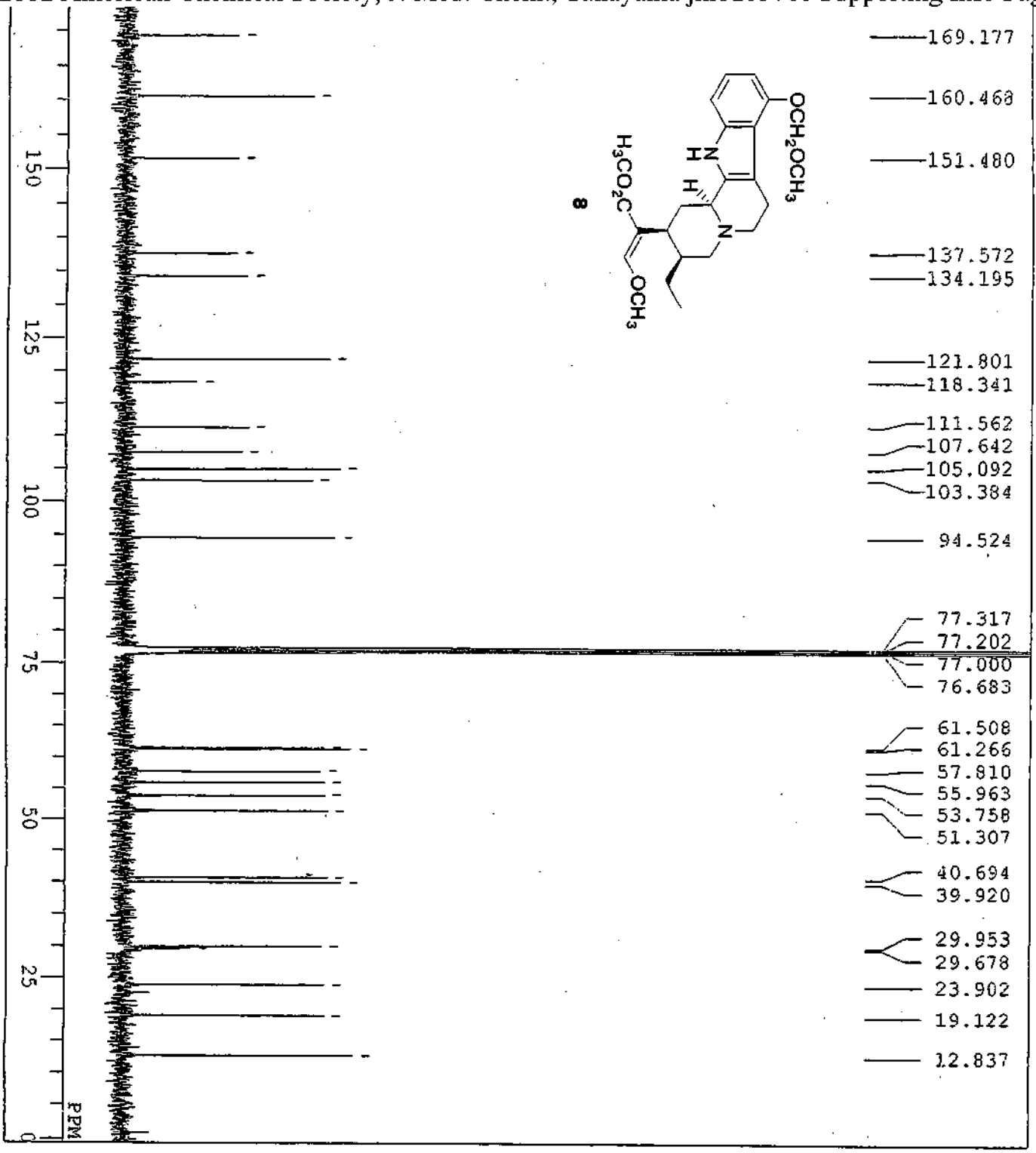


7-MAY-1999 18:21:40.32
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COMNT : MG-2-1-1 BCM
EXMOD : SINGL
IRMOD : BCM
POINT : 65536
FREQU : 30030.03 Hz
SCANS : 1200
DUMMY : 4
ACQTM : 2.1823 sec
PD : 1.5000 sec
RGAIN : 23
PW1 : 4.40 usec
OBNUC : 13C
OBFRQ : 125.65 MHz
OBSET : 127782.65 Hz
IRNUC : 1H
IRFRQ : 500.00 MHz
IRSET : 162410.00 Hz
IRATN : 511
IRAPW : 67.0 usec
IRBP1 : 30
IRBP2 : 6
IRAMS : 0
ADBIT : 16
CTEMP : 20.5 c
CSPED : 11 Hz
SLVNT : CDCL3
RESOL : 0.46 Hz
BF : 2.00 Hz
T1 : 0.00 %
T2 : 0.00 %
T3 : 90.00 %
T4 : 100.00 %
REFVL : 77.00 ppm
XE : 22763.09 Hz
XS : 1212.68 Hz
operator

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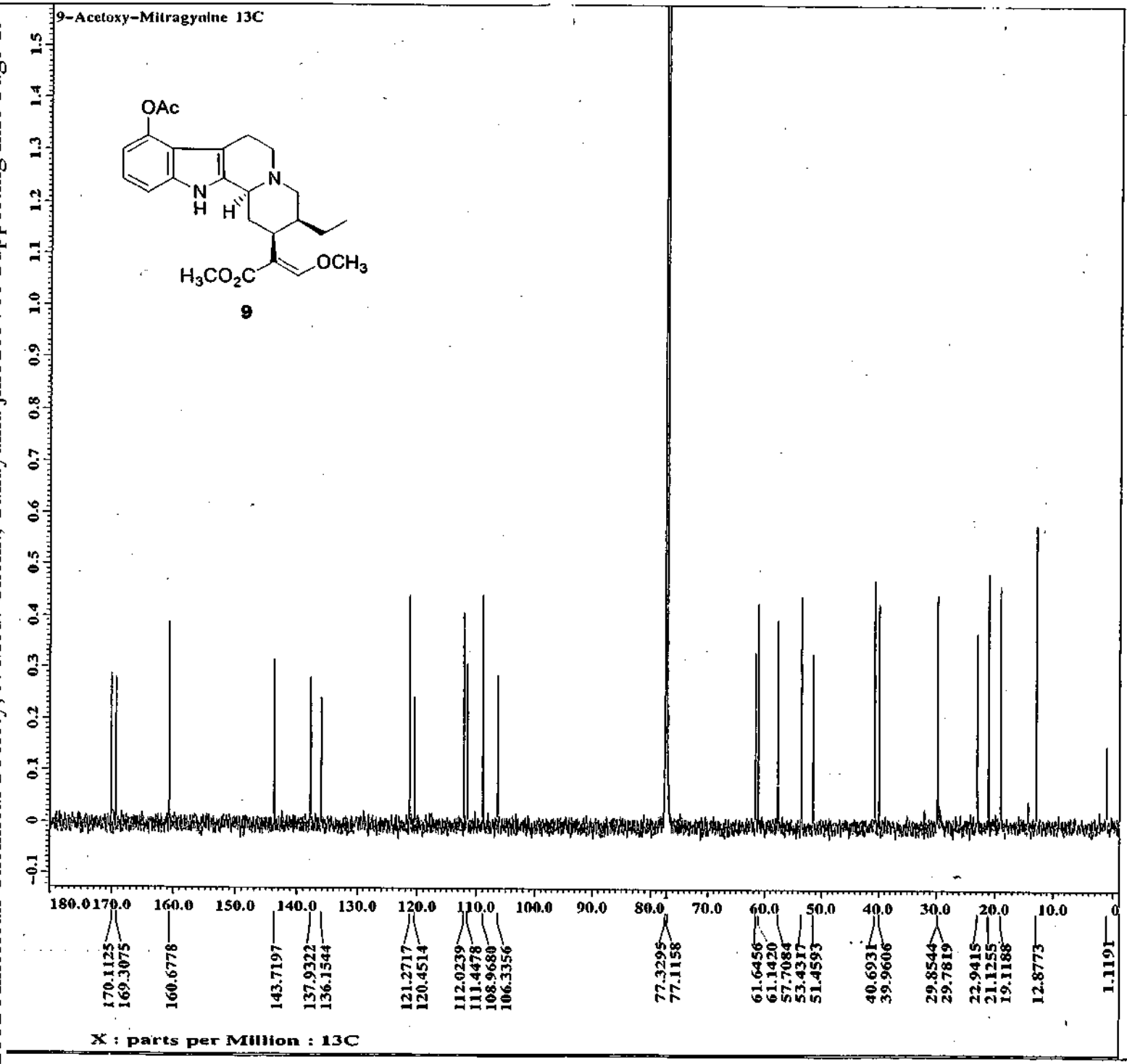


DFILE D:\SK\ISHIKAWA\SKHI-MG1
COMNT MG10-2-1BCM
DATIM
OBNUC 13C
EXMOD SINGL-BCM
OBFRQ 100.40 MHz
OBSET 130.00 KHz
OBFIN 5500.0 Hz
POINT 65536
FREQU 27100.3 Hz
SCANS 1120
ACQTM 2.418 sec
PD 1.300 sec
PW1 4.8 us
IRNUC 1H
CTEMP 29.6 c
SLVNT CDCL3
EXREF 77.00 ppm
BF 2.00 Hz
RGAIN 23

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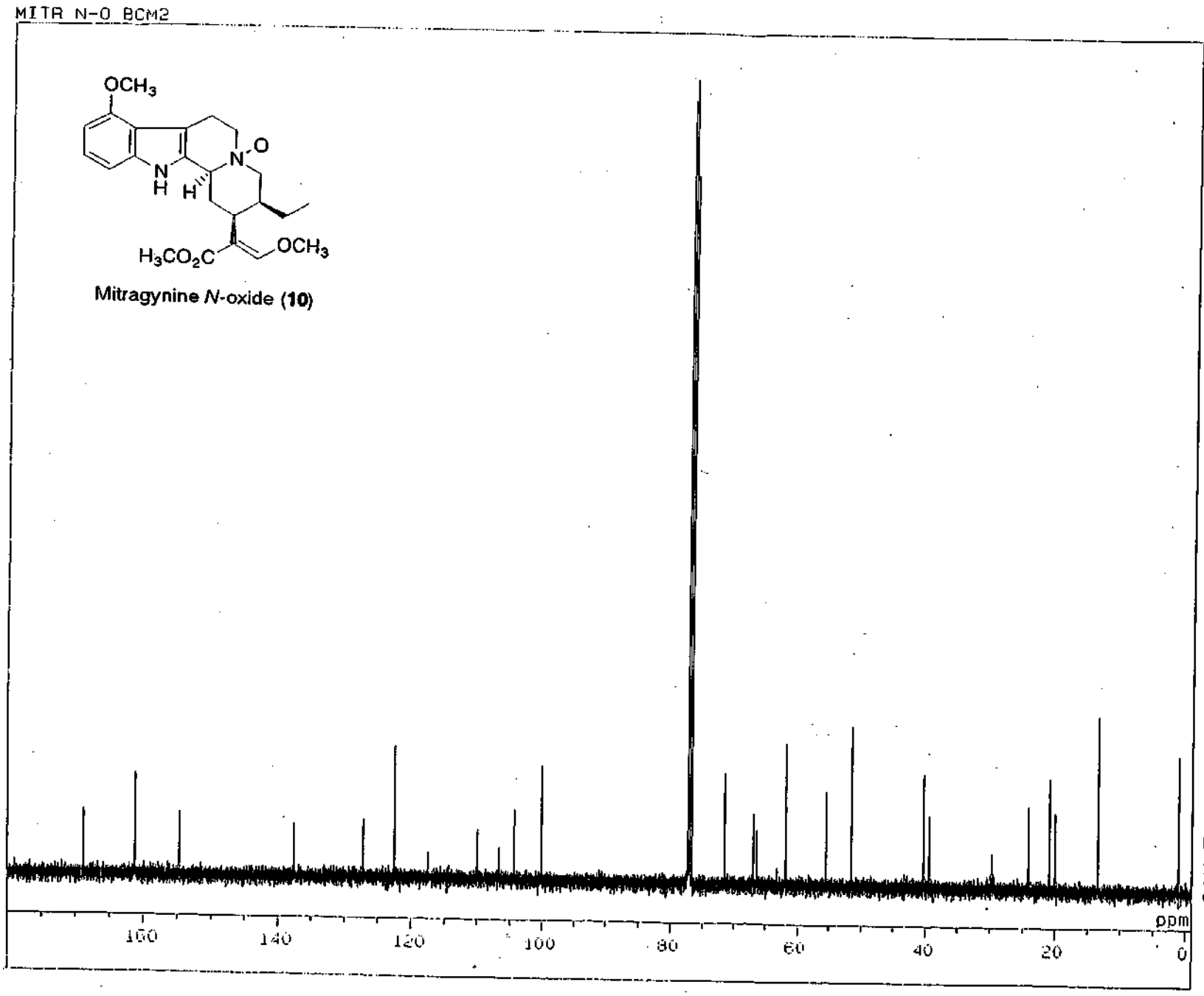


---- ACQUISITION PARAMETERS ----

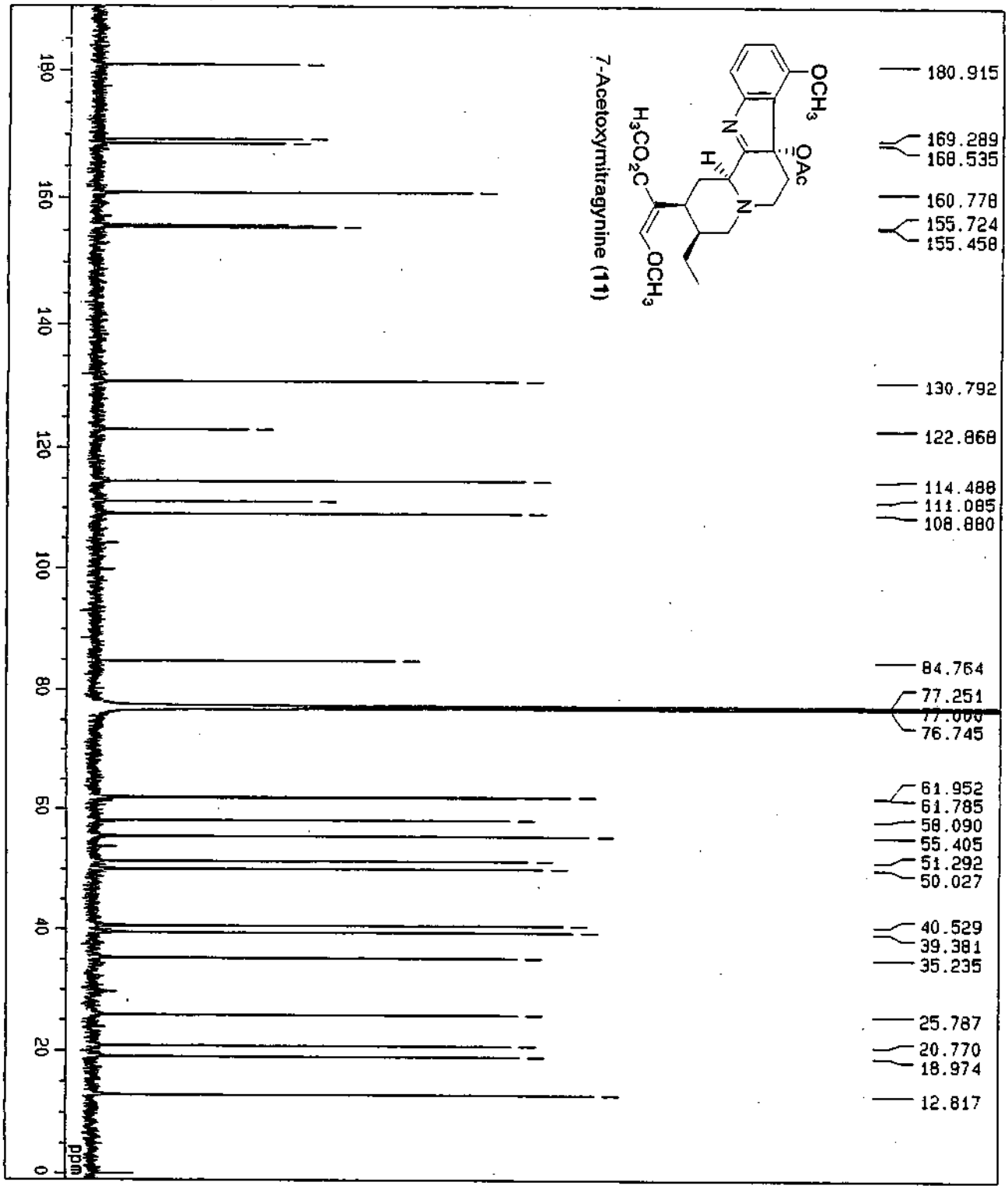
File Name	= SKISHIKAWA-9-OAC-MIT
Author	= JEOL LTD.
Sample ID	= SKISHIKAWA-9-OAC-MIT
Content	= 9-Acetoxy-Mitragynin
Creation Date	= 1-MAY-2001 19:03:07
Revision Date	= 1-MAY-2001 19:14:59
Spec Site	= ECP600
Spec Type	= DELTA_NMR
Data Format	= 1D COMPLEX
Dimensions	= X
Dim Title	= 13C
Dim Size	= 65536
Dim Units	= [ppm]
Actual_start_time	= 1-MAY-2001 19:03:08
Delay_of_start	= 1[s]
Digital_filter	= FALSE
End_time	= 1-MAY-2001 19:37:41
Experiment	= single_pulse_bil
Field_strength	= 14.09636928[T]
Filter_mode	= BUTTERWORTH
Filter_width	= 18.8641788[kHz]
Irr_code	= 194
Irr_domain	= 1H
Irr_freq	= 600.17530548[MHz]
Irr_noise	= WAUGH
Irr_offset	= 5.0[ppm]
Irr_pwidth	= 63[us]
Iterations	= 1
Local_time	= 1-MAY-2001 19:03:06
Obs_noise	= WAUGH
Obs_pwidth	= 1[us]
Probe_id	= 2692
Recvr_gain	= 27
Relaxation_delay	= 1.5[s]
Scans	= 960
Solvent	= CHLOROFORM-D
Spin_get	= 14[Hz]
Spin_lock_90	= 45[us]
Spin_lock_attn	= 11.5[dB]
Temp_get	= 25.2[dC]
X90	= 10.8[us]
X_acq_duration	= 1.736704[s]
X_angle	= 30[deg]
X_domain	= 13C
X_freq	= 150.9300327[MHz]
X_offset	= 110[ppm]
X_points	= 65536
X_prescans	= 4
X_pulse	= 3.6[us]
X_resolution	= 0.57581215[Hz]
X_sweep	= 37.73584906[kHz]
Tri90	= 10[us]
Tri_noise	= WALTZ
Tri_pwidth	= 1[us]
Qua90	= 10[us]
Qua_noise	= WAUGH
Qua_pwidth	= 1[us]

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Mitragynine *N*-oxide (**10**)



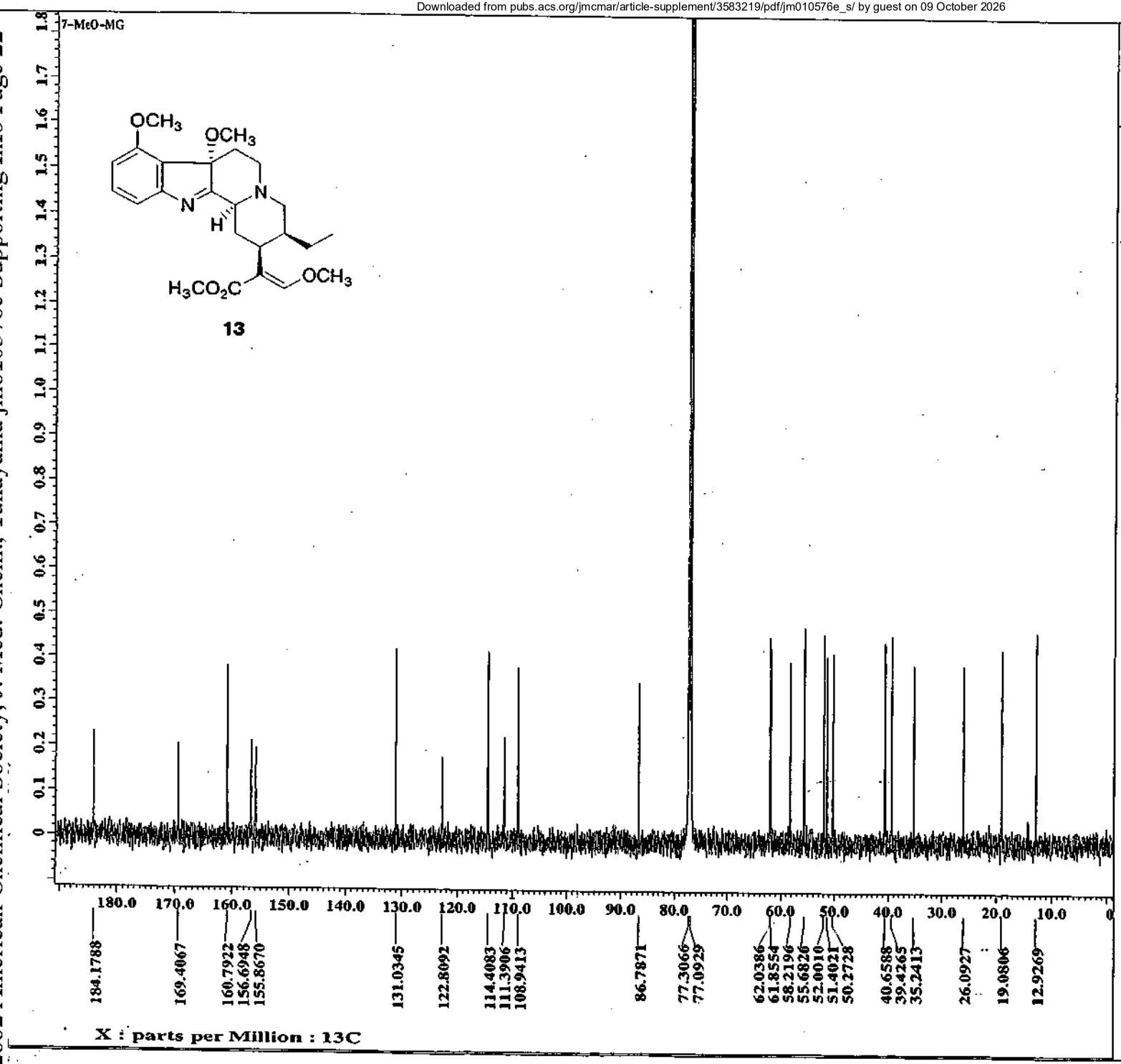
5-JUN-2001 20:14:32.01

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COMNT	7-OACMIT-BCM
EXMOD	SINGL
IRMOD	BCM
POINT	65536
FREQU	30030.03 Hz
SCANS	2800
DUMMY	4
ACQTM	2.1823 sec
PD	1.5000 sec
RGAIN	23
PW1	3.95 usec
OBNUC	13C
OBFRQ	125.65 MHz
OBSET	127782.65 Hz
IRNUC	1H
IRFRQ	500.00 MHz
IRSET	162410.00 Hz
IRATN	511
IRAPW	50.0 usec
IRBP1	30
IRBP2	5
IRRNS	0
ADBIT	16
CTEMP	21.9 c
CSPEO	13 Hz
SLVNT	CDCL3
RESOL	0.46 Hz
BF	3.00 Hz
T1	0.00 sec
T2	0.00 sec
T3	90.00 sec
T4	100.00 sec
REFVL	77.00 ppm
XE	24020.91 Hz
XS	584.69 Hz
operator	

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JEOL

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---- ACQUISITION PARAMETERS ----
File Name           = SKISHIKAWA-7MEO-MG-1
Author              = JEOL LTD.
Sample ID           = SKISHIKAWA-7MEO-MG
Content             = 7-MeO-MG
Creation Date       = 30-OCT-2001 14:27:14

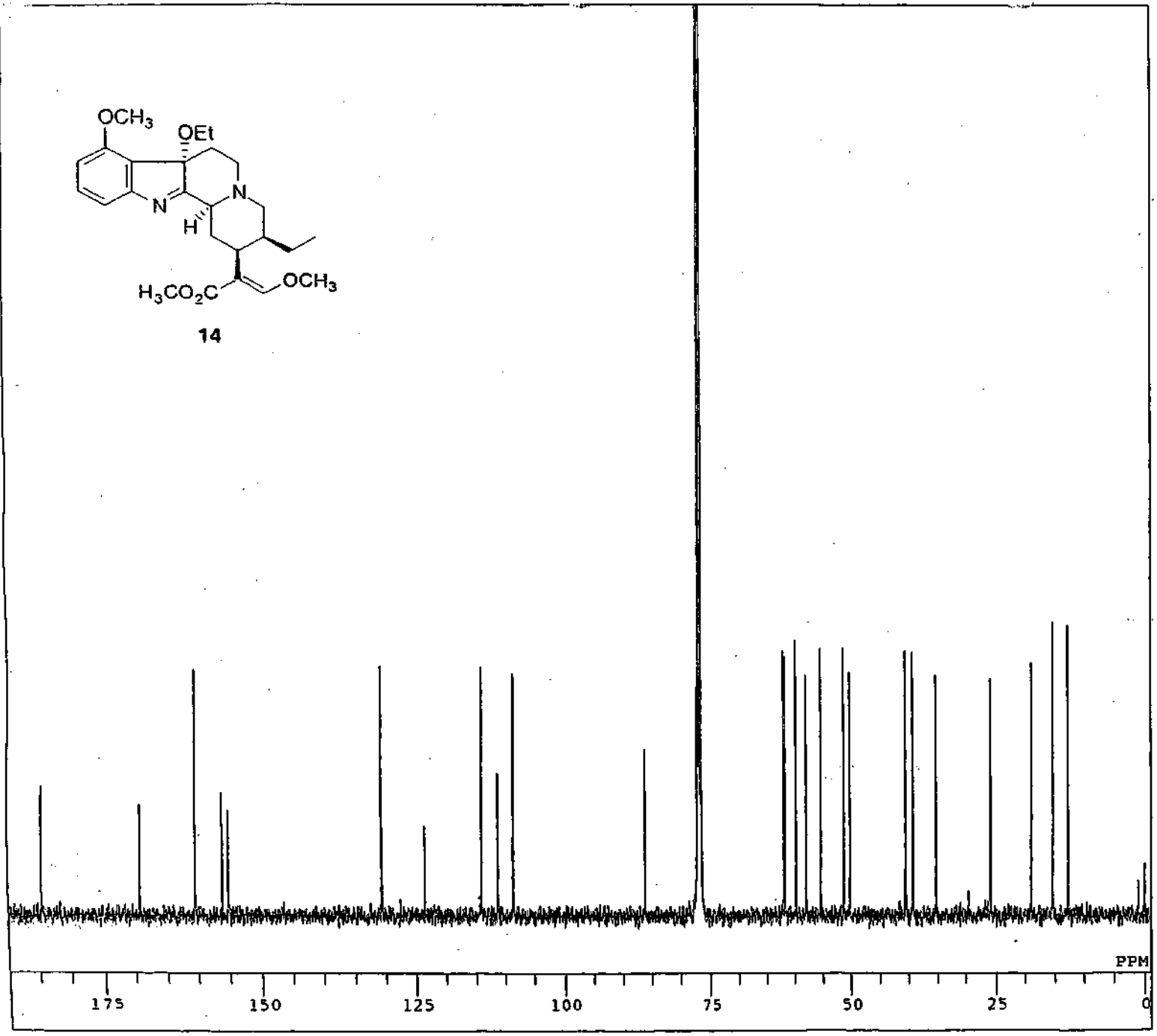
Revision Date       = 30-OCT-2001 14:34:16
Spec Site           = ECP600

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Dimensions          = X
Dim Title           = 13C
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Dim Units           = [ppm]
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Delay_of_start      = 1[s]
Digital_filter      = FALSE
End_time            = 30-OCT-2001 15:18:57
Experiment          = single_pulse_bil
Field_strength      = 14.09636920[T]
Filter_mode         = BUTTERWORTH
Filter_width        = 18.8641788[kHz]
Irr_code            = 194
Irr_domain          = 1H
Irr_freq            = 600.17530548[MHz]
Irr_noise           = WAUGH
Irr_offset          = 5.0[ppm]
Irr_pwidth          = 60[us]
Iterations          = 1
Local_time          = 30-OCT-2001 14:27:13
Obs_noise           = WAUGH
Obs_pwidth          = 1[us]
Probe_id            = 2692
Recvr_gain          = 26
Relaxation_delay    = 1.5[s]
Scans               = 1842
Solvent             = CHLOROFORM-D
Spin_get            = 15[Hz]
Spin_lock_90        = 45[us]
Spin_lock_attn      = 11.5[dB]
Temp_get            = 24.1[dC]
X90                 = 13[us]
X_acq_duration      = 1.736704[s]
X_angle             = 30[deg]
X_domain            = 13C
X_freq              = 150.9300327[MHz]
X_offset            = 110[ppm]
X_points            = 65536
X_prescans          = 4
X_pulse             = 4.33333333[us]
X_resolution        = 0.57581215[Hz]
X_sweep             = 37.73584906[kHz]
Tri90               = 10[us]
Tri_noise           = WALTZ
Tri_pwidth          = 1[us]
Qua90               = 10[us]
Qua_noise           = WAUGH
Qua_pwidth          = 1[us]
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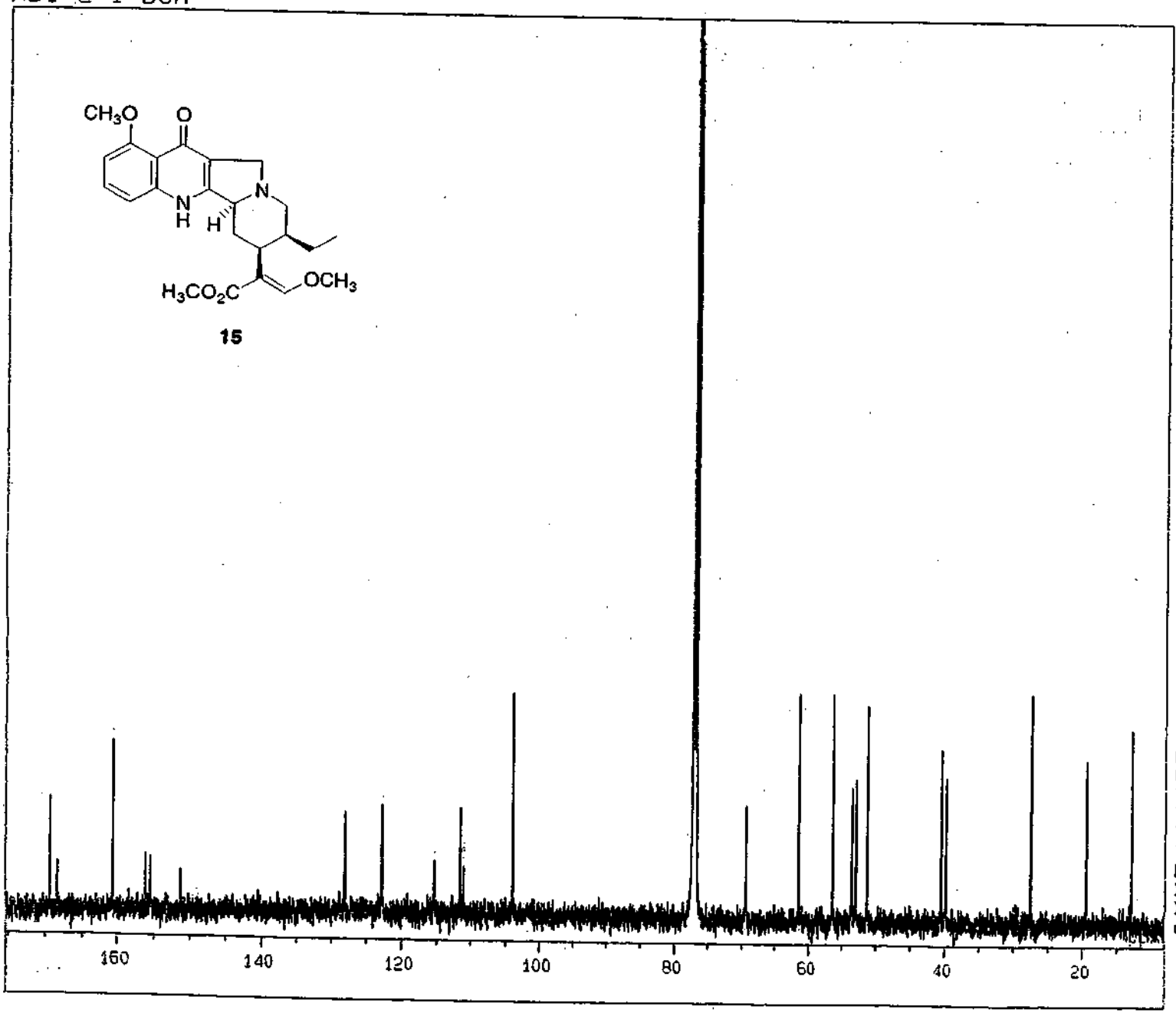
DFILE	D:\NMR-datas\Ishikawa\SKI
COMNT	7-OETMIT
DATIM	
OBNUC	13C
EXMOD	SINGL-BCM
OBFRQ	100.40 MHz
OBSET	130.00 KHz
OBFIN	5500.0 Hz
POINT	65536
FREQU	27100.3 Hz
SCANS	12400
ACQTM	2.418 sec
PD	1.300 sec
PW1	4.9 us
IRNUC	1H
CTEMP	23.3 c
SLVNT	CDCL3
EXREF	77.00 ppm
BF	2.00 Hz
RGAIN	23

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NB1-2-1 BCM

15



30-JUN-1998 13:30:24.94

DFILE : ALPHA
SFILE : SKKATO-NB1-2-1BCM

COMNT : NB1-2-1 BCM
EXMOD : SINGL
IRMOD : BCM
POINT : 65536
FREQU : 30030.03 Hz
SCANS : 2710
DUMMY : 4
ACQTM : 2.1823 sec
PD : 1.5000 sec
RGAIN : 23

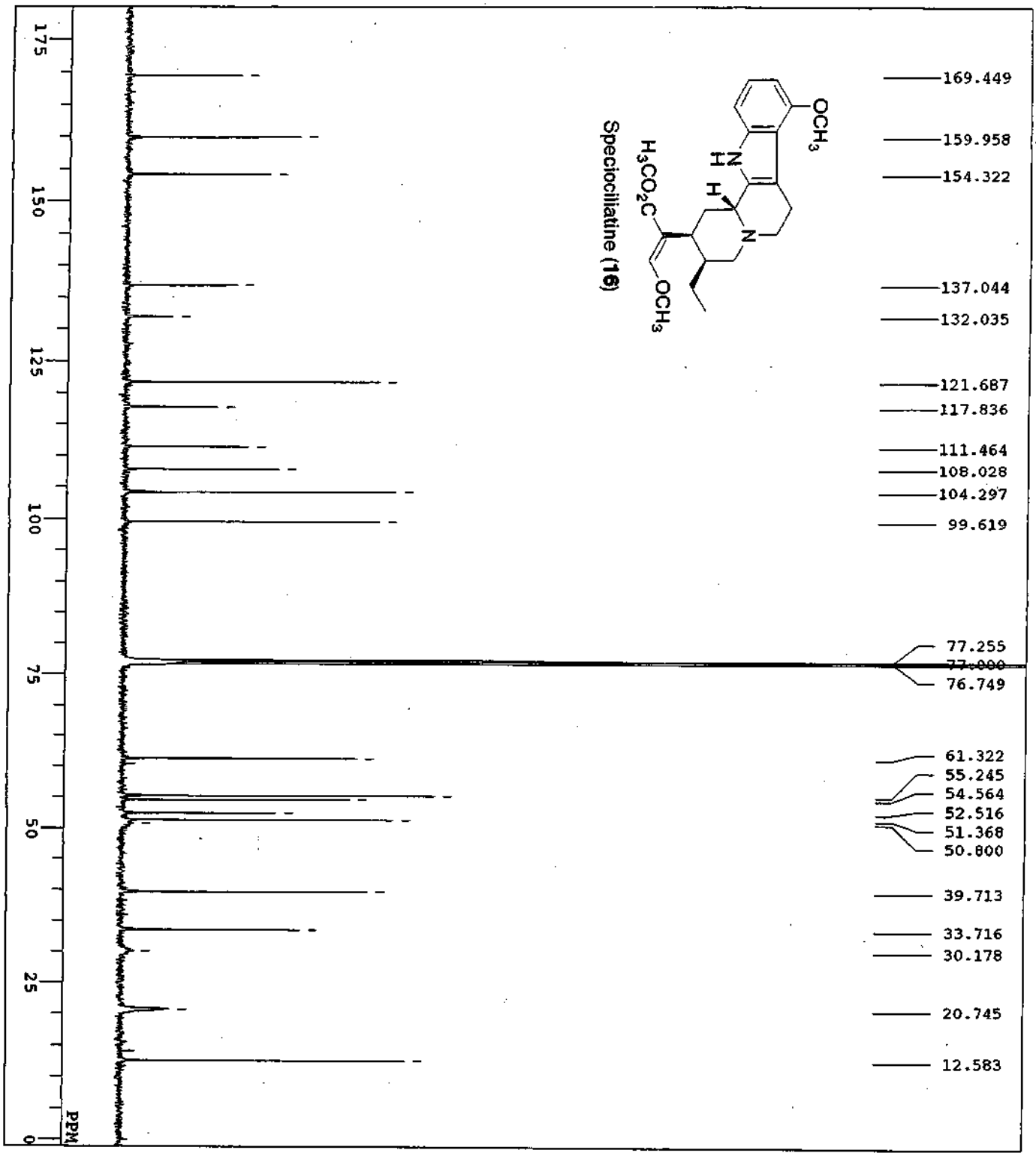
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OBSET : 127782.65 Hz

IRNUC : 1H
IRFRQ : 500.00 MHz
IRSET : 162410.00 Hz
IRATN : 511
IRRPW : 67.0 usec
IRBP1 : 30
IRBP2 : 6
IRRNS : 0

ADBIT : 16
CTEMP : 21.2 c
CSPED : 12 Hz
SLVNT : CDCL3

RESOL : 0.46 Hz
BF : 2.00 Hz
T1 : 0.00 %
T2 : 0.00 %
T3 : 90.00 %
T4 : 100.00 %
REFVL : 77.00 ppm
XE : 21013.14 Hz
XS : 918.28 Hz
operator



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DFILE
COMNT
DATIM
OBNUC                13C
EXMOD          SINGL-BCM      LQNOR   Àe
OBFRQ          125.65 MHz
OBSET          120.00 KHz
OBFTN          7782.6 Hz
POINT          65536
FREQU          30030.0 Hz
SCANS          1000
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CTEMP          21.5 c
SLVNT          CDCL3
EXREF          77.00 ppm
BF             3.00 Hz
RGAIN          23
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Mitragynine/Corynantheidine Pseudoindoxyls As Opioid Analgesics with Mu Agonism and Delta Antagonism, Which Do Not Recruit β-Arrestin-2

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Abstract

Natural products found in *Mitragyna speciosa*, commonly known as kratom, represent diverse scaffolds (indole, indolenine, and spiro pseudoindoxyl) with opioid activity, providing opportunities to better understand opioid pharmacology. Herein, we report the pharmacology and SAR studies both in vitro and in vivo of mitragynine pseudoindoxyl (**3**), an oxidative rearrangement product of the corynanthe alkaloid mitragynine. **3** and its corresponding corynantheidine analogs show promise as potent analgesics with a mechanism of action that includes mu opioid receptor agonism/delta opioid receptor antagonism. In vitro, **3** and its analogs were potent agonists in [^{35}S]GTPγS assays at the mu opioid receptor but failed to recruit β-arrestin-2, which is associated with opioid side effects. Additionally, **3** developed analgesic tolerance more slowly than morphine, showed limited physical dependence, respiratory depression, constipation, and displayed no reward or aversion in CPP/CPA assays, suggesting that analogs might represent a promising new generation of novel pain relievers.

Graphical abstract

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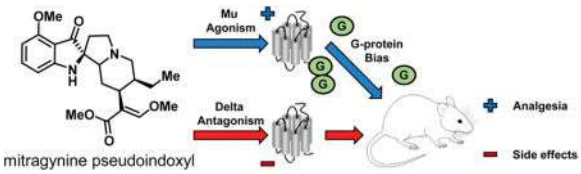
Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jmedchem.6b00748.
Experimental details and data (PDF)
Compound data (CSV)

Notes

The authors declare no competing financial interest.

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INTRODUCTION

Opioids, including morphine, are clinically used for the treatment of moderate to severe chronic pain. However, despite their proven efficacy, mu opioid receptor agonists have problematic side effects such as tolerance, physical dependence, and substance abuse.[1] Agonists selective for other opioid receptors produce analgesia, but with their own liabilities.[2–4] The ultimate goal of opioid-related drug development has been to design and synthesize potent antinociceptive agents that are devoid of adverse side effects. Many approaches have been taken over the years, starting with the development of partial agonists or mixed agonist/antagonists.[5–9] A more recent approach takes advantage of biased agonism, in which distinct downstream signaling pathways are activated by different agonists working through the same receptor.[10,11] It has been proposed that ligands biased against recruiting β-arrestin-2, or showing preference for activating specific G-protein-mediated signal transduction pathways, will demonstrate diminished side effects.[12,13] Oliceridine (TRV130)[14,15] is an example of a mu opioid receptor-biased agonist which has recently entered phase-III clinical trials, showing separation between antinociception and some opioid-related side effects. 6′-Guanidinonaltrindole (6′-GNTI),[16] 22-thiocyanatosalvinorin A (RB-64),[17] and two new classes of kappa opioid ligands from the Aube group have also recently been reported in the opioid literature as biased kappa opioid receptor agonists.[18,19]

Natural products have provided many lead compounds leading to the design of new pharmaceuticals. Natural products and their derivatives account for approximately 50% of approved drugs.[20] Morphine, the most commonly employed opioid, and thebaine, the structure on which the vast majority of semisynthetic opiates is based, are natural alkaloids found in the poppy plant, *Papaver somniferum*. While opioid chemistry has traditionally been dominated by thebaine-derived alkaloids isolated from poppy, there are a growing number of opioid natural products derived from structures other than the traditional morphinan scaffold and thus structurally not closely related to morphine. These include analogs of salvinorin A[21–23] such as herkinorin[24] and thiocyanatosalvinorin A,[17] which have been developed as mu and kappa opioid receptor-biased agonists, respectively, while some peptide analogs such as cyclo[Phe-D-Pro-Phe-Trp] (CJ-15,208)[25] are being developed as analgesics and medications against cocaine abuse (Figure 1). Mitragynine (indole core) (**1**) and its congeners isolated from the Southeast Asian plant *Mitragyna speciosa*, commonly known as kratom, are monoterpene indole alkaloids structurally not closely related to morphine.[26] In addition to its traditional use, kratom has become a quickly emerging substance of abuse. It is currently legal in many parts of the world, and kratom leaves are available for purchase over the Internet. Case studies of fatalities resulting from overdose have been published, although the risk posed by mitragynine remains uncertain, given that simultaneous use or contamination with other substances (including opioids) that may have

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been involved in the reported deaths.[27–30] Both mitragynine (**1**) and its naturally occurring oxidation product, 7-OH mitragynine (indolenine core) (**2**), are opioid antinociceptive agents that have been examined both in vitro and in vivo.[31–39] Mitragynine pseudoindoxyl (**3**), a rearrangement product of **2** with a spiro-pseudoindoxyl core, was first isolated in 1974 by Zarembo et al. as a microbial metabolite of **1** by the fungus *Helminthosporum sp.*[40] Yamamoto et al. reported that it acted nonselectively on mu- and delta opioid receptors, while its kappa opioid receptor affinity was negligible.[41] In later publications by Takayama et al., the in vivo supraspinal analgesic properties of **3** were briefly discussed.[39]

In this work, we report the in vitro and in vivo pharmacology and structure–activity relationships (SAR) of mitragynine pseudoindoxyl (**3**). We demonstrate for the first time that **3** and its C-9 substituted derivatives, corynantheidine pseudoindoxyls, are systemically active mixed mu opioid receptor agonist/delta opioid receptor antagonist compounds in vitro and produce potent antinociception in vivo. Characterization of **3** demonstrated an opioid-mediated analgesia devoid of any place-conditioning effects and a side effect profile far superior to clinically used mu opioid-based antinociceptive agents.

RESULTS

Chemistry

1 was extracted from dry kratom powder using a modified protocol reported by Ponglux et al.[42] Compounds **2** and **3** were synthesized from **1** as shown in Scheme 1.[39,43] To better understand the pharmacology of this template, SAR studies were carried out by modifying the C-9 and N-1 (indole nitrogen) positions. Six analogs with various substituents in the C-9 positions and 2 analogs at N-1 were synthesized. C-9-substituted corynantheidine pseudoindoxyl derivatives were synthesized starting from **2**. To gain access to the C-9 position on the pseudoindoxyl scaffold, **2** was converted to 9-hydroxycorynantheidine pseudoindoxyl (**4**) using $AlCl_3$ and ethanethiol in DCM. This intermediate was converted to its triflate (**5**) using triflic anhydride and pyridine, which was subsequently used as the precursor for further reactions.

The pseudoindoxyl of corynantheidine (**6**) was synthesized using palladium-catalyzed removal of the triflate ester by formic acid. The synthesis of the nitrile **7** was accomplished in a palladium-catalyzed reaction of **5** with $Zn(CN)_2$. Compounds **8** and **9** were obtained via Suzuki coupling reactions of **5** and the appropriate boronic acids. **3** was alkylated in the N-1 position with benzyl bromide and iodomethane to synthesize **11** and **12**, respectively.

In vitro Pharmacology

Initial investigations used in vitro radioligand binding assays with cell lines stably expressing murine opioid receptors (Figure 2, Table 1). Mitragynine (**1**) showed poor affinity at all opioid receptors, whereas 7-hydroxymitragynine (**2**) showed moderate affinity at the mu opioid receptor clone MOR-1, 5-fold higher than **1** and was considerably more potent at the expressed delta opioid receptor clone DOR-1 than **1**.[44] Mitragynine pseudoindoxyl (**3**) displayed the highest overall binding affinity for MOR-1 and DOR-1 (K_i 0.8 nM and 3 nM, respectively) while also showing a moderate affinity for KOR-1. These data suggest that

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conversion of the indole to indolenine ring to the spiro-pseudoindoxyl core dramatically increases affinity for opioid receptors. The binding affinities of **3** at MOR-1 and DOR-1 were comparable to the prototypic mu ligands morphine and DAMGO and delta ligands DPDPE and NTI. Alkyl substitution at the N-1 position of the template (compounds **11**, **12**) eliminated opioid affinity, suggesting that the unsubstituted indole NH is important for receptor binding. Substitutions at C-9, however, yielded potent derivatives. All six C-9-modified compounds (**4**, **6**–**10**) maintained the high affinity at both MOR-1and DOR-1 sites as observed with **3** previously. **2** and **3** were also screened across a panel of other nonopioid drug targets using the PDSP screening facility at NIMH.[45] **2** exhibited no affinity appreciable affinity at these receptors (K_i > 10 μM). **3** had poor affinity at α_{2A} adrenergic receptor, α_{2C} adrenergic receptor, and $5HT_7$ (Table S1).

In [^{35}S]GTPγS functional assays using opioid transfected cell lines, **1** was a partial agonist with moderate potency at MOR-1 and a weak antagonist at both DOR-1 and KOR-1 (Table 2). **2** was a partial agonist at MOR-1, 4-fold more potent than **1**, and a weak KOR-1 and DOR-1 antagonist.[44] Analog **3** was a potent full agonist at MOR-1 and an antagonist at both DOR-1 and KOR-1. Most C-9-modified derivatives (**4**, **6**, **7**, and **9**) were MOR-1 agonists and DOR-1 antagonists in functional assays with the exception of compound **8**, which was a dual MOR-1/DOR-1 agonist (Table 2).

Both DAMGO and endomorphin-2 effectively recruited β-arrestin-2 in CHO cells expressing MOR-1, as measured with the DiscoveRx PathHunter assay. In contrast, compounds **1**–**4** and **6**–**9** failed to recruit β-arrestin-2 at concentrations as high as 10 μM (Figure 3A). Both **2** and **3** reduced DAMGO-induced β-arrestin-2 stimulation in these cells in a concentration-dependent manner (Figure 3B)[44] consistent with their respective binding affinities at the receptor. Thus, **3** and its analogs potently stimulated [^{35}S]GTPγS binding without stimulating β-arrestin-2 recruitment. However, their antagonism of DAMGO stimulation of β-arrestin-2 recruitment revealed that they could still bind to both G-protein and arrestin functional receptor configurations.

Antinociception

The antinociceptive effect of mitragynine and its derivatives was evaluated in vivo in mice using the radiant heat tail flick assay (Table 1). After subcutaneous administration, **1** produced antinociception with an ED_{50} (and 95% CI) value of 166 mg/kg (101, 283), 66-fold less active than morphine. On the other hand, **2** was about 5-fold more potent than morphine and 350-fold more potent than **1** (Figure S1), similar to literature values.[32–34] Compound **3** was 1.5-fold more potent than morphine after intracerebroventricular administration (icv, Figures 4A and S2) and 3-fold more potent following subcutaneous (sc, Figures 4B and S4A and S4B) administration. Compound **3** has a shorter duration of antinociceptive effect than morphine with a peak effect at 15 min (Figure S3). Compound **3** proved equally active in CD1, C57BL/6, and 129Sv6 strains of mice subcutaneously (Figure S5). Furthermore, **3** also was active orally, with an ED_{50} (95% CI) value of 7.5 (4.3–13) mg/kg (Figure 4C). The C-9 derivatives (**4** and **6**–**10**) also produced antinociception following systemic administration, with ED_{50} values comparable to **3** (Table 1). Compound **3** was also active in the hot plate assay of antinociception, with an ED_{50} (95% CI) value of

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0.99 (0.75–1.3) mg/kg (Figures 4D and S4C), comparable to morphine (ED_{50} = 1.7 (1.3, 2.4) mg/kg) (Figure S4D).

Opioid Receptor Antagonism

Based on the high affinity and favorable mixed mu agonist/delta antagonist profile in vitro, we examined the activity of compound **3** in greater detail. Naloxone and the mu-selective antagonist β-FNA effectively reversed **3**-induced antinociception, whereas the delta-selective antagonist NTI and the kappa antagonist norBNI did not. Yohimbine, an α_2 antagonist, had no effect on the antinociception of **3** (Figure 5A).

To examine the selectivity of antinociception further, we used an antisense oligodeoxynucleotide mapping paradigm. The activity of the oligodeoxynucleotide antisense probes for MOR-1, KOR-1, and DOR-1 has been established previously.[46–48] Targeting exon 1 of MOR-1, the antisense oligodeoxynucleotide lowered the analgesic actions of morphine, reproducing earlier studies (Figure S6).[46] Similarly, the responses of **3** were lowered (Figure 5B). The specificity of the response was established by the inactivity of the control mismatch. Similarly, downregulation of exon 3 of DOR-1[48] and exon 2 of KOR-1[49] with antisense oligodeoxynucleotides attenuated antinociception produced by the prototypic delta agonist DPDPE and the prototypic kappa agonist U50,488H, in accordance with previous studies (Figure S6). However, these oligodeoxynucleotides targeting kappa and delta receptors did not alter the antinociception of **3** (Figure 5B).

The mu opioid receptor *Oprm1* creates an array of splice variants through alternative splicing with patterns conserved from rodents to humans.[50,51] The major sets of variants are full-length 7 transmembrane domain (7TM) variants associated with exon 1 (E1). A second set of variants of truncated six transmembrane domain splice variants (6TM) is generated by an alternative promoter associated with exon 11 (E11) of the *Oprm1* gene.[52,53] MOR-1 KO mice were used to establish the contributions of 7TM E1-MOR-1 and 6TM E11-MOR-1 variants to **3** antinociception.

Two different types of MOR-1 KO mice were utilized: exon 11 (E11) MOR-1 KO mice, which lack the 6TM E11splice variants of MOR-1 but retain expression of 7TM E1 splice variants of MOR-1, and total mu opioid receptor knockout in which both exons 1 and 11 were disrupted (E1/E11) to eliminate all 7TM and 6TM mu opioid receptor variants of the *Oprm1* gene. Morphine antinociception has previously been demonstrated to be independent of the E11-associated 6TM splice variants of MOR-1, maintaining full analgesic activity in the E11 KO,[54] but its antinociception was completely eliminated in E1/E11 MOR-1 KO mice,[55] suggesting 7TM E1-MOR-1 variant as the primary mechanism of analgesic action. Compound **3** showed similar antinociceptive responses as morphine. Compound **3** antinociception was similar in wild-type (ED_{50} = 0.83 mg/kg (0.37–1.9)) and exon 11 KO C57/BL6 mice (ED_{50} = 1.4 mg/kg (0.34–5.8)) in a tail flick assay (Figure 5D). However, antinociception of compound **3** was eliminated in E1/E11 MOR-1 KO mice (ED_{50} > 30 mg/kg), indicating a mu opioid receptor mechanism.[55] Taken together with the antisense results, these in vivo findings indicate that **3** analgesia is mediated by 7TM E1-MOR-1 receptors.

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Side Effect Profile

We next evaluated **3** in mouse models of antinociceptive tolerance, dependence, respiratory depression, and inhibition of GI transit (Figure 6). Mice developed antinociceptive tolerance to morphine after twice daily administration for 5 days (5 mg/kg/injection, sc). In contrast, antinociceptive tolerance to **3** developed far more slowly, with twice daily administration of **3** (at an equianalgesic dose as morphine) requiring 29 days instead of 5 days (Figure 6A). After 5 days, the morphine ED_{50} value was shifted 6-fold to 12.1 (7.6–19.4) mg/kg from 2.0 (1.2–3.3) mg/kg. In contrast, after 5 days, the ED_{50} value for **3** was shifted <2-fold to 1.1 (0.66–2.0) mg/kg. After 29 days, the ED_{50} value for **3** was shifted 6-fold (4.5 mg/kg (2.7–7.7) mg/kg) (Figure S7). An independent sample *t* test was used to test for the group difference between **3** and morphine on day 5. The mean difference of 53.97%MPE between the two groups was found to be statistically significant (p-value <0.0001) at the 5% level of significance. This difference is likely to range between 42.27 and 65.67% MPE as measured by a 95% confidence interval. To highlight **3**'s ability to sustain antinociception over repeated dosing, the composite areas under the curve (AUCs) were calculated using the trapezoidal rule on the mean response across 1–29 days for the **3** group and 1–5 days for the morphine group, respectively. The AUC for the morphine group was 205.9, while for the **3** group it was 6-fold larger, 1239.

Time action studies revealed that **3** has a shorter duration of action than morphine (Figure S3). Since the shorter duration of action of **3** led to a decreased drug exposure, we also examined tolerance in a different dosing paradigm in which **3** was given four times per day to provide similar drug exposures for morphine and **3**. In this paradigm, **3** failed to show a significant decreased effect over 5 days, whether examined at a fixed dose over the 5 days (Figure S8A) or dose–response curves at day 5 (Figure S8B). After 5 days, the ED_{50} value for **3** was shifted from 0.69 (0.46, 1.0) to 1.6 (0.97–2.6) mg/kg (Figure S8C), a shift similar to that observed when **3** was given twice a day (ED_{50} = 1.1 (0.66–2.0)) on day 5 (Figure S7).

Physical dependence was assessed in mice treated repeatedly with **3** at twice its ED_{50} dose (1.5 mg/kg) twice daily for 5, 22, or 29 days by administration of the opioid antagonist naloxone. Mice showed only minimal signs of withdrawal following administration of naloxone with 12 ± 4.1, 13 ± 4.4, and 14 ± 6.0 jumps on average, respectively (Figure 6B). These values were not significantly different from saline but differed significantly from mice treated with morphine for 5 days, which showed 77 ± 7.2 jumps (Figure 6B). In summary, compound **3** demonstrated limited antinociceptive tolerance and physical dependence in comparison with morphine following chronic administration.

Differences in effect between **3** and morphine extended to other opioid effects. Mu selective agonists inhibit gastrointestinal transit, a major component of constipation. At a dose twice its analgesic ED_{50} value (5 mg/kg, sc), morphine almost totally eliminated transit (Figure 6C). An equianalgesic dose of **3** (1.5 mg/kg, sc) also lowered gastrointestinal transit, but not nearly as much as morphine. The effect on GI transit plateaued, with a greater dose (4 mg/kg, sc) also showing no further inhibition (Figure 6C).

Morphine dose-dependently reduced the respiratory rate in mice, with a decrease of approximately 30% by a dose twice its analgesic ED_{50} value (5 mg/kg, sc) by approximately

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50% after a higher dose corresponding to 4-fold its analgesic ED_{50} value. In contrast, **3** showed no respiratory depression at ~2× its antinociceptive ED_{50} dose in C57BL/6 mice (1.2 mg/kg, sc). Although a higher dose (3 mg/kg, sc), transiently lowered the respiratory rate by approximately 15%, this was still significantly less than morphine (Figure 6D).

Compound **3** also failed to show either rewarding or aversive behavior in a conditioned place preference paradigm. In this study, morphine produced significant conditioned place preference (CPP; $F_{(4180)} = 5.62$, $p = 0.003$; two-way ANOVA with Tukey HSD post hoc test) and U50,488H produced conditioned place aversion (CPA), but **3** demonstrated neither preference or aversive behavior at doses 2-fold or 5-fold its analgesic ED_{50} value (n.s.; Tukey HSD post hoc test; Figure 6E). Overall, these results demonstrate that **3** produces potent opioid receptor-mediated antinociception both centrally and systemically, yet shows a separation of antinociception from some classic opioid side effects such as antinociceptive tolerance, dependence, and conditioned place preference. Furthermore, **3** shows a lower propensity to cause respiratory depression and constipation compared with the canonical opioid, morphine.

Modeling

In silico docking studies were carried out to unravel potential differences in receptor interactions of **1**–**3**. The results of in silico prediction of inhibitory constants are listed in Table 3 for all three mitragynine compounds and the opioid receptors. In general, ligands of the lowest energy complexes were located in the binding pocket observed in the crystal structures. Inhibitory constants (K_i) calculated for the lowest energy complexes follow the trend observed in the experiments, but the range of values is much more narrow compared to experimental data, suggesting much lower selectivity of both the ligands and the receptors. The lowest energy complexes, which are considered to reflect specific binding between **1**–**3**, and the receptors are shown in Figures 7 and S14. These in silico K_i values reproduce experimental data with much higher accuracy compared to the ones calculated for the lowest energy complexes in the first pass, where all nonspecific hits were included (for explanation, see the Experimental Section). Receptor side chains in contact with the bound **1**–**3** are also depicted in Figure 7 and listed in Table 3.

DISCUSSION

Although initially described in the scientific literature as early as 1974, very little was known about the pharmacology of **3** prior to this study. Its chemical structure suggested it as an excellent starting point for semisynthetic diversification. Thus, we subjected **3** to detailed pharmacological analysis. In opioid receptor-transfected CHO cell lines, **3** had a high affinity for MOR-1 and DOR-1 sites with a moderate affinity at KOR-1. After subcutaneous administration, **3** was a potent antinociceptive agent in two thermal pain models: radiant heat tail flick and hot plate. It was also active in the mouse tail flick assay when administered supraspinally and orally. We established the agonist selectivity of **3** through pharmacological and genetic approaches, demonstrating that its antinociception was mediated by mu opioid receptors and not kappa, delta, and/or α_2 adrenergic receptors. Mu antagonists attenuate the antinociception, but it was insensitive to kappa and delta antagonists. Clonidine, an α_2

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agonist, is a potent analgesic used for the treatment of various pain conditions.[56] Yohimbine, an α_2 antagonist, did not affect the antinociception of **3**, therefore, its antinociception is likely not related to adrenergic pathways. The nonreversal of analgesia also rules out a role of α_{2A} and α_{2C} adrenergic receptors in mediating analgesia of **3**, although the drug had some affinity for these receptors in our initial screening. Antisense downregulation of E1-MOR-1 variants attenuated antinociception, while downregulation of DOR-1 and KOR-1 failed to modify **3** antinociception. In mice lacking E11-MOR-1 variants (E11 MOR-1 KO mice) compound **3**, like morphine, still exhibited antinociception comparable to wild-type mice, while in mice lacking all mu opioid receptor splice variants (E1/E11 MOR-1 KO mice), **3** antinociception was completely eliminated. The results from antisense and KO mice experiments implicate traditional 7TM mu opioid receptor aka E1-MOR-1 variants in **3** antinociception.

Of interest, **3** signaling failed to recruit β-arrestin-2 alone and antagonized both DAMGO-induced β-arrestin-2 recruitment and stimulation of [^{35}S]GTPγS binding. To the best of our knowledge, **3** is the first example of a mixed activity mu opioid agonist/delta antagonist ligand which does not recruit β-arrestin-2. Prior evidence in the literature suggests that failure to recruit β-arrestin-2 and delta antagonism may both be successful in separating antinociception from unwanted side effects. Consistent with this, **3** displayed a robust antinociceptive effect in mice and without conditioned place preference or aversion. Two different dosing paradigms (two and four times per day) revealed the slow development of tolerance and a marked decrease in jumping following challenge with naloxone, which is characteristic of physical dependence. Furthermore, **3** showed no respiratory depression at twice its analgesic ED_{50} dose and far less constipation than morphine.

A number of mixed mu agonist/delta antagonist ligands have been reported in the literature. DIPP-NH_2[Ψ],[57] 5″-(4-chlorophenyl)-6,7-didehydro-4,5α-epoxy-3-hydroxy-17-methylpyrido[2′,3′:6,7]morphinan (SoRi20411),[5] and 14-alkoxy pyridomorphinans[58] are potent delta antagonists and mu agonists which produce analgesia with reduced tolerance when given supraspinally. It must be noted that a majority of these studies monitored the development of tolerance through the accepted practice of administering the ED_{80} antinociceptive dose of the test drug twice daily for 5–7 days. The current study tested this approach even more rigorously, yet still found significant reductions in antinociceptive tolerance following administration of **3** for 29 days. DIPP-NH_2[Ψ] also shows no physical dependence in treated mice. 4a,9-Dihydroxy-7a-(hydroxymethyl)-3-methyl-2,3,4,4a,5,6-hexahydro-1H-4,12-methanobenzofuro[3,2-*e*]isoquinolin-7(7aH)-one (UMB425)[59] and the cyclic peptide analog of KSK103 (C-terminal Ser(β-Glc)NH_2)[6] and ([Dmt^1]DALDA → CH_2CH_2NH ← TICP-[Ψ]), which connects Dmt^1-DALDA (mu agonist) with the delta antagonist TICP[Ψ] through a spacer, also show reduced acute antinociceptive tolerance compared to morphine when given systemically.[60] Administration of delta antagonists or induced downregulation of delta opioid receptors has been reported to prevent morphine tolerance without sacrificing analgesic potency.[61,62] Genetic disruption of the opioid system led to similar observations. In DOR-1 antisense knockdown mouse models, tolerance and acute dependence to morphine were eliminated,[63] while DOR-1 knockout animals did not develop antinociceptive tolerance to morphine.[64] The role of delta opioid receptors in

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blocking mu opioid-mediated CPP is still unclear.[65] CPP is retained unchanged or abolished depending on assay conditions in DOR-1 KO mice, while morphine reportedly was more rewarding in a study with β-arrestin-2 KO mice.[66] Additional study of this phenomenon should be possible as additional novel ligands with suitable MOR agonist/DOR antagonist activity profiles such as **3** become available.

Although GPCRs are mostly studied in relation to their ability to activate G-proteins, these transmembrane receptors are capable of recruiting β-arrestins to initiate separate cellular signal transduction pathways.[67] β-Arrestin-2 activation has been implicated in the mechanism of receptor desensitization and the occurrence of deleterious side effects.[68] Similarly to G-protein signaling, β-arrestin-2 activation is ligand and receptor dependent. Different ligands are able to stabilize GPCRs in a variety of conformations, resulting in the differential activation (bias) of G-protein and β-arrestin-2-mediated signaling pathways. Therefore, ligands interacting with the receptor do not simply have a linear effect on efficacy but also affect the functional quality of the downstream pathways.[69] Biased signaling of opioid receptors has been studied in detail by Bohn and co-workers. Compared with wild-type mice, morphine displayed enhanced antinociception and significantly attenuated respiratory depression and inhibition of GI transit in β-arrestin-2 KO mice.[70–72] These results suggest that a fully G-protein-biased opioid ligand that does not activate β-arrestin-2 signaling may be able to separate antinociception from some opioid adverse effects. An important example of G-protein-biased opioids is oliceridine, a synthetic mu agonist currently in clinical trials as an alternative to morphine and fentanyl for the treatment of chronic pain. Oliceridine is a potent analgesic that causes less respiratory depression and constipation than morphine at equianalgesic doses in humans.[15]

Taken together, these previous observations may explain why **3** is able to separate antinociception from typical opioid side effects resulting in low risk of developing tolerance while showing limited physical dependence, respiratory depression, and inhibition of GI transit. However, the relative contributions of its mu agonist/delta antagonist activity and its inability to recruit β-arrestin-2 to this advantageous pharmacological profile are not clear.

In order to investigate the SAR of the mitragynine pseudoindoxyl scaffold, semisynthetic analogs were made starting from **1** (C-9 analogs) and **3** (N-1 analogs). Receptor affinities were not significantly affected by modifications at the C-9 position, although 9-O-acetylation slightly lowered mu and delta affinities. Compounds **4** and **6**–**9** retained high, subnanomolar affinity at cells expressing MOR-1 with little change in affinity in DOR-1 cells. None of the derivatives stimulated β-arrestin-2 activation. According to previous literature reports, C-9 modifications of mitragynine altered the efficacy at mu receptors. Replacing the C-9 methoxy group with H yields corynantheidine, a mu antagonist, whereas C-9 O-demethylation yields 9-OH corynantheidine, a partial agonist in in vitro assays.[39,73] According to our studies, the SAR of the C-9-modified pseudoindoxyl scaffold is quite distinct. Various substituents can be tolerated at this position, maintaining full mu agonism. Neither C-9 O-demethylation (**4**) nor the removal of the methoxy group (**6**) affected the efficacies at mu as **4** and **6** were mu full agonists. However, the activity at delta receptors is differentially affected by varying substituents. Compounds **4**, **6**, **7**, **9**, **10** retained delta antagonism, while the 9-phenyl analog, **8**, was a delta agonist. **8** was a dual mu–delta agonist

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with similar intrinsic activity and potency at both receptors. Substitution and the introduction of bulky groups at N-1 (the indoxyl nitrogen) were not tolerated. Both the *N*-benzyl (**11**) and *N*-methyl (**12**) derivatives showed diminished affinities at all three opioid receptors compared to **3**. In vivo, the C-9 analogs were active after systemic administration. The 9-OH derivative (**4**) was more potent than **3**. Removal of the methoxy group (**6**) increased analgesic potency. The analgesic potencies of the corresponding C-9 mitragynine analogs (corynantheidine and 9-OH corynantheidine) are not reported in the literature. These results, in addition to the in vitro data, suggest that the SAR of compounds of the pseudoindoxyl scaffold differs from that of the natural *Mitragyna* alkaloids. Substitution of the C-9 methoxy group with –CN, phenyl, and furan-3-yl groups afforded products (**7**–**9**, respectively) roughly equipotent to **3**. Acetylation of **4** had a slightly negative impact on the analgesic potency (**10**).

In comparison to the mu opioid receptor-bound morphinans,[74–76] in silico modeling suggests the mitragynine derivatives (**1**–**3**) are likely to have a different binding pose. The salt bridge between Asp^{147} and the tertiary amine of the ligand and the participation of the phenolic OH in a water molecule-assisted hydrogen-bonded polar network of Tyr^{148}, Lis^{233}, and His^{297} were described as the main, conserved interactions between morphinan ligands and the binding pocket.[77] The presence of a salt bridge between Asp^{147} and the tertiary amine of **1** and its derivatives was a filtering criterion of docking results, therefore, it is present in the docked complexes of all three mitragynine compounds. The *β*-methoxy acrylate moiety in the mitragynine compounds docked in the mu receptor pocket occupied the same space as the phenol moiety of the morphinan scaffold in the crystallographic structures.[75,77]

The main difference between the mu receptor-bound **1**, **2**, and **3** is that the oxidation products (**2** and **3**) seem to participate in the polar network formed between Tyr^{148}, Lys^{233}, and His^{297} by contributing their methoxy (C-9) and methyl ester groups and replacing hydrogen-bond assisting water molecules observed in the crystal structure of the agonist-bound mu receptor (Table 3).[76] On the other hand, the beta-methoxy methyl acrylate moiety of **1** forms polar interactions with Gln^{124} and Tyr^{128} and does not take part in the aforementioned polar network (Figure S9 and Table 3). In addition to the salt bridge between Asp^{147} in the mu receptor crystal and the tertiary amine, it seems that there is a hydrogen bond present, formed with the 7-OH group of **2** (Figure S10). There is no significant difference in the relative orientation between the receptor-bound **2** and **3** (Figure S11). Differences in the calculated binding free energies and K_i values are most likely to emerge from the type and number of receptor contacts formed by these two compounds in those particularly similar docked orientations. Such differences in interactions are possibly due to the different heterocyclic scaffolds of **2** and **3**. Compared to its derivatives, **1** adopts a different orientation when bound to the delta receptor and forms fewer contacts with the residues constituting the binding pocket, resulting in a loss of affinity for this receptor. Admittedly, the relative orientation and the number and type of contacts formed between **2** and **3** with the delta receptor and the calculated in silico K_i values are highly similar, rendering it difficult to give an accurate explanation for the 1 order of magnitude difference in the experimentally determined binding affinities. A key difference between the two bound

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geometries is that the beta-methoxy methyl acrylate moiety of **2** is positioned in a hydrophobic environment of Ile^{277}, Leu^{300}, and Ile^{304}, (Figure S12 and Table 3), while in the case of **3**, this group is projected into a more hydrophilic part of the binding pocket lined with Asn^{131}, Trp^{274}, and His^{278} (Figure S13). Furthermore, the 7-OH group of **2** was found to interact with Asp^{128} (Figure S12) similarly to that observed when docked to the mu (Figure S10). In addition, the 9-methoxy group of **2** formed contact with the phenolic OH of Tyr^{129}. In the delta-bound **3** complex, both the 9-methoxy and the carbonyl groups of the pseudoindoxyl moiety were calculated to be in favorable position for hydrogen-bonding with Tyr^{129} (Figures 7 and S13).

CONCLUSIONS

In summary, we report for the first time the detailed in vitro and in vivo studies on mitragynine pseudoindoxyl. Mitragynine pseudoindoxyl is a mu agonist/delta antagonist opioid with a signaling bias for G-protein-mediated signaling pathways in vitro and which produced potent antinociception in vivo. Perhaps owing to its mixed mu agonism/delta antagonism activity, mitragynine pseudoindoxyl may avoid some of the major problems of opioid therapy, as we observed no reward or aversion and diminished antinociceptive tolerance, physical dependence, respiratory depression, and GI transit inhibition in mouse models. Upon chemical modification of this scaffold, key SAR features distinct from the mitragynine template were revealed. Among analogs modified at the C-9 position, compounds with differential efficacies within in vitro functional assays and improved in vivo potencies were identified. Docking studies to opioid receptors revealed the characteristic binding modes of mitragynine-type derivatives. It is hoped that these studies will contribute to improved understanding of the mechanism of action of compounds related to mitragynine, while holding the promise to provide novel antinociceptive drug candidates based on the mitragynine/corynantheidine pseudoindoxyl template that separates antinociception from the potential for abuse and other side effects due to their unique pharmacological properties. Observations reported in this paper and past studies on opioid ligands suggest a dual mechanism (mu agonism/delta antagonism coupled with β-arrestin2-nonrecruitment) which may account for the separation of side effects from antinociception seen with this template. The potential contributions from each of these two mechanisms will be explored in future studies on this template.

EXPERIMENTAL SECTION

Drugs and Chemicals

Opiates were provided by the Research Technology Branch of the National Institute on Drug Abuse (Rockville, MD). IBNtxA and [^{125}I]BNtxA were synthesized in our laboratory as previously described.[78–80] Na^{125}I and [^{35}S]GTPγS were purchased from PerkinElmer (Waltham, MA). Selective opioid antagonists were purchased from Tocris Bioscience. Miscellaneous chemicals and buffers were purchased from Sigma-Aldrich. Kratom “Red Indonesian Micro Powder” was purchased from Moon Kratom (Austin, TX).

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Mice

Male CD1 mice (20–32 g) were obtained from Charles River Laboratories, and C57BL/6J mice (20–32 g each) were obtained from Jackson Laboratories (Bar Harbor, ME). Exon-11 KO[79] and Exon-1/Exon-11 KO mice[55] were bred in our laboratory. All mice used throughout the manuscript were opioid naïve. All mice were maintained on a 12 h light/dark cycle with Purina rodent chow and water available ad libitum and housed in groups of five until testing. All animal studies were preapproved by the Institutional Animal Care and Use Committees of the Memorial Sloan Kettering Cancer Center or University of Florida, in accordance with the 2002 National Institutes of Health Guide for the Care and Use of Laboratory Animals.

Radioligand Competition Binding Assays

[^{125}I]IBNtxA binding was carried out in membranes prepared from Chinese Hamster Ovary (CHO) cells stably expressing murine clones MOR-1, DOR-1, and KOR-1, as previously described.[7–9,80] Binding was performed at 25 °C for 90 min. Binding in MOR-1/CHO was carried out in 50 mM potassium phosphate buffer with 5 mM $MgSO_4$ and 20 μg/mL protein, while binding in KOR-1/CHO and DOR-1/CHO was carried out in 50 mM potassium phosphate pH = 7.0 buffer and 40 μg/mL protein. After the incubation, the reaction was filtered through glass-fiber filters (Whatman Schleicher & Schuell, Keene, NH) and washed three times with 3 mL of ice-cold 50 mM Tris-HCl, pH 7.4, on a semiautomatic cell harvester. Nonspecific binding was defined by the addition of levallorphan (8 μM) to matching samples and was subtracted from total binding to yield specific binding. K_i values were calculated by nonlinear regression analysis (GraphPad Prism, San Diego, CA). Protein concentrations were determined using the Lowry method with BSA as the standard.[81]

[^{35}S]GTPγS Functional Assay

[^{35}S]GTPγS binding was performed on membranes prepared from transfected cells stably expressing opioid receptors in the presence and absence of the indicated compound for 60 min at 30 °C in the assay buffer (50 mM Tris-HCl, pH 7.4, 3 mM $MgCl_2$, 0.2 mM EGTA, and 10 mM NaCl) containing 0.05 nM [^{35}S]GTPγS; 2 μg/mL each leupeptin, pepstatin, aprotinin, and bestatin; and 30 μM GDP, as previously described.[82] After the incubation, the reaction was filtered through glass fiber filters (Whatman Schleicher & Schuell, Keene, NH) and washed three times with 3 mL of ice-cold buffer (50 mM Tris-HCl, pH 7.4) on a semiautomatic cell harvester. Filters were transferred into vials with 3 mL of Liquiscint (National Diagnostics, Atlanta, GA), and the radioactivity in vials was determined by scintillation spectroscopy in a Tri-Carb 2900TR counter (PerkinElmer Life and Analytical Sciences). Basal binding was determined in the presence of GDP and the absence of drug. Data were normalized to 1000 nM DAMGO, DPDPE, and U50,488 for MOR-1, DOR-1, and KOR-1 binding, respectively. EC_{50}, IC_{50}, and $\%E_{max}$ values were calculated by nonlinear regression analysis (GraphPad Prism, San Diego, CA).

β-arrestin-2 Recruitment Assay

β-arrestin-2 recruitment was determined using the PathHunter enzyme complementation assay (DiscoveRx, Fremont, CA) using modified MOR-1 expressed in CHO cells

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(DiscoveRx). Cells were plated at a density of 2500 cells/well in a 384-well plate as described in the manufacturer's protocol. The following day, cells were treated with the indicated compound for 90 min at 37 °C, followed by incubation with PathHunter detection reagents for 60 min. Chemiluminescence was measured with an Infinite M1000 Pro plate reader (Tecan, Männedorf, Switzerland). For the antagonist dose–response assay, the cells were incubated with the antagonist for 30 min at 37 °C prior to the addition of agonist. Following antagonist treatment, the cells were treated with 10 μM DAMGO for 90 min at 37 °C, and chemiluminescence was detected using the PathHunter detection reagents.

Antinociception

Tail flick antinociception was determined using the radiant heat tail flick technique using an Ugo Basile model 37360 instrument as previously described.[8,9] The intensity was set to achieve a baseline between 2 and 3 s. Baseline latencies were determined before experimental treatments for all mice. Tail flick antinociception was assessed quantally as a doubling or greater of the baseline latency, with a maximal 10 s latency to minimize damage to the tail. Data were analyzed as percent maximal effect, %MPE, and was calculated according to the formula: % MPE [(observed latency − baseline latency)/(maximal latency − baseline latency)] × 100. Compounds were injected subcutaneously (sc) or intracerebroventricularly (icv), and antinociception was assessed 15 min later at the peak effect. Intracerebroventricular dosing (icv) was carried out as previously described.[83] Briefly, the mice were anesthetized with isoflurane. A small incision was made, and synthetic opiate analog (2 ul/mouse) was injected using a 10 μL Hamilton syringe fitted to a 27 gauge needle. Injections were made into the right lateral ventricle at the following coordinates: 2 mm caudal to bregma, 2 mm lateral to sagittal suture, and 2 mm in depth. Mice were tested for antinociception 15 min post injection. For oral (po) studies, mice were fasted for 18 h with access to water before administering the drug by oral gavage. For the antagonism studies, β-FNA (40 mg/kg, sc) and norbinaltorphimine (norBNI, 10 mg/kg, sc) were administered 24 h before **3**. Naltrindole (NTI, 0.5 mg/kg, sc) was administered 15 min before **3**. Antinociception also was assessed using the hot plate test.[54] The hot plate (Ugo Basile 35100) consisted of a metal surface (55 °C) with a transparent plexiglass cylinder to contain the mouse. The latency to lick a hind paw or shake/flutter when the mouse was placed on the hot plate was measured, with a maximal latency of 30 s to avoid tissue damage. Baseline latencies were taken for each mouse prior to any drug administration. Mice were tested for analgesia with cumulative subcutaneous doses of the drug until the mouse can withstand the maximal latency. Once the mouse reached the maximal latency, the mouse was no longer given higher doses. In vivo experiments were evaluated using GraphPad Prism, San Diego, CA as described above.

Antisense Assays

Antisense (AN) and mismatch (MIS) oligodeoxynucleotides were designed based on the published sequences of the mouse mu opioid receptor gene (*Oprm1*), delta opioid receptor gene (*Oprd1*), and kappa opioid receptor gene (*Oprk1*) (Table 3). These probes have been previously described and validated.[49,84–86] Antisense oligodeoxynucleotide injection: Groups of mice received the stated antisense by icv administration (5–10 μg) or mismatch (5–10 μg) oligodeoxynucleotide icv under light isoflurane anesthesia on days 1, 3 and 5, as

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previously described.[85] Tail flick antinociception was tested on day 6. Control groups received no injection prior to testing. On test day, mice received **3** (1.5 mg/kg, sc), morphine (0.75 μg, icv), DPDPE (10 μg, icv), or U50,488H (5 mg/kg, sc). All experiments were performed 3× with similar results observed with each determination.

Respiratory Depression Assessment

Respiratory rate was assessed in awake, freely moving, adult male C57BL/6 mice with the MouseOx pulse oximeter system (Starr Life Sciences) as described previously.[79] Each animal was habituated to the device for 30 min and then tested. A 5 s average breath rate was assessed at 5 min intervals. A baseline for each animal was obtained over a 25 min period before drug injection, and testing began at 15 min postinjection and continued for a period of 35 min. Groups of mice ($n = 5$) were treated sc with either morphine (5 or 10 mg/kg) or **3** (1.2 or 3 mg/kg). Groups were compared with repeated-measures ANOVA followed by Tukey's multiple-comparison test.

GI Transit

Gastrointestinal transit was determined as previously described.[87] Animals received the indicated drug followed by a charcoal meal (2.5% gum tragacanth in 10% activated charcoal in water) by gavage. Animals were sacrificed 30 min later, and the distance traveled by charcoal was measured. Significance was determined by ANOVA followed by Tukey's multiple-comparison test.

Conditioned Place Preference/Aversion

Mice were conditioned with a counterbalanced place conditioning paradigm using similar timing as detailed previously.[8] The amount of time subjects spent in each of three compartments was measured over a 30 min testing period. Prior to place conditioning, the animals ($n = 95$) did not demonstrate significant differences in their time spent exploring the left (543 ± 13 s) vs right (571 ± 12 s) compartments ($p = 0.15$; Student's t-test), resulting in a combined preconditioning response of −0.1 ± 19 s. During each of the next 2 days, mice were administered vehicle (0.9% saline) and consistently confined in a randomly assigned outer compartment for 40 min, half of each group in the right chamber, half in the left chamber. Four h later, mice were administered morphine (10 mg/kg, ip), U50,488H (30 mg/kg, ip), cocaine (10 mg/kg, ip), or **3** (1.3 or 3.2 mg/kg, ip) and confined to the opposite compartment for 40 min. Conditioned place preference data are presented as the difference in time spent in drug- and vehicle-associated chambers and were analyzed via repeated measures two-way ANOVA with the difference in time spent on the treatment- vs vehicle-associated side as the dependent measure and conditioning status as the between groups factor. Where appropriate, Tukey's HSD or Sidak's multiple comparison post hoc tests were used to assess group differences. Effects were considered significant when $p < 0.05$. All effects are expressed as mean ± SEM.

In Silico Docking

Full sequence target structures of the human mu opioid and delta opioid receptors for docking studies were built and used as described elsewhere,[9] using crystal structures of the

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homologous murine opioid receptors[74,75] as templates (PDB codes: 4DKL and 4EJ4, respectively. The X-ray structure of the human kappa receptor (PDB code: 4DJH)[88] was used as docking target after missing side chains were added. Dockings were performed with the Autodock 4.2 software. Side chains in contact with the bound ligands observed in the crystal complexes of the mu, delta, and kappa opioid receptors were kept flexible as well as all ligand torsions. Mitragynine and its natural derivatives were docked using the Lamarckian genetic algorithm in an 80 × 80 × 80 Å grid volume with 0.375 Å spacing. This docking volume is large enough to cover the whole receptor region accessible from the extracellular side (Figure S14). In this sense, blind docking studies were performed, and 1000 dockings were done for all compounds and receptor models. The resultant ligand–receptor complexes were clustered and ranked according to the corresponding binding free energies, which were also used to calculate inhibitory constants according to the following equation: $\Delta G = RT\ln K_i$.

Chemistry

General Methods—All chemicals were purchased from Sigma-Aldrich Chemicals, and were used without further purification. Reactions were carried out in flame-dried reaction flasks under Ar. Reaction mixtures were purified by Silica Flash chromatography on E. Merck 230–400 mesh silica gel 60 using a Teledyne ISCO CombiFlash R_f instrument with UV detection at 280 and 254 nm. RediSep R_f silica gel normal phase columns were used. The yields reported are isolated yields. IR spectra were recorded on a Bruker Optics Tensor 27 FTIR spectrometer with peaks reported in cm^{-1}. NMR spectra were recorded on Bruker Avance III 500, Avance III 600 with DCH CryoProbe instruments. NMR spectra were processed with MestReNova software (ver. 10.0.2.). Chemical shifts are reported in parts per million (ppm) relative to residual solvent peaks rounded to the nearest 0.01 for proton and 0.1 for carbon ($CDCl_3$ ^1H: 7.26, ^{13}C: 77.3; CD_3OD ^1H: 3.31, ^{13}C: 49.0; DMSO-d_6 ^{13}C: 39.5). Peak multiplicity is reported as follows: s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet. Coupling constants (*J*) are expressed in Hz. Mass spectra were obtained at the MSKCC Analytical Core Facility on a Waters Acuity SQD LC-MS by electrospray (ESI) ionization. High-resolution mass spectra were obtained on a Waters Acuity Premiere XE TOF LC-MS by electrospray ionization. Accurate masses are reported for the molecular ion $[M + H]^+$. Purity (≥95%) was confirmed using HPLC: Waters 1525 Binary Pump, Waters 2489 UV– vis detector, Waters XBridge C18 column (5 μm × 150 × 4.6 mm), mobile phase: solvent A: water with 0.1% TFA; solvent B: acetonitrile with 0.1% TFA. Gradient: 5–95% acetonitrile/water. Flow rate: 1 mL/min.

Isolation of Mitragynine (1) from Mitragyna speciosa (Kratom)—Kratom “Red Indonesian Micro Powder” was purchased from Moon Kratom (Austin, TX). Mitragynine (**1**) was extracted from the powdered leaves by a modified method (added new step: petroleum ether extraction of the acidic aqueous phase) from that reported by Ponglux et al.[42] Kratom powder (450 g) was extracted by refluxing with MeOH (5 × 500 mL) for 40 min. The suspension was filtered after each extraction, and the solvent evaporated. The residue was resuspended in 20% acetic acid solution (2 L) and rinsed with petroleum ether (3 × 500 mL). The aqueous layer was then cooled on ice bath and basified (pH ~ 9) with 50% aqueous NaOH solution. The basified suspension was extracted with DCM (4 × 1 L).

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The combined organic layers were dried over Na_2SO_4 and filtered. The solvent was evaporated and the residue purified using flash column chromatography (gradient: 0–50% EtOAc in hexanes). The major constituent was **1** (yield 5.59 ± 0.59 g (1.24%); smaller quantities of speciogynine and paynantheine were also isolated.

(E)-Methyl-2-((2S,3S,12bS)-3-ethyl-8-methoxy-1,2,3,4,6,7,12,12b-octahydroindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (Mitragynine, 1)—IR (NaCl): 3363, 2950, 2796, 1698, 1643, 1570, 1508, 1435, 1310, 1275, 1255, 1148, 1106, 769, 734. ^{1}H NMR (600 MHz, Chloroform-*d*) δ 7.74 (s, 1H), 7.43 (s, 1H), 6.99 (t, J = 7.9 Hz, 1H), 6.90 (d, J = 8.0 Hz, 1H), 6.45 (d, J = 7.7 Hz, 1H), 3.87 (s, 3H), 3.72 (s, 3H), 3.71 (s, 3H), 3.18–3.08 (m, 2H), 3.06–2.99 (m, 2H), 3.00–2.93 (m, 1H), 2.94–2.90 (m, 1H), 2.57–2.42 (m, 3H), 1.83–1.75 (m, 2H), 1.62 (dt, J = 11.5, 3.2 Hz, 1H), 1.24–1.16 (m, 1H), 0.87 (t, J =7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 169.45, 160.75, 154.69, 137.41, 133.90, 121.98, 117.82, 111.67, 108.03, 104.37, 99.91, 61.74, 61.46, 57.94, 55.52, 53.98, 51.57, 40.87, 40.12, 30.14, 24.14, 19.28, 13.07. HRMS calcd for $C_{23}H_{30}N_2O_4$ (MH+), 399.2284; found 399.2285.

(E)-Methyl-2-((2S,3S,7aS)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7a,12b-octahydro indolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (7-OH Mitragynine, 2)—Mitragynine (**1**, 2.00 g, 5.02 mmol) was dissolved in acetonitrile (150 mL), then water (50 mL) was added. The resulting suspension was cooled to 0 °C, and the following solution was added slowly over the course of several minutes: PIFA (2.16 g, 1.1 equiv) in 22 mL acetonitrile. The reaction mixture was stirred at 0 °C for 1 h, then saturated aqueous $NaHCO_3$ solution was added, and the mixture extracted with EtOAc. The organic phase was rinsed with brine (60 mL) and dried over anhydrous Na_2SO_4, and then it was evaporated under reduced pressure. The residue was dissolved in DCM and purified using flash column chromatography (gradient: 0–75% EtOAc in hexanes). The fractions containing the product were evaporated to yield 1075 mg (57%) of **2** as a light brown amorphous powder. IR (NaCl): 3436, 2952, 1702, 1645, 1599, 1487, 1461, 1436, 1270, 1246, 1145, 1078, 795, 738. ^{1}H NMR (600 MHz, Chloroform-*d*) δ 7.44 (s, 1H), 7.34 (t, J = 8.0 Hz, 1H), 7.24 (d, J = 7.6 Hz, 1H), 6.78 (d, J = 8.3 Hz, 1H), 3.91 (s, 3H), 3.80 (s, 3H), 3.70 (s, 3H), 3.31 (dd, J = 11.1, 2.6 Hz, 1H), 3.03 (ddt, J = 11.5, 5.5, 2.8 Hz, 2H), 2.84–2.75 (m, 3H), 2.67 (ddd, J = 12.3, 4.3, 2.6 Hz, 1H), 2.53–2.46 (m, 1H), 1.98–1.93 (m, 1H), 1.87 (ddd, J = 14.6, 12.2, 4.3 Hz, 1H), 1.70–1.54 (m, 3H), 1.26–1.23 (m, 1H), 0.81 (t, J = 7.3 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 181.25, 169.44, 160.94, 156.02, 154.67, 131.50, 126.24, 114.46, 111.38, 109.65, 69.62, 62.00, 60.73, 58.32, 55.92, 51.54, 50.39, 40.67, 39.32, 36.22, 26.38, 19.10, 13.02. HRMS calcd for $C_{23}H_{30}N_2O_5$ (MH+), 415.2233; found 415.2248.

(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-4-methoxy-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (Mitragynine Pseudoindoxyl, 3)—7-OH-mitragynine (**2**, 200 mg, 0.48 mmol) was dissolved in dry toluene (6 mL), and $Zn(OTf)_2$ (350 mg, 2 equiv) was added. The reaction was stirred in a sealed tube for 2 h at 110 °C. To the cooled mixture were added 10 mL sat. aqueous $NaHCO_3$ solution and water (20 mL). Extracted with EtOAc (30 mL). The organic layer was rinsed with brine (20 mL) and dried over anhydrous Na_2SO_4. After evaporation of

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the solvent under reduced pressure, the residue was redissolved in DCM and purified using flash column chromatography (gradient: 1– 5% MeOH in DCM) to yield: 78 mg (39%) of **3** as a yellow amorphous powder. NMR was identical to that reported in the literature.[39] IR (NaCl): 3350, 2947, 2794, 1687, 1615, 1502, 1343, 1269, 1246, 1148, 1079, 757. ^{1}H NMR (600 MHz, Chloroform-*d*) δ 7.32 (t, J = 8.1 Hz, 1H), 7.27 (s, 1H), 6.40 (d, J = 8.1 Hz, 1H), 6.13 (d, J = 8.1 Hz, 1H), 5.13 (s, 1H), 3.89 (s, 3H), 3.66 (s, 3H), 3.62 (s, 3H), 3.15–3.07 (m, 2H), 2.76 (dt, J = 11.9, 3.5 Hz, 1H), 2.38–2.29 (m, 2H), 2.29–2.18 (m, 1H), 2.14 (dt, J = 10.2, 6.3 Hz, 1H), 1.93–1.84 (m, 1H), 1.63 (dt, J = 11.3, 6.8 Hz, 1H), 1.49 (d, J = 11.3 Hz, 1H), 1.18 (ddd, J = 13.2, 7.8, 2.9 Hz, 1H), 1.11 (dd, J = 11.3, 3.6 Hz, 1H), 0.84 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 199.73, 169.05, 162.27, 160.40, 158.74, 138.85, 111.85, 109.96, 103.95, 99.21, 75.37, 73.38, 61.61, 55.86, 54.96, 53.35, 51.36, 40.28, 38.57, 35.25, 23.95, 19.47, 13.11. HRMS calcd for $C_{23}H_{30}N_2O_5$ (MH+), 415.2233; found 415.2216.

(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-4-hydroxy-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-OH-Corynantheidine Pseudoindoxyl, 4)—7-OH mitragynine (**2**, 400 mg, 0.97 mmol) was dissolved in dry DCM (20 mL), then $AlCl_3$ (1.29 g, 10 equiv) was added. The mixture was cooled to °0 C, and ethanethiol (1.39 mL, 20 equiv) was added. The mixture was stirred at rt for 5 h. Water (30 mL) was slowly added, then it was separated from the organic layer. The organic layer was rinsed with brine (30 mL), then separated and dried over Na_2SO_4. Evaporated under reduced pressure. The residue was redissolved in DCM and purified using flash column chromatography (gradient: 1– 3% MeOH in DCM) to yield 342 mg (89%) of **4** as a bright yellow amorphous powder. IR (NaCl): 3211, 2945, 1697, 1628, 1513, 1451, 1348, 1247, 1148, 1120, 1082, 744. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.29 (m, 2H), 6.30 (d, J = 8.1 Hz, 1H), 6.15 (d, J = 8.0 Hz, 1H), 5.12 (s, 1H), 3.67 (s, 3H), 3.63 (s, 3H), 3.17–3.11 (m, 2H), 2.83–2.75 (m, 1H), 2.36–2.29 (m, 2H), 2.26–2.20 (m, 2H), 2.14 (dd, J = 11.2, 2.8 Hz, 1H), 1.98–1.90 (m, 1H), 1.69–1.57 (m, 2H), 1.51 (d, J = 11.2 Hz, 1H), 1.24–1.16 (m, 1H), 1.14–1.08 (m, 1H), 0.86 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 203.60, 169.03, 160.52, 160.05, 157.31, 140.24, 111.63, 108.97, 103.68, 102.27, 75.49, 72.99, 61.69, 54.93, 53.32, 51.41, 40.21, 38.63, 34.79, 23.97, 19.43, 13.07. HRMS calcd for $C_{22}H_{28}N_2O_5$ (MH+), 401.2076; found 401.2068.

(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-3-oxo-4-(trifluoromethylsulfonyloxy)-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-O-Trifluoro Methane Sulfonyl Corynantheidine Pseudoindoxyl, 5)—4 (200 mg, 0.5 mmol) was dissolved in dry DCM (15 mL), and pyridine (647 uL, 16 equiv) was added. Then the solution was cooled to −40 °C on a dry ice acetone bath, and the following solution was slowly added over 2–3 min: 5 mL DCM and triflic anhydride (340 uL, 4 equiv). The reaction was stirred for 1 h at −40 °C. After warming up to rt, the solution was purified using flash column chromatography immediately (gradient: 20–75% EtOAc in hexanes) to yield 233 mg (83%) of **5** as a brown amorphous solid. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.41 (t, J = 8.1 Hz, 1H), 7.29 (s, 1H), 6.84 (d, J = 8.3 Hz, 1H), 6.52 (d, J = 7.9 Hz, 1H), 5.58 (s, 1H), 3.68 (s, 3H), 3.63 (s, 3H), 3.15 (d, J = 9.7 Hz, 2H), 2.80 (dd, J = 12.5, 3.5 Hz, 1H), 2.35 (d, J = 7.4 Hz, 2H), 2.27 (d, J = 8.8 Hz, 2H),

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2.18–2.13 (m, 1H), 1.66–1.58 (m, 1H), 1.52 (d, J = 10.9 Hz, 1H), 1.22–1.16 (m, 2H), 0.85 (t, J = 7.3 Hz, 4H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 198.15, 171.40, 168.92, 161.46, 160.55, 145.34, 138.13, 111.73, 110.01, 73.58, 61.69, 60.62, 54.84, 53.65, 53.33, 51.38, 40.14, 38.33, 35.12, 23.84, 21.28, 19.41, 14.41, 13.00. HRMS calcd for $C_{23}H_{27}F_3N_2O_7S$ (MH+), 533.1569; found 533.1547.

(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizin]-7′-yl)-3-methoxyacrylate (Corynantheidine Pseudoindoxyl, 6)—5 (10 mg, 0.019 mmol) was dissolved in dry DMF (500 uL) in a sealed tube, and the following reagents were added: $Pd(OAc)_2$ (1.4 mg, 0.3 equiv), dppp (4 mg, 0.5 equiv), triethylamine (52.4 uL, 20 equiv), and formic acid (1 uL, 1.8 equiv). The mixture was stirred at 60 °C for 1h. The reaction mixture was diluted with EtOAc (10 mL) and washed with brine (5 mL) 5 times. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was redissolved in DCM and purified using preparative TLC (75% EtOAc in hexanes) to yield 4 mg (57%) of **6** as an amorphous solid. IR (NaCl): 3286, 2927, 2360, 1676, 1620, 1437, 1248, 1200, 1138, 755. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.55 (d, J = 7.7 Hz, 1H), 7.43–7.38 (m, 1H), 7.27 (d, J = 6.7 Hz, 1H), 6.85 (d, J = 8.3 Hz, 1H), 6.77–6.72 (m, 1H), 5.20 (s, 1H), 3.66 (s, 3H), 3.62 (s, 3H), 3.20–3.12 (m, 2H), 2.82–2.76 (m, 1H), 2.38–2.29 (m, 2H), 2.25 (s, 2H), 2.20–2.13 (m, 1H), 1.97–1.89 (m, 1H), 1.69–1.61 (m, 1H), 1.55–1.48 (m, 1H), 1.23–1.17 (m, 1H), 1.05 (s, 1H), 0.86 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 202.60, 169.02, 160.95, 160.51, 137.30, 124.61, 118.30, 111.85, 111.68, 75.22, 73.52, 61.67, 54.94, 53.42, 51.40, 40.20, 38.59, 35.13, 23.92, 21.29, 19.41, 13.06. HRMS calcd for $C_{22}H_{28}N_2O_4$ (MH+), 385.2127; found 385.2120.

(E)-Methyl-2-((1′S,6′S,7′S)-4-cyano-6′-ethyl-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-Cyano Corynantheidine Pseudoindoxyl, 7)—5 (20 mg, 0.038 mmol) was dissolved in dry DMF (500 uL) in a sealed tube, and $Pd(PPh_3)_4$ (4.3 mg, 0.1 equiv) and $Zn(CN)_2$ (8.8 mg, 2 equiv) were added. The reaction mixture was stirred at 80 °C for 3 h. After 3 h, the reaction was diluted with EtOAc (20 mL) and washed with brine 5 times. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was redissolved in DCM and purified using flash column chromatography (gradient: 20– 80% EtOAC in hexanes) to yield 10 mg (65%) of **7** as an amorphous brown solid. IR (NaCl): 3355, 2940, 2233, 1699, 1606, 1501, 1438, 1242, 993, 859, 1082, 776. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.47–7.43 (m, 1H), 7.28 (s, 1H), 7.07 (d, J = 8.4 Hz, 1H), 7.05 (d, J = 7.2 Hz, 1H), 5.43 (s, 1H), 3.68 (s, 3H), 3.62 (s, 3H), 3.15 (t, J = 10.1 Hz, 2H), 2.79 (dt, J = 12.8, 3.5 Hz, 1H), 2.40–2.33 (m, 2H), 2.31–2.27 (m, 1H), 2.23 (t, J = 12.1 Hz, 1H), 2.19–2.13 (m, 1H), 1.96–1.90 (m, 1H), 1.66–1.59 (m, 1H), 1.51 (d, J = 11.2 Hz, 1H), 1.23–1.16 (m, 1H), 1.07 (d, J = 12.7 Hz, 1H), 0.85 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 201.47, 169.06, 162.01, 160.51, 142.35, 137.85, 136.72, 129.43, 128.16, 127.88, 120.06, 111.79, 110.73, 75.10, 73.89, 61.70, 55.07, 53.59, 51.40, 40.28, 38.57, 36.87, 36.86, 35.53, 24.92, 24.08, 19.48, 13.08. HRMS calcd for $C_{23}H_{27}N_3O_4$ (MH+), 410.2080; found 410.2068.

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(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-3-oxo-4-phenyl-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-Phenyl Corynantheidine Pseudoindoxyl, 8)—5 (75 mg, 0.14 mmol) was dissolved in dry toluene (0.5 mL), and the solvent was removed under reduced pressure to ensure azeotropic removal of water residues. Dry methanol (1 mL) and dry toluene (1.5 mL) were added. To the resulting solution were added phenylboronic acid (19 mg, 1.1 equiv), K_2CO_3 (38.9 mg, 2 equiv), and $Pd(PPh_3)_4$ (8.1 mg, 0.05 equiv). The mixture was stirred at 80 °C for 2 h. The solvent was evaporated under reduced pressure, and the residue suspended in DCM, rinsed with water and brine (20 mL), then the organic layer was dried over Na_2SO_4 and was evaporated. Purified using flash column chromatography (gradient: 20–50% EtOAc in hexanes) to yield: 21 mg (32%) of **8** as a yellow amorphous solid. IR (NaCl): 3364, 2936, 2360, 1698, 1600, 1483, 1436, 1233, 1150, 759. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.52–7.49 (m, 2H), 7.43–7.34 (m, 4H), 7.30 (s, 1H), 6.82 (d, J = 8.2 Hz, 1H), 6.68 (d, J = 7.3 Hz, 1H), 5.26 (s, 1H), 3.69 (s, 3H), 3.63 (s, 3H), 3.15 (d, J = 10.0 Hz, 2H), 2.79 (dt, J = 13.0, 3.3 Hz, 1H), 2.34–2.25 (m, 3H), 2.21 (s, 1H), 2.17–2.10 (m, 1H), 1.95–1.88 (m, 1H), 1.69–1.63 (m, 1H), 1.53–1.47 (m, 1H), 1.24–1.18 (m, 1H), 1.14–1.09 (m, 1H), 0.86 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 201.47, 169.06, 162.01, 160.51, 142.35, 137.85, 136.72, 129.43, 128.16, 127.88, 120.06, 111.79, 110.73, 75.10, 73.89, 61.70, 55.07, 53.59, 51.40, 40.28, 38.57, 36.87, 36.86, 35.53, 24.92, 24.08, 19.48, 13.08. HRMS calcd for $C_{28}H_{32}N_2O_4$ (MH +), 461.2440; found 461.2422.

(E)-Methyl-2-((1′S,6′S,7′S)-6′-ethyl-4-(furan-3-yl)-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-Furyl Corynantheidine Pseudoindoxyl, 9)—The procedure described for the synthesis of **8** was used. Instead of phenylboronic acid, (furan-3-yl)boronic acid was employed. Yield: 81%. Compound **9** is a bright yellow amorphous powder. IR (NaCl): 3358, 2954, 2795, 2360, 2341, 1691, 1604, 1437, 1316, 1238, 1152, 796, 772. ^{1}H NMR (600 MHz, Chloroform-*d*) δ 8.49 (d, J = 1.4 Hz, 1H), 7.48 (d, J = 1.7 Hz, 1H), 7.40 (t, J = 7.8 Hz, 1H), 7.31 (s, 1H), 6.87 (d, J = 7.5 Hz, 1H), 6.84 (dd, J = 1.9, 0.9 Hz, 1H), 6.77 (d, J = 8.1 Hz, 1H), 5.22 (s, 1H), 3.70 (s, 3H), 3.65 (s, 3H), 3.22–3.14 (m, 2H), 2.81 (dt, J = 12.6, 3.7 Hz, 1H), 2.40–2.32 (m, 2H), 2.32–2.22 (m, 2H), 2.17 (dd, J = 11.5, 3.2 Hz, 1H), 2.00–1.92 (m, 1H), 1.75–1.65 (m, 1H), 1.57–1.51 (m, 1H), 1.23 (dtd, J = 15.1, 7.4, 2.7 Hz, 1H), 1.15–1.10 (m, 1H), 0.89 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 201.73, 169.03, 162.40, 160.45, 143.85, 142.48, 137.00, 132.42, 122.74, 118.06, 116.66, 111.77, 110.60, 110.18, 75.11, 74.00, 61.65, 55.07, 53.54, 51.38, 40.28, 38.61, 35.68, 24.01, 19.50, 13.11. HRMS calcd for $C_{26}H_{30}N_2O_5$ (MH+), 451.2433; found 451.2215.

(E)-Methyl-2-((1′S,6′S,7′S)-4-acetoxy-6′-ethyl-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (9-O-Acetyl Corynantheidine Pseudoindoxyl, 10)—4 (20 mg, 0.05 mmol) was dissolved in pyridine (0.5 mL), and acetic anhydride (80 uL) was added. The mixture was stirred at rt for 2 h. The solution was poured into sat. aqueous $NaHCO_3$ solution and extracted with DCM (30 mL). The organic layer was separated, rinsed with brine (10 mL), dried over Na_2SO_4, and evaporated under reduced pressure. The residue was redissolved in DCM and purified using flash column chromatography (gradient: 0–5% MeOH in DCM) to yield 13 mg (59%)

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of **10** as a bright yellow amorphous solid. IR (NaCl): 3393, 2956, 2874, 2787, 1761, 1688, 1628, 1503, 1239, 1217, 910, 764. ^{1}H NMR (500 MHz, Chloroform-*d*) δ 7.38 (t, J = 8.0 Hz, 1H), 7.28 (s, 1H), 6.71 (d, J = 8.2 Hz, 1H), 6.36 (d, J = 7.6 Hz, 1H), 5.73 (s, 1H), 3.68 (s, 3H), 3.62 (s, 3H), 3.20 (dd, J = 8.6, 3.0 Hz, 2H), 2.79 (dd, J = 12.6, 3.9 Hz, 1H), 2.36 (s, 3H), 2.33–2.22 (m, 3H), 2.22–2.14 (m, 1H), 1.95 (dt, J = 15.8, 6.3 Hz, 1H), 1.63 (dt, J = 13.0, 6.1 Hz, 1H), 1.53 (d, J = 11.6 Hz, 1H), 1.30–1.16 (m, 2H), 1.16–1.09 (m, 1H), 0.86 (t, J = 7.3 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 199.57, 169.14, 168.98, 161.57, 160.47, 148.07, 138.28, 113.07, 111.70, 110.70, 109.35, 75.71, 73.70, 61.67, 55.02, 53.49, 51.38, 40.25, 38.51, 35.16, 23.91, 21.01, 19.49, 13.08. HRMS calcd for $C_{24}H_{30}N_2O_6$ (MH+), 443.2182; found 443.2174.

(E)-Methyl 2-((1′S,6′S,7′S)-1-benzyl-6′-ethyl-4-methoxy-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (N-Benzyl Mitragynine Pseudoindoxyl, 11)—Compound **3** was dissolved in dry acetonitrile (0.5 mL), and NaH (6 mg, 10 equiv) was added. The resulting suspension was stirred at rt for 30 min, during which its color turned red. Benzyl bromide (7.2 uL, 2.5 equiv) was added, and the mixture stirred for 2 h at rt. The red color disappeared promptly after the addition of benzyl bromide. After the reaction time, the mixture was carefully poured into cold water (20 mL) and extracted with DCM (30 mL). The organic layer was rinsed with brine (5 mL), separated, dried over Na_2SO_4, and evaporated under reduced pressure. The residue was redissolved in DCM and purified using flash column chromatography (gradient: 0–5% MeOH in DCM) to yield: 7.8 mg (64%) of **11** as a bright yellow amorphous solid. IR (NaCl): 2940, 2794, 1694, 1610, 1497, 1337, 1265, 1239, 1078, 732. ^{1}H NMR (500 MHz, Chloroform-*d*) δ 7.42–7.28 (multiple overlapping peaks, 5H), 7.25–7.17 (multiple overlapping peaks, 2H), 6.08 (d, J = 8.1 Hz, 1H), 6.04 (d, J = 8.3 Hz, 1H), 5.35 (d, J = 17.4 Hz, 1H), 4.73 (d, J = 17.3 Hz, 1H), 3.90 (s, 3H), 3.70 (s, 3H), 3.64 (s, 3H), 3.13 (dd, J = 11.1, 2.2 Hz, 1H), 3.02 (t, J = 8.1 Hz, 1H), 2.78–2.72 (m, 1H), 2.33–2.18 (m, 4H), 2.09–2.02 (m, 1H), 1.94 (dt, J =13.7, 8.5 Hz, 1H), 1.66 (dt, J = 19.2, 6.8 Hz, 1H), 1.50 (d, J = 10.9 Hz, 1H),, 1.13 (d, J = 11.8 Hz, 1H), 0.86 (t, J = 7.4 Hz, 3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 199.55, 169.06, 161.60, 160.40, 158.81, 138.88, 138.82, 129.25, 129.02, 128.76, 128.64, 126.95, 126.51, 111.99, 108.49, 101.82, 98.07, 61.63, 55.89, 55.23, 53.92, 51.38, 47.61, 40.46, 38.67, 33.80, 32.06, 24.18, 19.82, 13.13. HRMS calcd for $C_{30}H_{36}N_2O_5$ (MH +), 505.2702; found 505.2726.

(E)-Methyl 2-((1′S,6′S,7′S)-6′-ethyl-4-methoxy-1-methyl-3-oxo-3′,5′,6′,7′,8′,8a′-hexahydro-2′H-spiro[indoline-2,1′-indolizine]-7′-yl)-3-methoxyacrylate (N-Methyl Mitragynine Pseudoindoxyl, 12)—The procedure described for the synthesis of **11** was used. Instead of benzyl bromide, iodomethane was employed. Yield: 58%. Compound **12** is a bright yellow amorphous powder. IR (NaCl): 2947, 2778, 2361, 1687, 1611, 1500, 1337, 1273. 1H NMR (600 MHz, Chloroform-*d*) δ 7.34 (t, J = 8.1 Hz, 1H), 7.28 (s, 1H), 6.25 (d, J =8.2 Hz, 1H), 6.05 (d, J = 8.0 Hz, 1H), 3.88 (s, 3H), 3.69 (s, 3H), 3.62 (s, 3H), 3.17 (s, 3H), 3.16–3.10 (m, 2H), 2.74 (dt, J = 13.0, 3.7 Hz, 1H), 2.34 (q, J = 8.9 Hz, 1H), 2.26 (dd, J = 11.3, 2.6 Hz, 1H), 2.21–2.12 (m, 2H), 2.08–2.03 (m, 1H), 1.92 (dt, J = 13.8, 8.7 Hz, 1H), 1.69–1.62 (m, 1H), 1.48 (dt, J = 11.1, 3.1 Hz, 1H), 1.28–1.23 (m, 1H), 1.20 (ddd, J = 13.3, 7.6, 2.8 Hz, 1H), 1.08 (dt, J = 13.0, 3.1 Hz, 1H), 0.84 (t, J = 7.4 Hz,

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3H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 199.87, 169.08, 162.01, 160.36, 158.90, 139.01, 112.04, 108.08, 100.37, 97.46, 78.27, 74.62, 61.59, 55.86, 55.14, 53.88, 51.34, 40.54, 38.72, 31.99, 30.02, 24.29, 19.71, 13.17. HRMS calcd for $C_{24}H_{32}N_2O_5$ (MH+), 429.2389; found 429.2393.

Supplementary Material

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ABBREVIATIONS USED

%MPE	percent maximal effect
6TM	six transmembrane
7TM	seven transmembrane
CHO	Chinese hamster ovary
CPA	conditioned place aversion
CPP	conditioned place preference
DAMGO	[D-Ala^2, *N*-MePhe^4, Gly-ol^5]-enkephalin
DCM	dichloromethane
DMF	*N,N*-dimethylformamide
DOR-1	a delta opioid receptor (*Oprd1*) clone
DPDPE	[D-Pen^2, D-Pen^5]Enkephalin
dppp	1,3-bis(diphenylphosphino)-propane
GDP	guanosine diphosphate
KO	knockout
KOR-1	a kappa opioid receptor (*Oprk1*) clone
MOR-1	a mu opioid receptor (*Oprm1*) clone

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norBNI	norbinaltorphimine
NTI	naltrindole
PIFA	[bis(trifluoroacetoxy)iodo]benzene

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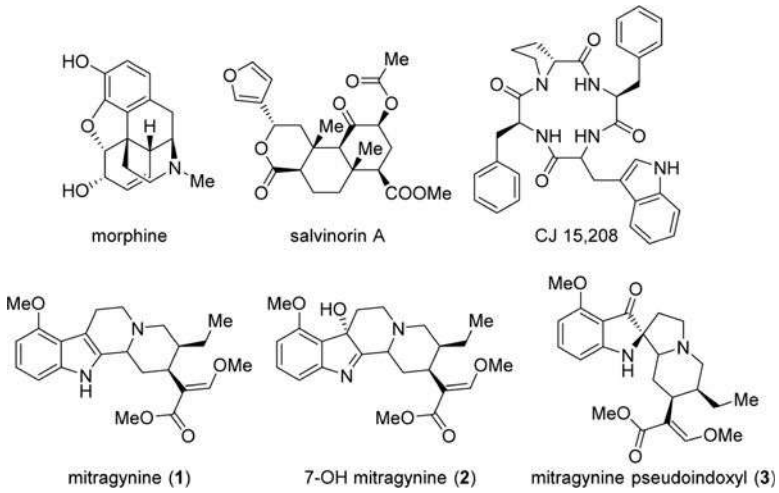


Figure 1.
Structure of several important opioid natural product analogs.

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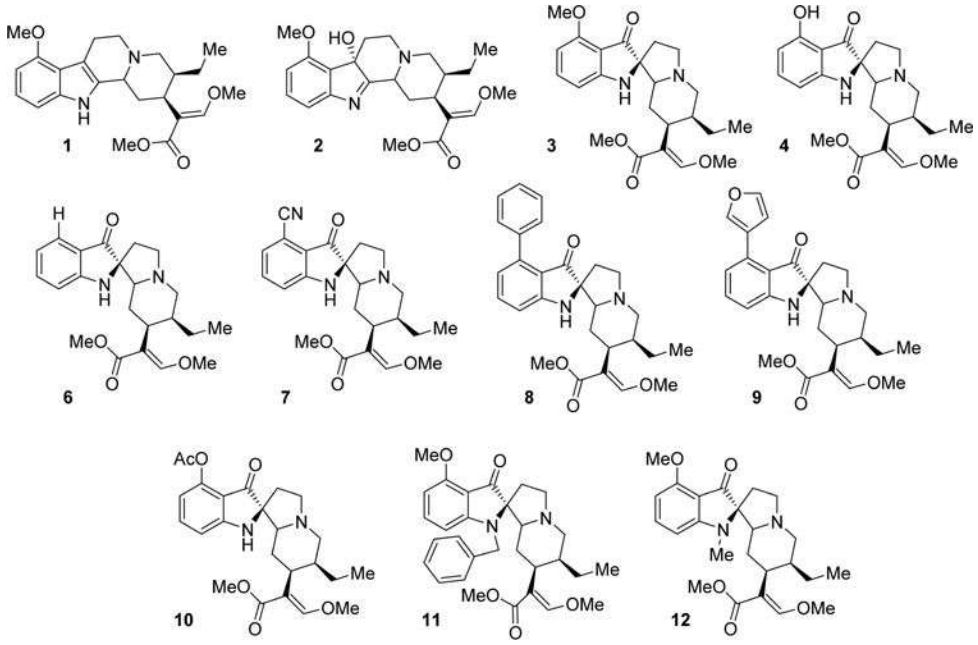


Figure 2.
Structure of the studied mitragynine analogs.

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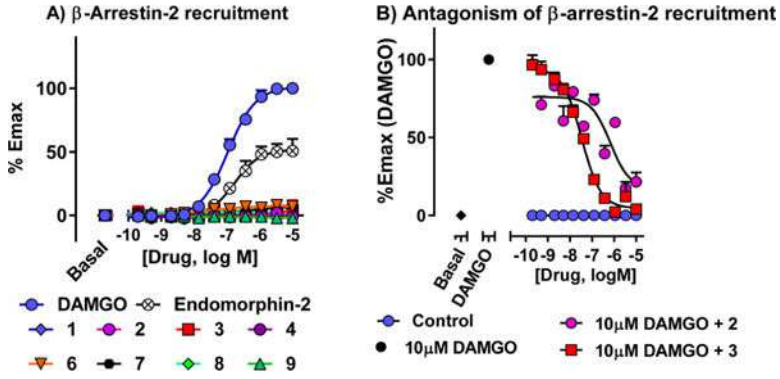


Figure 3.

β-arrestin-2 recruitment and antagonism of *β*-arrestin-2 recruitment. (A) *β*-arrestin-2 recruitment: *β*-arrestin-2 recruitment was determined using the DiscoveRx PathHunter enzyme complementation assay using modified MOR-1 in CHO cells. Compounds were found to be completely G-protein biased. (B) Antagonism of *β*-arrestin-2 recruitment: The same cells were incubated with the antagonist (**2**, **3**) for 30 min at 37 °C prior to the addition of agonist (10 *μ*M DAMGO) at MOR-1. Compounds **2** and **3** were able to antagonize *β*-arrestin-2 recruitment by DAMGO. IC_{50} values: **2**: 725 ± 292 nM; **3**: 34 ± 2 nM.

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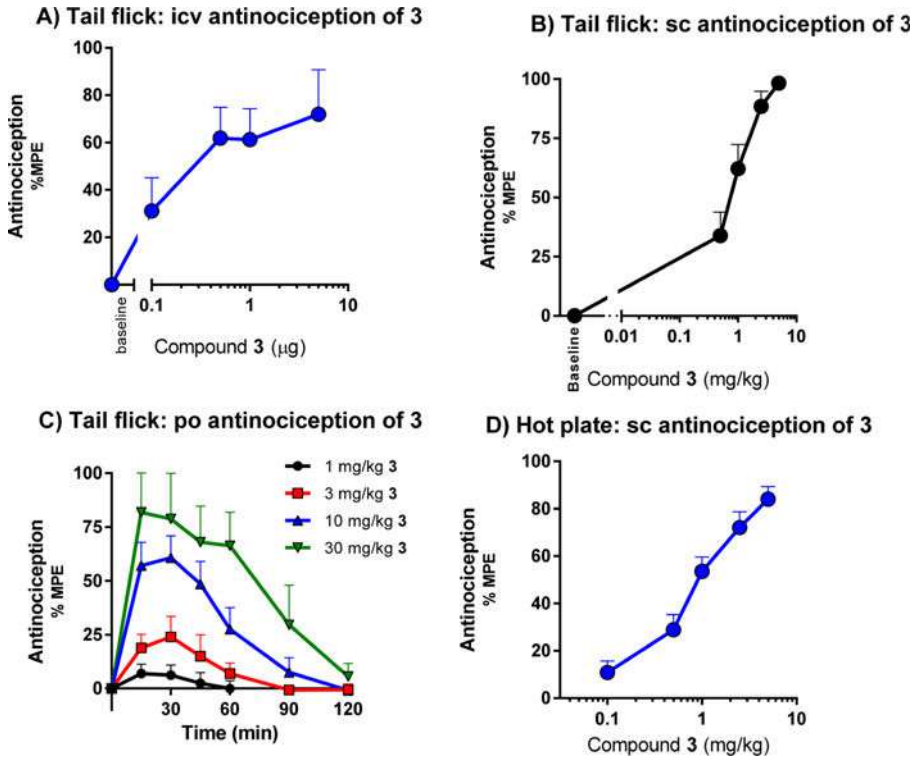


Figure 4.

Antinociception of compound **3** upon intracerebroventricular, subcutaneous, and oral administration. (A) Dose–response curves of antinociception of **3** and morphine given supraspinally in CD1 mice. Two independent determinations of the cumulative dose–response curves were performed on groups of mice (n = 5) for antinociception in the tail flick assay with **3** intracerebroventricularly. Animals were tested 15 min later at peak effect to generate the analgesic dose–response curve. Each point represents mean ± SEM for 10 mice. ED_{50} values (and 95% CI) were 0.38 (0.18- 0.81) μg for **3**. (B) Dose–response curves of antinociception of **3** given subcutaneously in CD1 mice. Three independent determinations of the cumulative dose–response curves were performed on groups of mice (n = 10) for antinociception in the tail flick assay, 30 mice in total. ED_{50} (and 95% CI) = 0.76 (0.56–0.83) mg/kg. (C) Time course of tail flick antinociception of **3** given orally in CD1 male mice. Groups of mice (n = 10) were given different doses of **3** orally by gavage and tested for analgesic response at the indicated time points. ED_{50} (and 95% CI) = 7.5 (4.3–13) mg/kg. (D) Hot plate. Groups of CD1 mice (n = 10) were assessed for antinociception of **3** at peak effect in two independent experiments (n = 20 total) in a cumulative dose–response paradigm. Analgesia was determined using a 55 °C hot plate, where the latency to respond with a hind paw lick or shake/flutter, whichever came first, was recorded. ED_{50} (and 95% CI) = 0.99 mg/kg (0.75–1.3).

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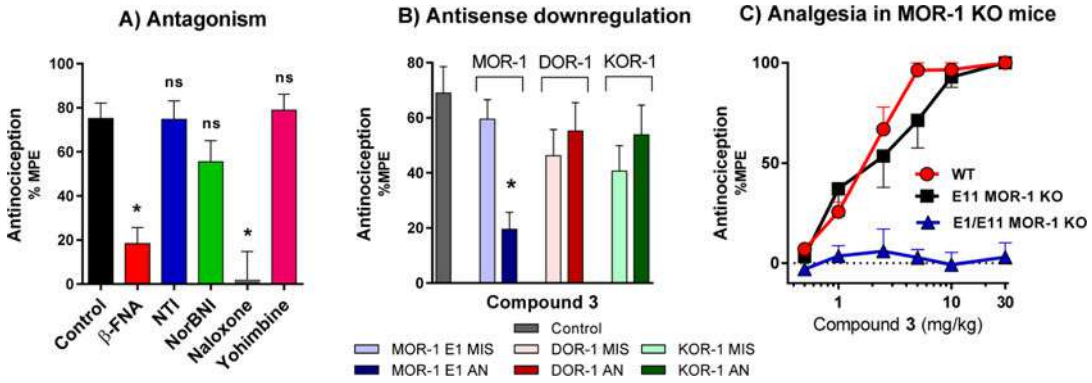


Figure 5.

Pharmacological and genetic reversal of antinociception of **3**. (A) Reversal of antinociception by selective antagonists. Groups of CD1 mice ($n = 10$) received **3** (1.5 mg/kg sc) and the indicated antagonist. β-Funaltrexamine (β-FNA; 40 mg/kg sc) and norbinaltorphimine (norBNI; 10 mg/kg sc) were administered 24 h before agonist testing. Naltrindole (NTI; 0.5 mg/kg sc), naloxone (1 mg/kg), and yohimbine (10 mg/kg) were administered 15 min before **3**. All antinociception testing was performed 15 min after the administration of **3**. Similar results were observed in two independent replications. **3** antinociception is insensitive to NTI, norBNI, and yohimbine, whereas antinociception is antagonized by β-FNA and naloxone (two-way ANOVA followed by Bonferroni post hoc comparisons test, $p < 0.05$). All values are expressed as the mean ± SEM. (B) Antisense oligodeoxynucleotide injection: Groups of mice ($n = 15$) received the stated antisense (5–10 μg) or mismatch (5–10 μg) oligodeoxynucleotide icv under light isoflurane anesthesia on days 1, 3, and 5. Tail flick antinociception was tested on day 6. Control groups received no injection prior to testing. On test day, mice received **3** (1.5 mg/kg, sc). All experiments were performed 3 times with similar results observed with each determination. Analgesic response of **3** was only affected in mu receptor downregulated mice (MOR-1 AN) (one-way ANOVA followed by Bonferroni post hoc comparisons test). *Significantly different from control ($p < 0.05$). Data for agonist controls are shown in the Figure S6 and sequences of AN and MIS oligos are shown in Table 4. (C) Antinociception of **3** in wild-type, exon 11 KO, and exon 1/exon 11 double KO C57 mice. Two independent determinations of the cumulative dose–response curves were performed on groups of mice ($n = 5$) for antinociception in the tail flick assay with **3** given subcutaneously. Compound **3** displayed similar antinociceptive effects in wild-type (ED_{50} = 0.83 mg/kg (0.37–1.9)) and exon 11 KO mice (ED_{50} = 1.4 mg/kg (0.34–5.8)), however, no antinociception was observed in exon 1/exon 11 double KO mice, suggesting that the antinociceptive effect of **3** is mediated by the E1MOR-1 variants.

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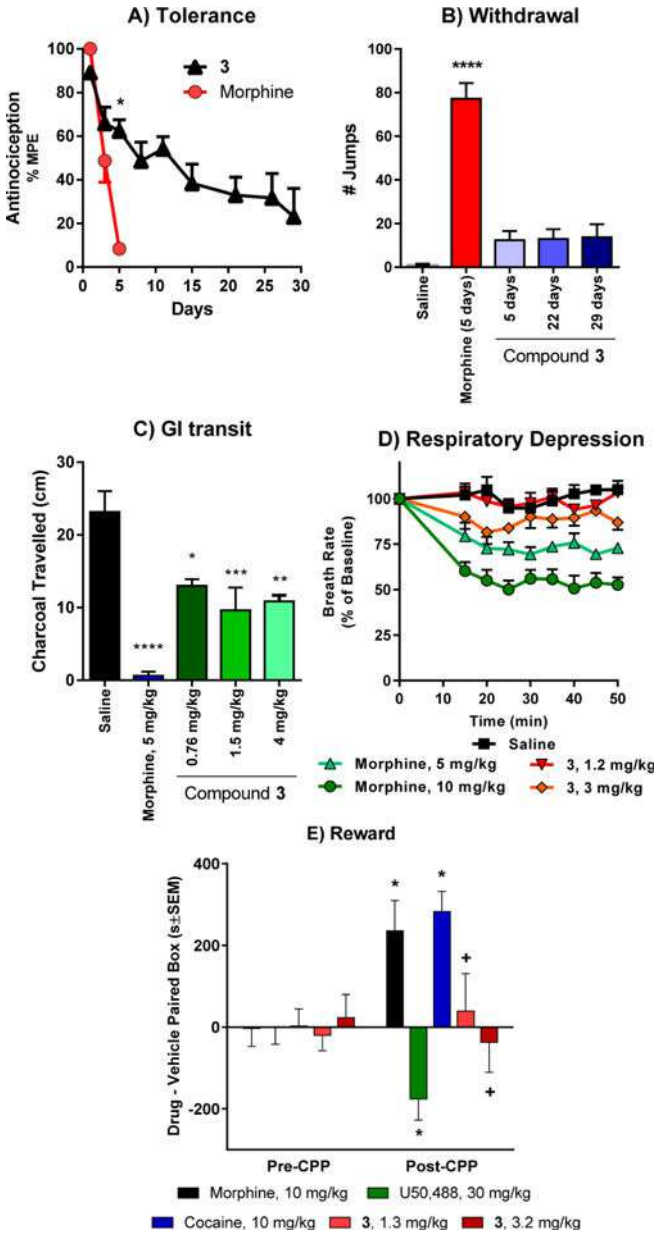


Figure 6.

Side effect studies with compound **3**. (A) Antinociceptive tolerance: Mice were dosed 2× daily with 2× antinociceptive ED_{50} with either morphine or **3** until they showed complete analgesic tolerance. **3** showed very slow onset of tolerance compared with morphine. *Significantly different from morphine ($p < 0.05$). The experiment was replicated at least twice with similar results. (B) Physical dependence: Groups of mice were dosed 2× daily with 2× antinociceptive ED_{50} with morphine or **3**. Separate groups of mice were used for the 5, 22, and 29 day treatment with **3** and the animals within each group were sacrificed following the experiment with naloxone. Animals were challenged with naloxone (1 mg/kg) on day 5 of the morphine group, and days 5, 22, and 29 of the **3** groups. Number of jumps was counted over a 15 min period postinjection. The response of mice treated with **3** on

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either day was not significantly greater than that of mice treated with saline. *Significantly different from saline. (One-way ANOVA followed by Dunnett's multiple comparison test, $p < 0.05$). (C) Gastrointestinal transit. Groups of mice ($n = 10$) received saline, morphine (5 mg/kg), or **3** (1.5 and 4 mg/kg) before receiving an oral dose of 0.2 mL of charcoal meal by gavage. Animals were sacrificed 30 min later, and the distance traveled by charcoal was measured. **3** lowered transit significantly compared with saline ($P < 0.05$) but less than morphine at both doses ($P < 0.05$) as determined by ANOVA followed by Tukey's multiple-comparison test. The inhibition of gastrointestinal transit seems to plateau even at doses ~5× higher than the antinociceptive ED_{50}. (D) Respiratory rate. Animals were randomly assigned to receive saline ($n = 5$), **3** (1.2 and 3 mg/kg, sc, $n = 5$ at each dose) or morphine (5 and 10 mg/kg, $n = 5$ at each dose). While morphine caused respiratory depression at both 2× and 5× antinociceptive ED_{50} dose (5 and 10 mg/kg, respectively), **3** did not depress respiratory rate at ~2× antinociceptive ED_{50} dose (1.2 mg/kg) and was not significantly different from saline at any time point, whereas morphine (5 mg/kg) decreased respiratory depression in comparison with **3** ($p < 0.05$) as determined by repeated-measures ANOVA followed by Tukey's multiple-comparison test. However, at ~5× antinociceptive ED_{50} dose (3 mg/kg), **3** showed signs of respiratory depression albeit significantly less than the equianalgesic dose of morphine (10 mg/kg) at any given time point (repeated measures ANOVA followed by Tukey's multiple-comparison test). (E) Conditioned place preference and aversion. Compound **3** alone did not produce conditioned place preference or aversion. After determination of initial preconditioning preferences, mice were place conditioned daily for 2 days with morphine (10 mg/kg/d, ip), U50,488 (30 mg/kg/d, ip), cocaine (10 mg/kg/d, ip), or **3** (1.3 mg/kg/d, ip and 3.2 mg/kg/d, ip). Mean difference in time spent on the drug-paired side ± SEM is presented ($n = 17–21$). * Significantly different from matching preconditioning preference ($p < 0.05$); + significantly different from cocaine, morphine, and U50,488 preference (two-way repeated measures ANOVA with Sidak's post hoc test).

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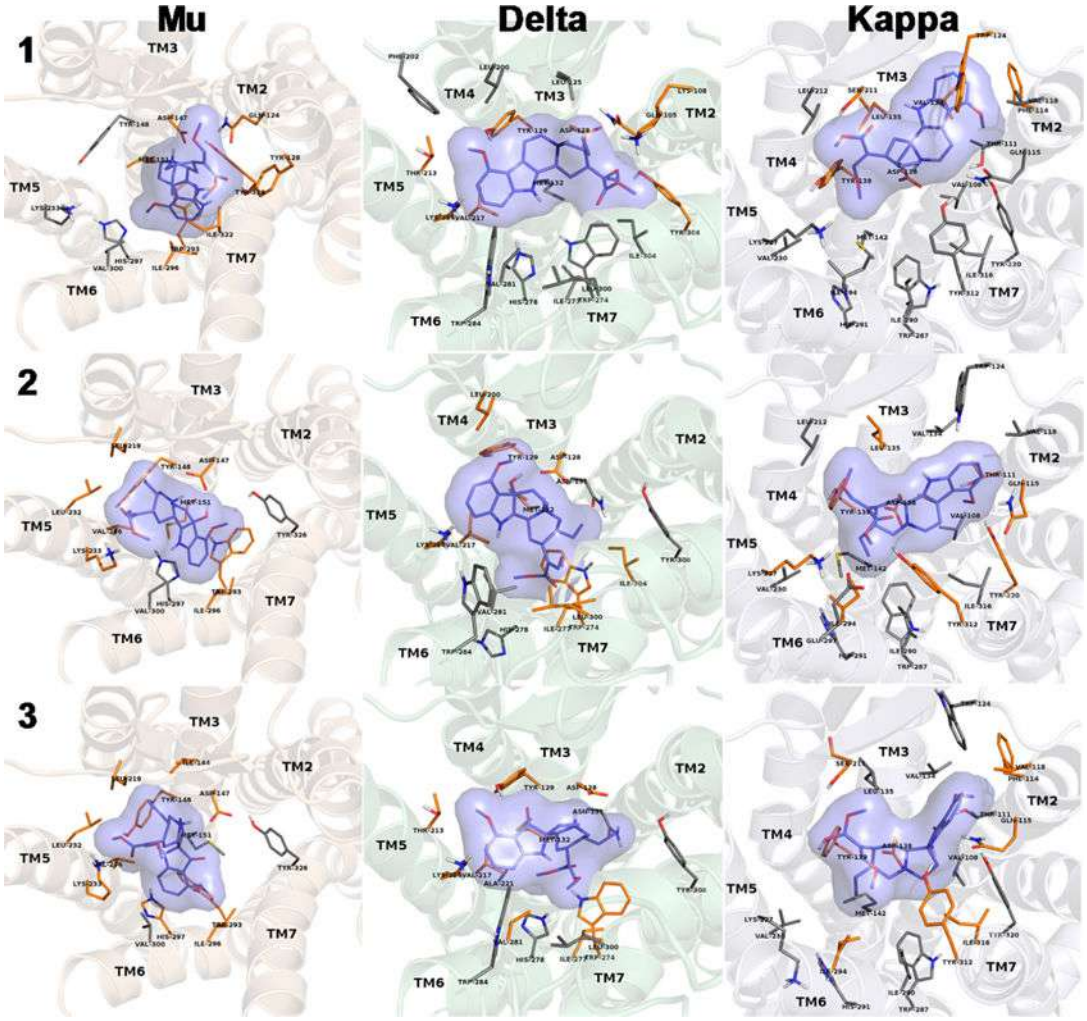
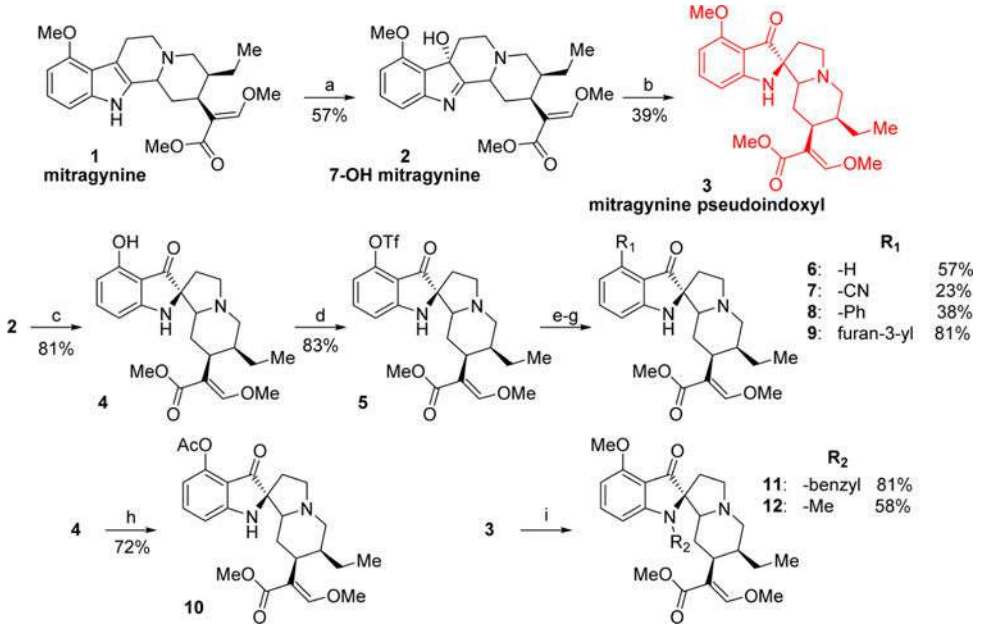


Figure 7.

Lowest energy docked complexes of specifically bound mitragynine derivatives (**1**–**3**) to opioid receptors. Binding pocket side chains which were identified in the crystal structures to take part in receptor ligand interactions and were kept flexible during docking are shown. Side chains which are in direct contact with the ligands are depicted in orange. Nonpolar hydrogens are omitted for clarity.

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Scheme 1. Synthesis of Mitragynine Derivatives 2–12[a]

[a]Reagents and conditions: (a) PIFA, H_2O, acetonitrile, 0 °C, 1 h; (b) $Zn(OTf)_2$, toluene, 110 °C, 2 h; (c) $AlCl_3$, EtSH, DCM, 0 °C, 5 h; (d) Tf_2O, pyridine, DCM, −40 °C, 1 h; (e, yielding **6**) $Pd(OAc)_2$, dppp, HCOOH, DMF, 60 °C, 1 h; (f, yielding **7**) $Zn(CN)_2$, $Pd(PPh_3)_4$, DMF, 80 °C, 2 h; (g, yielding **8** and **9**) phenylboronic acid (**8**) or 3-furanylboronic acid (**9**), $Pd(PPh_3)_4$, K_2CO_3, MeOH, toluene, 80 °C, 2 h; (h) Ac_2O, pyridine, rt, 1 h; (i, yielding **11** and **12**) benzyl bromide (**11**) or iodomethane (**12**), NaH, acetonitrile, rt, 2 h.

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Table 1

Receptor Affinities and Subcutaneous Antinociception in Mice

	affinity (K_i nM)[a]			
compd	**MOR-1**	**KOR-1**	**DOR-1**	**antinociception sc,[b] mg/kg (CI)**
1	230 ± 47	231 ± 21	1011 ± 49	166 (101–283)
2	37 ± 4	132 ± 7	91 ± 8	0.46 (0.39–0.71)
3	0.8 ± 0.2	24 ± 0.9	3.0 ± 1.3	0.76 (0.56–0.83)
4	1.4 ± 0.2	170 ± 61	6.1 ± 1.1	0.18 (0.16–0.20)
6	0.46 ± 0.01	19 ± 4.7	2.9 ± 0.29	0.24 (0.20–0.26)
7	0.5 ± 0.01	47 ± 3.3	2.4 ± 0.3	0.32 (0.26–0.41)
8	0.91 ± 0.06	51 ± 9.7	0.8 ± 0.13	1.0 (0.70–1.35)
9	0.94 ± 0.02	39 ± 11	1.5 ± 0.37	1.1 (0.83–1.38)
10	2.5 ± 0.6	31 ± 14	20 ± 1	0.38 (0.29–0.49)
11	249 ± 41	136 ± 8	258 ± 34	
12	375 ± 138	>1000	>1000	
DAMGO	3.3 ± 0.43[c]	–	–	
U50,488H	–	0.73 ± 0.32[c]	–	
DPDPE	–	–	1.39 ± 0.67[c]	
NTI	–	–	0.46 ± 0.32[c]	
norBNI	–	0.23 ± 0.03[c]	–	
morphine	4.6 ± 1.8[c]	–	–	2.5(1.8, 3.4)

[a] Competition studies were performed with the indicated compounds against ^{125}I-IBNtxA (0.1 nM) in membranes from CHO cells stably expressing the indicated cloned mouse opioid receptors. Results are presented as nM ± SEM from three independent experiments performed in triplicate.

[b] Cumulative dose–response curves were carried out on groups of CD1 mice (n = 10) using radiant heat tail flick assays with indicated compound at the indicated doses (sc), and antinociception was tested 15 min later at peak effect. Results from two independent experiments are shown as mean (95% Cl).

[c] Values from the literature.[78] "–" Denotes not determined or not applicable.

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Table 2

[^{35}S]GTPγS Functional Assays in Transfected Cell Lines

	[^{35}S]GTPγS functional assays[a]					
	MOR-1		KOR-1		DOR-1	
compd	EC_{50} (nM)	E_{max} (%)	EC_{50} (nM)	IC_{50} (nM)	EC_{50} (nM)	IC_{50} (nM)
1	203 ± 13	65 ± 2.8	–	>4 μM	–	>4 μM
2	53 ± 4	77 ± 5	–	2524 ± 552	–	691 ± 434
3	1.7 ± 0.1	84 ± 5	–	31 ± 3	–	61 ± 6
4	2.0 ± 0.1	124 ± 2	–	–	–	293 ± 129
6	1.4 ± 0.03	116 ± 2	–	252 ± 48	–	193 ± 48
7	0.7 ± 0.2	122 ± 2	–	721 ± 58	–	73 ± 3
8	1.4 ± 0.25	123 ± 5	–	171 ± 10	0.83 ± 0.37 (89 ± 3)[b]	–
9	1.5 ± 0.3	100 ± 1	–	202 ± 14	–	39 ± 5
10	3.9 ± 0.5	120 ± 6	–	–	–	–
11	–	–	–	–	–	–
12	–	–	–	–	–	–
DAMGO	19 ± 7	–	–	–	–	–
U50,488H	–	–	17 ± 6.1	–	–	–
DPDPE	–	–	–	–	10 ± 2.2	–
NTI	–	–	–	–	–	0.72 ± 0.11
norBNI	–	–	–	2.9 ± 0.14	–	–

[a]Efficacyata were obtained using agonist induced stimulation of [^{35}S]GTPγS binding assay. Efficacy is represented as EC_{50} (nM) and percent maximal stimulation (E_{max}) relative to standard agonist DAMGO (MOR-1), DPDPE (DOR-1), or U50,488H (KOR-1) at 1000 nM. To determine the antagonist properties of a compound, membranes were incubated with 100 nM of the appropriate agonist in the presence of varying concentrations of the compound

[b]Compound **8** is an agonist at DOR-1. Results are presented as nM ± SEM from three independent experiments performed in triplicate. "–" Denotes not determined or not applicable.

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Table 3

In Vitro and in Silico Binding Affinities and Specific Contacts of Mitragynine and Derivatives

	compd	in vitro K_i (nM)	in silico K_i (nM) nonspecific[a]	in silico K_i (nM) specific[b]	receptor side chains in polar contact with the ligand	receptor side chains in nonpolar contact with the ligand
mu	**1**	230 ± 47	23	569	Gln^{124}, Tyr^{128}, Asp^{147}, Tyr^{326}	Met^{151}, Trp^{293}, Ile^{296}, Ile^{322}
	2	37 ± 4	4.0	55	Asp^{147}, Lys^{233}, Trp^{293}	Tyr^{148}, Met^{151}, Leu^{219}, Leu^{232}, Ile^{296}
	3	0.75 ± 0.18	2.5	2.5	Asp^{147}, Tyr^{148}, Lys^{233}	Ile^{144}, Leu^{219}, Leu^{232}, Trp^{293}, Ile^{296}, His^{297}
delta	**1**	1011 ± 49	159	1565	Gln^{105}, Lys^{108}, Asp^{128}, Tyr^{129}, Tyr^{129}	Leu^{128}, Lys^{214} (aliphatic chain), Val^{217}
	2	90 ± 8	19	19	Asp^{128}, Tyr^{129}	Met^{132}, Lys^{214} (aliphatic chain), Val^{217}, Trp^{274}, Ile^{277}, Ile^{304}
	3	3 ± 1.3	17	17	Asp^{128}, Tyr^{129}	Met^{132}, Lys^{214} (aliphatic chain), Val^{217}, Trp^{274}, Ile^{277}, Val^{281}
kappa	**1**	231 ± 21	17	69	Asp^{138}, Tyr^{139}, Ser^{211}	Trp^{124}, Val^{134}, Leu^{135}
	2	131 ± 7	8.3	53	Thr^{111}, Asp^{138}, Tyr^{139}, Lys^{227}, Tyr^{312}, Tyr^{320}	Gln^{115}, Leu^{135}, Ile^{294}
	3	24 ± 9	3.2	23	Gln^{115}, Asp^{138}, Tyr^{139}, Ser^{211}, Tyr^{312}	Phe^{114}, Val^{118}, Ile^{294}, Ile^{316}

[a] Inhibitory constants calculated for the lowest energy docked complexes including nonspecifically bound poses.

[b] Inhibitory constants calculated for the lowest energy docked complexes excluding low energy poses which were regarded as false positives.

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Table 4

Sequences of Antisense (AN) and Mismatch (MIS) Oligodeoxynucleotides

target	antisense	mismatch control
MOR-1 exon 1	CGCCCCAGCCTCTTCCTCT	CGCCCCGACCTCTTCCCTT
DOR-1 exon 3	AGGGGAAGGTCGGGTAGG	GAGGAGAGGTGCGTGGAG
KOR-1 exon 2	CGCCCCAGCCTCTTCCTCT	CTCCGCGCTCTCACCCTCT

Regular Article

Asymmetric Total Syntheses of Mitragynine, Speciogynine, and 7-Hydroxymitragynine

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A number of alkaloids found in *Mitragyna* species belonging to the Rubiaceae family have been shown to have potent biological activity such as analgesic properties. Here, we report the asymmetric total syntheses of mitragynine, speciogynine, and 7-hydroxymitragynine, which are classified as corynantheine-type monoterpenoid indole alkaloids, isolated from *Mitragyna speciosa*. These syntheses were accomplished within 12 steps and in >11% total yield from commercial 3-(trimethylsilyl)propanal using an organocatalytic *anti*-selective Michael reaction and bioinspired transformations.

Key words monoterpenoid indole alkaloid; mitragynine; speciogynine; 7-hydroxymitragynine; bioinspired reaction

Introduction

Mitragyna speciosa, which belongs to the Rubiaceae family, is endemic to the tropics of Southeast Asia, where it is called "Kratom" in Thailand and "Biak Biak" in Malaysia.[1] Traditionally, the leaf of this plant has been used to alleviate the fatigue of workers laboring in the scorching heat and as a substitute for opium. Many chemists have been fascinated by its special biological activity, and isolation and structure elucidation studies of this plant have been conducted since the 1960s.[2–6] As a result, several corynantheine-type monoterpenoid indole alkaloids have been found with a methoxy group attached to the C-9 position on the indole ring (Fig. 1).

Mitragynine (**1**) is the major component of a base fraction of *Mitragyna speciosa* leaves, and speciogynine (**2**) has been found as the stereoisomer at the C-20 position. Recently, the *in vitro* and *in vivo* activity of **2** at serotonin receptors (5-HTRs) was reported; speciogynine (**2**) exhibited a high affinity for 5-HT1ARs and 5-HT2BRs.[7] 7-Hydroxymitragynine (**3**) was found as a minor component of the leaves; this molecule had a potent analgesic activity *via* opioid μ receptors (mitragynine (**1**) also showed mild analgesic activity).[6,8–16] In addition, it was discovered that this alkaloid maintains potent activity through oral administration.[9] This molecule is now widely recognized as a key component in folk medicine.

Because of their interesting biological activity, several total syntheses of these alkaloids have been reported.[10,17–22] Very recently, our group also achieved the total syntheses of the same class of corynantheine-type monoterpenoid indole alkaloids, corynantheidine and dihydrocorynantheine.[23] However, the efficiency of these total syntheses could be improved, specifically because the goal of previous total syntheses was to reproduce the biosynthetic transformations from strictosidine in the flask.

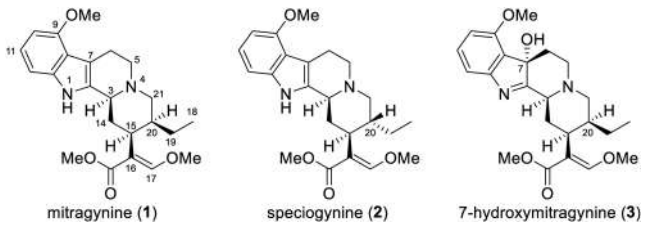


Fig. 1. Structures of Mitragynine, Speciogynine, and 7-Hydroxymitragynine

Here, we report the practical asymmetric total syntheses of mitragynine (**1**), speciogynine (**2**), and 7-hydroxymitragynine (**3**). These syntheses were achieved within 12 steps from commercially available materials with total yields of >11%.

Results and Discussion

Our retrosynthetic analysis of (−)-mitragynine (**1**) and (+)-speciogynine (**2**) is outlined in Chart 1. We envisioned that our previously developed bioinspired transformations toward the syntheses of corynantheine-type alkaloids could also reveal the syntheses of (−)-**1** and its C-20 epimer (+)-**2**. Thus, (−)-**1** and (+)-**2** were retrosynthesized to 9-methoxystrictosidine derivative **5**, *via* bioinspired skeletal transformations and stereoselective reduction of the common intermediate **4**. Preparation of key intermediate **5** was expected to be achieved by diastereoselective Pictet–Spengler cyclization using (*R*)-4-methoxy-α-cyanotryptamine (**6**) and secologanin derivative **7**.[24] Tryptamine derivative **6** was considered for the installation of a 4-methoxyindole unit to the known methyl (*R*)-*N*-benzyloxycarbonyl (Cbz)-azirizine-2-carboxylate (**8**).[25] For the concise synthesis of secologanin derivative **7**, our discovered *anti*-selective organocatalytic Michael reaction was selected as a key step to install all carbons of **7**.[26] Thus, butyraldehyde (**9**) and conjugated ene-yne compound **10**, prepared from commercially available 3-(trimethylsilyl)propanal (**11**), were assumed as optimal building blocks.

As shown in Chart 2, our synthesis commenced with the preparation of (*R*)-4-methoxy-α-cyanotryptamine (**6**). Synthesis of D-tryptophan derivative **13** was performed *via* a Lewis acid-mediated coupling reaction of 4-methoxyindole (**12**) with known chiral aziridine **8**.[27] After several trials, $Yb(OTf)_3$ effectively promoted the coupling reaction and the desired **13** was obtained as a major product (1.5 equivalent (equiv.) of $Yb(OTf)_3$; room temperature (r.t.). Hydrolysis and subsequent

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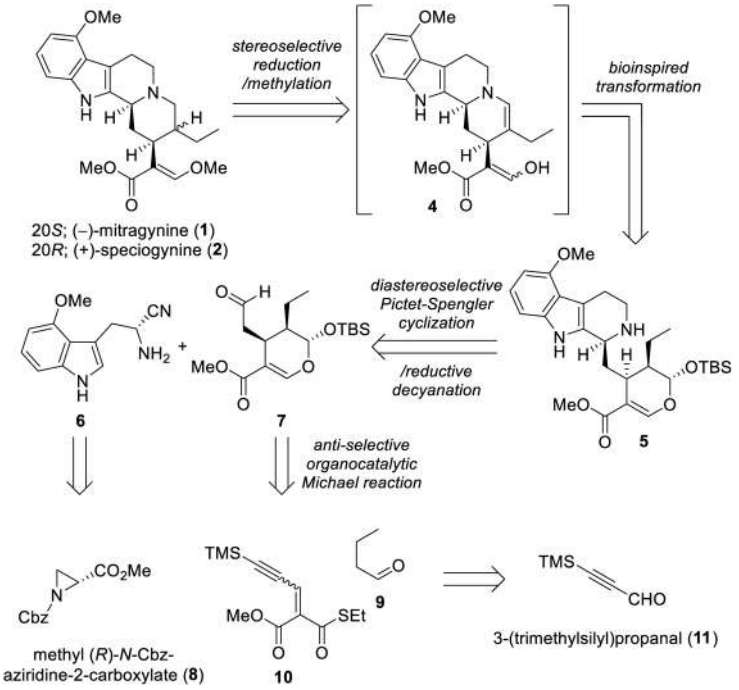


Chart 1. Retrosynthetic Analysis of (−)-Mitragynine and (+)-Speciogynine

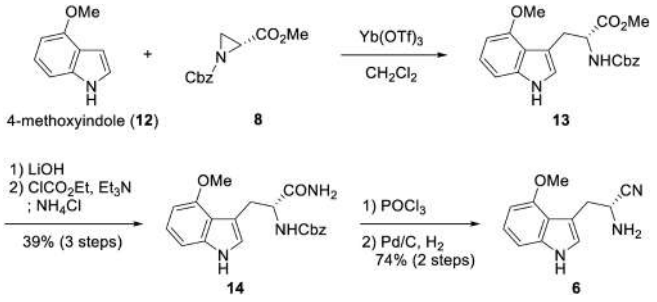


Chart 2. Preparation of (*R*)-4-Methoxy-α-cyanotryptamine

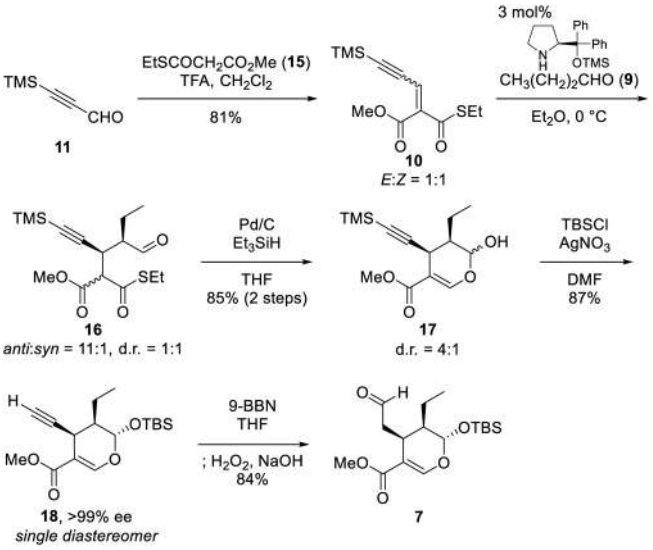


Chart 3. Preparation of Secologanin Derivative

amidation of the crude mixture of **13** gave primary amide **14** in gram quantities (39% yield over three steps). Then, amide **14** was converted to **6** *via* dehydration to construct a cyano group and deprotection of the Cbz group (74% yield over two steps).

As in our previous synthesis of monoterpenoid indole alkaloids, the preparation of secologanin derivative **7** commenced with the Knoevenagel condensation with aldehyde **11** and methyl 3-ethylthio-3-oxopropanoate (**15**)[28] in the presence of trifluoroacetic acid (TFA), affording ene-yne compound **10** (81%, *E*:*Z* = 1 : 1) (Chart 3). To install the C-15 chiral center and C-20 ethyl group in (−)-**1** and (+)-**2**, an *anti*-selective organocatalytic Michael reaction to form **16** was set up. Thus, when ene-yne compound **10** and butyraldehyde (**9**) were treated with 3 mol% diphenylprolinol trimethylsilyl ether catalyst in Et_2O at 0 °C (40 h), the *anti*-adduct **16** was obtained with excellent diastereoselectivity (*anti*:*syn* = 11 : 1). Subsequent Fukuyama reduction[29] to convert aldehyde from a thioester of **16** and spontaneous cyclization were carried out, affording optically active dihydropyran derivative **17** in 85% yield over two steps (4.5 equiv. Et_3SiH, 15 mol% of Pd/C, r.t.). Subsequent silylation of the hemiacetal hydroxy group in **17** was performed. Thus, when **17** was treated with $AgNO_3$ and *tert*-butyldimethylchlorosilane (TBSCl) in dry *N*,*N*-dimethylformamide (DMF), stereoselective silylation proceeded from the sterically less hindered site to afford α-oriented silyl ether **18** as a single isomer (87%). Enantiomeric excess was determined at this stage: >99% enantiomeric excess (ee) by chiral HPLC analysis. Finally, alkyne **18** was treated with hydroboration–oxidation conditions to afford secologanin derivative **7** in excellent yield (84%, 1.2 equiv. 9-borabicyclo[3.3.1]nonane (9-BBN), tetrahydrofuran (THF), r.t. then 5.0 equiv. H_2O_2, 1.0 equiv. NaOH, r.t.). All protocols in the synthesis of **7** were performed in gram-scale quantities.

With the desired synthons **6** and **7** in hand, we next attempted bioinspired transformations for the construction of a corynantheine-type scaffold (Chart 4). To prepare 9-methoxystrictosidine derivative **5**, our developed diastereoselective

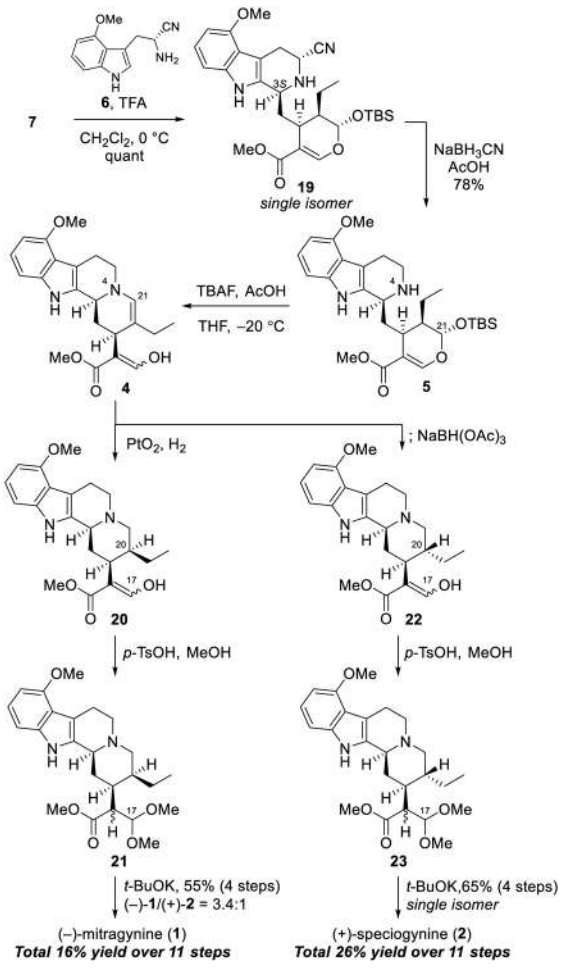


Chart 4. Total Syntheses of (−)-Mitragynine and (+)-Speciogynine

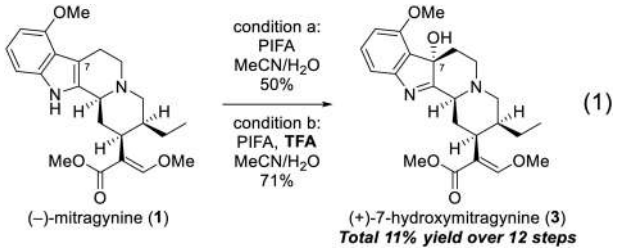
Pictet–Spengler cyclization/reductive decyanation sequences were examined. When a mixture of **6** and **7** was treated with 50 mol% TFA in CH_2Cl_2 at 0 °C (30 min), the desired 3*S*-isomer **19** was obtained as a single diastereomer in quantitative yield.[30] Reductive removal of the cyano group in **19** was performed by treatment with $NaBH_3CN$ in the presence of AcOH in MeOH, affording 9-methoxystrictosidine derivative **5** in excellent yield (78%).

Next, key bioinspired transformations triggered by removal of the silyl group in **5** were examined. Thus, **5** was exposed to a simple desilylation condition (2.0 equiv. tetrabutylammonium fluoride (TBAF), 2.0 equiv. AcOH, THF, –20 °C) to afford enamine **4** linked by N-4 and C-21. After confirming a full conversion to **4** from **5** by NMR spectroscopic analysis, the crude mixture of **4** was directly treated with 100 wt% PtO_2 under H_2 atmosphere.[17] As a result, kinetically reduced product **20**, which has the same C-20*S* stereochemistry as (−)-mitragynine (**1**), was preferentially obtained. To complete the total synthesis of (−)-**1**, modification of the enol hydroxy group at the C-17 position was performed. Thus, the crude mixture of **20** was treated with *p*-TsOH in MeOH to afford dimethoxy acetal **21**.[31] Finally, stereoselective E1cB elimination of the C-17 methoxy group of **21**, using *t*-BuOK in DMF,[17] afforded a separable mixture of (−)-mitragynine (**1**) and (+)-speciogynine (**2**) [(−)-**1**/(+)-**2** = 3.4 : 1, 55% yield over four steps].

Bioinspired transformations toward the selective synthesis of C-20*R* isomer (+)-speciogynine (**2**) were also demonstrated. After generating enamine **4** under the same conditions as for the synthesis of (−)-**1**, 2.0 equiv. $NaBH(OAc)_3$ was added in a one-pot protocol. In this case, thermodynamic controlled protonation at the C-20 position followed by reduction of a generated iminium ion afforded C-20*R* isomer **22** as a single diastereomer. Subsequent modification of the C-17 enol hydroxy group was carried out as in the synthesis of (−)-**1**, and the E1cB elimination reaction, following the formation of **23**, afforded (+)-speciogynine (**2**) in 65% overall yield within four steps from **5**.

In summary, practical total syntheses of (−)-mitragynine (**1**, total 16% over 11 steps from **11**) and (+)-speciogynine (**2**, total 26% over 11 steps from **11**) were accomplished.

Next, to prepare (+)-7-hydroxymitragynine (**3**), the direct C-7 oxidation of (−)-**1** was investigated (Eq. 1). Following our previously reported method, when freshly purified (−)-**1** was oxidized in aqueous acetonitrile by treatment with 1.0 equiv. [bis(trifluoroacetoxy)iodo]benzene (PIFA), (+)-**3** was obtained in moderate yield (condition a in Eq. 1, 50%).[32] After several trials toward further improvement of the chemical yield, it was determined that the addition of 2.0 equiv. TFA led to the effective production of (+)-**3** (condition b in Eq. 1, 71%). We suppose that TFA masked the reactive lone pair of the N-4 nitrogen in (−)-**1** as a salt, thereby suppressing undesired N-4 oxidation. As a result, the total synthesis of (+)-7-hydroxymitragynine (**3**) was accomplished with 11% overall yield within 12 steps from **11.**



(1)

Conclusion

In conclusion, we accomplished the total syntheses of three bioactive *Mitragyna* alkaloids: (−)-mitragynine (**1**), (+)-speciogynine (**2**), and (+)-7-hydroxymitragynine (**3**). The synthesis of secologanin derivative **7**, a key fragment in this synthesis, was carried out *via* an *anti*-selective organocatalytic Michael reaction using ene-yne compound **10** and simple aldehyde **9**, with excellent enantiomeric excess. Preparation of common intermediate **4** was achieved *via* bioinspired transformations of strictosidine derivative **5** triggered by the removal of a silyl group, which branched into (−)-**1** and (+)-**2** by stereoselective reduction. In addition, (+)-**3** was obtained through direct oxidation of (−)-**1** *via* a nitrogen-masking strategy. As a result, syntheses of (−)-mitragynine (**1**), (+)-speciogynine (**2**), and (+)-7-hydroxymitragynine (**3**) were realized within fewer than 12 steps and in >11% overall yield. Using the synthesized natural products and their derivatives, a study of their structure–activity relationships is currently under way in our laboratory.

Experimental

General Experimental Procedures All reactions were monitored by TLC using Merck 60 F254 precoated silica gel

plates (0.25 mm thick) and Fuji Silysia Chemical precoated amino-silica gel plates (0.25 mm thick). UV spectra were recorded in MeOH on a JASCO V-560 instrument. Specific optical rotations were measured using a JASCO P-2200 polarimeter. Circular dichroism spectra were recorded on a JASCO J-1100 spectrometer. Fourier transform (FT) IR spectra were recorded on a JASCO FT/IR-4700. ^{1}H- and ^{13}C-NMR spectra were recorded on a JEOL ECZ 400 (400 MHz ^{1}H-NMR, 100 MHz for ^{13}C-NMR), ECS 400 (400 MHz ^{1}H-NMR, 100 MHz for ^{13}C-NMR), JEOL ECX 500 (500 MHz ^{1}H-NMR, 125 MHz for ^{13}C-NMR), and ECZ 600 (600 MHz for ^{1}H-NMR, 150 MHz for ^{13}C-NMR) FT-NMR spectrometer instrument. Data for ^{1}H-NMR are reported as chemical shifts (δ ppm), multiplicity (s = singlet, d = doublet, t = triplet, dd = double doublet, ddd = double double doublet, dt = double triplet, q = quartet, m = multiplet, br = broad), coupling constant (Hz), integration, and assignment. Data for ^{13}C-NMR are reported as chemical shifts. The high-resolution mass spectra were recorded on a JEOL AccuTOF LC-plus JMS-T100LP. Flash chromatography was performed using Kanto Chemical silica gel 60N and Fuji Silysia Chemical amino-silica gel (SiO_2-NH, NH-DM2035). HPLC analysis was performed on a JASCO MD-4017; UV detection was monitored at an appropriate wavelength using Daicel Chiralcel OD-H (0.46 × 25 cm).

Preparation of Compound 14 To a solution of known aziridine **8** (1.0 g, 4.25 mmol) and 4-methoxyindole (**12**, 1.25 g, 8.50 mmol) in CH_2Cl_2 (17 mL) was added $Yb(OTf)_3$ (2.64 g, 4.25 mmol) under Ar atmosphere. The reaction mixture was stirred for 24 h at r.t. under Ar atmosphere in the dark. To the resulting mixture was added additional $Yb(OTf)_3$ (1.32 g, 2.13 mmol), followed by stirring for a further 48 h at r.t. The mixture was then quenched with saturated aqueous $NaHCO_3$ and the resulting mixture was filtered through a Celite pad, eluted with $CHCl_3$. The filtrate was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4 and then concentrated under reduced pressure. Flash chromatography (SiO_2, 22–28% AcOEt/*n*-hexane gradient) provided an inseparable regioisomer mixture of **13** (780 mg, r.r. = 6.7 : 1). To a solution of a mixture of **13** in aqueous THF (34 mL, THF/H_2O = 10 : 1) was added 2 M aqueous LiOH (3.06 mL, 6.12 mmol) at r.t. under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. before quenching with 2 M aqueous HCl (pH 2). Brine was added to the resulting mixture and the aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Obtained crude materials were dissolved in anhydrous THF (40.8 mL), then ethyl chloroformate (272 μL, 2.86 mmol) and triethylamine (853 μL, 6.12 mmol) were added at 0 °C under Ar atmosphere. The reaction mixture was stirred for 30 min at 0 °C. To the resulting mixture was added 1.0 M aqueous NH_4Cl (3.06 mL, 3.06 mmol) at 0 °C. The reaction mixture was stirred for an additional 20 min at 0 °C. It was then quenched with saturated aqueous $NaHCO_3$ and the aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 2%–5% MeOH/$CHCl_3$ gradient) afforded amide **14** (607.5 mg, 39% over three steps).

Compound **14**: White amorphous powder; $[\alpha]_D^{23}$ +2.6 (*c* 0.31, MeOH); IR (ATR) ν_{max} cm^{-1} 3448, 3338, 2945, 2480, 1674, 1620, 1589, 1525, 1502, 1423, 1356, 1313, 1257, 1153, 1095, 1039, 966, 910, 860, 775, 752, 735, 719, 694, 617; high resolution (HR)MS (electrospray ionization (ESI)) $[M + Na]^+$ Calcd. for $[C_{20}H_{21}N_3Na_1O_4]^+$: 390.1430, Found: 390.1435; UV (MeOH) λ_{max} 291, 281, 265, 220 nm; ^{1}H-NMR (400 MHz, $(CD_3)_2SO$) δ: 10.8 (br s, 1H), 7.34–7.19 (m, 5H), 7.03 (br s, 1H), 6.96–6.92 (m, 3H), 6.44 (d, J = 6.4 Hz, 1H), 4.95 (s, 2H), 4.23 (dt, J = 8.8, 4.4 Hz, 1H), 3.84 (s, 3H), 3.28 (dt, J = 14.4, 3.6 Hz, 1H), 2.98 (ddd, J = 14.4, 10.0, 3.2 Hz, 1H); ^{13}C-NMR (150 MHz, $(CD_3)_2SO$) δ: 174.2, 155.8, 154.1, 137.8, 137.1, 128.4, 127.7, 127.4, 122.2, 121.8, 117.0, 110.6, 105.0, 98.9, 65.2, 56.2, 55.1, 39.9, 39.8, 39.7, 39.5, 39.4, 39.2, 39.1, 29.3.

Preparation of (*R*)-4-Methoxy-α-cyanotryptamine (6) To a solution of amide **14** (608 mg, 1.65 mmol) in anhydrous pyridine (11 mL) was added phosphoryl chloride (355 μL, 3.80 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 50 min at 0 °C. MeOH was added at 0 °C to the resulting mixture and stirred for 10 min, followed by concentration under reduced pressure. The resulting mixture was dissolved in Et_2O, and the organic layer was washed with 1 M aqueous NaOH three times, and with brine, dried over Na_2SO_4, and then concentrated under reduced pressure. Flash chromatography (SiO_2, 1% MeOH/$CHCl_3$) gave a dehydrated compound (510 mg, 88%). To a solution of the dehydrated compound (500 mg, 1.43 mmol) in degassed 1,4-dioxane (10.4 mL) was added 10% Pd/C (152 mg, 0.143 mmol) at r.t. under Ar atmosphere. The resulting mixture was purged with a stream of hydrogen and the reaction mixture was stirred for 13 h at r.t. under H_2 atmosphere. The resulting mixture was filtered with a Celite pad with $CHCl_3$. The filtrate was concentrated under reduced pressure and the resulting residue was purified by flash chromatography (SiO_2, 60% AcOEt/*n*-hexane, then 5% MeOH/$CHCl_3$) to afford (*R*)-4-methoxy-α-cyanotryptamine (**6**, 260.0 mg, 84%).

(*R*)-4-Methoxy-α-cyanotryptamine (**6**): Pale pink amorphous powder; $[\alpha]_D^{23}$ –15.0 (*c* 1.96, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3363, 3194, 2914, 1741, 1587, 1547, 1510, 1431, 1356, 1257, 1167, 1132, 1086, 1053, 955, 895, 852, 814, 683, 623; HRMS (ESI) $[M–CN]^+$ Calcd. for $[C_{11}H_{13}N_2O_1]^+$: 189.1028, Found: 189.1025; UV (MeOH) λ_{max} 290, 281, 267, 220 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 8.22 (br s, 1H), 7.11 (t, J = 7.8 Hz, 1H), 7.00 (d, J = 2.4 Hz, 1H), 6.98 (d, J = 7.8 Hz, 1H), 6.52 (d, J = 7.8 Hz, 1H), 4.18 (t, J = 7.2 Hz, 1H), 3.93 (s, 3H), 3.33 (dd, J = 13.8, 7.2 Hz, 1H), 3.27 (dd, J = 13.8, 7.2 Hz, 1H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 154.4, 138.2, 123.3, 122.6, 122.5, 117.0, 110.0, 104.9, 99.8, 55.3, 45.4, 33.6.

Preparation of Compound 17 To the solution of an *E* and *Z* mixture of half thioester malonate derivative **10** (500 mg, 1.85 mmol) and diphenylprolinol trimethyl silyl ether (18.0 mg, 0.0555 mmol) in Et_2O (9.2 mL) was added *n*-butyraldehyde (**9**, 167 μL, 1.85 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 40 h at 0 °C. The resulting mixture was concentrated under reduced pressure. Crude ^{1}H-NMR revealed that *anti*-adduct **16** was the main product (diastereomer ratio of **16**; *anti:syn* = 11 : 1). The crude materials of **16** were solved to degassed THF (3.7 mL), then 10% Pd/C (197 mg, 0.185 mmol) and Et_3SiH (886 μL, 5.55 mmol) were added at 0 °C under Ar atmosphere. The reaction mixture was stirred for 4.5 h at r.t, whereafter an additional 10% Pd/C (98.4 mg, 0.0924 mmol) and Et_3SiH (443 μL, 2.77 mmol) were added at r.t. under Ar atmosphere. This was stirred for 5 h at r.t. before the addition of 1 M aqueous HCl (3.7 mL). The resulting

mixture was stirred for a further 18 h at r.t. before filtration with a Celite pad eluted with AcOEt, and partitioned between an AcOEt layer and an aqueous layer. The aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography (SiO_2, 5%–20% AcOEt/*n*-hexane gradient) afforded two diastereomers of dihydropyran **17** (444.5 mg, 85% over two steps, d.r. = 4 : 1).

Compound **17**: Yellow waxy solid; IR (ATR) ν_{max} cm^{-1} 3370, 2953, 1672, 1629, 1444, 1383, 1311, 1247, 1204, 1148, 1125, 1100, 1030, 1015, 981, 945, 908, 879, 835; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{14}H_{22}Na_1O_4Si_1]^+$: 305.1185, Found: 305.1145; ^{1}H-NMR (500 MHz, $CDCl_3$) δ: 7.53 (s, 1H), 7.46 (s, 1H), 5.36 (dd, J = 12.0, 1.5 Hz, 1H), 5.22 (t, J = 7.5 Hz, 1H), 4.85 (d, J = 7.5 Hz, 1H), 3.77 (s, 3H), 3.75 (s, 3H), 3.58 (d, J = 5.0 Hz, 1H), 3.55 (d, J = 5.0 Hz, 1H), 3.34 (d, J = 7.5 Hz, 1H), 1.85–1.71 (m, 3H), 1.64–1.53 (m, 3H), 1.05 (t, J = 7.0 Hz, 3H), 1.02 (t, J = 7.0 Hz, 3H), 0.14 (s, 9H), 0.12 (s, 9H); ^{13}C-NMR (125 MHz, $CDCl_3$) δ: 166.9, 166.7, 153.3, 152.5, 107.2, 106.3, 104.0, 103.7, 97.1, 96.3, 90.1, 97.9, 51.7, 51.6, 42.2, 39.9, 26.8, 23.6, 21.6, 20.4, 11.4, 11.0, 0.17 (3C), 0.12 (3C).

Preparation of Compound 18 To a solution of two diastereomers of dihydropyran **17** (1.0 g, 3.54 mmol) in dry DMF (11.8 mL) were added $AgNO_3$ (782 mg, 4.60 mmol) and TBSCl (800.5 mg, 5.31 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 1 h at r.t. under Ar atmosphere. Additional $AgNO_3$ (391 mg, 2.30 mmol) and TBSCl (400 mg, 2.66 mmol) were then added under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. Additional $AgNO_3$ (391 mg, 2.30 mmol) was then added under Ar atmosphere and the mixture was stirred for 15 min at r.t. The resulting mixture was directly filtered through a short plug of silica gel eluted with 20% AcOEt/*n*-hexane and the filtrate was added to saturated $NaHCO_3$. The aqueous layer was extracted three times with AcOEt. The combined organic layer was washed four times each with water and brine, dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography (SiO_2, 1% AcOEt/*n*-hexane) afforded **18** (1.0 g, 87%). Enantiomeric excess was determined to be over 99% by HPLC (Chiralcel OD-H column) (see Supporting Information). Results for compound **18**: 1.5% *i*-PrOH/*n*-hexane, 0.500 mL/min; major enantiomer t_R = 7.27 min, minor enantiomer t_R = 8.30 min.

Compound **18**: Colorless oil; $[\alpha]^{22}_D$ –235.6 (*c* 0.71, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3315, 3269, 2954, 2931, 2883, 2858, 1714, 1639, 1462, 1437, 1392, 1298, 1254, 1228, 1167, 1130, 1088, 1001, 949, 924, 879, 837, 781, 764, 679, 663, 638, 617; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{17}H_{28}Na_1O_4Si_1]^+$: 347.1655, Found: 347.1669; ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 7.48 (s, 1H), 5.18 (d, J = 8.4 Hz, 1H), 3.75 (s, 3H), 3.56 (dd, J = 5.2, 2.4 Hz, 1H), 2.15 (d, J = 2.4 Hz, 1H), 1.78 (m, 1H), 1.59 (m, 1H), 1.47 (m, 1H), 1.00 (t, J = 7.2 Hz, 3H), 0.93 (s, 9H), 0.15 (s, 3H), 0.15 (s, 3H); ^{13}C-NMR (100 MHz, $CDCl_3$) δ: 167.1, 153.9, 106.6, 97.6, 82.4, 71.4, 51.6, 42.5, 25.8, 20.3, 18.1, 11.1, –4.1, –5.1.

Preparation of Secologanin Derivative 7 To a solution of alkyne **18** (1.0 g, 3.1 mmol) in anhydrous THF (10.3 mL) was added 9-BBN (1.0 M in THF solution, 7.4 mL, 3.70 mmol) under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. The resulting mixture was quenched with H_2O (1.0 mL) and then stirred for an additional 10 min. To the resulting mixture were added 30% aqueous H_2O_2 (1.57 mL, 15.4 mmol) and 1.0 M aqueous NaOH (3.08 mL, 3.08 mmol) at 0 °C. This reaction mixture was stirred for 80 min at r.t. under Ar atmosphere. After the addition of saturated aqueous NH_4Cl, the aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine, dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography, repeated three times (SiO_2, 5%–50% AcOEt/*n*-hexane gradient then 10% MeOH/$CHCl_3$) afforded secologanin derivative **7** (887 mg, 84%).

Secologanin Derivative **7**: Colorless oil; $[\alpha]^{23}_D$ –141.1 (*c* 1.89, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 2954, 2931, 2887, 2858, 1703, 1633, 1466, 1439, 1385, 1362, 1309, 1281, 1254, 1167, 1138, 1101, 1080, 1007, 918, 874, 837, 781, 675, 640; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{17}H_{30}Na_1O_5Si_1]^+$: 365.1760, Found: 365.1769; ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 9.76 (dd, J = 3.2, 2.0 Hz, 1H), 7.49 (s, 1H), 5.06 (d, J = 7.6 Hz, 1H), 3.69 (s, 3H), 3.30 (dt, J = 7.6, 5.2 Hz, 1H), 2.48 (ddd, J = 15.2, 5.2, 2.0 Hz, 1H), 2.35 (ddd, J = 15.2, 7.6, 3.2 Hz, 1H), 1.74–1.59 (m, 2H), 1.15 (m, 1H), 0.98 (t, J = 7.2 Hz, 3H), 0.91 (s, 9H), 0.14 (s, 3H), 0.13 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 201.6, 167.7, 154.0, 108.5, 96.4, 51.4, 45.1, 43.6, 26.8, 25.8, 19.3, 18.1, 11.5, –4.2, –5.1.

Preparation of Compound 19 To a solution of secologanin derivative **7** (320 mg, 0.934 mmol), (*R*)-4-methoxy-α-cyanotryptamine (**6**, 201.1 mg, 0.934 mmol), and powdered MS 4 Å (1 g) in CH_2Cl_2 (18.7 mL) was added TFA (1.0 M in CH_2Cl_2, 467 μL, 0.467 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 30 min at 0 °C. The resulting mixture was filtered through a cotton plug with $CHCl_3$ and the filtrate was added to an excess amount of saturated aqueous $NaHCO_3$ solution at 0 °C. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 15% AcOEt/*n*-hexane) afforded **19** (504.3 mg, quantitative yield) as a single diastereomer.

Compound **19**: White amorphous powder; $[\alpha]^{23}_D$ –230.1 (*c* 0.85, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3328, 2928, 1688, 1630, 1572, 1509, 1463, 1437, 1384, 1357, 1312, 1279, 1254, 1207, 1167, 1137, 1102, 1029, 1018, 974, 938, 892, 871, 837; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{29}H_{42}N_3O_5Si_1]^+$: 540.2894, Found: 540.2859; UV (MeOH) λ_{max} 292, 225 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 7.82 (br s, 1H), 7.59 (s, 1H), 7.04 (t, J = 7.8 Hz, 1H), 6.91 (d, J = 7.8 Hz, 1H), 6.47 (d, J = 7.8 Hz, 1H), 5.26 (d, J = 9.0 Hz, 1H), 4.37 (dd, J = 4.2, 3.6 Hz, 1H), 4.07 (ddd, J = 10.8, 3.6, 1.8 Hz, 1H), 3.88 (s, 3H), 3.79 (s, 3H), 3.35–3.29 (m, 2H), 2.97 (dt, J = 12.0, 4.2 Hz, 1H), 1.79 (ddd, J = 12.6, 11.4, 3.0 Hz, 1H), 1.73–1.63 (m, 2H), 1.55 (td, J = 12.6, 2.4 Hz, 1H), 1.18 (m, 1H), 0.98 (t, J = 7.2 Hz, 3H), 0.93 (s, 9H), 0.19 (s, 3H), 0.18 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 169.3, 154.6, 154.4, 137.4, 132.4, 122.8, 120.0, 117.5, 109.0, 105.6, 104.3, 100.0, 97.2, 55.3, 52.2, 46.8, 44.4, 43.8, 36.9, 27.8, 26.9, 25.8, 20.2, 18.1, 11.1, –3.9, –4.9.

Preparation of 9-Methoxystrictosidine Derivative 5 Acetic acid (265 μL, 4.63 mmol) and sodium cyanoborohydride (582 mg, 9.26 mmol) were added to a solution of **19** (500 mg, 0.936 mmol) in MeOH (4.6 mL) at r.t. under Ar atmosphere. The reaction mixture was stirred for 24 h at r.t. To the resulting mixture were added additional acetic acid (265 μL, 4.63 mmol) and sodium cyanoborohydride (582 mg, 9.26 mmol) under Ar atmosphere. The reaction mixture was stirred for a further 36 h at r.t. The resulting mixture was quenched with

saturated aqueous $NaHCO_3$ solution at 0 °C. The aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine, dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials were filtered through a short plug of amino-silica gel (SiO_2-NH), eluted with 5% MeOH/$CHCl_3$, to remove an excess amount of sodium cyanoborohydride. Flash chromatography (SiO_2, 25% AcOEt/*n*-hexane then 20% MeOH/$CHCl_3$) afforded the 9-methoxystrictosidine derivative **5** (371.6 mg, 78%) with 80.5 mg of starting material **19**.

9-Methoxystrictosidine Derivative **5**; White amorphous powder; $[\alpha]_D^{23}$ –189.9 (*c* 0.93, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 2933, 2850, 1698, 1630, 1510, 1460, 1436, 1383, 1357, 1308, 1276, 1252, 1166, 1137, 1096, 1031, 1007, 975, 950, 927, 908, 872, 861, 836, 825; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{28}H_{43}N_2O_5Si_1]^+$: 515.2941, Found: 515.2955; UV (MeOH) λ_{max} 292, 223 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 8.33 (br s, 1H), 7.55 (s, 1H), 7.02 (t, J = 7.8 Hz, 1H), 6.94 (d, J = 7.8 Hz, 1H), 6.46 (d, J = 7.8 Hz, 1H), 5.23 (d, J = 8.4 Hz, 1H), 4.01 (dd, J = 9.0, 3.0 Hz, 1H), 3.88 (s, 3H), 3.74 (s, 3H), 3.29 (m, 1H), 3.05–2.95 (m, 4H), 1.92 (ddd, J = 13.8, 10.8, 3.6 Hz, 1H), 1,71–1.61 (m, 2H), 1.52 (ddd, J = 13.8, 10.2, 3.6 Hz, 1H), 1.27 (m, 1H), 0.99 (t, J = 7.2 Hz, 3H), 0.92 (s, 9H), 0.17 (s, 3H), 0.16 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 169.3, 154.8, 154.6, 137.3, 133.6, 122.3, 117.5, 109.8, 108.6, 104.5, 99.9, 97.3, 55.4, 51.8, 50.5, 44.4, 42.1, 36.7, 28.0, 25.8, 24.2, 20.1, 18.1, 11.6, –4.0, –5.0.

Synthesis of (−)-Mitragynine (1) To a solution of 9-methoxystrictosidine derivative **5** (150 mg, 0.291 mmol) in degassed anhydrous THF (2.9 mL) were added acetic acid (83.3 μL, 1.46 mmol) and tetrabutylammonium fluoride (1.0 M in THF solution, 583 μL, 0.583 mmol) at –20 °C under Ar atmosphere. The reaction mixture was stirred for 5 h at –20 °C. The resulting mixture was directly filtered through a short plug of silica gel eluted with 40% AcOEt/*n*-hexane. The residue of **4** was dissolved in degassed MeOH (14.6 mL) to which was added PtO_2 (150 mg, 100 wt%) at r.t. under Ar atmosphere. The resulting mixture was purged with a stream of hydrogen and the reaction mixture was stirred for 1 h at r.t. under H_2 atmosphere. The resulting mixture was filtered with a Celite pad with AcOEt, and the filtrate was concentrated under reduced pressure. The resulting residue of **20** was dissolved in degassed MeOH (14.6 mL) to which was added $CH(OMe)_3$ (52.7 μL, 0.481 mmol) and *p*-TsOH·H_2O (211 mg, 1.11 mmol), and the mixture was stirred at 70 °C for 16 h. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution, and was concentrated under reduced pressure. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials of **21** were dissolved in dry degassed DMF (5.8 mL) to which was added *t*-BuOK (98.1 mg, 0.874 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 45 min at r.t. under Ar atmosphere. The resulting mixture was quenched with water and the aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 25–45% AcOEt/*n*-hexane gradient then 5% MeOH/$CHCl_3$) afforded (−)-mitragynine (**1**, 48.7 mg, 42% over four steps) and (+)-speciogynine (**2**, 15.2 mg, 13% over four steps). All spectral data of the obtained (−)-mitragynine (**1**) and (+)-speciogynine (**2**) were identical to those of natural products.[33,34]

(−)-Mitragynine (**1**): Yellow amorphous powder; $[\alpha]_D^{24}$ –103.1 (*c* 0.57, $CHCl_3$) [lit. $[\alpha]_D^{24}$ –126 (*c* 1.2, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3359, 2926, 1697, 1622, 1567, 1508, 1461, 1434, 1350, 1273, 1253, 1188, 1170, 1146, 1104, 1078, 1020, 977, 903, 885, 862; HRMS (ESI) $[M+H]^+$ calculated for $[C_{23}H_{31}N_2O_4]^+$: 399.2284, found: 399.2291; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 320 (0), 291 (2.27), 283 (2.15), 273 (2.45), 261 (0), 239 (–8.18), 229 (–7.33), 219 (–11.32), 206 (0); UV (MeOH) λ_{max} 292, 224 nm; ^{1}H- and ^{13}C-NMR see supporting information.

(+)-Speciogynine (**2**): Yellow amorphous powder; $[\alpha]_D^{24}$ +17.0 (*c* 1.02, $CHCl_3$) [lit. $[\alpha]_D^{24}$ +26.8 (*c* 0.85, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3363, 2931, 2846, 2796, 2736, 1699, 1633, 1568, 1508, 1460, 1434, 1355, 1335, 1317, 1276, 1252, 1222, 1187, 1144, 1102, 1013, 989, 915, 894, 862, 834, 810; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{23}H_{31}N_2O_4]^+$: 399.2284, Found: 399.2273; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 309 (0), 291 (2.93), 281 (3.56), 269 (4.47), 253 (3.36), 237 (9.55), 230 (0), 219 (–20.57), 203 (0); UV (MeOH) λ_{max} 291, 225 nm; ^{1}H- and ^{13}C-NMR see supporting information.

Synthesis of (+)-Speciogynine (2) To a solution of 9-methoxystrictosidine derivative **5** (30 mg, 0.058 mmol) in degassed anhydrous THF (580 μL) were added acetic acid (16.7 μL, 0.291 mmol) and tetrabutylammonium fluoride (1.0 M in THF solution, 117 μL, 0.117 mmol) at –20 °C under Ar atmosphere. The reaction mixture was stirred for 5 h at –20 °C. To the resulting mixture was added sodium triacetoxyborohydride (24.7 mg, 0.117 mmol) at –20 °C. The reaction mixture was stirred for 2.5 h at –20 °C. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution and the aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine then dried over Na_2SO_4. The resulting residue of **22** was dissolved in degassed MeOH (2.9 mL), to which was added $CH(OMe)_3$ (10.5 μL, 0.0962 mmol) and *p*-TsOH·H_2O (42.1 mg, 0.221 mmol), and then stirred at 70 °C for 12 h. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution and then concentrated under reduced pressure. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials of **23** were dissolved in dry degassed DMF (1.2 mL) to which was added *t*-BuOK (19.6 mg, 0.175 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 40 min at r.t. under Ar atmosphere. The resulting mixture was quenched with water and the aqueous layer was extracted five times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 45% AcOEt/*n*-hexane) afforded (+)-speciogynine (**2**, 15.2 mg, 65% over four steps).

Synthesis of (+)-7-Hydroxymitragynine (3) To a solution of freshly purified (−)-mitragynine (**1**, 30 mg, 0.075 mmol) in MeCN (900 μL) and water (100 μL) was added TFA (11.5 μL, 0.151 mmol) at 0 °C, and the reaction mixture was stirred for 5 min at 0 °C. PIFA (32.4 mg, 0.0753 mmol) was added to the stirred mixture at 0 °C under Ar atmosphere. The reaction mixture was stirred for 20 min at 0 °C, followed by quenching with saturated aqueous $NaHCO_3$ solution. The aqueous layer was extracted three times with CH_2Cl_2. The combined organic layer was washed with brine, dried over Na_2SO_4, and

concentrated under reduced pressure. Flash chromatography (SiO_2-NH, 40% AcOEt/*n*-hexane) afforded (+)-7-hydroxymitragynine (**3**, 22.0 mg, 71%). All spectral data of the obtained (+)-7-hydroxymitragynine (**3**) were identical to those of natural product.[6,8)]

(+)-7-Hydroxymitragynine (**3**): Pale yellow amorphous powder; $[\alpha]_D^{23}$ +62.5 (*c* 0.50, $CHCl_3$) [lit. $[\alpha]_D^{23}$ +47.9 (*c* 0.55, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3420, 2947, 2872, 2835, 2811, 2745, 1698, 1644, 1597, 1486, 1460, 1435, 1376, 1266, 1237, 1187, 1143, 1104, 1074, 1018, 989, 959, 935, 920, 870, 841; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{23}H_{31}N_2O_5]^+$: 415.2233, Found: 415.2256; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 344 (0), 305 (0.53), 282 (0), 258 (0.82), 243 (0), 229 (−1.13), 214 (0); UV (MeOH) λ_{max} 307, 245, 220 nm; ^{1}H- and ^{13}C-NMR see supplementary materials.

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Conflict of Interest The authors declare no conflict of interest.

Supplementary Materials This article contains supplementary materials.

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Chemical composition and biological effects of kratom (*Mitragyna speciosa*): In vitro studies with implications for efficacy and drug interactions

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The safety and efficacy of kratom *(Mitragyna speciosa*) for treatment of pain is highly controversial. Kratom produces more than 40 structurally related alkaloids, but most studies have focused on just two of these, mitragynine and 7-hydroxymitragynine. Here, we profiled 53 commercial kratom products using untargeted LC–MS metabolomics, revealing two distinct chemotypes that contain different levels of the alkaloid speciofoline. Both chemotypes were confirmed with DNA barcoding to be *M. speciosa*. To evaluate the biological relevance of variable speciofoline levels in kratom, we compared the opioid receptor binding activity of speciofoline, mitragynine, and 7-hydroxymitragynine. Mitragynine and 7-hydroxymitragynine function as partial agonists of the human μ-opioid receptor, while speciofoline does not exhibit measurable binding affinity at the μ-, δ- or κ-opioid receptors. Importantly, mitragynine and 7-hydroxymitragynine demonstrate functional selectivity for G-protein signaling, with no measurable recruitment of β-arrestin. Overall, the study demonstrates the unique binding and functional profiles of the kratom alkaloids, suggesting potential utility for managing pain, but further studies are needed to follow up on these in vitro findings. All three kratom alkaloids tested inhibited select cytochrome P450 enzymes, suggesting a potential risk for adverse interactions when kratom is co-consumed with drugs metabolized by these enzymes.

According to the US Department of Health and Human Services, opioids were responsible for more than 42,000 deaths in the US in 2016, the highest in recorded history[1]. More than 40% of these deaths involved a prescription opioid. Some individuals who suffer from chronic pain are turning to other options, one of which is the plant known as kratom [*Mitragyna speciosa* (Korth.) Havil. (Rubiaceae)], a tropical tree native to peninsular Thailand, Myanmar, Malaysia and other countries in Southeast Asia[1,2]. Kratom has skyrocketed in popularity in western countries in the past decade; current estimates are that as many as 5 million individuals in the US use kratom on a regular basis[3]. There has been considerable controversy over the safety and efficacy of kratom use by US consumers. Citing safety concerns (one study reports that kratom has been implicated as at least partially involved in 91 deaths)[4], the United States Department of Agriculture (USDA) has made it their practice to confiscate shipments of kratom into the US. The US Drug Enforcement Administration (DEA) threatened to assign kratom as a schedule 1 controlled substance, which would make possession of kratom illegal. The DEA then suspended that decision in response to a backlash from some US consumers[5], who claim that it is a safer alternative to opioids for treatment of pain and/or opioid addiction[6]. The effectiveness of kratom for these purposes continues to be a hotly debated and politically charged topic, and one that requires rigorous scientific investigation.

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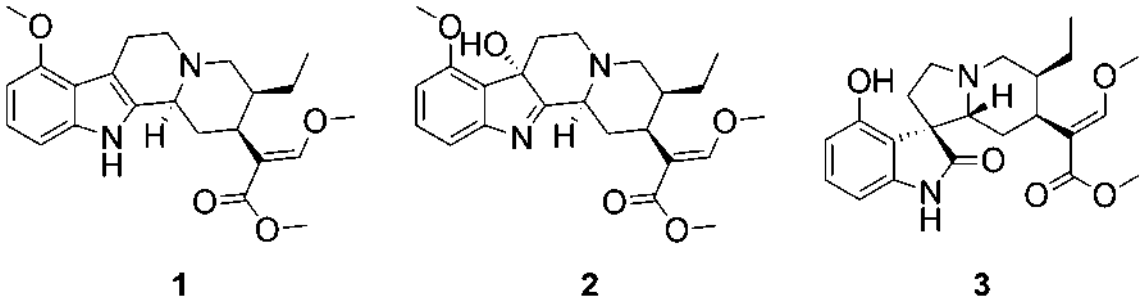


Figure 1. Structures of select alkaloids present in kratom (*Mitragyna speciosa*), mitragynine (**1**), 7-hydroxymitragynine (**2**) and speciofoline (**3**).

Widespread kratom use is a relatively new phenomenon in the US[6]. However, the medicinal use of kratom in southeast Asia has a long history. Kratom was first documented in the scientific literature in 1836[7], in a paper that described use of kratom leaves by the Malays as a substitute for opium. The isolation of mitragynine was reported in 1921 by Ellen Field[8], who ended the introduction of her paper with the following statements, "*According to Redley…….Mitragyne [sic] speciosa is used in Perak against the opium habit, whilst, according to Dr. P.P. Laidlaw, mitragynine is a local anaesthetic [sic]…*" Thus, controversy over the use and effectiveness of kratom was documented nearly 100 years ago. It is a controversy that continues today.

Consistent with the claim that kratom can be effective in the treatment of pain, extracts from this plant demonstrated opioid-receptor mediated analgesic effects in mouse model studies[9]. This activity has generally been attributed to alkaloids that the plant produces, of which mitragynine (**1**) and 7-hydroxymitragynine (**2**) (Fig. 1) have been the focus of multiple pharmacological investigations[10–13]. Mitragynine and 7-hydroxymitragynine both bind to the human μ-opioid and ƙ-opioid receptors (hMOR, hKOR) with nanomolar affinity, and function as partial agonists at the μ-opioid receptor and weak antagonists at ƙ-opioid and δ-opioid receptors[14,15]. 7-Hydroxymitragynine exhibits approximately fivefold greater affinity at the μ-opioid receptor compared to mitragynine. Upon receptor activation, mitragynine and 7-hydroxymitragynine exhibit functional selectivity for G-protein signaling, with no measurable recruitment of β-arrestin[14]. In antinociception assays, 7-hydroxymitragynine exhibits 40-fold greater potency than mitragynine and tenfold greater potency than morphine[16,17], whereas mitragynine is less potent than morphine in antinociceptive assays[18]. Combined administration of mitragynine with morphine increases antinociception compared with morphine alone and prevents the development of morphine tolerance[19]. In contrast to mitragynine, repeated administration of 7-hydroxymitragynine produces antinociceptive tolerance as well as cross-tolerance to morphine's antinociceptive action and induces physical dependence[13]. Furthermore, mitragynine does not exhibit abuse liability and decreases the reinforcing effects of morphine whereas 7-hyroxymitragynine demonstrates abuse liability and increased morphine self-administration in rats[10]. Interestingly, a recent study by Kruegel et al.[12] shows that mitragynine can be converted to 7-hydroxymitragynine both in vitro and in a mouse model, therefore, some of the in vitro activity attributed to mitragynine may in fact be due to the action of it metabolite 7-hydroxymitragynine.

Collectively, the scientific data on mitragynine and 7-hydroxymitragynine suggest that kratom tea might indeed be an effective alternative to opioids against pain. However, kratom products are sold to consumers under a variety of trade names with no data regarding the chemical composition of plant material being consumed. Kratom has been documented to produce variable levels of its more than 40 alkaloids, depending on genetic variation and differences in growth and processing conditions[6,20–22]. What are the potential health implications of the variability in chemical content of kratom preparations? As of yet, this question has been difficult to answer because mechanistic studies of kratom alkaloids have focused largely on isolated alkaloids without consideration of how these alkaloids are represented in kratom materials used medicinally. Herein, we sought to connect the dots between investigations of pure alkaloids and their relevance to medicinally used kratom. Towards this goal, we employed untargeted metabolomics to determine which alkaloids vary in content across commercial kratom products. We then isolated these relevant alkaloids and compared their biological effects in vitro, including activity at the μ-opioid receptors and inhibitory effects on CYP isoforms involved in the metabolism of opioids and other drugs. Our ultimate objective with this study was to capture the variability of commercial kratom products being employed by US consumers and to evaluate (based on vitro studies) the potential implications of this variability in terms of safety and efficacy.

Results

Commercial kratom products can be divided into two chemotypes with differing alkaloid profiles. We conducted untargeted metabolomics analysis of more than 50 commercial available kratom products (Table S1) to obtain a global comparison of chemical composition. A principal component analysis (PCA) plot of the resulting data (Fig. 2) reveals what appear to be two different groupings of the data. It was of interest to determine which chemical constituent(s) were responsible for these differences. To do so, we examined loadings and volcano plots (Fig. 3) and determined that the peak area of a constituent with *m/z* of 401.2064 and retention time of 2.70 varied across the two groups. This constituent was isolated and identified as the known kratom oxindole alkaloid speciofoline (**3**, Fig. 1) ($[M+H]^+ = 401.2071$, $\Delta = 1.7$ ppm) based on comparison of NMR, ECD, and mass spectrometry data with literature values[23–26] (Table S2, Figure S1 and S2). Several other

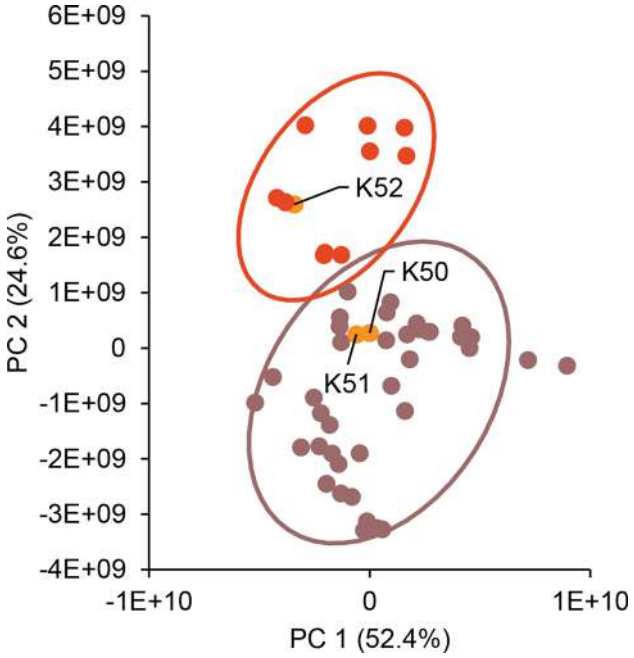


Figure 2. Principal component analysis (PCA) scores plot generated based on LC–MS metabolomics traces of kratom (*Mitragyna speciosa*) samples in Table 1. Each data point represents the average peak areas of triplicate extractions prepared for each sample. Samples selected for in-depth analysis herein (K50–K52) are highlighted in orange. Two groupings were evident from the initial analysis (indicated here with separate Hotelling's T^2 95% confidence intervals). The underlying cause for these apparent differences in the data was further explored with loadings plot and volcano plot data (Fig. 3).

ions were also observed to vary in content across the samples (Fig. 3), and based on their similar mass defect to indole alkaloids[27], are likely other more minor alkaloids of kratom. Structures of these other putative alkaloids were not determined in this study.

Based on the chemical data (Fig. 3), it appears that the commercial kratom products studied herein can be grouped into separate "high-speciofoline" and "low-speciofoline" chemotypes. To demonstrate this further, a comparison is shown of chromatographic peak area across all samples for the ions corresponding to mitragynine and speciofoline (Fig. 4). The mitragynine content in the commercial samples varied by fourfold, while the speciofoline content varied by more than 90-fold, with a minority of samples appearing as "high-speciofoline" outliers (Fig. 4).

To further explore the phenomenon of variable speciofoline levels in kratom samples, we utilized the isolated speciofoline to conduct quantitative analysis for representative high and low-speciofoline materials (Table 1). A limitation of working with commercial kratom samples is that morphological information that could be used to identify the plants is lost when they are powdered. To address this limitation, we obtained a young, living plant from a supplier in Ohio, which appeared based on morphological characteristics to be authentic *M. speciosa*. This sample was coded as K55 and was included in the quantitative analysis.

As expected, the quantities of alkaloids differ in the two commercial kratom products subjected to quantitative analysis (K51 and K52). Sample K51 contained mitragynine at a level of 11.45 ± 0.74 mg/g (1.145 ± 0.074%), and 7-hydroxymitragynine at a level of 0.056 ± 0.019 (0.0056 ± 0.002%). These values are consistent with prior literature[28], which found that mitragynine was detected in commercial kratom products at a concentration range between 1 and 6% of leaf content and that 7-hydroxymitragynine levels ranged from 0.01 to 0.04% of leaf content. In contrast, sample K52 contains lower levels of mitragynine (8.13 ± 0.95 mg/g) but, as expected based on the data in Fig. 4, has a much higher content of speciofoline (Table 1) than sample K51. Interestingly, the kratom plant used for morphological identification (K55) showed much lower abundance of mitragynine and 7-hydroxymitragynine than either of the commercial kratom samples (K51 and K52). The reason for this difference cannot be ascertained with the experimental design employed for this study, but may be related to differences in plant age and/or growing conditions for K55 as compared to the commercial kratom products. Because of the markedly low alkaloid levels (Table 1), the K55 kratom sample was employed only for genetic studies and was not further evaluated for pharmacological activity.

Genetic differences do not appear to explain the observed chemotypes in commercial kratom samples. There are 10 related species within the *Mitragyna* genus[6], including *Mitragyna speciosa* (Korth.) Havil., *Mitragyna diversifolia* (Wall. ex G. Don) Havil., *Mitragyna hirsuta* Havil., and *Mitragyna rotundifolia* (Roxb.) Kuntze. While these species can be distinguished by morphological characteristics in raw plant material, contamination or misidentification could occur during collection and could not be distinguished in powdered material such as that used in this study. The dramatic differences in alkaloid quantity in various kratom samples

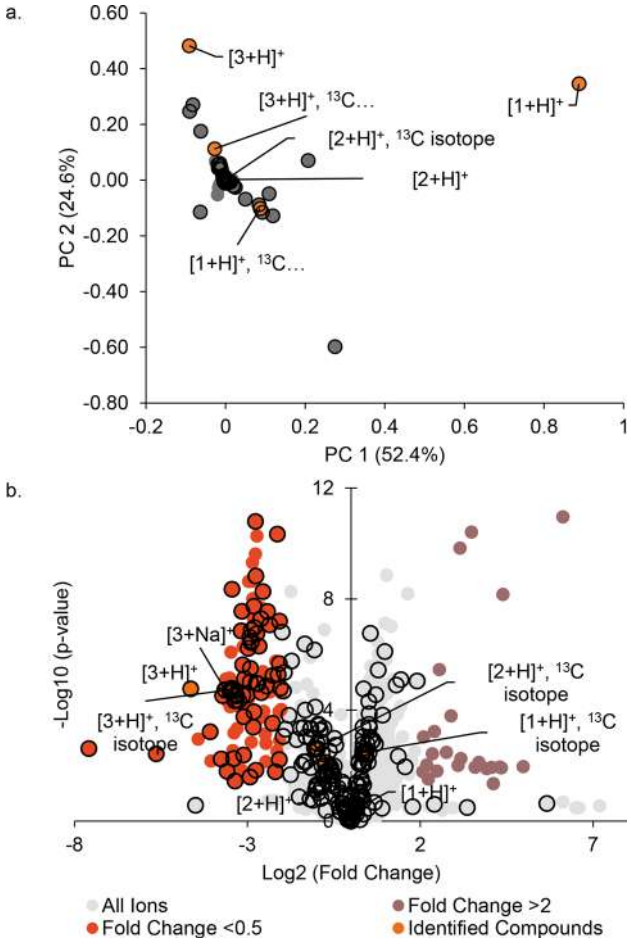


Figure 3. Plots of significant features based upon the PCA analysis. (**A**) Loadings plot, where the spatial arrangement of the features (unique *m/z*-RT pairings) corresponds to the distribution of the samples (i.e., scores plot, Fig. 2). Speciofoline (**3**) appears to be a feature that is responsible for discriminating between the two sample groupings along the y-axis (PC2). (**B**) Volcano plot, highlighting significant features of the metabolites based on statistical testing (p value < 0.05, fold change > 2 or < 0.5) between the two groupings, with the further a feature's position away from the origin (0, 0), the more significant the feature is. The speciofoline-rich group (red circles) yielded 120 significant features, while 19 features were significantly distinct in the non-speciofoline-rich grouping of commercial samples (purple circles). Speciofoline (**3**) was found to be a metabolite that differed significantly across sample groups, while mitragynine (**1**) and 7-hydroxymitragynine (**2**) were not significantly different between the two groups (orange circles). Other features predicted to be alkaloids based on mass defect (open circles). Labels on plot refer to the compound number and the type of ion observed (i.e.$[3+H]^+$ refers to the protonated molecule of speciofoline).

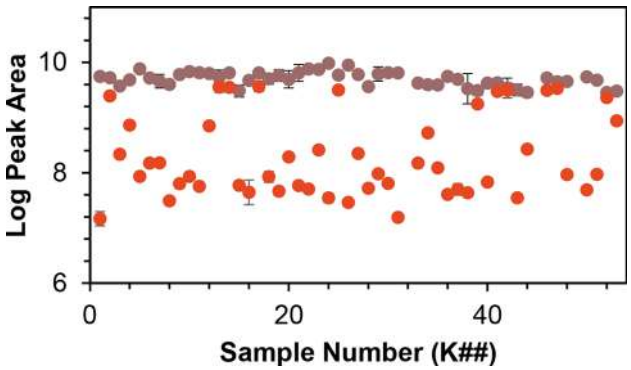


Figure 4. Comparison of mitragyine and speciofoline levels across samples of *M. speciosa*. The x-axis represents arbitrary codes assigned to the samples, as described in Table S1, and the y-axis represents the log of the peak area for the relevant selected ion trace in LC–MS analysis.

Sample code	Mitragynine (mg/g)[b]	7-Hydroxymitragynine (mg/g)[b]	Speciofoline (mg/g)[b]
K50	11.37 ± 0.83	0.053 ± 0.014	< LOQ[c]
K51	11.45 ± 0.74	0.056 ± 0.019	< LOQ[c]
K52	8.13 ± 0.95	0.0349 ± 0.0030	4.13 ± 0.45
K55	0.75 ± 0.14	< LOQ[d]	< LOQ[c]

Table 1. Alkaloid quantity in Kratom products. Samples K50, K51 and K52 are commercial kratom products that were purchased for this study. Sample K55 is a young cultivated kratom plant. Quantities were calculated by preparing methanolic extracts in triplicate and analyzing by LC–MS. [a]Quantity is denoted as mg of compound per g of dried kratom plant material. [b]Uncertainty calculated using standard deviation of triplicate extractions, each analyzed separately. [c]Limit of quantitation (LOQ) for speciofoline is 4.9 ng/mL (Table S3) and this sample contained speciofoline below that level. [d]LOQ for 7-hydroxymitragynine is 0.49 ng/mL (Table S3) and this sample contained 7-hydroxymitragynine below that level.

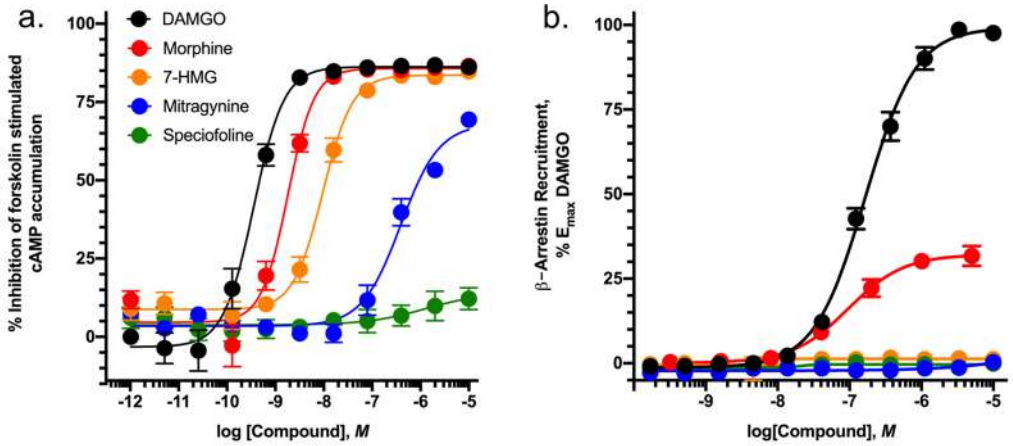
demonstrated in Table 1 and Fig. 4 led us to ask whether the samples reported to be *Mitragyna speciosa* might be different species or strains of *Mitragyna*. To address this question, we extracted DNA from three kratom samples with varying speciofoline content (K51, K52 and K55 in Table 1), and used the resulting data for DNA barcoding via BLAST search of *rbcL* + *matK* (core plant barcoding markers) and performed maximum likelihood and Bayesian phylogenetic analysis, of the combined *matK* + *trnH-psbA* and the separate nrDNA ITS regions. These regions were selected because they are designated as the official DNA barcodes for the plant kingdom[29–37].

Molecular identification of plant samples using a BLAST search of core plant loci (*rbcL* + *matK*) using the BOLDSYSTEMS database (version 4) showed that samples K51, K52 and K55 had ≥ 99% sequence similarity with *M. speciosa* (Figures S4–S6). Based on uncorrected p distances calculated for *trnH-psbA*, there was ≥ 99–100% sequence similarity with sequences of published *M. speciosa*. This means that samples K51, K52 and K55 were more similar to sequences of *M. speciosa* and more dissimilar to sequences of *M. diversifolia, M. hirsuta, M. rotundifolia*, and *M. parvifolia* (Table S5a and b). Maximum likelihood analysis of the partial *matK* + *trnH-psbA* fragments placed all three samples in a strongly supported clade (100% RAxML bootstrap support, 78% PhyML bootstrap and ≥ 95% Bayesian posterior probability support) with the published sequences for *M. speciosa* (Figure S7). Notably, this analysis included a partial *matK* sequence from a recently sequenced genome of *M. speciosa* (BioProject: PRJNA325670). Finally, phylogenetic analysis using the nrDNA ITS region also placed K51 and K52 in a strongly supported clade (≥ 84% PhyML bootstrap and ≥ 95% Bayesian posterior probability support) with *M. speciosa* sequences (Figure S8). However, the ITS analysis showed that K55 does not group with *M. speciosa*, but rather formed a cluster with other *Mitragyna* spp., albeit without any significant bootstrap support and/or Bayesian posterior probability support (Figure S8).

The results of DNA barcoding and phylogenetic analyses classify all three kratom products as *Mitragyna speciosa*, based on comparison with the published sequences for this species, except for the ITS analysis, which showed a conflicting outcome for sample K55. Thus, it appears that the observed differences in speciofoline content among the various kratom samples is not due to misidentification of species or substitution of a different *Mitragyna* species for *M. speciosa*. Notably, a number of other factors including plant age, growing conditions, and processing methods could all contribute to the observed differences in alkaloid level in the various kratom samples. Exploring these differences would require a different experimental design and could be the subject of future studies. For the purpose of the current investigation, the important finding is that dramatic differences in alkaloid content can be observed among different samples of what appear to be authentic (based on molecular data) *M. speciosa*. Such differences in alkaloid profile could have implications for biological activity, a possibility that we explore in the next sections.

Differences in alkaloid profile in kratom plant material is expected to have implications for opioid receptor-mediated activity. The alkaloid speciofoline is an oxindole, while the more commonly studied alkaloids mitragynine and 7-hydroxymitragynine are indoles (Scheme 1). Thus, we hypothesized that there might be biological implications to the observed variability in speciofoline levels across kratom extracts (Table 1). To explore this hypothesis, we tested mitragynine (**1**), 7-hydroxymitragynine (**2**) and speciofoline (**3**) for μ-opioid receptor inhibition of forskolin-stimulated cyclic-AMP (cAMP) accumulation and for their ability to recruit β-arrestin-2 in CHO-K1 cells stably expressing the μ-opioid receptor. Results indicate that DAMGO and morphine fully inhibit cAMP accumulation, mitragynine and 7-hydroxymitragynine partially inhibit cAMP accumulation, whereas speciofoline has no effect in this assay (Fig. 5A). Furthermore, DAMGO and morphine effectively recruit β-arrestin-2 in CHO cells (Fig. 5B), while mitragynine, 7-hydroxymitragynine, and speciofoline fail to recruit β-arrestin-2 at concentrations as high as 10 mM.

To further characterize the pharmacology of the kratom alkaloids, competitive radioligand binding assays were conducted with HEK293 cells stably expressing μ-, κ- and δ-opioid receptors to assess the affinity of the alkaloids at these receptors (Table 2). Mitragynine exhibited moderate affinity at μ- and κ-opioid receptors whereas 7-hydroxymitragynine showed strong affinity at the μ- opioid receptor (14X greater than mitragynine) and moderate affinity at the κ- and δ-opioid receptors (4X and 70X greater than mitragynine). In contrast, speciofoline did not exhibit appreciable affinity for any of the opioid receptors. Results for mitragynine and 7-hydroxymitragynine are consistent with previous literature reports[14,15]; however, this is the first published assessment of receptor binding of speciofoline at opioid receptors.



	cAMP accumulation		*β-arrestin recruitment*	
Compound	**EC_{50}**	**E_{max} %**	**EC_{50}**	**E_{max} %**
DAMGO	0.9 ± 0.2	86.2 ± 1.8	180.4 ± 37	100.0
Morphine	4.8 ± 1.0	86.9 ± 1.5	81.7 ± 36.5	30.7 ± 13.1
Mitragynine	398 ± 36.2	69.4 ± 1.4	--	--
7-Hydroxymitragynine	13.6 ± 0.9	85.9 ± 0.4	--	--
Speciofoline	4243 ± 2932	12.2 ± 3.5	--	--

Figure 5. Kratom alkaloid mediated inhibition of forskolin-stimulated cAMP accumulation and β-arrestin recruitment. (**A**) CHO-K1 cells stably expressing hMOR were incubated with the indicated concentrations of compounds as described in Methods. Data are expressed as the percent of forskolin-stimulated cAMP accumulation and represent mean ± SEM of 3–4 experiments. (**B**) β-arrestin-2 recruitment. CHO-K1 cells stably expressing the μ-opioid receptor and β-galactosidase linked β-arrestin-2 were incubated with the indicated concentrations of compounds as described in Methods. Data are expressed as the percent of maximal DAMGO mediated response and represent mean ± SEM of 3–4 experiments. E_{max} and EC_{50} values for cAMP accumulation and β-arrestin2 recruitment are listed in the table.

	Receptor binding affinity (Ki ± SEM, nM)		
Compound	**μ-opioid**	**ƙ-opioid**	**δ-opioid**
Mitragynine	238 ± 28	482 ± 29	*> 10,000*
7-Hydroxymitragynine	16 ± 1	113 ± 37	137 ± 21
Speciofoline	*> 10,000*	*> 10,000*	*> 10,000*
Morphine	1.50 ± 0.04	48.9 ± 5.0	*> 10,000*

Table 2. Opioid receptor affinities for kratom alkaloids. Competitive binding studies were performed against ^{3}H-DAMGO, ^{3}H-U69-593 and ^{3}H-deltorphin for μ-, ƙ-,and δ-opioid receptors respectively. Results are presented as nM ± SEM from a minimum of three replicates.

The differential binding affinity of mitragynine and 7-hydroxymitragynine at opioid receptors (Table 2) is consistent with previous literature[14,15]. Also consistent with literature reports[14,15] both mitragynine and 7-hydroxymitragynine function as partial agonists at the μ-opioid receptors and are functionally selective for G protein versus β-arrestin-2 signaling (Fig. 5). These effects are quite different from the binding and activity of typical opioids such as morphine (Fig. 5). The failure of mitragynine or 7-hydroxymitragynine to recruit β-arrestin-2 may be an important factor separating the analgesic effect from the adverse effects of typical opiates such as abuse liability, respiratory depression and constipation. However, 7-hydroxymitgraynine, but not mitragynine, exhibits abuse liability in rodent models of human drug-taking[18], suggesting that β-arrestin-2 signaling may not be a significant contributing factor to the addictive properties of opiates. In contrast, speciofoline does not bind at the opioid receptors and does not exhibit functional activity via the μ-opioid receptor. Future studies are warranted to characterize speciofoline and determine if the ligand may induce antinociception or other activities through non-opioid receptor systems, and to further characterize the pharmacological effects of kratom extracts and their isolated alkaloids in vivo.

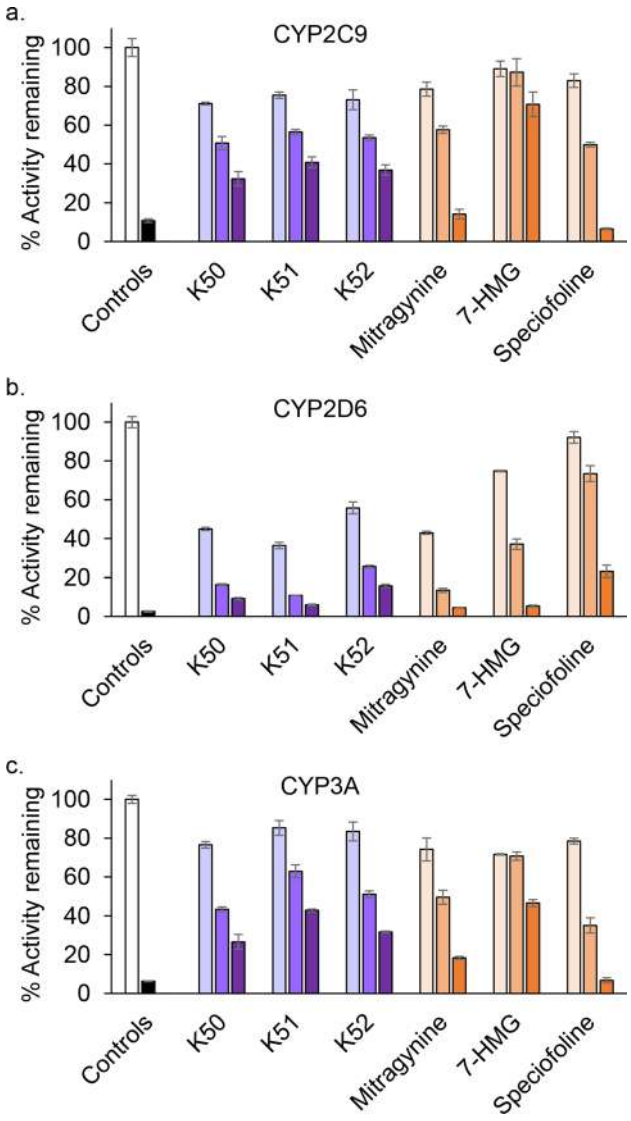


Figure 6. Kratom extracts (K50, K51, and K52), mitragynine, 7-hydroxymitragynine (7-HMG), and speciofoline showed concentration-dependent inhibition of cytochrome P450 (CYP) enzymes in human liver microsomes. Diclofenac 4′-hydroxylation (**A**), dextromethorphan *O*-demethylation (**B**), and midazolam 1′-hydroxylation (**C**) were used as measures of CYP2C9, CYP2D6, and CYP3A activity, respectively. Extracts were tested at 2, 10, and 20 µg/mL; alkaloids were tested at 1, 10, and 100 µM. Positive control inhibitors included sulfaphenazole (1 µM), quinidine (2 µM), and ketoconazole (0.1 µM) for CYP2C9, CYP2D6, and CYP3A activity, respectively. Control activities averaged 220, 70, and 1270 pmol/min/mg, respectively. Bars and error bars denote means and standard deviations, respectively, of triplicate incubations.

High-speciofoline and low-speciofoline kratom extracts have similar inhibitory effects on three major CYPs in vitro. The cytochromes P450 (CYPs) constitute a superfamily of oxidative enzymes that mediate the metabolism of approximately 50% of the top 200 most prescribed drugs[38]. Inhibition of these enzymes by xenobiotics, including drugs and botanical products (precipitants), can lead to increased systemic exposure to the object drug and potential adverse or toxic effects. Kratom extracts[39] and mitragynine[11,40] have been tested as inhibitors of several CYPs, including CYP1A2, CYP2C9, CYP2C19, CYP2D6, and CYP3A4. Tested extracts and mitragynine were generally stronger inhibitors of CYP2D6 activity relative to the other isoforms[39,40].

Mitragynine has been reported to inhibit several cytochrome P450s (CYPs)[11,39–41], which could lead to increased systemic exposure to co-consumed drugs, including opioids, and potential adverse effects. We evaluated whether the observed variability in alkaloid profile for kratom extracts results in altered inhibitory effects on three major CYPs, specifically CYP2C9 (Fig. 6A), CYP2D6 (Fig. 6B), and CYP3A (Fig. 6C). All three kratom extracts tested showed concentration-dependent inhibition of each CYP, with stronger effects on CYP2D6 compared to CYP2C9 and CYP3A. Effects on each CYP were similar between the high speciofoline (K52) and low speciofoline (K50 and K51) kratom samples (Fig. 6). Mitragynine and speciofoline showed similar inhibitory

effects towards CYP2C9 and CYP3A activity, whereas mitragynine showed stronger inhibition than speciofoline against CYP2D6 activity. Relative to mitragynine and speciofoline, 7-hydroxymitragynine was a weak inhibitor of CYP2C9 and CYP3A activity and a moderate inhibitor of CYP2D6 activity (i.e., inhibitory effects via mitragynine > 7-hydroxymitragynine > speciofoline).The concentration of mitragynine in the kratom extracts tested at 20 μg/mL was ~ 1 μM. The stronger inhibitory effects of the extracts compared to an equimolar concentration of isolated mitragynine indicated additional CYP inhibitors are present in the extracts. If the concentration of mitragynine in the liver upon kratom consumption nears the IC_{50} values estimated from the screening assay (~ 1 μM for CYP2D6; 10–100 μM for CYP2C9 and CYP3A), a CYP-mediated kratom-drug interaction is possible. In addition, because CYP3A is also expressed in the intestine, mitragynine concentrations in the intestine could well exceed 10 μM, increasing the possibility of CYP3A-mediated kratom-drug interactions. A comprehensive clinical pharmacokinetic study with a well-characterized kratom product is needed to determine the systemic exposure to mitragynine and other kratom alkaloids, as well as additional mechanistic in vitro studies, to ascertain the likelihood of pharmacokinetic kratom-drug interactions.

Discussion

Collectively, our chemical and in vitro studies support previous studies suggesting that the purported efficacy of kratom against pain is mediated at least in part by kratom alkaloids that function as biased ligands of the μ-opioid receptors. These findings further highlight the value of pursuing mitragynine and related alkaloids as an alternative strategy to treat pain, as, consistent with previous literature, our findings suggest that they have less abuse liability than classic opiates such as morphine. Future in vivo studies are necessary to further explore this possibility.

Our results also demonstrate that the identity and abundance of major alkaloids is highly variable in genetically similar kratom plant material, with some kratom products being dominated by the indole alkaloid mitragynine and others containing high levels of both mitragynine and the oxindole alkaloid speciofoline. On the basis of our studies, it is reasonable to predict that individuals who self-administer kratom tea to treat pain, addiction, or depression might achieve very different results depending on the alkaloid profile of the product that they use. Such differences in activity would be difficult to predict for consumers, given that the variability in alkaloid content is in no way reflected in the product labeling. Additional studies are warranted to understand the underlying causes for variability in alkaloid content in kratom plant material, and to evaluate the therapeutic implications of this variability.

Consistent with previous literature, our results raise concern about potential drug interactions that may occur when kratom is consumed concurrently with opiates or other drugs metabolized by the cytochrome P450s. Following up on these findings, future clinical studies to evaluate the safety and efficacy of kratom and its alkaloids would be of great benefit to the public that may employ this botanical for the treatment of pain, currently or in the future. In such studies, it would be useful to consider the potential differences in activity that would be observed by complex *M. speciosa* preparations as compared to their isolated alkaloids, and to account for potential differences in alkaloid profile in different kratom materials. Chemical profiling of samples prior to evaluating their biological activity is clearly crucial, and it is important that the biological studies be informed by the results of the chemical analyses. This study demonstrates how untargeted mass spectrometry metabolomics followed by targeted mass spectrometric quantitation can be effectively employed to generate chemical data valuable to inform biological evaluation. As such, our approach can serve as a model for the design of future studies.

Materials/methods

General. DAMGO, quinidine, ketoconazole, morphine sulfate, and alprazolam were purchased from MilliporeSigma (St Louis, MO). Mitragynine, 7-hydroxymitragynine, midazolam, 1′-hydroxymidazolam, sulfaphenazole, and NADPH were obtained from Cayman Chemical Company (Ann Arbor, MI). Diclofenac, 4′-hydroxydiclofenac, dextromethorphan, and dextrorphan were obtained from Toronto Research Chemicals, Inc. (Toronto, Canada). Potassium phosphate buffer salts were obtained from Fisher Scientific (Fair Lawn, NJ). Human liver microsomes were obtained from XenoTech, LLC (H0604, mixed gender, pool from 15 donors, lot no. 1010191; Kansas City, KS). All other chemicals and reagents used were analytical grade.

Acquisition of Kratom plant material. Fifty commercial products labeled as kratom were obtained from vendors within the US. A table with details about these products is available in the supporting information (Table S1). The commercial kratom products were obtained as dried powders and were various shades of grey-green, with the exception of product K49, which was obtained as a cut leaf. All plant material was stored dry at room temperature until time of analysis.

A cultivated plant of kratom *(Mitragyna speciosa*) (coded as K55) was generously donated to the project by Shon Lenzo. The plant was grown from the cutting of a 7-year-old tree from a strain labeled as "rifat." Prior to shipping to the University of North Carolina Greensboro, the cutting was grown indoors in a greenhouse under a 1000 W high pressure sodium (HPS) LED light for 6 months. Upon receipt, the plant was repotted and transferred to an outdoor location near the University of North Carolina Greensboro, where it was grown for several summer months. Leaves were cut from this plant, dried at room temperature and powdered for extraction using a Wiley Mill Standard Model No. 3 (Arthur Thomas Company). A sample of leaves from this plant, harvested at the coordinates 36°4′53.616″ N 79°47′6.18″ W, was pressed and submitted to the University of North Carolina Herbarium, with accession #670043 (sernecportal.org catalog number NCU00433756).

Isolation of speciofoline. Kratom plant material (K52, 1 kg) was extracted exhaustively with 2 L of $CHCl_3$/CH_3OH (1:1) and 50 mL of KOH (10%) by maceration over 24 h at room temperature. The mixture was filtered,

and the solvent was evaporated under reduced pressure. The dried extract was reconstituted in 1 N HCl solution and hexanes (1:1), transferred to a separatory funnel, and shaken vigorously. After removal of the hexanes phase, the pH of the aqueous phase was adjusted to 9.0 with concentrated NH_4OH solution. The basic phase was exhaustively extracted with $CHCl_3$, and the organic phase was evaporated to dryness to yield 1.0 g of the dried extract. Fractionation of the extract was conducted with normal phase flash chromatography using a silica column (24 g) and a gradient solvent system of hexane–acetone-CH_3OH over 52 min at a flow rate of 35 mL/min. In total, thirteen pooled fractions were obtained. Fraction 4 was subjected to preparative reversed phase HPLC over a Luna PFP(2) (pentafluorophenyl) column (Phenomenex, 250 mm × 21.2 mm, 5 µm) using a gradient of 70:30 to 100:0 CH_3OH–H_2O (10 mM of ammonium acetate in both phases) over 20 min at a flow rate of 20 mL/min. This process yielded 25.0 mg of speciofoline (95% purity based on UPLC-UV), the structure of which was solved by comparison of the calculated ($[M+H]^+$ $C_{22}H_{29}N_2O_5$, 401.2076) and measured (401.2067) *m/z* values and by comparison of key ^{1}H-NMR signals (Figure S1 and Table S2) and the ECD spectra (Figure S2) with those reported in the literature[25].

Extraction for metabolomics profiling. For each extraction of powdered kratom sample, 50 mg of dried kratom plant material was extracted with 5.0 mL of methanol in a 25 mL scintillation vial. Extracts were shaken at 150 rpm for 12 h and the powder was allowed to settle. The extracts were decanted into clean scintillation vials and dried under nitrogen. Extracts were resuspended in methanol containing 0.125 µg/mL of mitragynine-D_3 (MilliporeSigma, St. Louis, MO, USA) for analysis. All diluted extracts contained mitragynine-D_3 at a final concentration of 0.125 µg/mL as an internal standard to monitor the consistency of the instrument performance.

Ultraperformance liquid chromatography–mass spectrometry (UPLC-MS) analysis of Kratom extracts. Untargeted metabolomics analyses were conducted using an Acquity Ultra-Performance Liquid Chromatography system (Waters, Milford, MA, USA) coupled to a Q Exactive Plus Hybrid Quadrupole-Orbitrap mass spectrometer (Thermo Fisher Scientific, Waltham, MA). A Kinetex F5 column (Phenomenex, 50 mm × 1 mm, 2.6 µm) was employed with a flow rate of 0.6 mL/min and a column temperature of 40 °C. The starting conditions were 65% A, optima grade water with 0.1% formic acid and 2 mM ammonium formate, and 35% B, optima grade methanol with 0.1% formic acid and 2 mM ammonium formate. The gradient is as follows: a slight decrease to 55% A at 0.5 min which is held until 3.0 min, followed by a linear decrease to 20% A at 10 min down to 0% A at 11.0 min. The gradient returns to starting conditions at 12 min and that is held for 1 min. A 5 µL injection was employed for all samples using a 5 µL sample loop.

Mass spectrometry analysis utilized a heated electrospray ionization (HESI) source: spray voltage 3.5 kV, capillary temperature 275 °C, sheath gas 55 arbitrary units, auxiliary gas 15 arbitrary units, sweep gas 3 arbitrary units, heater temperature 450 °C, and an RF level of 50. A scan range of 150–900 m/*z* was employed in positive mode with a resolving power of 70,000 and an AGC target of 3E6. MS/MS data were collected on the standards using a normalized collision energy (NCE) of 35 V.

Metabolomics data processing. The UPLC-MS data were analyzed, aligned, and filtered using MZmine 2.28 software (https://mzmine.github.io/) with a slightly modified version of a previously reported method[42]. The following parameters were used for peak detection of the data: noise level (absolute value), 1×10^5 counts; minimum peak duration 0.5 min; tolerance for *m/z* intensity variation, 20%. Peak list filtering and retention time alignment algorithms were performed to refine peak detection. The *join algorithm* was used to integrate all the chromatograms into a single data matrix using the following parameters: the balance between *m/z* and retention time was set at 10.0 each, *m/z* tolerance was set at 0.001 or 5 ppm, and retention time tolerance was defined as 0.5 min. Once an aligned peak list was created, features (*m/z* and retention time pairs) present in the first blank sample of the run were deleted as a means of blank filtering. The peak areas for individual ions detected in the process replicates and analytical replicates were exported from the data matrix for further analysis. The peak list from MZmine was exported into Excel (Microsoft, Redmond, WA) for further filtering based on relative standard deviation between analytical replicates. Analytical replicates would be expected to have comparable profiles, and similar peak areas for each feature. Ions detected within the analytical replicates with disparate peak areas (based on an RSD cutoff of 25%) were assigned as artefacts of the instrument and excluded from the metabolomics analysis[43].

Principal component analysis (PCA) was performed using Sirius version 10.0 (Pattern Recognition Systems AS, Bergen, Norway). The 95% confidence intervals were calculated using Hotelling's T^2 with the R package 'car'[44] using an R script available from https://github.com/joshkellogg/Composite-score.

Identification and quantitative analysis of alkaloids in kratom plant material. Standard solutions of mitragynine and 7-hydroxymitragynine were purchased from MilliporeSigma (St. Louis, MO, USA) and supplied as methanolic solutions at a concentration of 100 µg/mL. An analytical standard for speciofoline was unavailable for purchase, therefore an aliquot of purified speciofoline isolated as part of this study was dissolved in methanol at a concentration of 100 µg/mL. A 100 µL aliquot of each standard solution was combined with 600 µL of methanol and 100 µL of water to make a stock solution containing 10 µg/mL of each alkaloid of interest. A standard calibration curve was prepared using the stock solution in two-fold dilutions from 5000 ng/mL to 4.9 ng/mL followed by a single 1:10 dilution of the lowest concentration in methanol. All samples were analyzed using the UPLC-MS method previously described. The selected-ion chromatograms for the calculated *m/z* values of the alkaloids of interest (mitragynine, $[M+H]^+ = 399.2284$; 7-hydroxymitragynine, $[M+H]^+ = 415.2233$; speciofoline, $[M+H]^+ = 401.2076$) within a mass error of 5 ppm were plotted, and a calibration curve was constructed as area of these chromatograms versus concentrations. Calibration parameters are provided in Table S3.

Identity of the alkaloids in the *M. speciosa* plant material was confirmed by matching retention time and m/z value with that of the standards. The alkaloid concentrations in the kratom extracts were determined using linear regression of these calibration curves. Peak areas were calculated using the Thermo Xcalibur QuanBrowser version 3.0.63 and statistical analysis was performed using RStudio version 1.2.1335.

Authentication of plant material. Authentication (assignment of genus and species) of kratom products K51, K52 (powdered material), and K55 (vouchered plant specimen) was accomplished using DNA barcoding and maximum likelihood phylogenetic analysis. The most widely recommended loci, the plastid intergenic spacer *trnH-psbA, rbcL*, *matK,* and the nuclear ribosomal internal transcribed spacers (ITS) were employed. These regions have been designated as the official DNA barcodes for the plant kingdom[29,30].

For DNA extraction from the plant material (K55), a whole leaf sample was ground to a fine powder with a sterile mortar and pestle using liquid nitrogen. Approximately 5 mg of plant powder from the plant (K55) or from commercial sources (K51, K52) was transferred to a bashing bead tube with DNA lysis buffer provided by Zymo Research Quick-DNA plant/seed miniprep DNA extraction kit. The powder in the bashing bead tube was further disrupted and homogenized in a Qiagen TissueLyser LT beadmill for five minutes. For K51 and K52, the samples were separately ground in bashing bead tubes, without treatment with liquid nitrogen. Genomic DNA was then extracted according to kit instructions.

Detailed methods for DNA extraction, PCR amplification, and Sanger sequencing are outlined previously[45]. The partial *rbcL* region was amplified and sequenced using primers rbcLa-F and rbcLa-R[30,46]. Details of PCR primers and thermocycler parameters for all fragments amplified are outlined in Supplementary Table S4.

For identification of kratom plant material via DNA Barcoding, two plastid regions (*rbcL* + *matK*) sequences were utilized in a combined analysis for plant identification by BLAST searching against the BOLDSYSTEMS database.

Uncorrected p-distances were calculated in Geneious, a bioinformatics desktop software package (https://www.biomatters.com) for sequences obtained from the *trnH-psbA* plastid region, which is considered as one of the most variable regions of angiosperm plants[47–50]. Comparisons were made using published sequence data from NCBI GenBank for *Mitragyna* spp., including *M. speciosa* and the newly obtained sequences in this study (Supplementary Table S5). The p-distances were obtained by dividing the number of nucleotide differences by the total number of nucleotides being compared. A cut off proxy of ≥ 98–100% was applied as a criterion to designate similar species; therefore, to be considered the same species based on *trnH-psbA* sequence comparison, the taxa being compared should have ≥ 98% sequence similarity (with 1–2% intraspecific variation or differences).

Maximum likelihood and Bayesian analysis was performed separately for both the combined analysis using *matK* + *trnH-psbA,* and the nrDNA ITS region to place the kratom samples (K51, K52, and K55) into a phylogenetic framework with authentic published sequences of *M. speciosa*[48–52] using methods outlined previously[53] (see Supplementary for details).

Nauclea officinalis (Rubiaceae) was used as an outgroup taxon in both *matK* + *trnH-psbA* and ITS region analysis, respectively. The sequence data obtained from samples K51, K52, and K55 are deposited in GenBank and accession numbers are provided in Table S6.

Cytochrome P450 (CYP) inhibition assay. Mitragynine and 7-hydroxymitragynine (purchased), speciofoline (the isolated alkaloid), and three methanolic kratom extracts (prepared from commercial products K50, K51, and K52; Table 1), were tested as inhibitors of CYP2C9, CYP2D6, and CYP3A activities using a cocktail approach[54]. Diclofenac (4 µM), dextromethorphan (4 µM), and midazolam (2 µM) were used as probe substrates for CYP2C9, CYP2D6, and CYP3A activities, respectively. Kratom extract (2, 10, 20 µg/mL) or alkaloid (1, 10, 100 µM) was incubated with human liver microsomes (0.05 mg/mL) in 100 mM potassium phosphate buffer, pH 7.4. Positive control inhibitors included sulfaphenazole (1 µM; CYP2C9), quinidine (2 µM; CYP2D6), and ketoconazole (0.1 µM; CYP3A). Final organic solvent (methanol) concentration in each incubation was limited to 0.8% v/v. After equilibrating the mixtures for 5 min at 37 °C, the reaction was initiated by addition of NADPH (1 mM final concentration) and incubated for 10 min (CYP2C9 and CYP2D6 activities) or 2 min (CYP3A activity) at 37 °C. Reactions were quenched with 2 volumes of cold methanol containing internal standard (alprazolam, 0.1 µM). The quenched mixtures were centrifuged at 2270 × *g* for 10 min, and the supernatants were subjected to UPLC-MS/MS analysis for quantification of 4′-hydroxydiclofenac, dextrorphan, and 1′-hydroxymidazolam. The percent of CYP inhibition in the presence of the test article was calculated relative to solvent control.

The CYP-mediated metabolites (4′-hydroxydiclofenac, dextrorphan, and 1′-hydroxymidazolam) were quantified using a QTRAP system (AB Sciex, Framingham, MA), operating in positive electrospray ionization mode, coupled to a Shimadzu Nexera X2 UHPLC (Shimadzu Corporation, Tokyo, Japan). Chromatographic separation was achieved using an Acquity C18 column (3 µm, 50 × 2.1 mm, Thermo Scientific, Waltham, MA) with a guard column, heated to 40 °C, and a binary gradient at a flow rate of 0.5 mL/min. The mobile phase consisted of 0.1% formic acid in water (A) and 0.1% formic acid in methanol (B). The following gradient was applied: 0–0.4 min, 10% B; 0.4–1.0 min, 10–95% B; 1.0–2.0 min, 95% B; 2.0–2.1 min, 95%-10% B; and 2.1–3.0 min, 10% B. The following multiple reaction monitoring transitions were used: *m/z* 312.0 → 231.0 (4′-hydroxydiclofenac), *m/z* 258.2 → 157.2 (dextrorphan), *m/z* 342.0 → 324.0 (1′-hydroxymidazolam), and *m/z* 309.1 → 281.0 (alprazolam). All analyte concentrations were quantified using MultiQuant software (version 2.1.1; AB Sciex; https://sciex.com/products/software/multiquant-software) by interpolation from calibration curves prepared from relevant matrix-matched standards over a range of concentrations (1.37–1000 nM).

cAMP accumulation assay. The cAMP Hunter eXpress OPRM1 CHO-K1 GPCR assay (Eurofins DiscoverX Corporation, St. Charles, MO, USA) was performed according to manufacturer's instructions. Briefly, cells were seeded in 100 µL cell plating reagent in 96 well plates and allowed to incubate at 37 °C (5% CO_2, 95% relative humidity) for 24 h. Medium was removed from cells, and cells were washed with 30 µL of cell assay buffer. Each compound was assessed using an 11 point fivefold serial dilution with a starting concentration of 10 µM. Aliquots of compound/forskolin solution were added to cells (final concentrations were 20 µM forskolin, and 0.4% DMSO) and incubated at 37 °C and 5% CO_2 for 30 min. Next, cAMP antibody reagent detection solution was added to each well according to manufacturer's instructions at room temperature and protected from light for 1 h. Next, enzyme acceptor solution was added and the solution allowed to incubate at room temperature and protected from light for 3 h. Luminescence was quantified using a SpectraMax iD5 multi-mode microplate reader with SoftMax Pro software (version 7.1, Molecular Devices, San Jose, CA, https://www.moleculardevices.com/products/microplate-readers/acquisition-and-analysis-software/softmax-pro-software). Data were normalized to vehicle and forskolin only control values and analyzed using nonlinear regression with Prism 8.4 (Graphpad, San Diego, CA).

β-arrestin recruitment assay. The PathHunter eXpress OPRM1 CHO-K1 β-Arrestin GPCR assay (Eurofins DiscoverX Corporation) was performed according to the manufacturer's protocol. Briefly, cells were seeded in 100 µL cell plating reagent in 96 well plates and allowed to incubate at 37 °C (5% CO_2, 95% relative humidity) for 48 h. Each compound was assessed using an 11 point fivefold serial dilution with a starting concentration of 10 µM. Aliquots of compound solution were added to cells (final concentrations were 20 µM forskolin, and 0.4% DMSO) and incubated at 37 °C and 5% CO_2 for 30 min. Detection solution was then added to cells and incubated at room temperature protected from light for 1 h. Luminescence was quantified using a SpectraMax iD5 multi-mode microplate reader with SoftMax Pro software (Molecular Devices, San Jose, CA). Data were normalized to control values and analyzed using nonlinear regression to determine maximal and EC_{50} values (Prism 8.4 for Mac; Graphpad San Diego, CA).

Binding affinity to µ-opioid receptors. HEK293 cells expressing human µ, δ- or κ-opioid receptors were washed in ice cold PBS and scraped from cell culture plates in cold 50 mM Tris–HCl buffer, pH 7.4. Cells were pelleted at 5200 × *g* for 10 min at 4 °C. Supernatant was discarded and the pellet was washed with more Tris–HCl buffer, sonicated, and centrifuged at 24,000 × g for 40 min at 4 °C. The pellet was re-suspended in Tris–HCl buffer, sonicated, and aliquoted for storage at − 80 °C until use. A Pierce BCA Protein Assay was utilized to determine protein concentration in the membrane, according to manufacturer's instructions. Membranes were evaluated in saturation experiments to determine the optimal concentration for radioligand binding and functional assays. A competitive binding assay was utilized to determine the K_i of each compound similar to previous reports[55,56]. For this, 25 µg of cell membrane was diluted in 50 mM Tris–HCl, pH 7.4 and pipetted into a 96 well plate. Test compounds were reconstituted in DMSO and added to the reaction plate.[^{3}H]-DAMGO, [^{3}H]-U69-593 or [^{3}H]-deltorphin (for µ-, ƙ-, and δ-opioid receptors respectively) was then added at a final concentration of 2–4 nM (Perkin Elmer, Waltham, MA, USA). The reaction was incubated for 1 h at room temperature prior to transfer onto GF/B filter plates (Perkin Elmer). The plates were washed 10 × with cold buffer and dried for 15 min at 50 °C. MicroScint20 was then added to each well, the plates were sealed, and the counts per minute were quantified on the TopCount NXT Microplate Scintillation counter (Perkin Elmer). The percent displacement of the radiolabeled compound was calculated as described[57]. Unlabeled DAMGO was used to determine non-specific binding control counts, while vehicle (DMSO) was added to determine total radioligand binding on each plate. A series of three-fold dilutions were used for each test compound starting at 10 µM. A control curve of naloxonewas included on each plate for internal control.

Data availability

Raw mass spectrometric data have been made publicly available through MassIVE (MassIVE ID: MSV000086288, https://doi.org/10.25345/C59V0X). All other data are available by request to the corresponding author.

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D.A.T., J.J.K., N.B.C., L.F.B., R.S.T., H.A.R., S.H.E., M.F.P., N.H.O. and N.B.C. wrote and edited the manuscript, D.A.T. coordinated the experimental work and prepared figures, J.J.K. and E.D.W. conducted the extractions and metabolomics analysis, M.K. performed quantitative analysis, L.F.B. performed isolation experiments, R.S.T. conducted the C.Y.P. inhibition assays, S.M. and S.E.H. evaluated receptor binding, T.N.G. provided experimental support for extract analysis and sample acquisition, HAR conducted genetic analysis, M.F.P., N.H.O., N.B.C. secured project funding, analyzed data, and supervised the experimental work.

Competing interests

The authors declare no competing interests.

Additional information

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Alkaloid biosynthesis in medicinal crop kratom (*Mitragyna speciosa*) varies with postharvest, genetic, and seasonal factors

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Introduction: Kratom (*Mitragyna speciosa*), a medicinally valuable ethnobotanical tree native to Southeast Asia, has traditionally been used to combat fatigue and enhance productivity. Recently, it has gained attention in North America and Europe for its potential therapeutic applications, particularly in pain management and opioid withdrawal, positioning it as a candidate for drug development. Postharvest processing is a critical stage that influences chemical transformations of bioactive compounds, yet its impact on kratom remains poorly understood.

Methods: This study investigated the effects of withering duration, drying temperature, cultivar, and season on kratom alkaloid composition and concentration. In Study I, leaves of cultivar Hawaii underwent four withering durations (0, 12, 24, 72 h) followed by drying at five temperatures (–40, 25, 40, 60, 80 °C). In Study II, leaves of cultivar MR-Malaysian were tested under two withering durations (0, 12 h) and two drying temperatures (25, 60 °C). Both studies were conducted across two seasons.

Results: Withering generally increased mitragynine concentrations by 14-65% (w/w) in 'Hawaii' and 3-8% in 'MR-Malaysian' in leaf alkaloid extracts. A 12-h withering followed by drying below 40 °C enhanced speciogynine and paynantheine in 'Hawaii' by 37-48% and 35-67%, respectively. Low drying temperatures preserved mitragynine, speciogynine, and paynantheine across cultivars. The average 7-hydroxymitragynine content in leaf alkaloid extracts ranged from 0.02-0.04% and was detected only in specific seasons, varying by cultivar, suggesting genotype-environment interactions.

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Discussion: This study demonstrates for the first time that kratom alkaloid composition and concentration are substantially influenced by genotype, season, and postharvest handling. These findings underscore the importance of optimizing postharvest processing strategies to enhance beneficial alkaloid profiles in kratom.

KEYWORDS

drying temperature, withering, indole alkaloid, mitragynine, 7-hydroxymitragynine speciogynine

1 Introduction

Kratom (*Mitragyna speciosa* Korth), a small-to-medium sized tropical tree indigenous to Southeast Asia (Figure 1A), has been consumed for centuries by native people as an ethnobotanical medicine to relieve pain and fatigue, treat diarrhea, enhance work productivity, and assist in opioid tapering and withdrawal alleviation (Suwanlert, 1975; Cinosi et al., 2015; Grundmann, 2017). The pharmacological properties of kratom are derived from indole alkaloids that induce non-opioid (dose-dependent stimulant properties) and opioid-like effects upon consumption (Grundmann et al., 2023). More than 40 monoterpene alkaloids have been identified in kratom, and a portion of those and their respective analogs are capable of binding opioid and adrenergic receptors throughout the body. The primary alkaloid, mitragynine, constitutes 0.7-38.7% of the alkaloid suite by mass and produces opioid-like antinociception effects upon binding μ-opioid receptors with partial agonistic activity (Váradi et al., 2016; Obeng et al., 2019). Its metabolite and structural analog,

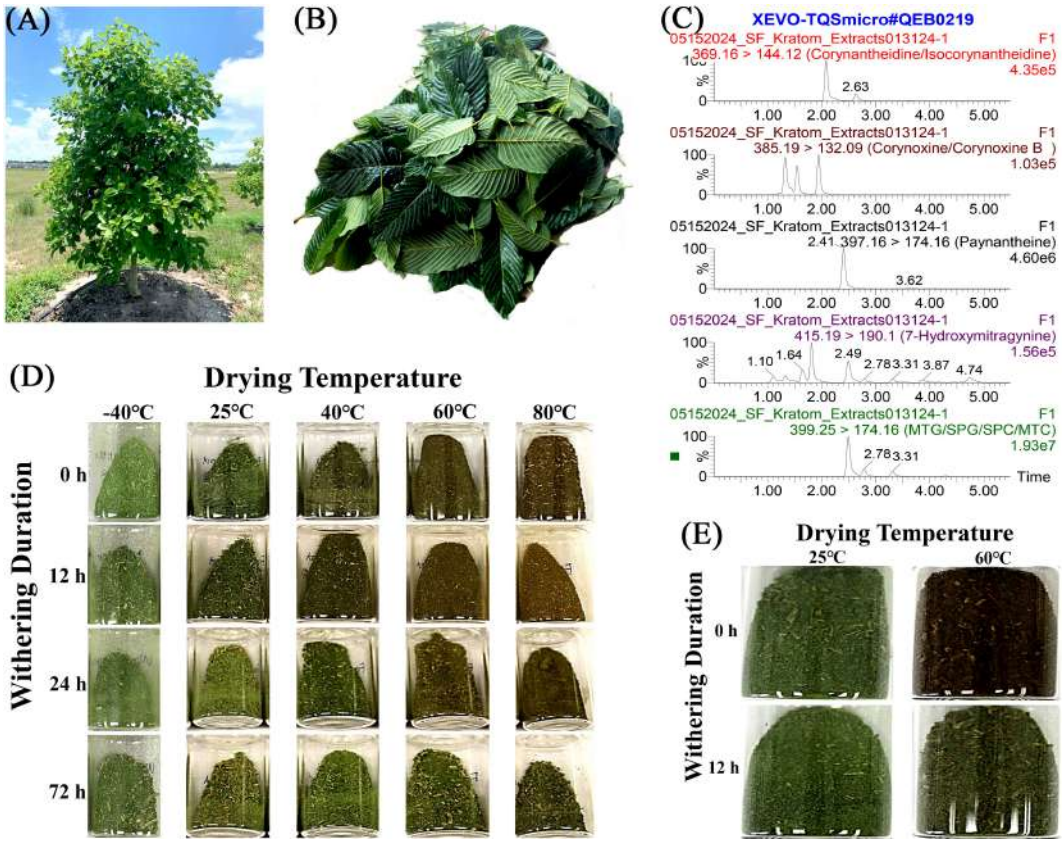


FIGURE 1
A process of postharvest treatments on collected kratom leaves for analysis of alkaloids. **(A)** Kratom tree; **(B)** Harvested bulk kratom leaves; **(C)** Representative LC-MS/MS chromatogram of sample #013124-1; **(D)** Powder color of cultivar Hawaii under each postharvest treatment from Study I; **(E)** Powder color of cultivar MR-Malaysian under each postharvest treatment from Study II.

7-hydroxymitragynine, possesses greater binding and functional activity at μ-opioid receptor than that of morphine; however, its concentration in nature is typically low (Horie et al., 2005; Hassan et al., 2013; Kruegel et al., 2016; Hemby et al., 2019; Obeng et al., 2019). Other minor alkaloids, such as paynantheine, speciogynine, and speciociliatine, possess similar serotonergic and adrenergic receptor binding (Obeng et al., 2019). Together, these alkaloids provide a comprehensive pharmacological effect obtained when consuming kratom products (León et al., 2021; Grundmann et al., 2023).

Over the past two decades, there has been a notable expansion of consumer interest in kratom beyond Asian borders into North America and Europe. Recent surveys indicated that 0.7-6.1% of the United States population was using kratom products, and the use was more prevalent among those with prior substance use disorder (Palamar, 2021; McCurdy et al., 2024). In Western countries, kratom is most commonly available as dried, ground leaf powder, oftentimes encapsulated or occasionally tableted for ease of consumption (Grundmann, 2017). Liquid extracts, resins, and soft gels usually contain high or sometimes excessive alkaloid concentrations. Such products are easily obtainable in North American markets (Kamble et al., 2021; Sharma et al., 2025). Contrarily, in Southeast Asia, kratom is primarily consumed as freshly harvested leaves by chewing directly or brewing in boiling water for 2-3 hours as a tea decoction. However, processed kratom products, like harvested leaves, crumbs, and powders, are also available in these regions (Suwanlert, 1975; Syarma et al., 2019).

The harvest and postharvest processing of kratom leaves are largely undocumented in the literature. Information gathered through the internet suggested that kratom is typically dried either directly after leaf harvesting, sorting, and washing or after leaf fermentation in Southeast Asia. Environmentally, kratom leaves can be dried either indoors, outdoors, or with a combination of both. Indoor drying avoids direct sunlight exposure and oftentimes takes place on shelved racks in darkened rooms equipped with fans to circulate air and accelerate evaporation. Outdoor drying is achieved by spreading the leaves on a sheet under direct sunlight where they are exposed to higher heat and stronger light. It is believed that the quality of the kratom product vary with different postharvest methods, thus resulting in different alkaloid profiles.

Contradictory results have been reported on the effect of drying temperatures on alkaloid concentration in other plant species. For example, Ademiluyi et al. (2018) suggested that the alkaloid in Moringa (*Moringa oleifera*) leaf decreased by 63% and 21% when drying under the sun and oven (40°C), respectively, compared to air drying at room temperature. Similarly, canadine from the medicinal forest herb goldenseal (*Hydrastis canadensis*) significantly decreased when drying temperatures increased beyond 30°C and its level decreased by more than 50% when dried under 54.5°C compared to freeze-drying (Zuiderveen et al., 2021). In contrast, compared to air drying (22-32° C), caffeine concentration in *Ilex guayusa* leaves was similar when dried under the sun (~34°C) but increased by 23% in the oven (46°C) (Manzano Santana et al., 2018).

Withering is a natural wilting process of leaves after they are removed from the plant. This is a critical step in tea production due to the limited capacity of leaf processing at peak harvest seasons. Typically, freshly harvested tea leaves are spread out on withering racks in sheds until they can be further processed. During this time, the leaves become soft due to moisture loss, resulting in internal chemical changes that shape the final product's flavor, aroma, and appearance. Since kratom is commonly consumed similar to a tea product in Southeast Asia, this practice may have been implemented unconsciously in the kratom postharvest process in Southeast Asia, although this is undocumented. Withering conditions and methods can significantly affect chemical composition of plant products. For example, Jia et al. (2023) found that sunlight and trough withering – compared to charcoal fire withering, which differed in temperature, duration, and turning intervals – resulted in significantly higher alkaloid concentrations in Wuyi rock tea. In a separate study, withering tea leaves for 18 hours at 15°C, 20°C, and 25°C resulted in an increase in caffeine concentration by 16%, 9%, and 10%, respectively, when compared to the initial caffeine concentration (Ye et al., 2018).

Most domestically available commercial kratom products contain moderate concentrations of mitragynine and low concentrations of 7-hydroxymitragynine (Sharma et al., 2019, 2025). In contrast, anecdotally, our kratom leaf samples possessed comparable concentrations of mitragynine but no 7-hydroxymitragynine, suggesting that the production of 7-hydroxymitragynine may be through postharvest processes. Thus, a better understanding of the postharvest environmental influence on kratom product quality is needed. The objective of this study was to assess the impact of postharvest drying temperatures and withering durations on the accumulation of fourteen alkaloids in leaves of two kratom cultivars harvested in two different seasons.

2 Materials and methods

2.1 Study I

2.1.1 Leaf materials

Bulk fresh leaves were harvested from 'Hawaii' kratom mother stocks (Figure 1A) on July 12 (hereafter referred to as summer) and November 15 (hereafter referred to as winter) 2022, respectively. Healthy, fresh, mature leaves (Figure 1B) were collected from similar positions on branches (2nd or 3rd pair of leaves from the meristem) throughout the trees to ensure uniform maturity and photosynthetic activity, thereby reducing potential variability in alkaloid concentration. Harvested leaves were placed into a collection bucket and mixed thoroughly before being randomly assigned to each postharvest treatment. Details on the maintenance and management of the 'Hawaii' mother stock are provided in the Supplementary Information.

2.1.2 Postharvest treatment and environmental control

Harvested composite bulk leaves were evenly divided into 48 samples for the summer harvest and 60 samples for the winter harvest. All samples were randomly assigned to one of four withering durations (0, 12, 24, and 72 hours). Leaves were laid flat in a single layer on air filters and placed in environmentally controlled withering rooms for the assigned durations. Each withering room was identical and

equipped with an individual air conditioning unit. Air temperature and relative humidity were monitored and recorded using a data logger (Elitech, San Jose, CA, United States) at 5-minute intervals. The withering rooms maintained an average air temperature of 25°C and a relative humidity of 55%, with no supplemental lighting.

Following withering, leaf samples from each withering duration group were further divided into four portions and subjected to drying treatments at either 25, 40, 60, or 80°C. A 25°C drying treatment was conducted using a forced air fan system within environmentally controlled rooms for six days, with an average air temperature of 25°C. The 40, 60, and 80°C drying treatments were achieved using thermo-controlled drying ovens (40GC, Qincy Lab Inc., Burr Ridge, IL, United States; SWN-46-6E, Wisconsin Oven Corp., East Troy, WI, United States; 4CEM, Precision Scientific Group, Chicago, IL, United States) for five days. For winter-harvested leaves, a freeze-drying treatment at -40°C was added as the fifth drying temperature. The freeze-drying treatment was performed using a freeze dryer (Harvest Right, Salt Lake City, Utah, United States). Detailed descriptions of all treatment combinations are listed in Table 1. Each treatment in both the summer and winter experiments included three replicate composite leaf samples.

TABLE 1 Summary of postharvest treatments used in Study I for cultivar Hawaii.

Study	Treatment	Withering duration	Drying temperature
I	1	0 hour	-40°C (only for winter harvest)
	2	0 hour	25°C
	3	0 hour	40°C
	4	0 hour	60°C
	5	0 hour	80°C
	6	12 hours	-40°C (only for winter harvest)
	7	12 hours	25°C
	8	12 hours	40°C
	9	12 hours	60°C
	10	12 hours	80°C
	11	24 hours	-40°C (only for winter harvest)
	12	24 hours	25°C
	13	24 hours	40°C
	14	24 hours	60°C
	15	24 hours	80°C
	16	72 hours	-40°C (only for winter harvest)
	17	72 hours	25°C
	18	72 hours	40°C
	19	72 hours	60°C
	20	72 hours	80°C

2.2 Study II

2.2.1 Leaf materials

A different kratom cultivar, Mitragynine-Rich (MR)-Malaysian, was used for this study. As in Study I, healthy, fresh, mature leaves were harvested from 'MR-Malaysian' kratom mother stocks, using the second or third pair of leaves from the apical meristem to ensure maturity and consistency. The harvested leaves were thoroughly mixed and randomly assigned to post-harvest treatments. Leaves were harvested twice in 2024: January 25 (hereafter referred to as winter) and June 12 (hereafter referred to as summer). Details on the maintenance and management of the 'MR-Malaysian' mother stock are provided in the Supplementary Information.

2.2.2 Postharvest treatment and environmental control

Similarly to Study I, harvested composite leaves were mixed thoroughly and evenly divided into 40 samples. All samples were subjected to one of two withering durations (0 or 12 hours). After withering, each group was further divided into two portions and dried at either 25°C or 60°C, with 10 leaf samples in each portion. A detailed description of the treatment combinations is provided in Table 2. The treatment setup was similar to Study I, except that a 25°C drying temperature was achieved using a food dehydrator (Weston PRO-2400, Weston, FL, United States) with a maximum temperature setpoint of 25°C.

2.3 Leaf alkaloid extraction and quantification

For both Study I and II, dried kratom leaves from each postharvest treatment were collected and ground into fine powders using a commercial grinder (BCG211, KitchenAid, St. Joseph, MI, United States). Subsequently, leaf alkaloids were extracted using the protocol described by Zhang et al. (2020) with slight modifications. Briefly, 0.5 g of fine-grinded kratom leaf powder samples were extracted with 3 mL 190 proof ethanol under water-bath sonication until exhaustion. The process was replicated three times to maximize alkaloid extraction. Quantification of alkaloids was performed using a Waters Xevo TQ-S Micro triple quadrupole mass spectrometer detector connected to Acquity Class I ultra performance liquid chromatography (UPLC) (Milford, MA, United States) and the chromatogram of a representative sample is provided in Figure 1C.

TABLE 2 Summary of postharvest treatments used in Study II for cultivar MR-Malaysian.

Study	Treatment	Withering duration	Drying temperature
II	1	0 hour	25°C
	2	0 hour	60°C
	3	12 hours	25°C
	4	12 hours	60°C

UPLC method, compound, and source parameters were the same as reported previously (Zhang et al., 2020). Alkaloid content (% w/w) was reported based on leaf extracts, as postharvest treatments can potentially affect extraction efficiency, making extract-based values more accurate.

2.4 Experiment design and statistical analysis

Both Study I and Study II were conducted using a completely randomized design, with three replications in Study I and 10 replications in Study II. Statistical analysis was performed using a restricted maximum likelihood mixed model analysis in JMP Pro 16 (SAS Institute, Inc., Cary, NC, United States). Unless otherwise noted, *post-hoc* mean separation tests were performed on values above lower limit of quantification (LLOQ) using Tukey's honest significant difference test. Statistical tests were considered significant if $P \leq 0.05$.

3 Results

3.1 Powder color

Kratom powder presented a reddish-brown color under high drying temperatures compared to a green color under -40°C and 25°C (Figures 1D, E). The color change was more substantial when the withering duration was short, such as 0 or 12 hours. Increasing the withering duration up to 72 hours preserved the greenness of the powder, especially under high drying temperatures.

3.2 Effects of withering duration on bioactive compounds

To facilitate interpretation of the results, the treatment without withering followed by drying at 25°C was considered as the baseline (control). Withering significantly increased mitragynine concentration in both studies. Overall, mitragynine concentration was elevated by 14-65% in 'Hawaii' leaf extract and 3-8% in 'MR-Malaysian' extract (Figure 2). Specifically, a 12-hour withering period followed by drying at -40, 25, 40, and 60°C increased mitragynine concentration by 117%, 38-123%, 16-61%, and 43-103% in 'Hawaii', respectively (Table 3). Extending the duration of withering from 12 to 24 and 72 hours did not further change the mitragynine concentration compared to the 12-hour treatment. Still, the concentrations remained significantly higher, or at least comparable, to those observed in non-withered leaf samples. Interestingly, withering duration had minimal impact on mitragynine concentration in 'MR-Malaysian', except for those harvested in the summer and dried at 60°C (Table 4).

For 7-hydroxymitragynine, a significant difference in the number of samples above LLOQ was observed in 'Hawaii', with 63% in summer-harvested leaves compared to 3% in winter-harvested

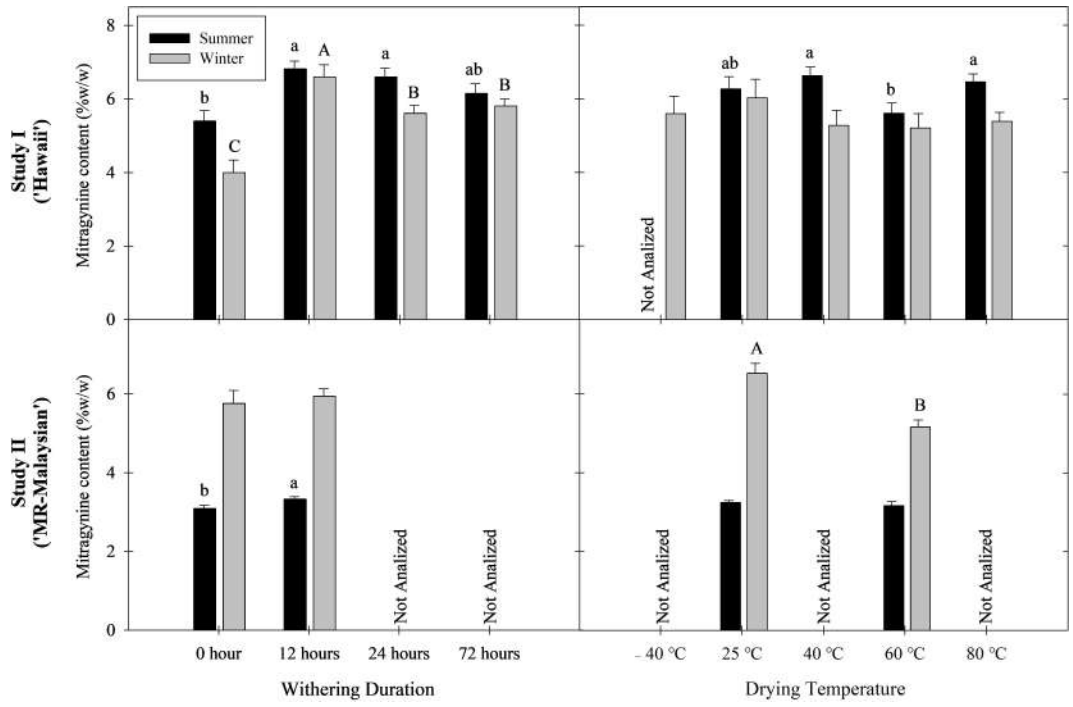


FIGURE 2
Mitragynine content in leaf alkaloid extracts across treatments in Study I (cultivar Hawaii) and Study II (cultivar MR-Malaysian). Lowercase letters denote summer treatment differences; uppercase letters denote winter differences. Treatments with different letters are significantly different according to Tukey's HSD test (Study I) and Student's t-test (Study II) at $P < 0.05$.

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TABLE 3 Mitragynine content (% w/w ± SE) in leaf alkaloid extracts from Study I (cultivar Hawaii) under postharvest treatments by season.

Withering duration	Mitragynine (%w/w)				
	— 40°C	25°C	40°C	60°C	80°C
Summer					
0 hour	NA	4.87 ± 0.36 **bAB**	6.19 ± 0.24 **A**	4.34 ± 0.12 **bB**	6.22 ± 0.38 **abcA**
12 hours		6.72 ± 0.39 **a**	7.18 ± 0.58	6.22 ± 0.28 **a**	7.15 ± 0.27 **ab**
24 hours		7.38 ± 0.60 **a**	6.89 ± 0.31	5.87 ± 0.35 **ab**	6.25 ± 0.22 **abc**
72 hours		6.10 ± 0.32 **ab**	6.26 ± 0.67	6.00 ± 0.66 **ab**	6.24 ± 0.66 **abc**
Winter					
0 hour	3.43 ± 0.28 **cB**	3.60 ± 0.43 **cB**	3.58 ± 0.33 **bB**	3.17 ± 0.11 **bB**	6.19 ± 0.68 **A**
12 hours	7.47 ± 0.40 **aAB**	8.04 ± 0.34 **aA**	5.77 ± 0.74 **aBC**	6.42 ± 0.33 **aABC**	5.31 ± 0.42 **C**
24 hours	5.29 ± 0.24 **b**	6.14 ± 0.24 **b**	6.26 ± 0.66 **a**	5.43 ± 0.41 **a**	4.93 ± 0.37
72 hours	6.21 ± 0.46 **ab**	6.35 ± 0.16 **b**	5.51 ± 0.73 **a**	5.83 ± 0.06 **a**	5.15 ± 0.17

Mean comparisons were conducted within each season. Lowercase letters indicate mean separation for withering duration; uppercase letters indicate mean separation for drying temperature. Means sharing different letters are statistically different by Tukey's HSD test at $P < 0.05$. NA = not analyzed.

TABLE 4 Key alkaloid content (% w/w ± SE) in leaf alkaloid extracts from Study II (cultivar MR-Malaysian) under postharvest treatments by season.

		Winter		Summer	
		25°C	60°C	25°C	60°C
Mitragynine	0 hour	6.67 ± 0.45 **a**	4.87 ± 0.26 **c**	3.25 ± 0.07 **a**	2.89 ± 0.14 **b**
	12 hours	6.41 ± 0.23 **ab**	5.47 ± 0.23 **bc**	3.26 ± 0.07 **ab**	3.39 ± 0.09 **a**
7-Hydroxymitragynine	0 hour	0.032 ± 0.001 **ab**	0.023 ± 0.001 **c**	Below LLOQ[x]	Below LLOQ
	12 hours	0.035 ± 0.001 **a**	0.029 ± 0.002 **bc**	Below LLOQ[y]	Below LLOQ
Speciogynine	0 hour	0.75 ± 0.03 **ab**	0.65 ± 0.03 **b**	0.45 ± 0.01 **b**	0.50 ± 0.01 **ab**
	12 hours	0.75 ± 0.03 **ab**	0.79 ± 0.04 **a**	0.44 ± 0.01 **b**	0.51 ± 0.02 **a**
Paynantheine	0 hour	1.04 ± 0.03 **a**	0.91 ± 0.02 **b**	0.65 ± 0.02	0.64 ± 0.02
	12 hours	1.01 ± 0.03 **ab**	1.02 ± 0.04 **ab**	0.64 ± 0.02	0.72 ± 0.02
Corynantheidine	0 hour	0.042 ± 0.005	0.038 ± 0.005	0.042 ± 0.003 **ab**	0.032 ± 0.005 **b**
	12 hours	0.045 ± 0.004	0.034 ± 0.004	0.052 ± 0.006 **a**	0.034 ± 0.002 **b**
Speciociliatine	0 hour	0.36 ± 0.03 **b**	0.43 ± 0.03 **ab**	0.55 ± 0.04	0.54 ± 0.03
	12 hours	0.46 ± 0.04 **ab**	0.52 ± 0.04 **a**	0.55 ± 0.05	0.65 ± 0.06
Mitraciliatine	0 hour	0.028 ± 0.003 **b**	0.033 ± 0.003 **ab**	0.054 ± 0.004	0.049 ± 0.004
	12 hours	0.045 ± 0.005 **a**	0.041 ± 0.004 **ab**	0.049 ± 0.006	0.059 ± 0.006
Isopaynatheine	0 hour	0.017 ± 0.002 **b**	0.019 ± 0.002 **ab**	0.028 ± 0.003	0.026 ± 0.002
	12 hours	0.027 ± 0.003 **a**	0.023 ± 0.002 **ab**	0.030 ± 0.006	0.031 ± 0.003
Corynoxine A	0 hour	0.053 ± 0.006 **a**	Below LLOQ	0.035 ± 0.003	0.045 ± 0.003
	12 hours	0.041 ± 0.004 **b**	Below LLOQ	0.040 ± 0.002	0.041 ± 0.003
Corynoxine B	0 hour	0.034 ± 0.005 **bc**	0.045 ± 0.005 **ab**	0.038 ± 0.003	0.049 ± 0.004
	12 hours	0.025 ± 0.003 **c**	0.056 ± 0.003 **a**	0.042 ± 0.002	0.040 ± 0.002
9-Hydroxycorynantheidine	0 hour	yes detection	no detection	NA	NA
	12 hours	yes detection	no detection	NA	NA

Mean comparisons were conducted separately for each alkaloid and season. Means sharing different letters are statistically different by Tukey's HSD test at $P < 0.05$. NA, not analyzed. LLOQ, Lower Limit of Quantification (limit of quantification = 0.005% in leaf alkaloid extract). [x]1 out of 10 samples were above LLOQ. [y]2 out of 10 samples were above LLOQ.

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leaves (Table 5). Almost all 'Hawaii' leaf samples with the harvested in the summer and dried at 25°C had above LLOQ levels of 7-hydroxymitragynine. Under 40 and 60°C, the number of samples with the above LLOQ level increased with the increase of withering duration. 7-hydroxymitragynine was above LLOQ in all samples after withering for 72 hours, regardless of the subsequent drying temperature. In contrast, for leaves harvested in the winter, the 7-hydroxymitragynine concentration in most of the 'Hawaii' leaf samples was below LLOQ. Additionally, no significant differences were noted among treatments in the average 7-hydroxymitragynine concentration. Interestingly, in 'MR-Malaysian', 7-hydroxymitragynine concentration was above LLOQ in almost all winter-harvested leaf samples, while they were all below LLOQ in summer-harvested samples, regardless of postharvest treatments (Table 4). The 7-hydroxymitragynine concentrations in 'MR-Malaysian' ranged from 0.023 to 0.035% in leaf extract, which were not significantly affected by the withering duration.

The withering duration also impacted the content of speciogynine, paynantheine, and speciociliatine in 'Hawaii' leaves. However, this effect was primarily observed at lower drying temperatures, with minimal variation occurring at drying temperatures above 40°C (Table 6). For speciogynine, a 12-hour withering period followed by drying at -40 and 25°C significantly increased its concentration by 48% and 37%, respectively. A 24-hour withering duration enhanced speciogynine concentration at 25°C, but not at other temperatures. Similarly, paynantheine concentration increased by 35-67% with a 12-hour withering period followed by drying at temperatures below 40°C, but the effect of withering duration was not observed at higher drying temperatures. For speciociliatine, under low drying temperatures (≤ 25°C), its concentration initially decreased as withering duration increased from 0 to 24 hours, then subsequently increased when withering duration was extended to 72 hours. However, this trend was not apparent at higher drying temperatures (> 40°C). Notably, a 12-hour withering period significantly reduced speciociliatine concentration by 0.36-fold at a drying temperature of 25°C, but this effect diminished once the withering duration exceeded 24 hours.

In contrast, a 12-hour withering period only significantly increased speciogynine concentration in 'MR-Malaysian' by 22% under a high drying temperature (60°C) in the winter-harvested samples (Table 4). Moreover, the concentrations of mitraciliatine and isopaynantheine increased by 61% and 59%, respectively, under 25°C in winter-harvested leaves, whereas the concentration of corynoxine A was reduced by 23%.

Additionally, withering duration did not have a noticeable effect on corynantheidine concentration in both cultivars, as well as on paynantheine, speciociliatine and corynoxine B concentration in 'MR-Malaysian' (Tables 4, 6). Furthermore, corynoxine A and B concentrations were below LLOQ in 'Hawaii'. No isospeciofoline, mitraphylline, or ajmalicine were above LLOQ in either cultivar (data not shown).

TABLE 5 Percentage of samples (%) with 7-hydroxymitragynine above LLOQ and 7-hydroxymitragynine content (% w/w ± $_{SE}$) in leaf alkaloid extracts from Study I (cultivar Hawaii) under postharvest treatments by season.

		Withering duration	— 40°C	25°C	40°C	60°C	80°C
% Samples Above LLOQ	Summer	0 hour	NA	100	0	0	67
		12 hours		67	0	33	0
		24 hours		100	67	33	0
		72 hours		100	100	100	100
	Winter	0 hour	0	33	0	0	0
		12 hours	0	0	0	0	0
		24 hours	0	0	0	0	0
		72 hours	0	0	0	0	33
7-hydroxymitragynine content (%w/w)	Summer	0 hour	NA	0.013 ± 0.001	Below LLOQ	Below LLOQ	0.039 ± 0.002
		12 hours		0.020 ± 0.001	Below LLOQ	Below LLOQ	Below LLOQ
		24 hours		0.020 ± 0.001	0.016 ± 0.006	Below LLOQ	Below LLOQ
		72 hours		0.020 ± 0.001	0.018 ± 0.001	0.021 ± 0.001	0.017 ± 0.003
	Winter	0 hour	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ
		12 hours	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ
		24 hours	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ
		72 hours	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ	Below LLOQ

Data for alkaloid content were only reported with more than 2 valid data points. NA, not analyzed. LLOQ, Lower Limit of Quantification (limit of quantification = 0.005% in leaf alkaloid extract).

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TABLE 6 Average speciogynine, paynantheine, corynantheidine, and speciociliatine content (% w/w ± SE) in leaf alkaloid extracts from Study I (cultivar Hawaii) under postharvest treatments.

Alkaloid name	Withering duration	— 40°C	25°C	40°C	60°C	80°C
Speciogynine	0 hour	0.65 ± 0.07 **bAB**	0.59 ± 0.06 **bB**	0.74 ± 0.06 **AB**	0.60 ± 0.06 **B**	0.79 ± 0.04 **A**
	12 hours	0.96 ± 0.06 **aA**	0.81 ± 0.03 **aAB**	0.72 ± 0.07 **B**	0.73 ± 0.03 **B**	0.73 ± 0.06 **B**
	24 hours	0.64 ± 0.04 **b**	0.80 ± 0.05 **a**	0.77 ± 0.04	0.61 ± 0.03	0.65 ± 0.03
	72 hours	0.73 ± 9.05 **ab**	0.72 ± 0.02 **ab**	0.63 ± 0.06	0.62 ± 0.02	0.67 ± 0.06
Paynantheine	0 hour	0.73 ± 0.07 **b**	0.79 ± 0.06 **b**	0.94 ± 0.09	0.8 ± 0.10	0.94 ± 0.05
	12 hours	1.22 ± 0.06 **a**	1.07 ± 0.02 **a**	0.95 ± 0.11	0.94 ± 0.04	0.95 ± 0.10
	24 hours	0.74 ± 0.04 **b**	1.00 ± 0.09 **ab**	0.98 ± 0.08	0.82 ± 0.07	0.82 ± 0.06
	72 hours	0.88 ± 0.06 **ab**	0.97 ± 0.04 **ab**	0.84 ± 0.07	0.84 ± 0.05	0.91 ± 0.10
Corynantheidine	0 hour	0.10 ± 0.01	0.08 ± 0.01	0.06 ± 0.01	0.05 ± 0.001	0.09 ± 0.01
	12 hours	0.08 ± 0.01	0.08 ± 0.01	0.10 ± 0.02	0.09 ± 0.02	0.07 ± 0.01
	24 hours	0.09 ± 0.03	0.10 ± 0.01	0.06 ± 0.01	0.08 ± 0.01	0.08 ± 0.01
	72 hours	0.10 ± 0.02	0.11 ± 0.01	0.09 ± 0.02	0.06 ± 0.002	0.09 ± 0.02
Speciociliatine	0 hour	0.99 ± 0.14 **abA**	0.78 ± 0.19 **abAB**	0.55 ± 0.16 **AB**	0.42 ± 0.14 **AB**	0.32 ± 0.03 **B**
	12 hours	0.45 ± 0.09 **ab**	0.28 ± 0.04 **c**	0.37 ± 0.15	0.34 ± 0.07	0.35 ± 0.10
	24 hours	0.34 ± 0.19 **b**	0.38 ± 0.12 **bc**	0.20 ± 0.06	0.50 ± 0.18	0.29 ± 0.11
	72 hours	1.02 ± 0.17 **aA**	0.91 ± 0.12 **aA**	0.47 ± 0.13 **AB**	0.32 ± 0.09 **B**	0.67 ± 0.24 **AB**

Mean comparisons were conducted within each alkaloid using data pooled from two seasons. Lowercase letters indicate mean separation for withering duration; uppercase letters indicate mean separation for drying temperature. Means sharing different letters are statistically different by Tukey's HSD test at $P < 0.05$.

3.3 Effects of drying temperature on bioactive compounds

Overall, low drying temperatures preserved mitragynine in kratom leaf extracts better than high drying temperatures. For example, the overall mitragynine concentration in leaves dried at 25°C was 3-26% higher than that dried at 60°C (Figure 2). The drying temperature had a more significant effect when leaves experienced no withering duration, and this influence diminished once the withering duration exceeded 24 hours (Tables 3, 4).

The greatest number of samples with 7-hydroxymitragynine above LLOQ in kratom 'Hawaii' was observed at a drying temperature of 25° C compared to other drying temperatures (Table 5). In contrast, drying temperatures had no significant effect on the number of samples with 7-hydroxymitragynine above LLOQ in 'MR-Malaysian' (Table 4). Additionally, in 'Hawaii', there were a few differences between the drying temperature treatments (Table 5). Conversely, in 'MR-Malaysian', 7-hydroxymitragynine concentration was 21-39% higher when dried at 25°C compared to drying at 60°C (Table 4).

For speciogynine, a drying temperature of 80°C resulted in the highest concentration when leaves were dried without withering in kratom 'Hawaii' (Table 6). In contrast, under a 12-hour withering period, drying at lower temperatures resulted in higher speciogynine concentration. Similarly, for paynantheine, although not significant, 'Hawaii' had the highest concentration under high drying temperatures when leaves experienced no withering. However, with a 12-hour withering period, leaves achieved the highest paynantheine concentration under low drying temperatures. Conversely, speciociliatine had the highest concentration under freeze-drying without withering, with its concentration being threefold higher compared to leaves dried at 80°C. Under no withering conditions, speciociliatine concentration generally decreased in 'Hawaii' as the drying temperature increased. However, in 'MR-Malaysian', its concentration typically increased, albeit not significantly, with the increase in drying temperature (Tables 4, 6).

Drying temperatures had little effect on corynantheidine in kratom 'Hawaii', whereas higher drying temperatures generally decreased its concentration in 'MR-Malaysian' (Tables 4, 6). Interestingly, in kratom 'MR-Malaysian', a high drying temperature at 60 °C significantly increased the accumulation of some oxindole alkaloids. For example, the concentration of corynoxine B increased by 32-124% in winter-harvested leaves, regardless of withering durations (Table 4). Additionally, when leaves were dried at 25°C, corynoxine A and 9-hydroxycorynantheidine were above the LLOQ in 100% and 90% of the samples, respectively. However, both compounds fell below the LLOQ in samples dried at 60°C. Furthermore, drying temperatures had little effect on mitraciliatine, isopaynantheine, and corynoxine A in kratom 'MR-Malaysian' (Table 4).

4 Discussion

Kratom products sold in Western markets are often categorized into different strains, commonly labeled as red, green, and white.

These classifications are most commonly based on the color of the powdered product rather than the actual leaf vein color and variety. Marketing narratives and anecdotal reports from kratom users frequently suggest that different strains produce distinct physiological effects. For instance, red kratom strains are often associated with anxiolytic and calming effects, while green strains are marketed as being more energizing, stimulating, and mood-enhancing (Huisman et al., 2023). Our study demonstrates that postharvest processing conditions significantly influence the final color of kratom powder, even when derived from the same batch of leaves. More importantly, our findings suggest that the powder color does not necessarily correlate with alkaloid composition, but rather serves as an indicator of the postharvest environment. For example, in kratom 'Hawaii', when leaves were dried immediately without a withering period, the key alkaloid contents, including mitragynine, speciogynine, and paynantheine, were similar between leaves dried at 25°C and 60°C. However, the visual appearance of the powder differed significantly, with samples dried at 60°C exhibiting a reddish hue, whereas those dried at 25°C retained a green color. These findings align with the survey results reported by Huisman et al. (2023), who found no significant differences in alkaloid profiles among six commercial kratom products labeled with different strains. Additionally, our study revealed that even when similar in color, kratom powder can present different alkaloid profiles. For example, when kratom leaves were dried at 25°C, both the 0-hour and 12-hour withering treatments produced green-colored powders. However, the key alkaloid concentrations, particularly mitragynine, speciogynine, and paynantheine, were significantly higher in leaves that underwent a 12-hour withering process compared to those with no withering. It is worth noting that although Huisman et al. (2023) did not observe any significant differences in alkaloid levels between strains, users often report distinct subjective experiences associated with different strains. These perceptions may be influenced by marketing strategies and user expectations (Huisman et al., 2023), or possibly by non-alkaloid constituents not assessed in this study. Collectively, these findings suggest that marketing kratom strains based on powder color can be misleading and may not accurately reflect product quality or chemical composition.

There were notable and at times opposing differences in alkaloid content in two kratom cultivars, Hawaii and MR-Malaysian, in response to environmental changes, which highlights the potential influence of genotype and seasonal variation on alkaloid biosynthesis. Withering duration significantly influenced mitragynine concentration in 'Hawaii', whereas it had a minimal effect on 'MR-Malaysian'. In contrast, drying temperature had a greater effect on 'MR-Malaysian' than on 'Hawaii'. As outlined in the Methods section, two cultivars were studied in both summer and winter, allowing for a comparison across seasonal conditions. Mitragynine concentrations in 'Hawaii' were generally higher during summer, while concentrations in cultivar 'MR-Malaysian' tended to be higher during winter. A similar seasonal contrast was observed with 7-hydroxymitragynine, a key metabolite of mitragynine. In 'Hawaii', 7-hydroxymitragynine was detectable only during the summer, whereas in 'MR-Malaysian', it was detectable only in the winter. Additionally, the alkaloids corynoxine A and B were consistently detected in 'MR-Malaysian' but were undetectable in 'Hawaii' regardless of seasons. This suggests that potential genetic differences occur in the biosynthesis of bioactive alkaloid compounds. Furthermore, greater variability in alkaloid concentration was observed in 'MR-Malaysian' between seasons compared to 'Hawaii', indicating that 'MR-Malaysian' may be more sensitive to subtle environmental or seasonal fluctuations. However, detailed environmental data were not collected for the mother stock's growing conditions across seasons in this study, limiting the ability to perform a more in-depth analysis of environmental influences during this period. Interestingly, Chear et al. (2021) also reported substantial variability in alkaloid profiles among naturally growing wild kratom trees in Malaysia, suggesting the presence of genetic differences within wild populations. This raises the possibility that kratom beverages consumed in Southeast Asia may often be derived from a mixture of genetically diverse leaves. Collectively, these results provide evidence that cultivar genetics may play a significant role in influencing the abundance and the seasonal dynamics of key alkaloids in kratom. Further research is warranted to investigate how specific environmental variables influence alkaloid expression in genetically distinct kratom cultivars.

The withering process, primarily driven by gradual water loss, imposes external stress on plant tissues, which can activate a cascade of physiological and biochemical signaling pathways. This stress response often leads to the enhanced biosynthesis of various secondary metabolites, including alkaloids, as part of the plant's adaptive defense mechanisms (Dai et al., 2025). Significant changes in primary metabolism also occur. Macromolecules, such as proteins, may undergo degradation, providing precursors and energy needed for the production of secondary metabolites. For instance, approximately 15% of chlorophyll content has been reported to degrade during withering (Sanyal, 2011). Additionally, moisture removal caused by withering can concentrate the cell sap, increasing enzyme activity and forming high molecular units from low molecular units (Deb and Jolvis Pou, 2016). These biochemical transformations create a favorable environment for the activation of specialized metabolism.

In the present study, we observed that the availability of key alkaloids, including mitragynine, speciogynine, paynantheine, and corynantheidine, generally increased with the increase of withering duration at ambient temperatures (25°C), particularly in kratom 'Hawaii'. This trend is consistent with findings in tea leaves where caffeine levels significantly increase by the end of the withering phase, likely due to the conversion of its precursor, theobromine, and enhanced activity of tea caffeine synthase under low-temperature conditions (Xia et al., 2017; Chen et al., 2020; Yu et al., 2020; Qi et al., 2024). Drawing on a simplified model of the kratom indole alkaloid biosynthesis pathway compiled from the current literature (Lopes et al., 2019; Kim et al., 2023; Schotte et al., 2023) (Figure 3), we propose that withering enhanced the synthesis of substances upstream of strictosidine aglycone, such as free amino acids, to enhance the overall indole alkaloid biosynthesis pathways in kratom. This hypothesis is based on our results that a 12-hour

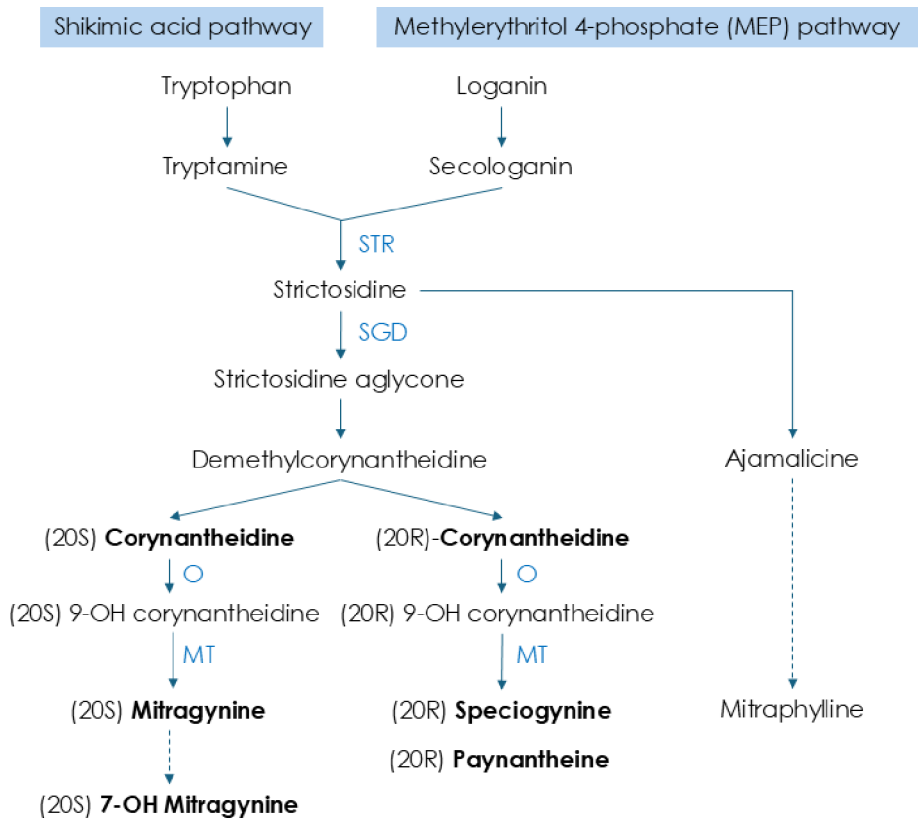


FIGURE 3
Biosynthesis pathways for key alkaloids in kratom. Summarized and simplified from Lopes et al., 2019; Kim et al., 2023; Schotte et al., 2023.

withering period yielded the highest quality kratom product. Beyond this duration, levels of some key alkaloids may begin to decline, potentially due to the depletion of precursors or the activation of feedback inhibition mechanisms. Similarly, in tea leaves, there was an increase in free amino acids during withering, attributing to proteolysis and the upregulation of protein degradation enzymes and amino acid synthases (Qi et al., 2024). Furthermore, our finding aligns with observations in tea, where alkaloid-related metabolic activity is most pronounced during the initial 6 hours of withering and gradually diminishes beyond 18 hours (Dai et al., 2025). Accordingly, tea's optimal chemical withering duration is approximately 14 hours and should not exceed 18 hours (Deb and Jolvis Pou, 2016; Owuor and Orchard, 1992; Omiadze et al., 2014). These collective findings indicate the critical role of withering duration in modulating alkaloid biosynthesis and provide insight into optimizing postharvest handling strategies for kratom.

It is important to note that in our study, the withering experiment was conducted using a single layer of kratom leaves, allowing for uniform exposure to ambient conditions. This methodology may differ from commercial practices, where leaves are typically stacked in layers during the withering process and require periodic manual turnovers to promote uniformity in drying. Several withering systems, including trough withering, solar withering, heated withering, indoor withering, and controllable light withering, have been developed over the years in the tea industry to optimize moisture removal and enhance the development of secondary metabolites and aroma (Qi et al., 2024). Similar systems are likely being adopted in the kratom industry in Southeast Asia to enhance product quality and consistency. Further research evaluating how different withering systems and stacking depths influence the biochemical dynamics of alkaloid biosynthesis in kratom is warranted.

In kratom 'Hawaii', several alkaloids, including mitragynine, 7-hydroxymitragynine, speciogynine, paynantheine, and corynantheidine, were found in the highest concentration in kratom powders dried at high drying temperatures without prior withering. However, when withering exceeded 12 hours, the greatest alkaloid concentrations were generally observed under lower drying temperatures. A similar trend was noted in kratom 'MR-Malaysian', where the highest alkaloid concentrations were typically achieved under lower drying temperatures. These patterns suggest that the biosynthesis of kratom alkaloids may be mediated by enzymes whose activities may be sensitive to both temperature and time. In a new equilibrium model of the enzyme thermal behavior proposed by Daniel and Danson (2010), enzymes exhibit higher optimal temperatures for activity at the onset of a biological process (time = 0), but this optimal temperature decreases as the process progresses. For instance, alkaline phosphatase activity peaked at approximately 315 K (41.9°C) at time zero but shifted to an

optimum of 305–310 K (31.9–36.8°C) after 3500 seconds. A similar trend was observed in *Caldocellulosiruptor saccharolyticus* β-glucosidase, where peak activity was initially recorded at 350 to 360 K (76.9 to 86.9°C) but shifted to 330 K (56.9°C) after 250 seconds. This supports our hypothesis that the enzymes involved in mitragynine and other kratom indole alkaloid biosynthesis may be influenced by both withering duration and drying temperature. As time progresses during postharvest processing, a lower temperature environment may generally be favorable for enzyme activity and alkaloid accumulation.

Mitragynine and its metabolite, 7-hydroxymitragynine, are the key psychoactive alkaloids in kratom that have attracted significant interest due to their pharmacological activities. Mitragynine is the predominant alkaloid in kratom, known to interact with opioid, adrenergic, and serotonergic receptors, and produces a range of pharmacological effects, including both sedative and stimulant properties, as well as anti-inflammatory and antinociceptive actions (Annuar et al., 2024). In our study, mitragynine concentrations in kratom powders processed from harvested leaves generally ranged from 0.5-1.5% in cultivar Hawaii and 0.2-1.4% in cultivar MR-Malaysian. In corresponding alkaloid extracts, concentrations range from 4–8.5% in 'Hawaii' and 2–8% in 'MR-Malaysian', depending on postharvest treatments and harvest seasons. These values are comparable to the mitragynine content found in native kratom trees from Thailand (0.4-2.2% in leaves; Leksungnoen et al., 2022, 2025) and Malaysia (0.9-1.8% in leaves (Chear et al., 2021), with differences likely attributed to environmental conditions, plant age, and genetic differences among cultivars.

In contrast, 7-hydroxymitragynine is a potent μ-opioid receptor agonist associated with strong analgesic activity but also with potential risks such as respiratory depression at higher doses, raising concerns regarding its abuse liability (Hemby et al., 2019; McCurdy et al., 2024). Although research remains limited, varying concentrations of 7-hydroxymitragynine have been reported in both kratom leaf samples and commercial kratom products. In commercial kratom products available in the US market (i.e., powders or crumbs from dried leaves), 7-hydroxymitragynine typically ranges from 0.01% to 0.21% (Sharma et al., 2025). Among native trees, 7-hydroxymitragynine concentrations have been reported to range from 0.003% to 0.012% in Thai kratom leaves and from 0.03% to 0.15% in dried leaves from Malaysian trees (Leksungnoen et al., 2022; Kikura-Hanajiri et al., 2009; Chear et al., 2021). In our previous studies, 7-hydroxymitragynine was not detected in kratom leaf samples, likely due to the use of a different cultivar that also produced low levels of mitragynine (Zhang et al., 2020, 2022). In the current study, the concentration of 7-hydroxymitragynine in the leaf alkaloid extract was 0.01–0.04% for 'Hawaii' and 0.02–0.04% for 'MR-Malaysian', corresponding to 0.002–0.004% and 0.003–0.006%, respectively, in the leaf material. Despite analyzing concentrated extracts, the calculated 7-hydroxymitragynine content in the dry leaves are 2- to 5-fold lower than the lowest 7-hydroxymitragynine concentrations reported in commercial kratom leaf products (Sharma et al., 2025) and more than 10-fold lower than those found in native Malaysian trees (Chear et al., 2021). Additionally, the highest 7-hydroxymitragynine concentration found in this study is unlikely to be clinically significant, especially considering that 7-hydroxymitragynine has limited dissolution when consumed through leaf material and exhibits low oral bioavailability following intragastric administration (Chiang et al., 2025). Collectively, these findings suggest that the germplasm we selected can achieve mitragynine levels comparable to those of Southeast Asian varieties when cultivated using appropriate techniques, such as optimized postharvest processing and seasonal harvesting, while maintaining significantly lower concentrations of 7-hydroxymitragynine, which reduces the potential for drug abuse. These results further reiterate that kratom should be consumed in its traditional, unadulterated form, such as fresh or dried leaf material, which naturally contains low amounts of 7-hydroxymitragynine, rather than in adulterated forms, such as concentrated extracts, that carry a higher risk of misuse (Grundmann et al., 2024).

The synthesis of 7-hydroxymitragynine remains unclear, and questions persist regarding whether it is naturally produced by the plant or formed during postharvest processing through oxidative transformations (Ganasan et al., 2024). Some studies have reported the presence of 7-hydroxymitragynine in "fresh leaves" (Chear et al., 2021; Karunakaran et al., 2024), however, these leaves were not preserved in liquid nitrogen immediately after harvest. As a result, ongoing biological and enzymatic activities may have influenced alkaloid biosynthesis prior to analysis. Therefore, further research using leaves preserved immediately after harvest is warranted to determine whether 7-hydroxymitragynine is synthesized by the plant itself or not.

Several studies support the hypothesis that 7-hydroxymitragynine is a postharvest oxidative or metabolic derivative of mitragynine, formed through processes such as oxidation or hydrolysis that may be influenced by environmental factors, including temperature and humidity, during storage and handling (Chear et al., 2021; Ganasan et al., 2024; Karunakaran et al., 2024). However, this does not fully explain our observation that 7-hydroxymitragynine was above LLOQ only during the summer months in cultivar Hawaii and only during the winter months in cultivar MR-Malaysian, regardless of postharvest treatments. Additionally, Kikura-Hanajiri et al. (2009) reported that kratom trees with low mitragynine production also had significantly lower levels of 7-hydroxymitragynine compared to those with high mitragynine production. This finding aligns with our previous results, where 7-hydroxymitragynine was below LLOQ in kratom trees that produced low levels of mitragynine (Zhang et al., 2020, 2022). Together, these findings suggest that kratom trees producing higher levels of mitragynine also tend to produce more 7-hydroxymitragynine. Collectively, we believe that the presence and concentration of 7-hydroxymitragynine in kratom products could be influenced by both endogenous biosynthesis within the plant, although no biochemical evidence of the responsible enzymes has been identified to date, and by the postharvest conversion of mitragynine. Genetic factors, potentially modulated by seasonal variation, likely play a more prominent role in determining its accumulation.

5 Conclusion

Kratom alkaloid biosynthesis was influenced by postharvest handling, genotype, and seasonal variation. Leaves withered for 12

hours and dried at low temperatures retained the highest levels of mitragynine and generally yielded greater concentrations of desirable alkaloids. Our findings challenge a common marketing misconception, demonstrating that the color of kratom powder is determined by postharvest processing conditions rather than alkaloid composition. The two cultivars examined in this study produced mitragynine levels comparable to those reported from trees grown in Southeast Asia but exhibited significantly lower levels of 7-hydroxymitragynine. Overall, the presence and concentration of 7-hydroxymitragynine in kratom products appear to result from a combination of endogenous biosynthesis and postharvest transformation.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

MZ: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. AL: Data curation, Investigation, Writing – original draft, Writing – review & editing. SK: Investigation, Writing – review & editing. AS: Resources, Writing – review & editing. BP: Conceptualization, Funding acquisition, Resources, Validation, Writing – review & editing. CM: Resources, Writing – review & editing. JC: Resources, Validation, Writing – review & editing.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fpls.2025.1653916/full#supplementary-material

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Orally Active Opioid μ/δ Dual Agonist MGM-16, a Derivative of the Indole Alkaloid Mitragynine, Exhibits Potent Antiallodynic Effect on Neuropathic Pain in Mice[S]

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ABSTRACT

(*E*)-Methyl 2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7*a*-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12*b*-octahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (7-hydroxymitragynine), a main active constituent of the traditional herbal medicine *Mitragyna speciosa*, is an indole alkaloid that is structurally different from morphine. 7-Hydroxymitragynine induces a potent antinociceptive effect on mouse acute pain through μ-opioid receptors. In this study, we developed dual-acting μ- and δ-opioid agonists MGM-15 and MGM-16 from 7-hydroxymitragynine for the treatment of acute and chronic pain. MGM-16 showed a higher potency than that of 7-hydroxymitragynine and MGM-15 in in vitro and in vivo assays. MGM-16 exhibited a high affinity for μ- and δ-opioid receptors, with K_i values of 2.1 and 7.0 nM, respectively. MGM-16 showed μ- and δ-opioid full agonistic effects in a guanosine 5′-*O*-(3-[^{35}S]thiotriphosphate) binding assay and in a functional test using electrically elicited guinea pig ileum and mouse vas deferens contractions. Systemic administration of MGM-16 produced antinociceptive effects in a mouse acute pain model and antiallodynic effects in a chronic pain model. The antinociceptive effect of MGM-16 was approximately 240 times more potent than that of morphine in a mouse tail-flick test, and its antiallodynic effect was approximately 100 times more potent than that of gabapentin in partial sciatic nerve-ligated mice, especially with oral administration. The antinociceptive effect of MGM-16 was completely and partially blocked by the μ-selective antagonist β-funaltrexamine hydrochloride (β-FNA) and by the δ-selective antagonist naltrindole, respectively, in a tail-flick test. The antiallodynic effect of MGM-16 was completely blocked by β-FNA and naltrindole in a neuropathic pain model. These findings suggest that MGM-16 could become a class of a compound with potential therapeutic utility for treating neuropathic pain.

Introduction

Morphine has been proposed for the treatment of neuropathic pain, but it has incomplete efficacy and dose-limiting adverse effects (Gilron et al., 2005). To develop additional analgesics, derivatives have been synthesized by simplification and introduction of substituents into morphine's chemical structure (Corbett et al., 2006). Thus, most morphine-derived opioid analgesics used clinically have μ-receptor agonist profiles. Although all three major types of opioid receptors (μ, δ, and κ) are able to mediate analgesia and antinociception, they

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ABBREVIATIONS: CHO, Chinese hamster ovary; DAMGO, [D-Ala^2, N-MePhe^4, Gly-ol^5]-enkephalin; DPDPE, [D-Pen^2, D-Pen^5]-enkephalin; β-FNA, β-funaltrexamine hydrochloride; GIT, gastrointestinal transit; 7-hydroxymitragynine, (*E*)-methyl 2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7*a*-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12*b*-octahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate; IGIT, inhibitory effects on gastrointestinal transit; MGM-9, (*E*)-methyl 2-(3-ethyl-7*a*,12*a*-(epoxyethanoxy)-9-fluoro-1,2,3,4,6,7,12,12*b*-octahydro-8-methoxyindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate; MGM-15, (*E*)-methyl 2-((2*S*,3*S*,7a*S*,12a*R*,12b*S*)-3-ethyl-7*a*-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate; MGM-16, (*E*)-methyl 2-((2*S*,3*S*,7a*S*,12a*R*,12b*S*)-3-ethyl-9-fluoro-7*a*-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate; norBNI, nor-binaltorphimine dihydrochloride; NTI, naltrindole; U69593, (5α,7α,8β)-(+)-*N*-methyl-*N*-[7-(1-pyrrolidinyl)-1-oxaspiro[4.5]dec-8-yl]-benzeneacetamide.

have different pharmacological activities. Recently, a prominent role of δ-opioid receptors in chronic pain such as neuropathic pain and inflammatory pain was reported from studies using mutant animals and selective agonists (Nadal et al., 2006; Nozaki et al., 2012). In a neuropathic pain rat model, δ-opioid receptor protein expression was increased compared with a control rat in the dorsal root ganglion (Kabli and Cahill, 2007). δ-Opioid receptor activation leads to decreased chronic pain but weakly influences acute pain, in contrast to μ-opioid receptor activation (Gaveriaux-Ruff et al., 2011). A novel strategy for pain management is to use dual-acting and/or mixed opioid agonists (Matsumoto et al., 2008; Cremeans et al., 2012). Therefore, opioid agonists, which act on not only δ- but also μ-opioid receptor subtypes, might be broad-spectrum analgesics useful to treat a variety of painful conditions.

The traditional herbal medicine *Mitragyna speciosa* has long been used in Thailand for its opioid-like effects (Burkill, 1935) and as a replacement for opium (Suwanlert, 1975). This medicinal herb contains many indole alkaloids (Takayama, 2004). (*E*)-Methyl 2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7a,12b-octahydroindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (7-hydroxymitragynine), a main active constituent of this plant, is an indole alkaloid that is structurally different from morphine (Horie et al., 2005). We have reported that 7-hydroxymitragynine induces a potent antinociceptive effect on mouse acute pain through the activation of μ-opioid receptors. It is approximately 14 times more potent than morphine when orally administered (Matsumoto et al., 2008). Furthermore, 7-hydroxymitragynine inhibited gastrointestinal transit (GIT) less potently than morphine at each equally antinociceptive dose in mice (Matsumoto et al., 2006a). We investigated the structural similarities between morphine and 7-hydroxymitragynine using molecular modeling techniques. However, we could not superimpose all three functional groups, i.e., a nitrogen atom, a benzene residue, and an oxygen atom on the benzene ring in their structures (Matsumoto et al., 2005a). Therefore, we developed novel opioid analgesics derived from 7-hydroxymitragynine that have different pharmacophore groups from morphine that bind to μ-opioid receptors. We found a novel dual-acting μ- and κ-opioid agonist, (*E*)-methyl 2-(3-ethyl-7a,12a-(epoxyethanoxy)-9-fluoro-1,2,3,4,6,7,12,12b-octahydro-8-methoxyindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (MGM-9), which has stronger antinociceptive and weaker rewarding effects than morphine (Matsumoto et al., 2008). 7-Hydroxymitragynine-related indole alkaloids have interesting pharmacological characteristics such as high oral potency and few side effects. Therefore, further investigation of the development of novel analgesics against acute and chronic pain is warranted.

In this study, we hypothesized that a dual-acting μ- and δ-opioid agonist derived from 7-hydroxymitragynine could not only induce potent antinociceptive effects against acute pain but also induce an antiallodynic effect against neuropathic pain. We synthesized the novel μ/δ-opioid dual agonists (*E*)-methyl 2-((2*S*,3*S*,7a*S*,12a*R*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7a,12,12a,12b-decahydroindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (MGM-15) and (*E*)-methyl 2-((2*S*,3*S*,7a*S*,12a*R*,12b*S*)-3-ethyl-9-fluoro-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7a,12,12a,12b-decahydroindolo[2,3-a]quinolizin-2-yl)-3-methoxyacrylate (MGM-16) and clarified their pharmacological profiles using in vitro and in vivo experiments under physiologic conditions. Furthermore, we examined the antiallodynic effects of MGM-16 in a mouse sciatic nerve ligation model. We found that the orally active μ- and δ-opioid receptor agonist MGM-16 produces potent effects against acute and neuropathic pain.

Materials and Methods

Preparation of MGM-15. $NaBH_4$ (4.3 mg, 0.11 mmol) was added to a stirred solution of 7-hydroxymitragynine (37.5 mg, 0.091 mmol) in dry MeOH (1.1 ml) at 0°C under an argon atmosphere. After 30 minutes, H_2O was added to the reaction mixture. The mixture was concentrated under reduced pressure and poured into saturated aqueous $NaHCO_3$ solution. The whole mixture was extracted with 5% MeOH/$CHCl_3$ three times. The combined extract was washed with brine, dried over $MgSO_4$, and evaporated to give a residue that was purified by silica gel column chromatography (ethyl acetate/*n*-hexane = 35:65). The purified compound was then crystallized from ethyl acetate to give 26.3 mg of MGM-15 (y. 70%).

MGM-15; m.p.: 219–223°C (ethyl acetate). UV (MeOH) l_{max} nm (log *e*): 288 (3.13), 277 (3.06), 239 (4.16), 232 (4.14), 214 (4.50). Infrared spectroscopy (KBr) n_{max} cm^{-1}: 3341, 2954, 1698, 1614, 1467, 1283. ^1H NMR (500 MHz, $CDCl_3$) *d* ppm: 7.40 (1H, s, H-17), 7.00 (1H, dd, *J* = 8.1, 8.1 Hz, H-11), 6.35 (1H, d, *J* = 7.9 Hz, H-12), 6.31 (1H, d, *J* = 8.2 Hz, H-10), 3.87 (1H, br.s, N_a-H), 3.83 (3H, s, 9-OCH_3), 3.80 (3H, s, 17-OCH_3), 3.70 (3H, s, 22-OCH_3), 3.47 (1H, br.s, H-2), 2.95 (2H, m, H-15 and H-21), 2.94 (1H, s, 7-OH), 2.53 (2H, m, H-5 and H-14), 2.25 (1H, ddd, *J* = 12.3, 12.3, 2.3 Hz, H-5), 2.16 (2H, m, H-3 and H-21), 2.03 (1H, d, *J* = 14.3 Hz, H-6), 1.92 (1H, ddd, *J* = 13.6, 13.6, 4.3 Hz, H-6), 1.79 (1H, m, H-19), 1.58 (1H, br.d, *J* = 11.3 Hz, H-20), 1.39 (1H, d, *J* = 12.8 Hz, H-14), 1.23 (1H, m, H-19), 0.85 (3H, dd, *J* = 7.3, 7.3 Hz, H_3-18). ^{13}C NMR (125 MHz, $CDCl_3$) *d* ppm: 169.1 (C-22), 160.3 (C-17), 155.9 (C-9), 149.9 (C-13), 129.3 (C-11), 121.3 (C-8), 111.8 (C-16), 105.2 (C-12), 101.9 (C-10), 77.1 (C-7), 69.8 (C-2), 61.6 (17-OCH_3), 61.6 (C-3), 58.5 (C-21), 55.1 (9-OCH_3), 51.2 (22-OCH_3), 50.7 (C-5), 40.7 (C-20), 40.2 (C-15), 35.2 (C-6), 28.5 (C-14), 19.1 (C-19), 13.1 (C-18). Electron ionization mass spectrometry (%) *m*/*z*: 416 (M^+, 61), 400 (96), 399 (100), 398 (97), 397 (78), 383 (41), 256 (64), 214 (84). Anal. Calcd for $C_{23}H_{32}O_5N_2$: C, 66.32; H, 7.74; N, 6.73. Found: C, 66.08; H, 7.77; N, 6.71. CD (*c* = 0.29 mM, MeOH, 24°C), Δε (λ nm): 0 (318), +1.4 (293), 0 (257), −1.0 (248), 0 (233), +0.2 (231), 0 (227), −7.7 (215), +0.1 (208).

Preparation of MGM-16. $NaBH_4$ (2.1 mg, 0.056 mmol) was added to a stirred solution of 10-fluoro-7-hydroxymitragynine (23.7 mg, 0.055 mmol) in dry MeOH (0.5 ml) at 0°C under an argon atmosphere. The reaction mixture was stirred for 30 minutes. After adding H_2O, the reaction mixture was poured into saturated aqueous $NaHCO_3$ solution. The whole mixture was extracted with 5% MeOH/$CHCl_3$ three times. The combined extract was washed with brine, dried over $MgSO_4$, and evaporated to give a residue that was purified by amino-silica gel column chromatography (ethyl acetate/*n*-hexane = 1:1) to give 22.6 mg of MGM-16 (y. 95%).

MGM-16; UV (MeOH) l_{max} nm (log *e*): 297 (3.13), 276 (3.10), 240 (4.22), 226 (4.11), 205 (4.60). Infrared spectroscopy (KBr) n_{max} cm^{-1}: 3364, 2947, 1701, 1626, 1490, 1284, 1240. ^1H NMR (500 MHz, $CDCl_3$) *d* ppm: 7.41 (1H, s, H-17), 6.76 (1H, dd, *J* = 12.8, 8.2 Hz, H-11), 6.28 (1H, dd, *J* = 8.5, 3.1 Hz, H-12), 4.01 (3H, d, *J* = 2.7 Hz, 9-OCH_3), 3.79 (3H, s, 17-OCH_3), 3.69 (3H, s, 22-OCH_3), 3.46 (1H, d, *J* = 3.1 Hz, H-2), 2.95 (2H, m, H-15 and H-21), 2.77 (1H, br.s, 7-OH), 2.52 (2H, m, H-5 and H-14), 2.25 (1H, m, H-5), 2.15 (2H, m, H-3 and H-21), 1.95 (2H, m, H-6), 1.76 (1H, m, H-19), 1.58 (1H, br.d, *J* = 11.3 Hz, H-20), 1.37 (1H, d, *J* = 12.8 Hz, H-14), 1.23 (1H, m, H-19), 0.84 (3H, dd, *J* = 7.3, 7.3 Hz, H_3-18). ^{13}C NMR (125 MHz, $CDCl_3$) *d* ppm: 169.0 (C-22), 160.3 (C-17), 148.4 (d, *J* = 237.4 Hz, C-10), 145.1 (C-13), 143.2 (d, *J* = 11.9 Hz, C-9), 126.8 (C-8), 116.4 (d, *J* = 21.5 Hz, C-11), 111.7 (C-16), 105.8 (d, *J* = 6.9 Hz, C-12), 77.4 (C-7), 70.0 (C-2), 61.6 (17-OCH_3), 61.4 (C-3), 61.2 (d, *J* = 7.8 Hz, 9-OCH_3), 58.4 (C-21), 51.2 (22-OCH_3), 50.5 (C-5), 40.7 (C-20), 40.1 (C-15), 35.2 (C-6), 28.4 (C-14), 19.0 (C-19), 13.1 (C-18).

CD (c = 0.26 mM, MeOH, 24°C), $\Delta\varepsilon$ (λ nm): 0 (337), +2.7 (297), 0 (255), −4.2 (236), −3.8 (227), −8.4 (214), +0.1 (207). Fast atom bombardment (nitrobenzyl alcohol) mass spectrometry m/z: 435 $[M+H]^+$. High-resolution fast atom bombardment (nitrobenzyl alcohol/polyethylene glycol) mass spectrometry: calcd. for $C_{23}H_{32}O_5N_2F$: 435.2295, found: 435.2301.

Drugs. The drugs used in this study were morphine hydrochloride (Takeda Chemical Industries, Osaka, Japan), naloxone hydrochloride (MP Biomedicals, Irvine, CA), gabapentin (Cayman Chemical Company, Ann Arbor, MI), naltrindole hydrochloride, nor-binaltorphimine dihydrochloride (norBNI), [D-Ala^2, N-MePhe^4, Gly-ol^5]-enkephalin (DAMGO), [D-Pen^2, D-Pen^5]-enkephalin (DPDPE), (5α,7α,8β)-(+)-*N*-methyl-*N*-[7-(1-pyrrolidinyl)-1-oxaspiro[4.5]dec-8-yl]-benzeneacetamide (U69593), cyprodime hydrobromide, and [Met]-enkephalin (Sigma-Aldrich, St. Louis, MO) and β-funaltrexamine hydrochloride (β-FNA; Tocris-Cookson, Bristol, UK).

For experiments with guinea pig ileum and mouse vas deferens, 7-hydroxymitragynine-related indole alkaloids and cyprodime hydrobromide were first dissolved in 100% dimethylsulfoxide to yield a 5 mM solution, and then subsequently diluted with distilled water. The other drugs were dissolved in distilled water. For subcutaneous administration, 7-hydroxymitragynine-related indole alkaloids were dissolved in 0.01 N hydrochloric acid. Other drugs were dissolved in saline. For oral administration and receptor-binding assays, MGM-15, MGM-16, 7-hydroxymitragynine, and gabapentin were dissolved in 0.01 N hydrochloric acid.

In the antinociceptive test, naloxone hydrochloride (2 mg/kg), naltrindole (3 mg/kg), norBNI (20 mg/kg), or β-FNA (40 mg/kg) was administered subcutaneously 15 minutes, 30 minutes, 24 hours, or 24 hours before drug injection, respectively. These protocols were described by Matsumoto et al. (2005a).

Animals. Male ddY strain mice (Japan SLC, Hamamatsu, Japan), male Hartley strain guinea pigs (Japan SLC), and male SD rats (Japan SLC) were used. Animals were housed in a temperature-controlled room at 24°C with lights on from 7:00 AM to 5:00 PM and free access to food and water. All experiments were performed in compliance with the *Guiding Principles for the Care and Use of Laboratory Animals* approved by the Japanese Pharmacological Society and the guidelines approved by the Ethics Committee on Animal Care and Animal Experimentation of Josai International University. The number of animals used was kept to the minimum necessary for a meaningful interpretation of the data, and animal discomfort was kept to a minimum.

Electrical Stimulation of Guinea Pig Ileum. Guinea pig ileum was dissected and placed in Krebs-Henseleit solution (in mM: NaCl, 112.08; KCl, 5.90; $CaCl_2$, 1.97; $MgCl_2$, 1.18; NaH_2PO_4, 1.22; $NaHCO_3$, 25.00; and glucose, 11.49). The ileum was placed under 1g tension in a 10-ml organ bath containing the nutrient solution. The bath was maintained at 37°C and continuously bubbled with a mixture of 95% O_2 and 5% CO_2. Tissues were stimulated by a platinum needle-ring electrode (Iwashiya Kishimoto Medical Instruments, Kyoto, Japan). After equilibration, the ileum was transmurally stimulated with monophasic pulses (0.2 Hz and 0.1-ms duration) by a stimulator (SEN-7203; Nihon Kohden, Tokyo, Japan). Contractions were isotonically recorded using a displacement transducer (TD-112S; Nihon Kohden). The effects of drug treatments on the twitch contractions evoked by transmural stimulation from the ring electrodes were examined. The height of the twitch response to transmural stimulation was measured before and after the drug challenge. Responses were expressed as contraction % of the twitch response to the transmural stimulation before the drug challenge as 100%.

Electrical Stimulation of Mouse Vas Deferens. Mouse vas deferens was dissected and placed in Krebs-Henseleit solution without $MgCl_2$. The tissues were placed under 0.2g tension in a 10-ml organ bath containing the nutrient solution. The bath was maintained at 37°C and continuously bubbled with a mixture of 95% O_2 and 5% CO_2. Tissues were stimulated by a platinum needle-ring (the ring was placed 20 mm above the base of a 5-mm-long needle) electrode. After equilibration, the tissues were transmurally stimulated with a train of 10 pulses of 0.5-ms duration with 2-ms intervals every 1 minute by a stimulator (SEN-7203). Contractions were isometrically recorded by using a displacement transducer (TB-651T; Nihon Kohden). The effects of drug treatments on the twitch contractions evoked by transmural stimulation from the ring electrodes were examined. The height of the twitch response to transmural stimulation was measured before and after the drug challenge. The responses were expressed as contraction percentage of the twitch response to the transmural stimulation before the drug challenge as 100%.

Receptor Binding. Guinea pig brain membrane protein (without cerebellum, 500 μg) fractions were incubated with 1 nM [^{3}H]DAMGO (50.0 Ci/mmol; American Radiolabeled Chemicals, St. Louis, MO) or 1 nM [^{3}H]U69593 (55.0 Ci/mmol; GE Healthcare, Little Chalfont, Buckinghamshire, UK) in a final volume of 1 ml of 50 mM Tris-HCl buffer. Rat brain membrane protein fractions were incubated with 2 nM [^{3}H]DPDPE (25.2 Ci/mmol; PerkinElmer Life and Analytical Sciences, Boston, MA) in a final volume of 1 ml of 50 mM Tris-HCl buffer. Nonspecific binding was measured by the inclusion of 10 μM naloxone. The incubation periods were 1, 2.5, and 1 hour for [^{3}H]DAMGO, [^{3}H]DPDPE, and [^{3}H]U69593, respectively, at 25°C. The reaction was terminated by rapid filtration under reduced pressure through glass fiber filters (Whatman GF/C, presoaked in 0.25% polyethyleneimine; Whatman, Clifton, NJ). After filtration, filters were washed three times with 3 ml of cold 50 mM Tris-HCl and counted in 4 ml of Ultima Gold scintillation cocktail (PerkinElmer Life and Analytical Sciences). The ability of unlabeled drugs to inhibit specific radioligand binding was expressed as the IC_{50} value, which is the molar concentration of the unlabeled drug necessary to displace 50% of the specific binding. Inhibition constants (K_i) of unlabeled compounds were calculated as described by Cheng and Prusoff (1973).

Guanosine 5′-*O*-(3-[^{35}S]Thiotriphosphate) Binding Assay. The agonist activities of the test compounds (MGM-15 and MGM-16) and DAMGO for μ-opioid receptor– or Met-enkephalin for δ-opioid receptor–positive controls were measured by radioactivities of guanosine 5′-*O*-(3-[^{35}S]thiotriphosphate) ([^{35}S]GTPγS). μ-Opioid receptor and δ-opioid receptor agonist activities were performed using a membrane fraction prepared from Chinese hamster ovary (CHO-K1) K1 cells expressing recombinant μ- and δ-opioid receptors. The membranes of cells stably expressing rat μ- and δ-opioid receptors were suspended with binding assay buffer [50 mM Tris-HCl (pH 7.4) containing EDTA (1 mM), $MgCl_2$ (12.5 mM), NaCl (100 mM), bovine serum albumin (0.5%), and GDP (3 μM)]. The membranes (10 μg) were incubated at 25°C for 1 hour in 200 μl of binding assay buffer with various concentrations of the test compound or positive control (0.1–100,000 nM) and 50 pM [^{35}S]GTPγS (specific activity, 1000 Ci/mmol; PerkinElmer Life and Analytical Sciences) in a 96-well plate. The reaction was terminated by filtration, and MicroScint-20 (PerkinElmer Life and Analytical Sciences) was added to the plate. Radioactivity in the samples was determined using a liquid scintillation analyzer. Nonspecific binding was measured in the presence of 10 μM unlabeled GTPγS. Agonist activation by each compound was expressed as the percentage stimulation, which was calculated as $(T_1 - T_0)/(T_2 - T_0) \times 100$, where T_0 is the nonspecific activity, T_1 is the [^{35}S]GTPγS activity in the presence of various concentrations of test compounds or positive control, and T_2 is the maximal stimulation of [^{35}S]GTPγS binding by the positive control.

All measurements were performed in duplicate. Three independent experiments were conducted in this study, and data are expressed as mean ± S.E.M. The EC_{50} values (a concentration that produces 50% of maximal stimulation of the [^{35}S]GTPγS binding) and E_{max} (percentage of maximal stimulation in the [^{35}S]GTPγS binding) were estimated from saturation analysis of agonist-stimulated [^{35}S]GTPγS binding using program XLfit (Microsoft, Redmond, WA).

Tail-Flick Test. Mice respond to a focused heat stimulus by flicking or moving their tail from the path of the stimulus, thereby exposing a photocell located in the tail-flick analgesia meter (Ugo Basile Tail-flick Unit 7360; Ugo Basile, Comerio, Italy) immediately

below the tail. The reaction time is automatically recorded. Before treatment with drugs, vehicle, or saline, the nociceptive threshold was measured three times, and the mean of the reaction time was used as the predrug latency for each mouse. A cutoff time of 10 seconds was used to prevent tissue damage.

Antinociception in the tail-flick test was quantified using the percentage of maximum possible effect and calculated as: % maximum possible effect = [(test latency – predrug latency)/(cutoff time – predrug latency)] × 100.

GIT. Mice were fasted, with water available ad libitum, for 18 hours before the experiments. Fifteen minutes after subcutaneous administration of 7-hydroxymitragynine, MGM-15, MGM-16, morphine, vehicle, or saline, a 0.25-ml charcoal meal (an aqueous suspension of 10% charcoal and 5% gum arabic) was orally administered. Fifteen minutes after oral administration of 7-hydroxymitragynine, MGM-15, MGM-16, or vehicle, and 30 minutes after oral administration of morphine or distilled water, a charcoal meal was orally administered. These time points were established by the time of peak effect for each drug and by the protocol as previously reported (Matsumoto et al., 2006a). Thirty minutes after administration of the charcoal meal, the animal was killed by cervical dislocation, and the small intestine from the pylorus to the ileocecal junction was carefully removed. Both the length of the small intestine from the pylorus to the ileocecal junction and the farthest distance to which the charcoal meal had traveled were measured. For each animal, the GIT was calculated as the percentage of the distance traveled by the charcoal meal relative to the total length of the small intestine. The inhibition of GIT (%) was calculated as: inhibition of GIT (%) = [(saline or vehicle GIT – drug GIT)/(saline or vehicle GIT)] × 100.

Neuropathic Pain Model. The mice were anesthetized with 3% isoflurane. We produced a partial sciatic nerve injury by tying a tight ligature with an 8-0 silk suture around approximately one-third to one-half the diameter of the sciatic nerve on the right side (ipsilateral side) under a light microscope (SD30; Olympus, Tokyo, Japan), as described previously (Seltzer et al., 1990; Malmberg and Basbaum, 1998). In sham-operated mice, the nerve was exposed without ligation. The sciatic nerve–ligated mice exhibit thermal hyperalgesia and mechanical hyperalgesia on the ipsilateral side, indicating the state of neuropathic pain hypersensitivity (Narita et al., 2008). In the present study, the antinociceptive assay was performed 7 days after partial sciatic nerve ligation.

Measurement of Mechanical Allodynia. The paw withdrawal threshold was determined as described by Chaplan et al. (1994). The mice were individually placed in suspended cages with wire mesh bottoms and allowed to acclimate to their environment for at least 30 minutes. The hind paw was stimulated with a series of calibrated (0.02–1.48 g) von Frey filaments applied to the plantar surface. The 50% paw withdrawal threshold that is nonparametrically distributed was determined using the up-down paradigm. Testing was initiated with 0.17-g hair, and a positive response was indicated by a withdrawal of the paw. The paw withdrawal threshold was measured repeatedly at the multiple time points indicated after administration of each drug.

Statistical Analysis. The data are expressed as the mean ± S.E.M. Statistical analyses were performed with the two-tailed Student's t test for comparison of two groups and by a one-way analysis of variance followed by a Bonferroni multiple-comparisons test for comparison of more than two groups. A P value < 0.05 was considered statistically significant. ED_{50} values and 95% confidence limits were determined using the Litchfield-Wilcoxon method (Litchfield and Wilcoxon, 1949). The statistical significance of differences between MGM-15 and MGM-16 in [^{35}S]GTPγS binding assay was assessed using two-way analysis of variance followed by Bonferroni multiple-comparison tests.

Results

Effects of 7-Hydroxymitragynine-Related Indole Alkaloids on Electrically Stimulated Twitch Contraction in Guinea Pig Ileum. The opioid agonistic activities of 7-hydroxymitragynine and semisynthetic compounds derived from 7-hydroxymitragynine (Fig. 1) were evaluated by measuring inhibition of the twitch contraction induced by electrical stimulation in the guinea pig ileum preparation (Table 1). MGM-15 and MGM-16 were prepared by $NaBH_4$ reduction of 7-hydroxymitragynine and 10-fluoro-7-hydroxymitragynine, respectively. MGM-15 and MGM-16 showed a higher potency than 7-hydroxymitragynine, based on their pD_2 values. The introduction of a fluoro group onto the C-10 position (MGM-16) led to a higher potency than MGM-15. MGM-16 showed an inhibitory effect as potent as that of the μ-selective agonist DAMGO.

Effects of MGM-15 and MGM-16 on Electrically Stimulated Twitch Contraction in Mouse Vas Deferens. To clarify the effect of MGM-15 and MGM-16 on δ-opioid receptors, we investigated the effect of MGM-15 and MGM-16 using mouse vas deferens (Table 2). MGM-15 and MGM-16 inhibited the electrically elicited mouse vas deferens contraction in a dose-dependent manner, as did 7-hydroxymitragynine and the δ-selective agonist DPDPE. MGM-16 showed a higher potency than MGM-15. MGM-16 showed an inhibitory effect as potent as that of DPDPE. The concentration-response curves for MGM-15 and MGM-16 were shifted to the right in the presence of the μ-opioid receptor antagonist cyprodime and κ-opioid antagonist norBNI (Fig. 2). The curves of MGM-15 and MGM-16 were further shifted to the right in the presence of cyprodime, norBNI, and the δ-selective antagonist naltrindole (Table 3).

Receptor Binding. The affinities of MGM-15 and MGM-16 for the three opioid receptor types were determined by evaluating the inhibition of radioligand binding to μ-, δ-, and κ-opioid receptors (Table 3). MGM-15 and MGM-16 had a relatively high affinity for μ- and δ-opioid sites. The [^{3}H]DAMGO and [^{3}H]DPDPE binding was displaced by MGM-15 and MGM-16 in a concentration-dependent manner in the membrane samples from guinea pig and rat brain, respectively (data not shown).The K_i value for MGM-15

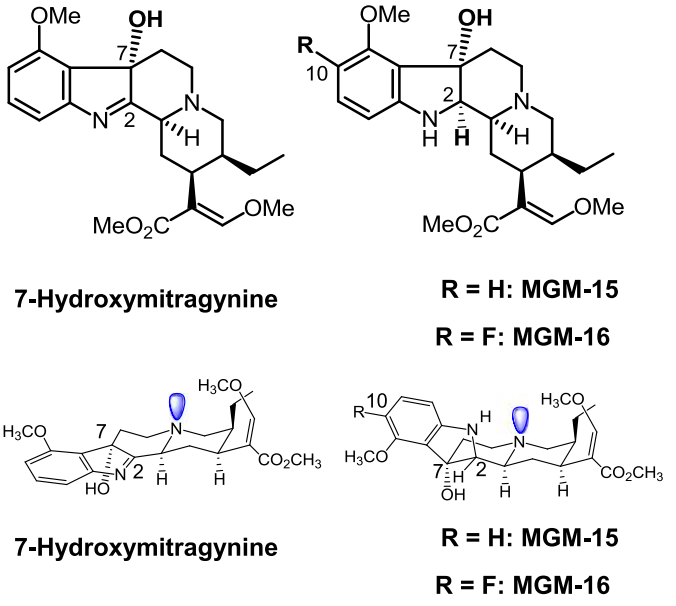


Fig. 1. Chemical structure and conformation of 7-hydroxymitragynine, MGM-15, and MGM-16.

TABLE 1

pD_2 values and maximum percentage inhibition of mitragynine derivatives in guinea pig ileum

Each value represents mean ± S.E.M. of data obtained from four to six animals.

Compound	pD_2 Value	Maximum Inhibition
		%
Morphine	7.16 ± 0.05	79.4 ± 5.7
MGM-15	8.26 ± 0.05	78.2 ± 3.4
MGM-16	8.81 ± 0.09	89.0 ± 2.3
7-Hydroxymitragynine	7.58 ± 0.12	79.6 ± 3.0
DAMGO	8.46 ± 0.06	87.2 ± 1.8

displacement of [^{3}H]DAMGO and [^{3}H]DPDPE binding to μ- and δ-opioid sites was 6.4 ± 0.30 and 16 ± 1.0 nM, respectively. The K_i value for MGM-16 displacement of [^{3}H]DAMGO and [^{3}H]DPDPE was 2.1 ± 0.028 and 7.0 ± 0.23 nM, respectively. MGM-15 and MGM-16 weakly displaced [^{3}H]U69593 binding to κ-opioid sites.

[^{35}S]GTPγS Binding Assay. We next investigated the ability of MGM-15 and MGM-16 to activate G-proteins in CHO-K1 cells expressing recombinant μ- and δ-opioid receptors (Fig. 3; Table 4). The μ-opioid agonist DAMGO or δ-opioid agonist [Met]-enkephalin produced a concentration-dependent increase in [^{35}S]GTPγS binding to the CHO-K1 cell membrane. MGM-16 also showed a concentration-dependent increase in [^{35}S]GTPγS binding to CHO-K1 cell membranes expressing recombinant μ- and δ-opioid receptors. In contrast, MGM-15 showed a smaller increase in the binding of [^{35}S]GTPγS than that of MGM-16.

Antinociception of MGM-15 and MGM-16 in Mouse Tail-Flick Test by Subcutaneous and Oral Administration. Antinociceptive effects of MGM-15 and MGM-16 were investigated in acute thermal pain tests in mice. MGM-15 and MGM-16 induced dose-related antinociceptive responses in tail-flick tests after subcutaneous and oral administration (Fig. 4). The effect peaked at 15–30 minutes after injection. The antinociceptive effect of MGM-15 was approximately 15 and 50 times more potent than that of morphine after subcutaneous and oral administration, respectively. The antinociceptive effect of MGM-16 was approximately 71 and 240 times more potent than that of morphine after subcutaneous and oral administration, respectively (Tables 5 and 6).

Next we investigated the effects of MGM-15 and MGM-16 on the inhibition of GIT (IGIT) and their antinociceptive effects in comparison with morphine and 7-hydroxymitragynine (Tables 5 and 6). The IGIT ED_{50} values of 7-hydroxymitragynine, MGM-15, and MGM-16 were slightly larger than their corresponding antinociceptive ED_{50} values. In the case of morphine, the IGIT ED_{50} value was much smaller than its antinociceptive ED_{50} value.

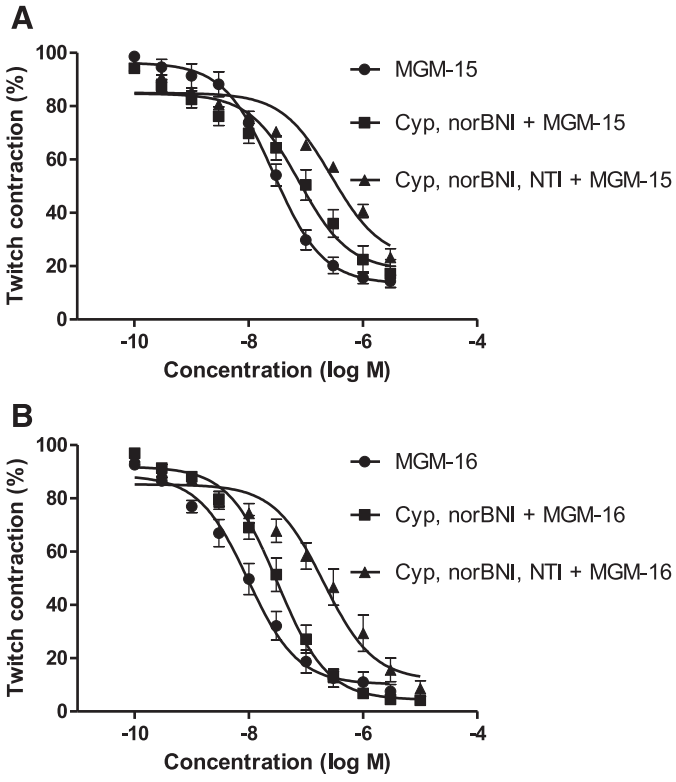


Fig. 2. Concentration-response curves of MGM-15 (A) and MGM-16 (B) for electrical stimulation–induced contraction in mouse vas deferens in the absence or presence of cyprodime (Cyp) and norBNI or Cyp, norBNI, and naltrindole (NTI). Responses are expressed as contraction % of the twitch contraction before agonist addition. Data represent mean ± S.E.M. of five animals.

Effects of Selective Antagonists on MGM-15 and MGM-16 Antinociception in Mouse Tail-Flick Test. To determine the opioid receptor type selectivity of MGM-15 and MGM-16 antinociception, mice were pretreated with selective opioid receptor antagonists in the tail-flick test (Fig. 5). The antinociceptive effect of MGM-15 and MGM-16 was completely blocked by the irreversible μ-opioid receptor selective antagonist β-FNA. These effects were partially and significantly blocked by the δ-opioid receptor selective antagonist naltrindole. The selective κ-opioid antagonist norBNI was ineffective against MGM-15- and MGM-16-induced

TABLE 2

pD_2 values and maximum inhibitory percent of DPDPE, MGM-15, MGM-16, and 7-hydroxymitragynine on electrical stimulation–induced contraction in mouse vas deferens in the absence or the presence of cyprodime (Cyp) and nor-BNI or Cyp, nor-BNI, and NTI

Each value represents mean ± S.E.M. of pD_2 value (maximum inhibitory percent) obtained from four or five animals.

Compound	pD_2 Value of:		
	Compound Alone	Compound + Cyp and norBNI	Compound + Cyp, norBNI, and NTI
DPDPE	8.16 ± 0.20 (89.3 ± 1.1)	7.80 ± 0.17 (90.7 ± 2.2)	6.29 ± 0.10 (90.5 ± 1.6)
MGM-15	7.56 ± 0.06 (85.7 ± 2.3)	7.10 ± 0.11 (82.3 ± 5.2)	6.30 ± 0.08 (78.8 ± 3.3)
MGM-16	8.08 ± 0.11 (92.9 ± 1.8)	7.50 ± 0.11 (95.8 ± 1.1)	6.52 ± 0.22 (91.4 ± 3.0)
7-Hydroxymitragynine	7.22 ± 0.13 (82.7 ± 3.9)	5.98 ± 0.16 (80.2 ± 3.3)	5.55 ± 0.12 (78.2 ± 4.8)

TABLE 3

K_i values for the inhibition of μ- and κ-opioid binding to guinea pig brain membranes and δ-opioid binding to rat brain by test compounds

The μ-binding sites were labeled with [^{3}H]DAMGO (1 nM), δ-binding sites with [^{3}H]DPDPE (2 nM), and κ-binding sites with [^{3}H]U69593 (1 nM). Data are expressed as the mean K_i value ± S.E.M. for three or four determinations performed in duplicate.

Compound	K_i of:		
	[^{3}H]DAMGO (μ)	[^{3}H]DPDPE (δ)	[^{3}H]U69593 (κ)
		nM	
Morphine	2.7 ± 0.24	40 ± 6.0	55 ± 4.1
MGM-15	6.4 ± 0.30	16 ± 1.0	55 ± 4.0
MGM-16	2.1 ± 0.028	7.0 ± 0.23	29 ± 1.5
DAMGO	1.2 ± 0.092	N.D.	N.D.
DPDPE	N.D.	1.2 ± 0.13	N.D.
U69593	N.D.	N.D.	0.66 ± 0.073

N.D., not determined.

antinociception. When these opioid antagonists were administered subcutaneously alone at the doses used in the present study, they did not produce any effects in the tail-flick test (data not shown).

Effect of MGM-16 on Thermal Hyperalgesia Induced by Sciatic Nerve Ligation in Mice. MGM-16 showed potent μ/δ-opioid dual agonistic activities and further

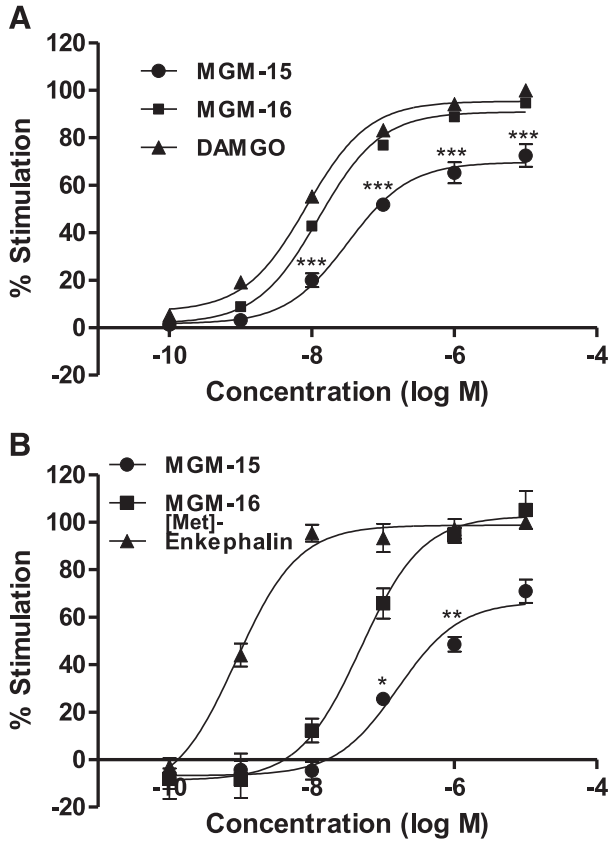


Fig. 3. (A) Concentration-response curves of MGM-15, MGM-16, and DAMGO for [^{35}S]GTPγS binding to CHO-K1 cells expressing recombinant μ-opioid receptors. (B) Concentration-response curves of MGM-15, MGM-16, and [Met]-enkephalin for [^{35}S]GTPγS binding to CHO-K1 cells expressing recombinant δ-opioid receptors. The membrane-bound [^{35}S]GTPγS was measured and expressed as % stimulation relative to the basal level. Each symbol represents the mean ± S.E.M. of three independent samples. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$ vs. MGM-16 group by Bonferroni correction.

TABLE 4

[^{35}S]GTPγS intrinsic efficacies and EC_{50} values of MGM-15, MGM-16, DAMGO, and [Met]-enkephalin in CHO-K1 cells expressing recombinant μ- and δ-opioid receptors

Each value represents the mean ± S.E.M. of three independent samples.

Compound	μ-Opioid Receptor		δ-Opioid Receptor	
	EC_{50}	E_{max}	EC_{50}	E_{max}
	nM		*nM*	
MGM-15	250 ± 61	72 ± 4.7	1492 ± 425	70 ± 4.9
MGM-16	18 ± 0.055	94 ± 1.8	61 ± 3.2	105 ± 8.1
DAMGO	8.0 ± 0.69	100	N.D.	N.D.
[Met]-enkephalin	N.D.	N.D.	1.3 ± 0.26	100

N.D., not determined.

demonstrated higher potency than MGM-15 in vitro and in an in vivo acute pain model. Thus, we investigated the antiallodynic effect of MGM-16 in a neuropathic pain model. Mice with partial sciatic nerve ligation exhibited marked neuropathic pain-like behavior on the ipsilateral side 7 days after the nerve ligation. We evaluated the antiallodynic effects induced by subcutaneous and oral administration of MGM-16 in sciatic nerve–ligated mice using von Frey filaments (Fig. 6). MGM-16 (0.1, 0.2, and 0.4 mg/kg) dose-dependently increased the ipsilateral paw withdrawal threshold in sciatic nerve–ligated mice, and maximal antihyperalgesic responses were seen at 15 or 30 minutes after subcutaneous administration of MGM-16 (Fig. 6A). MGM-16 (0.2 mg/kg) reversed the threshold to control level in sciatic nerve–ligated mice. With oral administration, MGM-16 (0.5, 1, and 2 mg/kg) dose-dependently increased the ipsilateral paw withdrawal threshold in sciatic nerve–ligated mice, and maximal antihyperalgesic responses were seen 30 minutes after administration (Fig. 6B). MGM-16 at 1 mg/kg and gabapentin at 100 mg/kg reversed the threshold to control level in sciatic nerve–ligated mice.

To investigate the contribution of the opioid receptor subtypes in the antiallodynic effect of MGM-16, sciatic nerve–ligated mice were pretreated with selective opioid receptor antagonists (Fig. 7). The antiallodynic effect of MGM-16 was completely blocked by β-FNA and naltrindole in the chronic pain model. The selective κ-opioid antagonist norBNI was ineffective against MGM-16-mediated antinociception.

Discussion

In the present study, we synthesized the dual-acting μ- and δ-opioid agonists MGM-15 and MGM-16 from 7-hydroxymitragynine for the treatment of acute and chronic pain. MGM-16 showed higher opioid agonist potency than that of 7-hydroxymitragynine and MGM-15 in vitro and in vivo. When administered orally, MGM-16 showed 100-fold-higher analgesic potency than that of gabapentin in sciatic nerve–ligated mice. Therefore, MGM-16 could be useful for the treatment of chronic pain such as mechanical allodynia. To our knowledge, this is the first study to investigate the antiallodynic effect of mitragynine-related indole alkaloids using a neuropathic mouse pain model.

The alkaloid 7-hydroxymitragynine was found to possess the most potent opioid agonistic effects among the constituents isolated from the Thai herbal medicine *M. speciosa* (Horie et al., 2005). We have previously synthesized opioid

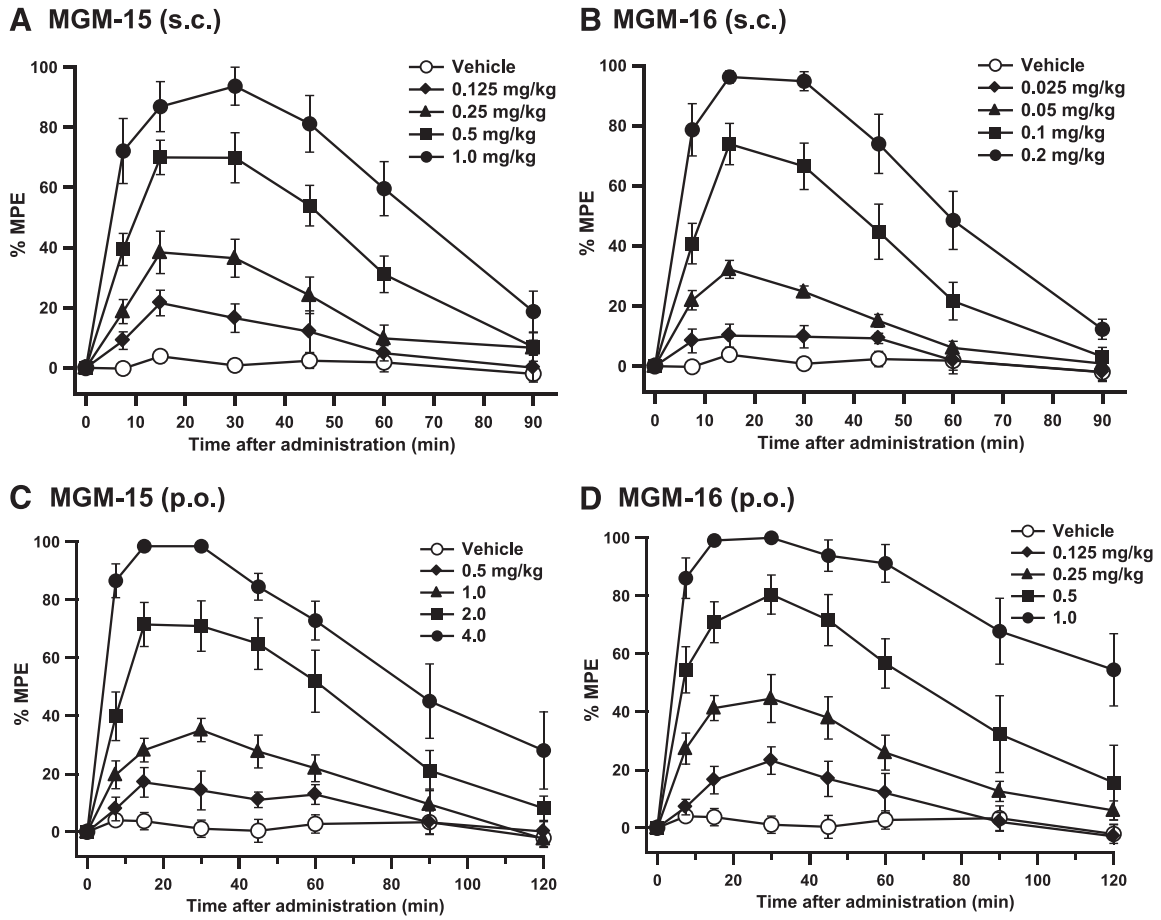


Fig. 4. Time course of the antinociceptive effects produced by subcutaneous (A and B) or oral (C and D) administration of MGM-15 (A and C) and MGM-16 (B and D) in the tail-flick test in mice. Each value represents mean ± S.E.M. of data obtained from eight mice.

analgesics from the indole alkaloid mitragynine (Takayama et al., 2002, 2006; Matsumoto et al., 2004, 2008; Takayama, 2004). We have surveyed these compounds for their opioid agonistic activities in vitro to elucidate the specific structure necessary for its pharmacophore to bind to opioid receptors. A nitrogen atom, a benzene residue, and an oxygen function play a significant role in producing opioid agonistic activity (Matsumoto et al., 2005a). The conversion of an indolenine moiety in 7-hydroxymitragynine (MGM-15) and 10-fluoro-7-hydroxymitragynine into an indoline derivative (MGM-16) led to an increase in agonistic potency. This indicates that the sp^3 carbon at the C2 position (Fig. 1), which spatially configured the benzene ring in the ligands, was more efficient at exerting the opioid activity.

Takayama et al. (2006) reported that the dimension or electronegativity of the functional group at the C10 position of mitragynine derivatives is important in eliciting opioid agonistic effects. Among those derivatives, the C10-fluorinated derivative, MGM-9, showed the highest potency (Takayama et al., 2006). MGM-16, the C10-fluorinated derivative of MGM-15, showed the most potent opioid agonistic effects among the derivatives tested previously (Takayama et al., 2002, 2006; Matsumoto et al., 2008). MGM-16 showed full agonistic properties on μ- and δ-opioid receptors. MGM-16 also showed higher affinities and intrinsic efficacies than MGM-15 on μ- and δ-opioid receptors in vitro. A benzene residue of 7-hydroxymitragynine-related indole alkaloids plays an essential role in producing opioid agonistic activity

TABLE 5

ED_{50} values of antinociceptive effects and IGIT produced by subcutaneous administration of morphine, 7-hydroxymitragynine (7-OHMG), MGM-15, and MGM-16 in mice

ED_{50} represents the median effective dose (mg/kg) (95% confidence limits).

Test	ED_{50} of:			
	Morphine	7-OHMG	MGM-15	MGM-16
	mg/kg			
Tail-flick	4.57 (3.12–6.69)	0.80 (0.48–1.33)	0.30 (0.18–0.50)	0.064 (0.040–0.10)
IGIT	1.07 (0.40–2.86)	1.19 (0.54–2.63)	0.35 (0.18–0.70)	0.089 (0.040–0.188)

TABLE 6

ED_{50} values of antinociceptive effects and IGIT produced by oral administration of morphine, 7-hydroxymitragynine (7-OHMG), MGM-15, and MGM-16 in mice

ED_{50} represents the median effective dose (mg/kg) (95% confidence limits).

Test	ED_{50} of:			
	Morphine	7-OHMG	MGM-15	MGM-16
	mg/kg			
Tail-flick	63.0 (37.2–106.8)	4.43 (1.57–6.93)	1.26 (0.84–1.88)	0.263 (0.165–0.420)
IGIT	11.7 (5.6–24.6)	7.50 (3.95–14.2)	2.20 (0.18–0.70)	0.451 (0.230–0.882)

for the specific structure necessary for its pharmacophore to bind to opioid receptors (Matsumoto et al., 2005a). It is speculated that the fluorine group at the C10 position on the benzene residue strengthens pharmacophore binding to opioid receptors because the lone fluorine pair easily forms hydrogen bonds with the receptor molecule (Takayama et al., 2006). Based on receptor-binding studies and tail-flick tests using selective antagonists, MGM-15 and MGM-16 showed similar opioid receptor type selectivity. Taken together, the findings show that fluorination at the C10 position increased the affinity to μ- and δ-opioid receptors but did not change the selectivity to the opioid receptor subtypes.

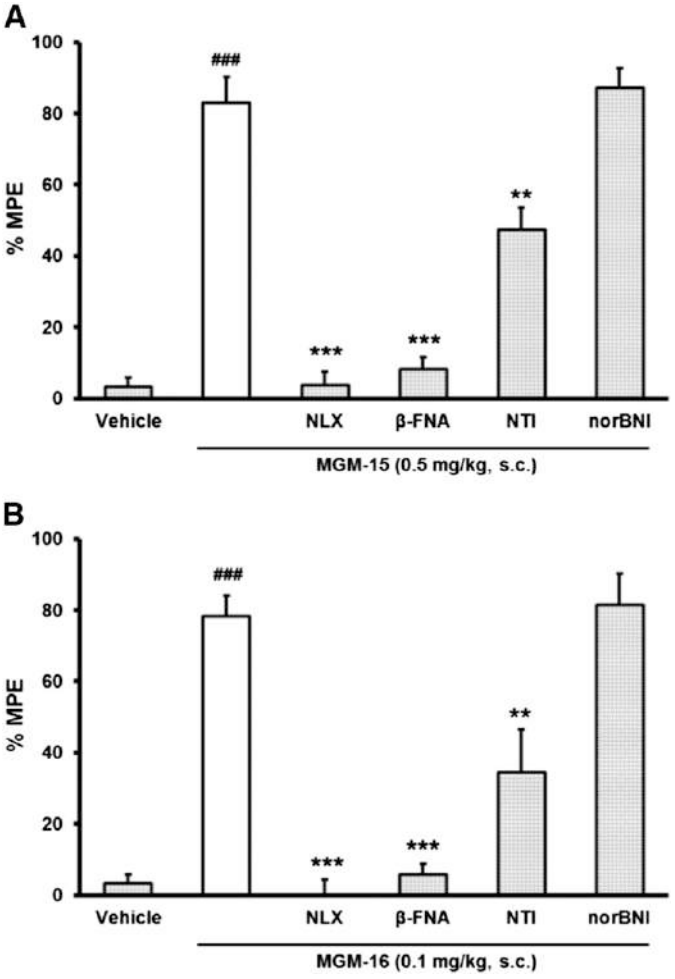


Fig. 5. Effects of opioid receptor antagonists on antinociception by subcutaneous administration of MGM-15 (A) and MGM-16 (B). The antinociceptive effects of MGM-15 and MGM-16 were determined in mice by the tail-flick test after subcutaneous administration of the following antagonists: β-FNA (40 mg/kg), naloxonazine (NLZ; 35 mg/kg), NTI (3 mg/kg), and norBNI (20 mg/kg). Measurements were performed 15 minutes after subcutaneous administration of MGM-15 (0.5 mg/kg) or MGM-16 (0.1 mg/kg) in the tail-flick test. Each value represents mean ± S.E.M. of eight or nine mice. $^{###}P < 0.001$ vs. vehicle-treated mice by Student's t test. $^{**}P < 0.01$; $^{***}P < 0.001$ vs. MGM-15 or MGM-16 by Bonferroni correction.

Subcutaneously and orally administered MGM-16 showed dose-dependent and antiallodynic effects in sciatic nerve–ligated mice. The antinociceptive effect of MGM-16 was completely and partially blocked by the μ-selective antagonist β-FNA and by the δ-selective antagonist naltrindole, respectively, in an acute pain model. The antiallodynic effect of MGM-16 was completely blocked by β-FNA and naltrindole in a chronic pain model. The contribution of δ-opioid receptors to the antiallodynic effect of MGM-16 in the neuropathic pain model was higher than in the acute pain test. As previously reported, upregulation of δ-opioid receptors is induced in a rat model of peripheral nerve injury to the dorsal root ganglion (Kabli and Cahill, 2007). δ-Opioid receptor knockout mice showed an increase in mechanical allodynia under neuropathic pain induced by partial sciatic nerve ligation (Nadal et al., 2006; Gaveriaux-Ruff et al., 2011). However, δ-opioid receptor knockout mice showed normal pain responses to acute pain. Furthermore, δ-opioid selective agonists induce clear antiallodynic effects in various animal models of neuropathic pain but do not show clear antinociceptive effects on acute nociception (Gaveriaux-Ruff et al., 2011). Thus, δ-opioid receptor activation leads to controlled mechanical allodynia. This indicates that the δ-opioid receptor agonistic properties of MGM-16 are involved in its antiallodynic effects in sciatic nerve–ligated mice.

Several studies showed the effectiveness of μ-opioid receptor agonists and demonstrated that μ-opioid receptors contribute to the control of mechanical allodynia in neuropathic pain (Mansikka et al., 2004; Finnerup et al., 2010). Indeed, μ-opioid receptor knockout mice showed an increase in mechanical allodynia under neuropathic pain induced by partial sciatic nerve ligation (Nadal et al., 2006; Gaveriaux-Ruff et al., 2011). Furthermore, μ-opioid receptor agonists attenuate mechanical allodynia in sciatic nerve–ligated mice (Mansikka et al., 2004). This suggests that the μ-opioid receptor contributes to alleviating mechanical allodynia. In this study, we found that the antiallodynic effect of MGM-16 was significantly blocked by a μ-opioid receptor antagonist. Therefore, the μ-opioid receptor agonistic activities of MGM-16 also contribute to its antiallodynic effect in partial sciatic nerve–ligated mice.

The traditional Thai herbal medicine *M. speciosa* has long been used in Thailand for its opioid-like effects (Burkill, 1935) and as a replacement for opium (Suwanlert, 1975). We have studied the pharmacological activities of mitragynine, a major

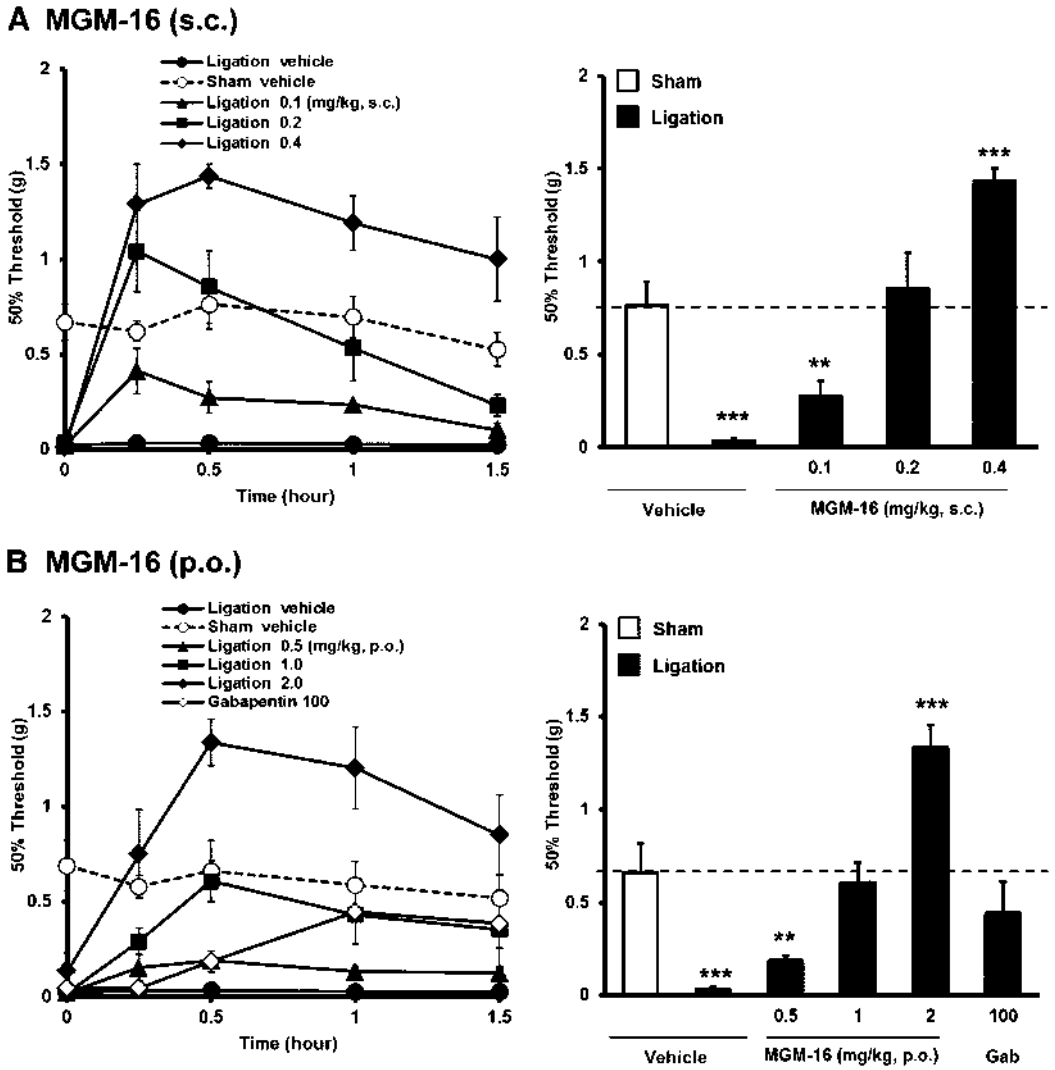


Fig. 6. Effects of MGM-16 on mechanical allodynia in partial sciatic nerve–ligated mice. MGM-16 was administered subcutaneously (A) and orally (B), and the paw withdrawal thresholds on the ipsilateral side were determined repeatedly. Gabapentin (Gab; 100 mg/kg) was administered orally as a positive control. Several doses of MGM-16 and 100 mg/kg gabapentin are presented at 0.5 and 1 hour after administration, respectively. Each value represents mean ± S.E.M. of six or seven mice. $^{**}P < 0.01$; $^{***}P < 0.001$ vs. vehicle by Bonferroni correction.

alkaloid of this herb (Watanabe et al., 1997; Matsumoto et al., 2005b), and 7-hydroxymitragynine, a main active alkaloid of this herb (Matsumoto et al., 2004, 2005a, 2006a). Recently, additional research has been performed on the opium-like effects of *M. speciosa* and mitragynine and the unique chemical and pharmacological properties of mitragynine-related indole alkaloids (Adkins et al., 2011; Raffa et al., 2013). We synthesized indole alkaloids such as the partial opioid agonist 9-hydroxycorynantheidine, the opioid antagonist corynantheidine, and the μ/κ-opioid dual agonist MGM-9. These compounds showed less rewarding effects, based on the structure-activity relationship study (Takayama, 2004; Matsumoto et al., 2006b, 2008). In this study, we synthesized a novel μ/δ-opioid dual agonist, MGM-16. MGM-16 showed opioid agonistic effects as potent as those of DAMGO and DPDPE in a functional assay using isolated guinea pig ileum and mouse vas deferens, respectively. Previously, there were no reports on G-protein action following the binding of mitragynine-related indole alkaloids to opioid receptors (Raffa et al., 2013). In this study, we show that MGM-16 dose-dependently increases GTPγS binding to CHO-K1 cells expressing recombinant μ- and δ-opioid receptors. We investigated the antinociceptive effects of MGM-16 ingested orally, based on the traditional usage of *M. speciosa* and the relevance of this route for clinical administration. Following oral administration, the antinociceptive effect of MGM-16 was about 240 times more potent than that of morphine in the mouse tail-flick test. Gabapentin, a synthetic analog of γ-aminobutyric acid, was initially developed as an anticonvulsant and later licensed for the management of postherpetic neuralgia (Gilron, 2007). In our results, gabapentin is effective for mechanical allodynia but very high doses (100 mg/kg p.o.) were needed to alleviate it, as reported previously (Kiso et al., 2008). Under this experimental condition, the antiallodynic effect of MGM-16 is approximately 100 times more potent than that of gabapentin in partial sciatic nerve–ligated mice. MGM-16 exerts a potential ability to treat chronic pain more effectively than existing drugs.

In conclusion, the present study demonstrated that MGM-16 derived from *Mitragyna* alkaloid 7-hydroxymitragynine

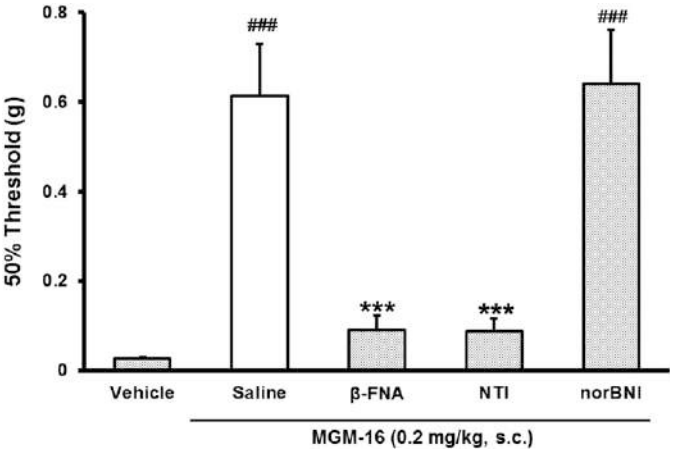


Fig. 7. Effects of opioid receptor antagonists on the antiallodynic effect of subcutaneous administration of MGM-16 in partial sciatic nerve–ligated mice. The effects of MGM-16 were determined after subcutaneous administration of the following antagonists: β-FNA (40 mg/kg), NTI (3 mg/kg), and norBNI (20 mg/kg). Measurements were performed 30 minutes after subcutaneous administration of MGM-16 (0.2 mg/kg) in partial sciatic nerve–ligated mice. Each value represents mean ± S.E.M. of six mice. $^{###}P < 0.001$ vs. vehicle by Student's t test. $^{***}P < 0.001$ vs. saline and MGM-16 by Bonferroni correction.

has potent μ/δ-opioid dual agonistic effects in vitro and in vivo. Because of its dual mechanism, MGM-16 has efficacy in a broad spectrum of acute and chronic pain models. This is the first study to report the antiallodynic effects of the alkaloids derived from mitragynine. MGM-16 may become a new class of seed compound with potential therapeutic utility for treating neuropathic pain.

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Authorship Contributions

Participated in research design: Matsumoto, Narita, Horie, Suzuki, Takayama.

Conducted experiments: Matsumoto, Muramatsu, Nakayama.

Contributed new reagents or analytic tools: Takayama, Kitajima, Misawa, Nakayama.

Performed data analysis: Matsumoto.

Wrote or contributed to the writing of the manuscript: Matsumoto, Horie, Devi, Tashima.

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The Chemistry of Kratom [*Mitragyna speciosa*]: Updated Characterization Data and Methods to Elucidate Indole and Oxindole Alkaloids

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Abstract

Two separate commercial products of kratom [*Mitragyna speciosa* (Korth.) Havil. Rubiaceae] were used to generate reference standards of its indole and oxindole alkaloids. While kratom has been studied for over a century, the characterization data in the literature for many of the alkaloids are either incomplete or inconsistent with modern standards. As such, full ^{1}H and ^{13}C NMR spectra, along with HRESIMS and ECD data, are reported for alkaloids **1**-**19**. Of these, four new alkaloids (**7**, **11**, **17**, and **18**) were characterized using 2D NMR data, and the absolute configurations of **7**, **17**, and **18** were established by comparison of experimental and calculated ECD spectra. The absolute configuration for the *N*(4)-oxide (**11**) was established by comparison of NMR and ECD spectra of its reduced product with those for compound **7**. In total, 19 alkaloids were characterized, including: the indole alkaloid mitragynine (**1**) and its diastereoisomers speciociliatine (**2**), speciogynine (**3**) and mitraciliatine (**4**); the indole alkaloid paynantheine (**5**) and its diastereoisomers isopaynantheine (**6**) and epiallo-isopaynantheine (**7**); the *N*(4)-oxides mitragynine-*N*(4)-oxide (**8**), speciociliatine-*N*(4)-oxide (**9**), isopaynantheine-*N*(4)-oxide (**10**), and epiallo-isopaynantheine-*N*(4)-oxide (**11**); the 9-hydroxylated oxindole alkaloids speciofoline (**12**), isorotundifoleine (**13**) and isospeciofoleine (**14**); and the 9-unsubstituted oxindoles corynoxine A (**15**), corynoxine B (**16**), 3-epirhynchophylline (**17**), 3-epicorynoxine B (**18**), and corynoxeine (**19**). With the ability to analyze the spectroscopic data of all of these compounds concomitantly, a decision tree was developed to differentiate these kratom alkaloids based on a few key chemical shifts in the ^{1}H and/or ^{13}C NMR spectra.

Graphical Abstract

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Supporting Information

The Supporting Information is available free of charge.

^{1}H NMR, ^{13}C NMR and HRESIMS data for compounds **1**-**19**. 2D NMR data (COSY, HSQC and HMBC) for new compounds **7**, **11**, **17** and **18**. ECD data for compounds **1**-**14** and **16**-**19**.

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The scientific literature on the chemical composition of kratom [*Mitragyna speciosa* (Korth.) Havil. Rubiaceae] dates back nearly 100 years, when the most studied alkaloid from this plant, mitragynine (**1**), was reported in 1921.[1] The introduction to that paper concludes with the following prescient statements: "*According to Redley……Mitragyne [sic] speciosa is used in Perak against the opium habit, whilst, according to Dr. P.P. Laidlaw, mitragynine is a local anaesthetic, which finding is of interest, since the alkaloid contains an ester grouping*." At that time, there was uncertainty on the use of kratom. Interestingly, a century later, this uncertainty continues, including in the scientific literature, the popular press, and even government regulatory bodies. Many questions persist such as: *does kratom ameliorate pain? can it be used as an opioid substitute? does it relieve opioid withdrawal symptoms? does its use precipitate trips to the emergency room? is kratom harmful, harmless, or somewhere in between?* Addressing these questions requires a thorough understanding of the chemical composition of kratom.

In the past decade there has been an explosion in the biomedical literature regarding at least some of the above questions, particularly case reports (Figure S1). There are reports that speak to its ability to address pain, broadly defined.[2–5] However, there are a similar number that suggest that it has opioid-like properties, and thus, may be an addiction liability.[6–15] Towards that end, the US Food and Drug Administration has threatened to classify kratom as a schedule I controlled substance more than once, although at the writing of this paper, it has not done so.[16–19]

As part of a project to study the potential for interactions between herbal medicines and drugs,[20–24] our team was charged with generating reference standards of the kratom alkaloids. In doing so, we found many challenges and opportunities, largely due to gaps in the literature. Extensive chemical studies have been performed over the last century, resulting in the isolation of at least 54 alkaloids from the plant (see Figure S2), starting with the isolation of the main alkaloid, mitragynine (**1**), in 1921 by Ellen Field.[1] More than four decades later, and over a series of three manuscripts, Becket *et al.* and Zacharias *et al.* worked on the isolation, structure elucidation, and absolute configuration of **1**.[25–27] Contemporaneously, the mitragynine diastereoisomers speciociliatine (**2**) and speciogynine (**3**) were reported.[28] A dozen years later, the fourth diastereoisomer, mitraciliatine (**4**), was

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described from the leaves of *M. speciosa*.[29] Like many alkaloids, a majority of isolation studies on new compounds from kratom were carried out between the 1960s and 1980s.[30–32] However, due to the growing popularity of this plant near the end of the last century, there was a renewed interest, resulting in the isolation of additional alkaloids (particularly minor constituents, which could be isolated due to advances in HPLC).[33, 34] Despite this, most of the pharmacological studies on constituents from kratom have focused on mitragynine (**1**) and 7-hydroxymitragynine (see Figure S1). In general, the tetracyclic indole and pentacyclic oxindole alkaloids isolated from *M. speciosa* embody a few variations, that, collectively, show the interrelatedness of these compounds (see Figure S2). Some key motifs of the 54 known compounds from *M. speciosa* include a hydroxy or methoxy group at C-9, unsaturation at C-3, C-5 or C-18, hydroxylation at C-7, and various configurations at C-3, C-7 and/or C-20 (see Figure S2).[30–32]

It was necessary to isolate standards for the alkaloids present in the plant, because even though there are over 50 kratom alkaloids reported, only 13 are commercially available (Figure S2). Moreover, in some cases we found that the quality of the commercial standards was not optimal, being mixtures, or were not even the correct compound.[35, 36] From a pharmacological testing point of view, this is a serious problem, especially for scientists that don't have the training or collaborations to verify structural identity, as it could perpetuate erroneous findings in the literature. For these reasons, we strove to develop a series of reference standards, being able to isolate and characterize 19 kratom alkaloids (**1**-**19**), including four new analogues (**7**, **11**, **17**, and **18**). In doing so, we quickly realized that the reported spectroscopy data for some of the known compounds were limited, as is the case for many natural products.[37–41] Thus, we took the opportunity to contribute by analyzing the NMR, HRESIMS, and ECD data for **1**-**19**, comparing the key differences between analogues.

RESULTS AND DISCUSSION

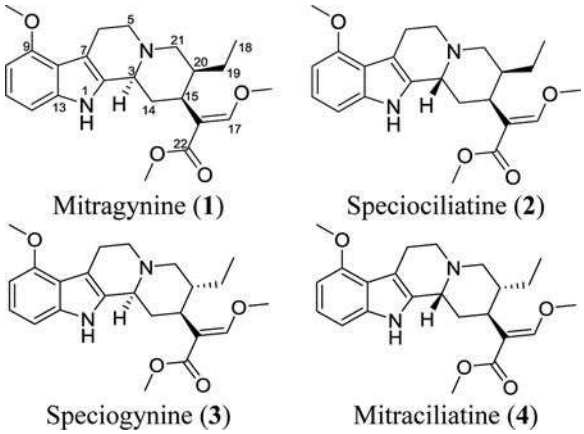
While there are some challenges associated with acquiring kratom plant material,[42] two different sources could be obtained on the kg scale, which were termed 'Green Maeng Da' and 'White Jongkong' by the suppliers. These samples were slightly different in appearance, with the former being ground leaf fragments (internally coded as K49) and the latter being a fine powder (internally coded as K52) (see Figure S6). The taxonomy of these was examined using a combination of molecular methods, specifically DNA barcoding and maximum likelihood phylogenetic analysis. For the purpose of molecular identification, the most widely recommended loci were analyzed, including the plastid intergenic spacer *trnH-psbA*, *matK*, and the nuclear ribosomal internal transcribed spacer (ITS), since these three regions have been designated as the official DNA barcodes for the plant kingdom.[43, 44] Of these, *trnH-psbA* is a widely used plastid marker, which displays a high rate of insertion and deletion and, as such, is widely used for plant barcoding studies.[45–47] All three regions indicated that samples K49 and K52 could be unequivocally identified as *Mitragyna speciosa*. Maximum likelihood analysis of the partial *matK* region placed both commercial samples in a strongly supported clade (100% bootstrap support) with the published sequences for *M. speciosa* (Figure S3). Importantly, this analysis included a partial *matK* sequence from a recently sequenced genome of *M. speciosa* (BioProject: PRJNA325670). In

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addition, a BLAST search of *matK* using the BOLDSYSTEMS database (version 4) showed that samples K49 and K52 had ≥ 99% sequence similarity with *M. speciosa* (Figures S4 and S5). As a confirmatory measure, maximum likelihood analysis using the ITS region also indicated that both samples formed a strongly supported clade (≥ 99% bootstrap support) with published sequence data of *M. speciosa* (Figure S6). Based on uncorrected p distances calculated for *trnH-psbA*, there was ≥ 99% sequence similarity with sequences of *M. speciosa*. This means that samples K49 and K52 were more similar to sequences of *M. speciosa* and more dissimilar to sequences of *M. diversifolia, M. hirsuta*, and *M. rotundifolia* (Tables S1 and S2). In short, three different methods all indicated that these materials were derived from *M. speciosa*. The sequence data were deposited in GenBank: *matK:* MT114409 and MT114408, *trnH-psbA*: MT114410 and MT114411, and ITS: MT111840, MT111841, MT111842, and MT111843.

Even though the sources of kratom (i.e. K49 and K52) were demonstrated to be *M. speciosa* based on DNA barcoding, chromatographic profiles generated via UPLC-HRESIMS (Figure S7) showed clear differences in the profile of secondary metabolites. For instance, mitragynine (**1**) was the most abundant alkaloid in the K49 materials, while the K52 materials contained higher levels of speciofoline (**12**) and related alkaloids. While those differences may be somewhat surprising, varied chemical composition for different sources of *M. speciosa* has been documented, where plant age, environmental factors, and processing conditions impact on the suite of kratom alkaloids.[48, 49] For example, in one published study the content of **1** derived from a Thai specimen was approximately five times higher than that derived from a Malaysian specimen.[50] Additionally, alkaloid differences between *M. speciosa* grown in Asia/Africa vs the United States have been noted.[49, 51] Thus, for the purpose of generating structurally diverse kratom alkaloid reference standards, we selected sample K49 for the isolation of indole-type alkaloids and sample K52 for the isolation of oxindole-type alkaloids using an alkaloid partitioning scheme. The indole alkaloids mitragynine (**1**), speciociliatine (**2**), speciogynine (**3**), mitraciliatine (**4**), paynantheine (**5**),[52] isopaynantheine (**6**)[29], epiallo-isopaynantheine (**7**), mitragynine-*N*(4)-oxide (**8**),[53] speciociliatine-*N*(4)-oxide (**9**),[53] isopaynantheine-*N*(4)-oxide (**10**),[54] and epiallo-isopaynantheine-*N*(4)-oxide **11** were isolated from the Green Maeng Da product (K49; see Figure S8), while the oxindole alkaloids speciofoline (**12**),[55] isorotundifoleine (**13**),[56] isospeciofoleine (**14**),[57] corynoxine A (**15**),[58, 59] corynoxine B (**16**),[59] 3-epirhynchophylline (**17**), 3-epicorynoxine B (**18**), and corynoxeine (**19**) were isolated from the White Jongkong product (K52; see Figure S9). Details for the structure elucidation for these compounds, including the identification of four new kratom alkaloids (i.e. **7**, **11**, **17**, and **18**) are presented below, using a suite of spectroscopic and spectrometric techniques (HRESIMS), including ECD. In doing so, a decision tree was developed that can be used to differentiate between these kratom alkaloids based on a few key signals in the ^{1}H and/or ^{13}C NMR spectra (Figure 1). This flow diagram was designed to be helpful for those with only a limited understanding of structure elucidation, focusing on key signals that serve to differentiate between the various kratom alkaloid chemotypes.

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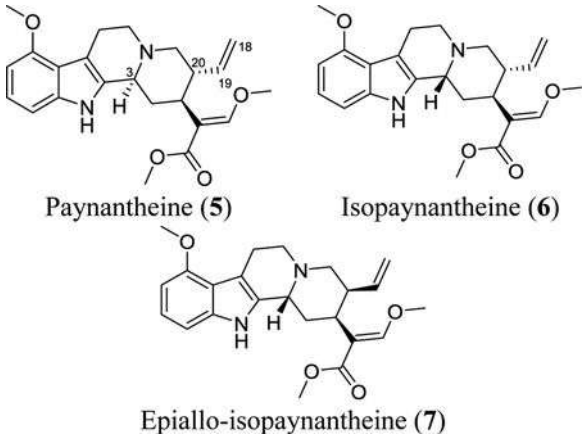


Indole Alkaloids: Mitragynine (1) and Its Diastereoisomers (2–4).

Four diastereoisomers with a molecular formula of $C_{23}H_{30}N_2O_4$ were isolated, specifically mitragynine (**1**), speciociliatine (**2**), speciogynine (**3**), and mitraciliatine (**4**), all of which were verified via HRESIMS data, where the protonated molecule displayed *m/z* 399.2274, 399.2273, 399.2274, and 399.2275 ions, respectively (Figures S10–S13). While the absolute amounts likely vary in different kratom plant materials, the quantity of compound **1** isolated was at least 20 times greater than that for each of the other diastereoisomers (i.e. >450 mg of **1** vs ~20 mg of **2**, **3**, or **4**). The ^{1}H and ^{13}C NMR data matched with those reported previously,[60] and key characteristics included: 1) eight sp^2 signals in the ^{13}C NMR spectrum assignable to the indole moiety (C-2 and C-7 to C-13); 2) two doublets ($\delta_{H\text{-}10}$ 6.44–6.49 and $\delta_{H\text{-}12}$ 6.87–7.01) and one triplet ($\delta_{H\text{-}11}$ 6.99–7.07), all corresponding to the indole moiety; 3) signals for the vinylic proton at C-17 (δ_C/δ_H 160.2–160.7/7.32–7.44); 4) three methoxy groups attached to C-9, C-17, and C-22 (δ_C/δ_H 51.1–61.9/3.66–3.89); and 5) the methyl group at C-19 (δ_C/δ_H 11.1–13.0/0.74–0.89). Importantly, a thorough analysis of the differences between the characteristic signals permitted the unequivocal identification of compounds **1**-**4**. For example, as Lee and colleagues described,[61] when a deshielded proton singlet assignable to H-17 appeared at δ_H 7.43 or 7.44 (i.e. compounds **1** and **2**, respectively), it was indicative of the α orientation of H-20. Alternatively, a more shielded chemical shift for H-17 (i.e. δ_H 7.32 and 7.35 for compounds **4** and **3**, respectively), indicated the β orientation for H-20.[61] Additionally, the orientation of H-20 could be confirmed orthogonally via the chemical shift of C-18, at δ_C of 12.5 or 13.0 (i.e. compounds **2** and **1**, respectively) for the α orientation and δ_C of 11.1 (i.e. for both **3** and **4**) for the β orientation. On the other hand, the presence of a broad singlet at δ_H 4.40 or 4.80 (i.e. compounds **2** and **4**) agreed with the β orientation for H-3, but a signal at δ_H 3.20 or 3.21 (i.e. compounds **1** and **3**) indicated the opposite orientation for the same proton (i.e., H-3α). Indeed, the $\Delta\delta_{H\text{-}3}$ values of the α and β orientations depend on the dihedral angle and the distance between H-3 and the nitrogen lone pair. An anti-coplanar arrangement, as seen in the axial H-3 of the *trans* fused system in compounds **1** or **3** (i.e., α orientation), leads to shielding relative to the syn-coplanar arrangement of the equatorial proton (i.e., β orientation) in compounds **2** or **4**, which have a dihedral angle of ~60° with the nitrogen lone pair. The donation of electron density into the antibonding orbital (i.e., n → σ*) of the axial C–H bond (i.e., α orientation) effectively results in the weakening of that bond and strong

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shielding of the proton (Figure S18).[62, 63] Key NMR signals for **1**-**4** are outlined (Table 1), and in addition, full 1D NMR spectra and Tables are reported in the Supporting Information (Figures S14–S17 and Table S4). Moreover, ECD spectra presented an additional approach for confirming the absolute configuration for these corynanthe-type alkaloids. It has been established that the Cotton effect between 250–300 nm could be used to establish the orientation of H-3, where a negative Cotton effect correlates with the β orientation (i.e., compounds **2** and **4**) and a positive Cotton effect for the α orientation for H-3 (i.e., compounds **1** and **3**; see Figure S19).[61]



Indole Alkaloids: Paynantheine and Its Diastereoisomers (Compounds 5–7).

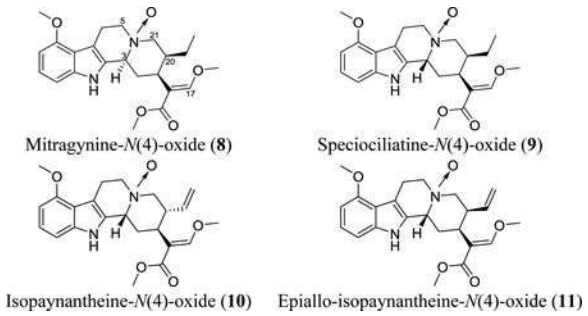
The structures of paynantheine (**5**) and isopaynantheine (**6**) were established by comparisons with literature data.[57] Their molecular formulae were confirmed by HRESIMS as $C_{23}H_{28}N_2O_4$, where the protonated molecule showed an *m/z* of 397.2119 and 397.2117, respectively (Figures S20 and S21). The NMR data of **5** and **6** resembled those for compounds **1**-**4**, sharing the previously enumerated characteristic signals for the indole ring. The main difference was the absence of signals for the C-20 ethyl moiety, and its replacement for a vinylic moiety, as evidenced by two doublets of doublets (δ_H 4.89–5.01) and a doublet of doublet of doublets or a doublet of triplets (δ_H 5.25–5.57), consistent with a Δ18(19) double bond ($\delta_{C\text{-}18}$ 115.7–116.7 and $\delta_{C\text{-}19}$ 137.9–139.4). As with **1**-**4**, the orientation of H-3 was deduced from chemical shift data, where the H-3α orientation was noted for **5** (δ_H 3.20), while the more deshielded shift in **6** (δ_H 4.73) indicated the H-3β orientation. These assignments were supported by the ECD spectra, where **5** and **6** had opposite Cotton effects at 250–300 nm. Specifically, a negative Cotton effect confirmed the β orientation (i.e. **6**) and a positive Cotton effect confirmed the α orientation (i.e. **5**) for H-3 (see Figure S32). Comparisons of the key NMR signals for compounds **5** and **6** are shown in Table 2, and the full 1D NMR spectroscopic data are reported in the Supporting Information (Figures S23–S24 and Table S5).

Compound **7** was isolated as an optically active yellow powder ($[\alpha]_D^{20}$ = +41.4, *c* 0.1 $CHCl_3$), with a molecular formula of $C_{23}H_{28}N_2O_4$, deduced by HRESIMS, where the protonated molecule showed an *m/z* of 397.2115 (Figure S22). The ^1H and ^{13}C NMR data of **7** resembled those of compounds **5** and **6**, sharing all the characteristic signals for an indole

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alkaloid with Δ18(19) unsaturation (see Tables 2 and S5, and Figures S23–S25). These observations, along with differences in chromatographic retention times (**5**: 7.5 min, **6**: 7.0 min, and **7**: 9.0 min; see Figure S34), suggested that **7** was a new diastereoisomer of **5** and **6**. The HSQC data, along with COSY and HMBC correlations, were used to elucidate the structure of **7** (Figures S26–S28). For example, three isolated spin systems (specifically, H-10/H-11/H-12, H_2-5/H_2-6, and H-3/H_2-14/H-15/H-20/H_2-21/H-19/H_2-18) and key HMBC correlations (i.e. H-3 to C-2/C-7, H-17 to C-15/C-22, H_2-18 to C-20, CH_3O-9 to C-9, CH_3O-17 to C-17, and CH_3O-22 to C-22) were used to confirm the skeleton for the proposed structure (Figure 2).

To establish the absolute configuration of **7**, several complementary lines of reasoning were used. For instance, the deshielded signal at 4.85 ppm suggested H-3β, since that was similar to the chemical shift for the same position in compounds **2** ($\delta_{H\text{-}3}$ 4.40), **4** ($\delta_{H\text{-}3}$ 4.80), and **6** ($\delta_{H\text{-}3}$ 4.73), all of which have been established to be H-3β. Moreover, biogenetic assumptions were used for the C-15 configuration, since all the indole alkaloids from *M. speciosa* (42 compounds, to date) have the H-15α orientation.[32] Those considerations narrowed the possibilities for compound **7** to either the pseudo (i.e., 3β, 15α, and 20β) or epiallo (i.e., 3β, 15α, and 20α) orientation.[61] The known compound, isopaynantheine (**6**), has the pseudo configuration, and thus, compound **7** is the 20-epimer of **6** (i.e. H-20α), yielding an overall configuration of epiallo; those considerations were used to suggest the trivial name epiallo-isopaynantheine (**7**). In addition, ECD spectroscopy was used as an orthogonal probe of the proposed configuration. The spectrum for the epiallo configuration (i.e. 3β, 15α, and 20α) was calculated using a time-dependent DFT (TDDFT) method at the B3LYP/6–31G+(d) level of theory, demonstrating concordance, including the aforementioned negative Cotton effect at ca. 275 nm that confirmed the H-3β orientation (Figure 3). The NOESY spectra of compounds **6** and **7** were compared as well (Figures S29–S31). However, these data were inconclusive, since clear differences could not be discerned. This could be explained based on atomic distances, which were measured for the most stable conformers; in all cases these distances were < 5 Å (Figure S31). Collectively, the amalgamation of NMR and ECD data, and biogenetic considerations were used to establish the (3*R*,15*S*,20*S*) absolute configuration of the new indole alkaloid, epiallo-isopaynantheine (**7**).



N(4)-oxides of Indole Alkaloids (8–11).

Compounds **8** and **9** had identical molecular formulae ($C_{23}H_{30}N_2O_5$), as determined by HRESIMS where the protonated molecules displayed an *m/z* of 415.2234 (**8**) and 415.2237

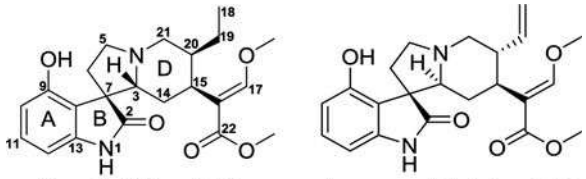
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(**9**) (see Figure S35 and S36). This indicated that **8** and **9** had one extra oxygen atom compared to the diastereoisomers **1**-**4**. ^{1}H and ^{13}C NMR spectra of both had strong similarities with the spectroscopic data reported for mitragynine-*N*(4)-oxide, sharing the characteristic NMR signals for the *N*(4)-oxide indole alkaloids ($\delta_{H\text{-}10}$ 6.43, $\delta_{H\text{-}11}$ 6.99–7.03, $\delta_{H\text{-}12}$ 6.92–6.95, $\delta_{C\text{-}17}/\delta_{H\text{-}17}$ 161.3–161.5/7.45–7.46, $\delta_{C\text{-}18}/\delta_{H\text{-}18}$ 11.8–12.7/0.85–0.93, $\delta_{C\text{-}21}/\delta_{H\text{-}21}$ 68.5–68.6/3.53–3.72, $\delta_{C\text{-}5}/\delta_{H\text{-}5}$ 65.8–65.9/3.51–3.98, and three methoxy groups δ_C/δ_H 51.6–62.1/3.61–3.86).[30] The main difference between the spectra for compounds **8** and **9**, relative to those discussed above for **1**-**4**, was deshielding of the signals attributed to H-3, H_2-5, and H_2-21, which was caused by a coordinate covalent bond in the *N*(4)-oxide group. There is an important distinction with the *N*-oxides, relative to the other kratom alkaloids, in that the $\Delta\delta_{H\text{-}3}$ value is small (~0.12 ppm) and does not correlate with the orientation of H-3. As noted in the discussion of **1**-**4**, the $\Delta\delta_{H\text{-}3}$ values of diastereoisomers mainly depend on the extent of interaction between the nitrogen free electron pair and the C–H antibonding orbitals. However, for the *N*-oxides, such an effect is not possible, since the electron pair is shared with the oxygen (i.e., $N^+{-}O^-$ group). For the *N*-oxides, the positive charge on the nitrogen has a strong electron-withdrawing inductive effect, resulting in decreased electron density regardless whether the α-protons are axial or equatorial. In both compounds, the deshielded signal for the vinylic proton at $\delta_{H\text{-}17}$ 7.45–7.46 indicated the α orientation for H-20, analogous to compounds **1** and **2**. In addition, H-15 was presumed to have the α orientation, based on the biogenetic considerations discussed for compound **7**. Therefore, **8** and **9** could have the allo (i.e., 3α, 15α, and 20α) or the epiallo (i.e., 3β, 15α, and 20α) configurations. This illustrates another advantage of analyzing the ECD spectra of kratom alkaloids, particularly the Cotton effect at ca. 275 nm. Compounds **8** and **9** had opposite effects at ca. 275 nm, where a positive Cotton effect correlated with the H-3α orientation (i.e. **8**) and a negative Cotton effect indicated the H-3β orientation (i.e. **9**). In turn, these data allowed us to assign these compounds as mitragynine-*N*(4)-oxide (**8**)[30] and speciociliatine-*N*(4)-oxide (**9**). This is the first report of NMR characterization data for **9**, and key NMR signals for **8** and **9** are collated in Table 3, with full 1D NMR data reported in the Supporting Information (see Figures S39 and S40 and Table S6).

A similar set of observations were used to assign the structures of compounds **10** and **11**, which were isolated as orange powders with a molecular formula of $C_{23}H_{28}N_2O_5$ as calculated by HRESIMS via a protonated molecule with an *m/z* of 413.2079 (**10**) and 413.2077 (**11**) (Figures S37 and S38). For instance, the NMR data for **10** and **11** were similar to those reported for isopaynantheine-*N*(4)-oxide (Table 3). These NMR data also resembled those of **8** and **9**, where the main difference was the lack of the C-20 ethyl group (**10** and **11**) that was replaced with a vinylic group (see Table 3). In both cases, the signals for the vinylic proton appeared at $\delta_{H\text{-}17}$ 7.30–7.31, which was similar to the shifts in compounds **6** and **7**. Again, ECD data were helpful, where the negative Cotton effects between at ca. 275 nm correlated with the H-3β orientation for both **10** and **11** (Figure S47). Additionally, a comparison of the ECD spectra of compounds **10** and **6** showed striking similarities; the same was true when comparing compounds **11** and **7** (see Figure S46). Therefore, compound **10** had the pseudo configuration (3β, 15α, and 20β; i.e., as seen in **6**) and **11** had the epiallo configuration (3β, 15α, and 20α; i.e., as seen in **7**). To confirm the absolute configuration of the new compound **11**, the method described by Shellard and

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colleagues[53] was used. Briefly, **11** was reduced by addition of H_2SO_4, and the reaction product should correspond to the indole alkaloid without the oxide. Indeed, the ^{1}H NMR spectrum for the reduced product of **11** matched the spectrum of **7** (Figure S46). Collectively, these data permitted the characterization of these compounds as isopaynantheine-*N*(4)-oxide (**10**) and the new kratom alkaloid, epiallo-isopaynantheine-*N*(4)-oxide (**11**). The spectroscopic data for both are reported in the Supporting Information (Figures S41–S45 and Table S6).



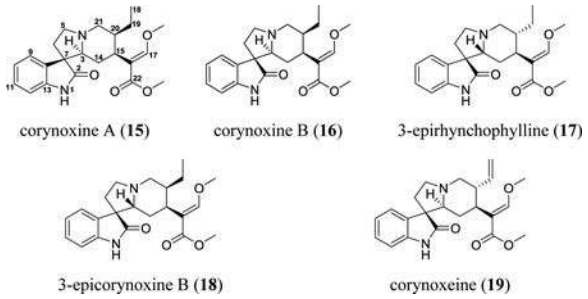
Oxindole Alkaloids: 9-Hydroxylated Oxindoles Like Speciofoline (12–14).

Compound **12** had a molecular formula of $C_{22}H_{28}N_2O_5$, based on the protonated HRESIMS molecule with an *m/z* of 401.2067 (Figure S48). Comparison of the NMR data with key NMR signals in the ^{1}H and ^{13}C spectra reported previously permitted the identification of this compound as speciofoline (**12**) (Table 4).[55, 56] This compound, which is one of the main alkaloids present in kratom (e.g. >55 mg isolated in this study), was first reported in 1963.[55] We have included the full spectroscopic data for **12** to serve as a reference point when analyzing the spectroscopic data for some of the more minor oxindole alkaloids. For instance, the C-ring (i.e., the pyrollidine ring) of **12** is connected to the γ-lactam (i.e., B-ring) through a C-7 spiro carbon, and this results in characteristic signals in the ^{13}C NMR spectrum at δ_C 180.2 and 57.3 for the carbonyl in the oxindole ring and the spiro C-7 carbon, respectively (Table 4); the lactam signal, in particular, serves to quickly differentiate between indole and oxindole alkaloids (see Figure 1). Other important NMR signals, which are similar to those discussed above for the indole alkaloids, include: 1) signals for the indole system ($\delta_{H\text{-}10}$ 6.35, $\delta_{H\text{-}11}$ 7.05 and $\delta_{H\text{-}12}$ 6.53); 2) a vinylic H-17 (δ_C/δ_H 159.8/7.38); 3) two methoxy groups at C-17 and C-22 (δ_C/δ_H 61.3/3.79 and δ_C/δ_H 51.7/3.65, respectively), and 4) a methyl group ($\delta_{C\text{-}18}/\delta_{H\text{-}18}$ 12.3/0.83). Key NMR signals for **12** are tabulated (Table 4), and the full 1D NMR spectroscopic data are reported in the Supporting Information (Figure S51 and Table S7).

Additionally, two other 9-hydroxylated oxindole alkaloids were isolated and characterized as isorotundifoleine (**13**) and isospeciofoleine (**14**). The molecular formula for both isomers was $C_{22}H_{26}N_2O_5$ as measured by HRESIMS based on protonated molecules with *m/z* values of 399.1909 and 399.1906, respectively (Figures S49 and S50). The structures were elucidated by comparing the NMR data with those reported previously (Table 4),[56, 57] and

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the full assignment of the ^{1}H and ^{13}C NMR spectroscopic data for both are reported in the Supporting Information (Figures S52 and S53, and Table S7); this is the first report of comprehensive NMR data for **13**. The NMR data for both **13** and **14** resembled those described above for speciofoline (**12**), with a major difference being the absence of the methyl group (i.e., C-18 in **12** at δ_C/δ_H 12.3/0.83), which was replaced by the Δ18(19) double bond (δ_C/δ_H 116.4/4.95–5.01 and δ_C/δ_H 138.6–138.8/5.43–5.49, respectively, for **13** and **14**). As with the indole alkaloids, ECD spectra were instrumental in assigning the configuration of these compounds, especially for analyzing the C-7 spiro center. According to the literature, the negative Cotton effect at 290 nm confirmed the (7*R*) configuration in compounds **12** and **13**, and the positive Cotton effect agreed with the (7*S*) configuration in compound **14**. Additionally, the negative Cotton effect around 250 nm confirmed the H-3β orientation in compounds **12** and **14**, whereas the opposite Cotton effect demonstrated the H-3α orientation in **13**. These results agreed with the rules for ECD spectra for the 9-hydroxylated oxindole alkaloids, and a calculated ECD spectrum was generated for **12** as a confirmatory measure (see Figure S54).[56, 64]



Oxindole Alkaloids: 9-Unsubstituted Oxindoles Such as Corynoxine A (15–19).

Two closely related isomers, **15** and **16**, were isolated with a molecular formula of $C_{22}H_{28}N_2O_4$, based on HRESIMS protonated molecules with *m/z* values of 385.2119 (i.e. **15**) and 385.2127 (i.e. **16**) (Figures S55 and S56). The compounds were characterized using ^{1}H and ^{13}C NMR data as corynoxine A (**15**) and corynoxine B (**16**), based on concordance with published data for synthetic standards of these compounds (Tables 5 and S8 and Figures S60 and S61).[65] In addition, the NMR data for **15** and **16** resembled those described for **12**, with the main difference being replacement of the C-9 phenolic group with an aromatic proton, which resulted in a pattern of two doublets and two triplets of doublets for the protons in the oxindole ring (H-9 to H-12, see Table 5). Both **15** and **16** had the allo configuration (i.e., 3α, 15α, and 20α), with opposite configurations at the spiro C-7, being (7*S*) for **15** and (7*R*) for **16**. As Trager and colleagues reported,[64] the configuration at C-7 can be deduced based on the chemical shift for the H-9 signal. For instance, when presuming the allo configuration, the presence of a deshielded signal for H-9 (δ_H 7.45) confirmed the (7*S*) configuration for **15** and the (7*R*) configuration in **16** due to the more shielded signal for H-9 (δ_H 7.19). While **15** was stable, an interesting challenge with compound **16** was its interconversion to **15** over a period of about 8 days in $CDCl_3$ (see Figure S62). As reported previously,[66] the epimerization process increases under acidic conditions via an intermolecular Mannich reaction, resulting in a ratio of ~70:30 of **15** and **16**, respectively (see Figure S62). As such, only the ECD spectrum of **15** is reported, supporting the (7*S*) and

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the allo configurations based on negative Cotton effects around 290 nm and 250 nm (Figure S76).

Compounds **17** and **18** were isolated as optically active powders ($[\alpha]_D^{23}$ −30 (*c* 0.1, $CHCl_3$) and $[\alpha]_D^{24}$ −46 (*c* 0.1, $CHCl_3$), respectively). Their molecular formulae were established as $C_{22}H_{28}N_2O_4$ based on the HRESIMS protonated molecule with *m/z* values of 385.2119 (i.e. **17**) and 385.2120 (i.e. **18**) (Figures S57 and S58). The ^1H and ^{13}C NMR spectra of these compounds were similar to those of **15** and **16** (Table 5). These data, along with their different retention times (**17**: 2.76 min and **18**: 2.87 min) (see Figures S57 and S58), suggested that **17** and **18** were new isomers of **15** and **16**. The HSQC data, along with COSY and HMBC correlations, were used to confirm the proposed structures. In both cases, three isolated spin systems (specifically H-9/H-10/H-11/H-12, H_2-5/H_2-6, and H-3/H_2-14/H-15/H-20/H_2-21/H_2-19/H_3-18) and key HMBC correlations (i.e. H-3 to C-2/C-7, H-17 to C-15/C-22, H_3-18 to C-20, CH_3O-17 to C-17, and CH_3O-22 to C-22) were used to confirm the 2D structures of **17** and **18** (see Figures 2 and S63–S65 and S69–S71). Similar to the discussion for **6** and **7**, the correlations observed in the NOESY spectra for **17** and **18** were not useful to determine the relative configuration for these compounds (Figures S67 and S73). The absolute configuration of these molecules was proposed based on ECD experiments, following the rules described for the 9-unsubstituted oxindole alkaloids. In particular, the related isomer rhyncophylline served as a reference, since the crystal structure for this compound has been published and the ECD data were well described.[64, 67] For instance, both **17** and **18** had the (7*R*) configuration, based on positive Cotton effects around 290 nm, and the H-3β orientation, as noted by positive Cotton effects around 250 nm (Figure S76). Using biogenetic reasoning for the H-15α orientation, along with the evidence observed in the ECD data, we deduced that compounds **17** and **18** should have the pseudo (i.e., 3β, 15α, and 20β) or epiallo (i.e., 3β, 15α, and 20α) configurations. To evaluate that presumption, the ECD spectra for the diastereoisomers at C-20 were calculated using a time-dependent DFT (TDDFT) method at the B3LYP/6–31G+(d) level of theory. The results showed that the calculated ECD spectrum for H-20β matched with the ECD spectrum for **17**, and similarly, the calculated ECD spectrum for an analogue with the H-20α orientation matched with the ECD spectrum for compound **18** (Figure 4). NMR data, biogenetic considerations, and the examination of ECD spectra were used to establish the absolute configuration of the new oxindole alkaloids, (3*R*,7*R*,15*S*,20*R*)-3-epirhynchophylline (**17**) and (3*R*,7*R*,15*S*,20*S*)-3-epicorynoxine B (**18**).

Compound (**19**) was isolated as a white powder, and its molecular formula was established as $C_{22}H_{26}N_2O_4$, based on the HRESIMS protonated molecule with an *m/z* of 383.1964 (Figure S59). The NMR data for this compound showed strong similarities with the data of **15**-**18**, where the main differences were the absence of the signals for the C-20 ethyl group, which was replaced by a vinylic group (Tables 5 and S8). The NMR data (Figure S75 and Table S8) matched with those reported for corynoxeine.[68] ECD data corroborated the H-3α orientation based on the negative Cotton effect around 250 nm, and the (7*R*) configuration based on the positive Cotton effect around 290 (Figure S76).

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To add complementary data to the structural characterization of the kratom alkaloids, an attempt was made to analyze compounds **1**, **6**, **7**, **15**, **17**, and **18** by VCD, so as to examine a range of both indole and oxindole alkaloids. However, in all cases the similarities between calculated and experimental data were low, ranging between 57.7 to 70.0% (Table S9). In contrast, prominent recent VCD results published in this *Journal* report similarities ranging between 74.1 to 98.5%.[69–73] In particular, we were surprised at the lack of concordance with the data for mitragynine (**1**), since it has been so well characterized, including by X-ray crystallography.[27] Thus, the purity of **1**, **6**, **7**, **15**, **17**, and **18**, were examined after the completion of the VCD studies, and indeed, it was obvious that the compounds decomposed during the experiments (see Figure S77). Based on the ^{1}H NMR spectra (data not shown), we hypothesize that epimers were formed, likely as a result of the VCD data being recorded at elevated temperatures in $CHCl_3$ for up to 18 hours; the epimerization potential of these compounds has been reported.[66] While disappointing, we determined that analysis of VCD spectra is not helpful for this class of compounds, at least with current technologies, due to their lack of stability under the analytical conditions.

In summary, 19 reference standards were isolated from kratom plant materials, including the two new indole alkaloids **7** and **11** and the two new oxindole alkaloids **17** and **18**. To characterize their structures, including absolute configuration, a combination of mass spectrometry and NMR and ECD spectroscopy were utilized. An added benefit of analyzing the NMR spectroscopy data for a suite of closely related compounds was that it facilitated the development of a decision tree (Figure 1). Using only a few key NMR signals, even a relative novice should be able to narrow the possible structures for at least these kratom alkaloids, and an expert is encouraged to examine a full suite of NMR data in relation to those reported in the Supporting Information. It is our hope that improvements in the structural characterization of the kratom alkaloids will result in more clarity in the biomedical literature on this interesting class of compounds.

EXPERIMENTAL SECTION

General Experimental Procedures.

Optical rotations were obtained in $CHCl_3$ using a Rudolph Research Autopol III polarimeter. UV and ECD spectra were acquired on a Varian Cary 100 Bio UV-Vis spectrophotometer and an Olis DSM 17 CD spectrophotometer (Olis), respectively. A Chiral*IR-2X* dual-source, dual-polarization-modulated FT-VCD spectrometer was used to carry out IR and VCD measurements. NMR experiments were conducted using either a JEOL ECA-500 NMR spectrometer operating at 500 MHz for ^{1}H and 125 MHz for ^{13}C or a JEOL ECS-400 NMR spectrometer equipped with a high sensitivity JEOL Royal probe operating at 400 MHz for ^{1}H and 100 MHz for ^{13}C; the residual solvent signals were utilized for referencing. HRESIMS was performed on a Thermo LTQ Orbitrap XL mass spectrometer equipped with an electrospray ionization source. UPLC was carried out on a Waters Acquity system with data collected and analyzed using Thermo Xcalibur software. HPLC was carried out using a Varian ProStar HPLC system equipped with ProStar 210 pumps and a ProStar 335 photodiode array detector (PDA), with data collected and analyzed using Galaxie Chromatography Workstation software (version 1.9.3.2). For preparative HPLC, a series of

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Phenomenex columns were used, including a Kinetex C_{18} (5 μm; 250 × 21.2 mm), a Luna CN (5 μm; 250 × 21.2 mm) and a Luna PFP (5 μm; 250 × 21.2 mm) all at a 20 mL/min flow rate. For UPLC, a Waters BEH C_{18} column (1.7 μm; 50 × 2.1 mm) was used with a 0.6 mL/min flow rate. Flash chromatography was performed on a Teledyne ISCO Combi*F*lash Rf 200 using various sizes of Silica Gold columns and monitored by UV and evaporative light-scattering detectors.

Plant Material, DNA Extraction, PCR and Sequencing for Molecular Identification.

The commercial kratom products used to prepare extracts [i.e. Green Maeng Da (K49, batch: M-1005, 4 kg) and White Jongkong (K52, batch: BN-1006, 1 kg)] were identified by DNA barcoding. The manufacturers reported these as both from Southeast Asia, with K49 being chipped leaf materials and K52 being a fine powder. Samples were handled using methods described previously for the DNA barcoding of commercial mushroom products.[74] Briefly, one scintillation vial was made from each product (see Figure S6), where the vial was filled with plant material from the top and middle layers of the commercial product, and these were labeled with product name, lot number, and batch number and stored at ambient temperature in a bench drawer. For DNA extractions, approximately 5 mg of plant powder was taken from the scintillation vial and transferred to a bashing bead tube with DNA lysis buffer provided by Zymo Research Quick-DNA plant/seed miniprep DNA extraction kit. DNA was extracted using procedures outlined in the Zymo Research plant DNA extraction kit. The plastid gene *matK* was PCR amplified with primer combination matK-xf and matK-MALP;[75, 76] for *psbA-trnH* primer combination *psbA – trnH*,[77] and for nuclear ITS, primer combination ITS-u1and ITS-u4[78] was used for PCR and sequencing. All PCRs were amplified on an Applied Biosystems Veriti thermal cycler using PuReTaq Ready-To-Go PCR Beads (GE Healthcare) with the above primers. The PCR reaction was carried out in 25 μL containing 5 μL template DNA, 2.5 μL BSA, 2.5 μL 50% DMSO, and 1 μL of each forward and reverse primer at a concentration of 10 μM. The rest of the volume was made up to 25 μL by adding molecular biology grade water from Fisher Scientific. PCR thermocycler parameters for all fragments are outlined (Table S3). The PCR products were then run on an ethidium bromide-stained 1% agarose gel (Fisher Scientific) along with a 1 kb DNA ladder (Promega) to estimate the size of the amplified band. PCR products were purified using a Wizard SV Gel and PCR Clean-up System. Sanger sequencing of the purified PCR products was performed at Eurofins Genomics (http://www.operon.com/default.aspx) using BigDye Terminator v3.1 cycle sequencing. The sequencing was accomplished bidirectionally using both strands with the same primers used for the PCR amplification. Sequences were generated on an Applied Biosystems 3730XL high-throughput capillary sequencer. For both sequencing reactions, approximately 15 μL of PCR template was used along with 2 μM sequencing primers. Sequences were assembled with Sequencher 5.3 (Gene Codes), optimized, and then corrected manually when necessary; the latter step was to ensure that the computer algorithm was assigning proper base calls. Each sequence fragment was subjected to an individual NCBI GenBank, Basic Local Alignment Search Tool (BLAST) search as well as the BOLDSYSTEMS database to verify identity.

BLAST Search and DNA Barcoding.—For identification of kratom plant material via DNA Barcoding, one plastid region (*matK*) was utilized for plant identification by BLAST

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searching against the BOLDSYSTEMS database version 4. In addition, the sequences of *matK* were compared against an authentic partial sequence of *matK* from the *Mitragyna speciosa* genome assembly (Center for Food Safety and Applied Nutrition (CFSAN), part of the USDA). Uncorrected p-distances were calculated in Geneious, a bioinformatics desktop software package (http://www.biomatters.com), for sequences obtained from the *psbA-trnH* plastid region, which is considered as one of the most variable regions of angiosperm plants, as well as for nuclear ITS region for the published sequence data from GenBank for *Mitragyna* spp.[79–81] including *M. speciosa* and the newly obtained sequences in this study. The p-distances were obtained by dividing the number of nucleotide differences by the total number of nucleotides being compared. An arbitrary cut off proxy of ≥ 98–100% was applied as a criterion to designate similar species; therefore, to be considered the same species based on *psbA-trnH* and ITS sequence comparison, the taxa being compared would have ≥ 98% sequence similarity.

Molecular Phylogenetic Analysis.—Maximum likelihood analysis was performed separately for both the nuclear ITS and the plastid *matK* region to place the kratom samples into a phylogenetic framework with published sequences of *M. speciosa*. Methods for maximum likelihood analysis have been outlined previously.[82] *Nauclea officinalis* (Rubiaceae) was used as an outgroup taxon in both nuclear ITS and *matK* analyses.

Extraction and Isolation of Kratom Alkaloids from Green Maeng Da.

The kratom plant material (Green Maeng Da, 4 kg; Figure S8) was extracted with 10 L of $CHCl_3$-CH_3OH (1:1) and 500 mL of 10% aqueous KOH by maceration over 24 h at room temperature. The mixture was filtered, and the solvent was evaporated under reduced pressure. The dried extract was reconstituted in a solution of 1M HCl and hexanes (1:1), transferred into a separatory funnel, and shaken vigorously. The hexanes phase was drawn off, the pH of the aqueous phase was adjusted to 9.0 with dropwise addition of concentrated NH_4OH, and the alkaloids were extracted as the free base with $CHCl_3$; after washing with neutral water, the organic phase was dried to yield 12 g of the alkaloid extract. This material was fractionated by normal phase flash chromatography using a silica column (120 g) and a gradient solvent system of hexanes-$CHCl_3$-CH_3OH at a flow rate of 85 mL/min over 67 min to yield 11 pooled fractions.

Fraction 4 (500 mg) was subjected to reverse phase HPLC using a CN column and a gradient system of 40:60 to 100:0 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min with a flow rate of 20 mL/min. This process yielded eight subfractions, and fraction 8 was identified as compound **1** (450.5 mg).

Fraction 9 (1.1 g) was fractionated by normal phase flash chromatography using a silica column (12 g) and a gradient system of hexanes-EtOAc-CH_3OH using a flow rate of 30 mL/min to generate four subfractions. Subfraction 2 (40 mg) was purified by preparative HPLC over a CN column using a gradient of 70:30 to 100:0 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min with a flow rate of 20 mL/min. This process yielded compounds **4** (13.1 mg) and **6** (15.20 mg). Subfraction 4 (800 mg) was subjected to flash chromatography using a silica column (12 g) via a gradient of $CHCl_3$-CH_3OH (10 mM of

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NH_4OAc in both phases) over 60 min with a flow rate of 30 mL/min to generate five fractions (F9–4_1 through F9–4_5). Fraction F9–4_3 was subjected to preparative HPLC over a Kinetex column using a gradient of 60:40 to 70:30 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 30 min with a flow rate of 20 mL/min; three fractions were collected and the first fraction was characterized as compound **2** (20.5 mg). Fraction F9–4_4 (300 mg) was subjected to flash chromatography using a silica column (4 g) and a gradient of hexanes-acetone-CH_3OH over 20 min at a flow rate of 18 mL/min to generate eight fractions. The second fraction was resolved by preparative HPLC over a CN column using a gradient of 50:50 to 100:0 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min with a flow rate of 20 mL/min; this process yielded 16.5 mg of compound **7**. The seventh fraction (16 mg) from F9–4_4 was subjected to preparative HPLC using a CN column and a gradient of 40:60 to 90:10 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min with a flow rate of 20 mL/min; two fractions were collected and the first fraction was characterized as compound **11** (2.3 mg). Fraction F9–4_5 (40 mg) was subjected to preparative HPLC over a Kinetex column with a gradient of 50:50 to 100:0 of CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 35 min with a flow rate of 20 mL/min; the third fraction was identified as **8** (10.5 mg).

Fraction 6 (680 mg) was subjected to a second fractionation using flash chromatography over a silica column (12 g) and a gradient of $CHCl_3$-CH_3OH (10 mM of NH_4OAc in both phases) over 60 min with a flow rate of 30 mL/min to generate five subfractions. Subfraction 1 (47 mg) was subjected to preparative HPLC over a CN column and a gradient system of 55:45 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 25 min; the first and second fractions were identified as **5** (9.83 mg) and **3** (18.33 mg), respectively.

Fraction 8 (1.3 g) was fractionated by flash chromatography using a silica column (12 g) with a gradient of $CHCl_3$-CH_3OH (10 mM of NH_4OAc in both phases) over 25 min with a flow rate of 30 mL/min to generate four subfractions. Subfractions 3 (12.8 mg) and 4 (7.0 mg) were purified by preparative HPLC over a CN column and a gradient system of 50:50 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 15 min at a flow rate of 20 mL/min to yield compounds **9** (5.7 mg) and **10** (4.5 mg), respectively.

Extraction and Isolation of Kratom Alkaloids from White Jongkong.

The plant material (White Jongkong, 1 kg; Figure S9) was extracted exhaustively using the same procedure described above, to obtain 1.0 g of the dried extract. Normal phase flash chromatography was used to separate the extract using a silica column (24 g) and a gradient system of hexanes-$CHCl_3$-CH_3OH over 52 min at a flow rate of 35 mL/min to generate 13 fractions. Fractions 3 (100.0 mg) and 4 (80.0 mg) were purified, separately, by preparative HPLC using a PFP column and a gradient of 70:30 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min at a flow rate of 20 mL/min to yield 38.2 mg of **15** (from fraction 3), and separately, 52.5 mg of **16** and 3.5 mg of **13** (from fraction 4).

Fraction 5 (80.5 mg) was subjected to preparative HPLC over a PFP column using a gradient of 70:30 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min at a flow rate of 20 mL/min to generate nine subfractions, and subfraction 4 was identified as compound **12** (55.0 mg).

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Fractions 6 (156.2 mg) and 7 (84.1 mg) were combined based upon their similar HPLC chromatographic profiles. The resulting fraction (240.3 mg) was resolved by flash chromatography over a silica column (12 g) using a gradient of hexanes-$CHCl_3$-CH_3OH over 25 min at a flow rate of 18 mL/min to generate eleven subfractions. Subfraction 2 (10.0 mg) was subjected to semipreparative HPLC over a PFP column using a gradient of 50:50 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min at a flow rate of 4.0 mL/min, yielding **17** (2.0 mg). Subfraction 4 (15.0 mg) was subjected to preparative HPLC over a PFP column using a gradient of 60:40 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min at a flow rate of 20.0 mL/min, yielding **14** (3.6 mg). Subfraction 5 (13.0 mg) was subjected to preparative HPLC over a PFP column using a gradient of 50:50 to 100:0 CH_3OH-H_2O (10 mM of NH_4OAc in both phases) over 20 min at a flow rate of 20.0 mL/min to generate four subfractions, where subfractions 2 and 3 were identified as **19** (1.7 mg) and **18** (1.1 mg), respectively.

Epiallo-isopaynantheine (7): Yellowish solid powder. $[\alpha]_D^{20}$ +41 (*c* 0.1, $CHCl_3$); UV (CH_3OH) λ_{max} (log ε) 233 (3.7), 339 (2.9) nm; ^1H and ^{13}C NMR see Table 2 and Table S5; HRESIMS *m/z* 397.2115 $[M + H]^+$ (calcd for $C_{23}H_{29}N_2O_4$, 397.2127).

Epiallo-isopaynantheine-N(4)-oxide (11): Orange powder. $[\alpha]_D^{23}$ +33 (*c* 0.1, $CHCl_3$); UV (CH_3OH) λ_{max} (log ε) 219 (3.2), 244 (3.9) nm; ^1H and ^{13}C NMR see Table 3 and Table S6; HRESIMS *m/z* 413.2077 $[M + H]^+$ (calcd for $C_{23}H_{29}N_2O_5$, 413.2076).

3-Epirhyncophylline (17): White powder. $[\alpha]_D^{23}$ −30 (*c* 0.1, $CHCl_3$); UV (CH_3OH) λ_{max} (log ε) 219 (3.3), 244 (3.6) nm; ^1H and ^{13}C NMR see Table 5 and Table S8; HRESIMS *m/z* 385.2119 $[M + H]^+$ (calcd for $C_{22}H_{29}N_2O_4$, 385.2127).

3-Epicorynoxine B (18): White powder. $[\alpha]_D^{24}$ −46 (*c* 0.1, $CHCl_3$); UV (CH_3OH) λ_{max} (log ε) 217 (3.4), 246 (3.7) nm; ^1H and ^{13}C NMR see Table 5 and Table S8; HRESIMS *m/z* 385.2120 $[M + H]^+$ (calcd for $C_{22}H_{29}N_2O_4$, 385.2127).

Computational Methods.

The minimum energy structures were built with Spartan'10 software (Wavefunction Inc., Irvine, CA, USA). The conformational analysis was performed using the Monte Carlo search protocol under the MMFF94 molecular mechanics force field.

For the ECD prediction, the resulting conformers were minimized using the DFT method at the B3LYP/6–311G+(2d,p) level of theory. Then, the time-dependent DFT (TDDFT) method at the B3LYP/6–31G+(d) level of theory was employed for ECD calculations. The calculated excitation energy (nm) and rotatory strength (R) in dipole velocity (Rvel) and dipole length (Rlen) forms were simulated into an ECD curve. All calculations were performed employing the Gaussian'09 program package (Gaussian Inc., Wallingford, CT, USA).

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Supplementary Material

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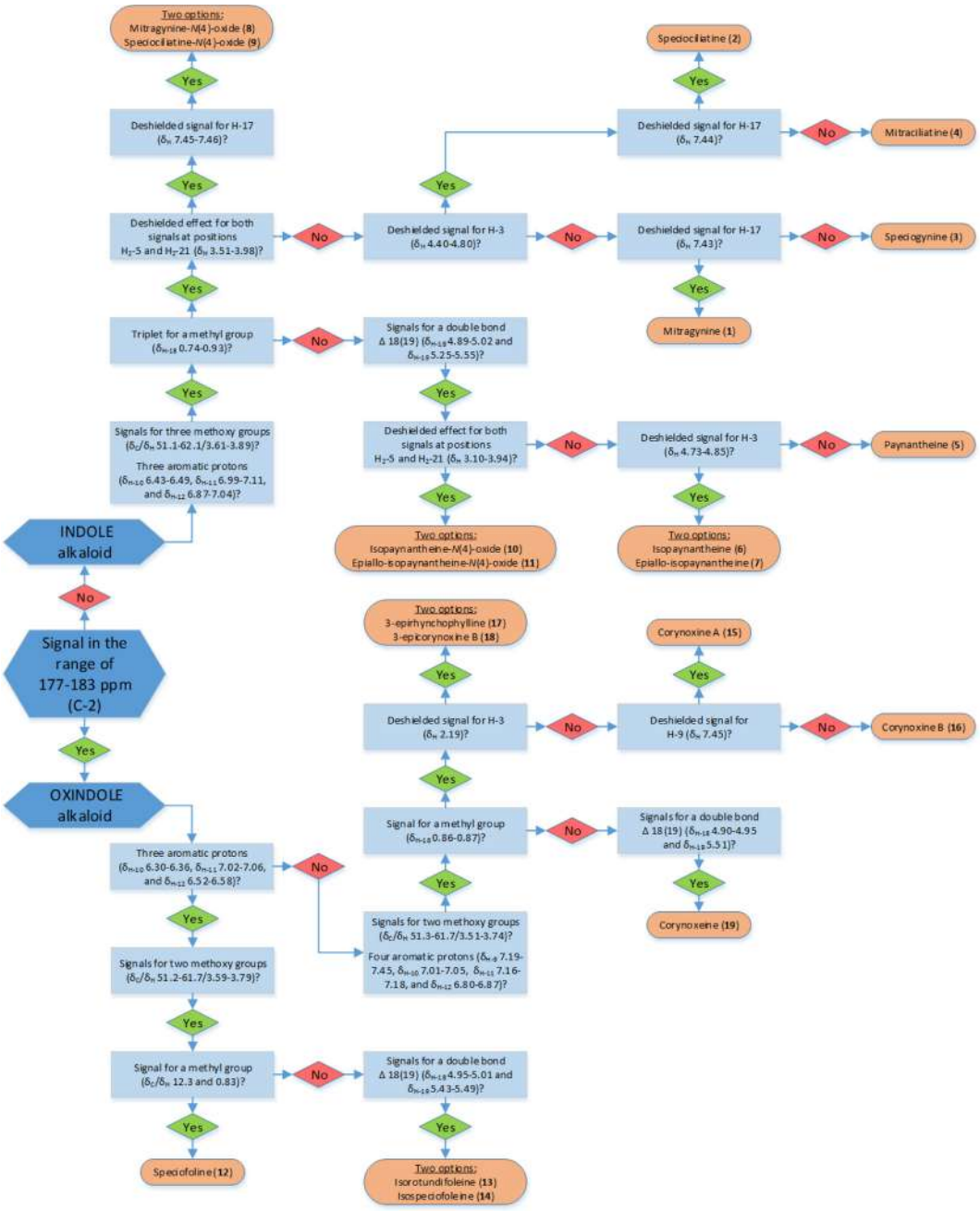


Figure 1.

Decision tree for differentiating between **1**-**19** based on key NMR signals. As a single starting point, the ^{13}C NMR signal for the lactam moiety is used to distinguish between oxindole (present) or indole (absent) kratom alkaloids. Once the broad structural class is determined, ^{1}H NMR data can be used via a flowchart as a guide to elucidate the structure of these alkaloids.

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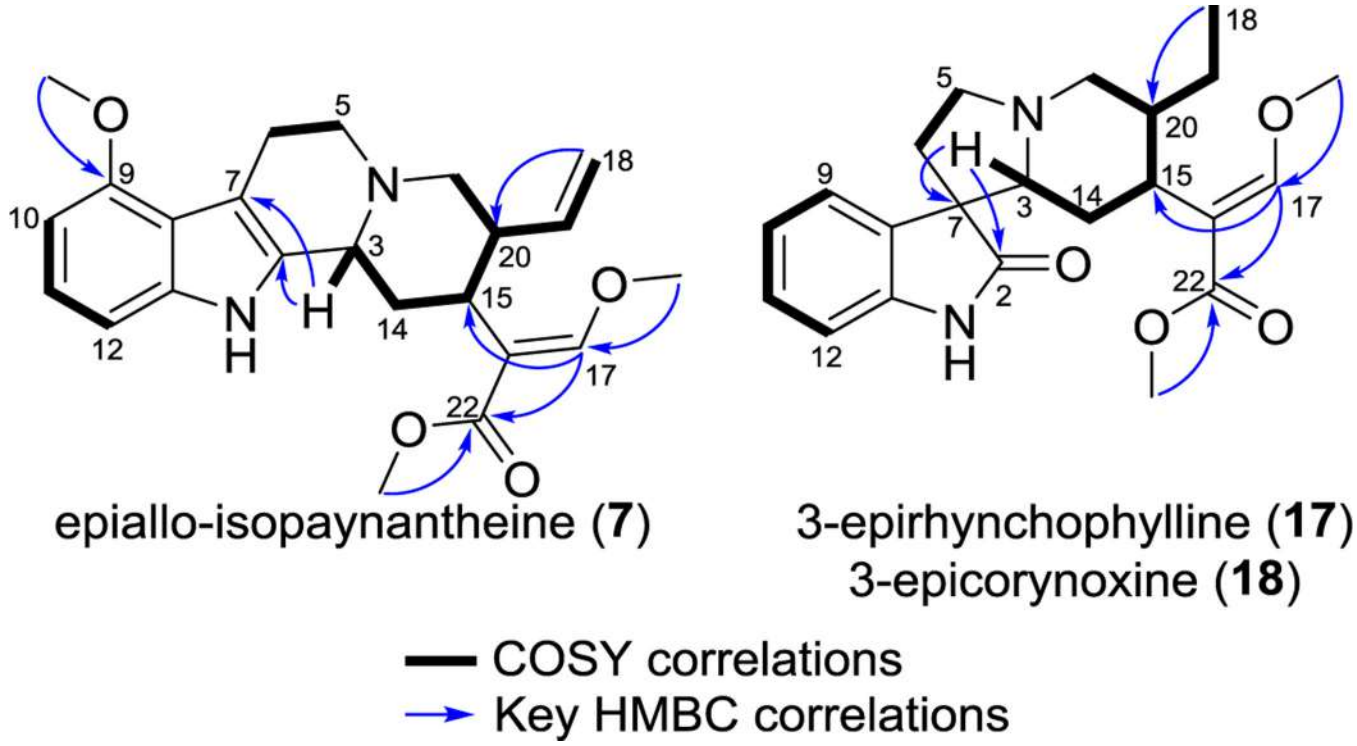


Figure 2.

Key correlations observed in the COSY and HMBC spectra for compounds **7**, **17**, and **18**. The latter two have the same 2D structure and thus the same correlations.

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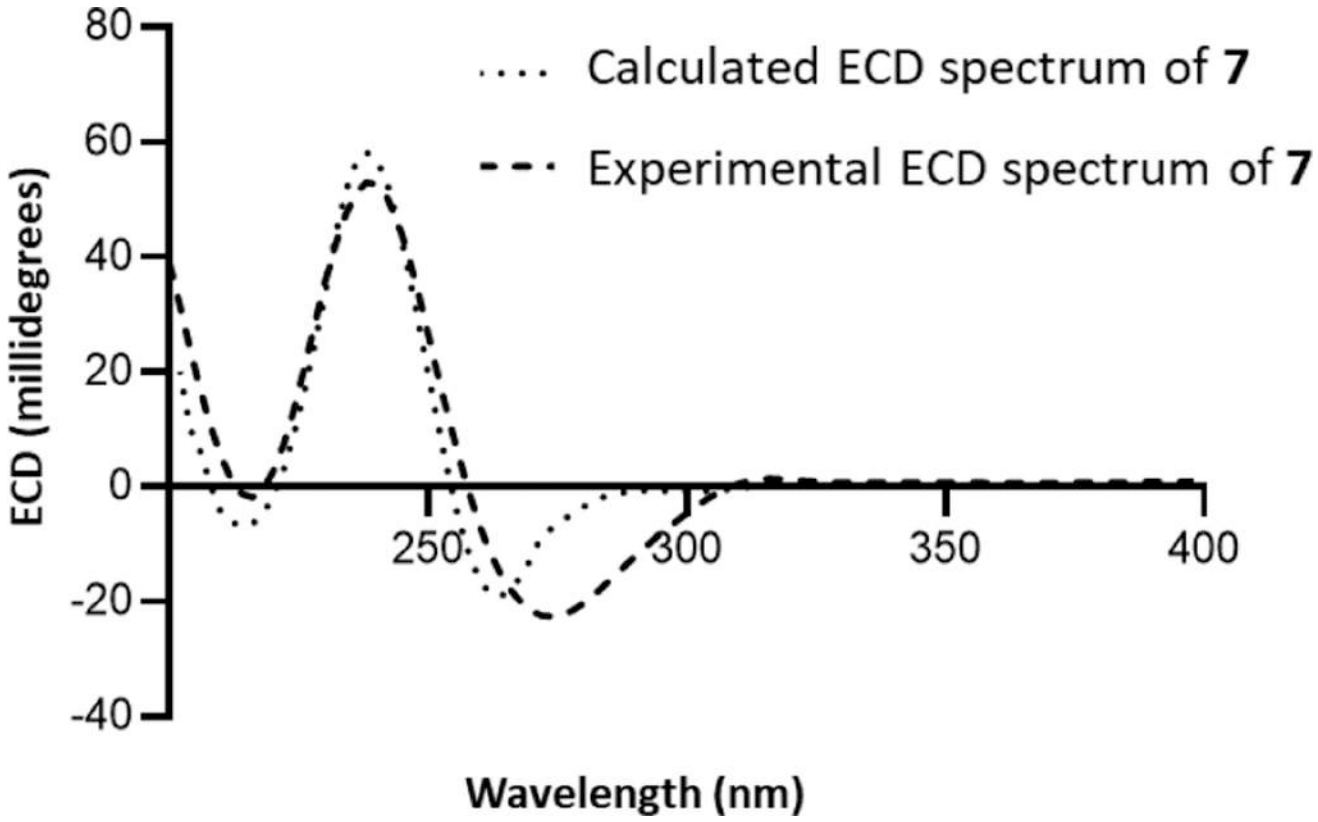


Figure 3.

Experimental ECD spectrum of **7** in CH_3OH at 0.2 mg/mL (dashed line) and calculated ECD spectrum of (3*R*,15*S*,20*S*)-**7** (dotted line).

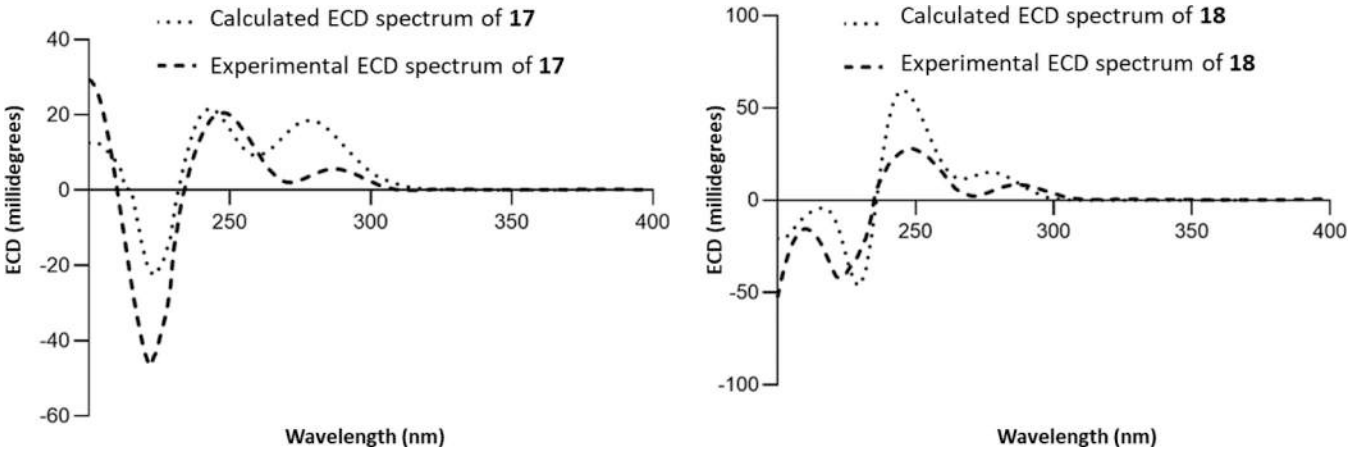


Figure 4.

A) Experimental ECD spectra of **17** in CH_3OH at 0.2 mg/mL (dashed lines) and calculated ECD spectrum of (3*R*,7*R*,15*S*,20*R*)-**17** (doted lines). B) Experimental ECD spectra of **18** in CH_3OH at 0.2 mg/mL (dashed lines) and calculated ECD spectrum of (3*R*,7*R*,15*S*,20*S*)-**18** (doted lines).

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Table 1.

Characteristic ^{1}H and ^{13}C NMR Data for Compounds 1–4 ($CDCl_3$, 400 and 100 MHz, Respectively)[a]

	mitragynine (1)			speciociliatine (2)			speciogynine (3)			mitraciliatine (4)		
position	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**
3	61.3	CH	3.20, d (8.4)	54.7	CH	4.40, bs	61.9	CH	3.21, m	53.9	CH	4.80, s
10	99.9	CH	6.45, d (7.7)	99.8	CH	6.47, d (7.7)	99.8	CH	6.44, d (7.8)	99.7	CH	6.49, d (7.6)
11	122.0	CH	7.00, t (7.9)	122.4	CH	7.02, t (8.0)	122.4	CH	6.99, t (7.9)	122.6	CH	7.07, t (7.9)
12	104.3	CH	6.90, d (8.1)	104.5	CH	6.91, d (8.1)	104.5	CH	6.87, d (8.0)	104.9	CH	7.01, d (8.0)
17	160.7	CH	7.43, s	160.6	CH	7.44, s	160.4	CH	7.35, bs	160.2	CH	7.32, s
18	13.0	CH_3	0.87, t (7.3)	12.5	CH_3	0.89, t (7.9)	11.1	CH_3	0.85, t (7.2)	11.1	CH_3	0.74, t (7.0)
9-OCH_3	55.5	CH_3	3.87, s	55.3	CH_3	3.88, s	55.4	CH_3	3.85, s	55.3	CH_3	3.89, s
17-OCH_3	61.7	CH_3	3.73, s	61.7	CH_3	3.78, s	61.9	CH_3	3.72, s	61.8	CH_3	3.77, s
22-OCH_3	51.5	CH_3	3.71, s	51.6	CH_3	3.66, s	51.1	CH_3	3.72, s	51.5	CH_3	3.68, s
NH			7.74, bs			8.00, bs			7.94, bs			8.98, bs

[a]See Table S4 for a full comparison for the ^{1}H and ^{13}C NMR data for **1–4** (Supporting Information).

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Table 2.

Characteristic ^{1}H and ^{13}C NMR Data for Compounds 5–7 ($CDCl_3$, 400 and 100 MHz, Respectively)[a]

	paynantheine (5)			isopaynantheine (6)			epiallo-isopaynantheine (7)		
Position	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**
3	60.1	CH	3.20, m	53.8	CH	4.73, bs	53.9	CH	4.85, bs
10	99.9	CH	6.45, d (7.8)	99.7	CH	6.49, d (7.6)	99.7	CH	6.49, d (7.8)
11	122.1	CH	7.00, t (7.9)	122.6	CH	7.07, t (7.9)	122.9	CH	7.08, t (7.9)
12	104.3	CH	6.88, d (8.0)	105.0	CH	7.02, d (8.1)	105.0	CH	7.01, d (8.1)
17	160.0	CH	7.33, s	160.2	CH	7.28, s	160.3	CH	7.28, s
18	115.7	CH_2	5.01, dd (17.3, 2.0) 4.96, dd (10.4, 2.1)	116.7	CH_2	4.96, dd (17.2, 1.8) 4.89, dd (10.2, 1.8)	117.2	CH_2	4.98, dd (17.3, 1.7) 4.91, dd (10.3, 1.8)
19	139.4	CH	5.55, dt (17.9, 9.3)	137.9	CH	5.29, ddd (18.0, 10.3, 8.3)	137.1	CH	5.27, ddd (18.0, 10.3, 8.3)
9-OCH_3	55.4	CH_3	3.88, s	55.3	CH_3	3.89, s	55.3	CH_3	3.88, s
17-OCH_3	61.7	CH_3	3.78, s	61.7	CH_3	3.76, s	61.7	CH_3	3.76, s
22-OCH_3	51.5	CH_3	3.69, s	51.4	CH_3	3.67, s	51.5	CH_3	3.67, s
NH			7.85, s			8.91, s			9.12, s

[a]See Table S5 for a full comparison for the ^{1}H and ^{13}C NMR data for **5–7** (Supporting Information).

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Table 3.

Characteristic ^{1}H and ^{13}C NMR Data for Compounds 8–11 ($CDCl_3$, 400 and 100 MHz, Respectively)[a]

	mitragynine-*N*(4)-oxide (8)			speciociliatine-*N*(4)-oxide (9)			isopaynantheine-*N*(4)-oxide (10)			epiallo-isopaynantheine-*N*(4)-oxide (11)		
position	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**
3	66.9	CH	5.16, s	66.0	CH	5.04, d (12.5)	69.2	CH	5.04, bs	69.6	CH	4.91, bs
10	99.7	CH	6.43, d (7.7)	99.8	CH	6.43, d (7.7)	99.9	CH	6.49, d (7.8)	99.8	CH	6.48 d (7.7
11	123.3	CH	7.03, t (7.9)	122.7	CH	6.99, t (7.9)	123.8	CH	7.11, t (7.9)	123.6	CH	7.10, t (7.9)
12	105.2	CH	6.95, d (8.1)	106.3	CH	6.92, bs	105.1	CH	7.03, d (8.2)	105.2	CH	7.04, d (8.1)
17	161.5	CH	7.45, s	161.3	CH	7.46, s	160.5	CH	7.31, s	160.5	CH	7.30, s
18	12.7	CH_3	0.93, t (7.2)	11.8	CH_3	0.85, t (7.4)	118.0	CH_2	5.02, dd (17.5, 1.0) 4.95, dd (10.2, 1.7)	117.8	CH_2	5.02, d (17.2) 4.94, dd (10.3, 1.6)
9-OCH_3	55.2	CH_3	3.85, s	55.3	CH_3	3.86, s	55.3	CH_3	3.88, s	55.3	CH_3	3.87, s
17-OCH_3	62.1	CH_3	3.80, s	61.9	CH_3	3.85, s	62.0	CH_3	3.80, s	62.0	CH_3	3.79, s
22-OCH_3	51.6	CH_3	3.61, s	51.8	CH_3	3.68, s	51.6	CH_3	3.68, s	51.6	CH_3	3.68, s
NH			9.46, s			-			8.84, s			-

[a]See Table S6 for a full comparison for the ^{1}H and ^{13}C NMR data for **8-11** (Supporting Information).

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Table 4.

Characteristic ^{1}H and ^{13}C NMR Data for Compounds 12–14 ($CDCl_3$, 500 and 125 MHz, Respectively)[a]

	speciofoline (12)			isorotundifoleine (13)			isospeciofoleine (14)		
Position	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**
2	180.2	C		177.7	C		179.1	C	
3	63.8	CH	3.06, dd (11.7, 3.6)	66.6	CH	3.39, dd (12.5, 3.6)	68.7	CH	2.68, dd (11.6, 3.2)
7	57.3	C		55.9	C		57.5	C	
9	154.6	C		155.1	C		154.6	C	
10	111.9	CH	6.35, d (7.6)	112.1	CH	6.30, dd (7.6, 0.8)	111.4	CH	6.36, dd (7.6, 0.7)
11	129.6	CH	7.05, t (8.0)	129.2	CH	7.02, dd (8.3, 7.7)	129.6	CH	7.06, dd (8.3, 7.7)
12	101.1	CH	6.53, d (8.3)	100.3	CH	6.52, dd (8.5, 0.8)	101.0	CH	6.58, d (8.3)
17	159.8	CH	7.38, s	159.8	CH	7.27, s	159.9	CH	7.19, s
18	12.3	CH_3	0.83, d (7.4)	116.4	CH_2	5.01, m 4.96, m	116.4	CH_2	4.97, m 4.95, m
19	24.2	CH_2	1.21, dqd (14.0, 6.8, 2.8)	138.8	CH	5.43, dt (18.7, 9.1)	138.6	CH	5.49, dt (17.3, 9.5)
17-OCH_3	61.3	CH_3	3.79, s	61.7	CH_3	3.79, s	61.5	CH_3	3.71, s
22-OCH_3	51.7	CH_3	3.65, s	51.3	CH_3	3.68, s	51.2	CH_3	3.59, s
NH			8.44, s			7.45,s			7.60, s

[a]See Table S7 for a full comparison for the ^{1}H and ^{13}C NMR data for **12-14** (Supporting Information).

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Table 5.

Characteristic ^{1}H and ^{13}C NMR Data for Compounds 15–19 ($CDCl_3$, 500 and 125 MHz, Respectively)[a]

	corynoxine A (15)			corynoxine B (16)			3-epirhynchophylline (17)			3-epicorynoxine B (18)			corynoxeine (19)		
position	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**	**δ_C**	**type**	**δ_H (*J* in Hz)**
2	182.4	C		182.3	C		181.2	C		182.2	C		181.0	C	
3	73.2	CH	2.41, dd (11.3, 2.7)	76.5	CH	2.24, bd (11.2)	77.4	CH	2.19, dd (11.4, 2.4)	77.0	CH	2.19, d (10.9)	75.2	CH	2.30, dd (11.3, 2.5)
9	125.0	CH	7.45, d (7.4)	123.2	CH	7.19, d (7.4)	123.3	CH	7.20, d (7.5)	123.4	CH	7.20, d (7.8)	123.5	CH	7.22, d (7.8)
10	122.5	CH	7.05, td (7.6, 1.0)	122.5	CH	7.01, td (7.5, 1.0)	122.5	CH	7.02, td (7.6, 1.0)	122.9	CH	7.02, td (7.5)	122.7	CH	7.05, td (7.6, 1.0)
11	127.4	CH	7.17, td (7.7, 1.3)	127.9	CH	7.16, td (7.7, 1.0)	127.8	CH	7.16, td (7.7, 1.2)	128.1	CH	7.17, t (7.7)	128.0	CH	7.18, td (7.7, 1.2)
12	109.5	CH	6.86, d (7.7)	109.5	CH	6.87, d (7.7)	109.1	CH	6.81, d (7.7)	109.6	CH	6.80, d (7.7)	109.2	CH	6.82, d (7.7)
18	13.0	CH_3	0.87, t (7.4)	13.4	CH_3	0.86, t (7.4)	13.5	CH_3	0.86, t (7.4)	13.3	CH_3	0.86, t (7.3)	115.6	CH_2	4.95, ddd (17.2, 2.01, 0.8) 4.90, dd (10.2, 2.1)
19	19.4	CH_2	1.10, dqd (15.7, 7.8, 2.9) 1.65, dq (14.3, 7.2)	19.3	CH_2	1.77, ddq (14.2, 11.1, 7.1) 1.18, m	19.3	CH_2	1.78, ddq (13.6, 11.2, 7.2) 1.18, dq (14.7, 7.7)	19.8	CH_2	1.80, m 1.17, dq (14.5, 7.3)	139.6	CH	5.51, dt (18.0, 9.1)
17-OCH_3	61.2	CH_3	3.51, s	61.6	CH_3	3.57, s	61.7	CH_3	3.67, s	61.7	CH_3	3.69, s	61.7	CH_3	3.74, s
22-OCH_3	51.3	CH_3	3.59, s	51.4	CH_3	3.61, s	51.4	CH_3	3.63, s	51.4	CH_3	3.63, s	51.4	CH_3	3.62, s
NH			8.40, s			8.91, s			7.62, s			7.33, s			7.53, s

[a]See Table S8 for a full comparison for the ^{1}H and ^{13}C NMR data for **15-19** (Supporting Information).

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Natural Products

Total Synthesis and Structural Plasticity of Kratom Pseudoindoxyl Metabolites**

Péter Angyal, Kristóf Hegedüs, Bence Balázs Mészáros, János Daru, Ádám Dudás, Anna Rita Galambos, Nariman Essmat, Mahmoud Al-Khrasani, Szilárd Varga,* and Tibor Soós**

Abstract: Mitragynine pseudoindoxyl, a kratom metabolite, has attracted increasing attention due to its favorable side effect profile as compared to conventional opioids. Herein, we describe the first enantioselective and scalable total synthesis of this natural product and its epimeric congener, speciogynine pseudoindoxyl. The characteristic spiro-5-5-6-tricyclic system of these alkaloids was formed through a protecting-group-free cascade relay process in which oxidized tryptamine and secologanin analogues were used. Furthermore, we discovered that mitragynine pseudoindoxyl acts not as a single molecular entity but as a dynamic ensemble of stereoisomers in protic environments; thus, it exhibits structural plasticity in biological systems. Accordingly, these synthetic, structural, and biological studies provide a basis for the planned design of mitragynine pseudoindoxyl analogues, which can guide the development of next-generation analgesics.

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Introduction

Pain management is one of the oldest challenges for medicine, and opioids targeting μ-opioid receptors (MORs) play a fundamental role in the treatment of moderate-to-severe acute and chronic pain. Although opioid agonists are still the most effective analgesics, there is a growing concern about the safety of their long-term administration owing to various adverse effects, such as respiratory depression, constipation, and addiction. Additionally, the current upsurge in opioid abuse, coupled with the spread of illicit synthetic opioids, has reached an epidemic level in the United States,[1] wreaking havoc on public health, society, and the economy. In addition to introducing new legislation and public education, the development of safer analgesics is a major goal in pain management.[2]

In this context, alkaloids from *Mitragyna speciosa* (commonly known as kratom) have recently emerged as a promising analgesic alternative for pain management with considerably fewer side effects than those associated with opioids.[3] Kratom is a tropical tree-like herb native to Southeast Asia that has been used for centuries for its distinctive psychotropic properties.[4] Kratom has been noted to have dose-dependent stimulant and opioid-like effects, and its illicit use to treat chronic pain and drug dependence is on the rise globally (>15 million estimated kratom users in the US alone).[5] The predominant natural products in kratom are indole alkaloids, mainly corynanthe-type monoterpenoids. While these mitragyna alkaloids are structurally significantly different from the morphine skeleton, they do display antinociceptive effects, which are supported by their ability to bind to opioid (i.e., MORs, DORs, KORs, NORs)[6] and other receptors (e.g., serotonin receptors).[7] Systematic investigations have also shown that the opioid agonistic effect of *Mitragyna speciosa* cannot be fully explained by that of its main alkaloid, mitragynine (**1**; 230 nM for MOR; Figure 1). Finally, more potent analgesics, which were the oxidative metabolites of mitragynine, were identified in in vitro assays. Accordingly, the main alkaloid metabolizes in vivo[8–10] to 7-OH-mitragynine (**2**; 37 nM for MOR), which is followed by a subsequent 1,2-semipinacol rearrangement in human plasma to the more potent mitragynine pseudoindoxyl (**3**; 0.8 nM for MOR, G-protein-biased MOR partial agonist). Nevertheless, hindering the completeness of the oxidative metabolic picture of mitragynine (**1**), another metabolite in addition to mitragynine

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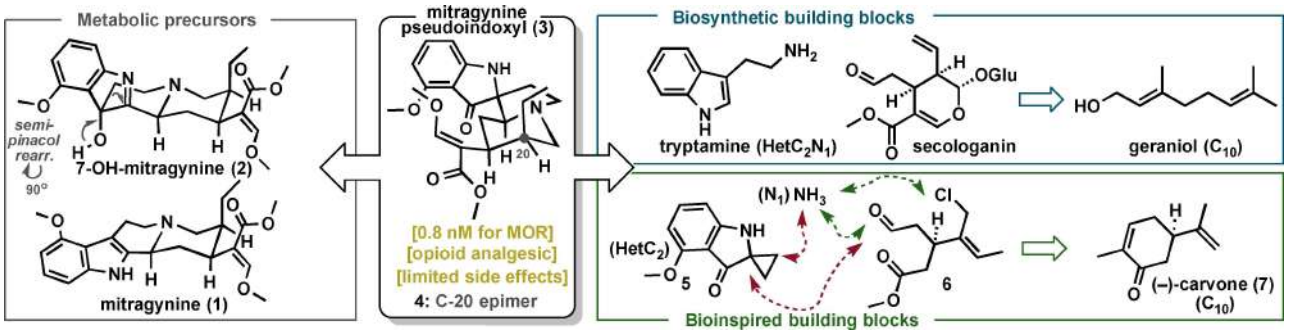


Figure 1. Introduction and comparison of distinct pathways toward pseudoindoxyl metabolites of kratom alkaloids.

pseudoindoxyl (**3**) was also detected in the semipinacol rearrangement step, but its structure and possible opioid effect have not been determined to date owing to its low occurrence in human plasma.[9]

Most importantly, however, animal studies have shown that mitragynine pseudoindoxyl (**3**) may avoid some of the key problems of current opioid therapies, as antinociceptive tolerance, physical dependence, and respiratory depression were not observed.[11] To gain insight into the underlying molecular mechanism of its distinct signaling profile, high-resolution cryo-electron microscopy (cryo-EM) structural studies on the MOR-Gi1 complex with **3** and molecular dynamics simulations have been performed.[12] Furthermore, combined double electron–electron resonance (DEER) and single-molecule fluorescence resonance energy transfer (smFRET) studies were recently conducted to investigate the molecular underpinnings of MOR activation and MOR mediated G protein signaling. Of interest, when bound to **3**, there was no shift toward active conformations of MOR, which affected its downstream signaling efficacy and selectivity.[13]

Although the syntheses of kratom's major indole alkaloids[4,14] and their oxindole derivatives[15] have already received considerable attention from the synthetic community, total syntheses of mitragynine pseudoindoxyl derivative **3** and kratom's other minor alkaloids (e.g., the recently reported speciogynine derivative **4**)[16,17] have not yet been revealed. Nevertheless, Sorensen and co-workers could synthesize the methoxy analogue of **3** by the ingenious construction of the challenging spiro-fused pseudoindoxyl system.[18] However, their interrupted approach based on the Ugi reaction was limited to highly electron rich aromatic derivatives and did not allow the synthesis of the natural product **3** itself. Thus far, access to mitragynine's oxidative metabolites **2** and **3** has relied exclusively on low-yielding biomimetic semisynthesis from mitragynine (**1**), which generates the pseudoindoxyl ring in a 1,2-semipinacol rearrangement.[11,19] Although these and related studies demonstrated the power of late-stage functionalization strategies (e.g., selective oxidation and site-selective aromatic C–H functionalization),[20,21] the attainable chemical space did not allow extensive structure–activity relationship (SAR) studies or deep-seated modifications of this unique pseudoindoxyl scaffold, thus hampering further advances in drug development.[22,23]

Given the enormous therapeutic potential and the lack of an enabling synthetic platform for SAR studies, we aimed to develop a scalable[24] and modular route to pseudoindoxyl opioid **3**. Of all the possible disconnections of its spirotetracyclic backbone, the one that maximizes convergence and skeletal simplification and whose forward synthesis is concise and potentially scalable was chosen. Figure 1 illustrates the process through which we disconnected mitragynine pseudoindoxyl (**3**) into two subunits, a dubbed "oxidized tryptamine" fragment **5** and a secologanin analogue[25] **6** that was anticipated to arise from (−)-carvone (**7**).[26] A cascade process was then devised that would i) strictly avoid the use of protecting groups and harness the natural reactivities of the two key building blocks and ii) forge the CD ring system and the critical stereogenic spirocyclic center. Furthermore, our strategy was based on a stereocenter derived from (−)-carvone (**7**), of which its stereochemical information could be propagated to diastereoselectively adjust two other stereocenters. The decision to implement this diastereoselective cascade relay was also underpinned by the computational analysis of the possible diastereomers of **3** (see the Supporting Information for details). It turned out that the thermodynamically most stable stereostructure was that of **3**, which was also the product of the biosynthetic pathway.[27] Finally, we believed that our secologanin analogue could further broaden the synthetic landscape, as it could provide access to both diastereomeric kratom pseudoindoxyls of the corynanthe type, *allo* (mitragynine) and *normal* (speciogynine) isomers, by selective hydrogenation of its olefinic bond.[28,29] As such, this synthetic development was inspired by the manner in which nature assembles mitragynine pseudoindoxyl (**3**), including rapid complexity generation through cascade reactions of tryptamine and secologanin analogues, step and redox economy, divergency and protecting-group-free assembly.[30] Herein, we report the first total synthesis of mitragynine pseudoindoxyl (**3**) and its diastereomer, speciogynine pseudoindoxyl (**4**).

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Results and Discussion

Our investigation began with the construction of the chiral secologanin-like building block **6** using (−)-carvone (**7**) as a chiral terpenoid starting material. To this end, we developed an operationally simple and scalable synthetic route that features a relay of chemoselective transformations (Figure 2).[31] First, the homologation of (−)-carvone (**7**) was achieved in a two-step procedure: Ene-type chlorination[32] of the electron-rich double bond with trichloroisocyanuric acid (TCICA) was followed by a copper-catalyzed C–C bond-forming step to form homocarvone (**8**). Notably, the latter step was selective, as no 1,2- or 1,4-addition products of enone were observed under optimized conditions. Then, two orthogonal oxidation reactions were carried out; a nucleophilic epoxidation selectively transformed the electron-deficient double bond into an epoxide, and a second ene-type chlorination formed allylic chlorine **9** with high *Z* selectivity, presumably due to *syn*-pentane strain.[33] After obtaining the desired intermediate **9** in an operationally simple way in a useful yield (39% yield over 4 steps, avg. 79%), two C–C bond cleavages remained to successfully remodel the initial terpene skeleton[34] and achieve the open-chain C_9 carbon core of the envisioned secologanin building block **6**. To prepare for these C–C bond-breaking reactions, mild acidic hydrolysis of the epoxide moiety was carried out at room temperature, which retained the sensitive allylic chlorine moiety. Then, the resulting mixture of diastereomeric ketone diols was exposed to an excess of sodium periodate to effect the required C–C cleavages and provide the open-chain carboxylic acid **10**. The handling of this multifunctional and reactive molecule **10**, however, turned out to be challenging owing to the presence of electrophilic and nucleophilic sites combined with its apparent ability to form cyclic acetals or esters. After careful experimentation, simple ester formation with diazomethane was found to address this problem, as it promoted clean methylation to complete the sequence without the need for further purifications (71% yield over 3 steps, avg. 89%). In summary, a scalable, protecting-group-free route to secologanin analogue **6** was achieved from (−)-carvone (**7**) in only seven steps. Each of the reaction steps was performed on a decagram scale, and only one purification step was involved, attesting to the robustness and potential further scalability of the sequence.

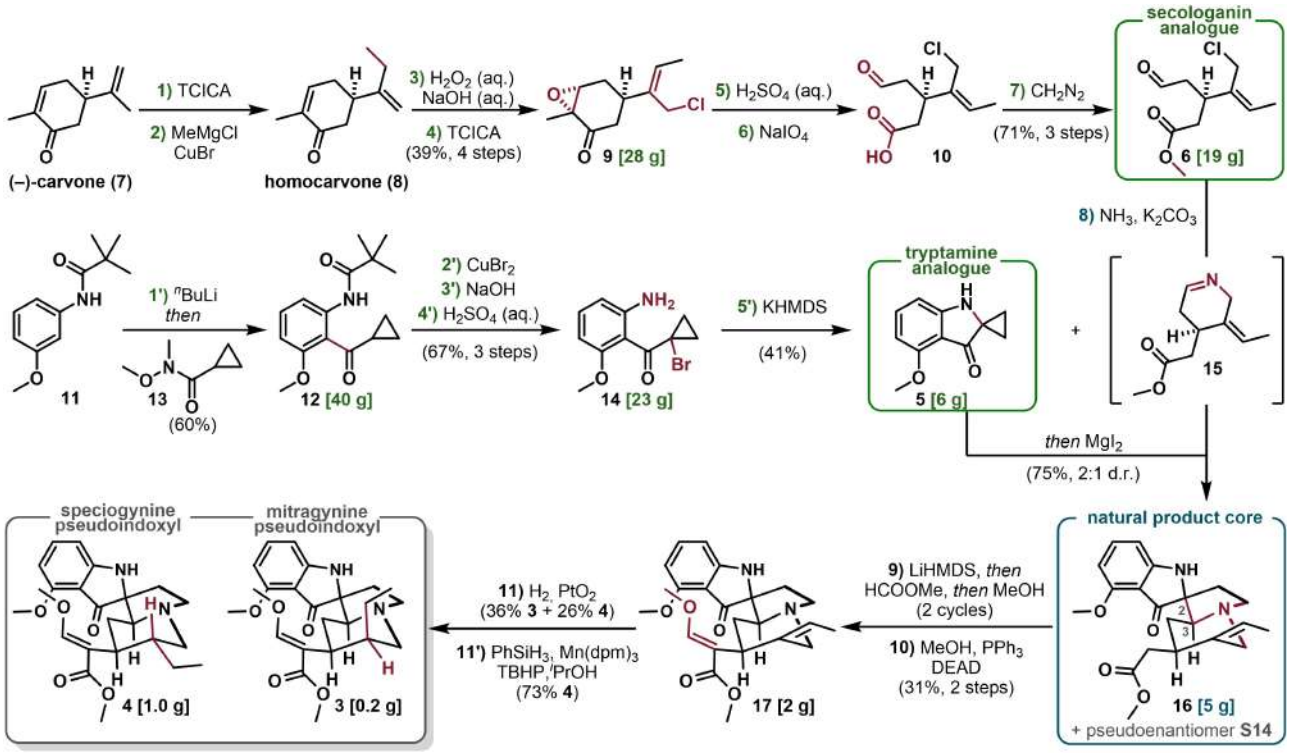


Figure 2. Bioinspired synthesis to form pseudoindoxyl metabolites on a gram scale. Reagents and conditions: 1) TCICA (0.33 eq.), $CHCl_3$, 5 to 25 °C; 2) CuBr (0.25 eq.), MeMgCl (1.0 eq.), THF, −65 °C; 3) H_2O_2 (aq.) (1.3 eq.), NaOH (aq.) (0.27 eq.), MeOH, −10 to 0 °C; 4) TCICA (0.24 eq.), $CHCl_3$, 5 to 23 °C; 5) 50% H_2SO_4 (aq.), THF, 5 to 23 °C; 6) $NaIO_4$ (3.0 eq.), THF/H_2O, 5 to 23 °C; 7) CH_2N_2, Et_2O, 5 to 10 °C; 1′) ^{n}BuLi (2.2 eq.) *then* **13** (1.0 eq.), THF, 5 to 25 °C; 2′) $CuBr_2$ (2.2 eq.), EtOAc, reflux; 3′) NaOH (2.0 eq.), MeOH, 25 °C; 4′) 25% H_2SO_4 (aq.), MeOH, 65 °C; 5′) KHMDS (2.0 eq.), THF, 5 to 23 °C; 8) **6** (2.2 eq.), NH_3 (1 atm), K_2CO_3 (exc.), $CaCl_2$ (exc.), CH_2Cl_2, 23 °C, *then* the addition of **5** (1 eq.) and MgI_2 (1 eq.) in THF, 23 °C; 9) LiHMDS (5.0 eq.), *then* HCOOMe (10 eq.), THF, −40 °C, *then* the addition of MeOH, 23 °C (2 cycles); 10) MeOH (2.0 eq.), DEAD (1.0 eq.), PPh_3 (1.0 eq.), THF, 23 °C; 11) H_2 (1 atm), $PtO_2 \cdot H_2O$ (50 m/m%), $NaHCO_3$ (exc.), TFE, 23 °C; 11′) $PhSiH_3$ (3.0 eq.), $Mn(dpm)_3$ (0.25 eq.), TBHP in decane (1.0 eq.), ^{i}PrOH, 23 °C. TCICA = trichloroisocyanuric acid, KHMDS = potassium bis(trimethylsilyl)amide, exc. = excess, LiHMDS = lithium bis(trimethylsilyl)amide, DEAD = diethyl azodicarboxylate, TFE = 2,2,2-trifluoroethanol, $Mn(dpm)_3$ = tris(2,2,6,6-tetramethyl-3,5-heptanedionato)manganese(III), TBHP = *tert*-butyl hydroperoxide.

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Next, we focused on synthesizing the spirocyclic cyclopropane–indoxyl building block **5**. Initial attempts to accomplish the synthesis of **5** on a gram scale were unsuccessful owing to the high reactivity and sensitivity of this compound (see the Supporting Information for details); at best, only milligram amounts of spiro-pseudoindoxyl derivative **5** were obtained. Upon extensive experimentation, we found an alternative synthetic tactic using mild enough conditions to provide **5** in sufficient quantities. As shown in Figure 2, commercially available aniline **11** was used to construct the reactive spirocyclic building block **5** in five steps. The synergistic effect[35] of the *meta*-positioned directing groups of **11** in directed *ortho*-metalation resulted in the construction of the contiguously functionalized 1,2,3-aromatic system **12** after acylation of the lithiated species with **13**, in 60 % yield on a decagram scale. Next, **12** was converted into anilino-α-bromo compound **14** through a dibromination/base-induced ring closure/acidic hydrolyzation sequence in 67 % yield, which set the stage for the last spiro-pseudoindoxyl formation step. After screening a variety of different reagents (see the Supporting Information for details), KHMDS was found to promote the formation of the desired tryptamine analogue **5** with good overall efficiency on a gram scale.

With ample quantities of the two building blocks **5** and **6**, we began to explore the key cascade relay sequence needed to construct the spirocyclic tetracyclic core found in mitragynine pseudoindoxyl (**3**). As both bioinspired analogues were endowed with reactive functionalities, it seemed less apparent that they could be woven efficiently and selectively together without using protecting groups. Nevertheless, we reasoned that the functional-group incompatibility could be addressed by developing a cascade relay sequence and leveraging a heterocyclic intermediate derived from secologanin analogue **6** in a cascade process. This fleeting cyclic imino compound **15** was considered to be a suitable substrate for the ring-opening and Mannich reaction cascade with **5** to produce the spirocyclic tetracyclic core of **3**.

However, this cascade-relay strategy was dependent on the feasibility of producing the unstable cyclic imine intermediate **15**. Although the synthesis and isolation of **15** were difficult owing to the formation of oligomeric and polymeric side products upon synthesis and storage, we were able to find appropriate basic and anhydrous conditions to form **15** with consistently high conversion and selectivity on a multigram scale (see the Supporting Information for details). Then, without an aqueous workup or removal of solvent, this unstable cyclic imine **15** was carried onwards for the ensuing cascade reaction with the cyclopropyl building block **5**. Gratifyingly, the formal [3+2] cycloaddition step[36] afforded the desired spirocyclic 5-5 pseudoindoxyl ring system at room temperature. Thus, there was no need to apply forcing reaction conditions reported by Pierce (i.e., 200 °C)[36] for the analogous intramolecular reaction or by Carreira[37] (i.e., 80 °C) for the conversion of less reactive oxindoles analogues. Accordingly, milder reaction conditions alleviated the problem of decomposition or isomerization[38] of **15** in our synthetic approach. Of the possible four diastereomers, the desired isomer **16** was obtained in 50 % yield, with its pseudoenantiomer **S14** (see the Supporting Information for the elucidated structure) in 25 % yield as an inseparable mixture (5 g scale). Although the installation of the two stereocenters was not fully diastereoselective, we anticipated that the forward reaction conditions would concomitantly epimerize those C2 and C3 stereocenters to the desired configurations. At least two important features of this cascade relay were noteworthy: i) Our imine formation/alkylation tactic enabled the facile synthesis of cyclic imines, which usually requires extensive redox and protecting-group manipulations;[39] and ii) the overall cascade sequence enabled rapid complexity generation and good convergence, as four bonds, two rings and two stereocenters were formed from two comparatively complex building blocks.[40]

With the pentacyclic core of mitragynine pseudoindoxyl (**3**) in hand, two seemingly straightforward tasks remained: the installation of the methyl enol ether moiety onto the acetyl side chain and stereoselective reduction of the alkene. To this end, an α-formylation protocol was adapted to our substrate; this protocol does not require preventive but only transient protection of the pseudoindoxyl NH. Accordingly, the double deprotonation of **16** with excess LiHMDS was followed by the addition of methyl formate and quenching with methanol; this produced the α-formylated product as a mixture of isomers (see the Supporting Information for details) and ensured the deprotection of the transient *N*-formyl group. Then, the selective methylation of the enol tautomer was explored using commonly used alkylation protocols (see the Supporting Information for details). However, these methods were plagued by poor and inconsistent yields owing to N-methylated side-product formation. To circumvent this problem, we used the Mitsunobu reaction, which allowed chemoselective O-alkylation and afforded **17** in good yield and quantity (gram scale). Finally, the stereoselective reduction of **17** was addressed to complete the total synthesis of the targeted mitragynine metabolite **3**, alongside its recently reported diastereomer, speciogynine pseudoindoxyl (**4**).

Many conditions and catalysts were screened to maximize the diastereoselectivity of the reduction process in both possible directions. We found that use of the Adam catalyst was necessary to provide our primary target, the *allo*-type mitragynine pseudoindoxyl (**3**), with modest diastereoselectivity. To accomplish the selective reduction of **17** to diastereoisomer **4**, a radical stepwise reduction was chosen to provide the thermodynamically more stable diastereoisomeric product. The HAT-type reduction, pioneered by Shenvi,[41] afforded the "normal-type" speciogynine pseudoindoxyl (**4**) in 73 % yield in a diastereoselective manner.

Although our primary goal was to access the kratom atypical opioid **3** in a scalable manner, this endeavor also provided new insight into its molecular properties. During the optimization of the last reduction step (see the Supporting Information for details), we noticed that this natural metabolite was labile under various reaction and isolation conditions, which was also manifested in dimin-

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ished yields. Prompted by these observations and having a useful quantity of **17**, we aimed to thoroughly study the reaction mixture of the last step and isolate minor or even trace components. These efforts ultimately resulted in the identification of two unknown diastereomers of mitragynine pseudoindoxyl (**18** in 5% yield and **19** in 4% yield), which most likely arose from a retro-Mannich/Mannich-type rearrangement of the spiro-pseudoindoxyl moiety (Figure 3A). We then identified conditions and methods that allowed the isolation and structural elucidation of those stereoisomers. The NMR structure elucidation of these isomers showed that they are spirocyclic diastereomers with *allo* and *epiallo* configurations; thus, the latter stereoisomer is the not-yet-known spiroindoxyl derivative of speciociliatine. Next, it was found that aprotic solvents impeded and polar protic solvents facilitated the isomerization and shifted the equilibrium (e.g., **3**:**18**:**19**≈6:1:3 in HFIP, 23 °C). Most importantly, we found that the thermodynamic equilibrium of the three stereoisomers (**3**:**18**:**19**≈7:2:1) could be reached within 20 min at 37 °C in Krebs–Ringer solution, which mimics physiological conditions (Figure 3B). Additional insight was obtained from the computational assessment of the relative free energies of the (protonated) isomers. First, it confirmed the experimental relative thermodynamic stabilities; second, it explained the observed experimental difference between protic and aprotic media, that is, protonation of the tertiary amine was found to reduce the energy difference between the isomers (see the Supporting Information for details).

While analogous and kinetically more stable spirooxindoles are known to exist in at least two isomeric forms,[42] only one structural isomer has been observed for spiro-pseudoindoxyl systems to date.[43] Although Takayama et al. hypothesized the formation of four possible stereoisomers 30 years ago in their pioneering work, only one pseudoindoxyl diastereomer was isolated after the 1,2-semipinacol rearrangement of the biogenetic precursor 7-OH-mitragynine (**2**).[19]

A critical question that remained at this stage concerned the opioid activity of these stereoisomers. Our computational studies of the stereoisomers suggested that the shapes of these molecules are significantly different (Figure 3C); thus, a profound difference in their biological activities might be expected. Nevertheless, on the basis of the observation of the rapid stereoisomerization phenomenon, we anticipated that all stereoisomers might show the same

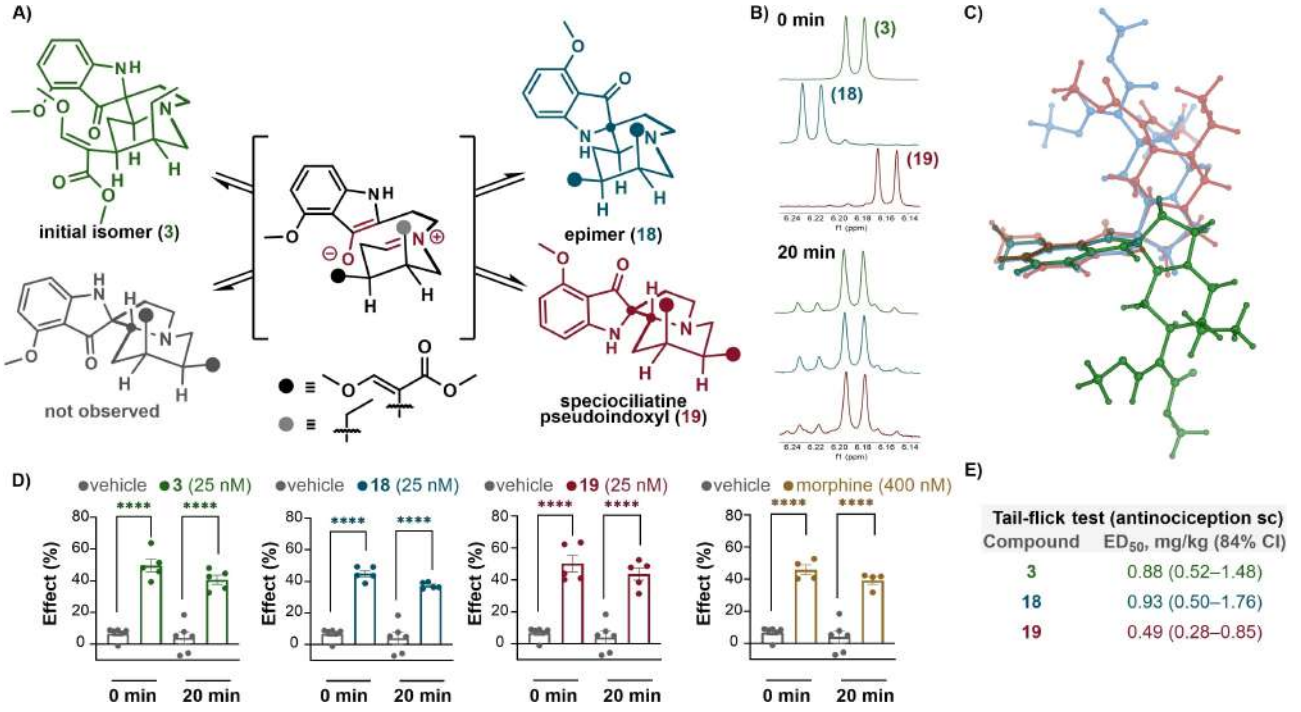


Figure 3. Isomerization of mitragynine pseudoindoxyl (**3**) and related studies. A) Retro-Mannich/Mannich isomerization and structure of the isomers obtained. B) NMR spectra: Single isomers **3**, **18** and **19** were introduced into physiological-like media, and their isomerization was monitored with ^{1}H NMR measurements, as illustrated by the shifts of characteristic aromatic signals (starting materials, 0 min (up) vs. 37 °C, pH = 7.4, 20 min (down)). C) Superimposed structures of the calculated lowest energy conformations of **3** (green), **18** (blue), and **19** (red). D) Results of the in vitro mouse vas deferens bioassay comparing the effect of single isomers: Test or reference compounds were added to the MVD immediately or 20 min after solution preparation. The effect on MVD smooth muscle contractions was assessed after 10 min. Statistical significance was accepted based on one-way ANOVA with the Tukey multiple comparison test, $p < 0.05$ (**** $p < 0.0001$). Data are presented as the mean ± S.E.M. of 5 or 6 tissues. E) Results of the in vivo radiant heat tail flick assay in mice comparing the effect of single isomers: The antinociceptive effect of test compounds was assessed 20, 30, 60 and 120 min after s.c. administration at various doses (see the Supporting Information for details). The antinociceptive potencies (ED_{50}) were calculated from linear dose response curves. The ED_{50} values and 84% confidence intervals were calculated at the most effective time point, namely, 20 min after s.c. administration.

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activity in biological systems. Thus, we initiated comparative in vitro and in vivo studies for each single stereoisomer. First, we evaluated the opioid-receptor-mediated effects of these isomers (**3**, **18**, and **19**) in the mouse vas deferens (MVD) assay[44,45] (Figure 3D). All compounds showed naloxone-sensitive inhibitory effects on MVD muscle contractions, with significantly higher potency than morphine. Furthermore, Student's t-test showed no difference in the inhibitory effect between stereoisomers administered immediately after solution preparation or 20 min later. Accordingly, this study suggests that the opioid effect of those isomers is related to their dynamic stereochemical equilibrium. Next, we examined the acute analgesic effect of these isomers. Thus, a thermal pain model, the radiant-heat tail-flick assay,[11,46] was selected and performed in the same mouse strain used for MVD experiments (Figure 3E). Every experiment comprised five series of measurements, one before and four after subcutaneous drug administration, carried out at 20, 30, 60 and 120 min intervals. The antinociceptive effect was assessed 20 min after s.c. drug administration since the antinociception of the compounds peaked at that time (see Supporting Information , Figure S35). Importantly, there was no significant difference between the ED_{50} values of the stereoisomers, as indicated by the overlap in 84 % confidence intervals. Accordingly, we found that mitragynine pseudoindoxyl (**3**) actually exists and acts as a dynamic ensemble of stereoisomers in the protic environment and, what is perhaps even more important, in biological systems. This finding has many important implications that are vital to the interpretation of previous results and can shape future drug developments: i) The elusive, unidentified 7-OH-mitragynine metabolite in human plasma described by McCurdy[9] is likely to be the above identified stereoisomer(s) of **3**; ii) mitragynine pseudoindoxyl (**3**) displays structural plasticity, and the evolving molecular ensemble could target additional receptors. As shown by biological and pharmaceutical investigations[47] for other ligands, an isomerization-induced conformational and structural heterogeneity may influence the ligand's affinity and efficacy to target receptors and affect its ADMET properties; iii) the structural plasticity of **3** might be important, or even critical for its low efficacy, G-protein-biased agonist activity. Additionally, assuming that the time window of the isomerization of **3** is comparable or smaller than that of the conformational change of MOR, the isomerization should be taken into consideration for molecular docking[48] and future drug development; iv) the local biological environment (e.g., pH, metal ions) can alter the potency of the pseudoindoxyl ensemble due to the shift of the equilibrium of stereoisomers; v) and the oxidative metabolism of speciociliatine to its pseudoindoxyl derivative (the C3 epimer of mitragynine)[49] is predicted to result in the same stereomeric mixture as described herein, which also implies an unusual convergent metabolic pathway for kratom alkaloids.

Conclusion

In summary, the first scalable and chiral-pool-based synthetic routes to members of kratom pseudoindoxyls have been developed by using oxidized tryptamine and secologanin analogues. The highly convergent strategy, the deliberate minimization of the use of protecting groups, and late-stage divergency allowed the rapid construction of targeted natural products **3** and **4** on a gram scale. While our synthetic route mainly benefited from effective strategy and tactics, several methodological advances were also achieved: i) The chemoselective homologation of carvone and a stereoselective ene-type chlorination formed the basis for a streamlined synthesis of our key secologanin building block; ii) a complexity-generating imine formation/alkylation/intermolecular [3+2] cascade relay was developed, which served as a cornerstone of our strategy; iii) Mitsunobu-type alkylation and HAT functionalization of late-stage pseudoindoxyl intermediates enabled the preparation of the natural products and illuminated the possibility of preparing diverse unnatural analogues not attainable by semisynthesis. As an unexpected spin-off of the synthetic efforts, we discovered that mitragynine pseudoindoxyl (**3**) exists and acts not as a single molecular entity but as a dynamic ensemble of stereoisomers in protic environments and, what is perhaps even more important, in biological systems. Accordingly, these synthetic, structural, and biological studies provide a basis for future functional modifications of mitragynine pseudoindoxyl (**3**), which can guide the development of next-generation analgesics.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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Keywords: Biological Activity · Chiral Pool · Medicinal Chemistry · Natural Products · Total Synthesis

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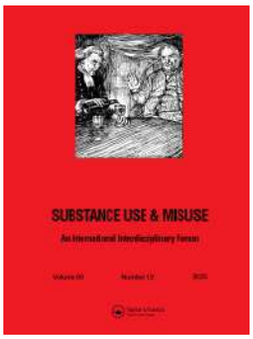
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A Descriptive Assessment of Products Containing the Opioid Receptor Stimulator Mitragynine Pseudoindoxyl

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ABSTRACT

Background: Potent semi-synthetic alkaloids that stimulate opioid receptors are being sold to consumers in gas stations, smoke shops, and online as the dietary supplement 'kratom'. *Objective*: We descriptively evaluated the characteristics of products containing mitragynine pseudoindoxyl. *Results*: Fifty-one unique products met our inclusion criteria. Seventy-one percent of products contained mitragynine pseudoindoxyl plus 7-hydroxymitragynine, 24% contained mitragynine pseudoindoxyl alone (24%), and 6% contained mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (with or without 7-hydroxymitragynine). Sixty-nine percent of products had a child appealing flavoring and/or scent, 63% of products were formulated using bright colors or had bright colors on their labeling, 12% of products had food pictures on the label (mainly representing the fruit or mint flavoring/scent), 6% of products had a cartoon mascot, and 6% of products had a food formulation (ice cream cone or chocolate bar). Only 18% of websites disclosed the products stimulated opioid receptors while 45% disclosed they could cause intoxication, tolerance, dependence, withdrawal, or addiction. *Conclusions*: There are many products that contain mitragynine pseudoindoxyl sold as dietary supplements and many have features that would appeal to children.

KEYWORDS
Kratom; mitragynine pseudoindoxyl; 7-hydroxymitragynine; opioid

Introduction

Fresh and dried leaves of the Mitragyna speciosa tree in Southeast Asia have been used as an herbal product called whole leaf kratom for energy boosting, relaxation, pain control, and opioid substitution for centuries (McCurdy et al., 2024). Mitragynine comprises as little as 12% and up to 66% of the total alkaloids in fresh Mitragyna speciosa leaves (whole leaf kratom) and primarily stimulates serotonin, alpha-1, and alpha-2 receptors while only indirectly and mildly stimulating mu-opioid receptors (Ponglux et al., 1994; Takayama et al., 1998). There are many other natural alkaloids in whole leaf kratom that reinforce the serotonin and alpha-2 agonist effects. When dried, some of the mitragynine (<2% of the total alkaloid content) degrades into the alkaloid 7-hydroxymitragynine (McCurdy et al., 2024).

Within the past two years, a proliferation of high dose semi-synthetic 7-hydroxymitragynine and mitragynine pseudoindoxyl products has occurred (White, 2024a). Preclinical studies show that these alkaloids potently stimulate mu-opioid receptors, are antagonized by opioid antagonists like naloxone and naloxonazine, and do not stimulate alpha-1 and alpha-2 adrenoceptors or serotonin receptors (Kruegel et al., 2019; Váradi et al., 2016; Yamamoto et al., 1999). It was previously shown in a rodent study that mitragynine use could mildly suppress respiration but had a ceiling effect preventing serious respiratory depression while 7-hydroxymitragynine suppressed respiration and did not have such a ceiling effect (Hill et al., 2022). A recent rodent study found that 7-hydroxymitragynine significantly suppressed rodent breathing frequency, tidal volume, and minute volume at higher doses, while mitragynine did not (Zuarth-Gonzalez et al., 2025). The impact of mitragynine pseudoindoxyl on respiratory depression was assessed in a rodent model and respiration was found to be unaffected at a low dose but reduced by 15% at a higher dose (Váradi et al., 2016). A recent letter to the editor from kratom researchers discussed that 7-hydroxymitragynine is a risky ingredient in products available to consumers without a prescription because of concerning animal studies showing physical dependence and withdrawal effects with a lack of human safety and efficacy trials (Smith et al., 2025). The same is true for mitragynine pseudoindoxyl as demonstrated by the development of physical dependence and withdrawal signs and symptoms in morphine addiction rodent models as evidenced by increased diarrhea, jumping, and wet shake frequency occurring when naloxone was administered or when treatment with this alkaloid was tapered (Wilson et al., 2021).

We sought to determine how many products for oral administration are sold over the internet that contain mitragynine pseudoindoxyl, what other alkaloids they contain, their serving sizes, whether they could be attractive to children, and what safeguards are available to prevent children from using them.

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Materials and methods

We performed a systematic search in the Google search engine using the terms "mitragynine pseudoindoxyl" or "Red-OH" (a pseudonym of mitragynine pseudoindoxyl) from January 10th to March 1st, 2025. Google was the only engine searched because it is used in over 88% of searches in the United States, and to reduce the identification of the same available products across multiple platforms (Oberlo, 2024). We did not use a VPN and conducted the search in the United States. Since the study was not on animal or human participants or their personal or medical data, it is exempt from IRB review.

We included all orally available products sold online that contained mitragynine pseudoindoxyl with or without other alkaloid ingredients. We found 21 distinct retailers selling at least one product containing mitragynine pseudoindoxyl alone or in combination with other alkaloids (Supplemental Table S1). We determined if the websites made shoppers check a box specifying they were over 18 or 21 years to enter and whether they required age verification before purchase.

We recorded all products meeting our criteria from each site. However, if a website had duplicative products that we discovered from a prior website, we only included the original website's data unless there was missing data that a subsequent website included. We recorded the alkaloids they contained, the suggested serving size, their enteral product formulations, whether the packaging was considered child-resistant (a child-resistant cap on a vial or bottle), whether the product labeling specified age restrictions to purchase, and the features of the product that would appeal to children. Since blister packing of over the counter or prescription drugs meets the criteria for being child resistant, we collected this information although we could not assure that the products did not have easy to open features.

We searched all the websites we identified and the labeling for each of the unique products they contained for the following terms: opioid, intoxication, impairment, dependence, tolerance, withdrawal, addiction, or habit-forming.

Table 1. Product formulations.

Active ingredient	Products found, No. (% of 51 total products)
Mitragynine pseudoindoxyl alone	
Chewable tablet	7
Liquid	5
Total	12 (23%)
Mitragynine pseudoindoxyl and 7-hydroxymitragynine	
Chewable tablet	30
Liquid	4
Ice cream cone	1
Chocolate bar	1
Total	36 (71%)
Mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (±7-hydroxymitragynine)	
Chewable tablet	2
Ice cream cone	1
Total	3 (6%)

No.: number.

Results

Supplemental Tables S2–S4 provide a synthesis of our raw data (Amsterdam Smoke, n.d.; CBD American Shaman, n.d.; Delta 8 Resellers, n.d.; Gala, n.d.; Great CBD Shop, n.d.; OPMS, n.d.; Pure Leaf Kratom, n.d.; The Powder Solutions, n.d.). Overall, we found 51 unique products that met our inclusion criteria (Amsterdam Smoke, n.d.; CBD American Shaman, n.d.; Delta 8 Resellers, n.d.; Gala, n.d.; Great CBD Shop, n.d.; OPMS, n.d.; Pure Leaf Kratom, n.d.; The Powder Solutions, n.d.). The largest group of products contained mitragynine pseudoindoxyl with 7-hydroxymitragynine (71%), followed by products with mitragynine pseudoindoxyl alone (24%), and then products with mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (with or without 7-hydroxymitragynine) (6%) (Table 1). Seventy-six percent of these products were chewable tablets, 18% were liquids, 4% looked like dried ice cream cones with ice cream, and 2% were chocolate.

Only one product (2%) came in a bottle with a child-resistant cap. Thirty-four products (67% of the total) came in a blister pack which may or may not be child resistant. The child resistant packaging from 4 products (8%) could not be determined, and 12 products (24%) were not child-resistant because they were: (1) in bottles with easy-open twist-off caps, (2) they were just wrapped or in an easy open box, (3) or because the label stated they were "not child-resistant." Overall, 37 products (73%) had an age restriction warning specified on their packaging, but 14 products (27%) did not.

We observed several product features that would appeal to children (Table 2). Sixty-nine percent of products had an appealing flavoring/scent [berry (raspberry, strawberry, wild berry), mint, watermelon, pink lemonade, candy apple, grape, citrus, mango, pistachio, and vanilla bean], 63% of product packaging or the tablets or liquid itself were formulated using bright colors, 12% of products had food pictures on their packaging (usually representing the flavoring/scent), 6% of products had a cartoon mascot, and 6% of products looked like an ice cream cone or chocolate bar. The company names include allusions to opioids including "Opia" for opioids, "Perks" for Percocet, "HiJoy" for the opioid euphoric effect, and "Lucid." Some child enticing product names included "Blue Magic Riz," "Purple Ease," "Purple Rein," "Purple Hornets," "Green Giants," "Yellow Jackets," and "The OG." While two of the eight retail websites (25%) used an honor code question of whether the searcher was over the age of 18 or 21 years old to enter, the rest (75%) did not. No site required proof of age to purchase a product.

Table 2. Product features that appeal to children.

Product feature	Products, No. (% of the 51 total products)
Flavoring/scent	35 (69%)
Mascot	3 (6%)
Formulation resembles food (i.e. ice cream cone, chocolate)	3 (6%)
Label features image of food	6 (12%)
Formulation is a bright color*	32 (63%)

No.: number.
*Excludes white, black, gray, tan, and brown.

Table 3. Website and/or label key words.

Opioid key word	Products, No. (% of 51 total products)
Opioid agonist or binds to opioid receptors	5 (9)
Produces "opioid-like effects"	2 (3)
Interacts with brain receptors like opioids	3 (6)
Other key words	**Products, No. (% of 51 total products)**
Intoxication	4 (8)
Impairment	4 (8)
Dependence	10 (20)
Tolerance	5 (10)
Withdrawal	5 (10)
Addiction	9 (18)
Habit-forming	3 (6)

Several products required splitting of a tablet or liquid to provide the suggested serving size. The tablet serving size may have been a full tablet, ½ tablet, or ¼ tablet (with the tablet not being scored for ¼ tablet splitting). Liquid serving sizes ranges from ½ bottles to a sixth of the bottle (with no liquid dosing aid to precisely measure the amount). Both ice cream cones products required splitting, either into halves or thirds with no scoring on the product to demarcate a serving. The chocolate bar listed a proper serving size as ½ a square.

We searched the website descriptions and the product labeling or packaging for the 51 unique products that they sold for our prespecified key terms with results summarized in Table 3. None of the product labels or packaging had the term "opioid" but 18% of the product descriptions on the websites did. Overall, 9% of the websites described that the products bind to opioid receptors, 3% said the products had opioid-like effects, and 6% said the products interact with brain receptors like opioids. Forty-five percent of products had at least one warning on the website product descriptions or the product labels. Specifically, 20% of products used the term dependence, 18% addiction, 10% tolerance, 10% withdrawal, 8% intoxication, 8% impairment, and 6% habit forming. The impairment term was followed by instructions not to operate heavy machinery.

Discussion

This is the first descriptive evaluation of products that contain the opioid stimulating semi-synthetic alkaloid mitragynine pseudoindoxyl with or without 7-, 8-, 11-hydroxymitragynine to our knowledge (McCurdy et al., 2024; White, 2024a; Wilson et al., 2021). These products are being sold over the internet as well as in gas stations, smoke shops, and convenience stores across the United States. However, given the semi-synthetic alkaloids that they contain, they do not meet the criteria to be a dietary supplement and are experimental drugs (McCurdy et al., 2024). FDA requirements for dietary supplements require companies to identify that they are new dietary ingredients if they were not in general use before 1994 and then wait until the required timeframe for FDA review before they can be sold (FDA, 2024a). That was not done with mitragynine pseudoindoxyl products before they were sold and mitragynine pseudoindoxyl is not an alkaloid found in fresh Mitragyna speciosa leaves (McCurdy et al., 2024). Dietary supplements also cannot make actual health claims unless an authoritative body has identified sufficient evidence to substantiate it, and the seller alerts the FDA that they plan to make a health claim and wait for the required timeframe for the FDA to review the claim (FDA, 2024b). These products are being touted for real health claims, such as pain relief without FDA notification. Tianeptine, which is a tricyclic antidepressant with potent opioid receptor stimulation, and these alkaloids have been termed "gas station heroin" because of their broad accessibility and ability to induce euphoria and opioid-like use disorders (Francis, 2024; White, 2024b). There are important parallels between tianeptine and the mitragynine pseudoindoxyl alkaloid products (Smith et al., 2025). Namely, the ability to access products in unlimited quantities with no medical oversight that potently stimulate mu-opioid receptors with data suggesting that they can cause physical dependence and withdrawal symptoms (Kruegel et al., 2019; McCurdy et al., 2024; Takayama et al., 1998; Váradi et al., 2016; Yamamoto et al., 1999).

We found that many of these products would appeal to children given their flavorings and scents, ice cream cone or chocolate bar formulations, cartoon mascots, and coloring of their tablets and/or packaging (Global Kratom Coalition, 2024; White, 2024b). Only one product (2%) clearly met the criteria for being child resistant. While most of the products are in blister packs which may or may not be child resistant, 23% of products are clearly not child resistant and could lead to inadvertent exposure if young children find them in a home. Although 73% specified that the products were not for use by children, young children may not be able to read the packaging. The child-friendly formulations and packaging may also entice teens (Soneji et al., 2019). Children with access to a credit card can purchase all these products with no age verification over the internet.

Some states ban "kratom" which includes bans on semi-synthetic alkaloids like 7-, 8-, or 11-hydroxymitragynine or mitragynine pseudoindoxyl. Nine states have specific Kratom Consumer Protection Laws that allow the sale of kratom products but specifically ban semi-synthetic 7-hydroxymitragynine or mitragynine pseudoindoxyl products and a few other states are currently considering such legislation (Global Kratom Coalition, 2025). In the 45 states without "kratom" bans, only Utah would consider an ice cream cone formulation of "kratom" to be illegal (Heflin, 2023). However, it is illegal in 14 states to sell cannabis products with this formulation. Furthermore, 13 states have limitations on images and words that can appear on cannabis product packaging to avoid appealing to children but have no such limitations for mitragynine pseudoindoxyl products (The Network for Public Health Law, 2022). The Federal Trade Commission and Food and Drug Administration have sent cease and desist letters to cannabis companies selling products that look like snacks or candy, but no letters have been sent to companies

selling mitragynine pseudoindoxyl products (Federal Trade Commission, 2023).

Many of the product serving sizes are not intuitive. The standard serving size ranges from ¼ to a whole tablet, two to six servings per bottle, two servings per chocolate square, and two to three servings per ice cream cone (Gala, n.d.; Great CBD Shop, n.d.; Pure Leaf Kratom, n.d.). The product suggesting ¼ tablet servings was not scored to allow precise splitting, the six serving liquid formulation did not contain a spoon or oral syringe dosing aid, and there were no instructions on how to split the asymmetric cone three ways (Gala, n.d.; Great CBD Shop, n.d.; Pure Leaf Kratom, n.d.). People not reading the packing carefully or children with limited reading comprehension may be prone to overconsumption.

Only 18% of product websites alluded to the opioid agonist nature of the products (McCurdy et al., 2024). Less than half of products alluded to the possibility of dependence, habit formation, addiction, tolerance, withdrawal, or intoxication. Mitragynine pseudoindoxyl and 7-hydroxymitragynine solely rely on opioid agonism for their pharmacologic effects and tolerance and addiction are important potential risks that should be disclosed (McCurdy et al., 2024).

Our assessment has some limitations. We were limited to the information that the online retailers placed on their websites. We did not visit brick and mortar stores to view products or purchase products to physically test them. By only searching the Google search engine we might have missed some products identified on other search engines or retail stores. Blister packs for over the counter and prescription products meet the criteria for being child-resistant but we did not physically purchase the mitragynine pseudoindoxyl products or test them to assure they did not have any easy open features or others that could render them not child resistant.

Conclusion

Products containing the semi-synthetic opioid receptor stimulating alkaloid mitragynine pseudoindoxyl, with or without 7-, 8-, or 11-hydroxymitragynine, are readily available through online retailers, even though they are not dietary supplements or approved drugs. Several of those products are designed in ways that may be alluring to small children and teens and do not have child resistant packaging. There are no firm restrictions for children with a credit card to purchase these products online. The serving sizes are not intuitive, risking an up to 4- to 6-fold increase in dosage by taking one tablet or the contents from an entire shot sized bottle. Since products with these alkaloids may cause physical dependence and withdrawal, preventing children from accessing these products is important and those risks should be disclosed but frequently are not.

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None.

Ethical approval

Not applicable.

Declaration of interest

C. Michael White, Pharm.D. is the inaugural chair of the nonprofit Kratom Consumer Advisory Council. He provides pharmacology and health policy expertise to the nonprofit group of kratom consumers who have either been harmed or helped by kratom products and develop position statements about protecting consumer safety and product transparency.

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Data availability statement

All source data is provided in Supplementary Tables.

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Article

Avoiding False Identification of 7-Hydroxymitragynine in Kratom Products Using a Multicriteria LC−MS Confirmation

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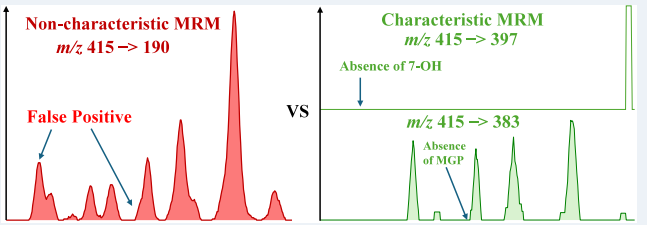
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ABSTRACT: Kratom (*Mitragyna speciosa*) products have become widely available in diverse commercial formulations, raising increasing analytical and regulatory interest due to the presence of potent alkaloids such as mitragynine (MG), 7-hydroxymitragynine (7-OH), and mitragynine pseudoindoxyl (MGP). The accurate identification of these compounds is analytically challenging because kratom matrices contain numerous structurally related alkaloids and isomeric species that produce similar mass spectral features. In this study, an ultrahigh-performance liquid chromatography−high-resolution mass spectrometry (UHPLC−HRMS) workflow was developed to characterize kratom alkaloids in authentic leaf materials and 38 commercial kratom products representing multiple formulations. Chromatographic separation combined with accurate mass measurement and diagnostic MS/MS fragmentation enabled reliable differentiation of MG, 7-OH, MGP, and related isomers. Multiple overlapping chromatographic peaks were observed at m/z 415.2227, demonstrating that reliance solely on accurate mass or commonly monitored transitions such as m/z 415 → 190 can lead to false-positive identification of 7-OH in complex botanical matrices. Application of the multicriteria confirmation strategy revealed substantial variability in alkaloid composition across commercial products, including the presence of 7-OH and MGP in several tablet and capsule formulations. Untargeted metabolomic profiling further distinguished kratom plant extracts, including the authentic leaf materials, from processed products enriched in specific alkaloids. These results highlight the importance of integrating chromatographic resolution with orthogonal MS/MS criteria for reliable identification of kratom alkaloids and provide an analytical framework for the characterization of complex botanical products.



KEYWORDS: *kratom, LC−MS, mitragynine, 7-hydroxymitragynine, mitragynine pseudoindoxyl*

INTRODUCTION

Kratom (*Mitragyna speciosa*) has rapidly emerged as a widely consumed psychoactive product in the United States. However, it is not approved by the U.S. Food and Drug Administration (FDA) as a drug, dietary supplement, or food ingredient,[1] and its regulatory status remains complex and evolving. Although kratom is not federally scheduled, several U.S. states, including Alabama, Arkansas, Indiana, Rhode Island, and Wisconsin, have enacted full bans, and additional jurisdictions have pending legislation to restrict or prohibit its sale and use. At the federal level, the FDA has issued multiple warnings regarding kratom's safety and has expressed intent to restrict or prohibit products containing 7-hydroxymitragynine (7-OH) due to toxicity concerns.[2,3] In December 2025, the U.S. Marshals, acting in coordination with the FDA, seized approximately 73,000 units of 7-OH products, including tablets, liquid shots, and gummies, valued at nearly $1 million, from firms in Missouri.[4] Meanwhile, the Drug Enforcement Administration (DEA) previously announced its intent to place kratom's major alkaloids, mitragynine and 7-OH, into Schedule I.[5] These regulatory uncertainties underscore the need for rigorous chemical characterization and analytical surveillance of kratom products currently available in the U.S. marketplace.

Compounding the regulatory uncertainty is the rapid expansion, diversification, and accessibility of kratom products to consumers. Kratom is sold in numerous forms, including live plants, raw plant powders, capsules, tablets, resin extracts, concentrated liquids, gummies, vape liquids, and "enhanced" or "extract" products, including those containing semisynthetic derivatives of 7-OH. These items are readily accessible both in-person and online, commonly sold in gas stations, smoke shops,

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convenience stores, and through e-commerce retailers with minimal to no age restrictions.[6] Many products are marketed as "natural," "herbal," or "wellness" supplements despite lacking standardization, quality control, or regulatory oversight. This variability increases the likelihood of exposure to highly heterogeneous and potentially modified products, presenting challenges for forensic analysis, toxicological interpretation, and regulatory monitoring. Accurate identification of kratom alkaloids within these complex matrices remains analytically challenging due to the presence of structurally similar and isomeric compounds.

Kratom's growing popularity is largely driven by perceptions of therapeutic benefit and in part due to its promotion as a natural psychoactive agent. Mitragynine, the most abundant natural alkaloid in kratom leaves, is currently under active investigation as an atypical opioid with reduced respiratory depression and gastrointestinal side effects compared to classical opioids.[7−9] Early data suggest that mitragynine-based therapeutics may offer safer alternatives for pain management and opioid withdrawal, with the potential to improve treatment adherence and completion rates in opioid detoxification programs.[9] At the community level, kratom is widely used informally to self-manage opioid withdrawal, reduce opioid consumption, and alleviate dependence symptoms. Among polydrug users, kratom use has also been associated with reduced methamphetamine use, supporting its potential harm-reduction value. Conversely, adverse effects, including death, have been reported for kratom alone as well as its use with other addictive substances, most notably conventional opioids.[10] Of increasing concern is the emergence of semisynthetic and chemically modified kratom products. While natural kratom leaves contain no or very low levels of 7-OH[11] and virtually no detectable mitragynine pseudoindoxyl (MGP), some commercial "extract," "enhanced," or tablet-based products contain high concentrations of these compounds.[12−14] These molecules exhibit substantially greater μ-opioid receptor potency, with MGP producing marked respiratory depression comparable to morphine. Such potency differences highlight critical pharmacological distinctions between natural kratom alkaloids and semisynthetic derivatives, which increase the risk of overdose, dependency, and toxicity associated with highly concentrated kratom products. Consumers purchasing products labeled as "kratom" or "7-OH kratom" may unknowingly ingest substances with pharmacological effects more closely aligned with potent synthetic opioids than with natural kratom. This discrepancy raises substantial concerns for consumer safety, misbranding, overdose risk, and toxicological interpretation. It also highlights a pressing regulatory problem: chemical modifications can fundamentally alter a product's risk profile, yet these products remain available on the commercial market with inconsistent and unclear enforceable standards.

Given the complexity of modern kratom products, robust analytical tools are essential for accurate identification, quantification, and differentiation of natural versus semisynthetic constituents. Multiple kratom alkaloids share structural features, stereochemistry, and fragmentation pathways, complicating mass spectral interpretation and quantification.[15−17] Diastereomeric and positional isomers, such as mitragynine, speciociliatine, speciogynine, and mitraciliatine, require chromatographic resolution for accurate identification. Liquid chromatography−mass spectrometry (LC−MS) has emerged as a reliable analytical platform for kratom alkaloid characterization owing to its high sensitivity, selectivity, and ability to resolve structurally similar indole alkaloids. LC−MS enables simultaneous identification and quantification of multiple kratom alkaloids, detection of semisynthetic analogues, and confirmation of product authenticity, making it indispensable for forensic casework, toxicology, regulatory monitoring, and product standardization.[17] However, accurate identification of kratom alkaloids remains analytically challenging because many indole alkaloids share similar or exact masses and fragmentation pathways. Despite widespread use, commonly monitored transitions such as m/z 415 → 190 for 7-OH or MGP lack structural specificity and may lead to false-positive identification in complex botanical matrices. To date, this limitation has not been systematically demonstrated. In this study, we provide direct chromatographic and MS/MS evidence of such misidentification and propose a multicriteria confirmation strategy to overcome this limitation.

We surveyed chemical profiles of 38 commercial kratom products, including plant powder, powder extract, capsules, tablets, gummies, and drink mix, and compared them with authentic kratom leaves using LC−MS. We identified the critical identification challenges for 7-OH and MGP and proposed a comprehensive approach to confidently confirm the identities of both highly concerning compounds in the commercial kratom products. Lastly, we describe the inclusion of multiple compound features for reliably distinguishing naturally sourced kratom products from chemically modified kratom products. The proposed identification framework relies on chromatographic resolution and diagnostic MS/MS fragments, making the approach applicable to a wide range of LC−MS platforms used in routine testing laboratories.

MATERIALS AND METHODS

Chemicals and Materials

LC−MS grade acetonitrile (ACN), methanol, formic acid, and HPLC grade water were obtained from Fisher Scientific, Inc. (Fair Lawn, NJ, USA). Mitragynine (MG), 7-hydroxymitragynine (7-OH), and mitragynine pseudoindoxyl (MGP) standards were purchased from Cayman Chemical (Ann Arbor, MI, USA). QuEChERS extract pouch was purchased from Agilent Technologies (Palo Alto, CA, USA).

Sample Preparation

A total of 39 commercial kratom products were obtained from the U.S. market, including 6 plant powder samples, 6 liquid plant extract samples, 9 tablet samples, 10 capsule samples, 4 gummy samples, 2 soft-gel samples, 1 sparkling water mix, and 1 drink-mix sample. In addition, one botanically authenticated kratom voucher specimen (AHP-Verified) was obtained from the American Herbal Pharmacopoeia (AHP) and is deposited in the herbarium collection of AHP. The details of these products are provided in Table S1. KM35 was a sparkling water mix sample containing kratom that was initially processed using the same preparation protocol as the drink powder samples. However, due to the low alkaloid concentrations in this sample, the applied dilution and extraction conditions were not suitable for reliable detection. Therefore, KM35 was excluded from the study.

Given the distinct physical and chemical characteristics of the matrices, matrix-specific sample-preparation procedures were employed for plant, gummy, tablet, and drink-mix samples, as previously described.[18] *Plant powder, tablet (crushed to fine powder with a mortar and pestle prior to extraction), powder from capsule, drink-mix and liquid plant extract samples*: One hundred milligrams of the sample were extracted with 10 mL of a 3:7 mixture of water and methanol with 2 mM glacial acetic acid by vortexing for 1 min, followed by vigorous shaking at 300 rpm for 20 min. After settling, 1 mL of the supernatant was filtered through a 0.22-μm nylon syringe filter. *Gummy and soft-gel samples*: One hundred milligrams of the finely chopped sample were extracted with 10 mL of water, followed by the addition of 10 mL of

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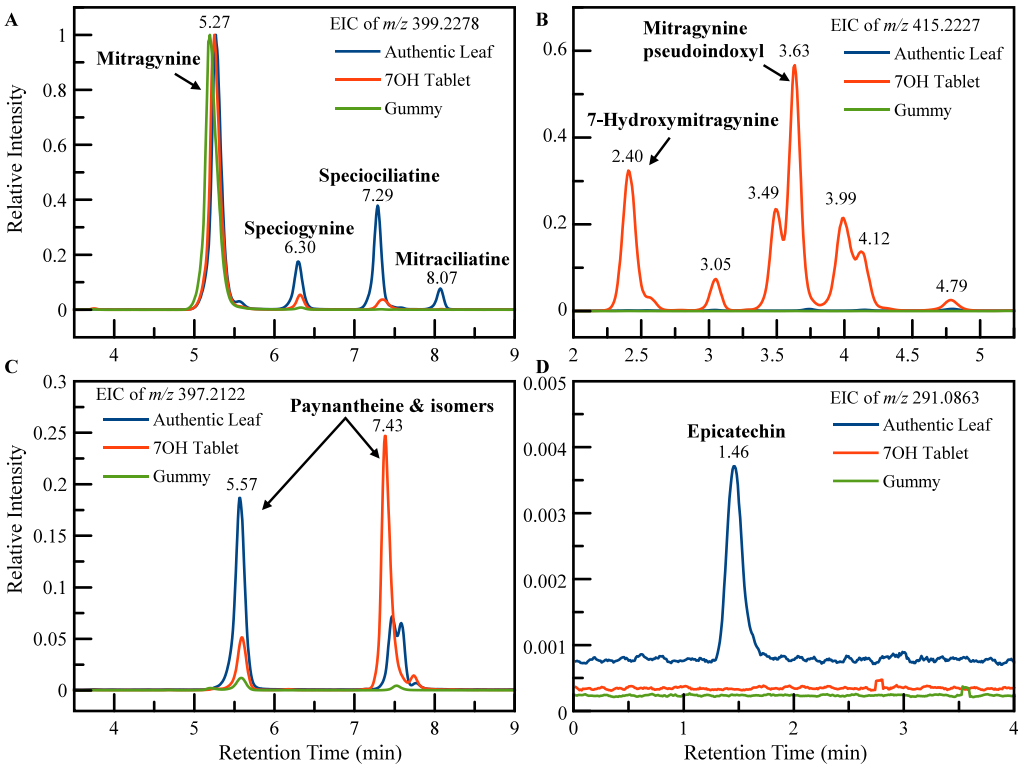


Figure 1. EICs of m/z 399.2284 (A), m/z 415.2227 (B), m/z 397.2122 (C), and m/z 291.0863 (D) for a representative authentic leaf sample (KM40), 7-OH tablet sample (KM03), and gummy sample (KM30). All chromatographic peak intensities were normalized to the peak height of MG within each sample.

ACN. The mixture was vortexed for 1 min and sonicated for 20 min. A QuEChERS extraction pouch (containing 4 g magnesium sulfate and 1 g sodium chloride) was then added, and the mixture was vortexed for an additional 1 min. After settling for 10 min, the ACN layer was collected and filtered. The samples were diluted 100-fold prior to LC–MS analysis. All samples were prepared and extracted in triplicate.

Unit-to-unit variability was evaluated by extracting 7-OH, MGP, and MG from five individual tablets (KM05, KM06, and KM18) or capsules (KM31 and KM33) (n = 5) using the aforementioned protocol.

LC–MS Instrumentation

All samples were analyzed using an ultrahigh-performance liquid chromatography–high-resolution mass spectrometry (UHPLC–HRMS) system consisting of a Vanquish UHPLC coupled to an Orbitrap Exploris 120 mass spectrometer (Thermo Scientific, Waltham, MA, USA). Chromatographic separation was performed on a Hypersil GOLD PFP (pentafluorophenyl) HPLC column (2.1 × 100 mm, 1.9 μm; Thermo Scientific, Waltham, MA, USA). The mobile phase consisted of water containing 0.1% formic acid (FA) and 2 mM ammonium acetate (mobile phase A) and methanol containing 0.1% FA and 2 mM ammonium acetate (mobile phase B). The gradient elution program was as follows: 0–0.5 min, 35–45% B; 0.5–3 min, 45% B; 3–10 min, 45 to 80% B; 10–11 min, 80 to 100% B; 11–11.8 min, hold at 100% B; 11.8–12 min, decrease to 35% B; and 12–14 min, hold at 35% B. The column temperature was maintained at 40 °C, with a flow rate of 0.3 mL/min and an injection volume of 2 μL. Full-scan MS and targeted MS^2 ($t\text{-}MS^2$) data were acquired using a heated electrospray ionization (H-ESI) source in positive mode over a mass range of m/z 150–900, with a resolving power of 60,000. The MS source parameters were set as follows: spray voltage, + 3.5 kV; ion transfer tube temperature, 350 °C; vaporizer temperature, 275 °C; sheath gas, 55 arbitrary units (Arb); auxiliary gas, 15 Arb; and sweep gas, 3 Arb. Targeted MS^2 scans were acquired with a resolution of 15,000 to monitor precursor ions at m/z 291.0869, 353.1865, 369.1814, 383.1971, 385.2127, 397.2127, 399.2284, 415.2233, and 417.2383 over a 14 min run without retention-time scheduling. These precursor ions were selected to represent key alkaloids and flavonoids commonly observed in kratom samples, including MG, 7-OH/MGP, mitraphylline, paynantheine, rhynchophylline, ajmalicine, corynoxeine, epicatechin, and their isomeric analogs.[20] Each sample extract was analyzed in triplicate.

Data Processing

In total, 351 LC–MS data objects were collected (39 samples × 3 extractions × 3 injections = 351). One data object (KM33, Extract No. 1, Injection No. 1) exhibited significant retention time drift due to inadequate HPLC equilibration and was therefore excluded from the data set. The LC–MS raw data (n = 350) were preprocessed with MZmine software (version 4.8.30)[19] for peak detection, alignment, peak peaking, and integration. Preprocessed data with 2784 features were generated and exported to a CSV file in Excel as a two-dimensional data matrix, including variable indices such as retention time, m/z (mass-to-charge ratio), peak area, and ion intensity. Potential alkaloids and secondary metabolites precursor ion features, including m/z 195.0882, 315.2324, 291.0869, 397.2127, 399.2284, 385.2127, 415.2227, 383.1971, 353.1865, and 369.1814, were selected and then used for statistical analysis.[20] Ions at m/z 195.0882, 315.2324 were also included to represent caffeine and cannabidiol (CBD), respectively, due to their presence in some kratom samples. Principal component analysis (PCA) score, loading plot, and box plots were generated with MetaboAnalyst 6.0.[21] The extracted ion chromatogram (EIC) was exported through Xcalibur 4.2 (Thermo Scientific, San Jose, CA, USA) and then plotted using MagicPlot 3.0.1 (Magicplot Systems, LLC, Saint Petersburg, Russia).

RESULTS AND DISCUSSION

Chromatography Profiles for the Authentic Leaf, 7-OH Tablet, and Gummy Samples

Representative EICs for a 7-OH tablet sample (KM03), a gummy formulation (KM30), and an authentic leaf-cut sample (KM40) are shown in Figure 1. All chromatographic peak

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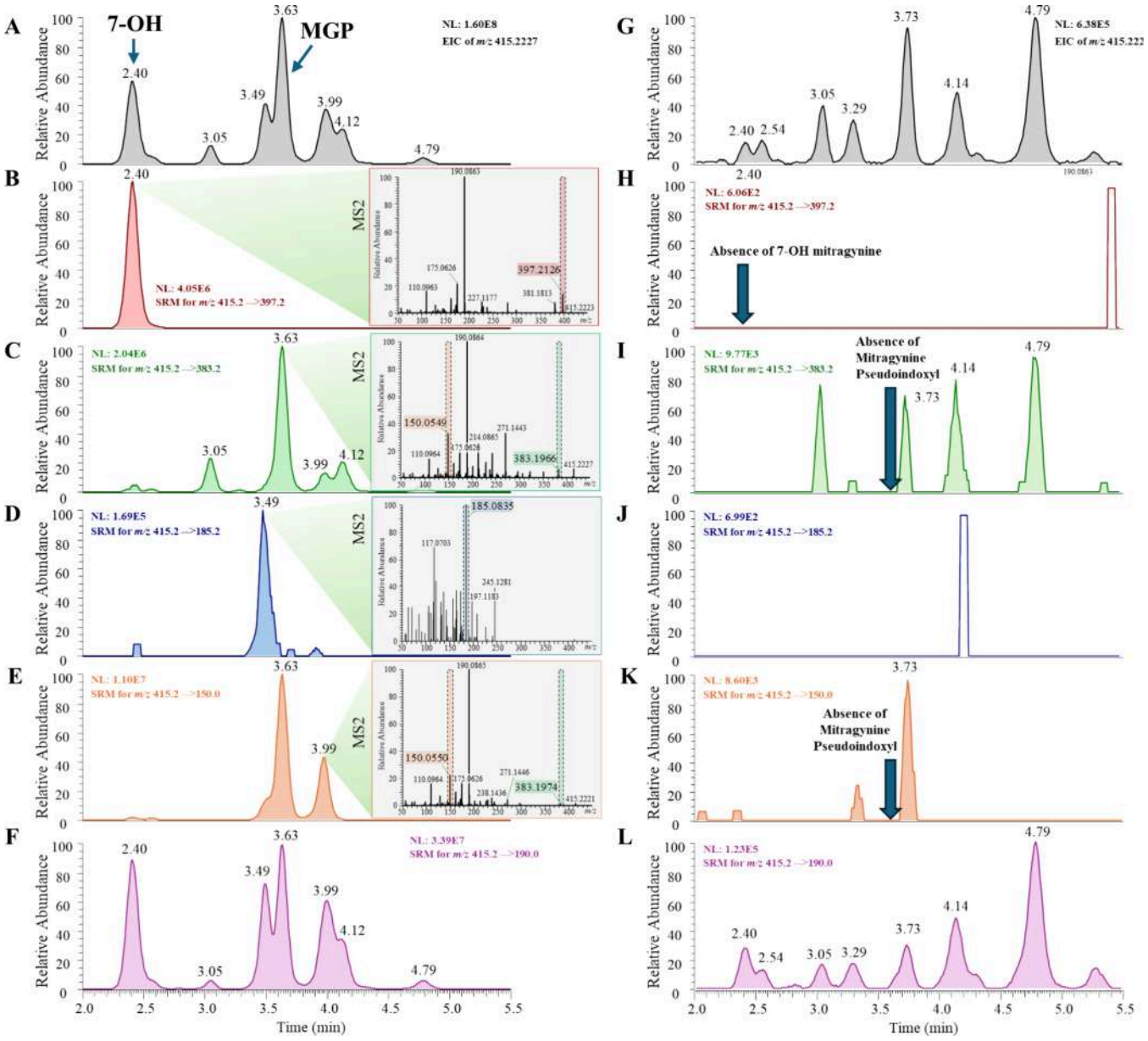


Figure 2. EIC of m/z 415.2227 for 7-OH/MGP isomers for tablet sample (KM03) (A) and authentic leaf sample (KM40) (G) with their respective selected ion monitoring (SRM) chromatograms in panels B−F and H-L. MS^2 spectra of 7-OH, MGP, and two isomeric peaks eluting at 3.49 and 3.99 min in the tablet sample (KM03) are embedded in their characteristic SRM chromatograms in panels B, C, D, and E, respectively, with characteristic fragment ions highlighted. Noncharacteristic SRM transition of m/z 415 → 190 for tablet sample (KM03) is shown in panel F. Corresponding SRM for the authentic leaf sample (KM40) are shown in panels H, I, J, K, and L respectively. All MS^2 spectra were collected via higher-energy collisional dissociation (HCD) at a normalized collision energy of 50.

intensities were normalized to MG to facilitate comparison of relative alkaloid distributions across matrices.

From the EIC of m/z 399.2284 (for $[C_{23}H_{30}N_2O_4+H]^+$, ± 0.001 Da), four alkaloid diastereomers, MG, speciogynine, speciociliatine, and mitraciliatine, were well resolved under the optimized chromatographic conditions (Figure 1A). In authentic leaf samples, speciogynine, speciociliatine, and mitraciliatine were present at substantial relative abundances (~18%, 40%, and 10% of MG peak height, respectively), consistent with previously reported authentic kratom leaf profiles.[22] In contrast, the 7-OH tablet product (KM03) exhibited MG as the dominant peak with markedly reduced relative abundances of the other diastereomers, while the gummy sample (KM30) displayed MG as the only detectable compound at this m/z value. These differences indicate a simplified alkaloid profile in certain processed commercial formulations relative to intact plant material. The chromatographic complexity was more pronounced at m/z 415.2227 (for $[C_{23}H_{30}N_2O_5+H]^+$, ± 0.001 Da) (Figure 1B). The precursor ion at m/z 415.2227 corresponds to multiple isomeric alkaloids, including 7-OH and MGP, as well as additional structurally related compounds. In the 7-OH tablet extract (KM03), seven distinct chromatographic peaks were detected within the 2−5 min retention window. Using authentic reference standards, two of these peaks were confirmed as 7-OH (2.40 min) and MGP (3.63 min). Notably, a peak at 3.49 min partially coeluted with MGP (3.63 min), highlighting the risk of misidentification in the absence of sufficient chromatographic resolution and orthogo-

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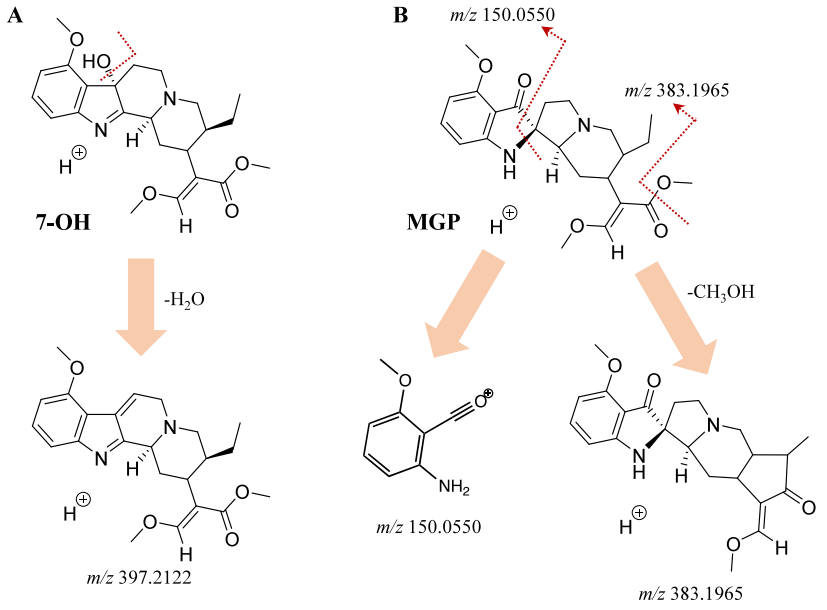


Figure 3. Formation of characteristic ions, including m/z 397.2122 for 7-OH (A) and m/z 150.0550 and m/z 383.1965 for MGP (B) in the MS2 spectrum via HCD.

nal confirmation. In contrast, neither 7-OH nor MGP was confirmed in the authentic leaf sample (KM40) or the gummy formulation (KM30) under the validated confirmation criteria described in the next section. Although several low-intensity peaks were observed at m/z 415.2227 in the authentic leaf extract, none matched both the retention time and MS/MS characteristics of authentic 7-OH or MGP (detailed discussion available in the next section and in Figure 2). From the EIC of m/z 397.2122 (for $[C_{23}H_{28}N_2O_4+H]^+$, ± 0.001 Da) (Figure 1C), two major peaks corresponding to paynantheine and related isomers were observed at 5.57 and 7.43 min. Both were present in authentic leaf and commercial products; however, relative intensities differed. The 5.57 min peak was more prominent in authentic leaf samples and less prominent in the 7-OH tablet product, suggesting an alteration in alkaloid distribution associated with processing or formulation. Epicatechin (m/z 291.0863 for $[C_{15}H_{14}O_6+H]^+$, error: −2.1 ppm, at 1.46 min) was detected exclusively in the authentic leaf sample (Figure 1D), consistent with the presence of native nonalkaloid phytochemicals in intact plant material and their reduction or removal in processed products.[23]

Collectively, the metabolite profiles for the authentic leaf cut samples showed strong similarities with the profiles in Figure S1, which was the reassembly of EIC profiles from the raw data of an authentic kratom sample (K-55B) previously published by Cech's group.[22] However, substantial variability was observed among the 39 commercial products surveyed, particularly in the distribution and relative abundance of alkaloids at m/z 415.2227, which will be discussed in detail in the next section.

Rigorous Confirmation of 7-OH and MGP

Accurate identification of 7-OH and MGP is critical for distinguishing intact botanical material from products potentially containing enriched or semisynthetic constituents. Accurate mass detection at m/z 415.2227 alone revealed multiple chromatographic peaks in the 7-OH tablet (Figure 2A) and authentic leaf extract samples (Figure 2G), underscoring the complexity of the kratom matrix.

MS/MS spectra were therefore examined for each detected peak. 7-OH exhibited a characteristic fragment ion at m/z 397.2126 (Figure 2B), corresponding to neutral loss of H_2O from the protonated molecule, consistent with the presence of a hydroxyl substituent ($[C_{23}H_{28}N_2O_4+H]^+$, mass error: 1 ppm, Figure 3A).[24] MGP displayed characteristic fragment ions at m/z 150.0550 ($[C_8H_7NO_2+H]^+$, mass error: −0.7 ppm) assigned as an even-electron fragment formed through rearrangement-assisted cleavage of the indole-containing moiety, and at m/z 383.1965 ($[C_{22}H_{26}N_2O_4+H]^+$, mass error: 0.3 ppm) via the neutral loss of methanol from the methyl ester group, respectively (Figures 2C and 3B). In contrast, the partially coeluting peak at 3.49 min produced a distinct fragment at m/z 185.0835 (Figure 2D), while the peak at 3.99 min exhibited fragmentation resembling MGP but with different retention behavior and relative ion intensity ratios (Figure 2E). Selected reaction monitoring (SRM) transitions targeting unique fragment ions enabled complete differentiation of MGP from the partially coeluting 3.49 min peak. These findings demonstrate that accurate mass measurement and monitoring of a single fragment ion are insufficient for unambiguous identification in complex matrices.

Importantly, much of the existing LC−MS/MS literature has monitored 7-OH using the transition m/z 415 → 190, as m/z 190 is frequently reported as a high-abundance product ion.[25−27] While m/z 190 is indeed the most intense fragment for 7-OH in our experiment, it is not structurally unique and may be produced by other closely related alkaloid isomers in kratom extracts. The transition m/z 415 → 190 provides no additional specificity to differentiate 7-OH/MGP alkaloid isomers, as all the peaks from EIC of m/z 415.2227 in Figure 2A are also observed in Figure 2F.

For the authentic leaf extract (KM40), a peak at 2.40 min was observed in the EIC at m/z 415.2227 (Figure 2G), which exactly matched the 7-OH retention time. Even when the m/z 415 → 190 transition was evaluated, a peak at approximately 2.4 min was still present in the authentic leaf extract (Figure 2L). However, this peak did not meet the defined confirmation criteria established with authentic standards, including reten-

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tion-time matching and the presence of characteristic secondary fragment ions, such as m/z 397.2126 (Figure 2H). If identification had relied solely on the high-resolution mass accuracy of m/z 415.2227 and/or uncharacteristic m/z 415 → 190 transition, even when retention time matched that of the authentic standard, this peak could have been misassigned as 7-OH, resulting in a false positive identification. These observations demonstrate that reliance solely on the m/z 415 → 190 transition and retention time matching, which has been overwhelmingly used in the existing literature, can lead to overestimation or misidentification of 7-OH in crude leaf matrices. Incorporation of chromatographic resolution, high-resolution mass accuracy, and secondary diagnostic fragment ions substantially increases analytical specificity and reduces the likelihood of false positive assignments.

The exclusion of MGP in the authentic leaf extract samples was confirmed by the absence of a chromatographic peak at 3.63 min under its characteristic m/z 415 → 383 transition (Figure 2I). Similarly, both compounds corresponding to the peaks at 3.49 and 3.99 min in the tablet sample (Figure 2D,E) were confirmed absent in the authentic leaf extract (Figure 2J,K). In addition, the peak intensities at m/z 415.2227 for the authentic leaf extract were approximately 250-fold lower in intensity than those observed in the 7-OH tablet extract. None of these peaks matched both the retention time of authentic 7-OH or MGP and the diagnostic fragment ions required for confirmation. These findings indicate that, under rigorous chromatographic and MS/MS confirmation criteria, 7-OH and MGP were not detected in the analyzed authentic leaf sample. While mitragynine was identified in all 39 samples, 7-OH was confirmed present in 3 capsules and all 9 tablet samples, and 8 out of 9 tablets and 2 capsules contained MGP. The combined use of retention time matching, accurate mass measurement, and diagnostic product ions is necessary in both confirming presence and avoiding false-positive identifications.

To our knowledge, this study represents the first systematic demonstration of the potential for false-positive identification of 7-OH in crude kratom matrices when relying solely on commonly monitored transitions, such as m/z 415 → 190, and retention-time matching. The analytical framework presented here provides improved specificity for distinguishing confirmed 7-OH and MGP from isomeric and coeluting interferences in complex botanical samples and consumable products. The proposed multicriteria confirmation strategy consists of (i) retention time matching with authentic standards, (ii) accurate mass measurement, and (iii) the presence of multiple diagnostic fragment ions. While accurate mass measurement is enabled by HRMS instrumentation, the identification specificity in this workflow is primarily driven by chromatographic resolution and structurally informative MS/MS fragments (e.g., m/z 397.2122 for 7-OH and m/z 150.0550 and 383.1965 for MGP). These fragment ions can be monitored using SRM transitions on triple quadrupole instruments, enabling implementation of this strategy in routine analytical laboratories without HRMS capability. The recommended confirmation criteria for 7-OH and MGP are summarized in Table 1. This broadens the practical applicability of the approach for laboratories performing regulatory, forensic, or quality control testing of botanical products.

Table 1. Confirmation Criteria for 7-OH and MGP

compound	RT (min)	precursor (m/z)	diagnostic fragments (m/z)	criteria
7-OH	2.40	415	397	RT + fragment
MGP	3.63	415	150 or 383	RT + fragment

Multivariate Metabolomic Profiling and Sample Classification

To complement the targeted alkaloid analysis, a metabolomic workflow was applied to evaluate whether overall metabolite feature distributions could distinguish vegetation leaf powders and powder extracts, including the authentic leaf materials, from commercial kratom products containing 7-OH and related constituents. High-resolution full-scan LC−MS data were processed to extract molecular features across all samples, followed by alignment, normalization, and filtering to remove background and low-reproducibility signals. PCA of the aligned feature matrix revealed clear separation between authentic leaf materials and 7-OH-enriched commercial products (Figure 4A). PC1 and PC2 explained 74.8% and 15.7% of the total variance, respectively. The authentic leaf samples clustered tightly and overlapped with commercial kratom herb extracts and powder samples, indicating relatively consistent metabolomic profiles characterized by a complex distribution of alkaloids and nonalkaloid phytochemicals. In contrast, the 7-OH tablet samples formed a distinct cluster, reflecting a simplified compositional profile dominated by mitragynine and 7-OH, with reduced representation of minor alkaloids and other secondary metabolites. The gummy formulation samples occupied an intermediate position in the score plot. While lacking detectable 7-OH under validated confirmation criteria, their metabolite profiles differed from authentic leaf materials, consistent with processing, formulation additives, and potential selective extraction of alkaloids. Loading analysis (Figure 4B) indicated that features corresponding to natural indole alkaloids such as MG, speciociliatine, and paynantheine isomer, contributed strongly to the separation of authentic leaf samples, whereas elevated 7-OH intensity was a primary contributor to the differentiation of tablet products. These findings demonstrate that multivariate metabolomic profiling provides complementary evidence supporting the compositional distinctions observed in targeted analyses.

To further characterize compositional variability, selected metabolite features were examined using box plot visualization across sample groups (Figure 5). Authentic leaf along with kratom herb extracts and powder samples displayed a broader distribution of minor alkaloid features, including diastereomeric compounds at m/z 399.2284 and related indole derivatives. In contrast, 7-OH tablet products exhibited significantly elevated relative abundance of features corresponding to 7-OH, with reduced variability among other minor alkaloids. This compositional contraction is consistent with selective enrichment or concentration of specific alkaloids relative to the native plant matrix. The gummy products showed lower total feature counts and reduced abundance of nonalkaloid phytochemicals, consistent with removal or degradation during processing and formulation. Importantly, features contributing most strongly to group separation were not limited to 7-OH alone. Rather, classification was driven by the combined distribution of multiple metabolite features (i.e., epicatechin), indicating that global compositional patterns provide a more robust framework

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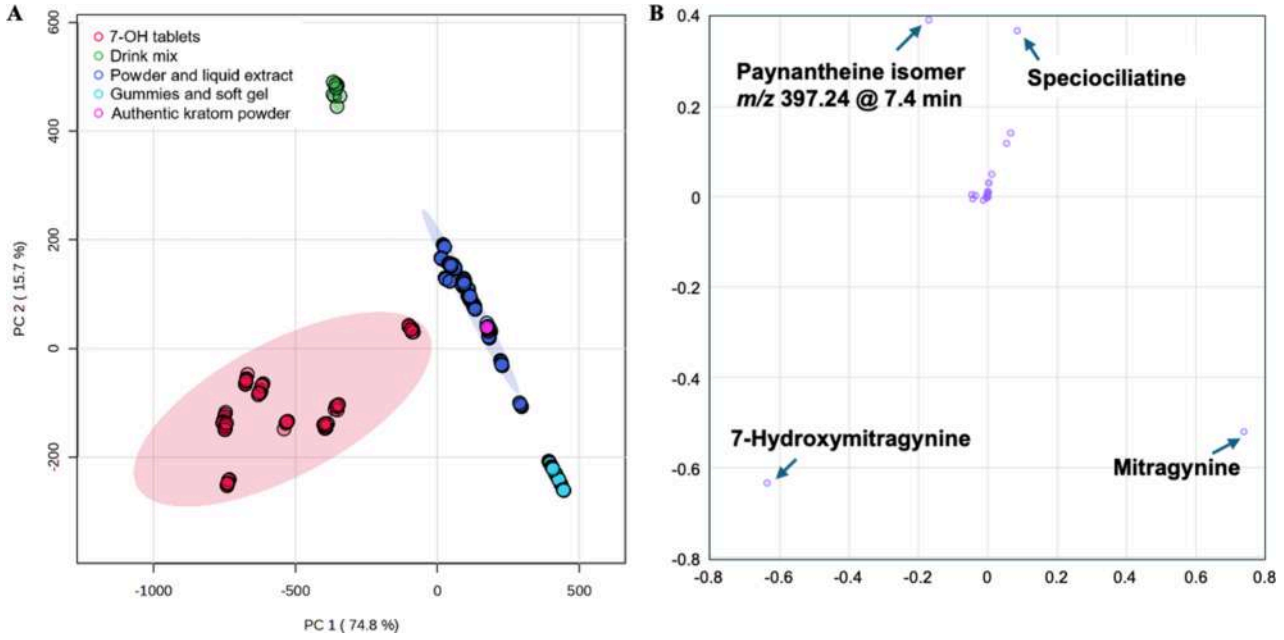


Figure 4. PCA score plot (A) and loading plot (B) for the kratom products based on 350 LC−MS runs.

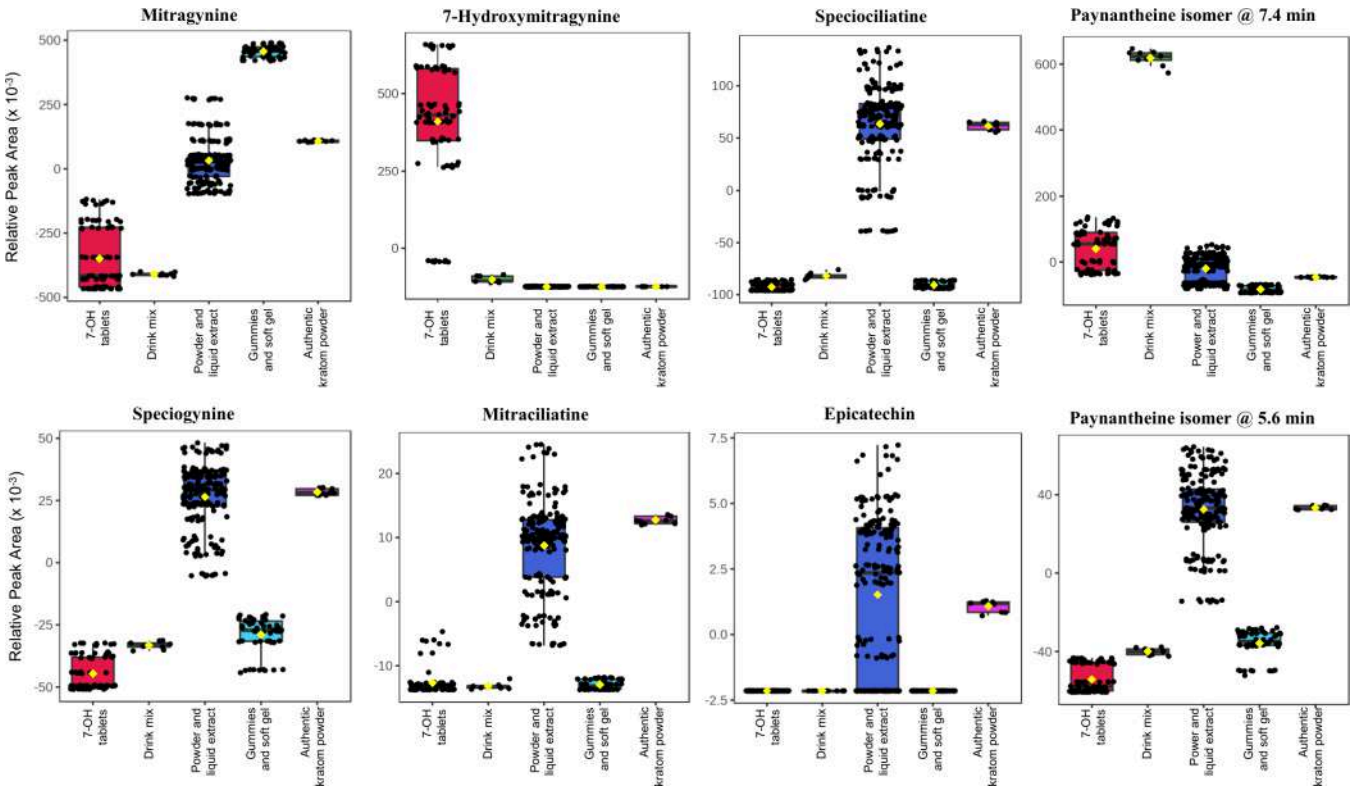


Figure 5. Relative peak areas of discriminative features for various groups of kratom samples. The peak areas were normalized to the sum of all the feature peak areas and then mean-centered.

for differentiation than reliance on a single marker compound. These results demonstrate that untargeted metabolomic profiling provides orthogonal evidence supporting the targeted LC−MS identification results.

Semiquantitative Analysis of 7-OH, MGP, and MG in the Samples

Semiquantitative analysis of MG, 7-OH, and MGP was performed using external calibration curves constructed for each analyte within the dynamic range of 0.1−5 μg mL^{-1}. Calibration curves for MG, 7-OH, and MGP exhibited good linearity with representative R^2 values exceeding 0.995 (Figure S2). These results indicate acceptable analytical performance for semiquantitative comparison across samples. The limits of detection (LOD) were determined by serial dilution of the standards until the signal intensity reached three standard deviations above the baseline noise. The LODs were estimated

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to be 1 ng mL^{-1} for 7-OH, 0.5 ng mL^{-1} for MGP, and 1 ng mL^{-1} for MG. The limit of quantitation (LOQ) was defined as the lowest calibration concentration (0.1 μg mL^{-1}), for which signal-to-noise ratios were substantially greater than 10, ensuring reliable quantification.

The concentrations of individual alkaloids in the samples were first calculated in units of mg g^{-1} using the following relationship:

$$\frac{P_0 - b_0}{m_0} \times \frac{V}{W} \times \frac{D}{100}$$

where P_0 represents the peak area of the target analyte in the sample chromatogram, b_0 and m_0 correspond to the intercept and slope of the calibration curve, respectively, V is the volume of the extracted solution (mL), W is the sample weight (g), and D represents the dilution factor. The calculated concentrations were subsequently converted to mg per serving using the serving size specified on the product label.

Among the 9 tablet products analyzed, 7-OH was detected in 7 samples, and approximately half of these contained levels 20−79% lower than those declared on the product labels (Table S2). Values in Table S2 reported as N.D. indicate concentrations below LOD, while N.Q. indicates concentrations detected but below LOQ. In addition, three capsule products (KM24, KM25, and KM26) were found to contain detectable levels of 7-OH. Notably, one capsule product (KM25) labeled as "made from 100% pure kratom leaves" contained 0.14 ± 0.03 mg per serving of 7-OH, whereas the remaining two capsule samples contained trace levels that were more than 10-fold below LOQ, preventing reliable quantification. Notably, detectable 7-OH was observed almost exclusively in tablet formulations, whereas plant powder, liquid extracts, gummies, and most capsule products contained no detectable 7-OH. This distribution further supports the hypothesis that highly processed formulations may involve enrichment or chemical modification of kratom alkaloids relative to the native plant matrix.

MGP was labeled on only one product (tablet sample KM03), yet it was detected in seven additional tablet products and two capsule products. In KM03, the measured MGP concentration was 3.8 ± 0.4 mg per serving, compared with a labeled value of 7.5 mg per serving. Three other tablet products (KM05, KM06, and KM08) contained 0.24 ± 0.01, 0.20 ± 0.01, and 0.28 ± 0.01 mg per serving, respectively. In the remaining products in which MGP was detected, concentrations were below the LOQ but were still confidently confirmed using the multitier identification criteria described earlier. These findings suggest that MGP may be more widely present in commercial kratom products than indicated by product labeling, though there is no requirement to list constituent profiles.

The MG content exhibited substantial variability across commercial products, ranging from not quantifiable (<LOQ) to 312 mg per serving (Table S2), whereas the authentic leaf-cut sample contained 6.3 mg g^{-1} of MG. Only one tablet product (KM03) reported an MG content on the label (7.5 mg per serving), and the measured value was consistent with this claim at 8.4 ± 0.4 mg per serving. In contrast, MG concentrations in the other 8 tablet samples were much lower, ranging from not quantifiable to 1.71 mg per serving, although their labels did not specify MG content. In contrast to the tablet products, which generally contained very low levels of MG, liquid extracts, gummies, and gel formulations exhibited substantially higher MG concentrations, in several cases exceeding 100 mg per serving. This variability indicates large differences in extraction efficiency, formulation concentration, and potential product standardization across commercial kratom products. In several products, particularly concentrated extract and gummy formulations, the measured MG content substantially exceeded the labeled amount (e.g., KM14, KM16, KM28, KM29, and KM30), suggesting inconsistent manufacturing practices or inaccurate labeling of kratom alkaloid content.

Unit-to-unit variability was assessed by analyzing five individual tablets or capsules and expressed as relative standard deviation (RSD), and the results are summarized in Table S3. The 7-OH content showed moderate tablet-to-tablet variability, with RSD values ranging from 10−17% (n = 5). In contrast, MGP exhibited greater variability, with RSD values of 24−27%. The variability of MG in capsule products was product-dependent, with one product showing relatively low variability (10% RSD) and another showing substantially higher variability (48% RSD). Overall, these results highlight significant heterogeneity in alkaloid content among individual units, particularly for MGP and certain capsule formulations, which may have implications for dosing consistency and potential toxicological risks in consumer use. It should be noted that the concentrations reported here may represent conservative estimates, as external calibration without matrix-matched standards can lead to underestimation due to ion suppression effects during LC−MS analysis. The quantitative results reported here should therefore be interpreted as semi-quantitative estimates intended to evaluate relative product composition rather than validated regulatory measurements. In complex dietary supplement matrices such as gummies, significant ion suppression has previously been observed, with reductions of up to 58% and 38% for tetrahydrocannabinol (THC) and tetrahydrocannabinolic acid (THCA), respectively, in cannabis products in comparable analytical systems.[18] Future studies will incorporate stable-isotope-labeled internal standards to improve quantitative accuracy and compensate for matrix effects. Taken together, the semiquantitative results demonstrate that commercial kratom products vary widely in alkaloid composition, both in absolute concentrations and in the relative distribution of MG, 7-OH, and MGP. This variability highlights the lack of standardization in the commercial marketplace and reinforces the need for robust analytical methods for product characterization and regulatory oversight.

Despite these analytical limitations, the detection of 7-OH at concentrations exceeding labeled values is noteworthy given the compound's ambiguous regulatory status across different U.S. jurisdictions. Similarly, the detection of MGP, even at trace levels, in products where it is not declared on the label raises concerns regarding product authenticity and manufacturing practices. MGP is known to possess significantly higher opioid receptor potency than morphine and is generally considered to arise from metabolic transformation or synthetic processes rather than natural biosynthesis in the plant.

These findings highlight the potential importance of quantitative determination of 7-OH and MGP in regulatory and forensic contexts. For example, the emergency rule issued under Ohio Administrative Code (OAC) 4729:9−1−01.1 in December 2025 classified synthetic kratom-related products as Schedule I controlled substances, while allowing kratom sold in its natural vegetation state (crude leaf products), provided that only trace levels of 7-OH are present.[28] However, the regulatory definition of "trace amounts" remains unclear. As kratom regulations continue to evolve, robust analytical methods

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capable of reliably identifying and quantifying potent alkaloids will be essential for law enforcement, regulatory oversight, and public health protection.

In addition to regulatory applications, the analytical framework described here may also support standardization of kratom botanical materials and pharmacological investigations, including studies on the metabolism and biological effects of kratom alkaloids.

CONCLUSION

This study provides a comprehensive LC−MS framework for the reliable identification and quantification of key kratom alkaloids in commercial products and botanical materials. Through systematic evaluation of chromatographic behavior and MS/MS fragmentation patterns, we demonstrate that reliance on commonly used transitions, particularly m/z 415 → 190, can lead to false-positive identification of 7-OH in complex kratom matrices. The integration of chromatographic resolution, accurate mass measurement, and diagnostic fragment ions enables confident differentiation of structurally related alkaloids, including MG, 7-OH, MGP, and closely related isomers. These findings highlight the importance of multicriteria LC−MS confirmation strategies when analyzing complex botanical products.

Application of this analytical workflow to a diverse set of commercial kratom products revealed substantial variability in alkaloid composition and identified multiple products containing 7-OH and MGP, including cases where these compounds were not disclosed on product labels. The quantitative results further demonstrate wide variation in MG content among products, underscoring the lack of standardization across the marketplace. Together, these observations illustrate how LC−MS can provide both targeted confirmation and broader compositional profiling for emerging botanical products. The multitier identification strategy presented in this work improves analytical specificity and may support regulatory testing, forensic investigations, and quality control of botanical materials. Because the confirmation strategy relies primarily on chromatographic separation and diagnostic fragmentation patterns rather than high-resolution mass measurements, the approach can be readily implemented on widely available tandem mass spectrometry platforms. As regulatory scrutiny of kratom and related products continues to evolve, robust MS-based analytical methods will play an increasingly important role in ensuring product authenticity, safety, and accurate chemical characterization. Future studies will focus on developing and validating robust quantitative LC−MS methods for the accurate determination of key kratom alkaloids in commercial products and botanical materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the JASMS Web site. The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jasms.6c00088.

Tables S1−S3: Kratom-related products information; Semiquantitative results for the 7-OH, MGP, and MG in the kratom-related products; Unit-to-unit variability of 7-OH, MGP, and MG in selected kratom products. Figure S1−S2: EIC of m/z 399.2284 (A), m/z 415.2227 (B), m/z 397.2122 (C), and m/z 291.0863 (D) for the authentic kratom sample (K-55B) reconstructed from raw LC−MS data previously published by Cech's group; Representative calibration curves for MG (A), 7-OH (B), and MGP (C) (n = 3) (PDF)

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Author Contributions

D.S. and E.M. performed the experiments. D.S., E.M., and Y.L. analyzed the data. Y.L. and M.Z. created the draft of the manuscript. R.U. and M.Z. conceptualized the study, acquired the samples, and edited the manuscript. M.Z. secured the resources and oversaw the project. All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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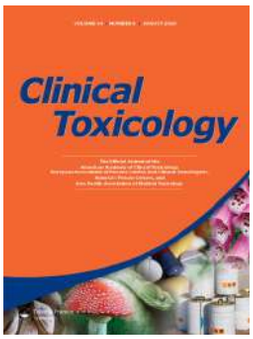
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Unlabeled kratom alkaloids detected in products marketed as kava

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ABSTRACT

Introduction: In October 2025, the Alabama Poison Information Center and local addiction medicine providers identified a cluster of cases involving liquid products marketed as containing kava associated with opioid-like withdrawal symptoms. Kava is not typically associated with physical dependence, raising concern for opioid-active substances.

Methods: Five unique retail products were submitted for laboratory analysis. Three underwent quantitative ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) testing for kavalactones, kratom alkaloids, and related semisynthetic derivatives, with additional screening for 260 substances. Two products underwent qualitative gas chromatography–mass spectrometry (GC–MS) testing for mitragynine and kavalactones.

Results: Qualitative GC–MS detected both mitragynine and kavalactones in one product, while another contained mitragynine with no detectable kavalactones. Quantitative UPLC–MS/MS identified both kavalactones and kratom alkaloids in three products, with mitragynine pseudoindoxyl predominating at concentrations of 0.16–0.21 mg per bottle.

Discussion: Detection of kratom alkaloids and related semisynthetic derivatives in products labeled as kava may explain the observed opioid-like withdrawal. Mitragynine pseudoindoxyl, which has greater opioid activity than mitragynine and 7-hydroxymitragynine, was identified as the predominant kratom-related compound.

Conclusion: Unrecognized exposure to opioid-active kratom compounds may occur through products marketed as kava, highlighting the need for improved surveillance and increased awareness.

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KEYWORDS
Kratom; kava; mitragynine; opioid; mitragynine pseudoindoxyl; dietary supplements

Introduction

Adulteration and mislabeling of botanical and dietary supplements, both intentional and unintentional, are well described in the literature, including reports of undisclosed ingredients and the addition of pharmacologically active compounds to enhance effects [1]. Limited regulatory oversight and routine testing make it difficult for consumers and providers to determine actual exposure.

In October 2025, the Alabama Poison Information Center, in collaboration with local addiction medicine providers, identified a cluster of cases involving liquid products labeled as containing *Piper methysticum* (kava) purchased from gas stations, convenience stores, and specialty shops. Calls involved patients aged 17–54 years reporting daily use with escalating doses and withdrawal symptoms upon cessation, prompting presentation to health care facilities. Several patients described symptoms partially resembling opioid withdrawal, and some experienced improvement after buprenorphine initiation.

Kavalactones are not typically associated with physical dependence; however, rare withdrawal-like symptoms have been described [2]. These products additionally claimed to contain *Uncaria tomentosa* (cat's claw) and/or *Picralima nitida* (akuamma), and

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neither substance has been clearly associated with clinically significant withdrawal. Published data on akuamma-related dependence or withdrawal are limited, with recent survey data suggesting a low risk of akuamma-related use disorder [3]. Given discordance between reported exposures and clinical presentation, and the absence of similar reports in the literature, informal surveillance of publicly available discussions suggested these products might contain kratom or related high-potency semisynthetic derivatives, such as 7-hydromitragynine and/or mitragynine pseudoindoxyl. Prior analyses of kratom-labeled products have demonstrated 7-hydroxymitragynine at concentrations unlikely to reflect native leaf composition, suggesting intentional addition or mislabeling [4,5]. Furthermore, these substances are not commonly included in standard toxicologic testing, which may complicate identification and delay treatment. We sought to evaluate these commercially available products for evidence of kratom alkaloids, related semisynthetic derivatives, and other illicit substances given the concern for unrecognized opioid exposure and potential legal and public health implications.

Methods

Selected products represented those most frequently referenced by patients and clinicians and available at local retail locations. Samples were unopened, maintained in original packaging, and submitted for laboratory analysis. Two independent analytical laboratories were utilized based on differences in testing capabilities and availability of methods. Three products underwent quantitative testing for kavalactones, kratom alkaloids, and kratom-related derivatives using ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) methods and were screened for 260 additional substances using the panel and approach described in the Sharma et al.'s protocol [6].

Two additional products underwent qualitative gas chromatography–mass spectrometry (GC–MS) testing, limited to detection of kavalactones and mitragynine based on available standards. This analysis was not quantitative and was not designed to comprehensively identify kratom alkaloids or related semisynthetic derivatives but allowed for rapid on-site analysis.

Results

None of the five products claimed to contain kratom alkaloids or related compounds. Of the five products analyzed, two underwent qualitative GC–MS analysis, which was limited to detecting mitragynine and kavalactones. One product contained detectable levels of both mitragynine and kavalactones. The other detected only mitragynine, with no detectable kavalactones despite the label claim. Three products underwent quantitative UPLC–MS/MS testing, which identified kavalactones as well as mitragynine, 7-hydroxymitragynine, and mitragynine pseudoindoxyl. The predominant kratom-related substance in UPLC-MS/MS-tested products was mitragynine pseudoindoxyl, with amounts ranging from 0.16 to 0.21 mg per bottle (Table 1). Targeted UPLC–MS/MS screening did not detect prescription or illicit opioids.

Discussion

Kratom alkaloids and related derivatives, including mitragynine pseudoindoxyl, were identified in products marketed as kava without disclosure on labeling, raising concern for intentional misrepresentation of widely available products. Kava products are marketed as calming supplements, and consumers may not anticipate opioid-active compounds. Such exposure may lead to escalating use, dependence, and withdrawal, and present diagnostic and management challenges for clinicians when patients report sole use of kava and have

Table 1. UPLC-MS/MS quantitative analysis of kava-labeled products.

Product name	Happy Hour Shot 777, mixed berry	Happy Hour Shot, BlueBurstPop	Boujee Bliss, strawberry banana
Label claim	Kavalactone extract 125 mg Akuamma 500 mg Cat's claw 750 mg	Kavalactone extract 125 mg Akuamma 500 mg Cat's claw 750 mg	*Piper methysticum* extract *Uncaria tomentosa* extract
Amount in each bottle (fl oz)	1.0	1.0	0.5
Total kratom alkaloids (µg/bottle)*	263 ± 9.6	258.5 ± 12.0	221.3 ± 4.0
Mitragynine (µg/bottle)	38.7 ± 1.9	34.3 ± 0.5	38.5 ± 1.1
7-hydroxymitragynine (µg/bottle)	10.5 ± 0.7	11.5 ± 0.7	3.41 ± 0.0
Mitragynine pseudoindoxyl (µg/bottle)	207.2 ± 6.9	206.5 ± 10.8	163.4 ± 2.6
Total kavalactones (mg/bottle)†	167.5 ± 1.2	158.1 ± 0.3	92.8 ± 0.1

Abbreviations: UPLC-MS/MS = ultra-performance liquid chromatography–tandem mass spectrometry, mg = milligram, fl oz = fluid ounce, µg = microgram.
*Kratom alkaloids tested for include: mitragynine, speciogynine, speciociliatine, mitraciliatine, 7-hydroxymitragynine, mitragynine pseudoindoxyl, corynantheidine, paynantheine, corynoxine, corynoxine B, isopaynantheine, mitraphylline, and speciofoline.
†Kavalactones tested for include: kavain, dihydrokavain, methysticin, dihydromethysticin, yangonin, desmethoxyyangonin, flavokavain A, and flavokavain B.

negative routine toxicology screens, potentially delaying recognition and treatment. These findings highlight the need for improved product transparency, public health messaging, clinician awareness, surveillance, and regulatory oversight to inform consumers of the potential for unlabeled opioid-active compounds in these products, particularly given the Schedule I status of kratom and related derivatives in many states.

Mitragynine is the primary alkaloid in *Mitragyna speciosa* (kratom), whereas 7-hydroxymitragynine is the primary active metabolite of mitragynine that can also be produced synthetically [7]. Mitragynine pseudoindoxyl is a related semisynthetic derivative formed through synthetic processes or *in vivo* from 7-hydroxymitragynine [7]. In preclinical studies, mitragynine pseudoindoxyl demonstrated substantially greater μ-opioid receptor potency than mitragynine and 7-hydroxymitragynine *in vitro*, and products containing higher concentrations of these compounds raise concern for increased opioid-like effects and risk of dependence and withdrawal [8]. Detection of a potent opioid receptor agonist provides a plausible explanation for the observed clinical presentations. Although the clinical significance of the detected concentrations remains uncertain, the predominance of mitragynine pseudoindoxyl supports concern for intentional adulteration or mislabeling.

Co-formulated kava and kratom products are available online and in states without kratom restrictions, and prior literature has evaluated this combination; however, the kratom studied was plant-derived and did not contain semisynthetic derivatives [9]. Additionally, the effects of kratom alkaloids and kavalactones may be altered through cytochrome P450-mediated pharmacokinetic interactions, as these substances share metabolic pathways and enzyme inhibition [10].

No additional substances of clinical concern, including prescription or illicit opioids, were identified; however, cat's claw and akuamma alkaloids were not included in the analytical panel, representing a limitation of this targeted approach. Additionally, only a limited number of products were tested, and selection was based on reported exposures rather than systematic market sampling, which may limit generalizability.

Conclusion

Kratom alkaloids and related semisynthetic derivatives were identified in products marketed as kava despite no disclosure on product labeling, raising concern for intentional adulteration or mislabeling and unrecognized exposure to opioid-active compounds in widely available retail products. Clinicians should consider exposure to kratom alkaloids or related compounds through products labeled as kava in patients presenting with opioid-like withdrawal, particularly when opioid use is denied. Continued surveillance, laboratory testing, and information sharing among poison centers, clinicians, law enforcement, and public health agencies are warranted to detect and address improperly labeled products marketed as "kava."

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Disclosure statement

CRM serves as an expert witness in kratom lawsuits for plaintiffs and defendants. The authors have no other competing interests to declare.

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Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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De Facto Opioids: Characterization of Novel 7-Hydroxymitragynine and Mitragynine Pseudoindoxyl Product Marketing

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Abstract

Background: Within the past year, the online sale of semi-synthetic products containing chemicals derived from mitragynine, kratom's primary alkaloid: 7-hydroxymitragynine (7-OH) and mitragynine pseudoindoxyl (MP) has emerged. Both are highly selective mu opioid receptor agonists undetectable or not present in fresh kratom leaves and may, as constituents of novel product formulations, pose public health risks.

Method: Between September 2024-February 2025 we abstracted from websites of vendors selling 7-OH and MP products information about these products, including formulation types, dose/serving size, cost, and effect, health, and drug claims

Results: We identified 304 7-OH and MP semi-synthetic products. Most (82.2%) were 7-OH-only products formulated as chewable/sublingual tablets, shots, or gummies; 14.5% were combination 7-OH-MP, and 3.3% MP-only. MP and 7-OH-MP combination products were almost uniformly marketed as "kratom" while 92.0% of 7-OH-only products were advertised as such. Across products' online marketing content, 73.3% made effect claims, most often promising increased focus and/or providing relaxation. Functional claims of pain and anxiety relief were made by 37.8% of products while 12.5% of made drug claims. The mean cost per recommended

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Declaration of Interest Statement

All authors have no competing interests to declare.

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dose/serving was $3.97; MP products had a mean cost closer to $5 per recommended dose/serving. Several products had names alluding to prescription opioids.

Conclusion: The increasing proliferation and availability of 7-OH and MP products is concerning as these products, and their chemical constituents, have never been studied in humans. Such formulations are not comparable to kratom leaf or extract products despite marketing material suggesting otherwise. Pending further investigation regulatory action should be undertaken.

Keywords

7-hydroxymitragynine; 7-OH, mitragynine pseudoindoxyl; kratom; opioid

1.0 Background

Kratom, a psychoactive botanical taken for centuries in Southeast Asia, is increasingly used in the West for its mild stimulatory and euphoric effects (Cinosi et al. 2015; Smith et al. 2023; Smith, Panlilio, Feldman, et al. 2024). Mitragynine, kratom's most abundant alkaloid that acts at adrenergic, opioid, dopamine, and serotonin receptors, may be responsible for many of kratom's neuropsychiatric effects. (León et al. 2021; Obeng et al. 2022; Ponglux et al. 1994). The chemical 7-hydroxymitragynine (7-OH) is not present in fresh kratom leaves but is an artifact of the drying process (Zhang et al. 2022, Smith, et al. 2024; Sharma, 2025); it is also formed by oxidation of mitragynine by hepatic and intestinal cytochromes. The binding affinity of 7-OH at the mu-opioid receptor (MOR) is 10-22 greater than morphine; 7-OH is further metabolized in humans to mitragynine pseudoindoxyl (MP), a chemical nearly as potent as fentanyl (Chakraborty et al. 2021; Kamble et al. 2020; Kruegel et al. 2016; 2019; Qu et al. 2023). Because 7-OH and MP are not found in fresh, native kratom leaf (Zarembo et al., 1974) products containing these ingredients manufactured through conversion of mitragynine can be considered semi-synthetic derivatives of mitragynine. The safety of 7-OH and MP in humans is unstudied.

Kratom products sold in the West include extracts and isolates, which differ from traditional decoctions made from boiling fresh leaves, a practice that limits the total alkaloids ingested per serving (Brown, Lund, and Murch 2017; Grundmann et al. 2023; Singh, Narayanan, and Vicknasingam 2016). Increasing global demand for kratom has coincided with growing diversity of kratom product formulations that include organic extracts, gummies, vapes, liquid shots, or tablets (Smith, Panlilio, Sharma, et al. 2024). In 2023-2024, however, novel semi-synthetic 7-OH and MP products appeared online with the monikers "7-OH," "7-hydroxy," or "7" and, in the case of MP, as "pseudo." These semisynthetic kratom alkaloids are overwhelmingly mislabeled or marketed as "kratom" even though the chemical composition is wholly distinct (Sharma 2025; Smith et al. 2025).

The emergence of high potency, non-prescription commercial products derived from kratom impels a better understanding of this semi-synthetic market. Therefore, we investigated the online prevalence of 7-OH and MP products; characterized products' dosing, cost, and formulation; and reviewed marketing claims including references to "kratom."

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2.0 METHODS

2.1 Data Collection

Between September 2024-February 2025, we monitored online 7-OH and MP marketing content to compile a product database. We identified vendor websites and products through browser searches using keywords “7-hydroxymitragynine,” “7-hydroxy,” “7-OH,” “Mitragynine pseudoindoxyl,” “pseudoindoxyl,” and “pseudo.” We counted products sold in varying sizes only once to avoid duplications in the database. Alternatively, some websites sold similar products (e.g., drink mix) under different flavors. If these contained slightly different formulations and/or made separate claims, we considered them distinct.

2.2 Product and Marketing Content

2.2.1 Product categorization was coded as: “7-OH,” “MP,” or “combination 7-OH and MP.”

2.2.2 Product information was obtained by scraping websites for: URL, brand name, product name, complete amount (mg) in full package, mg in a single dosing unit, mg in recommended dose/serving, online product description, product images, product cost per package, cost per serving, servings per package, customer reviews, and semi-synthetic substance(s).

2.2.3 Product type was classified: chewable/sublingual tablets, shots, gummies, drink mixes, vape pens, syrups, capsules, food products, sublingual strips, or powders.

2.2.3 Product claims were roughly classified: “general wellbeing claims,” “functional claims,” or “drug claims” and identified by reading product descriptions across websites and product packaging, including images when available. *“General wellbeing claims”* were statements linking the semi-synthetic product to beneficial general effects; products were categorized into the following emergent general wellbeing claims: boosting energy, increasing focus, improving wellbeing, improving mood, providing relaxation, and workout supplement. General wellbeing claims that we identified were similar, but not always equivalent, to those of a structure/function claim of a dietary supplement as defined by the U.S. Food and Drug Administration (Borchers, Keen, & Gershwin, 2016). *“Functional claims”* were statements that linked the product to more specific therapeutic-like outcomes; products were categorized into the emergent functional claims: pain/discomfort relief or management, and stress/anxiety reduction. *“Drug claims”* were statements indicating the product conferred some specific therapeutic outcome, such as stimulatory or opioid activity. Note that this claim is not isomorphic with disease claims, which dietary supplements are not permitted to make. We want to emphasize that our categorizations for product claims are not rigid and that, depending on specific marketing language, could correspond to one or more products claims. As such, our three types of “claims” provide only a crude method for categorizing the statements made online for novel 7-OH and MP marketing.

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2.3 Data Analysis

Quantitative analyses were done using Excel with findings summarized descriptively. We abstracted and present illustrative quotes but have removed brand names. These appear in the full Supplement 1 database; we strongly encourage readers to view this material.

3.0 RESULTS

We identified 304 products: 82.2% sold as 7-OH, 14.5% as combination 7-OH-MP, and 3.3% as MP. Note that text segments quoted in the results may exemplify more than one product claim type (each type was coded separately and quantified).

3.1 Product Type, Dosage, Claims, and Reviews

Table 1 shows marketed semi-synthetic products by type. Chewable/sublingual tablets were the most common (60.2%), followed by shots (20.7%). Websites stressed the ease of using semi-synthetics compared to kratom. For instance, a 7-OH chewable tablet is marketed as follows:

> "The chewable format is designed for convenience, allowing you to simply chew a tablet and savor the berry flavor. This eliminates the need for mixing or preparation… The compact size of each tablet makes them portable and perfect for on-the-go use, so you can carry them in your bag or pocket and take them whenever needed."

In this way, many websites marketed semi-synthetics as consumer-friendly formulations while highlighting that kratom products required effort to prepare.

Table 1 summarizes products' characteristics and websites and/or packaging that made specific claims or misleadingly presented their product as kratom. Most (73.4%) of products made what we considered to be *general wellbeing claims*, the most common being increased focus (58.2%), provides relaxation (46.7%), or boosts energy (39.5%). Products sold as 7-OH had a greater proportion of effect claims (76.8%) than MP (50.0%) or combination products (59.1%). Emblematic of the effect claims is one 7-OH vendor's description of the benefits customers might expect:

> "Enhanced Relaxation, Focus, Energy Motivation, Mood Elevation, Body Buzz, Euphoria, Discomfort Suppression, Apprehension Relief, Feeling Present, Stimulation, Arousal, Social Lubrication"

Functional claims were made by 37.8% of labels or websites, generally related to pain relief (26.3%) stress/anxiety reduction (21.4%) effects. One 7-OH product touted:

> "Each tablet contains 15mg of 7-Hydroxymitragynine, a compound renowned for its strong analgesic and mood-enhancing properties. This makes '[Brand] Kratom Chewable Tablets' an ideal choice for those seeking quick relief from pain, anxiety, or stress, as well as those looking to improve their overall sense of well-being."

Some branded their products to imply curative powers associated with prescriptions. For example, the brand names included terms such as "7rx," "Curevana," "Generic RX,"

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"Pain Crusher Rx." Many products were packaged in ways that mimicked FDA-approved medications (e.g., cough syrup, blister packs).

Few vendor websites and/or packaging made *drug claims* about their semi-synthetic product (12.5%). Most included content related to opioid pharmacology. For instance, one MP tablet product description explained to consumers that:

> "Pseudoindoxyl, the active compound in these tablets, is known for its potent analgesic properties, making it an excellent choice for those dealing with chronic pain or discomfort. It works by interacting with the body's opioid receptors, providing significant relief without the typical side effects associated with traditional opioids."

Another drug claim (which was interspersed with other claims) that we identified included the potential for sedation, wherein a 7-OH tablet vendor stated:

> "[A]t higher doses, 7-Hydroxymitragynine may induce sedation, offering a calming effect that can be particularly helpful for individuals suffering from insomnia. This sedative quality can promote better sleep and reduce tension, contributing to overall well-being."

Other products were also promoted for stimulant effects; one 7-OH chewable product was described as:

> "Unlike some stimulants, 7-OH delivers a smooth, sustained energy boost without the jitteriness or crash, ensuring a balanced and effective experience."

3.2 Misleading Marketing

Most products were misleadingly marketed as kratom or *Mitragyna speciosa* (93.1%). Mislabeling occurred in naming (e.g., 'Kratom 7-hydroxymitragynine Extract Chew'), packaging (e.g., 'Pure Kratom Alkaloid'), and/or in the descriptions (e.g. '[this shot] provides a powerful, convenient, and fast-acting way to enjoy the benefits of kratom'). One entitled "Benefits of 7-OH" elaborated:

> "[K]ratom's mood-enhancing qualities can alleviate stress and anxiety, fostering a sense of calm and positivity. Many find relief from discomfort through kratom's analgesic properties, which may help manage chronic pain conditions effectively… its relaxing effects can aid in achieving restful sleep, contributing to overall mental and physical rejuvenation. Kratom's versatility extends to its use as a tool for managing withdrawal symptoms from other substances, providing support during recovery processes. As a natural botanical, kratom offers a holistic approach to wellness, appealing to those seeking alternatives to synthetic medications."

3.3 Dosing and cost

Products contained a wide range (1-700 mg) of 7-OH and/or MP in a single dose/serving and per recommended dose/serving (Table 2). The average recommended dose/serving was 20mg. Compared to the other semi-synthetics, MP products had the lowest mean

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recommended dose (10.1mg). Irrespective of product, marketing focused on ease of dosing. One 7-OH vendor stated:

> "Every pouch includes four expertly pressed classic extract tablets. These tablets are scored down the center for easy dosing and contain two servings total. Each serving is 7.5mg of pure 7-OH extract, for 15mg total per tablet."

The mean cost per recommended dose was $3.97; MP products had a mean cost closer to $5 per recommended dose. Lastly, while some products had many customer reviews (up to 143 for 7-OH), most had zero or few reviews.

4.0 DISCUSSION

We identified widespread online marketing of semi-synthetic products containing ingredients which, if accurate, have high MOR binding affinity. We identified that 7-OH as the most common constituent of commercial semi-synthetic products. The difficulty and expense of isolating 7-OH from kratom leaves, but also the sheer amount of active ingredient, strongly suggests that these semi-synthetic products are manufactured in a laboratory from purified mitragynine. The chemical processes used to make 7-OH also likely places consumers at risk for inadvertent exposure to toxicity from synthetic reagents and reaction by-products. Novel 7-OH products, therefore, carry fundamentally different risks than those presented by the kratom market that existed in the US for nearly 20 years (Boyer et al. 2008; Smith et al. 2022).

The trend toward increasingly potent semi-synthetic chemicals based upon kratom alkaloids is concerning. Product descriptions frequently hinted at neuropsychiatric opioidergic effects with terms such as "buzz" or "euphoria" coupled with brand names that evoke opioids (e.g., Roxy, Opia, Hydroxie, Perks). Consumers may not appreciate that 7-OH products may induce tolerance and withdrawal symptoms if used regularly, and can, for some, include risks for developing addiction.

In some respects, the widespread availability of these semi-synthetic alkaloids is reminiscent of internet 'pharmacies' that sold oxycodone and other opioids in the 2000s without prescription. Just as the online pharmacies once did, online 7-OH vendors may use discreet packaging and direct-to-home shipping practices to obviate state laws intended to protect consumers. Whereas kratom products have been adulterated with hydrocodone and morphine (LeSaint et al., 2022), the potency of semi-synthetic kratom alkaloids may obviate the need for adulteration with opioids.

Marketing of these products poses problems. First, labeling 7-OH and MP products as "kratom" misleads consumers who intend to use a low-potency botanical supplement but instead ingest a high-potency opioid. Second, websites may persuade consumers that 7-OH and MP formulations are chemically pure and consistently manufactured; however, online photographs from some manufacturers themselves show several vials of powdered 7-OH, each a different shade in color that indicates differences in composition and purity. Inconsistencies in manufacturing and non-adherence to good manufacturing practices likely lead to variation in dose and composition of products, which, due to the

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MOR binding potency of 7-OH and MG, affects overdose risk and toxicity. Finally, the marketing material for some formulations (e.g., sublingual tablets) ignore their products' increased bioavailability which, when coupled with relatively shorter duration of action of 7-OH, heightens the risk of dependence or addiction (Maxwell et al. 2021; Sharma 2025). We conclude that the widespread availability of these semi-synthetics represents an unrecognized but growing public health threat

Our methodology has limitations. Products sold in brick-and-mortar stores were not included in this sample, although the location of sales would not affect the pharmacology under discussion (Hemby et al. 2019). We did not chemically analyze products or interview consumers; this work, however, is underway and beyond the scope of this exploratory project. We are confident that health risks of 7-OH and MG are real, as forensic toxicologists have identified 7-OH in postmortem samples (Osawa and Johnson 2025). Given the pace with which these products are emerging, further investigation and clinical awareness is warranted.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- Synthetic 7-hydroxymitragynine products are misleadingly marketed as kratom
- Synthetic mitragynine pseudoindoxyl products are misleadingly marketed as kratom
- Marketed products have high mu opioid receptor affinity and no human safety data
- Drug and health claims do not conform to supplement regulation but are legally sold
- Cost per recommended dose across these novel products averaged $3.97

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Table 1.

Characteristics of Products Sold as 7-Hydroxymitragynine (7-OH), Mitragynine pseudoindoxyl (MP), or Both

Characteristics	All Products (N=304)		Products Sold as 7-OH (n=250)		Products Sold as MP (n=10)		Products Sold as Both 7-OH and MP (n=44)	
	Freq	%	Freq	%	Freq	%	Freq	%
Product Type								
Chewable or Sublingual Tablet	183	60.20	140	56.00	5	50.00	38	86.36
Shot	63	20.72	55	22.00	4	40.00	4	9.09
Gummy	13	4.28	13	5.20	0	0.00	0	0.00
Drink Mix	12	3.95	12	4.80	0	0.00	0	0.00
Vape Pen	9	2.96	9	3.60	0	0.00	0	0.00
Syrup	7	2.30	6	2.40	1	10.00	0	0.00
Capsule	6	1.97	6	2.40	0	0.00	0	0.00
Sublingual Strip	6	1.97	6	2.40	0	0.00	0	0.00
Food Product	4	1.32	2	0.80	0	0.00	2	4.55
Powder	1	0.33	1	0.40	0	0.00	0	0.00
Website / Packaging Product Claims								
General Wellbeing Claims	223	73.36	192	76.80	5	50.00	26	59.09
Claims to Provide Increased Focus	177	58.22	152	60.80	2	20.00	23	52.27
Claims to Provide Relaxation	142	46.71	125	50.00	3	30.00	14	31.82
Claims to Boost Energy	120	39.47	101	40.40	2	20.00	17	38.64
Claims to Improve Wellbeing	114	37.50	97	38.80	2	20.00	15	34.09
Claims to Improve Mood	110	36.18	97	38.80	2	20.00	11	25.00
Claims to be a Workout Supplement	8	2.63	8	3.20	0	0.00	0	0.00
Functional Claims	115	37.83	100	40.00	3	30.00	12	27.27
Claims to Pain/Discomfort Relief	80	26.32	70	28.00	2	20.00	8	18.18
Claims to Reduce Anxiety/Stress	65	21.38	54	21.60	2	20.00	9	20.45
Drug Claims	38	12.50	33	13.20	2	20.00	3	6.82
Marketed/Defined as Kratom	283	93.09	230	92.00	10	100.00	43	97.73

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Table 2.

Dosage, Cost, and Number of Reviews for Products Sold as 7-Hydroxymitragynine (7-OH), mitragynine pseudoindoxyl (MP), or Both

	All Products (N=304)					Products Sold as 7-OH (n=250)					Products Sold as MP (n=10)					7-	
Characteristics	**Min**	**Max**	**Mdn**	**Mean**	**SD**	**Min**	**Max**	**Mdn**	**Mean**	**SD**	**Min**	**Max**	**Mdn**	**Mean**	**SD**	**Min**	**Max**
Total Package (mg)	2	7000	60	122	423	2	7000	58.00	120.48	465.03	15	240	50.00	70.20	64.94	30	400
Single Dose/ Serving (mg)	2	750	20	35	72	2	750	18.00	36.41	78.97	4	40	15.00	16.52	9.73	10	100
Recommended Dose/Serving (mg)	1	700	10	20	57	1	700	10.00	21.92	63.01	4	20	9.50	10.12	4.42	9	50
Cost of Package ($)	3.95	139.95	19.49	22.86	14.78	3.95	139.95	18.47	21.97	14.90	11.49	90.00	16.29	27.46	25.42	6.50	64.95
Cost Per Recommended Dose/Serving ($)	1.14	17.99	3.37	3.97	2.36	1.14	17.99	3.37	4.04	2.52	2.44	8.25	4.50	4.90	2.15	1.94	6.24
Number of Product Reviews	0	143	0	3	12	0	143	0.00	3.45	13.12	0	9	0.50	1.90	3.11	0	25

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SHORT COMMUNICATION

From Kratom to Semi-Synthetic Opioids: The Rise and Risks of MGM-15

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ABSTRACT

Kratom (*Mitragyna speciosa*), a plant native to Southeast Asia, has long been used for its stimulant and analgesic properties. 7-Hydroxymitragynine (7-HMG) is a potent and selective opioid agonist in vitro and demonstrates a potent opioid effect in living subjects, reversible by naloxone. It has been semi-synthesized into products that are readily available in retail and virtual shops. It is known that 7-HMG has earned the nickname "legal morphine," and has gained popularity among users seeking pain relief and/or a "high" comparable with prescription opioids. Medicinal chemistry efforts have led to synthetic 7-HMG derivatives such as MGM-15, where stereospecific saturation of the imine N(1)–C(2) double bond increases opioid receptor affinity and activity. Despite its higher in vitro opioid potency, MGM-15 is currently sold in the US for human consumption as a "research chemical" in tablet form, even though there is an absence of this being studied in humans and obviously no FDA approval. In this study, we analyzed commercially available MGM-labeled tablets using UPLC–MS/MS and subsequently evaluated the binding affinities of purified MGM-15 across multiple opioid receptors. Tablets contained an average of 10.9 ± 0.2 mg (10.7 to 11.2 mg) of MGM-15, with no naturally occurring kratom alkaloids or illicit substances detected. MGM-15 shows greater hMOR and hDOR binding affinities than 7-HMG, indicating the potential for higher opioid effects and risks, emphasizing the urgent need for more research to advise regulation and hopefully prevent misuse and harm.

1 | Introduction

The market for kratom (*Mitragyna speciosa* (Korth.) Havil.) is growing at a tremendous rate [1]. Freshly picked leaves have been traditionally used for their ethnopharmacological properties, including analgesic, antipyretic, antidiabetic, antidiarrheal, antitussive, antidepressant, and anxiolytic effects [2]. Kratom displays a polypharmacological profile, acting as a stimulant at low doses (2–4 g) and producing opioid-like effects at higher doses (5–8 g). This paradoxical dual activity makes kratom a unique and complex natural product of interest for pain management as well as for mitigating opioid addiction and withdrawal symptoms [3, 4]. Mitragynine (MTG) (Figure 1), the most abundant alkaloid found in native kratom plants, has long been believed to be responsible for many of the beneficial effects of kratom [5]. *M. speciosa* is widely used as a self-treatment for opioid use disorders, as a low-cost substitution/replacement for opioids, opioid withdrawal, and use disorders to other substances [4, 6]. As kratom products have gained popularity in the Western world, there has been

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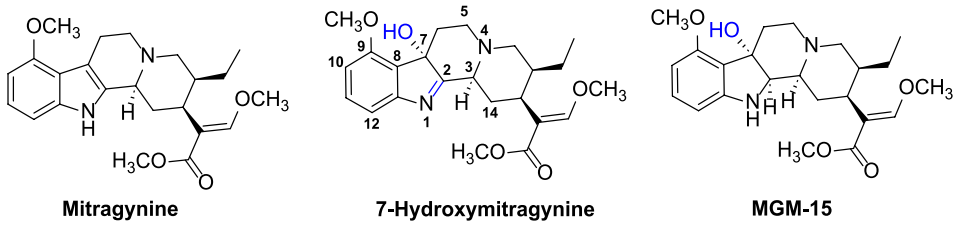


FIGURE 1 | Chemical structure of mitragynine, 7-hydroxymitragynine, and MGM-15.

a growing effort to explore the pharmacology of the alkaloidal constituents more thoroughly, including in vivo metabolites and semi-synthetic derivatives of the naturally occurring alkaloids.

7-Hydroxymitragynine (7-HMG) is an active metabolite of MTG [7]. 7-HMG is a potent opioid that exhibits significantly higher affinity for human μ-opioid receptors (hMOR) than MTG. In rat hot plate assays, 7-HMG demonstrated stronger analgesic effects than morphine, with an ED_{50} of 1.9 mg/kg versus 5.1 mg/kg for morphine, indicating its full agonist activity in living subjects. Unlike MTG, which reportedly acts as a biased partial agonist at hMOR, 7-HMG induces classical opioid side effects such as respiratory depression, tolerance, and cross-tolerance to morphine [8, 9]. 7-HMG is readily self-administered in rats and has demonstrated high abuse potential, with reinforcing effects mediated by hMOR. It has been shown that prior exposure to 7-HMG increases subsequent morphine use, suggesting it may enhance vulnerability to other opioids or act as a gateway drug. In contrast, MTG is not self-administered and reduces morphine intake, suggesting low abuse potential and possible therapeutic use [8]. Several 7-HMG-containing products are commercially available in the US market, but these potentially harmful products should not be classified as kratom. They are semi-synthetic derivatives produced through chemical oxidation of MTG-rich kratom extracts and have not undergone safety or efficacy evaluation in animals or humans [10]. In 2014, continuing the work of others who had previously modified and optimized 7-HMG derivatives, a Japanese research team reported the discovery and characterization of two novel 7-HMG analogs: MGM-15 and MGM-16. MGM-15 (Figure 1) was synthesized by reducing 7-HMG with $NaBH_4$ while MGM-16 was prepared similarly from 10-fluoro-7-HMG [11].

MGM-15 exhibited antinociceptive potency approximately 15- and 50-fold greater than morphine following subcutaneous and oral administration in mice, respectively. Similarly, MGM-16 demonstrated even higher potency, being approximately 71- and 240-fold more effective than morphine following subcutaneous and oral administration, respectively. Receptor binding studies revealed that MGM-16 has a high affinity for MOR and DOR, with receptor affinity (K_i) values of 2.1 and 7.0 nM, respectively, while MGM-15 showed K_i values of 6.4 nM (MOR) and 16 nM (DOR), indicating strong interaction with both receptor subtypes. These K_i values were derived from assays using membrane proteins isolated from rodent and pig brain tissue. MGM-16, a potent μ/δ-opioid dual agonist, showed superior efficacy to MGM-15 in vitro and in vivo, producing dose-dependent antiallodynic effects in sciatic nerve–ligated mice through both subcutaneous (0.1–0.4 mg/kg) and oral (0.5–2 mg/kg) administration, with peak effects at 15 to 30 min and full reversal of allodynia at 0.2 mg/kg (subcutaneous) and 1 mg/kg (oral), comparable with gabapentin (100 mg/kg). These derivatives were reported to be not only the most potent MOR agonists among all known MTG derivatives but also acted as full agonists at DOR [11]. This dual agonist activity highlighted the potential of these derivatives to treat acute, chronic, and neuropathic pain, as hDOR agonism has primarily been investigated for managing chronic pain rather than acute pain [12]. However, the targeting of DOR agonists has been abandoned by the pharmaceutical industry due to the high incidence of seizure occurrence [13].

Since these findings were published, no other primary research has been reported on these two novel derivatives. However, the synthesis of MGM-16 involves a complex, multistep process making large-scale production challenging compared to MGM-15 [14]. Conversely, MGM-15 is available for sale on multiple websites, typically marketed as "research purposes only" and accompanied by disclaimers stating the products are "not for human or animal consumption." Despite these warnings, many of these products are offered in a range of flavors and forms, including isolate powders, tablets, and liquid shots, offering customers multiple options to choose from. Currently, just a handful of vendors are selling MGM-15 products, most of which are labeled as "MGM." Increasing positive reviews on vendor websites and discussion about these products on online forums will inevitably lead to an increased demand for these products, similar to how 7-HMG products gained popularity. The aim of this study is to raise awareness about the availability of MGM products on the market and to conduct both qualitative and quantitative analyses to confirm the presence of MGM-15 and identify any other kratom alkaloids or potential adulterants followed by evaluation of their binding affinity at various receptors.

2 | Material and Methods

2.1 | Chemicals and Reagents

Tablets were purchased online from Ritually Pure (Niagara Falls, NY, USA). 7-HMG was synthesized from MTG as reported earlier [15]. Kratom alkaloids (MTG, mitraphylline, corynantheidine, paynantheine, corynoxine, corynoxine B, mitraciliatine, speciociliatine, and speciogynine) were isolated and purified in-house from a kratom alkaloid-rich extract as reported earlier [16]. Phenacetin was purchased from Sigma-Aldrich (St. Louis, MO, USA). LC–MS-grade acetonitrile, formic acid, acetic acid, ammonium acetate, and methanol were procured from Fisher Scientific (Fair Lawn, NJ, USA).

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2.2 | Experimental Procedure for the Synthesis of MGM-15

To a stirred solution of 7-HMG (150 mg, 0.36 mmol) in dry MeOH (6 mL), $NaBH_4$ (16.4 mg, 0.43 mmol) was added at 0°C under argon atmosphere. The reaction mixture stirred for 30 min (as monitored by TLC), H_2O was added to the reaction mixture followed by saturated aqueous $NaHCO_3$ solution (5 mL). The mixture was concentrated under reduced pressure; the crude residue was extracted with dichloromethane (3 × 30 mL). The combined extract was washed with brine, dried over $MgSO_4$, evaporated, and purified using flash column chromatography (gradient: 0%–40% EtOAc in *n*-hexanes), yielding 117 mg of MGM-15 (yield 78%) as a yellow solid (Figure 2). Previous literature report matches the NMR data (Figures S1 and S2) [11].

2.3 | Instrumentation and Chromatographic Conditions

Chromatographic analysis was carried out using a Waters Acquity I-Class Plus ultra-performance liquid chromatography (UPLC) system (Milford, MA, USA) coupled with a Xevo TQ-S Micro triple quadrupole mass spectrometer (MS/MS, Waters Corporation, Milford, MA, USA). Mass spectrometric detection utilized electrospray ionization (ESI) operated in positive ion mode. The target analytes were quantified using multiple reaction monitoring (MRM), ensuring high selectivity through the monitoring of specific precursor-to-product ion transitions. Instrument parameters, including capillary voltage, desolvation temperature, cone gas flow, and desolvation gas flow, were optimized to improve signal strength and detection sensitivity. The MRM transitions for MGM-15 and kratom alkaloids are presented in Table S1. Parent to product ion mass spectra of MGM-15 are presented in Figure S3.

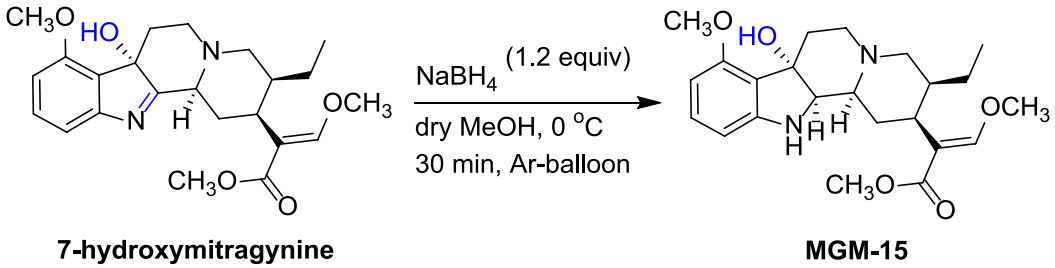
Chromatographic separation of MGM-15 and kratom alkaloids was achieved using a Waters Acquity BEH C18 column (1.7 µm, 2.1 × 100 mm) with gradient elution. The mobile phases consisted of aqueous 2.5 mM ammonium acetate buffer (pH 3.5) as solvent A and acetonitrile as solvent B. The optimized gradient elution program was as follows: 0–3 min at 30% B; 3–3.9 min, gradient from 30% to 40% B; 3.9–4.0 min at 40% to 70% B; 4.0–4.5 min at 70%; 4.5–4.6 min at 70% to 30% B; and 4.6–5.5 min at 30% B. The flow rate was maintained at 0.35 mL/min, the column temperature was set to 55°C, and the autosampler was kept at 10°C. The retention times of MGM-15 and the internal standard (IS, phenacetin) were 1.14 and 1.66 min, respectively. The injection volume used was 2 µL. The strong needle wash solution comprised water, acetonitrile, methanol, and isopropanol (1:1:1:1, v/v/v/v) with 0.1% formic acid. The weak needle wash solution consisted of acetonitrile, methanol, and water (1:1:2, v/v/v) with 0.1% formic acid [17].

2.4 | Preparation of Standard and Quality Control Samples

Stock solutions (1 mg/mL) of MGM-15 and kratom alkaloids were prepared in methanol, followed by the preparation of secondary stocks (50 µg/mL) and spiking stocks, which were further used to prepare calibration standards (CS) in the concentration range of 1 to 500 ng/mL for each analyte. Quality control (QC) samples were prepared similarly at four concentration levels: 1.0 ng/mL [lower limit of quantification (LLOQ)], 3.0 ng/mL [low quality control (LQC)], 150 ng/mL [medium quality control (MQC)], and 450 ng/mL [high quality control (HQC)]. Fresh CS and QC samples were prepared by adding 2.5 µL of spiking mix stock solution to 22.5 µL of blank methanol, and then vortex-mixed for 5 min using a BenchMixer (Benchmark, San Francisco, CA, USA). Methanol:water mixture (80:20, 100 µL) containing 5 ng/mL of IS was added to 25 µL of sample and vortexed for 5 min. These were filtered using a Multiscreen Solvinert 96-well 0.45-µm filter plate (Millipore, St. Louis, MO, USA) followed by centrifugation at 841 *g* for 5 min at 4°C and injected onto the LC–MS/MS for analysis. Data acquisition and processing were performed using MassLynx version 4.2.

2.5 | Method Validation for Quantitation of MGM-15

A UPLC-MS/MS method was partially validated for the quantification of MGM-15 in tablet formulations, following the guidelines outlined in the USFDA's "Analytical Procedures and Methods Validation for Drugs and Biologics" [17, 18]. Key validation parameters assessed included specificity, limit of detection (LOD), LLOQ, linearity, accuracy, and precision. Specificity was evaluated to ensure that MGM-15 could be accurately identified and quantified in the presence of other expected components or potential interferences. The LOD and LLOQ were established based on the signal-to-noise ratio and the consistency of repeated injections. To assess linearity, a calibration curve was plotted across the experimental range, and linear regression analysis was performed. Accuracy and precision were tested at four concentration levels: LLOQ, LQC, MQC, and HQC. Intra-day accuracy and precision were determined using six replicates per level ($n = 6$) within a single day, while inter-day performance



7-hydroxymitragynine **MGM-15**

FIGURE 2 | Synthesis scheme of MGM-15 from 7-hydroxtmitragynine.

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was assessed over three consecutive days using the same number of replicates. Precision was calculated as the percent relative standard deviation (%RSD).

2.6 | Quantitative Analysis of Commercially Available MGM Tablet

A total of three tablets were used for analysis with an average weight of 456.8 ± 3.6 mg. These were finely ground using a mortar and pestle. The resulting powder was transferred to a 100-mL volumetric flask and dissolved in methanol. It was further sonicated for 60 min to ensure complete extraction, followed by filtration. The filtrate was diluted 100,000-fold with methanol to fall within the linearity range, processed as described in Section 2.4, and analyzed using the UPLC-MS/MS system. These tablets, along with a sample of kratom leaf aqueous extract and a commercially available kratom product (Optimized Plant Mediated Solutions Gold capsules, Choice Organics, Los Angeles, CA, USA) were also analyzed for further confirmation if any phytoconstituent interfered with the analysis of MGM-15 [17, 19, 20].

These commercially available MGM-15 tablets were screened for illicit substances, prescription drugs, and cannabinoids using a gradient UPLC-MS/MS method following previously reported methods [19, 20]. A few of the target compounds included tianeptine, hydromorphone, morphine, midazolam, oxycodone, fentanyl, tramadol, cannabidiol, Δ9-tetrahydrocannabinol, buprenorphine, and a complete list of analytes is available in Table S2. This method was not validated for quantitative analysis but was used for targeted screening to identify compounds with mass transitions matching those of known illicit drugs, prescription medications, and/or cannabinoids. If any target analyte had been detected in an MGM tablet, a separate validated quantitative analysis would have been performed.

2.7 | In Vitro Receptor Binding Assays

The binding assay was performed using monoclonal opioid receptors expressed in Chinese Hamster Ovary (CHO) cell lines. MGM-15, 7-HMG, and MTG were screened for in vitro binding affinity at hMOR, hDOR, and human κ-opioid receptors (hKOR), as previously reported with some modifications [21]. [^{3}H] DAMGO, [^{3}H] U69593, and [^{3}H] DADLE radioligands were employed to label hMOR-CHO, hKOR-CHO, and hDOR-CHO, respectively. Saturation binding assays were conducted to determine the K_d and B_{max} of each radioligand at its corresponding receptor. Prior to these studies, membrane evaluations were conducted to identify the optimal protein concentration for each receptor. Briefly, 20 μg of protein was incubated with a radioligand in the presence or absence of compounds in an assay buffer (50-mM Tris, 3-mM $MgCl_2$, and 0.2-mM EGTA, pH 7.4). Specific (i.e., opioid receptor-related) binding at these opioid receptors was determined as the difference between total and nonspecific binding (NSB). NSB was determined using 10 μM of appropriate receptor-selective competitors: naltrexone for hMOR, U69593 for hKOR, and SNC80 for hDOR. Following incubation for 90 min at 30°C, the bound and unbound radioligands were rapidly separated by filtration under vacuum through glass fiber filters pre-soaked with 0.05% polyethyleneimine (PEI) and rinsed several times with an ice-cold washing buffer (50-mM Tris) using a 96-well plate cell harvester. The filters were dried for 20 min at 50°C and counted for radioactivity using a Microbeta2 PerkinElmer microplate reader (Ramsey, MN, USA) with a scintillation cocktail. The receptor binding assays were conducted in triplicate (two plates) and repeated at least twice on different days. Displacement binding curves were analyzed using nonlinear least-squares regression (GraphPad Prism, GraphPad Software, San Diego, CA, USA) to determine IC_{50} (concentration causing a half-maximal inhibition of control-specific binding) values, which were subsequently converted to K_i values using the Cheng–Prusoff equation.

3 | Results and Discussion

Compound parameters for MGM-15 were added to a previously reported UPLC–MS/MS-based analytical method available for the simultaneous quantification of kratom alkaloids and partially validated for MGM-15 [17]. No interference of kratom alkaloids was observed at the retention time of MGM-15 and vice versa. LOD was defined as the lowest concentration of MGM-15 spiked in a methanol–water mixture that produced a signal-to-noise (S/N) ratio of ≥3:1. LLOQ was established through a series of preliminary experiments, confirming the method's high sensitivity. At an LLOQ of 1 ng/mL, the signal-to-noise ratio exceeded the recommended threshold of 10:1, and both accuracy and precision met the acceptance criteria—within 20% of the nominal concentration and % relative standard deviation (RSD), respectively. The calibration curve was linear over the concentration range of 1 to 500 ng/mL, with correlation coefficients consistently greater than 0.99 in all analytical runs. The method's reliability and reproducibility were further validated through intra-day and inter-day assessments of accuracy and precision. For the LQC, MQC, and HQC, accuracy and precision (%RSD) remained within 15% of the nominal values. For the LLOQ, which allows slightly broader limits due to its low concentration, both parameters were within 20% (Table S3). These findings confirm that the method provides consistent, accurate, and precise quantification of MGM-15 across a wide range of concentrations and over multiple days. The analysis of commercially available MGM tablets confirmed the presence of MGM-15 and the absence of other kratom alkaloids or known adulterants, including 7-HMG. Each tablet contained 10.9 ± 0.2 mg of MGM-15. The representative chromatograms of MGM-15 for both reference standard and tablet product are presented in Figure S4. Screening for illicit substances, prescription drugs, cannabinoids, and kratom alkaloids revealed no contamination or substitution, suggesting that these tablets are chemically consistent with their labels.

A receptor binding assay was further employed to evaluate the affinity of MGM-15 at opioid receptors utilizing CHO recombinant cell lines expressing hMOR, hDOR, and hKOR and compared with MTG and 7-HMG. In contrast, Matsumoto et al. conducted their binding studies using guinea pig tissue homogenate membranes for MOR and DOR, and male Sprague–Dawley rat brain homogenate membranes for KOR [11]. These membranes express all receptors with multiple binding sites and enzymes that could interact with the radioligand or affect the

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TABLE 1 | Affinities ($K_i \pm$ SEM) of mitragynine, 7-hydroxymitragynine, and MGM-15 at opioid receptor subtypes.

Compound	hMOR (nM)	hKOR (nM)	hDOR (nM)
Mitragynine	299 ± 15	248 ± 45	2009 ± 281
7-Hydroxymitragynine	72 ± 11	136 ± 21	202 ± 51
MGM-15	28 ± 7	340 ± 79	59 ± 4

Abbreviations: hDOR, human δ-opioid receptor; hKOR, human κ-opioid receptor; hMOR, human μ-opioid receptor.

assay. Because the objective of this study was to determine if MGM-15 poses a potential risk to human health, binding affinities at opioid receptors were determined in human-based transfected cells expressing individual receptor subtypes.

7-HMG is itself a concern to public health due to its high potency at the hMOR, contributing to opioid-associated risks such as sedation, respiratory depression, and potential for abuse and dependence. However, MGM-15 could represent an even greater threat. MGM-15 binds to hMOR with a K_i of 28 ± 7 nM, suggesting it to be approximately 2.6-fold more potent than 7-HMG (Table 1). At hDOR, MGM-15 displays a K_i of 59 ± 4 nM, 3.4-fold higher affinity than 7-HMG. Although MGM-15 is slightly weaker at the hKOR with a K_i of 340 ± 79 nM compared with 7-HMG (136 ± 21 nM), its overall binding profile across multiple opioid receptors suggests a strong opioid-like effect. Although MTG, the main alkaloid in traditional kratom products, is a weak agonist at opioid receptors, approximately 4.2- and 10.7-fold weaker at hMOR, 10- and 34-fold weaker at hDOR, and 1.8-fold weaker than 7-HMG but 1.4-fold stronger than MGM-15 at hKOR, respectively. This indicates that MTG is a relatively weak partial agonist in vitro at these receptors and contributes to a significantly milder opioid effect profile compared to either 7-HMG or MGM-15.

Despite these pharmacodynamic properties, MGM-15 has not undergone FDA review for safety, efficacy, or toxicity in humans. The presence of MGM-15 in flavored, consumer-ready formats despite being labeled as "not for human consumption" suggests a loophole in current regulatory frameworks. With its high affinity for both hMOR and hDOR and increasing availability, MGM-15 poses a potential for misuse and public health risk comparable to, or even greater than, 7-HMG. FDA has already issued warning letters for products containing 7-HMG due to its pharmacological profile and potential for harm, similar regulatory scrutiny is warranted for MGM-15.

4 | Conclusions

In recent years, legislation and regulations on products containing kratom and their constituents have become increasingly more stringent. Kratom products, including 7-HMG, are completely banned in six US states and regulated in several others. Moreover, on July 15, 2025, the US Food and Drug Administration (FDA) issued warning letters to seven companies for the unlawful marketing of products containing 7-HMG. Recently, the FDA has initiated a significant move to safeguard public health by recommending the scheduling of 7-HMG products under the Controlled Substances Act, aiming to curb the risks posed by dangerous, illegal opioids [22]. This action highlights the FDA's increasing concern over the sale of powerful, novel opioid products being marketed to US consumers, both online and through retail outlets, such as smoke shops, gas stations, and convenience stores. Due to the decreasing accessibility of kratom and 7-HMG, vendors have found that newer derivatives may be exempt from these laws and regulations, allowing for the marketing and sale of products such as MGM-15.

MGM-15 is a potent, unregulated opioid compound now found in consumer products despite lacking FDA approval or safety data in humans, and this is the first time analysis of a consumer product containing this compound. Its increasing availability highlights the urgent need for regulatory oversight and market removal to protect public health. The potential Schedule 1 classification of 7-HMG by the FDA leaves an open door for MGM products to fill the void. Food products are required to have premarket approval by the FDA to ensure their safety, and MGM-15, along with other synthetic derivatives of kratom alkaloids (i.e., 7-HMG and mitragynine pseudoindoxyl), represents potentially highly dangerous products that would be recognized as unsafe under section 409(a) of the FD&C Act.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest. C.R.M. is an expert witness as a plaintiff and defendant in wrongful death kratom civil lawsuits.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** dta3952-sup-0001-Supplementary_Material.docx. ^{1}H-NMR spectrum of methyl (*E*)-2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (MGM-15). **Figure S2:** dta3952-sup-0001-Supplementary_Material.docx. ^{13}C-NMR spectrum of (*E*)-2-((2*S*,3*S*,7a*S*,12b*S*)-3-ethyl-7a-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12,12*a*,12*b*-decahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (MGM-15). **Figure S3:** Product-ion spectra of MGM-15. **Figure S4:** Representative MRM chromatograms of blank (A), at LLOQ level (1 ng/mL, B), commercially available tablet (C) and internal standard (D). **Table S1:** Mass spectrometer source parameters and retention times for MGM-15, kratom alkaloids and internal standard (IS). **Table S2:** Mass spectrometer source parameters for illicit compounds. **Table S3:** Intra-day and inter-day accuracy and precision data for the determination of MGM-15 using UPLC–MS/MS.

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A Pilot, Dose-Finding, Pharmacodynamic and Pharmacokinetic Study of Orally Administered Botanical Kratom

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Background: Kratom (*Mitragyna speciosa*) is a plant indigenous to Southeast Asia. This pilot study evaluated the pharmacodynamic (PD) effects, safety, and pharmacokinetics (PK) of kratom and several of its alkaloids.

Methods: Recreational polydrug users (8 participants/cohort; 6 active: 2 placebo, N = 40) completed the study. Participants had experience with opioids but were otherwise healthy. This study utilized a double-blind, between-subjects design where participants randomly received a single dose of placebo or kratom. The kratom used in the study had alkaloid levels representative of botanical kratom products (i.e., leaf) previously characterized in the literature and contained trace levels of 7-hydroxymitragynine (7-OH). The starting dose was 1 g and doses of 3, 8, 10, and 12 g were administered after safety reviews after each dose. After dosing, pupillometry and assessments of subjective effects were performed, and blood samples were collected. Safety assessments included adverse events (AE) monitoring, laboratory tests, vital signs, ECG assessments, physical examination findings, and assessment of suicidality.

Results: No deaths or serious adverse events (SAEs) occurred. Somnolence, vomiting, and nausea were the most common AEs reported. Kratom alkaloid concentrations showed generally orderly, dose-related effects. At doses ≥3 g, kratom produced pupillary constriction. Few dose-related effects were observed, although the 12 g dose of kratom produced increases on several subjective measures including ratings of "drug liking."

Conclusions: This study investigated the safety of single-sourced botanical kratom; the results may not be representative of other kratom-containing products. Kratom produced some opioid-like effects including pupillary constriction, and the 12 g dose produced effects commonly associated with drugs of abuse such as visual analog scale (VAS) ratings of drug liking, good effects, and high.

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Kratom (*Mitragyna speciosa*) is a wetlands tree native to Southeast Asia including Thailand, Indonesia, and Malaysia.[1] Kratom has been used for centuries in a variety of sociocultural contexts including the treatment of opioid withdrawal, pain relief, and for mood-enhancing effects.[1–3] In the United States, kratom began increasing in popularity in the 1990s. According to the National Survey on Drug Use and Health, an estimated 1.7 million Americans aged 12 and older used kratom in 2021.[4] Characterizing kratom use is complicated by a heterogeneous product landscape that includes teas, tablets, extracts, and whole leaf products with varying routes of administration and alkaloid content. Much of the data regarding the use patterns of kratom in the United States is derived from survey-based studies. For example, Garcia-Romeu et al. found that most regular kratom users reported using 1 to 3 g (49%) or 4 to 6 g (33.4%) per consumption.[5] In other survey studies, the self-reported average consumption of kratom powder was 4 to 5 g per serving with a range of 2.6 to 7.5 g.[6,7] When quantifying mitragynine levels consumed through the use of kratom products, individuals self-reported consuming an average of 31.3 mg of mitragynine/serving, corresponding to 78.3 to 134.6 mg of mitragynine per day.[8]

As of November 2025, no drug products containing kratom or its components have been approved by the US Food and Drug Administration (FDA) and FDA has concluded that:

> *"…kratom is a new dietary ingredient for which there is inadequate information to provide reasonable assurance that such ingredient does not present a significant or unreasonable risk of illness or injury and, therefore, dietary supplements that are or contain kratom are adulterated under section 402(f)(1)(B) of the FD&C Act. Further, FDA has determined that kratom, when added to food, is an unsafe food additive within the meaning of section 409; food containing an unsafe food additive, such as kratom, is adulterated under section 402(a)(2)(C)(i). On the basis of these determinations by FDA, kratom is not lawfully marketed as a dietary supplement and cannot be lawfully added to conventional foods."*[9]

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Despite the lack of regulatory oversight of kratom and kratom-related products, there has been substantial scientific interest in the pharmacology of kratom and its alkaloids. More than 40 alkaloids have been identified in kratom, the most abundant of which is mitragynine. Mitragynine accounts for ~66% of the total alkaloid content, and 4 additional alkaloids (speciociliatine, speciogynine, paynantheine, and corynantheidine) make up roughly 20% of the alkaloid content.[1,10] The minor alkaloids make up the remaining ~14% of alkaloid content of the leaf material and there are few data describing the pharmacological effects of these minor alkaloids.

Research efforts investigating kratom have focused primarily on its alkaloid constituents and mitragynine has been the subject of most of these studies given its abundance in the kratom in the tree. Mitragynine has affinity for, and binds to mu-opioid receptors.[11] Mitragynine also has affinity for a variety of receptor subtypes including adrenergic (α_{1a}, α_{1b}, α_{1d}, α_{2a}, α_{2b}, and α_{2c}), adenosine A_{2A}, dopamine D_2, and serotonin (5-HT_{1A}, 5-HT_{2B}, 5-HT_{2C}, and 5-HT_7) receptors.[12–14] The effects of mitragynine on mu-opioid receptors vary by the assay system utilized. For example, in vitro functional assays have characterized mitragynine as a full agonist,[15] partial agonist,[11,16] and as an antagonist.[17] In vivo assessments of antinociception using the hot plate test have shown that mitragynine functions as an agonist, producing a degree of antinociception similar to classic opioids such as morphine and oxycodone.[18,19] In contrast, mitragynine did not produce significant antinociception in hot plate tests utilizing rats.[17] Drug discrimination procedures have been used to examine the stimulus effects of mitragynine and model its subjective effects. In a drug discrimination study that trained rodents to discriminate mitragynine from vehicle, administration of morphine produced full substitution for the mitragynine cue.[20] In a second group of animals trained to discriminate morphine from vehicle, administration of mitragynine produced full substitution for the morphine stimulus cue (i.e., symmetrical generalization). Of note, the substitution of mitragynine for the morphine cue was dose-dependent: where the middle dose of mitragynine produced full generalization to the morphine cue but the low and highest doses examined did not.[20] Although the drug discrimination results from Harun and colleagues seem to be in keeping with mitragynine's mu-opioid receptor binding profile, disparate drug discrimination results have been reported. For example, in separate groups of rats trained to discriminate either mitragynine or morphine from saline, mitragynine failed to produce full substitution for the morphine stimulus cue, and morphine did not substitute for the stimulus effects of mitragynine.[17] Interestingly, a variety of other opioids did substitute for the mitragynine stimulus cue including fentanyl, nalbuphine, and buprenorphine.[17] The authors hypothesized that the incomplete substitution of mitragynine for morphine (and vice versa) may be due to rate-limiting effects, and that the stimulus effects of mitragynine are dose-related and may be mediated by more than mu-opioid agonist effects alone.

The rewarding properties of mitragynine have also been examined using conditioned place preference (CPP). In CPP studies with isolated mitragynine, rats demonstrated a place preference for mitragynine that was similar to stimulants and morphine, suggesting its rewarding properties.[21,22] A follow-up study using naloxone suggested that although opioid receptors mediated the acquisition of CPP, they were less involved in CPP expression after it was acquired.[23] In contrast, when the reinforcing effects of mitragynine were examined directly using self-administration procedures, mitragynine was not self-administered in rats previously trained to self-administer morphine.[24]

Although data characterizing the pharmacology of the minor alkaloids are sparse, 7-hydroxymitragynine (7-OH) has been the subject of numerous research efforts. 7-OH is typically reported to account for $<0.05\%$ of dried kratom leaf mass, a substantially lower proportion than mitragynine.[25] 7-OH has been postulated to be a post-harvest artifact of kratom and is also formed as a metabolite of mitragynine in humans.[26] 7-OH has received considerable attention due to its greater affinity and activity at mu-opioid receptors relative to mitragynine and the other kratom alkaloids.[11,27] Importantly, 7-OH is self-administered in preclinical models, demonstrating its reinforcing effects and potential for abuse.[24]

Despite the extensive literature on kratom and its alkaloids using in vitro, in vivo, and epidemiological data to characterize its pharmacology and use patterns, prospective and properly controlled clinical studies are sparse. For example, Trakulsrichai and colleagues conducted a prospective study of the pharmacokinetics (PK) of kratom by administering kratom tea to regular kratom tea users. However, there was no placebo or positive control comparator, and subjective effects (i.e., abuse-related measures) were not assessed. In addition, the dose range of kratom was relatively limited and only 3 mitragynine equivalent doses were examined: 6.25, 9.96, and 11.5 mg.[28] Vicknasingam and colleagues examined the effects of kratom tea on pain tolerance using the cold pressor test. The authors found an increase in analgesia, though only a single dose of kratom tea was administered and mitragynine (1.6 mg/kg) was the only alkaloid measured in the tea.[29] Tanna et al.[30] also assessed the PK profile of a 2 g tea, whereas Huestis et al.[31] performed a double-blind, dose-escalation study of 0.5 to 4 g encapsulated dried leaf kratom powder administered orally to assess its PK profile. Other clinical assessments of kratom have allowed participants to self-administer their own kratom products in uncontrolled, outpatient settings without active or placebo comparators.[32–34] Given the dearth of rigorously controlled clinical studies of kratom, along with an absence of abuse-related outcome measures, the objective of this single ascending-dose study of botanical kratom was to generate pilot data on its behavioral effects and safety profile and to inform the design of a future human abuse potential (HAP) study.

MATERIALS AND METHODS

Study Design

This single-center, single-dose, randomized, adaptive, double-blind, placebo-controlled study consisted of 5 single ascending-dose cohorts. Dose levels could be adjusted to higher or lower doses following a review of the safety, tolerability, and pharmacodynamic (PD) data from previous cohorts by a Safety Review Committee (SRC). The SRC consisted of the principal investigator at the investigational site, the study manager, and personnel from FDA's scientific team. The study was conducted at Altasciences

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company in Overland Park, Kansas. The study received ethical approval from an Institutional Review Board and was performed under an Investigational New Drug Application (IND).

Kratom

The kratom used in the study was obtained from Sun Distribution, Super Organics. The kratom was powdered, raw leaf, and administered in 500 mg, size 00 capsules. To meet the chemistry, manufacturing, and controls (CMC) requirements for the IND, repeated stability testing was performed and the kratom met USP guidelines for heavy metals, microbial load, pesticides, mycotoxin, residual solvents, and moisture content. The kratom product was analyzed for alkaloidal content using a validated method UPLC-MS/MS as reported earlier,[35] and stability studies were performed at room temperature in a stability chamber (25°C and 60% RH) for 13 months. Kratom product was found stable (acceptable limit, 90% of initial content) and the alkaloid content seemed to be consistent with that reported in the published literature (Table 1).[8,36] For each cohort, matching placebo capsules containing 500 mg of microcrystalline cellulose were administered.

Quantitative Analysis Using UPLC-MS/MS

Two separate bioanalytical methods were developed and validated for quantitative analysis of mitragynine and 7-hydroxymitragynine and speciogynine, speciociliatine, and paynantheine at Alta Sciences and University of Florida, respectively. In brief, mitragynine and 7-hydroxymitragynine were quantified using AB Sciex API5500 triple quadrupole mass spectrometer coupled with Shimadzu Nexera X2 ultra high-performance liquid chromatography system (UPLC-MS/MS) for a linearity range of 0.5 to 500 and 0.2 to 100 ng/mL, respectively. Analysis was conducted in multiple reaction monitoring (MRM) mode using precursor to product ion mass transitions of *m/z* 415.22 > 397.30, 399.23 > 238.10, 418.24 > 400.20, and 402.25 > 238.10, respectively, for 7-hydroxymitragynine, mitragynine, 7-hydroxymitragynine *d3*, and mitragynine *d3*, respectively. 7-hydroxymitragynine *d3* and mitragynine *d3* were used as an internal standard for 7-hydroxymitragynine and mitragynine, respectively. Chromatographic separation was achieved using a gradient flow (0.6 mL/min) of mobile phase consisting of methanol and water containing 0.5% formic acid and a Raptor biphenyl (100 ×2.1 mm, 2.7 μm) column. UPLC-MS/MS analysis of speciogynine, speciociliatine, and paynantheine was carried out using Waters Acquity Class I Plus UPLC coupled with a Waters Xevo TQ-S Micro triple quadrupole mass spectrometer for a linearity range of 1-250 ng/mL in human plasma. The chromatographic separation was achieved using Waters Acquity Premier CSH C18 (1.7 μm, 2.1×10 mm) with Vanguard fit guard column using the mobile phase consisting of aqueous ammonium acetate buffer, (2.5 mM, pH-3.5 ± 0.1) (A) – acetonitrile (B) with a gradient elution. The column and autosampler temperatures were kept at 55 and 4 °C, respectively. The mobile phase was delivered at a flow rate of 0.5 mL/min, and the injection volume was set to 2 μL. The source temperature was 150°C, Ion spray voltage was set at 500 V, desolvation temperature was 500 °C, desolvation gas flow was 900 L/h, and the cone gas flow was 50 L/h. The mass spectrometer was operated in positive ion mode and detection of the ions was performed in MRM mode, monitoring the transition of *m/z* 399.25 precursor ion $[M+H]^+$ to the *m/z* 174.16 product ion for speciogynine and speciociliatine (collision energy 32 V and cone voltage 60V), *m/z* 397.16 precursor ion $[M+H]^+$ to the *m/z* 174.16 product ion for paynantheine (collision energy 30 V and cone voltage 58 V), *m/z* 402.21 precursor ion $[M+H]^+$ to the *m/z* 177.06 product ion for internal standard (collision energy 30 V and cone voltage 26V) (mitragynine-*d3*). Both bioanalytical methods were validated for selectivity, specificity, accuracy, precision, calibration and range, carryover, matrix effect, dilution integrity, and stability following FDA/ICH M10 guidelines for bioanalytical method validation.

Participants

A total of 40 healthy male and female, nondependent, recreational polydrug users with prior opioid experience were included in the study (8 subjects in each of the 5 cohorts). Recreational drug users were defined as those who had used opioid drugs for recreational purposes (i.e., for psychoactive effects) at least 10 times in the subject's lifetime and at least once in the last 12 weeks from screening. Inclusion criteria included a history of recreational use of 2 or more perception-altering (eg, lysergic acid diethylamide (LSD), kratom, cannabis, dronabinol, ketamine, phencyclidine, dextromethorphan, 3,4 methylenedioxymethamphetamine (MDMA), mescaline, psilocybin, tryptamine derivatives or ring-substituted amphetamines with perception-altering effects) or stimulant drugs (eg,

TABLE 1. Alkaloid Composition and Stability of Kratom Capsules (25°C/60% RH)

	Timepoint (mo)					
Alkaloid*	**0**	**1**	**2**	**3**	**6**	**13**
Mitragynine	5.22 ± 0.65	5.08 ± 0.35	4.90 ± 0.51	5.22 ± 0.46	5.07 ± 0.71	5.38 ± 0.32
Speciogynine	0.99 ± 0.12	0.96 ± 0.07	0.90 ± 0.10	0.97 ± 0.08	0.92 ± 0.13	0.94 ± 0.07
Speciociliatine	2.02 ± 0.24	1.98 ± 0.14	1.98 ± 0.21	1.94 ± 0.17	1.98 ± 0.26	1.82 ± 0.12
Mitraciliatine	0.29 ± 0.03	0.28 ± 0.02	0.29 ± 0.03	0.31 ± 0.02	0.29 ± 0.04	0.25 ± 0.02
7-hydroxymitragynine	0.06 ± 0.01	0.06 ± 0.01	0.03 ± 0.01	0.03 ± 0.00	BLLOQ	0.04 ± 0.01
Paynantheine	1.34 ± 0.16	1.32 ± 0.09	1.26 ± 0.13	1.33 ± 0.12	1.28 ± 0.18	1.19 ± 0.08
Corynantheidine	0.14 ± 0.02	0.13 ± 0.01	0.13 ± 0.01	0.13 ± 0.01	0.13 ± 0.02	0.14 ± 0.01
Corynoxine A	0.05 ± 0.01	0.05 ± 0.01	0.05 ± 0.00	0.04 ± 0.00	0.04 ± 0.01	0.04 ± 0.00
Corynoxine B	BLLOQ	BLLOQ	BLLOQ	BLLOQ	BLLOQ	0.03 ± 0.00
Mitraphylline	BLLOQ	BLLOQ	BLLOQ	BLLOQ	BLLOQ	BLLOQ

*Alkaloids content(s) are expressed as the milligram (mg) amount per 500 mg capsule.
BLLOQ indicates below the lower limit of quantitation (1 ng/mL equivalent to 32 ng/capsule); RH, relative humidity.

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cocaine, amphetamine, methamphetamine, methylphenidate, methcathinone, and other synthetic cathinones) on at least 5 occasions in the participant's lifetime. Polydrug recreational users with prior opioid experience were selected to ensure familiarity with a broad range of psychoactive drug effects. This population is considered appropriate and face-valid for evaluating the pharmacodynamic profile of escalating kratom doses, given its reported spectrum of activity ranging from stimulant-like effects at low doses to opioid-like sedation and dissociative/hallucinogenic effects at higher doses.[37–39] Experienced recreational polydrug users also represent the typical participant population enrolled in HAP studies, which this SAD study was designed to inform.[40] All participants provided written informed consent form and agreed to use appropriate contraception methods (eg, abstinence, systemic contraceptives, intrauterine device, double-barrier method, etc.).

Exclusion criteria included: (1) Difficulty swallowing capsules; (2) females who were lactating or pregnant; (3) self-reported sensitivities to kratom; (4) significant history of disease including gastrointestinal, liver or kidney disease, cardiovascular, pulmonary, hematologic, neurological, psychiatric, gastrointestinal, endocrine, immunologic, ophthalmologic, or dermatologic disease; (5) smoking or a history of heavy smoking; (6) excessive caffeine intake; (7) other physiological diseases or abnormalities identified at screening, including ECG abnormalities and QT prolongation; and (8) additional criteria, including a history of substance use disorder and intake of kratom in the 14 days before study participation.

Procedures

After successful screening, participants were confined to the clinical site from the day before drug administration until 48 hours after drug administration. Subjects were randomized 6:2 (active: placebo) to kratom or placebo in each cohort.

Kratom or placebo was administered under double-blind conditions. Study treatments were administered in the morning, 30 minutes after a standardized high-fat breakfast (fed state) after an overnight fast of at least 10 hours. This approach was chosen to increase tolerability, facilitate dose escalation, and mitigate nausea and vomiting previously reported under fasted conditions.[30] Capsules were administered with ~240 mL of water at room temperature. Given the large number and size of capsules required for certain cohorts (range of 2 to 24 size 00 capsules), an additional 240 mL of water was allowed if required. Time of dosing was set equal to the time when the first capsule was administered to the subject. The dosing procedure had to be completed within 5 minutes. Dosing procedures completed up to 2 minutes outside the allowed time window were not considered protocol deviations but were documented. Start and end times of dosing and the volume of water consumed were recorded. All capsules were swallowed whole and not chewed or broken.

Safety

Safety assessments included adverse event (AE) monitoring, clinical laboratory tests, vital signs (blood pressure, pulse, respiratory rate, oxygen saturation, and body temperature), ECG assessments, physical examination findings, and assessment of suicidality using the Columbia-Suicide Severity Rating Scale. At the investigator's discretion, additional safety assessments could be performed as needed to ensure subject safety.

Pharmacokinetic Assessments

Blood samples for PK measurements were collected predose and at 0.25, 0.5, 1, 1.5, 2, 2.5, 3, 4, 6, 8, 10, 12, 24, and 48 hours, post-dose. Samples were collected from an intravenous catheter or direct venipuncture, if necessary. The time of PK blood sample collection was calculated relative to the time of treatment administration. PK measurements of the study included plasma concentrations of mitragynine, 7-hydroxymitragynine, speciogynine, speciociliatine, and paynantheine using validated bioanalytical methods (see Supplemental Digital Content 1, http://links.lww.com/JCP/B19 and 2, http://links.lww.com/JCP/B24).

Pharmacodynamic Assessments

Before completing computerized PD measures, all participants underwent a standardized training and practice session. Participants who seemed to have difficulty differentiating between bipolar and unipolar VAS (eg, making errors such as selecting a score of 50 to indicate the absence of a drug effect on a unipolar scale) or difficulty distinguishing between "at this moment" and "overall" measures received additional training focused on these distinctions.

Participants were monitored to ensure that PD assessments were completed appropriately. Reasonable attempts were made to rouse participants who fell asleep during testing. If a participant was unable to complete PD assessments in a timely manner before the next required procedure or timepoint due to an AE or other clinical concern, the PD assessment at that timepoint was aborted.

Pupillometry

Data from a series of frames were used in the calculation of pupil size. Measurements were collected using a handheld electronic pupillometer (NeurOptics VIP-300 Pupillometer System, CA) under mesopic lighting conditions. For each participant, the same eye was used for all assessments when feasible. The pupillometry assessment was completed immediately before the administration of other PD measures at each prespecified timepoint.

Visual Analog Scales (VAS)

All VAS were scored on a 0 to 100 scale and measured at predetermined time points. The VAS were structured as either bipolar or unipolar scales, depending on the subjective effect being measured. Bipolar scales asked about the neutrality, direction, and intensity of a subjective opinion, whereas unipolar scales only asked about the extremity or intensity of a subjective effect. When VAS were administered as bipolar scales, a neutral point equal to 50 was included in the scale [eg, drug liking, overall drug liking (ODL), take drug again (TDA), drowsiness/alertness VAS]. The neutral point reflected a state whereby a subject was experiencing neither negative nor positive effects (eg, neither dislike nor like the effects of the drug) and was labeled with an anchor, such as "neither like nor dislike." When VAS were administered as unipolar scales, anchors were presented using text such as "not at all" (score = 0) to "extremely" (score = 100; eg, good, bad, high, and any drug effects VAS). Unipolar scales did not include a neutral point but rather, a rating of "0" reflected the complete absence of a subjective effect, whereas a rating of "100"

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reflected the maximum presence of a subjective effect (eg, no good effects = 0, extremely good effects = 100). Scales that referred specifically to drug (eg, drug liking, good drug effects, bad drug effects, and any drug effects VAS) were not administered at predose.

Bowdle Visual Analog Scale

An adapted computerized version of the Bowdle VAS, consisting of 13 unipolar items, was administered to assess perceptual changes. Each item was scored from 0 ("not at all") to 100 ("extremely") with lower scores indicating fewer perceptual and psychedelic effects.

The individual items were:

1. My body or body parts seemed to change their shape or position (BODY)
2. My surroundings seemed to change in size, depth, or shape (SURROUNDINGS)
3. The passing of time was altered (TIME)
4. I had feelings of unreality (REALITY)
5. It was difficult to control my thoughts (THOUGHTS)
6. The intensity of colors changed (COLORS)
7. The intensity of sound changed (SOUND)
8. I heard voices or sounds that were not real (VOICES)
9. I had the idea that events, objects, or other people had particular meaning that was specific for me (MEANING)
10. I had suspicious ideas or the belief that others were against me (SUSPICIOUS)
11. I felt anxious (ANXIOUS)
12. I felt high (HIGH)
13. I felt drowsy (DROWSY)

Responses to items 1, 2, 3, 5, 6, and 7 were summed to produce a "subjective external perceptions" composite score. Response to items 4, 8, 9, 10, and 11 were summed to produce a "subjective internal perceptions" composite score. Items 12 and 13 were not included as they were administered as part of the individual VAS items. If a response to one of the items was missing, the associated composite score was not calculated.

Drug Similarity VAS

Drug Similarity VAS items provided an estimate of how similar the test drug was to drug classes with which participants were familiar. On a unipolar scale ranging from "not at all similar" to "very similar," participants were asked to compare a drug they had taken previously to the test drug. A drug use history was completed during screening and used to create a subject-specific drug similarity measure. Drugs that a subject had not used often (< 2 lifetime uses of a given drug), were not included in their questionnaire.

Addiction Research Center Inventory (ARCI)

The Addiction Research Center Inventory (ARCI)[41–43] Morphine-Benzedrine Group (MBG), Amphetamine (A), Benzedrine Group (BG), Pentobarbital-Chlorpromazine-Alcohol Group (PCAG), and lysergic acid diethylamide (LSD) scales were administered alongside the basic VAS measures.

The 49-item ARCI was a shortened version (49 true-false items) compiled by Martin and colleagues from the original 550-item ARCI.[41–43] This version contained 5 scales, which measured the following effects: Euphoria—MBG scale; Stimulant effects—Amphetamine scale and BG scale; Dysphoria—LSD scale; and Sedation—PCAG scale. Subjects indicated their responses by selecting "false" or "true." One point was given for each response that agreed with the scoring direction on the scale (i.e., true items received a score of 1 if the answer was "true;" a score of 0 was given when the answer was opposite to the scoring direction).

The following scales were administered:

- Euphoria: Morphine-Benzedrine Group (MBG) scale
- Stimulant effects: Amphetamine (A) scale, Benzedrine Group (BG) scale
- Dysphoria: Lysergic acid diethylamide (LSD) scale
- Sedation: Pentobarbital-Chlorpromazine-Alcohol Group (PCAG) scale

Statistical Analyses

For measures collected within 24 hours, the peak responses [maximum (E_{max}) or minimum (E_{min}), as appropriate] were the endpoints. The primary pharmacodynamic endpoint was E_{max} of drug liking VAS. Key secondary endpoints included Emax scores of high VAS, overall drug liking VAS, and take drug again VAS. Inferential statistics using the aforementioned peak responses were performed on the primary and key secondary endpoints. Because of the large number of endpoints and the large number of comparisons in the study, for controlling overall type I error rate only descriptive statistics were calculated for other endpoints.

ANOVA model was prespecified for the statistical analyses for the primary and key secondary endpoints. The 1-sided Dunnett test at 5% significance level was used for the comparisons between each dose of kratom and pooled placebo.[44] For the sensitivity analysis, Hodges-Lehmann estimator was used to estimate the difference between each dose of kratom and pooled placebo.[45,46] The 1-sided Mann-Whitney *U* test (or called Wilcoxon rank sum test)[47,48] at 5% significance level was used to test the null hypothesis that: "the observations produced by kratom and pooled placebo have the same distribution" versus the alternative hypothesis: "the distribution of observations produced by kratom is shifted to the right of that of pooled placebo." The Benjamin-Hochberg procedure[49] was used to adjust multiplicity for the comparisons between each dose of kratom and pooled placebo for each endpoint. As the dosing escalating study is exploratory in nature, the type I error rate adjustment was not performed for the multiplicity among endpoints.

RESULTS

Safety Analyses and Adverse Events

All 40 enrolled participants in the study were included in the safety analysis population. Overall, 18 of the 40 participants (45.0%) that received kratom or placebo experienced a total of 38 treatment-emergent adverse events (TEAEs) and 33 of the TEAEs (86.8%) were considered related to study drug (Table 2). The most commonly reported TEAEs were somnolence, vomiting, and nausea. Somnolence occurred in 1 of 6 participants (16.7%) receiving kratom 8 g, 3 of 6 (50.0%) receiving kratom 10 g, and 2 of 10 (20.0%) receiving placebo. Vomiting occurred in 2 of 6 participants (33.3%) in each of the kratom 8 g and 12 g cohorts, and in 1 of 6 (16.7%) in the kratom 10 g cohort. Nausea occurred in 2 of 6 participants (33.3%) receiving kratom 8 g, and in 1 of 6 (16.7%) in each of the kratom 10 g and 12 g cohorts. Nausea and vomiting

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TABLE 2. Summary of Treatment-Emergent Adverse Events by MedDRA Preferred Term and System Organ Class

System Organ Class and MedDRA Preferred Term	Kratom 1 g (N = 6) 2 Capsules	Kratom 3 g (N = 6) 6 Capsules	Kratom 8 g (N = 6) 16 Capsules	Kratom 10 g (N = 6) 20 Capsules	Kratom 12 g (N = 6) 24 Capsules	Pooled Placebo (N = 10)	Overall (N = 40)
Subjects with at least one TEAE *	1 (16.7)	1 (16.7)	5 (83.3)	4 (66.7)	3 (50.0)	4 (40.0)	18 (45.0)
Ear and labyrinth disorders	0	0	0	1 (16.7)	0	0	1 (2.5)
Tinnitus	0	0	0	1 (16.7)	0	0	1 (2.5)
Gastrointestinal disorders	0	0	3 (50.0)	2 (33.3)	2 (33.3)	0	7 (17.5)
Dry mouth	0	0	0	1 (16.7)	0	0	1 (2.5)
Nausea	0	0	2 (33.3)	1 (16.7)	1 (16.7)	0	4 (10.0)
Vomiting	0	0	2 (33.3)	1 (16.7)	2 (33.3)	0	5 (12.5)
General disorders and administration site conditions	0	0	0	1 (16.7)	1 (16.7)	0	2 (5.0)
Chills	0	0	0	1 (16.7)	0	0	1 (2.5)
Feeling hot	0	0	0	0	1 (16.7)	0	1 (2.5)
Infections and infestations	0	1 (16.7)	0	0	0	0	1 (2.5)
Viral upper respiratory tract infection	0	1 (16.7)	0	0	0	0	1 (2.5)
Investigations	0	0	0	0	0	1 (10.0)	1 (2.5)
Blood pressure systolic increased	0	0	0	0	0	1 (10.0)	1 (2.5)
Musculoskeletal and connective tissue disorders	0	1 (16.7)	0	0	0	0	1 (2.5)
Neck pain	0	1 (16.7)	0	0	0	0	1 (2.5)
Nervous system disorders	1 (16.7)	0	2 (33.3)	4 (66.7)	1 (16.7)	3 (30.0)	11 (27.5)
Dizziness	0	0	0	2 (33.3)	0	0	2 (5.0)
Headache	0	0	1 (16.7)	0	1 (16.7)	1 (10.0)	3 (7.5)
Paresthesia	1 (16.7)	0	0	0	0	0	1 (2.5)
Presyncope	0	0	1 (16.7)	0	0	0	1 (2.5)
Somnolence	0	0	1 (16.7)	3 (50.0)	0	2 (20.0)	6 (15.0)
Psychiatric disorders	0	0	2 (33.3)	2 (33.3)	0	0	4 (10.0)
Anxiety	0	0	0	1 (16.7)	0	0	1 (2.5)
Euphoric mood	0	0	1 (16.7)	1 (16.7)	0	0	2 (5.0)
Irritability	0	0	1 (16.7)	0	0	0	1 (2.5)
Renal and urinary disorders	0	0	1 (16.7)	0	0	0	1 (2.5)
Leukocyturia	0	0	1 (16.7)	0	0	0	1 (2.5)
Skin and subcutaneous tissue disorders	0	0	0	0	1 (16.7)	1 (10.0)	2 (5.0)
Hyperhidrosis	0	0	0	0	1 (16.7)	0	1 (2.5)
Pruritus	0	0	0	0	0	1 (10.0)	1 (2.5)

Data are reported as n (%).

*Each TEAE was counted only once for each subject within each MedDRA SOC and PT.

MedDRA indicates Medical Dictionary for Regulatory Activities; N, number of subjects who received the specified treatment; n, number of subjects in a category; PT, preferred term; SOC, system organ class; TEAE, treatment-emergent adverse event.

were not observed in the pooled placebo group. No serious adverse events or deaths were reported, and no participants discontinued the study due to TEAEs.

Pupillometry

Mean (SE) pupil diameter over time at each dose level of kratom and placebo is illustrated in Figure S1 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20). Minimal fluctuations in pupil diameter were observed for placebo and kratom 1 g. In contrast, kratom doses of 3, 8, 10, and 12 g produced decreases in pupil diameter (i.e., constriction), with an apparent dose-related effect. Mean maximum pupil constriction over time was 0.9 mm for kratom 1 and 3 g, 2.2 mm for kratom 8 g, 1.9 mm for kratom 10 g, and 2.4 mm for kratom 12 g compared with 1.1 mm for placebo. Summary statistics for maximum pupil constriction are presented in Table 3.

Dunnett test[44] was used to test the difference in means between each dose of kratom and pooled placebo for maximum pupil constriction. The results showed that the means of maximum pupil constrictions produced by 8 and 12 g of kratom were significantly smaller (i.e., produced more pupil constriction) than that in the pooled placebo group ($P < 0.05$).

Pharmacokinetic Results

All doses of kratom produced a measurable plasma concentrations of the targeted alkaloids.

Mitragynine

Plasma concentration-time profiles for mitragynine are shown in Figure 1, and PK parameters are summarized in Table 4. After a single oral administration of kratom under fed conditions, mitragynine was absorbed into the systemic

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TABLE 3. Summary Statistics for Maximum Pupil Constriction (mm)

Parameter	Kratom 1 g (N = 6)	Kratom 3 g (N = 6)	Kratom 8 g (N = 6)	Kratom 10 g (N = 6)	Kratom 12 g (N = 6)	Pooled Placebo (N = 10)
Mean (SE)	−0.9 (0.2)	−0.9 (0.2)	−2.2 (0.3)	−1.9 (0.3)	−2.4 (0.4)	−1.1 (0.3)
Median	−0.7	−0.8	−2.2	−1.7	−2.3	−0.8
Min, max	−1.9, 0.3	−1.5, −0.5	−3.0, −1.2	−3.1, −1.3	−3.9, −1.1	−2.6, 0

circulation with a median time to reach peak plasma concentration (T_{max}) ranging from 3.0 to 5.0 hours across cohorts 1 to 5. Mean observed peak plasma concentration (C_{max}) ranged from 41.7 to 378.5 ng/mL with increasing kratom doses from 1 to 12 g. The mean estimated area under concentration-time curve up to the last timepoint ($AUC_{0\text{-}T}$) ranged from 305.4 to 3271.2 h·ng/mL. An increase in mitragynine C_{max}, and $AUC_{0\text{-}T}$ was noted with an increased dose of kratom; however, dose-proportionality could not be concluded because of the limited number of subjects in parallel dosing cohorts.

7-Hydroxymitragynine

Concentration-time profiles for 7-hydroxymitragynine (7-OH) are shown in Figure 2, with PK parameters provided in Table 4. After a single oral administration of kratom under fed conditions, 7-OH produced a median T_{max} ranging from 2.8 to 6.0 hours for cohorts 1 to 5. The mean observed C_{max} ranged from 6.7 to 58.4 ng/mL with increasing kratom doses from 1 to 12 g. The mean estimated $AUC_{0\text{-}T}$ ranged from 53.5 to 574.3 h·ng/mL. An increase in 7-OH C_{max}, and $AUC_{0\text{-}T}$ was noted with an increased dose of kratom; however, dose-proportionality could not be concluded because of the limited number of subjects in parallel dose cohorts.

Paynantheine

Plasma concentration-time profiles for paynantheine are shown in Figure S2 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20), with PK parameters provided in Table 4. Paynantheine produced a median T_{max} ranging from 2.99 to 5.04 hours for cohorts 1 to 5. The mean observed C_{max} ranged from 7.7 to 66.7 ng/mL with increasing kratom doses from 1 to 12 g. The mean $AUC_{0\text{-}T}$ estimate ranged from 47.0 to 727.6 h·ng/mL. An increase in paynantheine C_{max}, and $AUC_{0\text{-}T}$ was noted with increased dose of kratom; however, dose-proportionality could not be concluded because of the limited number of subjects in parallel dose cohorts.

Speciogynine

Plasma concentration-time profiles for speciogynine are shown in Figure S3 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20), with PK parameters provided in Table 4. Speciogynine had a median T_{max} ranging from 2.54 to 5.04 hours for cohorts 1 to 5. The mean observed C_{max} ranged from 6.3 to 58.9 ng/mL after administration of kratom doses from 1 to 12 g. The mean estimated $AUC_{0\text{-}T}$ ranged from 53.4 to 819.2 h·ng/mL. An increase in speciogynine C_{max}, and $AUC_{0\text{-}T}$ was noted with increased doses of kratom; however, dose-proportionality

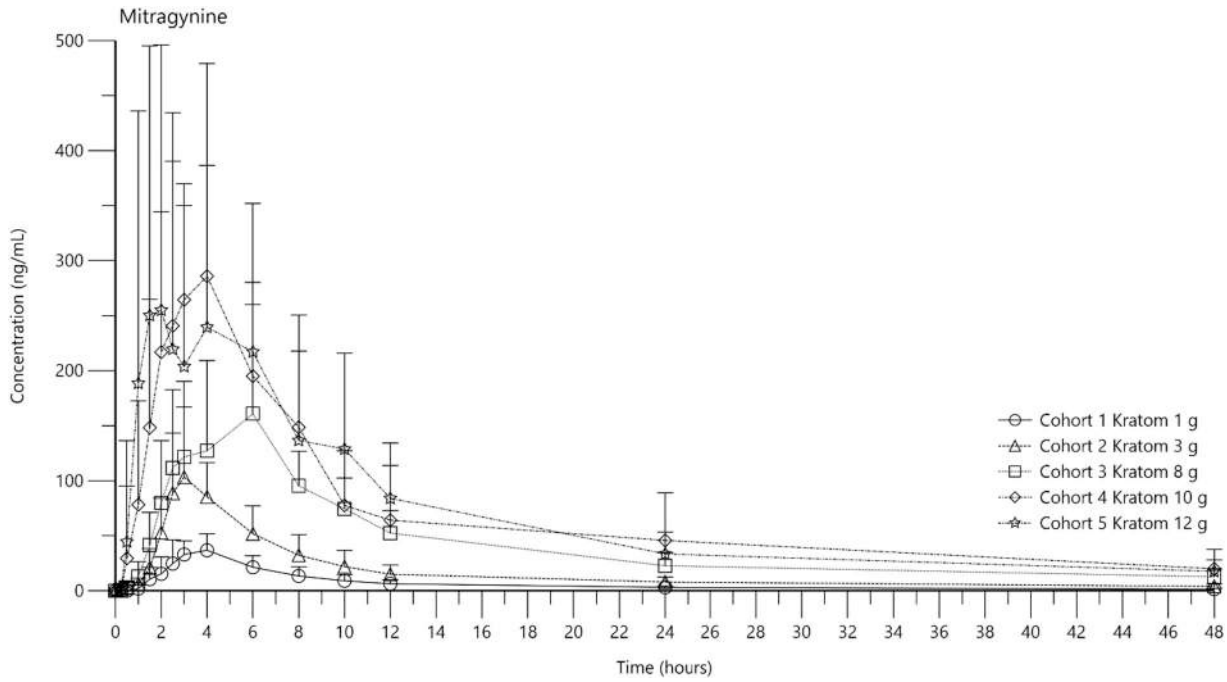


FIGURE 1. Plasma concentration-time profiles of mitragynine (Mean ± SD) by cohort in healthy participants (N = 6, each). The inset displays the 0 to 2 hours profile.

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TABLE 4. Pharmacokinetic Parameters of Mitragynine and 5 Additional Alkaloids After a Single-Dose Oral Administration of Kratom to Healthy Participants

Parameter* (Unit)	Cohort 1 (1 g) N = 6	Cohort 2 (3 g) N = 6	Cohort 3 (8 g) N = 6	Cohort 4 (10 g) N = 6	Cohort 5 (12 g) N = 6
Mitragynine (mg)	9.8-10.7	29.4-32.3	78.4-86.1	98.0-107.6	117.6-129.1
C_{max} (ng/mL)	41.7 (35.4)	118 (39.7)	197.5 (44.7)	318.1 (51.4)	378.5 (48.9)
T_{max} (h)	4.0 (2.5-6.2)	3.0 (2.5-4.0)	5.0 (2.5-6.1)	3.5 (2.5-4.0)	2.0 (1.0-6.0)
$AUC_{0\text{-}T}$ (h·ng/mL)	305.4 (37.9)	784.8 (48.5)	1942 (38.0)	3254 (68.0)	3271 (48.2)
λ_Z (1/h)	0.042 (21.5)	0.038 (14.6)	0.04 (24.5)	0.034 (29.5)	0.045 (22.1)
T_{half} (h)	17.28 (21.3)	18.64 (15.2)	17.47 (32.4)	22.07 (30.2)	15.9 (17.6)
7-hydroxymitragynine (mg)†	0-0.12	0-0.36	0-0.96	0-1.2	0-1.4
C_{max} (ng/mL)	6.7 (32.4)	17.8 (48.9)	34.0 (32.9)	48.4 (41.3)	58.4 (38.9)
T_{max} (h)	4.0 (2.5-6.2)	3.5 (3.0-4.0)	6.0 (3.0-10.0)	4.0 (3.0-4.0)	2.8 (1.5-6.0)
$AUC_{0\text{-}T}$ (h·ng/mL)	53.5 (40.4)	132.9 (64.5)	412.6 (37.2)	459.1 (49.3)	574.3 (33.9)
Metabolite:parent ratio (%)	17.5	16.9	21.2	14.1	17.6
Paynantheine (mg)	2.38-2.68	7.14-8.04	19.04-21.44	23.8-26.8	28.56-32.16
C_{max} (ng/mL)	7.7 (34.3)	19.0 (26.5)	29.7 (49.0)	66.7 (54.8)	65.8 (52.0)
T_{max} (h)	4.02 (2.5-6.18)	2.99 (2.5-4.0)	5.04 (3.0-6.14)	3.0 (2.5-4.0)	3.29 (1.5-6.01)
$AUC_{0\text{-}T}$ (h·ng/mL)	47.0 (40.3)	132.9 (33.1)	320.2 (41.8)	725.2 (62.4)	727.6 (54.7)
λ_Z (1/h)	0.13 (44.4)	0.077 (100.3)	0.043 (26.2)	0.041 (41.9)	0.049 (21.1)
T_{half} (h)	6.33 (50.2)	15.26 (60.2)	16.86 (23.8)	19.07 (35.0)	14.7 (20.5)
Speciogynine (mg)	1.8-1.98	5.4-5.94	14.4-15.84	18-19.8	21.6-23.76
C_{max} (ng/mL)	6.3 (30.4)	15.8 (27.2)	23.2 (43.5)	58.9 (60.3)	55.5 (55.9)
T_{max} (h)	4.02 (2.5-6.18)	3.0 (2.99-4.0)	5.04 (3.0-10.02)	4.0 (3.0-6.0)	2.54 (1.5-6.01)
$AUC_{0\text{-}T}$ (h·ng/mL)	53.4 (36.0)	154 (28.6)	341 (43.3)	819.2 (63.7)	790 (57.4)
λ_Z (1/h)	0.086 (69.6)	0.052 (45.7)	0.039 (16.9)	0.038 (37.6)	0.047 (19.4)
T_{half} (h)	11.06 (50.3)	15.91 (42.8)	18.09 (17.0)	20.26 (34.9)	15.20 (19.3)
Speciociliatine (mg)	3.64-4.04	10.92-12.12	29.12-32.32	36.4-40.4	43.68-48.48
C_{max} (ng/mL)	39.1 (30.0)	106.6 (22.4)	168.9 (40.3)	388.1 (56.0)	409.7 (47.3)
T_{max} (h)	4.02 (3.0-8.04)	5.0 (3.0-9.99)	8.07 (3.0-12.0)	5.0 (4.0-8.0)	5.06 (2.0-10.01)
$AUC_{0\text{-}T}$ (h·ng/mL)	573.9 (49.7)	1910.9 (24.9)	3359.2 (37.4)	7904.8 (60.6)	7449.8 (49.6)
λ_Z (1/h)	0.072 (33.3)	0.059 (24.8)	0.056 (41.1)	0.043 (18.7)	0.062 (16.6)
T_{half} (h)	11.17 (52.6)	12.55 (30.5)	14.04 (35.1)	16.39 (18.0)	11.41 (14.8)

*Doses are approximate based on measured alkaloid levels in capsules (Table 1).
†Some capsules were verified to have levels of 7-hydroxymitragynine below the limit of quantitation.
λ_Z indicates elimination rate constant; $AUC_{0\text{-}T}$, area under concentration-time curve up to last timepoint; C_{max}, peak plasma concentration; T_{half}, elimination half-life; T_{max}, time to reach C_{max}.

could not be concluded because of the limited number of subjects in parallel dose cohorts.

Speciociliatine

Plasma concentration-time profiles for speciociliatine are shown in Figure S4 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20), with PK parameters provided in Table 4. Median T_{max} ranged from 4.02 to 8.07 hours for cohorts 1 to 5. Mean C_{max} ranged from 39.1 to 409.7 ng/mL with increasing kratom doses from 1 to 12 g. The mean estimated $AUC_{0\text{-}T}$ ranged from 573.9 to 7904.8 h·ng/mL. An increase in speciociliatine C_{max}, and $AUC_{0\text{-}T}$ was noted with increased doses of kratom; however, dose-proportionality could not be concluded because of the limited number of subjects in parallel dose cohorts.

Pharmacodynamic VAS Results

Statistical Analysis on the Primary and Key Secondary Endpoints

In the tables and figures, K1, K3, K8, K10, K12, and P denote 1, 3, 8, 10, and 12 g doses of kratom, and pooled placebo, respectively. Table 5 summarizes the mean, SD, minimum (Min), the first quartile (Q1), median (Med), the third quartile (Q3), and maximum (Max) rating for the primary and key secondary endpoints for all 6 treatments.

As shown in Table S1 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20), mean E_{max} values produced by 8 and 12 g kratom had at least 10 points difference from the placebo group for the primary and all key secondary endpoints. A noticeable difference in median values between each dose of kratom and pooled placebo was found for the comparison between kratom 12 g and pooled placebo for all primary and key secondary measures except E_{max} of Take Drug Again VAS. The median TE_{max} for the primary endpoint was calculated and occurred ~1 hour post-dose for 12 g kratom. Figure S5 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20) shows the mean dose-response curves for these 4 endpoints. The mean dose responses produced by kratom 10 g for the primary and key secondary endpoints were all smaller than those produced by kratom 8 g. We plotted the heat maps for the individual time-course response profiles for 8, 10, and 12 g doses of kratom for the primary measure of drug liking VAS in Figure S6 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20).

Among the 6 participants who received kratom 8 g, one participant (ID 35) reported a drug liking VAS score of 100 at every timepoint, including the first assessment (i.e., 15 min after administration). Without including this subject, kratom 8 g would not show any meaningful effect on assessments of drug liking. Thus, it is possible the 12 g dose was the only dose to produce meaningful changes on subjective measures of drug effects. Figures S7A and S7B

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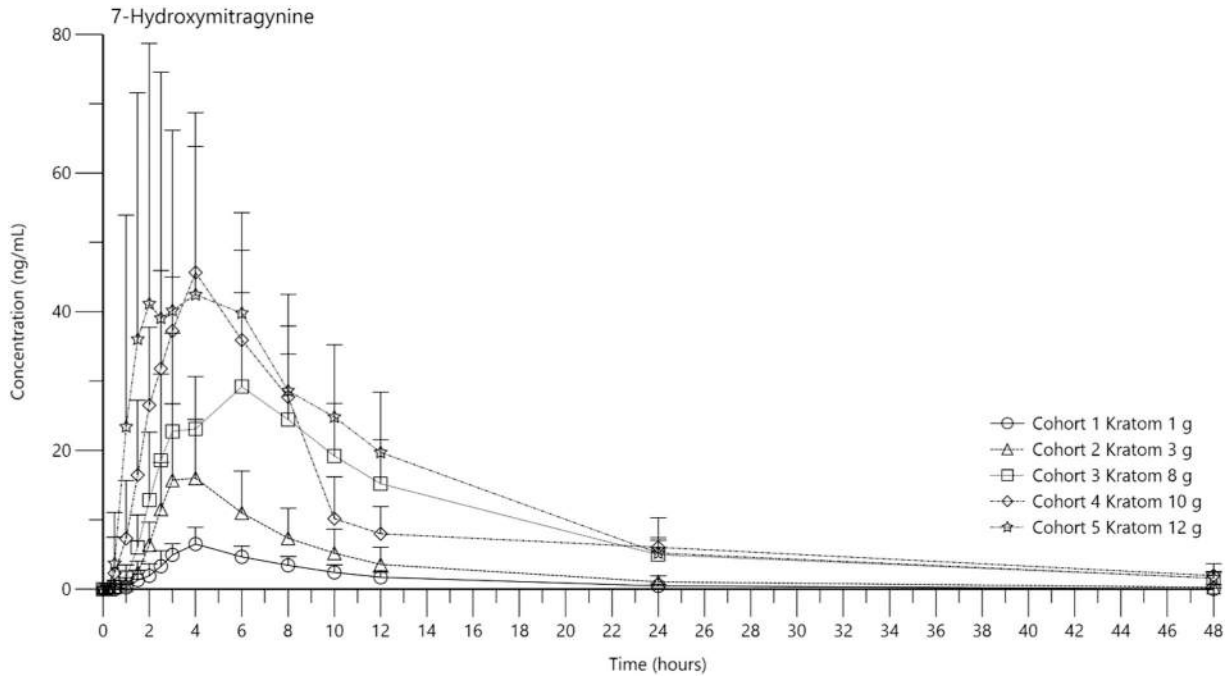


FIGURE 2. Plasma concentration-time profiles of 7-hydroxymitragynine (mean ± SD) by cohort in healthy participants (N = 6, each). The inset displays the 0-2 hours profile.

(Supplemental Digital Content 3, http://links.lww.com/JCP/B20) display the mean time-course profiles for drug liking VAS and High VAS.

In human abuse potential studies, the placebo range is typically considered to be between 40 and 60 for drug liking VAS. Figure S7A and S7B (Supplemental Digital Content 3, http://links.lww.com/JCP/B20) shows that the mean time-course profiles are bounded between 40 and 60 except the mean liking score of kratom 10 g that is below 40 eight hours post dose. For High VAS, the mean time-course profile for kratom 12 g is > 20 from 1.5 to 6 hours post dose. The peak mean response to kratom 12 g (33.7) occurred 6 hours post dose. As no data were collected between hours 6 and 8, it is possible, albeit unlikely, that the time to peak mean response occurs at hour 7.

Figure S8 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20) shows mean values with SDs for overall drug liking VAS and take drug again VAS 12 and 24 hours post dose for each treatment. Even though the means produced by pooled placebo were below 50, the means produced by kratom for these 2 endpoints were below, or close to 60.

Inferential Statistics

An ANOVA model was used to test treatment differences for the primary and key secondary endpoints. To control for multiplicity, Dunnett's test was used to compare the means of kratom groups to pooled placebo. Significant treatment differences were demonstrated only for drug liking Emax and High Emax. The analysis results are presented in Table S1 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20).

The mean differences between kratom 12 g and pooled placebo were 18.5 and 45.9 for drug liking E_{max} and High E_{max}, respectively. The analysis results show that the mean of kratom 12 g is significantly larger than that of pooled placebo at the significance level of 5% for these endpoints. The limitation of this inferential analysis is that the sample size for each treatment was very small. The assumptions of the ANOVA model were not met.

As a sensitivity analysis, Hodges-Lehmann estimator was used to estimate the difference between each dose of kratom and pooled placebo, the Wilcoxon Rank Sum test was conducted for each comparison, and Benjamini-Hochberg procedure was used for multiplicity adjustment. In the sensitivity analysis a significant result was only demonstrated in the comparison between kratom 12 g and pooled placebo for High Emax. However, clearly the difference in distributions between 8, 10, and 12 g doses of kratom and pooled placebo does not differ only in location. Therefore, the results from both the Dunnett test and Wilcoxon Rank Sum test should be interpreted with caution.

Descriptive Statistics for other Endpoints

Descriptive statistics are provided for 8 VAS endpoints: Emax scores of any effects, bad effects, good effects, feeling drunk, bowdle external perceptions composite, bowdle internal perceptions composite, and E_{min} scores of drowsiness/alertness, and relaxation/agitation, and 5 ARCI Emax endpoints: A, BG, LSD, MBG and PCAG, as well as Drug Similarity VAS. Table S2 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20) summarizes mean and SE for these endpoints produced by 8, 10, and 12 g doses of kratom and pooled placebo.

For kratom 12 g, VAS assessments of any effects, good effects, psychedelic effects (as measured through the Bowdle VAS), and bad effects were larger than those produced by 8 and 10 g doses of kratom and pooled placebo. Sedative effects (including PCAG) and LSD

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TABLE 5. Summary Statistics for Primary and Key Secondary Pharmacodynamic Endpoints

Endpoint: E_{max}	TRT	N	Mean (SD)	Min	Q1	Med	Q3	Max
Drug liking VAS	P	10	50.2 (0.4)	50.0	50.0	50.0	50.3	51.0
	K1	6	50.7 (1.6)	50.0	50.0	50.0	51.0	54.0
	K3	6	51.7 (4.1)	50.0	50.0	50.0	52.5	60.0
	K8	6	60.2 (20.0)	50.0	50.0	50.0	70.8	100.0
	K10	6	52.2 (2.4)	50.0	50.0	51.5	54.5	56.0
	K12	6	68.7 (22.6)	50.0	50.0	60.0	95.0	98.0
High VAS	P	10	0.4 (1.0)	0.0	0.0	0.0	0.3	3.0
	K1	6	0.5 (1.2)	0.0	0.0	0.0	0.8	3.0
	K3	6	1.2 (2.9)	0.0	0.0	0.0	1.8	7.0
	K8	6	10.8 (12.1)	0.0	0.0	9.0	21.3	28.0
	K10	6	8.5 (13.0)	0.0	0.0	3.5	16.0	34.0
	K12	6	46.3 (34.6)	0.0	22.5	37.0	83.5	94.0
Overall drug liking VAS	P	10	45.1 (15.5)	1.0	50.0	50.0	50.0	50.0
	K1	6	50.2 (0.4)	50.0	50.0	50.0	50.3	51.0
	K3	6	50.0 (0.0)	50.0	50.0	50.0	50.0	50.0
	K8	6	61.8 (20.5)	50.0	50.0	50.0	78.3	100
	K10	6	55.7 (32.4)	2.0	37.3	55.0	79.8	100
	K12	6	67.0 (19.5)	50.0	50.0	63.0	85.0	94.0
Take drug again VAS	P	10	45.9 (16.4)	0.0	49.8	50.0	50.0	60.0
	K1	6	53.3 (7.7)	50.0	50.0	50.0	55.5	69.0
	K3	6	50.0 (0.0)	50.0	50.0	50.0	50.0	50.0
	K8	6	65.2 (19.7)	50.0	50.0	59.0	79.8	100
	K10	6	49.8 (42.0)	0.0	0.8	56.0	89.5	100
	K12	6	61.0 (21.0)	42.0	48.0	50.0	83.0	95.0

E_{max} indicates peak response expressed as the maximum response; K, kratom dose; Max, maximum; Med, median; Min, minimum; *P*, placebo; Q1, first quartile; Q3, third quartile; (1 = 1 g, 3 = 3 g, 8 = 8 g, 10 = 10 g, 12 = 12 g); TRT, treatment; VAS, visual analog scale.

(dysphoria) were also larger for kratom 12 g than those produced by other doses of kratom and pooled placebo.

The drug similarity unipolar VAS items provide an estimate of how similar the drug classes with which drug users have familiarity with are compared with the test drug. On a 100 mm scale ranging from "Not at all similar" to "Very similar," participants are asked to compare the drug they received that day to a drug they previously declared having familiarity/experience with taking.

Drugs that a participant has not personally experienced often enough to use as a standard of comparison (< 2 lifetime uses of a given drug), are not included in the questionnaire. Most subjects had experience with benzodiazepines, nicotine, opioids, and THC. One subject in the kratom 10 g group and 3 subjects in the placebo group did not have experience with nicotine. One participant in the kratom 8 g group did not have experience with benzodiazepines.

From the summary statistics of Drug Similarity scores for benzos, caffeine, cocaine, ketamine, LSD, MDMA, methadone, nicotine, opioids, PCP, placebo, speed, Sudafed, and THC, Table 3 (Supplemental Digital Content 3, http://links.lww.com/JCP/B20) displays summary statistics for participants choosing benzodiazepines, opioids, and THC which produced larger means compared with other choice of drugs by participants in 8, 10 and 12 g doses of kratom groups. Placebo ratings are also included. Although all participants in the pooled placebo group correctly recognized that placebo was not a benzodiazepine or opioid, ~75% of these participants assigned a Drug Similarity score of 100 to THC.

DISCUSSION

This study was designed to evaluate the safety and tolerability, PK, and PD of single ascending oral doses of kratom in healthy, nondependent recreational polydrug users with opioid experience and to inform dose selection for a future human abuse potential study of kratom. Overall, using the specific botanical kratom sourced for the study and in a controlled clinical setting, kratom was generally well-tolerated in healthy subjects after oral doses of 1, 3, 8, 10, and 12 g. No serious adverse events (SAEs) or deaths occurred and no participants discontinued due to adverse events (AEs). The most frequently observed AEs were somnolence, nausea, and vomiting, which were observed at doses ≥8 g. The tolerability findings were likely attributable to kratom exposure rather than capsule burden, as placebo-treated participants who received an equivalent number of matched capsules did not exhibit similar AEs. However, the small sample size (n = 6 per dose) was insufficient to thoroughly characterize the safety profile of the kratom used in the study, and the study was not powered to capture rare or infrequent AEs.

Kratom doses of 8, 10, and 12 g produced measurable pupillary constriction relative to placebo. These data seem consistent with the reported agonist and partial agonist effects of kratom alkaloids (eg, mitragynine, speciociliatine, and 7-OH) on mu-opioid receptors. The magnitude of pupillary constriction produced by kratom was relatively mild, with a mean maximum constriction of −2.4 (0.38) mm at the 12 g dose. Although limited by cross-study comparisons, other investigations have reported larger reductions in pupil diameter after administration of classic opioids (eg, ≥ 3.0 mm after 40 mg oxycodone).[50,51] Although pupillary constriction may be considered a sign of mu-opioid receptor activation, many of the kratom alkaloids also exhibit affinity for mu-opioid receptors, as well as adrenergic, serotonergic, dopamine, and adenosine receptors, and their intrinsic efficacy at these receptor subtypes varies.[12–14,52,53] Moreover, the efficacy and activity of the alkaloids at these receptor subtypes can

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vary, giving rise to the complex pharmacology of kratom that has led some authors to describe kratom as an "atypical opioid."[54]

Generally, kratom doses of 1 and 3 g were similar to placebo across subjective measures and did not show effects on VAS assessments of drug effects. Compared with placebo, increased at-the-moment drug liking and high, and overall drug liking, as well as a desire to take the drug again were observed for kratom 12 g. Kratom doses of 8 and 12 g showed increases in endorsements of positive subjective effects with 12 g showing the highest responses compared with placebo for most measures. However, across the primary and key secondary endpoints, only the 12 g dose showed a statistically and clinically significant difference compared with placebo for drug liking and High E_{max}. The 12 g dose was also associated with increased ratings of feeling drunk, and good drug effects compared with placebo. These ratings were generally ≥10 points above placebo, suggesting they were clinically significant and may be indicative of an abuse potential. Of interest, the 10 g dose did not show a similar degree of increase, despite being the intermediate dose of 8 and 12 g. Consistent with these findings, the 10 g dose showed a trend of pupillary constriction that was not as pronounced as the 8 or 12 g doses. Bad drug effects were highest for the 12 g dose compared with placebo. The mean E_{max} of bad drug Effects produced by the 12 g dose was 25.5. Compared with placebo, the 8 and 12 g doses were associated with increased drowsiness and 8, 10, and 12 g doses were associated with increased relaxation. When examining psychedelic effects, only the 12 g dose showed elevated Bowdle VAS scores (mean External Perceptions 80.8; Internal Perceptions 67.3). Taken together with the AE profile (which did not show notable increases in the 10 g dose compared with 8 and 12 g), these data do not support the hypothesis that negative effects suppressed subjective responses at the 10 g dose. Rather, the apparent elevation in effects at 8 g seems to have been driven by a single participant (ID 001-035). This subject produced maximum ratings of drug liking at every timepoint, including the 15-minute post-dose time period where absorption of kratom is expected to have been minimal. This underscores the need for caution when interpreting data from small cohorts and highlights the potential value of qualification phases to identify and manage atypical response patterns in future studies.

The ARCI responses showed that the 12 g dose produced higher ratings of PCAG (sedatives) and LSD (dysphoria) compared with placebo. When assessed for similarity with known drugs of abuse, all doses of kratom were rated as being similar to THC (minimum similarity score = 82, maximum similarity score = 100). Kratom doses of 8, 10, and 12 g were also reported to produce effects similar to benzodiazepines and opioids. Irrespective of dose, kratom was not found to produce effects similar to drugs with stimulant or psychedelic-like properties including caffeine, cocaine, LSD, ecstasy, phencyclidine, or amphetamines.

Despite the limitations in the study due to few subjects in each dose cohort, plasma mitragynine, 7-hydroxymitragynine, paynantheine, speciogynine, and speciociliatine were measurable at every dose, and increased with dose as assessed by the extent of absorption after single doses of kratom from 1 to 12 g. Notable increases in the coefficient of variation were also observed with the administration of 10 g kratom for all compounds, with a general increase with increasing dose. Future studies may incorporate a larger sample size to decrease the variation observed in the PK data.

Peak plasma concentrations (C_{max}) of speciociliatine increased with dose as assessed by peak concentrations across the 1 to 12 g range. Plasma mitragynine, 7-hydroxmitragynine, paynantheine, and speciogynine increased in a slightly less than dose-proportional manner as assessed by peak concentration (C_{max}) after single doses of kratom from 1 to 12 g. The lack of dose-related increases in (C_{max} and AUC) was seen in doses above 8 g, and continued with subsequent doses of 10 and 12 g under fed conditions. These data suggest there is saturation in absorption of these compounds. It is important to note that the number of capsules also increased with increasing doses. Given the large number of capsules necessary to administer the higher doses (eg, 20 and 24 size 00 capsules for the 10 and 12 g doses), it is possible the large volume of botanical plant matter inhibited or slowed absorption.

Median times to peak concentration for mitragynine, 7-hydroxymitragynine, paynantheine, speciogynine, and speciociliatine were all similar, with median T_{max} values ranging from 2.00 to 8.07 hours. We note that the T_{max} values generally did not occur before the 2-hour timepoint and did not coincide with peak effects on subjective measures. The interaction of PK and PD effects is complex. Although reviews have found that maximum ratings of dug liking generally occur no later than peak plasma concentrations T_{max} or C_{max} alone may not be predictive of abuse potential.[55] The average range of elimination for mitragynine and speciociliatine was similar between cohorts 1 to 5, with values ranging from 11.17 to 22.07 hours. There were only 2 additional blood sample measurements after 12 hours at 24 hours and the last sample at 48 hours. This sampling schedule is limited in accurately determining the elimination rates for all kratom alkaloids in this study. With these limitations considered the average range of elimination half-life values for 7-hydroxymitragynine, paynantheine and speciogynine were not similar between cohorts 1 to 5, with values ranging from 6.69 to 19.07 hours. Increases in average range of elimination generally correlated with increased dose. Notably, other studies of encapsulated dried kratom powder found differences in T_{max} values, producing peak mitragynine concentrations 1 to 1.3 hours after administration and 7-OH T_{max} values 1.2 to 1.8 hours after administration.[31] However, the dose range was lower (0.5 to 4 g) and participants received kratom under fasting conditions. The present study administered kratom in the fed state, after a high-fat meal and preclinical data have demonstrated food effects in the bioavailability of mitragynine.[56] Interestingly, when examining the C_{max} values at identical (1 g) doses, broadly similar values were obtained, although there was substantial variability in our study due to the small sample size.[31]

An additional consideration when interpreting findings is the composition of kratom and the method of administration. Botanical formulations can vary greatly in the composition of the alkaloids that contribute to their pharmacological effects. Although mitragynine is the most abundant alkaloid isolated from kratom, minor alkaloids such as 7-hydroxymitragynine (which exhibits greater opioid potency than mitragynine and morphine)[37,38] and speciociliatine (which also has greater opioid activity than mitragynine)[13] may exert disproportionate influence on abuse-related and safety outcomes. The specific composition of the kratom used in this study contained ~5 mg of mitragynine per capsule (1% by weight) and low levels of 7-hydroxymitragynine (ranging from undetectable amounts to 0.07 mg/capsule). The kratom used in our study had alkaloid values similar to what has been reported in the literature;[8,10,36] however, it may be anticipated that different kratom

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formulations with different alkaloid compositions or amounts may have variable pharmacological effects. The kratom sold in the United States is commonly in the form of dry plant material filled into capsules. In contrast, kratom use in Asian countries is commonly consumed as an herbal tea or chewed. Variability in composition, preparation, and administration may all affect bioavailability and exposure. In this study, dry plant material was filled into oral capsules, as would be commonly available in the United States. Future studies may examine various strains or the effects of isolated alkaloids in characterizing the safety, PK, and PD effects of kratom.

In summary, in the current study, orally administered botanical kratom was well-tolerated with the most commonly reported AEs observed at the highest doses (eg, 8-12 g) and classified as mild in severity (i.e., grade 2 or lower). Doses ≥ 8 g produced modest pharmacodynamic effects, including pupillary constriction, and all doses produced measurable plasma concentrations of key kratom alkaloids. Kratom alkaloid exposure was generally dose proportional and the time to maximum plasma concentrations ranged from 2 to 4 hours. The highest dose of kratom (12 g) was associated with increases in a variety of subjective effects measures associated with abuse potential, including statistically significant increases in drug liking and also on a positive measure of drug effects "high" relative to placebo. When subjects completed drug similarity measures, kratom effects were reported as resembling those of THC, and being somewhat similar to opioids and benzodiazepines.

The small sample size, between-subjects design, and absence of a qualification phase to reduce variability in response measures (eg, inappropriate responding on VAS measures) limit conclusions regarding the subjective effects and abuse potential of kratom. Appropriately powered studies are required to further characterize kratom's PD profile, safety, and its relative potential for abuse. Finally, although the botanical kratom used in the study was well-characterized, it is unknown how the kratom used in our study compares to the variety of kratom-related products available in the commercial marketplace. Although the kratom used in this study had an alkaloid profile consistent with botanical products reported in the literature, it seems to have a substantially different alkaloid composition relative to newly emergent "spiked" or enhanced kratom-related products in the commercial marketplace.[57–59] Further studies are necessary to more thoroughly characterize the abuse potential of botanical kratom and determine the effects of the various alkaloids on its pharmacological effects.

AUTHOR DISCLOSURE INFORMATION

C.M. has served as an expert witness in several kratom lawsuits. B.S. and D.K. are full-time employees of Altasciences. D.M. is a part-time consultant of Altasciences. The remaining authors declare no conflicts of interest.

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Disclaimer: The opinions expressed in this article are those of the authors and should not be interpreted as the position of the US Food and Drug Administration.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analyzed during the current study are publicly available at clinicaltrials.gov under ID # NCT06072170.

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Natural Products

Elevated 7-Hydroxymitragynine Levels Found in Products Misbranded as Kratom

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Abstract

Background: The leaves of *Mitragyna speciosa*, commonly called kratom, have been consumed for centuries for their energizing and analgesic properties. Kratom has gained popularity in North America, with an estimated 16 million American consumers. Recently, a new product has emerged, labeled "kratom extract" but purportedly composed of high concentrations of 7-hydroxymitragynine, an oxidation product of mitragynine, the primary alkaloid found in kratom.

Objective: 7-hydroxymitragynine, a potent mu-agonist, may pose a serious public health risk. Commercial products labeled as containing 7-hydroxymitragynine were evaluated to determine whether the constituent was present and in what concentration.

Method: The concentrations of 7-hydroxymitragynine and mitragynine were determined by a previously validated HPLC-DAD method, AOAC *Official Method*[SM] **2017.14**. The chromatographic profiles of products were also compared to authentic kratom leaf.

Results: All products were described on their label as containing "kratom extracts." The concentration of mitragynine ranged from 2.0 to 6.0 mg/g dry weight, substantially lower than typically found in kratom leaves or extracts. 7-hydroxymitragynine was determined at 22–75 mg/g dry weight, 5–28% higher than the label claims. The chromatographic profiles of all products were inconsistent with kratom leaf, having exceptional levels of 7-hydroxymitragynine, lacking the other major naturally occurring alkaloids, and exhibiting peaks not observed in leaf.

Conclusions: The products were verified to contain very high levels of 7-hydroxymitragynine, achievable only by synthetic means. These products differ significantly from authentic native kratom leaf and are not kratom extracts, as labeled, but rather synthetic derivatives that constitute unapproved new drugs being unlawfully marketed. There is an urgent need to differentiate kratom, which has been safely consumed for centuries, from 7-hydroxymitragynine, which may pose a significant risk to public health.

Highlights: Traditional kratom leaf is distinguished from products found to contain high concentrations of 7-hydroxymitragynine, only achievable by synthetic means. These unapproved new drugs contained other unidentified constituents, also not present in the native kratom leaf. The presence of concentrated 7-hydroxymitragynine products in the market masquerading as kratom is misleading, unlawful, and potentially dangerous to the public.

Kratom is the common name attributed to the deciduous tree *Mitragyna speciosa* (Korth.) Havil., a tropical tree that is part of the Rubiaceae family, sharing a lineage with coffee and indigenous to Southeast Asia. Historically, kratom leaves were chewed or brewed to mitigate fatigue and alleviate pain, particularly among manual laborers in the hot, tropical environments of countries such as Thailand and Malaysia (1–3). More recently, kratom has gained popularity in North America, with an estimated 16 million Americans reportedly consuming kratom for a mild energy boost, to improve mood, and to alleviate stress and minor pain (2, 4–7).

Kratom leaf contains a broad spectrum of plant secondary metabolites including alkaloids, flavonoids, polyphenolic compounds, triterpenoids, triterpenoid saponins, monoterpenes, and secoiroids (2). The alkaloids are recognized as the primary active compounds, with leaves containing several alkaloids representing 0.5–3% of the total weight (8–10). The primary alkaloid is mitragynine, reported to comprise 51–82% of the total alkaloid composition weight (8–12) and on a percent weight basis ranging from <1.0% but usually not >2.5% (10, 13). The other pharmacologically active alkaloids typically found within the leaf are paynantheine, speciogynine, and speciociliatine, accounting for most of the remaining alkaloid composition.

Mitragynine is metabolized in the human body to some degree into 7-hydroxymitragynine (7-HMG), an active human metabolite (14). Although initially thought to be present in native kratom leaf material, it is now recognized that 7-HMG is generated during drying and processing as an oxidative byproduct of mitragynine, observed only in post-harvest kratom leaf at trace levels (0.003–0.04%, w/w) (15–17). The pharmacology of 7-HMG drastically differs from mitragynine and the other natural kratom alkaloids. Unlike mitragynine, which acts on multiple neurotransmitter systems in the human body, 7-HMG exerts effects similar to

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conventional opiates, with studies showing it to have a binding affinity that is up to 22 times higher than morphine at the μ-opioid receptor (18–22). A recent study has reported 7-HMG induced significant respiratory depression comparable to morphine, which was reversed by naloxone, and further they found mitragynine increased respiratory frequency and is unaffected by naloxone (23). These findings highlight the critical differences between the naturally occurring kratom alkaloid mitragynine and the semisynthetic product 7-HMG.

As kratom products have surged in popularity within the United States, a new class of products has emerged that claim exceptionally high concentrations of 7-HMG, while still being labeled as kratom or "kratom extract" (11, 24). A recent survey found that 178 brands of products sold online were labeled as containing 7-HMG with claims of potentiating the product's analgesic or euphoric effects and positioning these products as "enhanced" versions of traditional kratom (25). Given that only trace levels of 7-HMG are reported in native kratom leaf, these products would have to be synthetically created in order to achieve the levels of 7-HMG claimed (11, 24). To date, no published study has verified the veracity of 7-HMG product claims, despite the potential for carrying a substantially higher risk profile than kratom. Moreover, the mislabeling of these products as kratom is misleading to consumers, who may believe they are ingesting a botanical product with a long history of safe use.

Herein we report the analysis of a selection of readily available so-called kratom products in the U.S. market that claim elevated levels of 7-HMG. Using a previously validated official AOAC First Action *Official Method*^{SM} high-performance liquid chromatography (HPLC) method (26, 27), we aim to quantify mitragynine and 7-HMG content, to allow for an accurate comparison with traditional kratom leaf standards. The described method met all AOAC standard method performance requirements for determining mitragynine and 7-HMG in kratom finished products with RSD values of ≤1% for both analytes, and recoveries between 94 and 99% (26).

Our findings are discussed in the context of the potential risks of concentrated 7-HMG products and their legal and health ramifications. In doing so, we hope to contribute to the growing discourse on kratom's safety profile and encourage regulatory interventions that differentiate kratom from synthetic unapproved drugs and protect consumers from potentially misleading product claims.

Experimental

Test Materials

Tea-cut kratom leaf was obtained from a supplier in Indonesia. Commercial products marketed as having elevated levels of 7-HMG were purchased from U.S. retailers. A total of five product brands were obtained. For Brand 1, three separate packages with the same lot number were obtained from the same vendor, and for Brand 2 two packages with the same lot number were obtained from the same vendor. All products were stored in the original packaging at room temperature until analysis. A summary of the products with their label claim is shown in Table 1.

Reference Standards

Certified reference standards for mitragynine ($C_{23}H_{30}N_2O_4$), FW of 398.495 g/mol CAS 4098-40-2, and 7-hydroxymitragynine ($C_{23}H_{30}N_2O_5$), FW of 414.49 g/mol CAS 174418–82-7, were obtained from Chromadex Standards (Longmont, CO, USA). The purity of the standards was confirmed using quantitative nuclear magnetic resonance (qNMR).

Reagents

HPLC grade methanol, acetonitrile, formic acid, and ammonia hydroxide and ACS grade acetic acid were purchased from VWR (Mississauga, ON, Canada). LC-MS grade ammonium bicarbonate was purchased from MilliporeSigma Canada Ltd (Oakville, ON, Canada). HPLC water was obtained from a Nanopure filtration unit.

Solutions

The following solutions were prepared before analysis.

(a) *Reference standard and liquid sample diluent.*—100% methanol.
(b) *Extraction solvent.*—0.5M acetic acid in 70% methanol.
(c) *HPLC Mobile phase A.*—5.0 mM ammonium bicarbonate buffer, pH 9.50.
(d) *HPLC Mobile phase B.*—Acetonitrile.

Sample Preparation

Analysis was performed using a previously validated method, AOAC *Official Method of Analysis* **2017.14**, for the determination of mitragynine and 7-HMG in raw materials and finished products (27). Sample preparation and working standard concentrations were modified to account for the higher concentrations of 7-HMG expected to be found in the commercial products.

For kratom leaf tea-cut material, 100.0 ± 5.0 mg was weighed into a 50 mL polypropylene centrifuge tube, and 10 mL of extraction solvent was added. The tube was capped and mixed using a vortex mixer for 30 seconds and then shaken on a wrist action shaker for 30 minutes. Approximately 5 mL of the solution was then filtered through a 0.45 µm PTFE filter into a test tube. Approximately 1.5 mL of the filtered solution was transferred into an HPLC vial for analysis.

For commercial product samples, a single tablet was removed from the product packaging and weighed to determine the product tablet weight. The tablet was then crushed in a mortar and pestle until a fine powder was produced. The powder was transferred to a preweighed 50 mL polypropylene centrifuge tube. The amount of powder transferred into the tube was determined and

Table 1. Summary of products analyzed and their label claim

Product	Label claim	Tablet weight, mg
Brand 1 Package 1	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	632.1
Brand 1 Package 2	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	651.7
Brand 1 Package 3	15 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 7.5 mg of 7-HMG	632.4
Brand 2 Package 1	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	350.6
Brand 2 Package 2	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	350.1
Brand 3	<14 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering <7 mg of 7-HMG	754.5
Brand 4	18 mg of 7-HMG per tablet claimed on front of package 15 mg of 7-HMG per tablet claimed on back of package	257.0
Brand 5	18 mg of 7-HMG per tablet; 1 serving = ½ tablet delivering 9 mg of 7-HMG	655.5

recorded, and 20 mL of the extraction solution was added to the tube. The tube was capped and mixed using a vortex mixer for 30 seconds and then shaken on a wrist action shaker for 30 minutes. Approximately 5 mL of the solution was then filtered through a 0.45 μm PTFE filter into a test tube. The solution was then subjected to a 1/10 dilution by adding 1 mL of the filtered solution to 9 mL of extraction solution. The diluted solution was mixed by shaking. Approximately 1.5 mL of the diluted solution was transferred to an HPLC vial for analysis.

Standard Preparation

Stock solutions were prepared by weighing out 5 mg of each chemical standard into separate 10 mL volumetric flasks. The flasks were diluted to volume with extraction solvent. Working standard solutions were prepared by serial dilution with extraction solvent as the diluent. For mitragynine, five working standard solutions ranging from 1 to 50 mg/L were prepared. For 7-HMG, five working standard solutions ranging from 1 to 150 mg/L were prepared. All working standards were analyzed by HPLC to serve as external calibration curves.

HPLC Analysis

All samples and standards were analyzed in accordance with the HPLC conditions described in AOAC **2017.14**, the previously validated method. In brief, separation was achieved on a Kinetex® EVO C18 100 Å, 150 × 4.6 mm, 5 μm particle size column (Phenomenex, Torrance, CA) using an Agilent 1200 Series Liquid Chromatograph equipped with binary pump and degasser, temperature-controlled column compartment, autosampler, flow cell, and diode-array detector (Agilent, Mississauga, ON, Canada). The column temperature was set at 25°C, the injection volume was 5 μL, and the flow rate was 1.5 mL/min. The instrument was set for a gradient binary solvent system, as described in Table 2. The detector was set to monitor at 226 nm (4 nm bandwidth).

Quantification of Alkaloids in Samples

The mitragynine and 7-HMG content for each of the samples was determined using the external calibration curves prepared for each of the analytes. The concentrations for each alkaloid in the tablet samples were first quantified in mg/g using the following equation:

$$\frac{P_0 - b_0}{m_0} \times \frac{V}{W} \times \frac{D}{1000}$$

P_0 = peak area of target analyte in sample chromatogram, b_0 = y-intercept of calibration curve for the alkaloid, m_0 = slope of calibration curve for the alkaloid, V = volume of test solution in mL, W = weight of sample in g, and D = dilution factor.

The amount of each alkaloid in each sample per serving size was then determined by multiplying the result from the equation and the weight of one serving as per the product label.

Table 2. HPLC gradient program

Time, min	% Solvent A	% Solvent B
0.0	70	30
17.0	35	65
17.9	35	65
18.0	70	30

Results and Discussion

Employing OMA AOAC **2017.14** for the determination of mitragynine and 7-HMG within kratom leaf materials and the products labeled as "kratom extracts" but purported to have high levels of 7-HMG ensured precision and accuracy in the analysis (26, 27). The mitragynine and 7-HMG content found in each product is presented in Table 3. The results are expressed as mg of alkaloid per serving based on label directions for serving size and mg of 7-HMG per tablet. The most abundant alkaloid in the kratom leaf material is mitragynine at 11.8 mg/g or 1.18% (w/w), which is within the reported levels typically found in leaf (10, 13). 7-HMG was not detected in leaf material, with a detection limit of <0.004% (w/w) for the analytical method (26). In contrast, mitragynine was not the most abundant alkaloid in the products analyzed in this study, with all but one product containing <0.6% (w/w). The most abundant alkaloid detected in all products tested was 7-HMG, with concentrations ranging from 22 to 75 mg/g. These concentrations were 5 to 28% higher than the levels claimed on the labels of these products. One product, Brand 4, tested had differing levels of 7-HMG listed on the front and back labels with neither value reflecting the actual level of 7-HMG determined. These findings show that consumers cannot rely on label statements being made on these products and would ingest levels of 7-hydroxymitragynine significantly higher than expected.

As discussed above, 7-HMG is a human metabolite of mitragynine and is found occasionally at trace levels in kratom leaf postharvest, as a byproduct of oxidation. To achieve levels of 7-HMG as found in these commercial products would require the addition of exogenous 7-HMG or chemical conversion of mitragynine to the oxidation product. Given that mitragynine in kratom leaf is typically <1.0% and usually not exceeding 2.5% on a dry weight basis and the determined levels of 7-HMG in these products are significantly greater, it is likely that highly concentrated extracts or an isolate of kratom alkaloids is used as a substrate for the synthetic transformation. Regardless, any processes used to elevate 7-HMG content would essentially create a new synthetic product that would be so different from natural kratom leaves that it could no longer be considered "kratom" in any meaningful sense of the term.

Differences in these 7-HMG products from typical kratom leaf and leaf extracts are also demonstrated through their chromatographic profiles. Figure 1 shows chromatograms obtained from authentic kratom leaf and one of the 7-HMG products analyzed using the described HPLC method. The kratom leaf material shows a typical kratom leaf profile with mitragynine as the most abundant alkaloid and the presence of the other major kratom alkaloids, namely speciogynine, panynantheine, and speciociliatine. The absence of a peak corresponding to 7-HMG is also typical of properly harvested and stored kratom leaf material.

In contrast, the most prominent peak in the 7-HMG product chromatogram is 7-HMG, with a much smaller mitragynine peak. The other common kratom alkaloids are conspicuously absent. More alarming is the presence of peaks not observed in the native leaf material, indicating the presence of compounds not naturally occurring in kratom that are likely byproducts of the process undertaken to synthetically convert mitragynine to 7-HMG. Consumers of these products may also be exposed to potentially toxic compounds with completely unknown risk profiles.

Another significant issue observed in this study was the lack of consistency both across the brands and also within products. Figure 2 shows a chromatographic overlay of all products analyzed and shows significant variance among the different

Table 3. Mitragynine and 7-HMG content found in commercial products

Product ID	Tablet weight, mg	Mitragynine content, mg/g	Mitragynine per serving, mg	7-HMG content, mg/g	7-HMG per serving, mg	7-HMG Label claim, mg/tablet	7-HMG content, mg/tablet	7-HMG % label claim
Brand 1–1	632.1	4.0	1.2	28.0	8.9	15	17.7	118%
Brand 1–2	651.7	2.7	0.8	29.0	9.4		18.9	125%
Brand 1–3	632.4	5.6	1.8	25.3	8.0		16.0	107%
Brand 2–1	350.6	2.9	0.5	53.9	9.5	18	18.9	105%
Brand 2–2	350.1	3.2	0.5	55.6	9.8		19.5	108%
Brand 3	754.5	2.1	0.8	22.4	8.5	<14	16.9	121%
Brand 4	257.0	14.6	3.8	74.7	19.2	18 (front of package) 15 (back of package)	19.2	107% (front) 128% (back)
Brand 5	655.5	2.1	0.7	29.9	9.8	18	19.6	109%

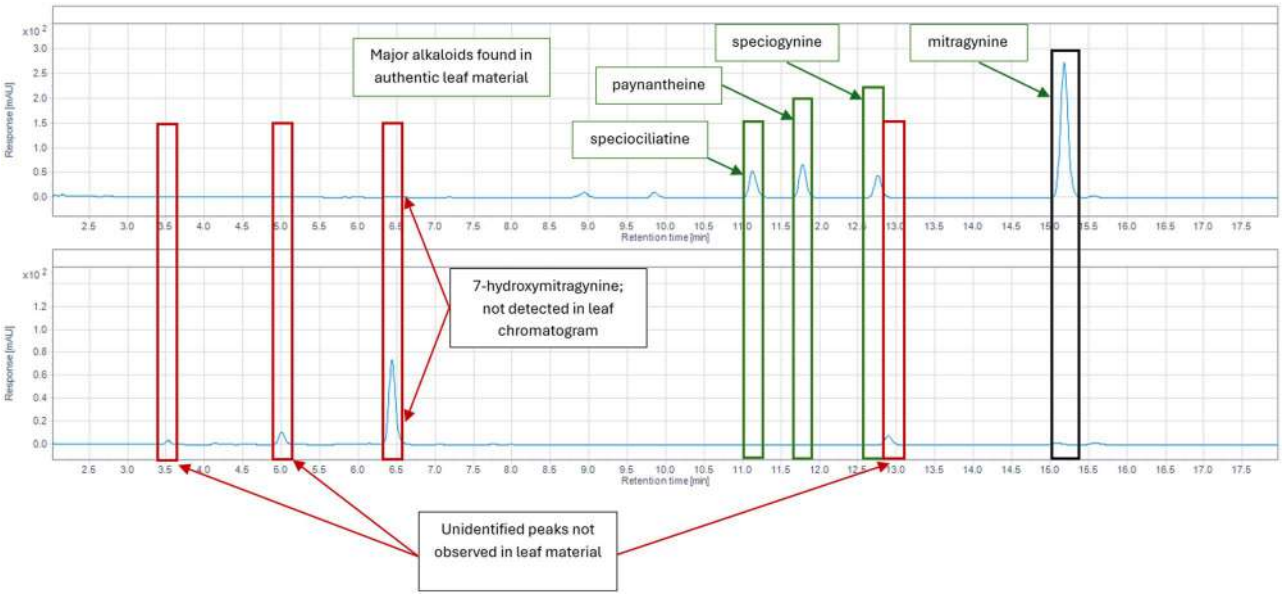


Figure 1. HPLC chromatograms of a leaf material (top) and 7-HMG product (bottom). Prominent peaks in each chromatogram are labeled.

products. All chromatograms exhibit 7-HMG as the most prominent peak; however, the presence and abundance of other peaks, both known and unknown, differ despite all products being labeled as "kratom leaf extracts." The observed differences among the products could be related to either the varying substrates used as the synthetic starting material or inconsistencies in the process itself leading to variable synthetic side- and end-products. Figure 3 shows the chromatograms obtained from analysis of the three packages of Brand 1 purchased together and shows a disturbing lack of within-product consistency. This is also evidenced in Table 3, which shows the tablets in Brand 1 contained 16.0–18.9 mg 7-HMG and 1.6–3.6 mg mitragynine per tablet. This demonstrates a lack of consistency in manufacturing and a lack of compliance with good manufacturing practices, as required by law for the safety of consumers.

Marketing these products as "kratom" is both false and misleading. Consumers unfamiliar with kratom or those relying on the package labels may erroneously purchase the products believing they are receiving a traditional kratom product, which has been associated with a long history of use in Southeast Asia (28, 29). There are an estimated 3–16 million users in the United States (30–32) and the available toxicology, clinical, and epidemiolocal studies indicate traditional kratom use has a low risk profile with respect to acute and chronic use (9, 33–37). However, this low-risk safety profile is attributed to kratom leaf and fluidized kratom leaf extracts, such as teas and decoctions, and cannot be extended to products that fall outside that definition. Consumers of 7-HMG products could be unknowingly exposing themselves to unapproved drug products with potentially higher abuse liability and unknown risks.

As discussed in the introduction, the pharmacology of 7-HMG differs from mitragynine in that it predominantly acts on opioid receptors, rather than multiple systems in the human body (18–20, 22). When available in minimal amounts and balanced with the full spectrum of alkaloids and other plant secondary metabolites, as found in traditional kratom leaf preparations, the risk of 7-HMG has never been an apparent issue. Based on ED_{50} values and minimum effective doses observed in several studies, exposure to 7-HMG for a typical kratom serving is expected to be below effect levels (9). These 7-HMG products, however, contain high concentrations of 7-HMG, do not contain the full spectrum of kratom alkaloids found within kratom leaf, and contain unknown synthetic byproducts for which there is no established safety profile. The bottom line is that these are synthetic products whose

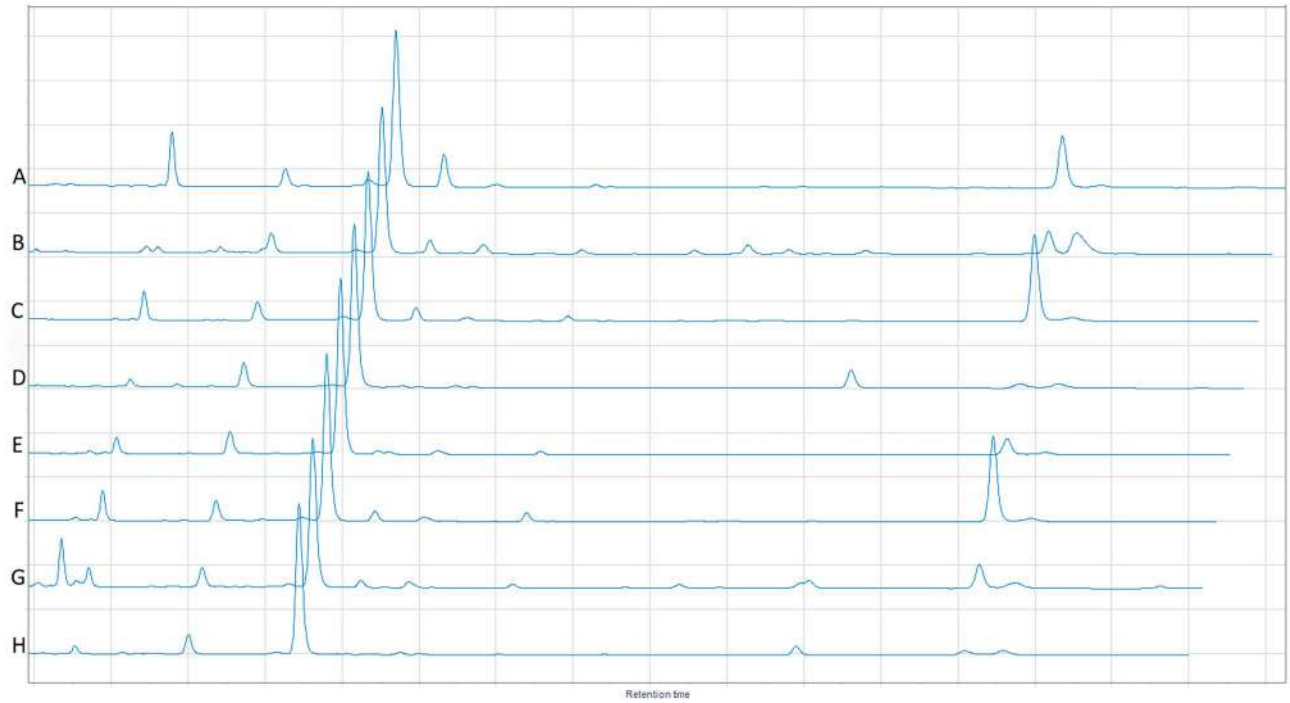


Figure 2. Stacked overlay of all eight 7-HMG products analyzed in this study. The overlaid chromatograms are identified as follows: Brand 1 Package 1 (A), Brand 1 Package 2 (B), Brand 1 Package 3 (C), Brand 2 Package 1 (D), Brand 2 Package 2 (E), Brand 3 (F), Brand 4 (G), and Brand 5 (H). A 1% time offset between the chromatograms has been used for better visualization of peaks.

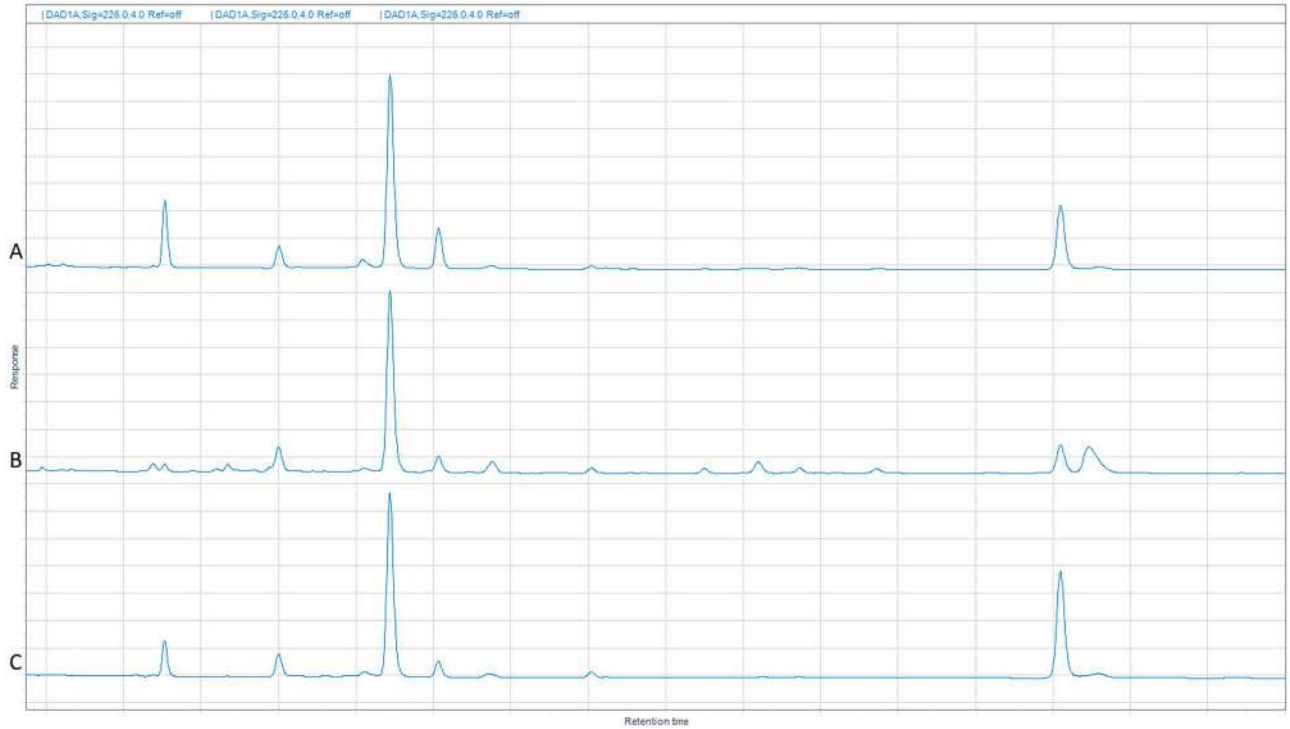


Figure 3. Chromatograms obtained from the analysis of three different packages of the same 7-HMG product. The overlaid chromatograms are identified as follows: Brand 1 Package 1 (A), Brand 1 Package 2 (B), and Brand 1 Package 3 (C). All packages were purchased from the same vendor at the same time.

chemical profiles no longer resemble kratom leaves and cannot be considered equivalent chemically, functionally, or biologically to traditional kratom preparations.

It has been proposed that the amount of 7-HMG should be limited in products, because it is not naturally occurring in native kratom leaf and is found only at trace levels in processed post-

harvest leaf material (38). This is a reasonable approach and would serve to help consumers differentiate kratom products from the synthetic unapproved drugs that have been masquerading as kratom. Another consideration that must not be ignored is that as long as 7-HMG products marketed as "kratom" are not removed from commerce, the adverse events associated with these products may be incorrectly attributed to kratom. This could negatively impact the regulation of kratom because policymakers and regulators may not differentiate between traditional kratom formulations and these unapproved synthetically produced drugs. Future considerations should include approaches for further differentiating traditional kratom leaf products from synthetic products post-consumption beyond mitragynine, which is often used as a biological marker postmortem (28).

Conclusions

This study shows that marketed 7-HMG products contain very high levels of 7-HMG and lack the chemical features of kratom leaf. To achieve these levels, a chemical conversion of mitragynine to 7-HMG is required, a process which for the products tested creates additional, unidentified compounds also not native to kratom leaf. These synthetic products were also found to be inconsistent with respect to 7-HMG content and chemical profile, demonstrating an alarming lack of control in manufacturing. 7-HMG products are misrepresented as kratom products and pose a serious concern due to the lack of proper manufacturing, characterization, and safety data. The marketing of these products is unlawful, and it is in the best interest of public health that these unapproved, synthetic products are removed from the market.

CRediT Author Statement

Paula N. Brown (Conceptualization [Lead], Funding acquisition [Lead], Investigation [Equal], Methodology [Lead], Project administration [Lead], Resources [Lead], Supervision [Lead], Writing—original draft [Equal], Writing—review & editing [Lead]), Michael Chan(Conceptualization [Supporting], Formal analysis [Lead], Investigation [Equal], Project administration [Supporting], Supervision [Supporting], Writing—original draft [Equal]), Xiaohui Zhang(Data curation [Equal], Formal analysis [Supporting], Investigation [Supporting], Methodology [Equal]), and Thomas Brendler(Conceptualization [Supporting], Investigation [Supporting], Writing—review & editing [Supporting])

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Conflict of Interest

P.B. and T.B. serve as scientific experts and regulatory consultants on botanical dietary ingredients, including kratom. No clients had any contribution or input into this study. M.C. and X.Z. have nothing to disclose.

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ORIGINAL ARTICLE

Suspected Adulteration of Commercial Kratom Products with 7-Hydroxymitragynine

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Abstract
Introduction Kratom (*Mitragyna speciosa*), a plant native to Southeast Asia, has been used for centuries for its stimulant and opium-like effects. Mitragynine and 7-hydroxymitragynine, exclusive to *M. speciosa*, are the alkaloids primary responsible for Kratom's biologic and psychoactive profile, and likely contribute to its problematic use. We purchased several commercially available Kratom analogs for analysis and through our results, present evidence of probable adulteration with the highly potent and addictive plant alkaloid, 7-hydroxymitragynine.
Methods A simple and sensitive LC-MS/MS method was developed for simultaneous quantification of mitragynine and 7-hydroxymitragynine in methanol extract of marketed Kratom supplements.
Results We found multiple commercial Kratom products to have concentrations of 7-hydroxymitragynine that are substantially higher than those found in raw *M. speciosa* leaves.
Conclusions We have found multiple packaged commercial Kratom products likely to contain artificially elevated concentrations of 7-hydroxymitragynine, the alkaloid responsible for *M. speciosa's* concerning mechanistic and side effect profile. This study describes a unique form of product adulteration, which stresses the importance of increased dietary supplement oversight of Kratom-containing supplements.

Keywords Kratom · 7-Hydroxymitragynine · Mitragynine · *Mitragyna speciosa* · Drugs of abuse

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Background

Kratom (*Mitragyna speciosa*), a member of the coffee family, is indigenous to Southeast Asia. Kratom has been utilized for thousands of years to enhance work productivity, cultural ceremonies, for medicinal purposes, and as a substitute for ethanol or opium [1–4]. Kratom contains a multitude of alkaloids, many of which are responsible for the biological effects of the plant [5]. Despite being criminalized in foreign countries with punishments of up to 70 years or more [1, 6], Kratom use has become increasingly popular in the USA, where it is still legal to possess and consume in most states [7–11].

M. speciosa contains over 40 different alkaloids which comprise 0.5–1.5 % of the plant matter [12, 13]. Of these, the most clinically consequential are mitragynine and 7-hydroxymitragynine, compounds which are found exclusively in *M. speciosa* [6, 14, 15]. Mitragynine, a major indole-containing constituent similar in structure to yohimbine [13], accounts for up to 66 % of the plant's alkaloid content [15, 16]. It produces opioid-like effects predominantly via mu- and delta-opioid receptor agonism [5, 13, 17] in addition to modulation of the descending serotonergic and noradrenergic

pathways [18]. In vitro characterization of mitragynine receptor binding in the central nervous system reveals the complexity with which Kratom acts (see Table 1) [10]. In regard to its antinociceptive actions, mitragynine is one third as potent as morphine and three times as potent as codeine [18, 19].

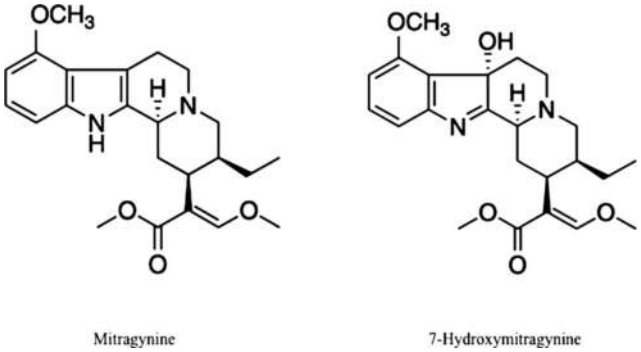
A minor alkaloid constituent of Kratom, 7-hydroxymitragynine, was first described in 1994 and is structurally identical to mitragynine except the addition of a hydroxyl group at the C7 position (see Fig. 1) [16]. Because auto-oxidation of mitragynine into 7-hydroxymitragynine occurs [16], this minor constituent may be an innate natural product or arise from mitragynine metabolism within the *M. speciosa* plant. Accounting for roughly 2 % of the plant's alkaloid content [16], 7-hydroxymitragynine is an opioid receptor agonist (like mitragynine) but demonstrates potent mu and kappa receptor selectivity [12]. This alkaloid is the major contributing factor for Kratom's analgesic properties, demonstrating opioid receptor affinity up to 17 times that of morphine [20–22]. It may also contribute to problematic Kratom use, which has been reported numerous times in the literature [10, 23–26]. 7-Hydroxymitragynine's role in Kratom abuse is supported by Matsumoto et al.'s findings (2005), which demonstrate development of tolerance, cross-tolerance to morphine, and physical dependence in 7-hydroxymitragynine treated mice [27]. Well established is the knowledge that morphine tolerance and physical dependence are secondary to mu-opioid receptor agonism [28, 29]. Therefore, 7-hydroxymitragynine, a potent mu-opioid receptor agonist, is likely to be a major contributing factor to the addictive potential of Kratom.

We purchased several commercially available Kratom analogs for analysis and, through our results, present evidence of adulteration with the highly potent and addictive plant alkaloid, 7-hydroxymitragynine.

Table 1 Central nervous system receptor binding data for mitragynine [10]

Percentage inhibition of radioligand binding by mitragynine at selected receptor systems
Adenosine A2A 65.66
Adrenergic (alpha 2) 61.92
Dopamine D2s 54.22
Opioid, mu 89.52
Opioid, kappa 90.21
Opioid, delta 7.00
Serotonin, 5HT2C 58.77
Serotonin, 5HT7 64.41
Dissociation constants for opioid receptor binding
Mu receptor 204 ± 26 nM
Delta receptor 2250 ± 120 nM
Kappa receptor 455 ± 47 nM

For contextual purposes, the rank order of binding affinity at the mu opioid receptor from highest to lowest is as follows: fentanyl [108] > 7-hydroxymitragynine [20] > mitragynine [21] > morphine [21]



Mitragynine 7-Hydroxymitragynine

Fig. 1 Structures of mitragynine and 7-hydroxymitragynine

Methods

Materials and Reagents

Mitragynine and 7-hydroxymitragynine (purity ≥98 %; IP grade) as free base (purity >98 %) were synthesized in-house. Mitragynine was isolated from dried leaves of *M. speciosa* as described by Ponglux et al. [16]. Synthesis of 7-hydroxymitragynine was performed in-house as reported earlier by Takayama et al. and Ponglux et al. [16, 21]. Purities of mitragynine and 7-hydroxymitragynine were determined by 1H nuclear magnetic resonance (NMR), 13C NMR, high performance liquid chromatography, elemental analysis, and high-resolution mass spectrometry. Ondansetron (internal standard [IS]) as ondansetron hydrochloride dihydrate was procured from AChemTek Inc. (Worcester, MA, USA). LC-MS grade acetonitrile, methanol, water, and ammonium acetate were purchased from Fisher Scientific (Fair Lawn, NJ, USA). Kratom supplements as Phoria^{tm} (Miami, FL, USA) red, Phoria^{tm} green, Phoria^{tm} regular, Phoria^{tm} Borneo white vein, Phoria^{tm} Borneo red vein, Phoria^{tm} Borneo green vein, Phoria^{tm} maeng da blue lotus, Phoria^{tm} maeng da kava, Green vein extra strength Kratom shot (Orbit distributors, Houston, TX, USA), and Viva Zen (Vivazen distribution, Salt Lake City, UT, USA) were purchased from a local market.

Preparation of Standards and Quality Control Samples

The primary stock solutions (1 mg/mL) of mitragynine and 7-hydroxymitragynine were separately prepared by dissolving the requisite amount in acetonitrile. The primary stock solutions were further diluted with acetonitrile for the preparation of working stock solutions (WS) (100 μg/mL). Combined spiking stock solutions (CSSs) containing 10, 7.5, 4.5, 2.5, 1.25, 0.25, 0.125, and 0.05 μg/mL for calibration standards (CS) and 9, 5, 0.1

and 0.05 μg/mL for quality control (QC) standards of each analyte were prepared by appropriately diluting the WS in acetonitrile. The CS containing 1, 2.5, 5, 25, 50, 90, 150, and 200 ng/mL of mitragynine and 7-hydroxymitragynine were prepared by diluting 2 μL of the individual CSS to 100 μL with methanol. The QC standards (N = 6, each) were also prepared in replicates containing 1 ng/mL (lower limit of quantification [LLOQ]), 2 ng/mL (low quality control [LQC]), 100 ng/mL (medium quality control [MQC]), and 180 ng/mL (high quality control [HQC]) by the same dilution scheme. WS containing 100 μg/mL of the IS was prepared by appropriate dilution in acetonitrile. All the stocks (MS, WS, and CSS solutions) were stored at 4 °C and vortex mixed. The stability of stock solutions of mitragynine and 7-hydroxymitragynine was assessed by comparing analytical standards prepared in acetonitrile form stored stock solutions (6 months at 4 °C) with freshly prepared stock solutions. Both mitragynine and 7-hydroxymitragynine were found chemically stable in stock solutions and the variance between the two less than 10 %.

Equipment and Conditions

The analysis of mitragynine and 7-hydroxymitragynine was performed using acquity ultra-performance liquid chromatography (Waters, Milford, MA, USA) equipped with triple quadrupole Micromass Quattro Micro™ (Waters, Milford, MA, USA) detector. The standard electrospray ionization (ESI) in the positive mode using multiple reaction monitoring (MRM) using parent ion to daughter ion transitions from m/z 399.39 → 174.37, 415.34 → 190.4, and 294.37 → 170.40 with collision energy 32, 30, and 28 eV was employed for the analysis of mitragynine, 7-hydroxymitragynine, and IS, respectively. The cone voltage was set to 34, 30, and 28 V for mitragynine, 7-hydroxymitragynine, and IS, respectively. The source parameters, viz. capillary voltage, extractor voltage, RF lens, source temperature, desolvation temperature, cone gas flow, and desolvation gas, were 4 kV, 4 V, 0.2 V, 120 °C, 400 °C, 20 psi, and 800 psi, respectively. Dwell time for both analytes was set for 200 ms. Nitrogen was used as the nebulizer and cone gas, while argon was employed as collision gas.

A BEH C18 column (1.7 μm, 2.1 mm × 50 mm; Milford, MA, USA) was used. Optimum separation of the analytes was achieved by gradient elution started with pump A (0.1 % acetic acid in water) and pump B (0.1 % acetic acid in acetonitrile) supplying 90 and 10 % of the mobile phase components, respectively. The concentration of the mobile phase component in pump A decreased linearly to 60 % up to 5 min and kept constant up to 6 min followed by a linear increase to 90 % by 6.5 min and maintained up to 8.5 min. The flow rate of mobile phase was 0.2 mL/min. The temperature of peltier-tray was 10 °C, while column oven temperature was set to 30 °C. MassLynx software version 4.1 (Waters, Milford, MA, USA) was used for control of the equipment, data acquisition, and analysis.

Sample Preparation

The requisite amount of capsule formulations (Phoria[tm] red, Phoria[tm] green, Phoria[tm] regular, Phoria[tm] Borneo white vein, Phoria[tm] Borneo red vein, Phoria[tm] Borneo green vein, Phoria[tm] maeng da blue lotus, and Phoria[tm] maeng da kava) was weighted and soaked in methanol (IS; 10 ng/mL) for a concentration of 1 mg/mL. For liquid drinks, a 100 μL of formulation (Green vein extra strength Kratom shot and Viva Zen) was mixed up to 1 mL of methanol (IS; 10 ng/mL). The methanol extract was vortexed for exactly 10 min on BenchMixer (Benchmark, USA) at 2500 rpm. After vortex mixing, methanol extract was centrifuged at 13,000 rpm for 10 min. The supernatants (10 μL) were further diluted up to 1 mL with methanol (IS; 10 ng/mL). The methanol extracts were vortex mixed followed by centrifugation. The supernatants were injected onto the column for analysis.

Results

Product ion spectra and MRM chromatograms of mitragynine, 7-hydroxymitragynine, and IS are depicted in Figs. 2 and 3, respectively. As shown in Table 2, mitragynine was quantified as 9.7–19.0 μg/mg in capsule formulations; however, it was 190.7–396.4 ng/μL in liquid drinks. The 7-hydroxymitragynine was also present in all the methanol extracts of tested formulations for the concentration range of 93.0–593.2 ng/mg and 1.96–2.51 ng/μL in capsule and drink, respectively.

Discussion

We found multiple commercial Kratom products to have concentrations of 7-hydroxymitragynine that are substantially higher than those found in raw *M. speciosa* leaves [16, 27, 30]. Our findings strongly suggest adulteration of commercial Kratom products with 7-hydroxymitragynine, a plant alkaloid with potent mu-opioid receptor activity, a practice that increases the abuse liability and addictive potential of Kratom products. For example, concentrations of mitragynine and 7-hydroxymitragynine in raw *M. speciosa* leaves range between 23.6–24.0 μg/mg and 114–134 ng/mg, respectively [30]. Comparison of naturally occurring 7-hydroxymitragynine concentrations with the Phoria products listed in Table 2 demonstrates striking deviation from what would be expected. The

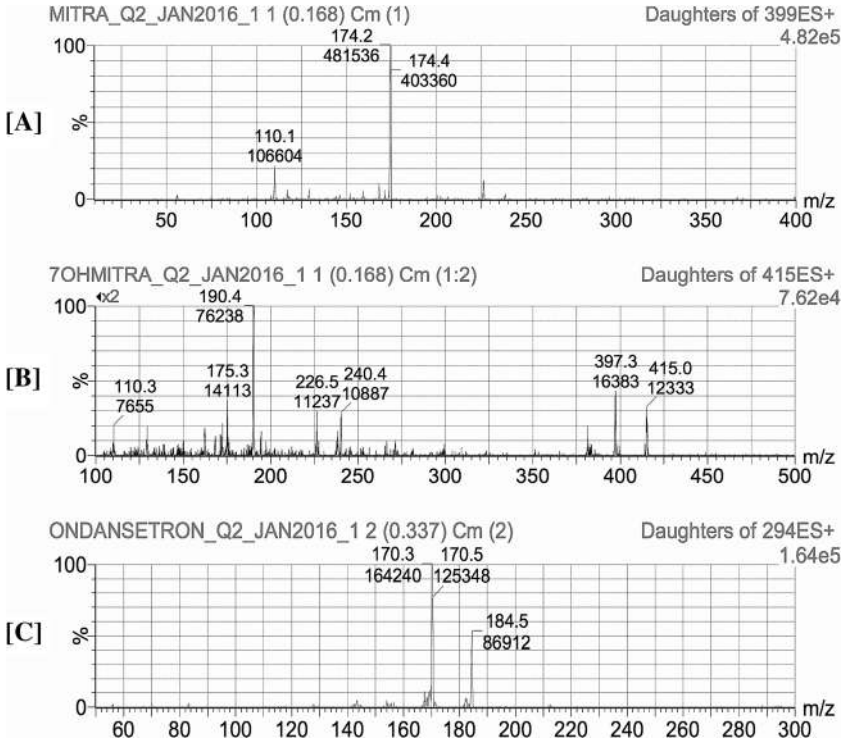


Fig. 2 Product ion spectra of **a** mitragynine (*m/z* 399.39), **b** 7-hydroxymitragynine (415.34), and **c** internal standard (*m/z* 294.37)

historically "minor" alkaloid constituent 7-hydroxymitragynine was found to be 109–520 % more concentrated in 7/8 capsule formulations studied. This research also highlights the need for regulations to prevent boosting the concentration of addictive

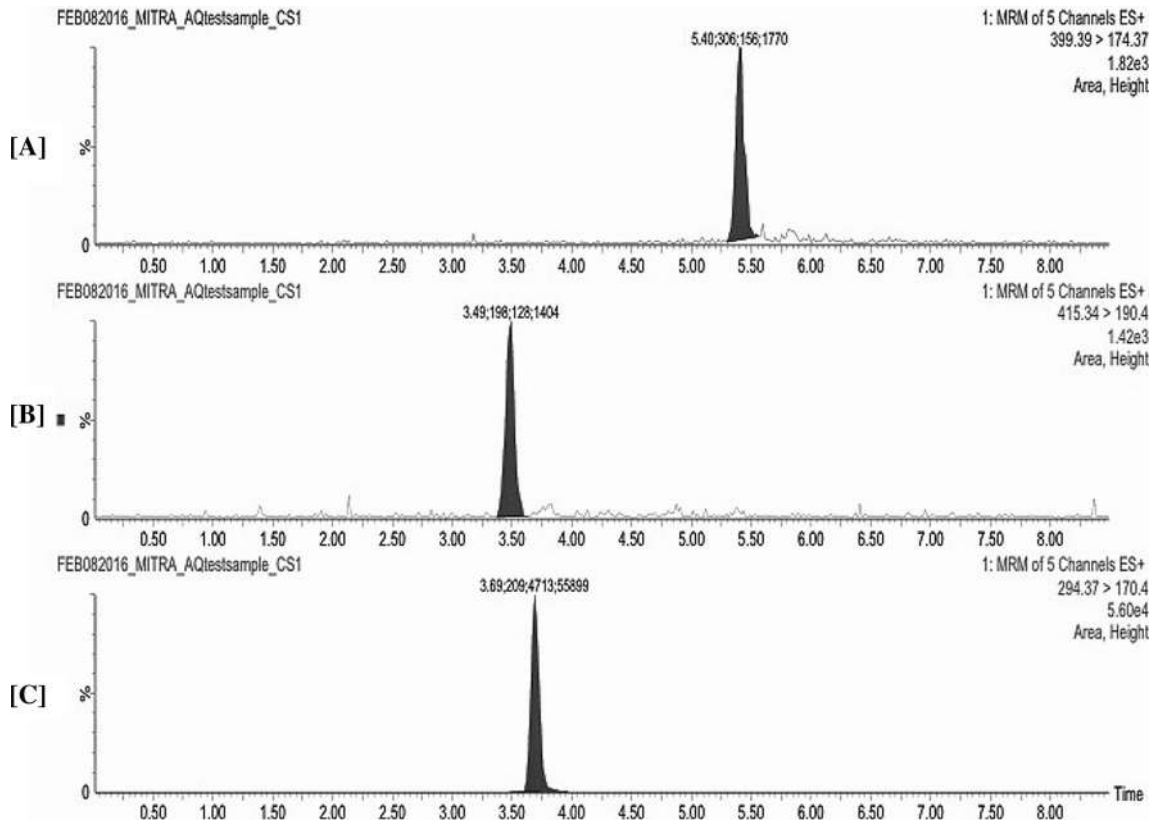


Fig. 3 Representative MRM chromatogram of **a** mitragynine (1.0 ng/mL; LLOQ), **b** 7-hydroxymitragynine (1.0 ng/mL; LLOQ), and **c** IS (10 ng/mL)

Table 2 Concentration of mitragynine and 7-hydroxymitragynine in naturally occurring Kratom leaf and marketed Kratom supplements

Brand name	Concentration	
	Mitragynine	7-Hydroxymitragynine
Natural Kratom leaf [30]	23.8 μg/mg (range 23.6–24.0)	124.0 ng/mg (114.0–134.0)
Phoria^{tm} Borneo white vein	18.3 μg/mg	593.2 ng/mg
Phoria^{tm} red	18.5 μg/mg	410.8 ng/mg
Phoria^{tm} green	11.7 μg/mg	378 ng/mg
Phoria^{tm} Borneo red vein	14.9 μg/mg	346.2 ng/mg
Phoria^{tm} maeng da kava	10.9 μg/mg	300.8 ng/mg
Phoria^{tm} maeng da blue lotus	9.7 μg/mg	146.7 ng/mg
Phoria^{tm} Borneo green vein	17.5 μg/mg	146.7 ng/mg
Phoria^{tm} regular	19.0 μg/mg	93.0 ng/mg
Kratom shot (liquid formulation)	190.7 ng/μL	2.51 ng/μL
Green vein extra strength (liquid formulation)	396.4 ng/μL	1.96 ng/μL

chemicals naturally occurring in Kratom, a unique form of supplement adulteration.

Kratom is sold on the internet and in specialty stores (i.e., head shops) as capsules, tablets, powder, concentrated extract, gum, or raw leaves for chewing or brewing tea [6, 13, 31]. Reported benefits include anticancer and antioxidant effects [32, 33], antiinflammatory properties [34], appetite reduction, glycemic control, treatment of chronic pain [35, 36], antidepressant actions [37], antidiarrheal and antimalarial properties [1, 38], treatment of ethanol [36, 39] or opioid withdrawal [1, 40, 41], and antibacterial activity [33]. Due to its stimulant and opioid-like properties, Kratom is also consumed for analgesic and recreational purposes [26].

Kratom produces dose-dependent clinical effects [3, 26, 42]. Historically, low doses of 1–5 g dried leaves produced stimulant effects which have been likened to those of cocaine [31]. At higher doses, opioid-like effects predominate [26]. In the past, low dose has been generally defined by users as 1–5 g of raw dried leaves and high dose as 5–15 g [11, 43]. As of 2014, however, the appearance of "super grade" strains of Kratom has redefined the amount of product needed to produce clinical effects. Low and high dosing are now considered by users to be 1–4 g and 4–8 g of dried leaf, respectively [43].

Adverse effects experienced by Kratom users are associated with chronicity of use. Short-term effects mimic those of standard opioid medications and include nausea, dizziness, constipation, itching, or sexual dysfunction [14, 44]. Chronic use results in insomnia, anorexia, weight loss, facial hyperpigmentation, dry mouth, polyuria, psychosis, or addiction [1, 45, 46]. Infrequent findings associated with chronic Kratom use include seizure or coma [10, 24, 47, 48], acute respiratory distress syndrome [49], hypothyroidism [50], intrahepatic cholestasis or toxic hepatitis [38, 48, 51], cardiotoxicity or dysrhythmia [52], and hypertension or nephrotoxicity [38]. Fatalities have been reported in the setting of therapeutic or supratherapeutic levels of coingestants such as sympathomimetics [53, 54], benzodiazepines and over the counter cold medications [55], antidepressants [56], muscle relaxers, and opioids [57, 58].

Supplement Regulation

Kratom is classified as a dietary supplement by the Federal Dietary Supplement Health and Education Act of 1994 and is defined as "a product intended for ingestion that contains a 'dietary ingredient' intended to add further nutritional value to (supplement) the diet" [59]. These dietary ingredients may include vitamins, minerals, amino acids, herbs or other botanicals or their concentrates, metabolites, constituents, or extracts [59]. Unlike FDA-approved drugs, laws surrounding dietary supplement regulation are lax. Adulteration is an exemplary repercussion of such laxity. Well documented are instances of diuretics, stimulants, anorectics, or oxidative uncouplers contained in weight loss supplements [60–74], steroids in performance enhancers [75–80], and phosphodiesterase 5 inhibitors contained within male enhancement products [72, 74, 81–93]. At least nine people died in Sweden following exposure to Kratom adulterated with *O*-desmethyltramadol, a potent opioid analgesic. All nine patients had pulmonary congestion and/or edema on autopsy, suggestive of respiratory depression [57]. Because pure Kratom does not appear to produce respiratory failure, these fatalities were likely due to *O*-desmethyltramadol alone and/or synergism with mitragynine or 7-hydroxymitragynine.

Legal Status

Thailand was the first country to criminalize Kratom under the Kratom Act of 1943 [6]. Today, Kratom is listed as a category V substance under the Thai Narcotics Act, similar

to cannabis and psychotropic mushrooms, with penalties of up to 1- or 2-year imprisonment for possession and production or disbursement of the substance, respectively [94, 95]. Since the Kratom Act, *M. speciosa* has been federally regulated in countries such as Denmark, Latvia, Lithuania, Poland, Romania, Sweden, Myanmar, Malaysia, Australia, and New Zealand [6]. In the USA, some cities and states have banned or are in the process of banning Kratom alkaloids due to health risks [96–102].

While Kratom is not yet regulated under the Controlled Substances Act in the USA, the federal government has steps to limit its use. The FDA, while classifying Kratom as a dietary supplement, also recognizes it as a "new dietary ingredient" as there is no evidence that it was sold as a supplement in the USA prior to October 15, 1994 [103]. A new dietary ingredient is only approved for marketing after a history of safe use or other evidence suggesting reasonable safety of the ingredient is documented [104]. Because Kratom has no evidence of reasonable safety, the FDA considers it adulterated and as a result has placed import bans on Kratom-containing supplements as of February 2014 [103, 105]. The FDA, with the help of US Marshals, has continued to seize Kratom-containing supplements on the grounds of product adulteration with an unapproved new dietary ingredient [106]. Additionally, the DEA has classified Kratom as a "drug and chemical of concern," given the lack of identified legitimate medical use and the potential risk to those who abuse it [107].

Limitations

Because 7-hydroxymitragynine is a product of mitragynine auto-oxidation, we considered whether the alkaloid is simply an artifact of testing procedures or of plant metabolism. As noted in our methods, stock solutions of mitragynine and 7-hydroxymitragynine are chemically stable over a period of 6 months. Likewise, no conversion of mitragynine into 7-hydroxymitragynine occurs in chloroform or passing through silica gel column chromatography; these findings suggest that 7-hydroxymitragynine is not an artifact of isolation [16]. The presence of normal mitragynine concentrations argues strongly that the observed supranormal 7-hydroxymitragynine concentrations arise from adulteration of the commercial products we analyzed, not variation in plant metabolism. We also recognize that it is possible, though unlikely, that the analyzed commercial Kratom strains are derived from *M. speciosa* plants with above average 7-hydroxymitragynine levels. While the alkaloid content of *M. speciosa* has been shown to vary based on geographic location and month of the year, the main indole alkaloid contents (i.e., mitragynine and 7-hydroxymitragynine) remain relatively stable compared to its oxindole counterparts, which show much greater variability [15]. To our knowledge, 7-hydroxymitragynine content has never been reported to exceed 2 %.

Conclusion

We have found multiple packaged commercial Kratom products to contain artificially elevated concentrations of 7-hydroxymitragynine, the alkaloid responsible for *M. speciosa*'s concerning mechanistic and side effect profile [20–22]. The amount of 7-hydroxymitragynine exceeded that found in naturally occurring material by up to 500 %. The recognition that 7-hydroxymitragynine is itself a metabolite further supports the notion of excessive concentrations being due to artificial addition of this psychoactive substance. Although the FDA already considers Kratom-containing supplements to be adulterated [103], federal regulations surrounding possession and use of Kratom are lacking. This study describes a unique form of product adulteration, which stresses the importance of increased dietary supplement oversight of Kratom-containing supplements.

Compliance with Ethical Standards

Conflict of Interest Authors AL, AS, CM, and BA declare that they have no conflict of interest. Authors KB and EB provide medico-legal consultation and receive royalties from UpToDate. Author EB also participates in an NIH-funded research on drugs of abuse.

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Quantitative analysis of 7-hydroxymitragynine in commercial kratom products and its stability under chemical and physiological conditions

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ABSTRACT

7-Hydroxymitragynine (7-OH), a minor alkaloid in kratom (*Mitragyna speciosa* (Korth.) Havil.) (Rubiaceae) and an active metabolite of mitragynine, is a critical transient precursor to mitragynine pseudoindoxyl, which exhibits >2-fold μ-opioid receptor–biased agonism and improved isoform selectivity relative to its precursors. Although present only in trace amounts in kratom leaves, a recent survey identified hundreds of consumer products with semisynthetic 7-OH and mitragynine pseudoindoxyl. Their high potency, ease of access, and risk of addiction have prompted the US FDA to recommend scheduling 7-OH as a Schedule I controlled substance. To address this public health crisis and the associated analytical challenges, a comprehensive analysis of commercially available kratom products was performed. The results revealed substantial inconsistencies between the quantities claimed on the products' labels and the actual 7-OH amount detected in several of these products. Along with 7-OH and mitragynine pseudoindoxyl, several (over)oxidized byproducts were detected, providing crucial insights into their synthetic origins. Analysis of >98% of 7-OH-labeled products indicated a semisynthetic origin, with labeled doses ranging from 0.001 to 33.6 mg per serving. Furthermore, consistent with the prior report, 7-OH degrades into the potentially toxic byproduct 3-dehydromitragynine under simulated gastric conditions. These findings underscore significant safety concerns over oral 7-OH products and necessitate validated analytical monitoring to guide regulatory oversight and consumer protection.

1. Introduction

Mitragynine (**1**) is the principal monoterpene indole alkaloid of *Mitragyna speciosa* (Korth.) Havil. (Rubiaceae), commonly known as kratom. 7-Hydroxymitragynine (7-OH, **2**) is an oxygenated analogue of mitragynine and is often reported as a minor constituent in post-harvest leaf material. Although kratom has been used for centuries, 7-OH was first identified in 1994 as a minor constituent of the plant (Ponglux et al., 1994). Subsequent research has established that 7-OH occurs at trace levels in kratom leaves and is also formed as an active metabolite of **1**, kratom's primary alkaloid. In biological systems, 7-OH (**2**) is produced by hepatic oxidation and exhibits a significantly higher binding affinity for μ-opioid receptors (MOR) than mitragynine itself (Kruegel et al., 2019). The resulting metabolic relationship is particularly relevant in postmortem biological fluids, where the ratio of mitragynine to 7-OH is significantly lower (32.4 to 14.8) (Osawa and Johnson, 2025) compared to ratio reported for original kratom powdered samples (53.84 to 49.32) (Horniakova et al., 2025). While confirming substantial *in vivo* conversion of mitragynine to 7-OH, direct comparison of metabolite ratios across biological and product-based studies should be interpreted cautiously due to differences in sample matrices and analytical context.

Furthermore, 7-OH serves as a key intermediate in the acid-mediated rearrangement of mitragynine, which leads to the formation of mitragynine pseudoindoxyl (**3**) and related spiro compounds. Recent studies

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indicate that **3** exhibits higher affinity for MOR than its precursors, 7-OH and mitragynine (Chakraborty et al., 2021; Kamble et al., 2020; Takayama et al., 2002). Hence, biotic oxidative metabolism of **1** into **3** via 7-OH is likely to influence kratom's complex pharmacological effects and potential abuse liabilities (Kamble et al., 2020).

Although fresh kratom leaves contain negligible amounts of 7-OH and **3**, a recent survey identified over 300 consumer products containing these compounds, suggesting that many are likely semisynthetic rather than enriched from minor kratom alkaloids (Hill et al., 2025). Most of these products (82.2%) contained only 7-OH and were sold as chewable or sublingual tablets, shots, and gummies. An additional 14.5% contained both 7-OH and **3**, while only 3.3% contained **3** only. Almost all products containing **3** or a combination of **3** and 7-OH were marketed as "kratom." In contrast, 92% of products containing 7-OH were sold as a standalone ingredient, a significant discrepancy given the low natural abundance of 7-OH in kratom leaf material.

In July 2025, the U.S. Food and Drug Administration formally recommended that 7-OH be classified as a Schedule I controlled substance, a category shared with heroin. In addition to the recommendation, the FDA issued warning letters (U.S. Food and Drug Administration, 2025a, b) to several companies, directing them to stop selling products containing 7-OH. These actions reflect growing public health concerns surrounding the proliferation of highly concentrated, often synthetic, 7-OH products that have received heightened media and regulatory attention. The concentration of 7-OH in "enhanced" products sold online and in convenience stores (Francis, 2024) is significantly higher than levels observed in kratom leaves, and such differences may pose serious inadvertent health risks, including addiction and overdose.

Recent literature has also reported the emergence of a subset of products formulated to contain elevated levels of 7-OH in whole extracts or to enrich specific oxygenated mitragynine-related compounds (Kampmeyer, 2025; Samee et al., 2024). Accurate quantification of 7-OH is therefore essential not only for consumer safety but also crucial for regulators to enforce the law and for scientists to assess the chemical and biological effects of 7-OH. To make the regulatory process more efficient, future efforts should include the quantification of 7-OH congeners, such as **3** and other isobaric compounds (Fig. 1), which are known to interact with opioid receptors with superior potency. Fig. 1

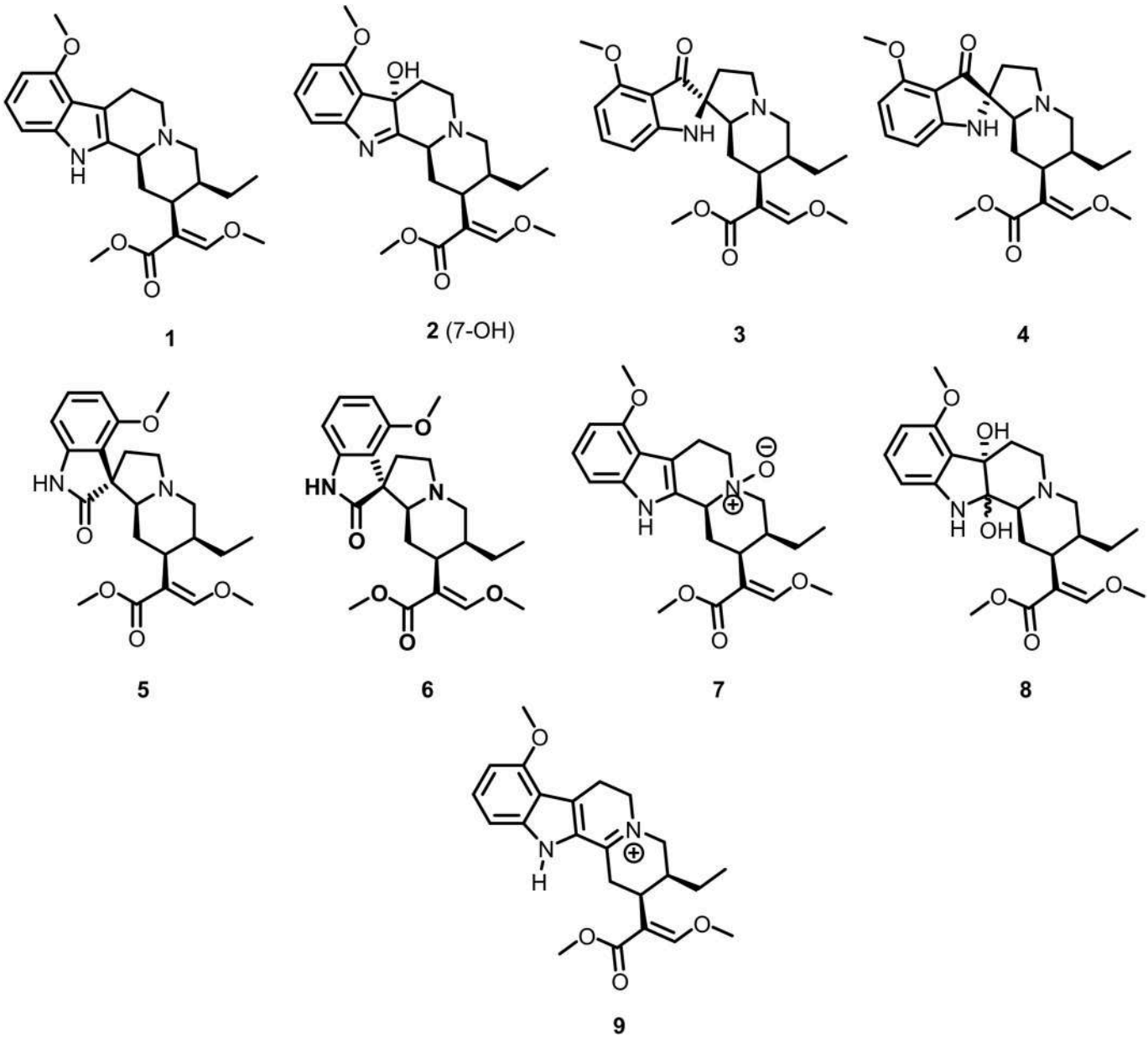


Fig. 1. Structures of mitragynine (**1**), related oxygenated alkaloids (**2**–**8**), and dehydromitragynine (**9**) studied in this work.

shows the structures of mitragynine, related oxygenated mitragynine alkaloids (**2**–**8**), and dehydromitragynine (**9**) investigated in this study. A comprehensive analytical approach would save time and resources by proactively identifying and characterizing semisynthetic substances in various kratom-based consumer products.

In addition to deliberate formulation or enrichment practices, the inherent chemical stability of 7-OH may also influence the composition and chemical profiles of commercial products. Oxidative, acidic, or basic environments can promote degradation or structural rearrangement of mitragynine derivatives, potentially complicating analytical interpretation if not properly considered (Angyal et al., 2023; Chear et al., 2021; Ganasan et al., 2025; Grundmann et al., 2024; Mudge and Brown, 2017; Takayama et al., 2002). Consequently, evaluation of the stability of 7-OH under selected chemical and physiologically relevant conditions is important for distinguishing pre-existing constituents in commercial products from compounds formed during storage, handling, or analysis.

In this study, several commercially available kratom products and specific "7-hydroxy" products, which claim a high percentage of 7-OH , were purchased and analyzed. The results provide evidence of adulteration with structurally related alkaloids as impurities. Furthermore, the stability of 7-OH under chemical and physiological conditions was also investigated to support the interpretation of commercial product profiles.

2. Results

The widespread proliferation of concentrated 7-OH products, such as tablets, gummies, and drink mixes, indicates an elevated concentration of the alkaloid that is not found in naturally occurring kratom. Such an elevated concentration indicates a semisynthetic origin, likely through the oxidative chemical conversion of mitragynine isolates or whole kratom extracts. As only trace amounts of 7-OH are naturally present in kratom leaves, direct extraction from chemically unaltered, full-spectrum extracts would not be economically feasible.

To assess the overall quality of commercial kratom products, especially those marketed with elevated levels of 7-OH , 28 out of 35 products were sourced via online vendors, and seven products (4656-4662) were purchased from a gas station in Oxford, MS, and analyzed using UHPLC-PDA-MS (Table 1). The sample set consisted of seven powders, six capsules, eight liquids, one soft gel, and 13 tablets. To facilitate interpretation of the analytical data, products were grouped by formulation and labeling claims into four functional categories: (i) conventional kratom powders and capsules, (ii) liquid or tincture formulations marketed as kratom extracts, (iii) highly concentrated solid products labeled as "7-OH" or similar terminology, and (iv) other miscellaneous products (Table 2). This classification allowed direct comparison of alkaloid content and artifact profiles across product classes rather than individual products alone. With reference standards and calibration curves, detection limits as low as 50 ppb were achieved using the current analytical methodology, which is expected to enable sensitive and accurate quantification of three specific analytes of interest, mitragynine (**1**), 7-OH (**2**), and mitragynine pseudoindoxyl (**3**). Among the products, 22 claimed the presence of 7-OH, two claimed **3**, and 14 claimed **1**. Additional details pertinent to analytical procedures were outlined under the materials and methods section.

2.1. Quantitative measurement of key kratom alkaloids

2.1.1. *Mitragynine (1) content*

Mitragynine , the principal alkaloid in kratom, was listed on the labels of 14 out of 35 kratom products analyzed. The content of **1**, as claimed on the manufacturers' label, ranged from ≤7.8 mg per serving in capsule formulations to 40.8 mg per serving in powder-based products. However, analytical testing detected **1** in all 35 products (Table 1). Mitragynine concentrations varied widely by formulation type: solid formulations ranged from 0.02 to 46.3 mg per serving, while liquid formulations ranged from 3.0 to 116 mg per serving.

Discrepancies between labeled claims and analytical concentrations of mitragynine were significant across several evaluated products, with multiple instances exceeding 150% of the stated value. For example, product #4656PR, which specified a range of 20.8–22.5 mg of mitragynine per serving, was determined to contain 40.3 mg upon analysis, which is 186% of the mean labeled claim. Furthermore, dosage forms significantly influenced the concentrations of **1**, with calculated daily intakes ranging from <10 mg to >198 mg based on recommended labeling instructions. Conversely, powder-based kratom products contained quantities closer to their labeled claims (e.g., product #4806 PR claimed 40.8 mg of mitragynine and was determined to contain 40.7 mg). Nevertheless, anomalies in mitragynine's content variability across different formulations highlight the crucial need for rigorous analytical monitoring to ensure label accuracy and safeguard consumer health.

2.1.2. *Mitragynine pseudoindoxyl (3) content*

The spiro indole alkaloid, **3**, a metabolite with substantially higher MOR agonist activity than either mitragynine or 7-OH, was detected in 33 of the 35 kratom products analyzed. Quantifiable levels were found in most products, with concentrations ranging from 0.01 mg (#4803 PR) to 0.5 mg (#4831PR, 4838 PR) per serving in solid formulations and 0.01 mg (#4659PR, 4660PR) to 8.4 mg (#4814PR) per serving in liquids. Three products (#4656PR, 4661PR, and 4662PR) contained **3** at concentrations below the limit of quantification (LOQ), whereas **3** was not detected in products #4846PR and 4812PR.

Although **3** was frequently detected during analysis, it was specified on the labels of only two products, which claimed 9.0 mg and 10.0 mg per serving, respectively, highlighting significant anomalies in ingredient transparency and quality control within the commercial kratom market.

2.1.3. *7-OH (2) content*

7-OH , an oxidative metabolite of mitragynine with superior MOR agonist activity, was declared on the labels of 22 out of 35 evaluated kratom products (most of these products were selected based on 7-OH content). Labeled claims for 7-OH exhibited substantial variance, ranging from ≤0.045% (~0.45 mg/g) in powder formulations to 12.5 mg per serving in tablets. However, analytical quantification detected 7-OH in 34 of 35 products, with measured concentrations ranging from 0.01 to 8.4 mg per serving in solid dosage forms and 0.001 to 0.2 mg/mL in liquid formulations; one product (4842PR) contained no detectable **2**. Calculated maximum daily intakes based on label instructions varied widely, from 0.001 to 33.6 mg. Furthermore, 7-OH concentrations in these formulations significantly exceeded levels typically found in natural dried leaf material (~0.01–0.04% w/w, corresponding to ~0.1–0.4 mg/g) (U.S. Food and Drug Administration, 2025a,b), suggesting that enrichment or targeted alkaloid preparation occurred during manufacturing.

Analysis of 7-OH content revealed significant discrepancies between measured quantities and manufacturer label claims. In products specifying 7-OH content, measured concentrations were consistently lower than advertised, typically ranging from 32 to 81% of the stated amount. Product #4839PR exhibited the lowest levels (32%), while #4832PR showed the highest (81%). Notably, detectable levels of **2** were identified in several products lacking explicit label claims, such as #4814PR (0.1 mg per 10 mL serving).

Alkaloid profiling of 35 commercial kratom products revealed substantial chemical heterogeneity. Of these, 12 products exhibited alkaloid profiles dominated by 7-OH and various structurally related artifacts that are not characteristic of authentic *M. speciosa* leaf extracts. These products consistently displayed chemical signatures indicative of semisynthetic modification rather than simple solvent extraction and enrichment.

The identity of these artifacts was confirmed using reference standards, which identified a consistent suite of oxidation products across

Table 1
Comparison of the label information with the detected quantities of 7-OH (**2**), mitragynine pseudoindoxyl (**3**), and mitragynine (**1**) in commercial kratom samples.

Product code #	Labeled ingredient and amount (mg) per serving size			Label recommended serving size	Label maximum daily servings	Measured ingredients per serving size (mg) (SD[a])			Measured quantity of ingredient in maximum recommended daily servings determined (mg)			Measured *vs* labeled ingredients per serving size (%)		
	7-OH	**3**	Mitragynine			7-OH	**3**	Mitragynine	7-OH	**3**	Mitragynine	7-OH	**3**	Mitragynine
4839PR	10.0	NA[b]	NA	1/2 tablet	1 tablet	3.2 (0.2)	0.3 (0.1)	1.7 (0.04)	6.3	0.5	3.4	31.5	NA	NA
4833PR	7.5	NA	NA	1/2 tablet	1 tablet	4.9 (0.1)	0.2 (0.1)	0.1 (0.001)	9.7	0.3	0.1	64.6	NA	NA
4838PR	7.5	NA	NA	1/2 tablet	1 tablet	5.3 (0.1)	0.5 (0.03)	0.4 (0.03)	10.6	1.0	0.8	70.6	NA	NA
4837PR	12.5	NA	NA	1/4 tablet	1 tablet	8.4 (0.9)	0.2 (0.01)	0.3 (0.02)	33.6	0.7	1.2	67.2	NA	NA
4843PR	7.5	NA	NA	1/2 tablet	NA	6.0 (0.3)	0.3 (0.1)	0.1 (0.001)	NA	NA	NA	79.5	NA	NA
4844PR	10.0	NA	NA	1/2 tablet	NA	6.6 (0.4)	0.1 (0.03)	0.02 (0.001)	NA	NA	NA	66.3	NA	NA
4835PR	10.5	NA	NA	1/2 tablet	2 tablets	7.6 (0.6)	0.2 (0.2)	0.2 (0.1)	30.5	1.0	0.6	72.6	NA	NA
4831PR	7.5	NA	NA	1 tablet	2 tablets	6.0 (0.7)	0.5 (0.3)	1.0 (0.04)	12.1	1.0	2.0	80.1	NA	NA
4834PR	9.0	NA	NA	1/2 tablet	1/2 tablet	5.3 (1.0)	0.1 (0.01)	0.2 (0.03)	5.3	0.1	0.2	59.1	NA	NA
4836PR	9.0	NA	NA	1/2 tablet	1 tablet	5.2 (0.5)	0.2 (0.1)	0.3 (0.1)	10.4	0.3	0.5	57.8	NA	NA
4832PR	7.5	NA	NA	1/2 tablet	1 tablet	6.1 (0.9)	0.2 (0.1)	0.2 (0.003)	12.2	0.3	0.4	81.0	NA	NA
4845PR	7.5	NA	NA	1/4 tablet	NA	5.2 (0.1)	0.1 (0.1)	0.3 (0.4)	NA	NA	NA	69	NA	NA
4842PR	NA	9.0	NA	1/2 tablet	1/2 tablet	ND[c]	3.9 (0.02)	1.1 (0.01)	ND	3.9	1.1	NA	43.2	NA
4814PR	NA	10.0	NA	10 mL	20 mL	0.1 (0.001)	8.4 (0.04)	3.0 (0.1)	0.1	16.8	6.0	NA	84	NA
4846PR	NA	NA	10 mg/mL	NA	NA	0.001[d] mg/mL (0.001)	ND	13.6 [d]mg/mL (0.2)	NA	NA	NA	NA	NA	136
4656PR	<0.04%	NA	20.8-22.5	2.5 mL	2.5 mL	0.001 (0.01)	DUL[e]	40.3 (0.1)	0.001	DUL	40.3	NA	NA	186
4657PR	NA	NA	20.0	28 mL	28 mL	0.03 (0.1)	0.03 (0.002)	30.8 (0.04)	0.03	0.03	30.8	NA	NA	154
4658PR	NA	NA	20.0	14 mL	14 mL	0.2 (0.2)	0.1 (0.001)	45 (0.02)	0.09	0.03	22.4	NA	NA	112
4659PR	NA	NA	NA	1 capsule	2 capsules	0.01 (0.02)	0.01 (0.002)	23.7 (0.2)	0.02	0.01	47.4	NA	NA	NA
4660PR	NA	NA	NA	7.5 mL	7.5 mL	0.07 (0.04)	0.01 (0.01)	116 (0.1)	0.07	0.01	116	NA	NA	NA
4661PR	NA	NA	NA	5 mL	15 mL	0.02 (0.1)	DUL	66 (0.1)	0.1	DUL	198	NA	NA	NA
4662PR	<1.0	NA	<60	5 mL	NA	0.1 (0.01)	DUL	84.4 (0.2)	NA	NA	NA	NA	NA	NA
4808PR	NA	NA	NA	5 capsules	10 capsules	0.08 (0.003)	0.1 (0.002)	31 (0.2)	0.2	0.2	61.9	NA	NA	NA
4810PR	NA	NA	NA	5 capsules	10 capsules	0.1 (0.02)	0.1 (0.002)	33.8 (0.2)	0.2	0.2	67.6	NA	NA	NA
4811PR	NA	NA	NA	5 capsules	10 capsules	0.1 (0.001)	0.1 (0.001)	42.3 (0.2)	0.2	0.2	84.6	NA	NA	NA
4812PR	NA	NA	40.0	1 soft gel	NA	0.1 (0.1)	ND	38.4 (0.4)	NA	NA	NA	NA	NA	96
4809PR	NA	NA	NA	5 capsules	10 capsules	0.1 (0.004)	0.1 (0.002)	46.3 (0.5)	0.2	0.2	92.7	NA	NA	NA
4806PR	0.07	NA	40.8	1 teaspoon = 2.4 g	2 teaspoons	0.1 (0.004)	0.1 (0.001)	40.7 (0.03)	0.2	0.2	81.3	143	NA	99.7
4800PR	<0.045%	NA	1.55%	NA	NA	0.003[d] % (0.003)	0.003[d] % (0.001)	0.9[d] % (0.02)	NA	NA	NA	NA	NA	55.5
4802PR	≤0.35	NA	≤10.5	¼ teaspoon	NA	0.03 (0.01)	0.02 (0.002)	10.3 (0.2)	NA	NA	NA	NA	NA	NA
4805PR	0.17	NA	36.48	1 teaspoon = 2.4 g	2 teaspoons	0.1 (0.01)	0.1 (0.004)	37.3 (0.4)	0.2	0.2	74.6	70.6	NA	102.3
4804PR	0.10	NA	40.56	1 teaspoon = 2.4 g	2 teaspoons	0.1 (0.01)	0.1 (0.001)	39.5 (0.2)	0.2	0.2	79	120	NA	97.3
4807PR	0.36	NA	28.8	1 teaspoon = 2.4 g	2 teaspoons	0.1 (0.001)	0.1 (0.001)	30.7 (0.1)	0.2	0.2	61.5	33.3	NA	106.7
4801PR	≤0.315	NA	≤9.1	¼ teaspoon	NA	0.02 (0.003)	0.02 (0.001)	8.9 (0.1)	NA	NA	NA	NA	NA	NA
4803PR	≤0.27	NA	≤7.8	1 capsule	1 capsule	0.03 (0.01)	0.01 (0.001)	9.4 (0.3)	0.03	0.01	9.4	NA	NA	NA

[a] SD, standard deviation of measurement
[b] NA, not available on product label
[c] ND, not detected
[d] Serving size not specified; data calculated as mg/mL or %
[e] DUL, detected, but under the limit of detection (LOD: 50 ng/mL)

Table 2
Summary of kratom product formulations, labeling claims, and characteristic alkaloid profiles.

Product category	Number of products (n)	Typical label claims	Dominant alkaloid profile	Artifacts detected
Powders	7	Kratom leaf	Dominant: Mitragynine and trace 7-OH	No synthetic artifacts detected
Capsules	6	Kratom leaf	Dominant: Mitragynine and trace 7-OH	No artifacts
Liquids	7	Kratom leaf extract	High mitragynine ± low to moderate 7-OH	No artifacts
	1	Mitragynine pseudoindoxyl shot from *M. speciosa* extract	High mitragynine pseudoindoxyl	No artifacts
Tablets	12	7-OH Mitragynine from *M. speciosa* leaf	High 7-OH ; frequent **3**	Multiple oxidation/ rearranged compounds
	1	Pseudoindoxyl-super alkaloid	High mitragynine pseudoindoxyl	No artifacts
Soft gels	1	Kratom leaf extract	Mitragynine with detectable 7-OH	No synthetic artifacts detected

multiple samples: mitragynine pseudoindoxyl (**3**) and its isomer (**4**), mitragynine oxindoles (**5** and **6**), mitragynine *N* (4) oxide (**7**), 2,7-dihydroxymitragynine (**8**), 2,7-dihydroxypaynantheine, and their corresponding *N*-oxides, and 3-dehydromitragynine (**9**). Compound **9**, a previously reported metabolite of mitragynine (**1**), has been characterized by its distinct pharmacological and toxicological profile (Chakraborty et al., 2021).

Unlike mitragynine or 7-OH , **9** does not display opioid receptor-mediated analgesic activity *in vivo*. However, it demonstrates significant toxicity in C57BL/6J mouse models. Subcutaneous administration of **9** at doses exceeding 50 mg/kg resulted in nearly 100% lethality. Reported LD_{50} values include 48.4 mg/kg (95% CI 71.27–342.3) in 129Sv6 mice and 74 mg/kg (95% CI 48.08–120.5) in CD1 mice (Chakraborty et al., 2021). These data confirm that **9** is toxic across multiple strains through a mechanism independent of opioid receptor activation.

Although specific cytotoxicity assays for **9** in human cell lines are currently limited, broader works on mitragynine metabolites show moderate cytotoxic effects in neuroblastoma (SH-SY5Y), lymphoblastoid (MCL-5), colon adenocarcinoma (Caco-2), and nasopharyngeal carcinoma (NPC/HK-1) lines (Karunakaran et al., 2022; Saidin et al., 2015).

As summarized in Table 1, products in conventional powder and capsule formats exhibited alkaloid profiles dominated by mitragynine, with 7-OH detected only at trace or low levels, consistent with the natural composition of *M. speciosa*. In contrast, products marketed as "7-OH" or "high-potency" formulations displayed markedly elevated concentrations of **2**, frequently accompanied by **3** and additional oxidized or rearranged alkaloid derivatives. These chemical patterns were consistent within their product categories and were notably absent in traditional kratom formulations. A comprehensive summary of formulation types, labeling claims, dominant alkaloids, and artifacts is presented in Table 2, which complements the quantitative findings presented in Table 1.

2.1.4. Chemical profiles across commercial kratom products

Beyond the primary alkaloids quantified in Table 1, several minor alkaloids and oxidative artifacts were identified in a specific subset of products. The distribution of these compounds was formulation dependent. As summarized in Table 3, products marketed as "7-OH" or "high-potency" formulations consistently exhibited a suite of structurally related artifacts. These included mitragynine pseudoindoxyl (**3** and **4**) isomers, oxindoles (**5** and **6**), *N*-oxide (**7**), and advanced oxidation products such as 2,7-dihydroxyanalog (**8**) and 3-dehydromitragynine (**9**). In contrast, these artifacts were notably absent in conventional powders, capsules, and most liquid formulations. The systemic co-occurrence of these artifacts within specific product classes suggests that they arise from shared chemical processing pathways, likely semi-synthetic oxidation rather than natural botanical variability.

2.2. Stability of 7-OH (2)

Given the identification of mitragynine pseudoindoxyl (**3**) and its isomer as artifacts in certain commercial kratom products, we conducted stability studies on 7-OH (**2**). The stability of compound **2** across various conditions was evaluated, including protic and aprotic solvent systems, as well as its pH-dependent stability in simulated gastric fluid and simulated intestinal fluid (Fig. 2). These experiments were designed to model the chemical stability of **2** and the potential formation of additional artifacts following the oral consumption of 7-OH enriched kratom products.

Ultra-high-performance liquid chromatography coupled with photodiode array detection (UHPLC-PDA) was employed to assess the chemical stability of 7-OH (**2**). Dissolution of **2** in methanol resulted in a progressive depletion of the parent compound, concomitant with the emergence of several structurally related alkaloid congeners. These findings indicate nascent structural rearrangements occurring within the protic environment. A comparable degradation pattern was observed in methanol containing 0.1 N ammonia, a solvent system frequently utilized by commercial suppliers of **2** as a reference standard. This observation underscores the inherent instability of **2** in protic environments, which may confound both analytical quantification and interpretation of biological outcomes. Given this instability, we expanded our stability assessment of 7-OH in aqueous buffer systems (pH 4.0, 7.0, and 10.0) and in simulated gastric and intestinal fluids to better mimic oral exposure conditions and compared these results with its stability in an aprotic solvent (acetonitrile). This comprehensive evaluation was designed to elucidate the degradation kinetics and potential structural transformations of 7-OH under physiologically relevant conditions. Samples were collected at 24 h intervals over 8 days. The resulting degradation and formation profiles for each condition are summarized in Fig. 2.

3. Discussion

Expressing 7-OH (**2**) levels as a simple percentage is scientifically debatable, given its status as a minor and highly variable constituent in natural kratom materials (Grundmann et al., 2024; Lydecker et al., 2016; Samee et al., 2024). As an active metabolite of mitragynine (**1**) (Basiliere and Kerrigan, 2020; Kamble et al., 2019; Kruegel et al., 2016), 7-OH is susceptible to abiotic oxidative conversion from its precursor during post-harvesting processes, including drying, storage, and alkaloid enrichment (Karunakaran et al., 2025; Leksungnoen et al., 2022; Sharma et al., 2019). Furthermore, while not yet explicitly reported, specific cytochrome P450 enzymes within the *M. speciosa* could potentially facilitate the biotic oxidation of **1** to produce **2**, mirroring the established human metabolic conversion of mitragynine. However, elevated levels of 7-OH in certain kratom products likely reflect intentional chemical transformation rather than natural occurrence. Although **2** is a transient metabolite of mitragynine, its absolute concentration represents the most direct and toxicologically-relevant metric for assessing consumer exposure and associated safety risks.

Due to the structural relationship between mitragynine and 7-OH ,

Table 3
Artifacts detected across analyzed kratom products claiming to contain 7-OH.

Product code #	3	4	Oxindoles (5)/ (6)	7	Undefined *M*-NOX	Over-oxidation products† (8)	2, 7-Dihydroxypaynantheine	Paynantheine *N*-oxides (tentative)	9
4839PR	✓	✓	✓	✓	✓	✓	✓	✓	✓
4833PR	✓	✓	✓	✓	ND	ND	✓	ND	✓
4838PR	✓	✓	✓	✓	✓	ND	✓	ND	✓
4837PR	✓	✓	✓	✓	ND	ND	✓	ND	✓
4843PR	✓	✓	✓	✓	ND	ND	✓	ND	✓
4844PR	✓	✓	✓	✓	✓	ND	✓	ND	✓
4835PR	✓	✓	✓	✓	✓	✓	✓	ND	✓
4831PR	✓	✓	✓	✓	✓	ND	✓	✓	✓
4834PR	✓	✓	✓	✓	✓	✓	✓	ND	✓
4836PR	✓	✓	✓	✓	ND	✓	✓	✓	✓
4832PR	✓	✓	✓	✓	ND	ND	✓	ND	✓
4845PR	✓	✓	✓	✓	✓	✓	✓	✓	✓

√ = Detected; ND=Not detected.

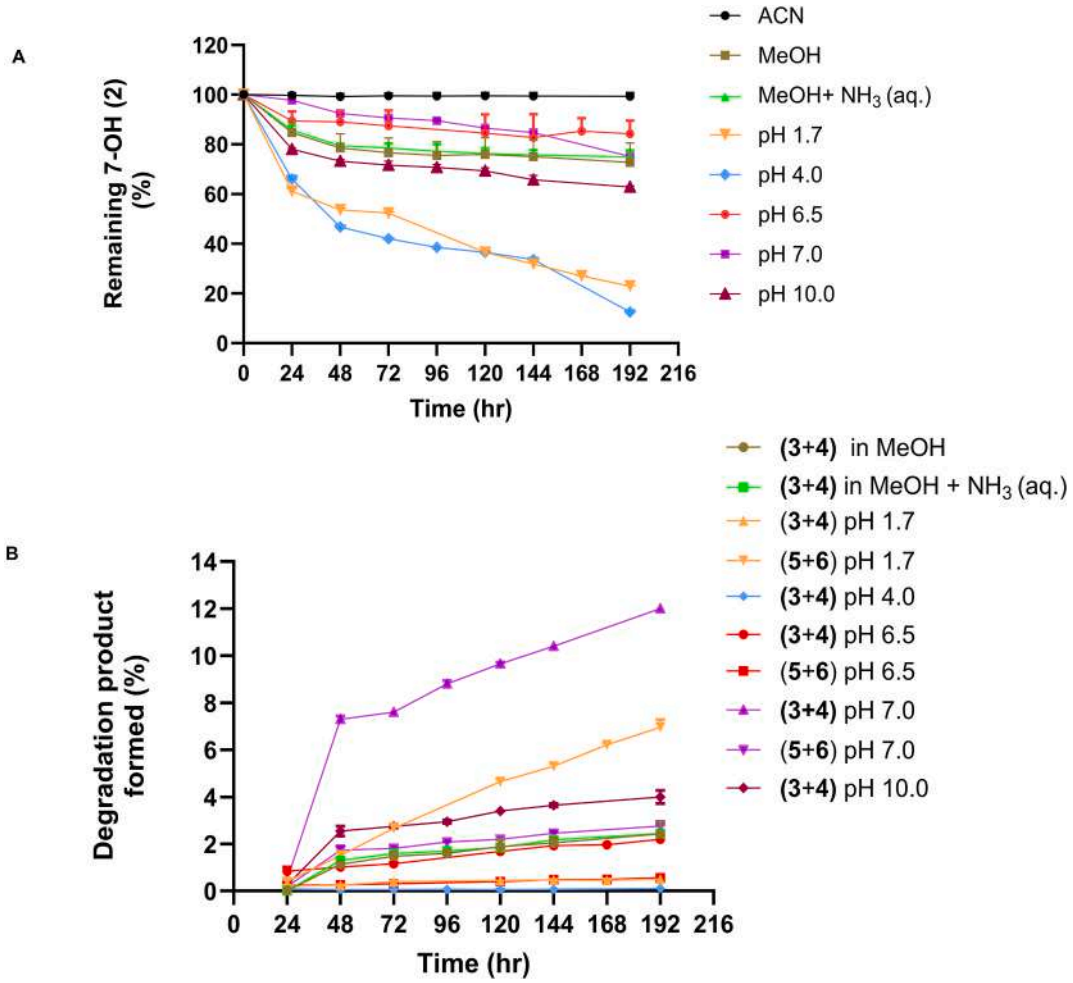


Fig. 2. Time-dependent stability profile of 7-OH under various conditions. (A) Time- and condition-specific degradation of 7-OH under nonprotic and protic solvents along with chemical and physiological conditions; (B) Formation of 7-OH byproducts under different solvents and pH conditions, including physiological conditions.

and oxidative (abiotic or biotic) mechanisms in the conversion of **1** to **2**, the mitragynine :7-OH concentration ratio could serve as a critical diagnostic metric. By normalizing the concentration of minor alkaloid relative to its more abundant precursor, mitragynine, the ratio could provide a standardized framework for distinguishing natural alkaloid profiles from those altered by chemical fortification or semisynthetic modification. Furthermore, the resulting ratio metric enables consistent evaluation of chemical profiles across diverse kratom formulations, regardless of absolute quantities of **1** or **2**. When applied to two previously published data sets on mitragynine and 7-OH (Kikura-Hanajiri et al., 2009; Lydecker et al., 2016), the ratio proves to be a consistent metric across different datasets and timeframes. For instance, a ratio-based analysis of two previous studies (Kikura-Hanajiri et al., 2009; Lydecker et al., 2016) published a few years apart yielded comparable average values of 93 and 85, demonstrating the metric's stability and reliability. In the present study, a notably higher mitragynine :7-OH ratio was observed for the powdered leaf samples and capsules containing powdered leaf. The higher ratio suggests that the

7-OH content in these samples is low, which could be attributed to two main factors: immediate processing and analysis that limited air oxidation of mitragynine, or plant varieties naturally containing a lower 7-OH content (Fig. 3). In contrast, liquid kratom products exhibited a wide range of mitragynine :7-OH ratios, indicating considerable variability and inconsistency in their chemical composition, plausibly due to dissolved oxygen and the presence of antioxidant preservatives.

Tablet products marketed as "7-OH" exhibited a distinctly low mitragynine :7-OH ratio, with values below 0.55. This extremely low ratio indicates a high concentration of 7-OH and a minimal presence of its precursor, mitragynine. Such a composition is not typical of kratom's natural alkaloid profile, indicating that the products were likely produced via economically feasible methods using the more abundant mitragynine as a precursor (Fig. 3).

The chemical profiles observed in these products differ substantially from those expected for traditional kratom leaf material, particularly with respect to elevated levels of oxygenated mitragynine derivatives. Similar findings have been reported in recent studies analyzing the chemical composition of kratom-labeled products, which documented high 7-OH content and additional alkaloids inconsistent with naturally occurring kratom profiles (Alsbrook et al., 2025; Brown et al., 2026; Ganasan et al., 2025; Hill et al., 2025). Analysis of 12 commercial products containing 7-OH revealed consistent artifacts across all samples. The identified artifacts included compounds **3**–**6**, mitragynine *N*(4)-oxide (**7**), **8**, 2,7-dihydroxypaynantheine, their corresponding *N*-oxides, and the potentially toxic (Chakraborty et al., 2021) 3-dehydromitragynine (**9**).

Despite product labeling that claims extraction from *M. speciosa* leaf or aerial parts, the presence of over-oxidation products, such as **8** and its related *N*-oxide, indicates that the oxidation process used to synthesize 7-OH is non-selective. The consistent detection of the potentially toxic **9** is particularly concerning, as it implies that a post-synthesis enrichment step may involve strong acidic conditions.

The pronounced *in vivo* activity of 7-OH observed in these products is particularly concerning in light of recent pharmacokinetic and toxicological studies demonstrating significant biological effects and adverse outcomes associated with 7-OH exposure (Chiang et al., 2025; Pullman et al., 2025; Reif et al., 2025; Zuarth Gonzalez et al., 2025). These reports underscore the importance of accurate quantitative analysis and clear product labeling to support consumer safety and informed regulatory oversight.

The findings from the commercial product analysis prompted a broader investigation into the stability of 7-OH (**2**), which revealed a dependence on both solvent polarity and pH. Compound **2** remained highly stable in acetonitrile, consistent with previous reports (Karunakaran et al., 2025). However, a significant transformation was observed in the methanol or a methanol containing 0.1 N ammonia solvent system, the latter commonly employed by commercial vendors for supplying **2** as an analytical reference standard. These specific conditions were selected to elucidate degradation pathways under laboratory-relevant conditions and are not intended to represent solvents typically used in commercial manufacturing or storage. Under these conditions, a progressive depletion of 7-OH was observed, concomitant with the formation of isobaric artifacts, specifically mitragynine pseudoindoxyls (**3** and **4**). Under alkaline conditions (pH 10.0 buffer), approximately 20% of **2** was depleted within 48 h, with a corresponding formation of pseudoindoxyls (**3** and **4**), approximately 3%; both degradation and isobaric analyte levels remained largely stable thereafter.

Interestingly, even under neutral pH 7.0 buffer conditions, 7-OH exhibited instability and followed first-order degradation kinetics (rate constant (k) = 0.001358 h^{-1}, R^2 = 0.95). Within eight days, >20% of the initial concentration of **2** was consumed, with concurrent formation of ~10% of pseudoindoxyls (**3**/**4**) and ~3% of mitragynine oxindoles (**5**/**6**).

Notably, under acidic conditions, 7-OH was highly unstable. Within eight days, incubation in a pH 4.0 buffer or Fasted-State Simulated Gastric Fluid (FaSSGF, pH 1.7) led to near-complete 7-OH depletion, with **9** identified as the sole byproduct. The rapid loss of stability observed in FaSSGF, mirroring the pH 4.0 results, necessitates further

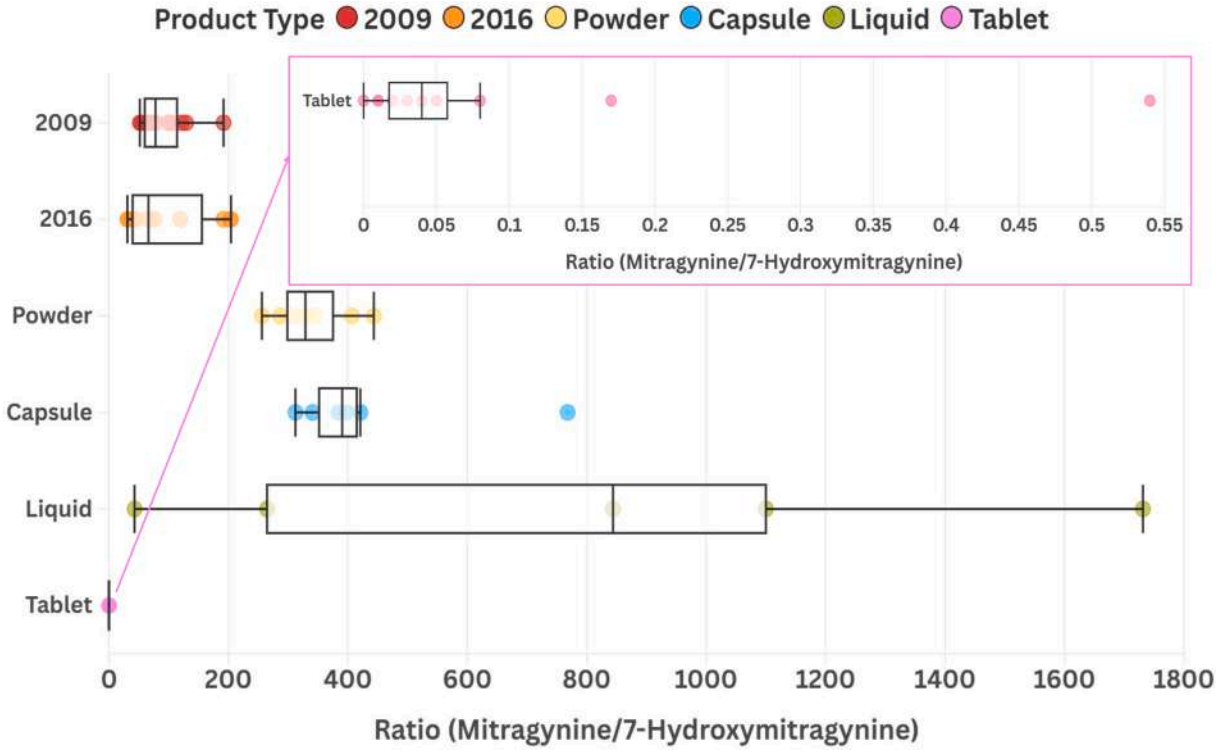


Fig. 3. Comparison of the mitragynine:7-OH ratio in various kratom products with previously reported data (Kikura-Hanajiri et al., 2009; Lydecker et al., 2016). The ratio observed in tablet samples is shown in the inset.

kinetic analysis to characterize the acid-catalyzed degradation mechanism.

The degradation kinetics of 7-OH in FaSSGF (pH 1.7) and at pH 4 were analyzed using zero- and first-order models, with the latter constrained to reflect complete degradation. Although both models yielded comparable R^2 values (~0.978 at pH 1.7 and ~0.899 at pH 4.0), the residual analysis was decisive (Fig. 4). The systematic curvature in the zero-order residuals indicated model misspecification, whereas the first-order residuals were randomly distributed around zero. Therefore, first-order kinetics were selected as the appropriate model, suggesting that the degradation of 7-OH is concentration-dependent under these acidic conditions.

The degradation followed first-order kinetics ($k = 0.005637\ h^{-1}$, $t_{1/2}$ = 123.0 h, 95% CI 111.3 to 136.9 h at FaSSGF (pH 1.7) and $k = 0.006904\ h^{-1}$, $t_{1/2}$ = 100.4 h, 95% CI 79.83 to 131.8 h at pH 4.0) and resulted in the proportional formation of **9** (Fig. 5). The apparent rate constant (k) and half-life ($t_{½}$) indicate faster degradation under stronger acidic conditions (pH 1.7) than at pH 4.0, aligning with acid-catalyzed transformation behavior. The transformation of 7-OH to **3** and **4** is particularly noteworthy, as these compounds are reported to be potent MOR agonists with reduced hyperlocomotion compared to **1** or 7-OH (Chakraborty et al., 2021), highlighting the pharmacological significance of the observed conversion. Following acidic degradation of 7-OH, we observed substantial conversion to **9** (Fig. 5). Given the higher toxicity of **9** in animal studies, its cellular impact is likely to be at least as significant as that of mitragynine, although direct comparative data are needed. The potential for conversion of kratom alkaloids like 7-OH into the toxic **9** under acidic or physiological conditions merits attention for further research. Ultimately, the prevalence of 7-OH in kratom products

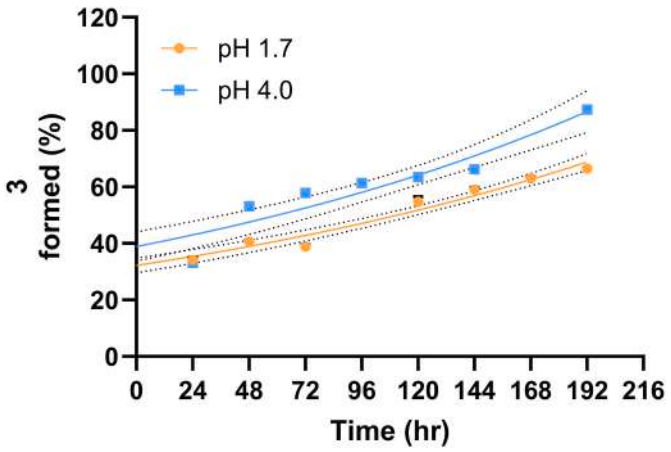


Fig. 5. Formation of 3-dehydromitragynine (**9**) from 7-OH in FaSSGF (pH 1.7) and at pH 4.0 buffer.

will raise substantial health concerns, especially in contexts of high exposure or poor-quality control. Due to its conversion to a toxic byproduct under physiological conditions, regulatory oversight should consider monitoring and limiting 7-OH in commercial kratom preparations.

This study has several limitations. First, the sample size was restricted, with only one product per brand analyzed, and intra-brand batch variability was not assessed. In addition, the analyzed products represent only a subset of the diverse kratom marketplace available via

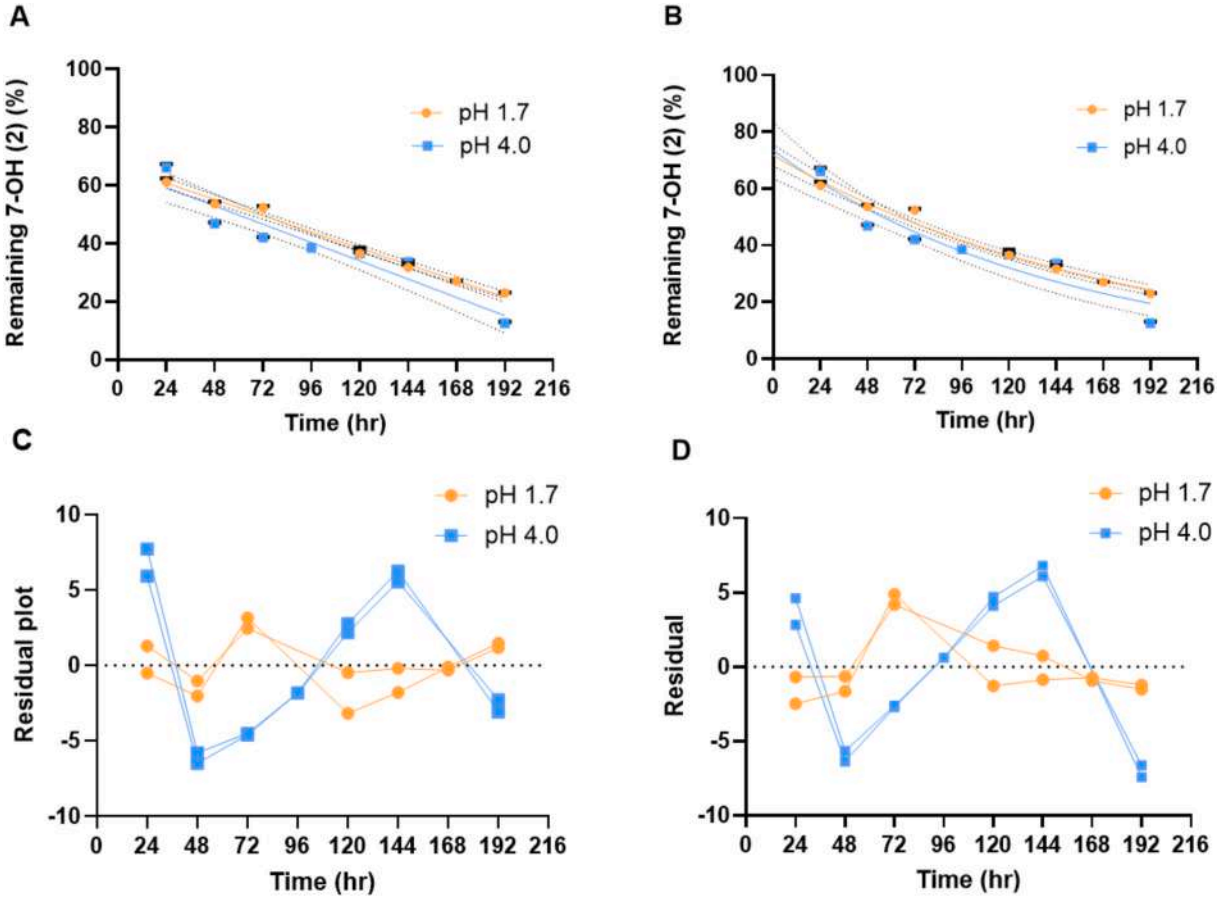


Fig. 4. Degradation kinetics of 7-OH in FaSSGF (pH 1.7) and pH 4.0 buffer with model residuals. (A) Zero-order linear regression of concentration *versus* time; (B) First-order non-linear regression using a one-phase exponential decay model (plateau constrained to zero); (C) Residual plot for the zero-order model (residuals *versus* time); (D) Residual plot for the first-order model (residuals *versus* time).

e-commerce platforms. Importantly, although all products were marketed within the kratom supplement category and frequently labeled as containing *M. speciosa*, our chemical profiling indicates that many products, particularly those labeled as containing 7-OH or **3**, are not representative of chemically unmodified *M. speciosa* leaf material and are best interpreted as semisynthetic formulations derived from mitragynine or enriched extracts. Therefore, these findings primarily apply to commercial products marketed as kratom that may be adulterated or chemically modified, and their broader relevance to other dietary supplement categories remains to be established.

A critical finding of the stability assessment was that 7-OH undergoes approximately 20% degradation within 24 h in methanol, whereas it remains chemically stable in acetonitrile. Although all commercial product analyses in this study were conducted within 12 h of sample preparation, a portion of the minor artifacts detected in methanolic extracts may reflect solvent-induced degradation rather than the original constituent profile of the products. Consequently, acetonitrile is recommended as the extraction and analytical solvent to ensure accurate quantification and minimize artifacts arising from protic degradation.

4. Conclusion

Our comprehensive analysis of commercial Kratom products and subsequent stability studies indicate that formulations marketed as 7-OH (**2**) are unlikely to represent simple botanical extracts. The consistent detection of multiple structurally related compounds, including over-oxidized byproducts and the potentially toxic metabolite **9**, suggests a semisynthetic origin for many of these products. Furthermore, the pronounced instability of 7-OH under selected chemical and physiologically relevant conditions raises significant concerns regarding product shelf life and the potential for consumer exposure to more potent isobaric mitragynine pseudoindoxyls (**3**–**4**) and oxindoles (**5**–**6**). In addition, a substantial conversion of 7-OH to **9** was observed, which raised concerns about the potential for toxic transformation under acidic or physiological conditions. The conversion of 7-OH to the more potent MOR agonist **3**, and the presence of both compounds in commercial kratom products, suggest that their sale in venues like gas stations and e-commerce outlets is inconsistent with the definition of a supplement originating from a botanical source. Collectively, these findings support strengthened product surveillance and validated analytical monitoring of concentrated 7-OH /**3** products and associated byproducts, including **9**, to improve consumer safety and informed regulatory oversight.

5. Experimental section

5.1. Chemicals, alkaloid standards, and supplement products

Mitragynine (**1**) was isolated from botanically verified *M. speciosa* at the National Center for Natural Products Research (NCNPR), University of Mississippi, and used as starting material to produce the following critical chemical standards, 7-OH (**2**), and compounds **6**–**9**. Mitragynine pseudoindoxyl (**3**) (Item No. 20033) was purchased from Cayman Chemical (Ann Arbor, MI, USA). The purity of all reference compounds was greater than 95%. Their identity was confirmed by NMR spectroscopy (Tables S1 and S2), high-resolution mass spectrometry (HRMS), and comparison with prior literature values (Chear et al., 2021; Houghton and Said, 1986; Sharma et al., 2025; Shellard et al., 1978; Smith et al., 2019). HPLC-grade acetonitrile, methanol, and formic acid (Fisher Scientific, Suwanee, GA, USA) were used for extraction and HPLC analysis. Water for the mobile phase was purified using a Milli-Q water purification system (Millipore). Potassium bromide, Oxone®, and *m*-chloroperbenzoic acid were purchased from Sigma-Aldrich (St. Louis, MO, USA). Reagent-grade anhydrous dichloromethane, chloroform, ethyl acetate, and acetic acid were purchased from Fisher Scientific (Suwanee, GA, USA). Twelve products (#4831PR-4839PR and 4843PR-4845PR) were obtained through an online survey of the '7-hydroxy' kratom market for the current study, conducted between July 2025 and August 2025. Selections were based on product availability and quick access, without a focus on any specific vendor or manufacturer.

5.2. Synthesis of key mitragyna alkaloids

5.2.1. 7-Hydroxymitragynine (7-OH, 2)

To a solution of mitragynine (120.1 mg, 0.301 mmol) in acetone (10 mL) in a 50 mL round-bottom flask, saturated aqueous $NaHCO_3$ (8 mL) was added, and the mixture was cooled to 0 °C. A solution of Oxone® (185 mg, 0.602 mmol) in water (5 mL) was added dropwise over 25 min, and stirring was continued. After 20 min, the reaction was diluted with water (50 mL) and extracted with EtOAc (15 mL × 3) in a separatory funnel. The combined organic layers were collected, washed with water (30 mL), saturated brine solution (15 mL), dried over anhydrous Na_2SO_4, and concentrated under vacuum (Kruegel et al., 2019). The resulting residue was subjected to flash chromatography (silica gel) and eluted with hexanes/EtOAc (0 - 70 %), yielding 7-OH product (69.6 mg, 0.168 mmol, 56%) as a pale-yellow solid. $[\alpha]_D^{25} = +57$ (*c* 0.3, $CHCl_3$); ESIMS *m/z* 415.2233 $[M+H]^+$ (calcd 415.2228); ^1H and ^{13}C NMR spectroscopic data: Tables S1 and S2.

5.2.2. 2,7-Dihydroxymitragynine (2,7-DOH (8))

To an ice-cold solution of mitragynine (40 mg, 0.1 mmol) in 5 mL acetonitrile/water (10:1) solvent, 500 µL of 0.01 M KBr (0.6 mg, 0.005 mmol) was added. After 10 min, Oxone® (37 mg, 0.12 mmol) was added in one portion, stirred for 20 min at 0 °C, and gradually warmed to room temperature. After 5 h, the reaction was quenched with saturated $NaHCO_3$ (5 mL), followed by saturated Na_2SO_3 (5 mL). The mixture was transferred to a separatory funnel and extracted with EtOAc (10 mL × 3). The organic layers were combined, washed with saturated brine (5 mL), dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced vacuum (Xu et al., 2019). The resulting crude product was subjected to silica gel chromatography with an isocratic mixture of hexanes/acetone/NH_4OH (7:3:0.05) to yield **8** as a viscous liquid (6.9 mg, 0.016 mmol, 16.8 %). $[\alpha]_D^{25} = -76$ (*c* 0.26, $CHCl_3$); ESIMS *m/z* 433.2333 $[M+H]^+$ (calcd 433.2339); ^1H and ^{13}C NMR spectroscopic data: Tables S1 and S2.

5.2.3. 3-Dehydromitragynine (9)

To a 25 mL scintillation vial containing ice-cold 7-OH (25.2 mg, 0.061 mmol) in anhydrous dichloromethane (500 µL) under argon, trifluoroacetic acid (TFA, 25 µL, 0.322 mmol) was added at 0 °C. The reaction mixture was stirred at 0 °C for 2 h, then quenched by further dilution with dichloromethane (5 mL) and concentrated under vacuum (Chakraborty et al., 2021). The resulting residue was redissolved in acetonitrile/water (1:3 v/v, 1 mL) and dried using SpeedVac to yield **9** as a pale-yellow solid (23.5 mg, 0.059 mmol, 97 %). $[\alpha]_D^{25} = +184$ (*c* 0.28, $CHCl_3$); ESIMS *m/z* 397.2123 $[M]^+$ (calcd 397.2122); ^1H and ^{13}C NMR spectroscopic data: Tables S1 and S2.

5.2.4. Mitragynine oxindole B (6)

To a 25-mL round-bottom flask containing an ice-cold, stirred solution of THF/AcOH/H_2O (1:1:1, v/v/v, 3 mL), mitragynine (110 mg, 0.276 mmol) and KBr (3.28 mg, 0.028 mmol) were added. After stirring for 10 min, Oxone® (102 mg, 0.331 mmol) was introduced in three equal portions at 5-min intervals. Following the addition, the ice bath was removed, and the reaction mixture was allowed to warm gradually to room temperature. After 12 h, an additional batch of Oxone® (50 mg, 0.162 mmol) was added, and stirring continued for 1 h. The reaction was quenched by the addition of saturated $NaHCO_3$ (10 mL) and saturated Na_2CO_3 (10 mL) solutions, followed by extraction with EtOAc (20 mL). The organic fraction was separated, and the aqueous phase was further extracted with EtOAc (20 mL × 3). The combined organic fractions were

washed with brine, dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure (Xu et al., 2019). The resulting residue was purified by silica gel-based flash column chromatography with gradient eluents (MeOH/$CHCl_3$, 1:100 to 1:20) to yield **6** as a pale-yellow solid (13.7 mg, 0.033 mmol, 12%). $[\alpha]_D^{25} = -92$ (*c* 0.4, $CHCl_3$); ESIMS *m/z* 415.2229 $[M+H]^+$ (calcd 415.2233); ^1H and ^{13}C NMR spectroscopic data: Tables S1 and S2.

5.2.5. *Mitragynine-N(4)-oxide (7)*

To a stirred solution of mitragynine (51.7 mg, 0.13 mmol) in anhydrous dichloromethane (1.0 mL) under argon atmosphere, *m*-chloroperbenzoic acid (25.0 mg, 0.16 mmol) was added at −50 °C. The reaction mixture was stirred for 30 min, then poured into a chilled 5% $NaHCO_3$ solution, and extracted with $CHCl_3$ (15 mL × 3). The combined organic layer was dried over anhydrous Na_2SO_4 and evaporated under reduced pressure (Takayama et al., 2002). The residue was purified by silica gel column chromatography (eluent: 5% MeOH in $CHCl_3$), affording **7** as a pale-yellow solid (9.5 mg, 0.023 mmol, 17.7%). $[\alpha]_D^{25} = -37$ (*c* 0.26, $CHCl_3$); ESIMS *m/z* 415.2227 $[M+H]^+$ (calcd 415.2233); ^1H and ^{13}C NMR spectroscopic data: Tables S1 and S2.

5.3. *Preparation of commercial M. speciosa product samples for analysis*

Samples were analyzed across various dosage forms, including tablets, capsules, powders, softgels, and liquids. Solid samples were prepared in their respective form. For tablets, five were weighed, pulverized with a mortar and pestle, and pooled. Capsule samples were prepared by weighing five capsules, opening them, pooling their contents, and triturating the mixture. Soft gel samples were prepared by accurately weighing 5 units, after which the ends of each capsule were manually cut with a knife to release the liquid content. Powder samples (approximately 1.0 g) were homogenized using a mixer mill to obtain a fine powder. All commercial kratom products were finely ground/mixed to ensure homogeneity.

For each solid sample, 100 mg was weighed and suspended in 2 mL of methanol. The methanolic solution was vortexed and sonicated for 30 min at room temperature and then centrifuged for 15 min at 10,000 rpm. The supernatant was transferred to a 10 mL volumetric flask. This extraction procedure was repeated four times, and the combined supernatants were brought to a final volume of 10 mL with methanol.

For liquid samples, a 1.0 mL aliquot of formulation (#4656-4658, 4660-4662, 4814, 4846) was mixed with 1.0 mL methanol, vortexed for 30 s, and sonicated for 30 min. The mixture was vortexed again and then centrifuged for 10 min at 4000 rpm. The clear supernatant was then used for analysis. Prior to LC–MS analysis, a 2.0 mL aliquot was filtered through a 0.45 μm PTFE membrane filter. The first 1.0 mL of filtrate was discarded, and the remaining filtrate was collected in an LC vial. All samples were prepared in duplicate to ensure reproducibility and were further diluted 10-to 50-fold as needed. All samples were freshly prepared and immediately analyzed without storage.

5.4. *Chromatographic conditions*

5.4.1. *Ultra high-performance liquid chromatography-photodiode array Mass spectrometry (UHPLC-PDA-MS) for alkaloids analysis*

Ultra-high-performance liquid chromatography-photodiode array-mass spectrometry (UPLC-PDA-MS) analysis was performed using a Waters Acquity UPLC™ system (Waters Corp., Milford, MA, USA) to separate and identify compounds. The system, which included a quaternary solvent manager, a sample manager, a column compartment, and a PDA detector, was controlled by Waters Empower 3 software. A Waters Acquity UPLC HSS C18 column (150 mm × 2.1 mm I.D., 1.8 μm) with a dedicated guard column (Waters Vanguard LC-18 (2.1 mm × 5 mm)) provided the chromatographic separation at 40 °C, while the samples were kept at 15 °C. The mobile phase, consisting of water (A) and acetonitrile (B) (both with 0.1 % formic acid), was run through a gradient elution at a flow rate of 0.21 mL/min, beginning at 10 % B for 0 min, progressing to 55 % B over 15 min, and subsequently to 100 % B over 17 min. To minimize carryover between runs, a strong and weak needle wash was used, and the column was re-equilibrated following a brief wash step. Finally, a 1 μL sample was injected. Peak identification was confirmed using a combination of mass spectrometry data, retention time matching, and UV-Vis spectral comparison with authentic reference standards.

Detection was performed using a UV detector coupled to a mass spectrometer operating in positive-ion mode. Mass spectrometer conditions were optimized to achieve maximal sensitivity. The probe temperature, capillary voltage, and cone voltage were set at 600 °C, 0.8 kV, and 15 V, respectively. Nitrogen was used as the drying gas. The mass range was set to *m/z* 50–1000.

The method provided excellent baseline resolution for mitragynine , 7-OH , **6**, **9**, and **3**. Peak identification was confirmed by comparing retention times, UV spectra, and mass data with those of reference standards. Specific detection wavelengths were used for quantification: 246 nm for mitragynine , 7-OH , and 233 nm for **3**.

5.4.2. *Method validation*

The UHPLC-PDA method was validated for the quantification of mitragynine , 7-OH , and mitragynine pseudoindoxyl in *Mitragyna* samples. Detection wavelengths were selected based on the UV absorption maxima of each analyte: 246 nm for mitragynine and 7-OH , and 233 nm for **3**, as determined from PDA spectra. Calibration curves were constructed over seven concentration ranges of 0.5–100 μg/mL for mitragynine and 7-OH , and 0.2–100 μg/mL for **3**, with linear regression yielding correlation coefficients (R^2) ≥ 0.99 for all analytes. Limits of detection (LOD) and quantification (LOQ) were determined using signal-to-noise ratios of approximately 3:1 and 10:1, respectively, at the respective wavelengths. The LOD and LOQ were 0.05 μg/mL and 0.2 μg/mL, respectively, providing sufficient sensitivity to reliably detect and quantify these compounds at low concentrations relevant to *Mitragyna* samples. Precision, expressed as intra- and inter-day relative standard deviation (%RSD), ranged from 0.4 to 3% for all analytes. Accuracy, assessed via recovery studies at 5 μg/mL for three commercial products (#4658 PR, 4801 PR, 4839 PR) in duplicate, ranged from 92 to 106%. Specificity was confirmed through analysis of blanks, standards, and spiked samples, verifying both the absence of interfering peaks and the purity of the analyte peaks. Robustness was confirmed as minor variations in column temperature (±2 °C), flow rate (±0.1 mL/min), mobile phase composition, or injection volume did not significantly affect results. Overall, the UHPLC-PDA method is specific, linear, precise, accurate, and robust for the simultaneous quantification of mitragynine, 7-OH, and mitragynine pseudoindoxylin various kratom product matrices.

5.5. *Kinetics of 7-OH (2) degradation*

The degradation kinetics of 7-OH were assessed in fasted-state simulated gastric fluid (FaSSGF) (pH 1.7) and buffer at pH 4.0 by measuring the percentage of the compound remaining at predetermined time intervals (24, 48, 72, 96, 120, 144, 168, and 192 h). Data were analyzed using GraphPad Prism version 10.6.0 (San Diego, CA, USA). Two kinetic models were evaluated: (i) a zero-order model, applied using linear regression of concentration versus time, and (ii) a first-order model, applied using non-linear regression with a one-phase exponential decay equation. For the first-order analysis, the plateau was constrained to zero, reflecting the complete degradation of the compound. Since sample collection began at 24 h, no experimental data were available for t = 0. Therefore, the initial concentration (C_0) was estimated by back-extrapolation of the fitted degradation curve obtained from the subsequent time points.

Credit author statement

Bharathi Avula: Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Syed Kashif Ali:** Synthesis and characterization of mitragyna alkaloids. **Kiran K. Tatapudi:** Method development, validation, analysis, data acquisition, data validation, writing – review & editing. **Pankaj Pandey:** Writing – original draft, Software, Methodology, Formal analysis. **Lochana Cheepilla:** Writing – review & editing, Methodology, Formal analysis. **Islam Husain:** Writing – review & editing, Methodology, Formal analysis. **Shabana I. Khan:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Kumar Katragunta:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Zulfiqar Ali:** Characterization of alkaloids, Writing – review & editing. **Ikhlas A. Khan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Amar G. Chittiboyina:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. A Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.phytochem.2026.114871.

Data availability

Data will be made available on request.

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Assessment of Abuse Potential-Related Effects of Oral Dried Kratom Leaf Powder in Healthy Normal Participants Following Single and Multiple Daily Doses

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ABSTRACT

Kratom is increasingly used in the United States for energy and self-management of pain and mood. While its active alkaloid, mitragynine, has partial μ-opioid receptor agonist activity, clinical data on its abuse potential remain limited. This study evaluated abuse-related outcomes following single and multiple doses of dried kratom leaf powder in healthy kratom-naive adults in a randomized, double-blind, placebo-controlled study. Healthy participants ($n = 116$) received a single oral dose or 15 consecutive daily doses of Mitra-Leaf kratom capsules (4 cohorts: 500–4000 mg dried leaf powder; equivalent to 6.65–53.2 mg mitragynine; $n = 12$–13 per cohort, $n = 49$ total) or placebo ($n = 67$). Subjective effects were assessed using the Drug Effects Questionnaire (DEQ). Opioid withdrawal symptoms were evaluated using the Subjective Opioid Withdrawal Scale (SOWS) and Clinical Opiate Withdrawal Scale (COWS). Abuse-related treatment-emergent adverse events (ARTEAEs) were recorded. DEQ responses were dose-related, except between the middle two dose levels with statistically significant increases in maximum effect and area-under-the-curve compared to placebo observed only at the highest single dose. DEQ responses decreased following repeated dosing. No participants met the threshold for clinically meaningful opioid withdrawal based on COWS or SOWS. ARTEAEs occurred more frequently at higher doses but were generally mild and resolved without sequelae. No serious adverse events or deaths were reported. Oral administration of kratom at tested doses was well tolerated and associated with modest, dose-related subjective effects, minimal withdrawal symptoms, and low ARTEAE rates. Findings support further research on kratom's safety and therapeutic potential, while emphasizing the importance of dose control and product standardization.

1 | Introduction

Kratom is a botanical supplement derived from the leaves of the *Mitragyna speciosa* (Korth.) tree native to Southeast Asia that has been used there for centuries as traditional medicine, and now more recently in the United States and other western countries for various health-related reasons, including to alleviate pain, reduce fatigue, boost mood, and anecdotally to manage opioid withdrawal [1–4]. As kratom consumption in the United States has become more widespread, it has gained significant attention from the public, regulatory authorities, and researchers—both for its potential benefits and due to questions pertaining to its safety and potential for abuse [3, 5], but well-conducted clinical studies examining both aspects of kratom use are needed.

The primary active alkaloid present in kratom is mitragynine, which is likely responsible for many of its effects. In animal drug discrimination studies, mitragynine most robustly generalizes

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to alpha-adrenergic acting lofexidine [6] and it has agonistic effects at adrenergic, serotonergic, D2-dopaminergic, and adenosine receptors [7–11]. Mitragynine is a partial agonist of the human μ-opioid receptor and is a competitive antagonist at κ- and δ-opioid receptors. It has lower potency and affinity for the μ-opioid receptor than prototypical opioids such as morphine and low beta arrestin ("biased G-protein") effects, which likely contribute to its lower respiratory depressant effects and abuse potential [12, 13].

Although mitragynine is the most abundant naturally occurring alkaloid in kratom leaves, there are more than 40 additional alkaloids with varying levels of types of activity that may also contribute to the effects experienced by consumers [14–17]. 7-Hydroxymitragynine, another alkaloid found in significantly lower amounts in kratom, has much stronger morphine opioid receptor binding and pharmacologic effect than most other kratom alkaloids and metabolites, but it is more appropriately considered a mitragynine metabolite than kratom constituent [8, 18, 19].

Taken together, the overall pharmacology and mechanisms of action of mitragynine and kratom generally help increase the understanding of the safety profile and diverse reasons for kratom use reported by consumers. For example, whereas mitragynine (5.6 and 10 mg/kg, intravenous [i.v.]) exhibits respiratory stimulating activity in rats, in the same study morphine produced substantial depression [20]. In a study involving oral administration of mitragynine and oxycodone (by gastric lavage) following an FDA recommended protocol for assessing respiratory effects by blood gas assessment, a broad range of mitragynine doses produced no significant reductions in blood oxygen or life threatening effects at any dose, whereas oxycodone produced profound blood oxygen reductions and deaths at the two highest doses recommended by FDA [21].

Analogously, whereas morphine and heroin maintained robust self-administration in the i.v. rat model, mitragynine was similar to saline with very low rates of self-administration [22, 23]. Due to this unique pharmacology, it is unsurprising that kratom and mitragynine-containing products are being consumed for pain relief, self-management of opioid and other drug withdrawal, and management of substance use disorders [2]. Reasons for use also include improved quality of life, increased productivity, and self-management of stress, mood, and sleep issues [3, 24–26].

The second most frequently reported reason for kratom consumption, after self-management of pain, found in a US nationally representative survey, was to boost energy. This is consistent with the findings of other surveys in which people reported use of kratom in place of or in addition to caffeinated beverages to boost energy and for its useful "instrumental" effects analogous to commonly reported reasons for caffeinated beverage consumption [25, 26]. This finding is not surprising given the botany of *Mitragyna speciosa*, which belongs to the Rubiaceae family that includes coffee trees [27, 28].

It is important to understand the abuse potential of this substance given its use for a wide array of reasons. A complete picture of abuse potential relies on pharmacology, animal self-administration models, clinical studies, and real-world evidence. An assessment of the abuse potential of kratom according to the eight factors of the Controlled Substances Act was published [9] and concluded that kratom did not pose a significant public health risk warranting Controlled Substance Act control, and to the contrary, it was in the interest of public health that kratom not be banned from licit sales and possession. This was also the conclusion of the Assistant Secretary of Health of the US Department of Health and Human Services [29].

Physical dependence and withdrawal associated with kratom are also generally milder and qualitatively distinct from that produced by opioids in animal and human studies [30]. In humans, there are several case reports of withdrawal symptoms due to cessation of kratom [31, 32], and online surveys of kratom users indicate that anywhere between 10% and 20% of respondents report possible kratom-related withdrawal symptoms such as craving, fatigue, irritability, and low energy [2, 33–35]. Importantly, these symptoms were typically rated as mild and not significantly impacting daily life [36, 37]. For comparison, amongst people who regularly consume caffeine, approximately 8% meet diagnostic criteria for caffeine use disorder [38] and between 35.9% and 46.6% experience some form of withdrawal upon cessation [39, 40].

The reported effects of kratom vary widely and depend on dose, frequency and history of use, and alkaloid content of the product used [41]; therefore, carefully controlled investigations of standardized kratom products are needed. The objective of this study was to evaluate potential adverse effects relied upon by the US FDA for assessing abuse potential and physical dependence and instruments validated for opioid withdrawal assessment in a single-dose exposure condition and a "chronic" condition of 15 consecutive daily oral doses of dried kratom leaf powder in healthy adults.

2 | Methods

2.1 | Ethical Practices

This study was conducted in accordance with International Conference on Harmonisation Good Clinical Practice (ICH GCP) guidelines and ethical principles that have their origin in the Declaration of Helsinki. Protocols were approved by an Independent Ethics Committee and Health Canada before eligibility screening. Written informed consent was obtained from each participant before any study-related procedures were performed.

2.2 | Study Participants

Healthy, nonsmoking adults (aged 18–55 years) with a body mass index between 18.5 and 29.9 kg/m^2 inclusive were enrolled. Female participants were of non-childbearing potential or were non-pregnant and non-lactating. Appropriate birth control was utilized by all participants throughout the study, with females of childbearing potential testing negative on pregnancy tests and not breastfeeding and having no intention to get pregnant or breastfeed throughout the study. Participants were required to be either naïve to kratom or to have not reported using it for

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a minimum of 1 year. The presence of gastrointestinal, hepatic, renal, cardiovascular, respiratory, autoimmune, endocrine, neurological, or active psychiatric diseases or positive serology results for HIV, HBsAg, or HCV or a history of cancer were exclusionary. Also exclusionary were an acute illness within 2 weeks of the single dose, a positive urine drug test at screening, a history of drug or alcohol abuse in the past 12 months, a Leeds Dependence Questionnaire score ≥ 21, or consumption of ≥ 2 alcoholic beverages a day. Hepatic or renal dysfunction with alanine aminotransferase (ALT) or aspartate aminotransferase (AST) ≥ 2× upper limit of normal (ULN) and serum creatinine or blood urea nitrogen (BUN) ≥ 1.5× ULN were evaluated by the principal investigator for inclusion or exclusion. Participation in another research study or blood donation within 4 weeks prior to single dosing was prohibited. Participant recruitment and stratification was employed to balance the ratio of sex between active and placebo groups. Medications that induced enzymes were not permitted for 14 days or 7 half-lives of Day 0 or within 7 days if not an enzyme inducer. Participants were required to appear at the study site for 31 visits during the active portion of the study for dosing and other assessments.

2.3 | Study Design

The present study was conducted between date and date at a single site (NutraSource, Guelf, Ontario, Canada). There were two phases in the study: a single-dose (SD) phase and a multiple-dose (MD) phase. Each phase investigated the safety, tolerability, pharmacokinetics, and central nervous system effects of encapsulated Mitra-Leaf powder (NP Pharma). Pharmacokinetic results are published elsewhere [42]. All visits and dosing occurred on site. Participants attended 1 screening visit to assess eligibility, which occurred up to 28 days prior to enrollment, and 1 training visit, which occurred up to 14 days prior to dosing day and familiarized the participant with the study questionnaires.

Single-dose phase: The SD phase was double-blind, randomized, and placebo controlled. Four groups of 12–13 healthy participants were enrolled and received a single oral dose of 6.65 mg, 13.3 mg, 26.6 mg, and 53.2 mg mitragynine contained in 500, 1000, 2000, and 4000 mg kratom raw leaf powder in 1, 2, 4, or 8 capsules, respectively, with associated placebo groups ($n = 67$). The placebo group is larger than the active group because this manuscript provides data from one formulation of kratom tested in a larger clinical study including other formulations not described here; thus, there were matching placebo groups for all formulations. Matched placebo capsules contained inactive ingredients and were identical in fill volume, weight, and appearance to the active ingredient in the investigational product. See Table 1 for detailed sample sizes. Administration of each successive dose depended on the safety and tolerability of previous doses.

Multiple-dose phase: The MD phase was double-blind, randomized, and placebo controlled, utilizing the same participants as the SD phase. The MD phase initiated 10 days after dosing in the SD phase, and pharmacokinetic testing was conducted to ensure complete washout of investigational product had occurred. Participants then received a daily single oral dose of 500, 1000, 2000, and 4000 mg kratom raw leaf for 15 consecutive days. The associated placebo groups ($n = 67$) utilized matching capsule appearance, weight, and number. Participants were not re-randomized; thus, they received multiple administrations of the same dose administered to them in the SD phase.

2.4 | Investigational Product Formulation and Administration

Kratom capsules were labeled according to ICH GCP and Health Canada guidelines and stored at room temperature (20–25 °C) and relative humidity ≤ 65%. The product contains between 0.8 and 1.4% mitragynine weight/leaf powder weight and met US Pharmacopeia (USP) standards for impurities (i.e., no impurities at or above levels accepted for dietary supplements). Each capsule consisted of 500 mg kratom leaf powder containing 6.65 mg of mitragynine. Matched placebo capsules contained inactive ingredients and were identical in fill volume, weight, and appearance as the active ingredient in the investigational product.

2.5 | Outcome Measures

2.5.1 | Subjective Drug Effect

Participants were queried on a 100 mm visual analog scale asking "Do you feel a drug effect right now?" This was asked on Day 0 (after SD administration) and Day 24 (after the last [i.e., 15th] dose of the MD phase) at the following time points: pre dose/baseline, and at 15-min intervals up to 3 h post-dosing, then hourly to 12 h post-dosing (i.e., 15 min, 30 min, 45 min, 1 h, 1 h 15 min, 1 h 30 min, 1 h 45 min, 2 h, 2 h 15 min, 2 h 30 min, 2 h 45 min, 3 h, 4 h, 5 h, 6 h, 7 h, 8 h, 9 h, 10 h, 11 h, and 12 h).

2.5.2 | SOWS and COWS

The SOWS is a self-administered scale for grading opioid withdrawal symptoms. It contains 16 symptoms that are rated on a scale of 0 (not at all) to 4 (extremely), with a total score range of 0 to 64. It was administered on Day 7 (7 days after SD), prior to dosing and 4, 8, and 12 h after dosing on the last day of MD phase (i.e., after 15 days of consecutive dosing), and 4 times per day for 3 days after the last dose of the MD phase. The COWS consists of 11 opioid-withdrawal signs rated by a trained observer on a scale from 0 to 4 (total scale range, 0–44, where 5–12 represents mild symptoms). It was administered on Day 7 (7 days after SD), prior to dosing and 12 h after dosing on Day 24 (last day of MD phase), and once per day for 3 days after the last dose of the MD phase.

Adverse Events: Adverse events (AEs) were not specifically queried during the study. When spontaneously reported, relatedness, time since last dose, severity, and outcome were all recorded. An AE was defined as any undesirable symptom or occurrence experienced by a participant during a clinical study, including worsening of a pre-existing condition recorded at Screening. Severity was characterized according to the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use Guidelines. Briefly, an AE was considered mild if it met the following criteria: transient or easily tolerated symptoms, minimal discomfort,

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TABLE 1 | Participant demographics.

	Single-dose phase					Multiple-dose phase				
	Mitragynine/dried kratom leaf powder (mg, PO)									
	0/0	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**	**0/0**	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**
Participants (*n*)	67	12	12	12	13	60	11	11	10	12
Sex										
Female	33	6	4	4	7	28	6	4	3	6
Male	34	6	8	8	6	32	5	7	7	6
Race										
American Indian	3	0	0	0	2	2	0	0	0	2
Asian	10	2	5	1	0	9	2	5	0	0
Black	12	0	0	4	1	10	0	0	3	1
Unknown	2	0	0	0	1	1	0	0	0	1
White	40	10	7	7	9	38	9	6	7	8
Ethnicity										
Hispanic	6	2	1	0	3	4	2	1	0	3
Not Hispanic	61	10	11	12	10	56	9	10	10	9
Age (mean, years)										
Mean	32.7	30.8	32.9	35.8	33.2	33.4	31.3	33.9	37.3	33.9
(SD)	(9.9)	(9.3)	(9.4)	(11.4)	(9.6)	(10.0)	(9.5)	(9.2)	(12.1)	(9.7)
BMI (mean, kg/m^2)										
Mean	25.4	23.1	25.3	26.2	23.4	25.5	22.8	25.4	26.1	23.8
(SD)	(2.7)	(2.7)	(2.6)	(3.4)	(2.9)	(2.7)	(2.6)	(2.7)	(3.7)	(2.8)

Note: Numbers here include all participants who received at least a single dose of investigational product (active or placebo) during each study phase. Participants completed both phases sequentially.
Abbreviations: BMI, body mass index; PO, *per os* (oral); SD, standard deviation.

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no interference with daily activities, and no or minimal medical intervention required. It was considered moderate if it met any of the following criteria: greater discomfort, interferes with usual daily activities, or may require medical intervention or treatment. A severe AE was characterized by any of the following: incapacitated or prevented normal daily activities, marked symptoms or significant clinical concern, or required intensive treatment or hospitalization.

If an AE's onset was on or after the first dose or the pre-existing medical condition was aggravated in severity or frequency following study intervention, it was treated as treatment-emergent adverse event (TEAE). The frequency (number of events and number of participants) and incidence (percentage of participants) of each TEAE in each active formulation and pooled placebo groups were reported if they occurred in more than 1 participant. All AEs were followed to resolution. A complete summary of AEs is available [43], but the focus of this publication is to discuss potentially abuse-related AEs; therefore, those are presented herein. Abuse-related AE terms were selected according to the Recommended List of Abuse-related AEs to Assess Abuse Potential, as presented at the Cross-Company Abuse Liability Counsel 2023 Meeting [44]. Withdrawal-related TEAEs were characterized according to definitions in Kosten, Baxter [45].

2.6 | Data Analysis

2.6.1 | Safety Analyses

These were performed for all participants who received at least one dose of study intervention in both SD and MD phases. These analyses considered AEs, which were monitored throughout the study, and which included any undesirable symptom or occurrence experienced by a participant during the study. Only AEs that were treatment-emergent and suspected to be related to the study treatment (excluding pre-existing conditions) were considered for these analyses (see supplemental materials for TEAEs regardless of suspected relation to study treatment). Analyses also separately considered AEs on the recommended list of abuse-related AEs to assess abuse potential as presented at the Cross-Company Abuse Liability Counsel (CCALC; see Supplemental Table 1 [44]). The number and percentage of participants who experienced an AE are presented for each group.

2.6.2 | DEQ

Area-under-the-curve (AUC) and maximum effect (Emax) for DEQ were compared. Because scores were not normally distributed (right-skewed), Mann–Whitney *U* tests were performed. Each dose was compared to pooled placebo, and scores from the SD phase were compared to the MD phase within each dose. Because a large number of tests were performed (four doses across two phases [SD and MD] compared to placebo for 8 tests, and four doses compared to themselves across phases for an additional four tests, or 12 tests total for AUC and 12 tests total for Emax), Bonferroni corrections were made so that the alpha levels for these tests were $0.05/12 \approx 0.004$. As a sensitivity test, the same tests were also considered without Bonferroni correction.

2.6.3 | COWS and SOWS

Mean COWS and SOWS total scores were compared to placebo for each dose. Because COWS and SOWS total scores were not normally distributed (right-skewed), Mann–Whitney *U* tests were performed. Because a large number of tests were performed (four doses across 6 time points for a total of 24 tests for COWS and four doses across 6 time points for a total of 24 tests for SOWS), Bonferroni corrections were made so that the alpha levels for these tests were $0.05/24 \approx 0.002$ for COWS and for SOWS. As a sensitivity test, the same tests were also considered without Bonferroni correction.

3 | Results

3.1 | Participants

A total of 116 participants were enrolled in the study and received at least a single dose of investigational product: 49 received active ingredient in the investigational product (12 participants each received the three lowest doses, 13 received the highest dose) and 67 received placebo. Participant demographics are shown in Table 1.

3.2 | Subjective Drug Effects

Overall, DEQ scores increased with increasing dose (except between the middle two dose levels). The data show that mean Emax and AUC DEQ values after the last day of the MD phase are lower than the DEQ values after a single dose. There were three participants who withdrew from the study before the last day of the MD phase that had above-average DEQ scores during the SD phase. This may have in part contributed to the observed reductions from SD to the MD phase in mean DEQ Emax and AUC. However, for most participants with DEQ results in both SD and MD phases, the DEQ maximum value and AUC were still higher after dosing in the SD phase. Importantly, because three participants with above-average DEQ responses withdrew prior to completion of the MD phase, the interpretation of reduced subjective effects over time must be limited. DEQ findings are shown in Table 2.

Mann–Whitney *U* tests indicated that there was no detectable difference in AUC at any dose or phase compared to placebo, nor at any dose between phases, when the Bonferroni correction was applied ($56.5 \leq Us \leq 651.5$; $ps > 0.004$). However, AUC was greater in the highest dose group (53.2 mg, mitragynine) compared to placebo for SD phase when the Bonferroni correction is *not* considered (84.3 vs. 16.4; $U = 651.5$; $p = 0.005$).

Mann–Whitney *U* tests indicated that Emax was statistically significantly greater in the highest dose group (53.2 mg, mitragynine) for single dose compared to placebo even after applying the Bonferroni correction (27.6 vs. 6.6; $U = 666.5$; $p = 0.002$). All other comparisons between mitragynine and placebo for Emax were not statistically significant, regardless of whether the Bonferroni correction was considered ($175 \leq Us \leq 485$; $ps > 0.05$). Mann–Whitney *U* tests also indicated that there was no detectable difference in Emax at any dose between phases when the

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TABLE 2 | Do you feel a drug effect right now?

	Single-dose phase					Multiple-dose phase (after 15 doses)				
	Mitragynine/dried kratom leaf powder (mg, PO)									
	0/0	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**	**0/0**	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**
Participants (*n*)	67	12	12	12	13	56	11	11	9	9
Emax										
Mean	6.6	2.9	18.5	11.0	27.6*	2.6	1.7	6.4	1.7	9.4
(SD)	(11.7)	(3.3)	(25.7)	(19.4)	(30.7)	(6.9)	(2.1)	(9.2)	(4.3)	(10.1)
AUC										
Mean	16.4	5.0	59.8	10.8	84.3	6.3	3.9	22.7	1.6	14.5
SD	(38.4)	(6.4)	(112.9)	(19.8)	(157.5)	(15.8)	(4.6)	(39.5)	(3.3)	(15.4)

Note: Numbers here include all participants who completed dosing for both phases. Participants completed both phases sequentially. Drug effect was queried after the first administration and after the last investigational product administration (i.e., after the 15th dose in the multiple dose phase).
Abbreviations: PO, *per os* (oral); SD, standard deviation.
*Statistically significantly different from placebo (0/0) after Bonferroni correction.

Bonferroni correction was applied ($57.5 \leq U\text{s} \leq 84.5$; $p\text{s} > 0.004$). However, Emax was greater in the second highest dose group (26.6 mg, mitragynine) for single dose compared to multiple dose when the Bonferroni correction is *not* considered (11.0 vs. 1.7; $U = 84.5$; $p = 0.03$).

3.3 | COWS and SOWS

Mann–Whitney *U* tests indicated that there was no detectable withdrawal at any dose or timepoint as measured by the COWS with or without a Bonferroni correction ($164.0 \leq U\text{s} \leq 449.5$; $p\text{s} > 0.002$ [with Bonferroni]; $p\text{s} > 0.05$ [without Bonferroni]). Similarly, Mann–Whitney *U* tests, with and without a Bonferroni correction, found no detectable withdrawal at any dose or timepoint as measured by the SOWS ($p\text{s} > 0.0007$ and $p\text{s} > 0.05$, respectively).

3.4 | Adverse Events

There were no serious adverse events (SAEs) or deaths in this study. Full listings of all AEs by incidence and frequency for the SD and MD phase are provided in Supplemental Tables 2 and 3, respectively. In the SD phase (Table S2), the proportion of participants experiencing at least one TEAE was generally higher in active treatment groups compared with placebo, although this pattern was not strictly dose-dependent. TEAEs were reported in 25.0%–76.9% of participants across active dose groups, compared with 29.9% in the pooled placebo group. The highest incidence was observed in the 53.2/4000 dose group (76.9%), while lower-dose groups demonstrated more variable rates (25.0%–41.7%). Overall, these findings suggest an increased likelihood of TEAEs with active treatment relative to placebo, particularly at higher doses.

In the MD phase (Table S3), a clearer separation between active treatment and placebo was observed. The proportion of participants reporting at least 1 TEAE ranged from 18.2% to 83.3% across active dose groups, compared with 41.7% in the placebo group. Notably, the highest TEAE incidence was again observed in the highest dose group (53.2/4000; 83.3%), while intermediate dose groups (e.g., 13.3/1000) also demonstrated relatively high rates (72.7%). Lower-dose groups showed more modest or comparable rates relative to placebo. Overall, TEAE incidence appeared to increase with dose in the MD phase, with several active treatment groups exceeding the placebo rate.

Abuse-related and withdrawal-related TEAEs are discussed below.

3.4.1 | Single-Dose Phase

Twenty-two participants reported 28 ARTEAEs in the SD phase. One episode of somnolence and 1 episode of anxiety were determined to be unrelated to investigational product; therefore, 20 participants with 26 ARTEAEs were suspected to be related to investigational product (See Table 3). All ARTEAEs in the SD phase were mild ($n = 25$, 96.2%) or

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TABLE 3 | Abuse-related treatment-emergent Adverse Events in the Single Dose Phase.

	Mitragynine/dried kratom leaf powder (mg, PO)					
	0/0, *n* = 67	**6.65/500, *n* = 12**	**13.3/1000, *n* = 12**	**26.6/2000, *n* = 12**	**53.2/4000, *n* = 13**	**Total *N* = 116**
Participants with ARTEAE (*n*; %)	6 (9.0%)	0 (0.0%)	4 (33.3%)	3 (25.0%)	7 (53.8%)	20 (17.2%)
Participants with no ARTEAE (*n*; %)	61 (91%)	12 (100%)	8 (66.7%)	9 (75.0%)	6 (46.2%)	96 (82.8%)
Number of ARTEAEs	9	0	4	4	9	26
System organ class	**Reported frequency**					
Preferred term	**No. (%)**					
General disorders and administration site conditions	**4 (44.4%)**	**0 (0.0%)**	**2 (50.0%)**	**2 (50.0%)**	**4 (44.4%)**	**12 (46.2%)**
Feeling abnormal	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	0 (0.0%)	3 (11.5%)
Feeling of relaxation	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	2 (22.2%)	5 (19.2%)
Feeling drunk	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (22.2%)	2 (7.7%)
Energy increased	2 (22.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (7.7%)
Nervous system disorders	**4 (44.4%)**	**0 (0.0%)**	**0 (0.0%)**	**1 (25.0%)**	**3 (33.3%)**	**8 (30.8%)**
Somnolence	4 (44.4%)	0 (0.0%)	0 (0.0%)	1 (25.0%)	3 (33.3%)	8 (30.8%)
Psychiatric disorders	**1 (11.1%)**	**0 (0.0%)**	**2 (50.0%)**	**1 (25.0%)**	**2 (22.2%)**	**6 (23.0%)**
Euphoric mood	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	1 (11.1%)	4 (15.4%)
Hypervigilance	0 (0.0%)	0 (0.0%)	1 (25.0%)	0 (0.0%)	1 (11.1%)	2 (7.7%)

Abbreviations: ARTEAE, abuse-related treatment-emergent adverse event; No., number; PO, *per os* (oral).

moderate ($n = 1$, 3.8%) in severity. The most commonly reported ARTEAEs in the SD phase were "somnolence" and "feeling of relaxation." The proportion of participants experiencing at least one ARTEAE was higher in active treatment groups compared with placebo and demonstrated a general increase with dose. Abuse-related TEAEs were reported in 0.0% of participants in the lowest active dose group (6.65/500), compared with 9.0% in the placebo group. Incidence increased at higher doses, with 33.3% and 25.0% of participants affected in the 13.3/1000 and 26.6/2000 groups, respectively, and reaching 53.8% in the highest dose group (53.2/4000). Overall, 17.2% of participants experienced at least one ARTEAE. These findings indicate a dose-dependent increase in abuse-related events, particularly at higher exposures, with rates exceeding placebo at all but the lowest dose.

3.4.2 | Multiple-Dose Phase

Twenty-two participants reported 154 ARTEAEs in the MD phase. One episode of somnolence was determined to be unrelated to investigational product; therefore, 22 participants with 153 ARTEAEs were suspected to be related to investigational product (see Table 4). All ARTEAEs in the MD phase were mild (153, 100.0%) in severity. The most commonly reported ARTEAEs in the MD phase were "feeling of relaxation" and "hypervigilance."

A similar but more pronounced dose-related pattern was observed in the MD phase. ARTEAEs were reported in 16.7% of participants receiving placebo. No ARTEAEs were observed in the 6.65/500 and 26.6/2000 dose groups, whereas 36.4% of participants in the 13.3/1000 group and 66.7% in the 53.2/4000 group experienced at least 1 ARTEAE. Overall, 21.2% of participants reported ARTEAEs. These results suggest a greater separation between active treatment and placebo at higher doses under repeated dosing conditions, with the highest dose group demonstrating the greatest incidence of abuse-related events.

3.4.3 | Follow-Up Phase

Only one withdrawal-related TEAE was reported during the follow-up phase; 1 participant receiving 1000 mg kratom (13.3 mg, mitragynine) reported 1 instance of mild diarrhea approximately 2 days after stopping kratom intake, which resolved in less than a day without any action taken.

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TABLE 4 | Abuse-related treatment-emergent adverse events in the multiple-dose phase.

	Mitragynine/dried kratom leaf powder (mg, PO)					
	0/0, *n* = 60	**6.65/500, *n* = 11**	**13.3/1000, *n* = 11**	**26.6/2000, *n* = 10**	**53.2/4000, *n* = 12**	**Total, *N* = 104**
Participants with ARTEAE (*n*; %)	10 (16.7%)	0 (0.0%)	4 (36.4%)	0 (0.0%)	8 (66.7%)	22 (21.2%)
Participants with no ARTEAE (n; %)	50 (83.3%)	11 (100%)	7 (63.6%)	10 (100%)	4 (33.3%)	82 (78.8%)
Number of ARTEAEs	53	0	31	0	69	153
System organ class	**Reported frequency**					
Preferred term	**No. (%)**					
General disorders and administration site conditions	33 (62.3%)	0 (0.0%)	14 (45.2%)	0 (0.0%)	38 (55.1%)	85 (55.6%)
Feeling abnormal	17 (32.1%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	3 (4.3%)	22 (14.4%)
Feeling of relaxation	16 (30.2%)	0 (0.0%)	12 (38.7%)	0 (0.0%)	12 (17.4%)	40 (26.1%)
Feeling drunk	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	23 (33.3%)	23 (15.0%)
Nervous system disorders	12 (22.6%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	7 (10.1%)	21 (13.7%)
Somnolence	12 (22.6%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	7 (10.1%)	21 (13.7%)
Psychiatric disorders	8 (15.1%)	0 (0.0%)	15 (48.4%)	0 (0.0%)	24 (34.8%)	47 (30.7%)
Euphoric mood	3 (5.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (2.0%)
Hypervigilance	3 (5.7%)	0 (0.0%)	15 (48.4%)	0 (0.0%)	15 (21.7%)	33 (21.6%)
Anxiety	1 (1.9%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	9 (13.0%)	10 (6.5%)
Irritability	1 (1.9%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.7%)

Abbreviations: ARTEAE, abuse-related treatment-emergent adverse event; No., number; PO, *per os* (oral).

4 | Discussion

This randomized, placebo-controlled study evaluated the central nervous system effects associated with abuse potential following single and multiple daily oral doses of dried kratom leaf powder in healthy adults. The study found that kratom was well tolerated across all dose levels, with no SAEs reported. Abuse-related subjective and objective outcomes were low overall and, when present, were most prominent at the highest tested dose (53.2 mg mitragynine). Importantly, short-term administration of standardized kratom leaf powder in this sample of healthy adults did not produce clinically significant opioid withdrawal as measured by COWS/SOWS and produced modest subjective effects at the highest dose tested.

These findings are broadly consistent with preclinical and epidemiological evidence suggesting a relatively low abuse liability for mitragynine and kratom products, particularly in comparison to classical opioids. In animal models, mitragynine has not supported self-administration and does not substitute for morphine or heroin, while human survey data indicate relatively mild and infrequent withdrawal symptoms among those who abruptly discontinue moderate use; although, this is dose-related, with persons consuming higher amounts more frequently reporting more withdrawal [34]. Consistent with this prior work, the present study demonstrated that while participants endorsed some subjective effects, particularly at the highest dose, these effects were modest in magnitude and diminished with repeated exposure. Likewise, only a single mild withdrawal-related AE (diarrhea) was observed during the follow-up phase, and neither COWS nor SOWS scores differed significantly from placebo at any dose or timepoint.

Across both phases, the highest incidence of ARTEAEs consistently occurred in the highest dose group, supporting a dose–response relationship. In contrast, lower dose groups showed variable and in some cases lower incidence than placebo, particularly in the MD phase. Taken together, these findings indicate that abuse-related AEs are more likely to occur at higher doses and with repeated administration while remaining relatively infrequent at lower doses.

The dose-dependent increase in ARTEAEs, particularly in the highest dose group, may reflect pharmacodynamic activity at the μ-opioid receptor, where mitragynine acts as a partial agonist [7]. Nonetheless, the nature and severity of these AEs, most commonly somnolence, feeling of relaxation, and hypervigilance, differ from those typically observed with full μ-opioid receptor agonists such as hydrocodone or oxycodone

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[46] which often include euphoria, respiratory depression, constipation, sleepiness, etc. This is consistent with similar findings from other surveys and reviews of kratom use where consumers generally reported symptoms as mild to moderate [25, 34]. Furthermore, the diminishing subjective drug effects observed between the SD and MD phases may suggest mild tolerance development or a desensitization of effect over time, although participants did not experience significant withdrawal effects when the study concluded and discontinued use of the investigational product.

The DEQ responses in particular offer important insight. Only the highest dose group demonstrated statistically significant increases in Emax and AUC compared to placebo following single administration. After 15 days of daily administration, DEQ responses were lower across all dose groups, further supporting reduced sensitivity or subjective valuation of the kratom product with repeated exposure. Importantly, because three participants with above-average DEQ responses withdrew prior to completion of the MD phase, the interpretation of reduced subjective effects over time must be limited.

Notably, while some participants reported symptoms classified as abuse-related (e.g., hypervigilance), these occurred at low frequencies and were not associated with sustained or escalating responses. The absence of a clear withdrawal syndrome, either through clinical observation (i.e., COWS) or participant self-report (i.e., SOWS), suggests limited potential for physical dependence under these specific test conditions (i.e., short-term administration in healthy kratom-naïve individuals).

This study has several limitations. First, while the study enrolled healthy participants without recent kratom use, findings may not generalize to individuals with a history of regular kratom consumption or polysubstance use. Second, the study duration, although longer than many human abuse potential studies, may not fully capture the potential risks associated with chronic, higher dose kratom use that may be observed in some real-world settings (in one survey, participants reported use of single doses up to 15 g of kratom) [34].

5 | Conclusion

The absence of SAEs, the mild nature of most ARTEAEs, and the lack of a significant withdrawal syndrome provide support for continued investigation into kratom's utility as a dietary supplement. At the same time, the dose-related emergence of some abuse-related effects highlights the importance of dose control and product standardization in both clinical and consumer settings.

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Conflicts of Interest

This study was supported by NP Pharma, currently doing business as Della Terra Pharmaceuticals. NP Pharma, currently doing business as Della Terra Pharmaceuticals, reviewed a draft of the manuscript. NP Pharma sponsored a trip for Marion Coe to present preliminary findings at the 63rd Annual Meeting of the American College of Neuropsychopharmacology (December 8–11, 2024) in Phoenix, Arizona. Through Pinney Associates, Marion Coe, Daniel Wang, and Jack Henningfield consult to NP Pharma. In addition to NP Pharma, in the last 36 months, Pinney Associates has consulted to the American Kratom Association, Holistic Alternative Recovery Trust, and Botanic Tonics LLC (the last uncompensated) on kratom science and regulatory issues and studies. Through Pinney Associates, in support of the American Kratom Association, JH has commented on kratom pharmacology, public health, and regulatory needs for state hearings and for a judicial hearing to address questions raised by the court on these same matters. Through ALA BioPharma Consulting, Thomas Hudzik consults to NP Pharma. Through JB PharmaConsult, JB consults to NP Pharma. Through Huestis and Smith Toxicology LLC, Marilyn Huestis consults to NP Pharma. JH, through Pinney Associates, and MH are paid expert witnesses in legal cases related to kratom.

Data Availability Statement

Research data are not shared.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Recommended list of abuse-related AEs to assess abuse potential as presented at the Cross-Company Abuse Liability Counsel (CCALC; Galati, S. (2023). Identifying Relevant Adverse Events of Interest and Recommendation for Analysis and Presentation of Data in the NDA Submission. Cross-Company Abuse Liability Council (CCALC) Meeting. Rockville, MD). **Table S2:** Treatment-emergent adverse events in the single-dose phase. **Table S3:** Treatment-emergent adverse events in the multiple-dose phase.

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

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Severe Early-Onset Withdrawal Following Intentional Use of Mitragynine Pseudoindoxyl: A Case Report and Emerging Clinical Considerations

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Abstract

Kratom is increasingly used in the United States for pain, mood, anxiety, and opioid substitution. Traditional kratom produces mild, delayed withdrawal, but semisynthetic derivatives such as mitragynine pseudoindoxyl (MP) are highly potent μ-opioid receptor agonists with δ-opioid receptor antagonism, leading to rapid-onset, full-spectrum opioid-like withdrawal. We describe a 34-year-old man with long-term remission from heroin use disorder who presented with suicidal ideation and severe opioid-like withdrawal despite no recent opioid use. He progressed from powdered kratom to 7-hydroxymitragynine tablets and then MP tablets, averaging nine 20mg doses daily, including nocturnal use. On presentation, he exhibited chills, tremors, diaphoresis, gastrointestinal distress, restlessness, with a Clinical Opiate Withdrawal Scale score of 31, blood pressure 168/107 mmHg, and heart rate 115 bpm. Supportive management with clonidine, hydroxyzine, gabapentin, loperamide, and antiemetics over 72-96 hours led to symptom resolution. Naltrexone, including a Vivitrol injection, was initiated without precipitating withdrawal, and the patient remained abstinent at one-month follow-up, despite a brief noneuphoric lapse. This case highlights MP as an emerging high-potency kratom derivative that poses unique diagnostic and management challenges. Clinicians should recognize its severe withdrawal profile, the limitations of standard toxicology, and the potential role of naltrexone in relapse prevention once an opioid-free interval is established.

Categories: Psychiatry, Substance Use and Addiction
Keywords: 7-hydroxymitragynine, addiction psychiatry, kratom, mitragynine pseudoindoxyl, opioids

Introduction

Kratom (Mitragyna speciosa) has gained substantial popularity in the United States as a self-management tool for pain, anxiety, mood symptoms, and as an opioid replacement [1,2]. Although kratom has been used for centuries in Southeast Asia with relatively low concern for severe toxicities and liabilities, the contemporary U.S. market has evolved in ways that diverge substantially from traditional botanical preparations [3]. To better characterize this transformation, we introduce a conceptual "three waves" framework describing the progression of kratom products in the United States. The first wave consisted primarily of ground leaf powders with alkaloid profiles comparable to those of the natural plant. The second wave emerged with the adoption of solvent-based extraction techniques designed to concentrate specific alkaloids naturally present in kratom [4]. The current third wave represents a marked departure from botanical kratom: manufacturers are now producing laboratory-derived compounds by chemically modifying kratom's primary alkaloid, mitragynine (MG), into isolated, highly potent derivatives. These products are frequently formulated into pills or concentrated preparations and marketed as "kratom," despite being chemically and pharmacologically distinct from traditional plant material [5].

MG is a partial μ-opioid receptor (MOR) agonist with atypical G-protein-biased signaling. Withdrawal from kratom containing traditional MG concentrations typically has a delayed onset and a milder clinical course compared to classical opioids. Hepatic CYP3A metabolism converts a portion of MG to 7-hydroxymitragynine (7-OHMG), and small amounts form naturally in the dried leaf. Further oxidative transformation yields mitragynine pseudoindoxyl (MP), a semisynthetic derivative that is not naturally present in kratom [6]. In receptor binding assays, MG demonstrates relatively low MOR affinity (Ki ~709 nM range), whereas 7-OHMG exhibits substantially higher affinity (Ki ~78 nM). MP shows even greater MOR potency and intrinsic efficacy (Ki ~1.5 nM) and near-full agonist activity in preclinical systems, exceeding morphine in functional assays [7,8]. In addition, MP exhibits δ-opioid receptor (DOR) antagonism. Although MG and 7-OHMG also demonstrate DOR antagonism, MP is distinguished by its markedly greater MOR intrinsic efficacy, approaching full agonism in preclinical systems. The combination of high MOR activation with concurrent DOR blockade may alter downstream signaling dynamics. This pharmacological profile may contribute to dysphoria and unusually intense withdrawal syndromes [9].

Although MP was initially identified only as a potential adulterant in some kratom preparations, it is now being intentionally sold as a stand-alone product or included in highly concentrated "extract" products

labeled as kratom. In many cases, vendors explicitly label these products as containing MP or other high-potency alkaloids and may specify the dose per unit; however, labeling practices are inconsistent across manufacturers, and product composition is not independently verified. These laboratory-derived preparations have dramatically different pharmacological effects and risk profiles relative to traditional plant material. Labeling, when available, does not list MP amounts contained in the product. Clinical recognition is challenging because standard toxicology screens do not detect MP, leaving clinicians to rely on patient history and symptomatology. This case illustrates the presentation of MP-associated withdrawal, which appears more rapid in onset, more severe, and more consistent with full-spectrum opioid withdrawal than withdrawal from traditional kratom products. Awareness of these distinctions is increasingly important as clinicians encounter patients exposed to manipulated, high-potency “kratom” analogs. Notably, in this case, concurrent naltrexone use appeared to blunt the expected euphoric effects of MP, providing additional insight into its opioid-mediated mechanism of action.

Case Presentation

A 34-year-old man with a history of depression, yet off psychotropics, presented to the emergency department with suicidal ideation and depressive symptoms and was subsequently admitted to the inpatient psychiatry unit. On admission, he endorsed no recent opioid exposure, and this was confirmed by a negative urine toxicology screen. This was a routine hospital urine immunoassay panel, which detects natural opioids as well as common synthetic opioids such as oxycodone, buprenorphine, methadone, and fentanyl. He also reported a history of remission from opioid use disorder for six years. Despite this, he was experiencing chills, cold flashes, restlessness, piloerection, diaphoresis, mild body aches, and nausea. Over the next two hours, these symptoms evolved into severe body aches, tremors, pronounced restlessness, diarrhea requiring frequent bathroom use, and two episodes of vomiting. Vital signs demonstrated blood pressure of 168/107 and a heart rate of 115. Routine laboratory studies, including electrolytes, kidney and liver function tests, were unremarkable. The patient denied any illicit drug use but reported regular "kratom" consumption, noting that similar withdrawal-like symptoms occurred at home if dosing intervals exceeded five to six hours. He had last used the product seven hours before the presentation.

The patient reported initiating kratom use five years before, approximately one year after achieving heroin remission, initially using powdered kratom (one teaspoon in the morning, two in the evening), often without experiencing withdrawal for several weeks. About one year ago, he began using “7-hydroxy” (7-OHMG) tablets, which he perceived as more potent for mood and euphoria. Within one month, his dose escalated from two 15mg tablets daily to 30mg three to four times daily, and three months before admission, he switched to a 65mg tablet formulation, taking four to six times daily, including nocturnal doses. He reported mild chills and discomfort upon awakening when doses were delayed.

Approximately three months before this admission, he began experimenting with MP tablets, initially taking two 8mg tablets, which he found more potent than 7-OH. He subsequently transitioned to a different brand of MP tablets, taking approximately 20mg per dose, averaging nine doses per 24 hours, including one or two nocturnal doses. This pattern caused marital stress due to financial strain, decreased productivity at work, lethargy, and disengagement at home. The patient also reported involvement in a recent motor vehicle accident while using MP. Attempts to reduce dose or frequency were unsuccessful due to extreme discomfort described as “crawling out of my skin.” Despite awareness of potential risks from online forums, he was unable to stop using and expressed interest in “getting off these things” while inpatient.

The patient’s Clinical Opiate Withdrawal Scale (COWS) score was 31 [10]. He was treated with supportive and symptomatic measures, including clonidine 0.2mg every six hours, hydroxyzine 50mg every six hours as needed for anxiety, Compazine 5 mg every six hours as needed for nausea, loperamide 2mg every four hours as needed, and gabapentin 600mg four times daily. His symptoms peaked over the first 24 hours and gradually improved over 72-96 hours. COWS scores remained elevated in the moderately severe range for the first 12 hours, with continued need for all pro re nata (PRNs) during the first two days. By day 2, the COWS score decreased to 10, and by day 3, it was 8, with minimal need for PRNs.

By day 5, the patient was symptom-free. Paroxetine was initiated based on prior response for depressive symptoms, and a long-term sobriety plan was discussed, including naltrexone therapy. On day 6, an oral naltrexone test dose was administered without precipitating withdrawal, followed by a naltrexone depot (Vivitrol) injection on day 7 before discharge. He was referred to outpatient addiction services for ongoing monitoring and support.

Two weeks after discharge, following a stressful event, the patient ingested three 20mg MP tablets remaining at home. He reported fatigue but no euphoria. At one-month follow-up, he remained abstinent from MP and continued engagement with outpatient addiction support. A chronology of the patient's kratom use, progression, withdrawal, and management is depicted in Figure *1*.

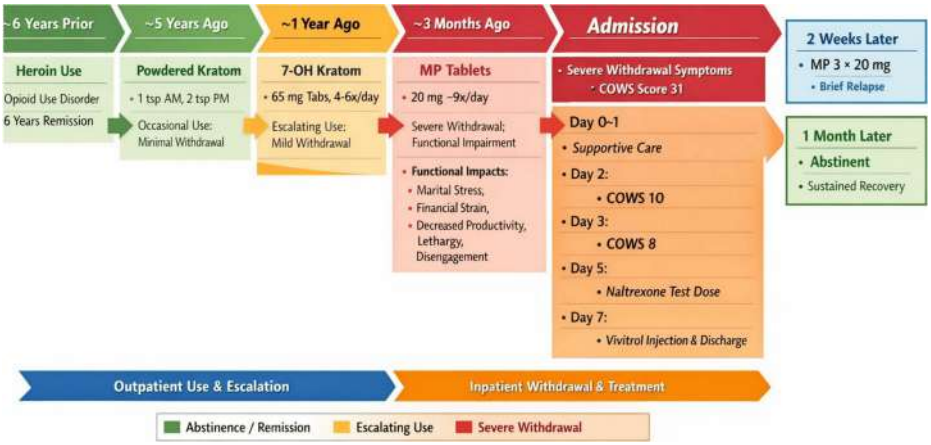


FIGURE 1: Chronology of kratom use, addiction progression, withdrawal, and maintenance treatment

COWS: Clinical Opiate Withdrawal Scale

Discussion

MP is now commercially available both as a stand-alone compound and as a concentrated ingredient within some products marketed as "kratom." This development marks a significant shift within the unregulated kratom marketplace, reflecting the emergence of semisynthetic, high-potency derivatives that differ substantially from traditional botanical preparations [11]. Compared with MG, MP demonstrates markedly greater potency, high MOR occupancy, and DOR antagonism. These properties offer a mechanistic explanation for the rapid onset, severity, and full-spectrum opioid-like withdrawal observed, features not previously typical of classic kratom withdrawal. However, confounding factors should be acknowledged: the patient had been using 260-390 mg/day of 7-OHMG with established interdose withdrawal prior to transitioning to 180 mg/day of MP. Without complete characterization of 7-OHMG vs. MP pharmacology, including oral bioavailability and blood-brain barrier penetration in humans, it is unclear whether 180 mg/day of MP truly represents greater total opioid exposure than the prior 7-OHMG regimen. Thus, the severity of withdrawal may reflect cumulative dependence across his kratom alkaloid use rather than being solely attributable to MP's pharmacologic profile.

These pharmacologic distinctions underscore that MP withdrawal represents a fundamentally different clinical entity. Whereas traditional kratom withdrawal is typically delayed and comparatively mild, MP's pharmacokinetic characteristics, including rapid central nervous system penetration and high receptor occupancy at nanomolar concentrations in preclinical models, result in an earlier onset and more intense opioid-like effects than seen with botanical kratom alkaloids. In this respect, the clinical syndrome of MP withdrawal may more closely resemble that of short-acting prescription opioids, which are characterized by rapid absorption and a relatively brief elimination half-life, leading to earlier and more pronounced withdrawal symptom onset. Such presentation necessitates closer monitoring and supportive care aligned with established opioid withdrawal management protocols [12]. This includes hydration, autonomic stabilization, antiemetics, symptomatic medications, and, when clinically appropriate, opioid agonist therapy [13].

Recognition is further complicated by the fact that routine urine drug screens do not detect MP or related semisynthetic alkaloids. At present, detection is limited to specialized mass spectrometry assays available through select research laboratories. This creates a diagnostic challenge and reinforces the need for clinicians to obtain a detailed substance use history, including inquiry about online vendors, "research chemicals," and products advertised as "extracts," "enhanced," or "potent kratom derivatives."

The growing online presence of MP, often marketed explicitly for its strong psychoactive and opioid-like effects, raises substantial public health and harm-reduction concerns. Some users intentionally seek more potent alternatives to kratom, while others may unknowingly consume MP in adulterated or mislabeled products. Early recognition of MP-related withdrawal is, therefore, essential to prevent unnecessary diagnostic workups, avoid misclassification as primary opioid use disorder, and guide appropriate supportive care.

The regulatory landscape surrounding kratom alkaloids and their derivatives is rapidly evolving. In July 2025, the FDA recommended federal scheduling of 7-OHMG due to concerns regarding potency and opioid-like pharmacology. While MP and related semisynthetic derivatives are not uniformly scheduled, increasing regulatory scrutiny of high-potency kratom constituents may alter product availability and marketplace

behavior. Awareness of this shifting context is important, as changes in regulation can influence formulation practices, adulteration patterns, and patient exposure risk.

Clinicians should incorporate MP and other semisynthetic kratom derivatives into the differential diagnosis when encountering patients with early-onset, intense opioid-like withdrawal despite reporting only "kratom" use. These compounds are not detected on routine toxicology screens; confirmatory testing typically requires specialized mass spectrometry-based assays that are not widely accessible, making clinical recognition essential. Related MG analogs may present with similar intoxication and withdrawal syndromes. Awareness of these emerging products may help avert iatrogenic complications and ensure accurate, mechanism-informed management. Reporting suspected cases to poison centers, public health entities, or surveillance programs is critical to improving detection and monitoring of this evolving class of psychoactive substances.

An additional clinically relevant observation in this case was the absence of euphoria when the patient ingested three 20 mg MP tablets approximately two weeks after receiving extended-release naltrexone. Despite MP's potent MOR agonism, the lack of subjective opioid effects is pharmacologically plausible. Naltrexone exhibits exceptionally high affinity for the MOR, with reported Ki values in the subnanomolar range (approximately 0.11 nM), indicating very tight receptor binding and slow dissociation [9]. As a competitive antagonist, naltrexone can, therefore, maintain substantial receptor occupancy and prevent agonist-mediated signaling even when potent opioids are subsequently introduced. The patient's experience suggests that extended-release naltrexone may effectively block the reinforcing effects of MP in a manner analogous to other opioid agonists. Although this observation derives from a single case and the precise receptor pharmacodynamics of MP in humans remain incompletely characterized, it raises the possibility that antagonist therapy may represent a viable relapse-prevention strategy for individuals dependent on high-potency kratom derivatives.

However, this pharmacologic property also carries clinical risk. Because MP acts as a potent MOR agonist, administration of naltrexone in an individual actively using MP could precipitate severe withdrawal. As with other opioids, initiation of antagonist therapy should, therefore, only occur after a confirmed opioid-free interval and careful clinical assessment to minimize the risk of precipitated withdrawal.

Conclusions

MP represents a highly potent, semisynthetic kratom derivative whose clinical effects and withdrawal profile differ substantially from traditional botanical kratom. As availability expands through both online vendors and brick-and-mortar retail settings (e.g., smoke shops, vape shops, and gas stations), intentional use appears to be increasing, and some individuals may unknowingly encounter MP in adulterated or mislabeled products. Its rapid-onset, full-spectrum opioid-like withdrawal highlights the need for clinicians to recognize MP's unique pharmacology, the limitations of standard toxicology testing, and the evolving landscape of manipulated "kratom" products. Incorporating MP into the differential diagnosis for severe or early-onset withdrawal in patients reporting kratom use is essential. Continued case reporting will help inform clinical practice, enhance surveillance of emerging products, and guide harm-reduction and public health responses.

Additional Information

Author Contributions

All authors have reviewed the final version to be published and agreed to be accountable for all aspects of the work.

Concept and design: Corneliu N. Stanciu

Acquisition, analysis, or interpretation of data: Corneliu N. Stanciu

Drafting of the manuscript: Corneliu N. Stanciu

Critical review of the manuscript for important intellectual content: Corneliu N. Stanciu

Disclosures

Human subjects: Informed consent for treatment and open access publication was obtained or waived by all participants in this study. **Conflicts of interest:** In compliance with the ICMJE uniform disclosure form, all authors declare the following: **Payment/services info:** All authors have declared that no financial support was received from any organization for the submitted work. **Financial relationships:** All authors have declared that they have no financial relationships at present or within the previous three years with any organizations that might have an interest in the submitted work. **Other relationships:** All authors have declared that there are no other relationships or activities that could appear to have influenced the submitted work.

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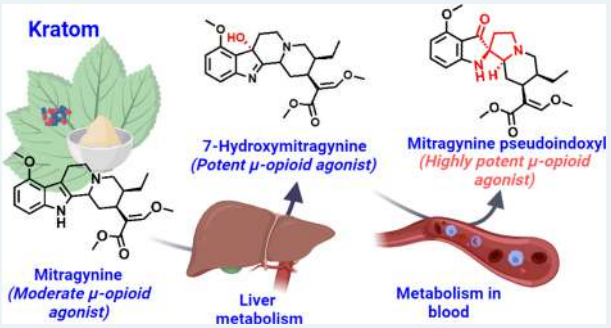
Metabolism of a Kratom Alkaloid Metabolite in Human Plasma Increases Its Opioid Potency and Efficacy

Shyam H. Kamble, Francisco León, Tamara I. King, Erin C. Berthold, Carolina Lopera-Londoño, Kanumuri Siva Rama Raju, Aidan J. Hampson, Abhisheak Sharma, Bonnie A. Avery, Lance R. McMahon, and Christopher R. McCurdy*

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ABSTRACT: Kratom is widely consumed in the United States for self-treatment of pain and opioid withdrawal symptoms. Mitragynine is the most abundant alkaloid in kratom and is a μ-opioid receptor agonist. 7-Hydroxymitragynine (7-HMG) is a mitragynine metabolite that is a more potent and efficacious opioid than its parent mitragynine. 7-HMG contributes to mitragynine's antinociceptive effects in mice, but evidence suggests it may also have a higher abuse potential. This *in vitro* study demonstrates that 7-HMG is stable in rodent and monkey plasma but is unstable in human plasma. Surprisingly, in human plasma 7-HMG is converted to mitragynine pseudoindoxyl, an opioid that is even more potent than either mitragynine or 7-HMG. This novel metabolite is formed in human plasma to a much greater extent than in the preclinical species tested (mouse, rat, dog, and cynomolgus monkey) and due to its μ-opioid potency may substantially contribute to the pharmacology of kratom in humans to a greater extent than in other tested species.



KEYWORDS: *mitragynine, 7-hydroxymitragynine, Mitragyna speciosa, mitragynine pseudoindoxyl, plasma stability, metabolism*

Mitragynine is the most abundant and pharmacologically well-characterized alkaloid (~ 66% of total alkaloidal content)[1,2] contained in the psychoactive substance kratom (*Mitragyna speciosa* Korth.). Mitragynine interacts with several receptors[3] including weak activity as a μ-opioid receptor agonist.[4] 7-Hydroxymitragynine (7-HMG) is a major mitragynine metabolite that in rodent models significantly contributes to mitragynine's analgesic and behavioral pharmacology.[5] 7-HMG is 22-fold more potent than mitragynine as a μ-opioid receptor agonist,[6] and in the electrically stimulated guinea pig ileum twitch assay, it is 13- fold more potent than morphine (46-fold greater than mitragynine).[7] In mice, mitragynine's antinociceptive properties can mostly be largely accounted for by its conversion to 7-HMG,[8] and while rats do not self-administer mitragynine, they will self-administer 7-HMG.[9] If rodent and human mitragynine metabolism is similar, then these data would suggest that 7-HMG may have a significantly higher abuse potential than that of mitragynine. However, different species often show significant quantitative and/or qualitative differences in drug metabolism, so following mitragynine/kratom consumption human exposure to 7-HMG may differ from that of rodents. One example of the differences between species is how rodents predominantly metabolize morphine to its inactive morphine 3-*O*-glucuronide, while humans also form morphine 6-*O*-glucuronide, a metabolite that is a more potent opioid agonist than its parent morphine.[10]

The current study aims to increase understanding of human systemic exposure to opioidergic mitragynine metabolites by comparing the plasma stability of 7-HMG in human and model animal species and elucidating the structures and activity resulting from 7-HMG metabolism.

RESULT AND DISCUSSION

Plasma Stability of 7-HMG. To determine which nonhuman animal model best predicts human exposure to 7-HMG, we evaluated the stability of 7-HMG in pooled mouse, rat, dog, monkey, and human plasma (Figure 1) under standard *in vitro* conditions (incubation of 1 μM 7-HMG in respective species plasma for 120 min at 37 °C). 7-HMG stability varied markedly across species, with high stability observed in mouse, rat, and monkey plasma (>80% 7-HMG

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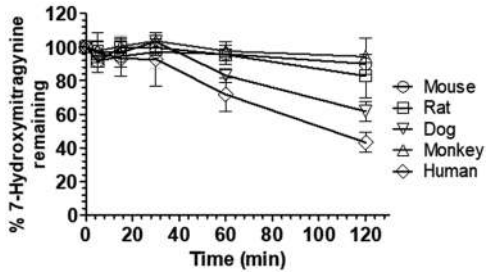


Figure 1. Stability of 7-HMG in mouse, rat, dog, cynomolgus monkey, and human plasma. The data represented as a percentage of 7-HMG remaining versus time in cross-species plasma). All values are plotted as mean ± SD (n = 3).

remaining after 120 min incubation) and intermediate stability in dog plasma (>61% 7-HMG remaining after 120 min). 7-HMG was the least stable in human plasma (~40% 7-HMG remaining after 120 min) with a half-life ($T_{1/2}$) of 98.7 min as shown in Figure 1.

Plasma is known to catalyze numerous hydrolytic drug metabolizing reactions, particularly ester cleavage, such as the β-methoxyacrylate moiety found in 7-HMG. Therefore, we next evaluated the stability of 7-HMG in human plasma in the presence and absence of a protease inhibitor cocktail comprised of 4-(2-aminoethyl)benzenesulfonyl fluoride hydrochloride, aprotinin, bestatin, E-64, EDTA, leupeptin, and pepstatin A. Together this cocktail is sufficient to inhibit most serine proteases, cysteine proteases, metalloproteases, and calpain proteases. The chemical degradation of 7-HMG was also assessed in a 50 mM phosphate buffer (pH 7.4) as a control. In phosphate buffer alone, 7-HMG was stable, and in human plasma, the protease inhibitors somewhat reduced the rate of 7-HMG degradation (68% remaining at 120 min vs 46% in untreated human plasma; Figure 2). These data

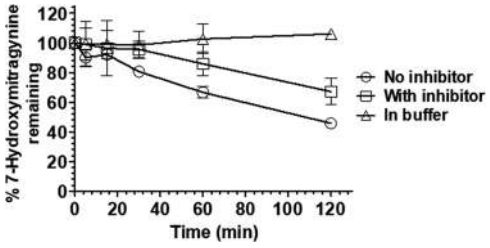


Figure 2. Stability of 7-HMG in human plasma in the presence and absence of protease cocktail inhibitor and 50 mM phosphate buffer pH 7.4 control. The data are represented as a percentage of 7-HMG remaining versus time in either human plasma with or without protease cocktail inhibitor treatment and blank phosphate buffer pH 7.4. All values are plotted as mean ± SD (n = 3).

suggested that 7-HMG was being enzymatically degraded. However, the liquid chromatography high-resolution mass spectrometry (LC-HRMS) analysis failed to identify a plasma metabolite that corresponds to a 7-HMG hydrolysis product with an expected m/z of 401.2071 (Figure S1).

Following incubation of 7-HMG with human plasma, two unknown peaks (unknown-1 and -2) with the same nominal mass (m/z 415.2227) and same daughter ions peaks (m/z 190.0863 and 226.1438) were observed. The abundance of these metabolites increased in a manner proportional to the incubation time of 7-HMG in human plasma (Figure 3). Relative to the conversion in human plasma, the percent

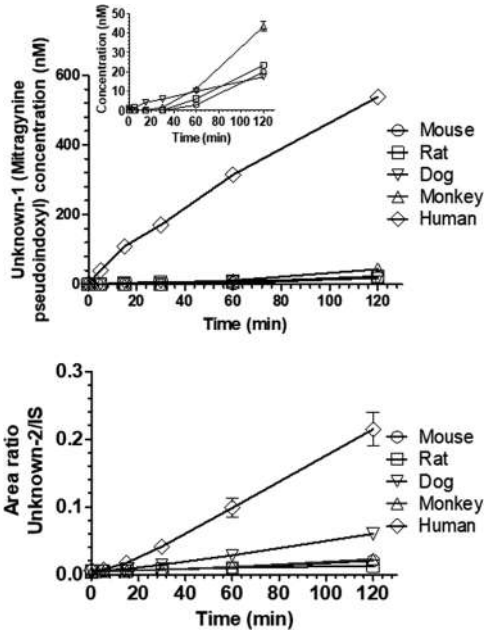


Figure 3. Unknown-1 (mitragynine pseudoindoxyl) and unknown-2 formation upon 7-HMG incubation with the mouse, rat, dog, cynomolgus monkey, and human plasma. Two unknown peaks (Unknown-1 and -2) were detected having mass transitions similar to that of 7-HMG. These are represented as the concentration of unknown-1 (mitragynine pseudoindoxyl, top) and ratio of unknown-2 to internal standard (IS) peak area (bottom) versus time. The inset in the top panel represents the concentration of unknown-1 (mitragynine pseudoindoxyl) in species other than human. All values are plotted as mean ± SD (n = 3).

conversion values of 7-HMG to unknown-1 in mouse, rat, dog, and monkey plasma were 3.7 ± 0.0, 4.4 ± 0.3, 3.2 ± 0.0, and 8 ± 0.0% (mean ± SD), respectively.

Similarly, relative to the conversion in human plasma, the percent conversion values of 7-HMG to unknown-2 in mouse, rat, and monkey plasma were 9.5 ± 0.7, 5.9 ± 0.3, and 10.6 ± 1%, respectively, and 28.0 ± 2.1% in dog plasma. These data demonstrate that human plasma predominantly metabolized 7-HMG to unknown-1. The degradation of 7-HMG in dog plasma to unknown-1 and -2 explains somewhat the contribution toward the total degradation, while other degradation products in dog plasma remain to be identified. Likewise, whether the conversion of 7-HMG in human plasma was exclusively to unknown-1 and -2 remains to be identified. A similar MS2 fragmentation pattern (Figure S2) was observed for unknowns-1 and -2 and 7-HMG suggesting the same molecular structure, i.e., unknown--1 and -2 are structural isomers of 7-HMG.

Isolation, Purification, and Structure Elucidation of Unknown-1 Metabolite. Multiple incubations of 7-HMG with human plasma were necessary to obtain a sufficient amount of unknown-1 for analysis using fractionation, preparative thin-layer chromatography (TLC), and high-performance liquid chromatography (HPLC) techniques. The relative abundance of unknown-2 was almost 10-fold lower than that of unknown-1 in human plasma, so it was not further evaluated due to the challenges in obtaining enough compound for further characterization. The ^{1}H NMR spectrum of unknown-1 (Figure S3) showed an unusual aromatic first-order (large chemical shift separations and clear

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coupling pattern) spin-system AMX, where the aromatic protons at δ_H 6.42 ppm (d, J = 8.1 Hz, 1H) and 6.15 ppm (d, J = 8.0 Hz, 1H) were low-shifted in comparison with those of 7-HMG at δ_H 7.20 ppm (d, J = 7.8 Hz, 1H), and 6.73 ppm (d, J = 7.8 Hz, 1H). Additional ^{1}H NMR signals of unknown-1 revealed the presence of one aromatic methoxy substituent, the typical signals for the β-methoxy acrylate group, and aliphatic protons, similar to those for 7-HMG (Figure S4), suggesting that the structural changes are located in the ring B of the corynanthe skeleton (Figure S4). The main differences between unknown-1 and 7-HMG were the absence of a quaternary carbon on 7-HMG at δ_C 81.0 ppm (Figure S5) and the presence of the highly deshielded carbon on unknown-1 at δ_C 199.5 ppm (Figure S6), suggesting conversion of the quaternary hydroxylated carbon to a carbonyl carbon. The presence of the quaternary carbon at δ_C 75.0 ppm is consistent with a ring B rearrangement, resulting in a spiro linkage between rings B and C of the corynanthe core (Figure S4). Analysis of the ^{1}H−^{1}H correlation spectroscopy (COSY), heteronuclear single quantum coherence spectroscopy (HSQC), heteronuclear multiple bond coherence (HMBC), and selective nuclear Overhauser effect (NOE) data (Figures S7−S10) confirmed the difference between ring B of unknown-1 and 7-HMG. Taken together, these data suggest a semipinacol−pinacolone rearrangement, similar to the biosynthetic pathway for oxindoles on corynanthe skeletons.[11] By comparing the NMR data for unknown-1 with those of reported corynanthe alkaloids, we concluded that unknown-1 was likely to be mitragynine pseudoindoxyl. We then synthesized a mitragynine pseudoindoxyl standard using the Varadi et al. procedure[12] and demonstrated that the standard and unknown-1 exhibited identical NMR spectra, UPLC (retention times), and HRMS data (Figures S11−S13). Therefore, we conclude that unknown-1 is mitragynine pseudoindoxyl (Figure 4). Furthermore, we used the synthetic

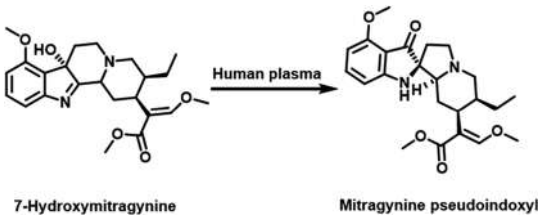


Figure 4. Novel metabolic conversion of 7-HMG to mitragynine pseudoindoxyl in human plasma. The incubation of 7-HMG in human plasma at 37 °C in 5% CO_2 incubator at 100 rpm for 2 h resulted in the metabolic conversion of 7-HMG to mitragynine pseudoindoxyl.

standard of mitragynine pseudoindoxyl to quantify the percentage of mitragynine pseudoindoxyl formed via the plasma incubation of 7-HMG. The percentages of mitragynine pseudoindoxyl formed at the end of 120 min of incubation of 7-HMG in mouse, rat, dog, monkey, and human plasma were 2.0 ± 0.0, 2.4 ± 0.2, 1.7 ± 0.0, 4.3 ± 0.2, and 53.8 ± 1.6%, respectively. Also, we evaluated the stability of mitragynine pseudoindoxyl in the plasma of each tested species using experimental conditions similar to those used for 7-HMG. Mitragynine pseudoindoxyl was found to be stable in the plasma of each species, confirming that the conversion of 7-HMG to mitragynine pseudoindoxyl is irreversible (Figure S14).

Plasma enzymes are known to catalyze hydrolytic and chiral inversion reactions, but to the best of our knowledge semipinacol rearrangement reactions such as those observed here have not been previously described to occur in mammalian plasma. Furthermore, mitragynine pseudoindoxyl biosynthesis has not even been reported in *Mitragyna speciosa*,[1] although Zarembo et al. have reported that some fungi can catalyze the semipinacol rearrangement of mitragynine into mitragynine pseudoindoxyl.[13]

The discovery that 7-HMG is converted in mitragynine pseudoindoxyl in human but not rodent plasma not only is unexpected but also may significantly alter our conclusions about the translatability of mitragynine findings using rodent models to the human condition.[8,9,14] While 7-HMG is more potent and efficacious than mitragynine in its ability to activate μ-opioid receptors, mitragynine pseudoindoxyl is yet more potent and at least as efficacious as 7-HMG. In GTPγS functional assays of μ-opioid activation, mitragynine pseudoindoxyl has been reported to be 31-fold more potent than 7-HMG and 119-fold greater than mitragynine (EC_{50} 1.7 ± 0.1 nM, E_{max} 84 ± 5%).[12] In the electrically stimulated contraction of guinea pig ileum model, mitragynine pseudoindoxyl is 20-fold more potent than morphine at inhibiting contractions and 100-fold more potent than mitragynine.[15] Previously, questions have been raised about whether the observed pharmacological activity in humans following kratom administration is due to mitragynine or mediated by the metabolite, 7-HMG. The systemic exposure of 7-HMG when calculated as percentage ratio of area under the curve of 7-HMG to mitragynine following mitragynine oral dosing of female beagle dogs[16] was found to be 12.6 ± 1.6%. This suggests that a reasonable amount of 7-HMG is present in systemic circulation which could be further metabolized to mitragynine pseudoindoxyl in plasma; however, the absolute conversion rate of 7-HMG to mitragynine pseudoindoxyl is unknown. These findings raise the possibility that opioid-like effects may be mediated by a combination of mitragynine, 7-HMG, and mitragynine pseudoindoxyl. The abuse potential of mitragynine pseudoindoxyl has not been well characterized. However, 7-HMG has been reported to have high abuse potential, and given that mitragynine pseudoindoxyl is an even more potent opioid agonist, it is likely to also be highly abusable.[9]

The metabolism of mitragynine to 7-HMG in humans is known to be catalyzed by the enzyme CYP3A4,[2,8] but the mechanism of this newly discovered conversion into mitragynine pseudoindoxyl in plasma is unknown. Additionally, microsomal metabolism of mitragynine did not result in the formation of mitragynine pseudoindoxyl in mouse, rat, dog, monkey, or human liver microsomes. It is possible that this plasma metabolism may be catalyzed by a genetically polymorphic enzyme or enzymes, which would potentially lead to substantial variability in an individual's relative exposure to 7-HMG and mitragynine pseudoindoxyl following administration of a defined kratom dose. Given the difference in potency between these two metabolites, such differential exposure could result in population variability in the pharmacological and adverse effects of kratom consumption. In the present study, conversion of 7-HMG to mitragynine pseudoindoxyl has been identified, but the extent of systemic exposure will depend on the rates of 7-HMG biosynthesis and its further conversion to mitragynine pseudoindoxyl, which remain to be defined. Assuming similar rates of 7-HMG formation across species, our *in vitro* plasma stability data

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suggest that systemic human exposure to 7-HMG in humans may be lower than that in nonhuman species tested.

7-HMG is substantially metabolized by rat and monkey liver microsomes ($T_{1/2}$ = 15 and 26 min, respectively, Figure S15), but metabolism in mouse, dog, and human liver microsomes was much slower ($T_{1/2}$ > 60 min, Figure S15). These systems only establish the phase I metabolism of 7-HMG, but the data suggest that liver metabolism of 7-HMG in rats and monkeys is much greater than that in plasma. Furthermore, comparing the 7-HMG rate of metabolism in humans (plasma and liver microsomes), the plasma metabolism of 7-HMG could form substantial amounts of mitragynine pseudoindoxyl. Studies comparing the intrinsic clearance of 7-HMG in plasma and hepatocytes (phases I and II metabolic systems present) would help us to gauge the relevance of plasma metabolism of 7-HMG in the respective species and could be used to identify species having a metabolic profile similar to humans for further translational characterization.

CONCLUSION

This study has demonstrated the potential for human plasma to form mitragynine pseudoindoxyl. Further research is required to elucidate whether the same mechanisms also reside in other tissue types, potentially resulting in significant local exposure to this highly potent opioid.

EXPERIMENTAL SECTION

Plasma Stability of 7-HMG. The plasma stability of 7-HMG (synthesized in-house from mitragynine) was performed using heparinized pooled non-Swiss albino mouse (Innovative Research, Novi, MI), Sprague−Dawley rat (GeneTex, Irvine, CA), beagle dog (BioIVT, Hicksville, NY), cynomolgus monkey (BioIVT, Hicksville, NY), and human plasma (with sodium citrate as an anticoagulant; Innovative Research, Novi, MI). 7-HMG was spiked in a respective species preincubated plasma sample (n = 3) at a final concentration of 1 μM and kept in a 5% CO_2 incubator shaker (100 rpm) maintained at 37 °C. The concentration of organic solvents was kept below 0.5% (v/v).[17,18] The aliquots (25 μL) were withdrawn at 0, 5, 15, 30, 60, and 120 min. The reaction was quenched by mixing the aliquots with 4 volumes acetonitrile containing 10 ng/mL phenacetin as IS. The quenched samples were vortex-mixed for 5 min, and these samples were then filtered through 0.45 μm pore size 96-well filtration plate. The filtrate was then subjected to ultraperformance liquid chromatography tandem mass spectrometry analysis (UPLC-MS/MS). The sample analysis was performed using a Waters Acquity UPLC system coupled with Waters Xevo TQ-S micro mass spectrometer (Waters, Milford, MA) using MassLynx V4.1 software (Waters, Milford, MA). Chromatographic separations were performed on a UPLC column Acquity, BEH, C18, 1.7 μm, 2.1 × 50 mm; maintained at 50 °C temperature. The mobile phase consisted of solvent A (10 mM ammonium acetate with 1% acetic acid in water) and solvent B (acetonitrile) delivered at a flow rate of 0.35 mL/min with a linear gradient as follows: 25% B until 3 min, 25−40% B over 3.9 min, 40−70% B from 3.9 to 4 min, decreased to 25% B from 4 to 4.6 min, and maintained at 25% B until 5.5 min. The MS was operated in the positive electrospray mode at a capillary temperature of 450 °C, source temperature of 150 °C, capillary voltage of 0.5 kV, desolvation gas flow of 900 L/h, and cone gas flow of 50 L/h. The analysis was conducted in multiple reaction monitoring modes (MRM, m/z 415.2 > 190.1). Enalapril for rodents and procaine or disulfiram for nonrodent plasma were used as positive controls to assess the enzyme activity. The positive control results were comparable to published reports.[19,20]

Human Plasma Stability of 7-HMG with or without Protease Inhibitors. 7-HMG at 1 μM was incubated with or without a protease inhibitor cocktail treatment and in 50 mM phosphate buffer pH 7.4 for 120 min at 37 °C in a CO_2 incubator. Aliquots (25 μL) were withdrawn at 0, 5, 15, 30, 60, and 120 min and mixed with 4 volumes of acetonitrile containing IS. The quenched reaction mixtures were vortex-mixed for 5 min and then filtered through a 0.45 μm 96-well filtration plate. The filtrate was subjected to UPLC-MS/MS analysis as described above. Procaine was used as a positive control to assess esterases enzyme activity. The percentage of parent compound remaining was calculated as the percentage ratio of analyte peak to the IS peak area ratio at time t to that at 0 min.

Isolation, Purification, and Structure Elucidation of Unknown-1 Metabolite. To identify the chemical structure of the unknown-1 metabolite, multiple incubations of 7-HMG (1 mM) with undiluted human plasma (1 mL) for 24 h at 37 °C were performed to obtain enough amount of metabolite unknown-1. These incubation reactions were mixed and quenched with 4 volumes of acetonitrile. The metabolite unknown-1 was then isolated, purified using several chromatographic techniques, and chemical structure elucidation was performed by 1D and 2D NMR analyses. The details of isolation, purification, and structural elucidation of unknown-1 (mitragynine pseudoindoxyl) and its chemical synthesis are described in the Supporting Information.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsptsci.0c00075.

Experimental procedures and HPLC, NMR, and HRMS characterization data for mitragynine pseudoindoxyl (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

7-HMG, 7-hydroxymitragynine; $T_{1/2}$, half-life; LC-HRMS, liquid chromatography high-resolution mass spectrometry; TLC, thin-layer chromatography; HPLC, high-performance liquid chromatography; CYP3A4, cytochrome P450 3A4; NMR, nuclear magnetic resonance; MRM, multiple reaction monitoring mode.

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Human Mitragynine and 7-Hydroxymitragynine Pharmacokinetics after Single and Multiple Daily Doses of Oral Encapsulated Dried Kratom Leaf Powder

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Abstract: Kratom leaves, consumed by millions worldwide as tea or ground leaf powder, contain multiple alkaloids, with mitragynine being the most abundant and responsible for most effects. Mitragynine is a partial μ-opioid receptor agonist and competitive antagonist at κ- and δ-opioid receptors; however, unlike morphine, it does not activate the β-arrestin-2 respiratory depression pathway. Due to few human mitragynine data, the largest randomized, between-subject, double-blind, placebo-controlled, dose-escalation study of 500–4000 mg dried kratom leaf powder (6.65–53.2 mg mitragynine) was conducted. LC-MS/MS mitragynine and 7-hydroxymitragynine plasma concentrations were obtained after single and 15 daily doses. Mitragynine and 7-hydroxymitragynine C_{max} increased dose proportionally, and AUC was slightly more than dose proportional. The median mitragynine T_{max} was 1.0–1.3 h after single and 1.0–1.7 h after multiple doses; for 7-hydroxymitragynine T_{max}, it was 1.2–1.8 h and 1.3–2.0 h. Steady-state mitragynine concentrations were reached in 8–9 days and 7-hydroxymitragynine within 7 days. The highest mean mitragynine $T_{1/2}$ was 43.4 h after one and 67.9 h after multiple doses, and, for 7-hydroxymitragynine, it was 4.7 and 24.7 h. The mean 7-hydroxy-mitragynine/mitragynine concentration ratios were 0.20–0.31 after a single dose and decreased (0.15–0.21) after multiple doses. These mitragynine and 7-hydroxymitragynine data provide guidance for future clinical kratom dosing studies and an interpretation of clinical and forensic mitragynine and 7-hydroxymitragynine concentrations.

Keywords: mass spectrometry; metabolism; analytical toxicology; mitragynine; kratom; 7-hydroxymitragynine

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1. Introduction

The kratom (*Mitragyna speciosa* Korth.) tree is a Southeast Asian tropical evergreen in the Rubiacaeae family related to the coffee tree [1]. The oral consumption of kratom leaves dates back hundreds of years in Southeast Asia, where the leaves are typically chewed fresh, brewed into a tea, or processed into tar-like extracts [2]. Millions consume kratom worldwide as food, dried leaf powders in capsules, concocted into beverages or teas, or as processed extracts [3–5].

Multiple alkaloids are present in kratom including mitragynine, speciogynine, speciociliatine, paynantheine, and other indoles and oxindoles [6–8], with their content varying based on geographic location, seasonal variation, and plant part used [9–11]. Mitragynine (IUPAC methyl (16E)-9,17-dimethoxy-16,17-didehydro-20β-corynan-16-carboxylate) is the most abundant alkaloid present in kratom and is responsible for many of its effects, although research into the contributions of kratom's other alkaloids is increasing [12–15]. Mitragynine is metabolized to 7-hydroxymitragynine (IUPAC methyl (2E)-2-[(2S,3S,7aS,12bS)-3-ethyl-7a-hydroxy-8- methoxy-1,2,3,4,6,7,7a,12b-octahydroindolo[2,3-a]quinolizin-2-yl]-3-methoxyprop-2-enoate), which is not found in measurable amounts in fresh kratom leaves [10,16]. Chemical structures

for both mitragynine and its metabolite 7- hydroxymitragynine are shown in Figure 1. Mitragynine and 7-hydroxymitragynine are partial agonists of the human μ-opioid receptor and competitive antagonists at κ- and δ-opioid receptors [17]. Mitragynine has a lower affinity and potency than morphine at the μ-opioid receptor and is unable to induce comparable phosphorylation and GTPγS stimulation [18]. Unlike full μ-opioid agonists, such as morphine and fentanyl, mitragynine does not activate the β-arrestin-2 pathway implicated in the adverse effects of μ-opioid agonists including respiratory depression and constipation [19,20]. Reduced respiratory depression of mitragynine compared to codeine was reported more than 50 years ago [14].

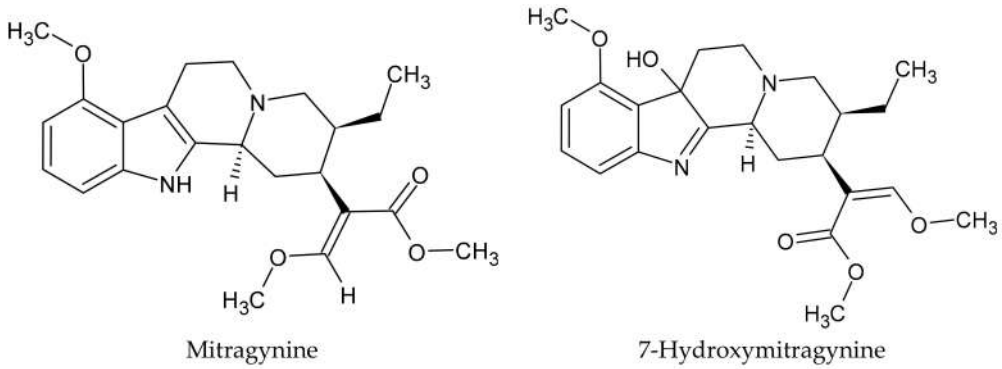


Figure 1. Chemical structures of mitragynine and 7-hydroxy-mitragynine.

Furthermore, mitragynine and 7-hydroxymitragynine are also bound to adrenergic (α_1 and α_2) and serotonergic (5-HT_{1A} and 5-HT_{2B}) receptors, producing a different profile of effects than prototypical opioids [16,21–23]. Currently, based on this unique pharmacology, there is substantial interest in the development of synthetic medicinal compounds built on the chemical scaffolding of kratom alkaloids for analgesia, namely, for the treatment of opioid withdrawal and opioid use disorder [17,19,24].

Despite the widespread availability and use of kratom, there are few controlled administration studies of mitragynine and 7-hydroxymitragynine exposure from well-characterized kratom products. In the only controlled kratom administration study prior to 2022, plasma mitragynine concentrations were determined in 10 chronic male kratom users after drinking different low kratom tea doses for 7 days [25]. More recently, mitragynine and 7-hydroxymitragynine pharmacokinetics were reported in five participants over 120 h after a single 39 mg mitragynine dose in 2 g kratom tea [26]. These authors also investigated the drug interactions of the same single kratom dose to 12 subjects [27].

Due to the dearth of controlled mitragynine and 7-hydroxymitragynine concentration data and the increasing intake of kratom around the world, pharmacokinetic data are needed for designing future kratom studies and for interpreting mitragynine and 7-hydroxymitragynine concentrations in clinical and forensic investigations. The primary objective of this study was to determine the pharmacokinetics of single doses (SDs), and, for the first time, 15 consecutive daily oral ascending doses (MDs) of dried kratom leaf powder in healthy adults. Pharmacokinetic data obtained after single and multiple dosing included the following: the time course of mitragynine and 7-hydroxymitragynine concentrations over 10 days after a single dose and 23 days after multiple dosing, dose-proportionality, times to reach steady state, half-lives and 7-hydroxy-mitragynine/mitragynine ratios.

2. Results

2.1. Participants

Twelve participants received active leaf powder (except 13 for the highest dose, due to one participant being replaced). A minimum of one SD or one MD pharmacokinetic parameter, in most cases, the C_{max}, was required for inclusion in the SD or MD pharmacokinetic dataset. If participants voluntarily withdrew or were withdrawn from the study due to

adverse events, or failed to follow study protocols and were withdrawn during the SD or MD phases of the study, they did not have MD pharmacokinetic data. The pharmacokinetic study for MD occurred after the last MD, resulting in a smaller number of subjects with MD pharmacokinetic data. Table 1 includes participants' demographic data.

Table 1. Participants' demographic data.

Mitragynine Dried Kratom Leaf Powder	**6.65 mg 500 mg**		**13.3 mg 1000 mg**		**26.6 mg 2000 mg**		**53.2 mg 4000 mg**	
	SD	**MD**	**SD**	**MD**	**SD**	**MD**	**SD**	**MD**
# Subjects	12	11	12	8	12	9	13	9
Gender	6 F 6 M	6 F 5 M	4 F 8 M	3 F 5 M	4 F 8 M	3 F 6 M	7 F 6 M	4 F 5 M
Race								
AI	0	0	0	0	0	0	2	2
Asian	2	2	5	3	1	0	0	0
Black	0	0	0	0	4	2	1	1
Unknown	0	0	0	0	0	0	1	1
White	10	9	7	5	7	7	9	5
Ethnicity	2 H 10 NH	2 H 9 NH	1 H 11 NH	1 H 7 NH	0 H 12 NH	0 H 9 NH	3 H 10 NH	3 H 6 NH
Mean age (STD) years	30.8 (9.3)	31.3 (9.5)	32.9 (9.4)	33.1 (9.8)	35.8 (11.4)	38.4 (12.2)	33.2 (9.6)	32.4 (7.8)
BMI Kg/m^2	23.1 (2.7)	22.8 (2.6)	25.3 (2.6)	25.3 (2.7)	26.2 (3.4)	25.7 (3.8)	23.4 (2.9)	23.9 (3.1)

SD, single-dose phase; MD, multiple-dose phase; AI, American Indian or Alaska Native; H, Hispanic; NH, not Hispanic; STD, standard deviation; BMI, body mass index.

2.2. Bioanalysis

The analytical method for mitragynine and 7-hydroxymitragynine was fully validated according to the FDA Guidance for Industry (May 2018) and EMA Guideline on Bioanalytical Method Validation EMEA/CHMP/EWP/192217/2009 Rev.1 Corr. 2, effective February 2012. Blood was collected with K_2-EDTA anticoagulant and drawn at specified time points before centrifuging and removing plasma within 2 h. The sample pretreatment consisted of protein precipitation of 100 µL of plasma, and the internal standards were mitragynine-D3 and 7-hydroxymitragynine-D3. Mitragynine and 7-hydroxymitragynine were identified and quantified using reversed-phase ultra-high-performance liquid chromatography/tandem mass spectrometry with MS/MS detection. Carryover was not observed at the highest calibrator concentration. Table 2 includes the most important validation data for mitragynine and 7-hydroxymitragynine. Figure 2 includes LC-MS/MS chromatograms of a blank, the lower limit of quantification, and an authentic plasma sample collected at 1.33 h after one participant received 13.3 mg mitragynine in 1000 mg of dried kratom leaf, achieving 46 ng/mL mitragynine and 7.3 ng/mL 7-hydroxymitragynine plasma concentrations.

Table 2. Validation data for the LC-MS/MS quantification of mitragynine and 7-hydroxymiragynine in human plasma.

Validation Parameter	**Mitragynine**	**7-Hydroxymitragynine**
Linearity	0.5–500 ng/mL	0.5–100 ng/mL
Lower limit of quantification	0.5 ng/mL	0.5 ng/mL
Quality control (QC) concentrations	0.5, 1.5, 250, and 375 ng/mL	0.5, 1.5, 50, and 75 ng/mL
Within-run accuracy (% bias)	−11.7–11.2%	−9.7–13.6%

Table 2. *Cont.*

Validation Parameter	Mitragynine	7-Hydroxymitragynine
Within-run precision (%CV)	1.4–10%	1.2–7.4%
Between-run accuracy (% bias)	−0.6–2.2%	2.4–10.4%
Between-run precision (%CV)	3.9–9.3%	4.2–10%
Internal standard normalized matrix factors for low and high QC	1.00–1.02 (CV 0.7–2.2%)	0.995–1.04 (CV 1.0–2.0%)
10-fold dilution integrity %bias (%CV)	−1.7% (3.0%)	−3.0 (2.8%)
Plasma stability for low and high QC		
22 °C for 23 h (% bias)	−0.6–4.4%	4.4–5.8%
−20 °C for 404 days (% bias)	6.7–9.1%	−1.5–−3.1%
−80 °C for 404 days (% bias)	8.4–8.9%	9.2–11.6%
4 freeze-thaw cycles (% bias)	0.4–5.3%	4.6–10.3%
Autosampler stability 4 °C 241 h (% bias)	−2.9–0.1%	−2.9–−3.8%

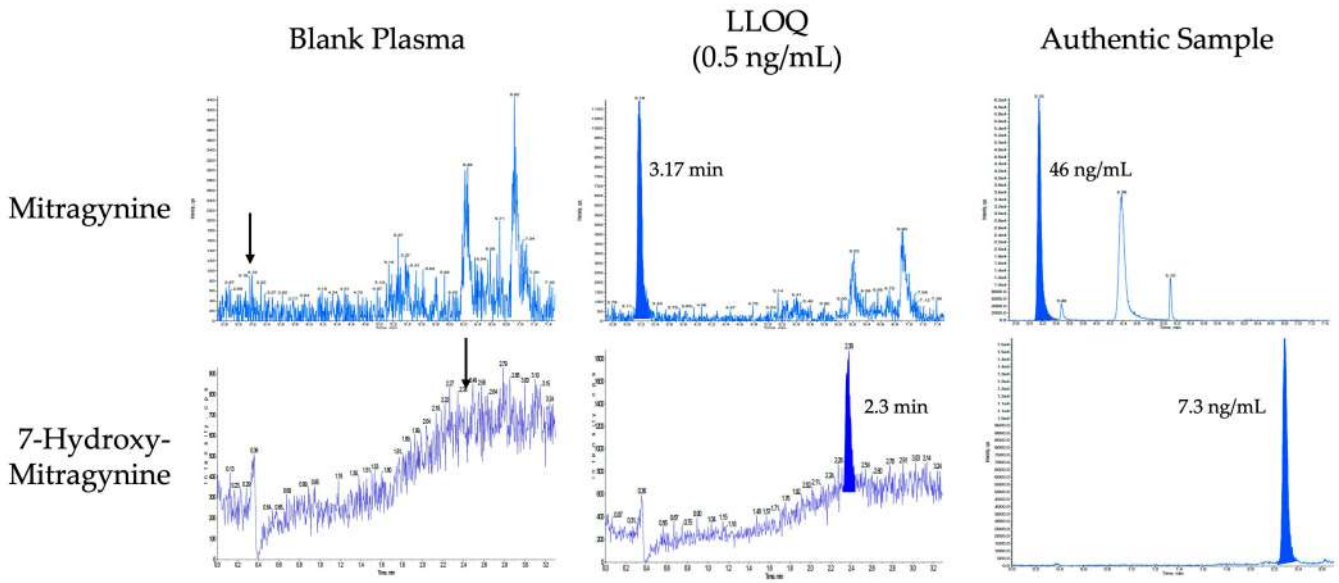


Figure 2. LC-MS/MS chromatograms of blank plasma, plasma fortified at the lower limit of quantification (0.5 ng/mL) for mitragynine and 7-hydroxymitragynine and an authentic plasma sample collected at 1.33 h after the participant received 13.3 mg of mitragynine in 1000 mg of dried kratom leaf, achieving plasma concentrations of 46 ng/mL of mitragynine and 7.3 ng/mL of 7-hydroxymitragynine. Arrows in the blank chromatogram indicate retention time of mitragynine and 7-hydroxymitragynine.

2.3. Pharmacokinetics

Figure 3 illustrates the mean ± standard deviation plasma concentration profiles of mitragynine and 7-hydroxymitragynine after SD and MD oral 500, 1000, 2000, and 4000 mg encapsulated dried kratom leaf powder doses containing 6.65, 13.3, 26.6, and 53.2 mg of mitragynine, respectively.

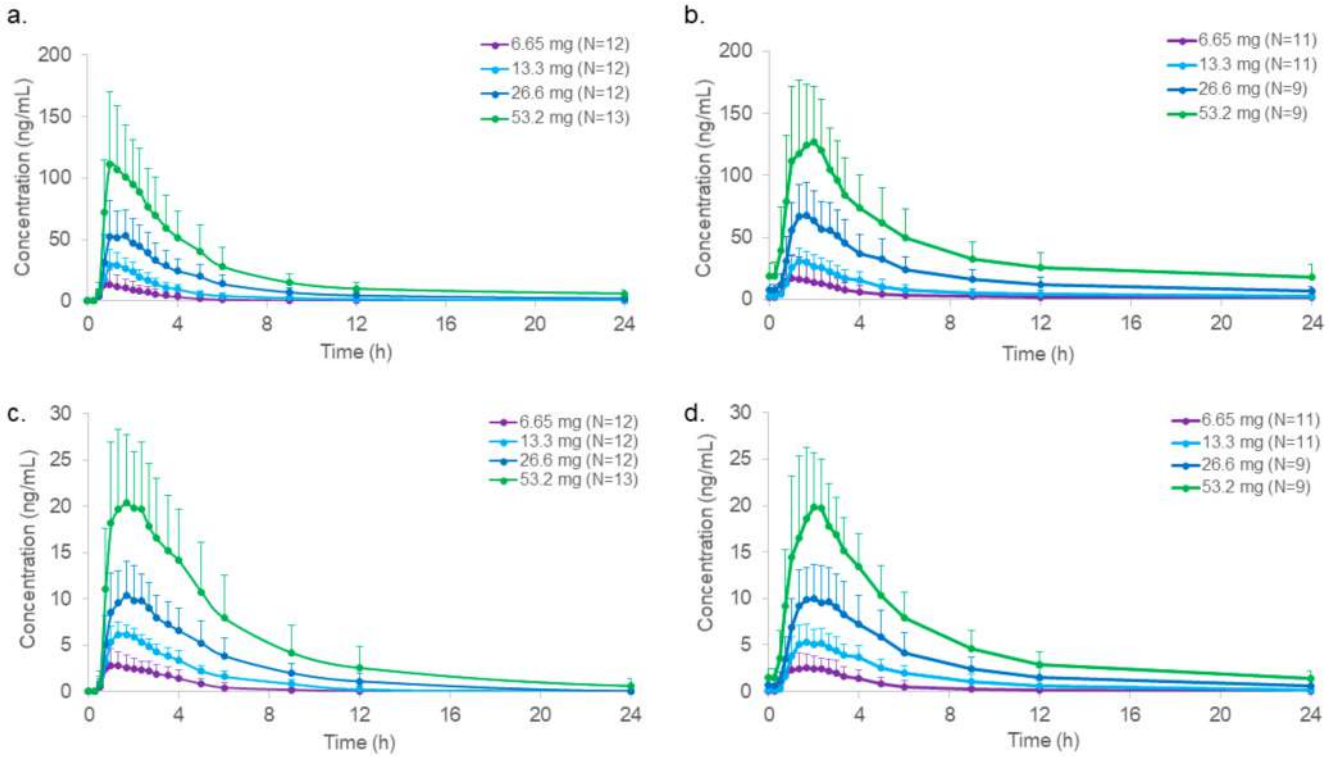


Figure 3. Mean plasma concentration profiles of (**a**) mitragynine (including standard deviation bars) after a single dose (SD); (**b**) mitragynine after 15 multiple doses (MD); (**c**) 7-hydroxymitragynine after an SD; (**d**) 7-hydroxymitragynine after 15 MD mitragynine for all four dried kratom leaf powder doses. Each dose was administered to a different cohort of healthy male and female subjects.

Table 3 contains mean (standard deviation) and median (range) plasma mitragynine pharmacokinetic parameters after SD and MD kratom leaf powder. C_{max} and AUC increased with each SD and MD escalation. The SD median T_{max} was fairly consistent from 1.0 to 1.3 h over these doses and similar to mitragynine MD 1.0–1.7 h T_{max}. The median terminal half-life and time to last quantifiable concentration generally increased with the dose since plasma concentrations were detectable over a longer period. The highest mitragynine median $T_{1/2}$ was 42.9 h after the highest 53.2 mg SD, and 61.2 h after the 26.2 mg MD. Fluctuation of the mitragynine concentrations across the dosing interval after MD at steady state ranged from 3.3 to 5.6, with lower values observed at higher doses. Dose proportionality of mitragynine was demonstrated based on C_{max} and $C_{max,ss}$ during SD and MD. However, $AUC_{0\text{-}Tlast}$ and $AUC_{0\text{-}tau,ss}$ did not fulfill the proportionality criterion, with increases slightly higher (1.42 SD and 1.33 MD) than dose proportionality predicted.

Table 3. Mean (standard deviation) and median (range) plasma pharmacokinetic parameters for mitragynine after administration of increasing single (SD) and 15 multiple doses (MD) of mitragynine in dried kratom leaf powder.

		Cohort 1 6.65 mg	**Cohort 2 13.3 mg**	**Cohort 3 26.6 mg**	**Cohort 4 53.2 mg**
Single dose		*n* = 12	*n* = 12	*n* = 12	*n* = 13
C_{max} (ng/mL)	mean (STD) median (range)	17.1 (15.2) 12.1 (1.8–61.5)	32.1 (12.2) 29.0 (16.0–58.1)	57.0 (25.1) 57.4 (20.4–90.0)	125 (51.8) 130 (34.2–204)
T_{max} (h)	median (range)	1.0 (0.8–2.7)	1.3 (0.8–1.7)	1.3 (0.8–5.0)	1.3 (0.8–2.0)

Table 3. *Cont*.

		Cohort 1 6.65 mg	**Cohort 2 13.3 mg**	**Cohort 3 26.6 mg**	**Cohort 4 53.2 mg**
T_{last} (h)	median (range)	12.0 (4.0–71.0)	23.4 (12.0–72.2)	71.6 (0.8–120)	120 (23.7–169)
$AUC_{0\text{-}24}$ (h*ng/mL)	mean (STD) median (range)	41.4 (32.4) 30.6 (3.6–111)	108 (40.6) 101 (52.8–183)	244 (121) 280 (3.3–375)	558 (250) 648 (155–970)
$AUC_{0\text{-}Tlast}$ (h*ng/mL)	mean (STD) median (range)	45.8 (42.2) 30.6 (3.6–137)	120 (57.7) 101 (52.8–237)	305 (164) 357 (3.3–480)	837 (466) 839 (179–1746)
$AUC_{0\text{-}inf}$ (h*ng/mL)	mean (STD) median (range)	52.8 (51.1) 34.8 (4.7–162)	136 (72.3) 108 (58.1–288)	364 [1] (154) 417 [1] (98.4–517)	908 (520) 903 (191–1951)
$AUC_{0\text{-}Tlast}$/ $AUC_{0\text{-}inf}$	median (range)	0.91 (0.74–0.96)	0.92 (0.76–0.95)	0.92 (0.87–0.95)	0.93 (0.88–0.96)
$T_{1/2}$ (h)	mean (STD) median (range)	8.5 (11.8) 3.7 (1.5–35.0)	15.8 (16.3) 10.5 (2.4–53.9)	29.3 [1] (17.5) 28.2 [1] (4.4–55.2)	43.4 (23.3) 42.9 (8.3–84.5)
CL/F (L/h)	mean (STD) median (range)	278 (369) 191 (40.9–1424)	123 (58.1) 123 (46.2–229)	102 [1] (79.8) 63.7 [1] (51.4–270)	94.0 (84.1) 58.9 (27.3–278)
Vz/F (L)	mean (STD) median (range)	1349 (808) 1017 (549–2998)	1892 (1013) 1786 (615–4438)	2875 (871) 2544 (1733–4084)	3788 (1287) 3314 (2104–6173)
Multiple doses		n = 11	n = 8	n = 9	(n = 9)
$C_{max,ss}$ (ng/mL)	mean (STD) median (range)	21.4 (18.9) 16.2 (2.8–60.1)	33.5 (7.43) 35.1 (21.8–43.1)	74.6 (22.8) 73.0 (42.4–109)	143 (50.4) 155 (64.3–215)
$T_{max,ss}$ (h)	median (range)	1.0 (0.8–2.3)	1.7 (1.0–4.0)	1.7 (1.0–3.0)	1.7 (1.0–3.0)
$T_{last,ss}$ (h)	median (range)	23.9 (4.0–240)	95.4 (23.5–239)	168 (71.5–240)	169 (72.4–383)
$AUC_{0\text{-}tau,ss}$ (h*ng/mL)	mean (STD) median (range)	85.1 (84.1) 60.1 (5.5–263)	175 (79.8) 163 (93.4–354)	457 (191) 554 (144–655)	958 (393) 1021 (227–1444)
$T_{1/2,ss}$ (h)	mean (STD) median (range)	25.7 [2] (24.0) 14.6 [2] (2.4–64.4)	44.1 (24.2) 40.6 (8.9–78.5)	67.9 [3] (20.6) 61.2 [3] (46.6–99.6)	51.1 (15.9) 48.6 (31.3–83.8)
CL/F (L/h)	mean (STD) median (range)	215 (333) 111 (25.3–1200)	87.4 (32.0) 81.5 (37.6–142)	74.6 (48.0) 48.0 (40.6–185)	74.7 (61.7) 52.1 (36.8–234)
Vz/F (L)	mean (STD) median (range)	2980 (1606) 2843 (738–5482)	4755 (2072) 4033 (1825–7872)	6020 [3] (3552) 5139 (2945–14,135)	4855 (2392) 4595 (2311–10,634)
$C_{av,ss}$ (ng/mL)	mean (STD) median (range)	3.6 (3.5) 2.5 (0.2–11.0)	7.4 (3.3) 6.9 (4.0–14.8)	19.3 (8.1) 23.4 (6.2–27.9)	39.9 (16.6) 42.9 (9.3–60.9)
C_{trough} (ng/mL)	mean (STD) median (range)	1.40 (1.79) 0.9 (0–5.3)	2.74 (1.61) 2.5 (0.7–5.8)	7.70 (3.71) 8.8 (2.3–12.3)	20.5 (12.2) 21.3 (3.4–39.7)
Fluctuation	mean (STD) median (range)	6.54 (2.28) 5.6 (4.5–11.8)	4.66 (1.44) 5.0 (2.5–6.8)	3.94 (1.41) 3.4 (2.2–6.8)	3.53 (1.37) 3.3 (2.0–6.7)

[1] n = 11; [2] n = 10; [3] n = 8; STD, standard deviation; CL/F, apparent clearance; Vz/F, apparent volume of distribution.

The mean (standard deviation) and median (range) plasma pharmacokinetic parameters for 7-hydroxymitragynine after SD and MD oral mitragynine in kratom leaf powder are shown in Table 4. 7-hydroxymitragynine C_{max} and AUC were lower than those of

mitragynine but also increased in a dose-appropriate manner after SD and MD. The median 7-hydroxymitragynine T_{max} was similar to mitragynine at 1.2–1.8 h and 1.3–2.0 h after SD and MD, respectively. Apparent increases in median $T_{1/2}$ were seen with increasing doses as for mitragynine, with a median (range) $T_{1/2}$ of 4.0 h (1.7–11.4) and 9.1 h (2.2–71.6) after the highest SD and MD, respectively. 7-hydroxymitragynine MD fluctuation was 3.4–9.8, with lower values at higher doses. 7-hydroxymitragynine dose proportionality was confirmed based on C_{max} and $C_{max,ss}$ after SD and MD but was slightly greater than dose-proportional (1.18-fold) based on $AUC_{0\text{-}Tlast}$ after SD and 1.32-fold after MD based on $AUC_{0\text{-}tau,ss}$. Multiple concentration peaks were observed in some participants more than 4 h after SD and MD, suggesting that food consumption 4 h after kratom leaf powder dosing might have contributed to these later concentration peaks.

Table 4. Mean (standard deviation) and median (range) plasma pharmacokinetic parameters for 7-hydroxymitragynine after increasing single (SD) and 15 multiple doses (MD) of mitragynine in dried kratom leaf powder.

		Cohort 1 6.65 mg	**Cohort 2 13.3 mg**	**Cohort 3 26.6 mg**	**Cohort 4 53.2 mg**
Single dose		*n* = 12	*n* = 12	*n* = 12	*n* = 13
C_{max} (ng/mL)	mean (STD) median (range)	3.6 (1.9) 3.6 (1.5–8.6)	6.6 (1.1) 7.0 (4.8–7.8)	10.8 (3.7) 11.0 (4.3–16.0)	22.7 (7.7) 21.7 (12.5–38.6)
T_{max} (h)	median (range)	1.2 (0.8–2.7)	1.5 (1.0–2.3)	1.8 (0.8–5.0)	1.7 (1.0–2.3)
T_{last} (h)	median (range)	5.0 (3.5–9.0)	9.0 (6.0–12.1)	12.0 (0.75–12.0)	23.3 (9.0–71.6)
$AUC_{0\text{-}24}$ (h*ng/mL)	mean (STD) median (range)	9.7 (5.4) 9.0 (2.9–19.3)	25.8 (6.7) 25.3 (14.7–37.6)	47.7 (22.4) 54.3 (0.7–78.8)	125 (69.7) 115 (40.2–319)
$AUC_{0\text{-}Tlast}$ (h*ng/mL)	mean (STD) median (range)	9.7 (5.4) 9.0 (2.9–19.3)	25.8 (6.7) 25.3 (14.7–37.6)	47.7 (22.4) 54.3 (0.7–78.8)	130 (84.2) 115 (40.2–380)
$AUC_{0\text{-}inf}$ (h*ng/mL)	mean (STD) median (range)	11.2 (5.6) 10.5 (4.1–21.0)	28.8 (7.6) 28.0 (16.4–41.8)	58.2 [1] (20.4) 64.5 [1] (27.6–85.7)	136 (86.6) 123 (41.6–391)
AUC_{Tlast}/ $AUC_{0\text{-}inf}$	median (range)	0.85 (0.71–0.92)	0.90 (0.87–0.92)	0.91 (0.81–0.95)	0.96 (0.93–0.97)
$T_{1/2}$ (h)	mean (STD) median (range)	1.7 (0.5) 1.7 (1.0–2.4)	2.9 (0.9) 2.7 (1.5–4.5)	3.2 [1] (0.8) 3.4 (1.9–4.5)	4.7 (2.7) 4.0 (1.7–11.4)
Multiple doses		*n* = 11	*n* = 8	*n* = 9	*n* = 9
$C_{max,ss}$ (ng/mL)	mean (STD) median (range)	3.2 (1.5) 3.3 (1.4–5.6)	5.9 (1.5) 5.6 (4.3–8.1)	11.3 (3.5) 12.0 (6.9–17.4)	21.8 (6.1) 20.9 (13.3–31.7)
$T_{max,ss}$ (h)	median (range)	1.3 (0.8–3.0)	1.7 (1.0–4.0)	1.7 (1.3–3.0)	2.00 (1.3–3.5)
$T_{last,ss}$ (h)	median (range)	5.0 (3.5–12.0)	12.0 (6.0–23.9)	23.5 (9.0–47.6)	48.3 (9.0–121)
$AUC_{0\text{-}tau,ss}$ (h*ng/mL)	mean (STD) median (range)	10.1 (6.8) 7.2 (3.2–22.9)	28.0 (13.7) 27.3 (13.5–55.7)	67.8 (31.3) 67.7 (25.0–119)	133 (45.0) 128 (105–161)

Table 4. *Cont.*

		Cohort 1 6.65 mg	Cohort 2 13.3 mg	Cohort 3 26.6 mg	Cohort 4 53.2 mg
$T_{1/2,ss}$ (h)	mean (STD)	2.4 (1.3)	3.8 (2.3)	8.3 (5.0)	24.7 (24.7)
	median	1.9	3.2	9.0	9.1
	(range)	(1.1–5.3)	(1.5–8.4)	(2.1–16.7)	(2.2–71.6)
$C_{ave,ss}$ (ng/mL)	mean (STD)	0.4 (0.3)	1.2 (0.6)	2.9 (1.3)	5.5 (1.9)
	median	0.3	1.1	2.9	5.4
	(range)	(0.1–1.0)	(0.6–2.3)	(1.0–5.0)	(1.6–8.2)
C_{trough} (ng/mL)	mean (STD)	0.0 (0.0)	0.1 (0.3)	0.6 (0.6)	1.7 (1.0)
	median	0	0	0.6	1.6
	(range)	(0–0)	(0–0.7)	(0–1.3)	(0–3.2)
Fluctuation	mean (STD)	8.8 (2.2)	5.6 (2.1)	4.3 (1.8)	4.1 (1.7)
	median	9.8	5.3	3.4	3.7
	(range)	(5.9–11.3)	(2.7–8.9)	(2.7–6.9)	(2.3–8.2)

[1] n = 11; STD, standard deviation.

2.4. Ratios of 7-Hydroxymitragynine/Mitragynine Plasma Concentrations

7-hydroxymitragynine/mitragynine concentration ratios are shown in Figure 4. Mean ratios were 0.20–0.29 based on C_{max} and 0.21–0.31 based on $AUC_{0\text{-}24}$ after a SD, with the highest ratios observed after the lowest 6.65 mg mitragynine dose. A similar pattern was seen with MD, with mean ratios of 0.16–0.21 based on $C_{max,ss}$ and 0.15–0.18 based on $AUC_{0\text{-}tau,ss}$. The ratio was generally higher after SD compared to MD and higher at the lower doses.

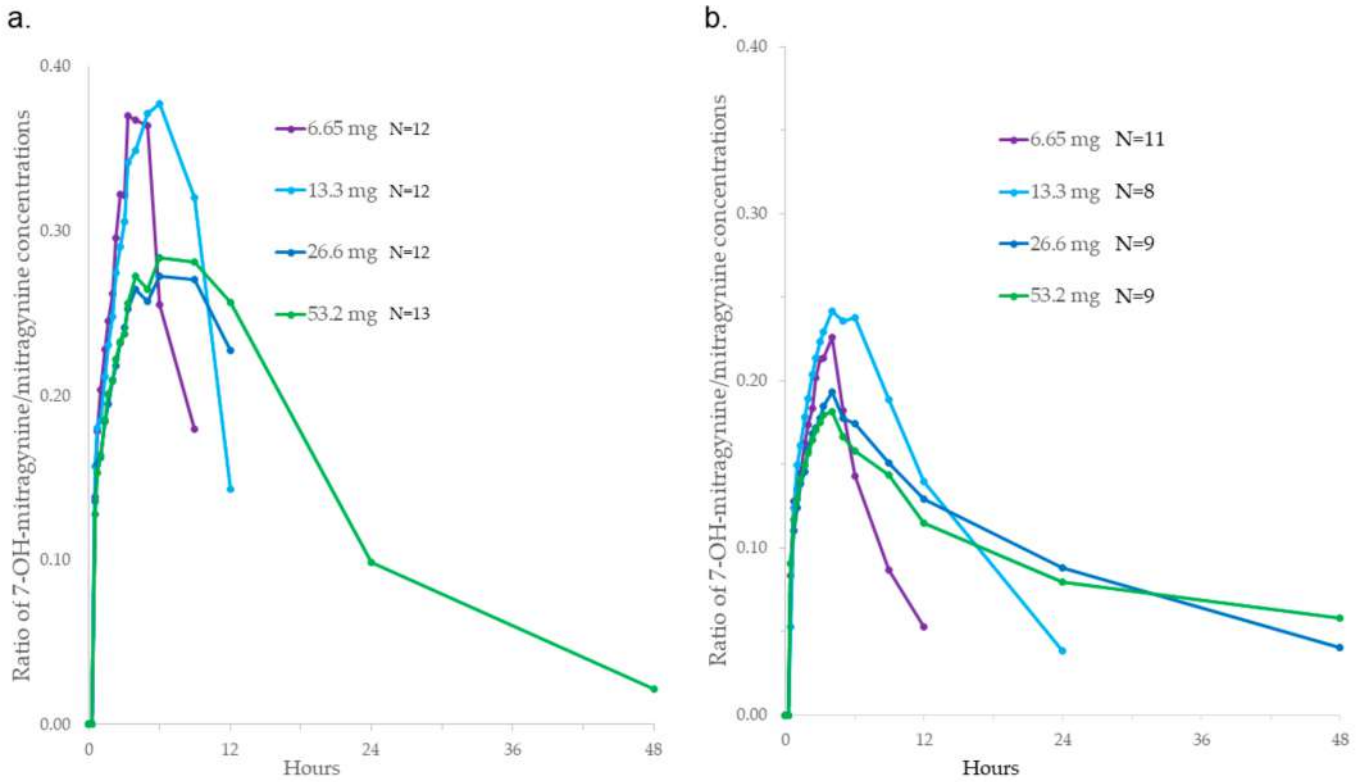


Figure 4. Mean ratios of 7-hydroxymitragynine/mitragynine plasma concentrations after (**a**) single (SD) and (**b**) multiple (MD) oral 6.65–53.2 mg mitragynine doses in dried kratom leaf powder.

2.5. Accumulation of Mitragynine and 7-Hydroxymitragynine during MD

Accumulation of mitragynine and 7-hydroxymitragynine was assessed by comparing $C_{max,ss}$ to C_{max} and $AUC_{0\text{-}tau,ss}$ to $AUC_{0\text{-}24}$ for the MD and SD, respectively. For mitragy-

nine, the accumulation was low to moderate across doses with C_{max} ratios of 1.1–1.3 and AUC ratios of 1.6–1.9. Corresponding ranges for 7-hydroxymitragynine were lower: 0.9–1.0 and 1.0–1.3, indicating no or low 7-hydroxymitragynine accumulation after MD.

2.6. Time to Reach Steady State for Mitragynine and 7-Hydroxymitragynine during Multiple Dosing

Figure 5 illustrates the accumulation in mitragynine and 7-hydroxymitragyine after multiple dosing over four increasing kratom leaf powder doses. Based on the trough concentrations determined each MD day, the time to reach steady state for mitragynine was 8–9 days. For 7-hydroxymitragynine, the time to reach steady state was 7 days based on the two highest doses, as the number of trough samples above the assay quantification limit was insufficient for the two lowest doses.

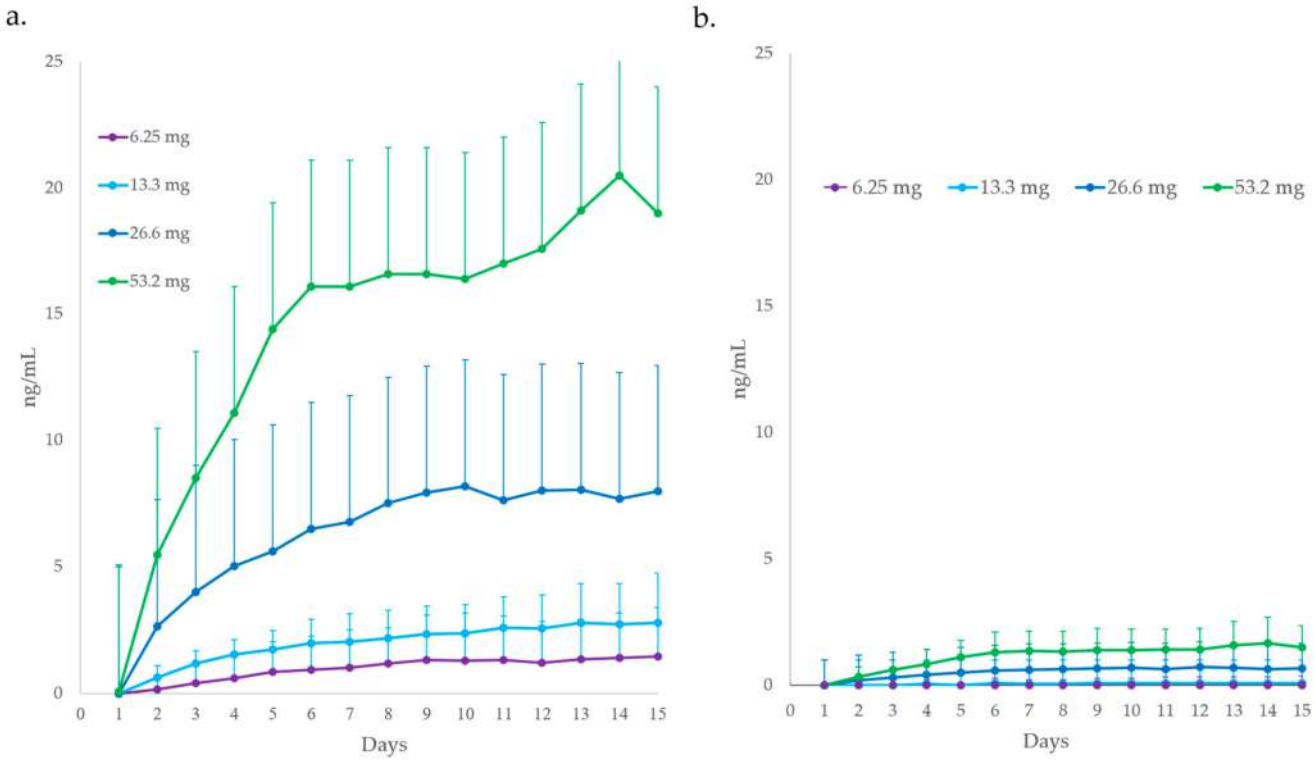


Figure 5. Trough plasma (**a**) mitragynine and (**b**) 7-hydroxy-mitragynine concentrations (including standard deviation bars) on days 1–15 of consecutive 6.65–53.2 mg mitragynine doses in dried kratom leaf powder during the attainment of steady-state concentrations.

3. Discussion

This study was the first to provide extensive mitragynine and 7-hydroxymitragynine pharmacokinetic data in a controlled setting after increasing kratom leaf powder SD and MD. Intensive blood sampling up to 10 days after SD and 23 days after MD ensured well-characterized pharmacokinetic exposure parameters and terminal half-lives. Concentrations of 7-hydroxy-mitragynine originated primarily from mitragynine metabolism because its content in the kratom leaf powder was <0.01%.

In the current study, median mitragynine T_{max} after SD and MD ranged from 1.0 to 1.7 h, with only slightly longer T_{max} for 7-hydroxymitragynine (1.2–2.0 h). In two recent studies with kratom tea, the median T_{max} for both compounds was 1.0 h [26,27]. The slightly longer T_{max} in the present study may be due to the encapsulated kratom leaf powder formulation. Concentration–time curves for both analytes showed an initial rapid descent after C_{max} for about 6 h and then a much slower terminal phase.

The individual mitragynine SD C_{max} of 16.0–90.0 ng/mL (24 subjects) following 13.3–26.6 mg mitragynine was similar to the range reported by Trakulsrichai [25] of 18.5–105 ng/mL (9 subjects) following 10–23.6 mg mitragynine in kratom tea. However, subjects in the latter study received different low doses (6.3–11.5 mg) of kratom tea for 7 days prior to the pharmacokinetics profile day and were not asked to cease other kratom use. The contribution of this additional kratom consumption to the reported mitragynine concentrations is unclear.

Compared to the recent SD kratom tea investigations [26,27], the current study documented higher mitragynine exposure. The geometric mean C_{max} and $AUC_{0\text{-}24}$ were 119 nM (approximately 47.5 ng/mL) and 388 h*nM (approximately 155 h*ng/mL) in six male and six female subjects following 39 mg mitragynine [27]. Our geometric mean results in 12 subjects were 51.0–113 ng/mL for C_{max} and 169–493 h*ng/mL for $AUC_{0\text{-}24}$ across 26.6–53.2 mg mitragynine. 7-hydroxymitragynine exposure was similar between the two studies: C_{max} and $AUC_{0\text{-}24}$, in the kratom tea study, were 31 nM (approximately 12.9 ng/mL) and 151 h*nM (or approximately 62.7 h*ng/mL) compared to the current study, with 10.2–21.6 ng/mL for C_{max} and 34.2–110 h*ng/mL for $AUC_{0\text{-}24}$ across the dose range of 26.6–53.2 mg mitragynine. Differences in the liquid kratom tea and solid encapsulated leaf powder formulations likely impacted absorption.

Between-subject variability for C_{max} and $AUC_{0\text{-}24}$ was quite high after SD and MD. For mitragynine, CV% was generally 30–60%, increasing up to 99% at the lowest 6.65 mg dose. For 7-hydroxymitragynine, between-subject variability was slightly lower at 20–55%, with the highest variability of 67%, again, at the lowest dose.

Mitragynine and 7-hydroxymitragynine terminal half-lives generally increased with increasing doses, as concentrations were measurable for longer at higher doses compared to lower doses. However, increases in terminal half-life were also observed between single and multiple dosing, which will be explored in future investigations.

The same issue of low concentrations after lower doses also affected $AUC_{0\text{-}inf}$, in some cases underestimating $AUC_{0\text{-}inf}$ and overestimating CL/F. For example, the mean CL/F of mitragynine for the lowest single 6.65 mg dose was much higher (278 L/h) than for the higher doses (94–123 L/h).

Accumulation factors for mitragynine comparing $C_{max,ss}$ and $AUC_{0\text{-}tau,ss}$ during MD at steady state with C_{max} and $AUC_{0\text{-}24}$ after a SD were low to moderate, ranging from 1.1 to 1.3 and 1.6–1.9, respectively. Corresponding ranges of 0.9–1.0 and 1.0–1.3 were observed for 7-hydroxymitragine, indicating no or only low accumulation after MD. These results contradict the observed long terminal $T_{1/2}$, suggesting that higher accumulation ratios should have been found. Sahin and Benet [28] address the issue of the overprediction of accumulation from extended terminal half-lives and define the operational multiple dosing half-life ($t_{1/2,op}$) as being equal to the dosing interval at a steady state, where the maximum concentration at the steady state is twice the maximum concentration found for the first dose. It would be useful to define an operational or MD $T_{1/2}$ that is more relevant than the terminal $T_{1/2}$ for predicting the accumulation of mitragynine and 7-hydroxymitragynine after different dosing regimens.

7-hydroxymitragynine to mitragynine ratios were always higher after SD than MD, and the highest ratios were consistently after lower doses. Across the concentration–time profile, ratios generally increased to a maximum and then decreased over time. The decrease reflects the shorter half-life of the metabolite compared to the parent compound. The range of the median C_{max} ratios observed at the top two doses was 0.18–0.21, which was similar to the median of 0.27 previously reported in a small number of subjects [27].

The CYP3A4 enzyme plays a major role in the metabolism of mitragynine to 7- hydroxymitragynine based on in vitro studies using human liver microsomes and S9 fractions [29]. Since CYP3A enzymes are the predominant phase 1 metabolism enzymes in the liver and intestine of dogs and men [30], it is of interest to compare the metabolic ratios of 7-hydroxymitragynine to mitragynine between dogs and humans. In one nonclinical study [31], 5 mg/kg mitragynine was administered orally to five beagle dogs, from which plasma concentration profiles were obtained for mitragynine and 7-hydroxymitragynine up to 24 h post-dose. The ratios of metabo-

lite to parent compound were 11.3% and 12.6% for C_{max} and AUC, respectively, about one-half of the ratios observed in the current human study.

The limitations of this study include the limited range of doses that were prescribed by the reviewing ethics committee and Health Canada. Future clinical studies will expand the administered dose range. The long terminal mitragynine $T_{1/2}$ requires an adequate washout in pharmacokinetic studies between treatments. After a SD, mitragynine concentrations were measurable in some subjects 10 days after dosing; however, in all cases, concentrations were <5% of C_{max}. The FDA guideline [32] recommends a washout of at least five half-lives or until concentrations in each subject are ≤5% of C_{max}. Here, a 10-day washout was adequate for a single kratom leaf powder dose containing up to 53.2 mg mitragynine; however, for the MD regimens, at least a 14-day washout period is recommended. After 15 MD, a steady state was reached in 7–9 days for both analytes. Visual inspection of the trough concentrations (Figure 5) suggested that, after day 10, a further slight increase in trough concentrations appears to occur at the highest 53.2 mg dose. It would be prudent to monitor trough concentrations over a longer period for higher doses to confirm steady state attainment.

In conclusion, a wealth of pharmacokinetic data were obtained from this largest kratom leaf powder administration study, providing much-needed new insights into the pharmacokinetic characteristics of mitragynine and 7-hydroxymitragynine over a range of increasing SD and MD. These data are of considerable help for the planning and design of future clinical kratom studies and, importantly, in interpreting clinical mitragynine and 7-hydroxymitragynine concentrations.

4. Materials and Methods

4.1. Study Design

A randomized, between-subject, double-blind, placebo-controlled, dose-escalation, single-site pharmacokinetic study of encapsulated dried kratom leaf powder was carried out for 10 days after SD, 15 days MD, and 23 days follow-up in healthy adults, with 12 participants receiving active kratom leaf powder per dose. There were 31 clinic visits over 47 days; participants fasted (≥10 h) before SD and during the last MD pharmacokinetic visits (Figure 6). Standardized meals were provided 4 and 10 h after the morning dose.

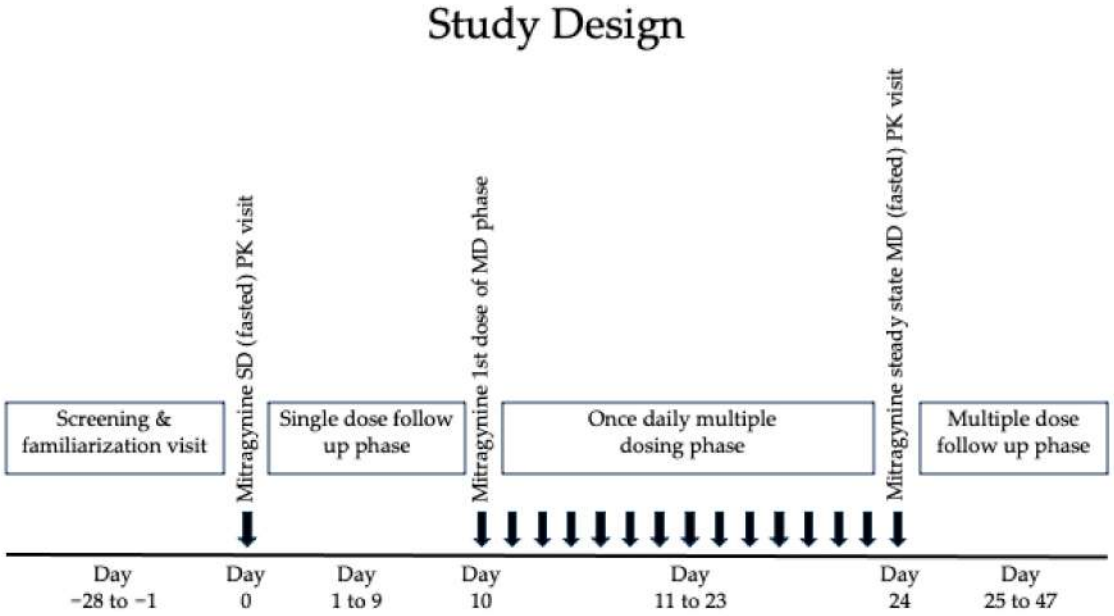


Figure 6. Study design showing dosing and in-person clinical visits (single dose (SD) follow-up phase days 1, 2, 3, 5, 7, and pre-dose day 10; 15 multiple doses (MDs), follow-up phase days 25, 26, 27, 29, 31, 34, 40, and 47), including two intensive pharmacokinetic (PK) days after a single oral mitragynine dose and on the last of 15 consecutive 6.65–53.2 mg daily mitragynine doses of dried kratom leaf powder.

The study was conducted in accordance with the Declaration of Helsinki and approved by the Advarra Institutional Review Board (protocol code 00048457 on 20 June 2022). The Advarra Institutional Review Board is organized and operates in compliance with the US and Canadian regulations and policies governing research with human subjects. Advarra's IRB is registered with the FDA and OHRP; IRB Organization (IORG) Number: 0000635, FWA Number: 00023875, IRB Registration Number: 00000971. Advarra is fully accredited by the Association for the Accreditation of Human Research Protection Programs (AAHRPP). The Advarra Institutional Review Board and Health Canada approved the study. Informed consent was obtained from all subjects involved in the study.

4.2. Study Population

Non-smoking healthy males and females of 18–55 years with BMI $\geq$ 18.5 and $\leq$29.9 kg/m^2 participated. Participants self-reported being either kratom-naïve or had no kratom use for $\geq$12 months prior to the first study visit. Female participants of childbearing potential had negative pregnancy tests and were not breastfeeding; appropriate birth control was utilized by all participants throughout the study. No medications were permitted within 7 days (14 days for enzyme inducers) or 7 half-lives of day 0, whichever was longer. Randomization was stratified by sex for males/females per dose.

Participants were excluded if there was a known allergy to product ingredients or known genetic polymorphism of CYP4503A4, CYP2D6, or CYP1A2. Fasting was required 10 h prior to the familiarization visit and prior to the pre-dose blood sample on days 0 and 10–24. The presence of gastrointestinal, hepatic, renal, cardiovascular, respiratory, autoimmune, endocrine, neurological, or active psychiatric diseases was exclusionary.

Participants were excluded if there was an acute illness within 2 weeks of the single dose, a positive urine drug test at screening, a history of drug or alcohol abuse in the past 12 months, a Leeds Dependence Questionnaire score $\geq$ 21, or a consumption of $\geq$2 alcoholic beverages a day. Positive serology results for HIV, HBsAg, or HCV or a history of cancer were exclusionary. Participants were non-smokers who had not used nicotine patches or gums for >3 months prior to the first dose and had veins suitable for cannulation/repeated venipuncture. Caffeine intake was restricted for 24 h prior to the two pharmacokinetic study days and 12 h prior and 5 h post each visit; alcohol and cannabinoids were restricted for 24 h prior to and 12 h post days 1–25 and day 47. Hepatic or renal dysfunction with ALT or AST $\geq$ 2X upper limit of normal (ULN) and serum creatinine or BUN $\geq$ 1.5X ULN were evaluated by the principal investigator. Participation in another research study or blood donation within four (4) weeks prior to single dosing was prohibited.

4.3. Dosing

The kratom dried leaf powder (MitraLeaf^{TM}) was manufactured by Johnson Foods, LLC., Kratom capsules were labeled according to ICH-GCP and Health Canada guidelines and stored at room temperature (20–25 °C) with relative humidity $\leq$65%. A large batch of dried kratom leaf was mixed in an industrial powder mixer to homogenize the powder. The powder was weighed and encapsulated, and quality control was performed to ensure adequate weight in the capsules. To determine the concentration of mitragynine in each capsule, Avomeen, the laboratory performing the LC-MS/MS analysis of the kratom powder, emptied 50 capsules of powder, mixed the powder well, and weighed an aliquot of the powder for analysis. The powder was extracted and analyzed using LC-MS/MS for mitragynine and 7-hydroxymitragynine. The Avomeen Certificate of Analysis showed 13.3 mg mitragynine in 1000 mg of the dried kratom leaf powder. Four cohorts of 12 different subjects received one of four increasing doses of 6.65 mg, 13.3 mg, 26.6 mg, and 53.2 mg mitragynine contained in 500, 1000, 2000, or 4000 mg leaf powder in 1, 2, 4, or 8 capsules, respectively, and 6 subjects at each dose received placebo. After 10 days of monitoring after the SD, 15 consecutive ascending doses of 6.65 mg, 13.3 mg, 26.6 mg, and 53.2 mg mitragynine in 500, 1000, 2000, or 4000 mg of leaf powder were administered, respectively, to participants. 7-hydroxymitragynine concentrations were <0.01% kratom leaf powder.

Interim safety analyses included laboratory assessments (hemoglobin, liver, and kidney function tests, adverse events (AEs), and an assessment of the Clinical Opiate Withdrawal Scale (COWS) and Subjective Opiate Withdrawal Scale (SOWS) after the completion of each dose cohort prior to the decision by medical staff to proceed to the next higher dose).

4.4. Blood Collection Timeline for Single and Multiple Dose Pharmacokinetic Study

Blood collections (total 484 mL) for the mitragynine and 7-hydroxymitragynine concentrations occurred prior to and 0.25, 0.5, 0.75, 1, 1.33, 1.67, 2, 2.33, 2.67, 3, 3.5, 4, 5, 6, 9, and 12 h after the single dose and after the 15th multiple dose. Participants returned to the clinic 24, 48, and 72 h—and 5, 7, and 10 days—after the SD for the daily post-dose blood collection. Participants fasted for ≥10 h prior to the pre-dose blood collection. Standardized meals were provided 4 and 10 h after the morning dose. Follow-up visits for MD occurred 1, 2, 3, 5, 7, 10, 16, and 23 days later, with a single blood collection in the morning. The follow-up phase began 24 h after the last MD, day 25, and ended on day 47 (end of study, EOS).

4.5. Pharmacokinetics

Pharmacokinetic parameters utilized standard non-compartmental methods (SAS version 9.4). Peak concentration ($C_{max}/C_{max,ss}$), time to reach C_{max} ($T_{max}/T_{max,ss}$), time of last measurable concentration ($T_{last}/T_{last,ss}$), area-under-the-curve (AUC) from dosing to 24 h ($AUC_{0\text{-}24h}/AUC_{0\text{-}tau,ss}$) and 72 h ($AUC_{0\text{-}72}$), last observable concentration ($AUC_{0\text{-}Tlast}/AUC_{0\text{-}Tlast,ss}$), infinity ($AUC_{inf}$), terminal phase half-life ($T_{1/2}/T_{1/2,ss}$), terminal phase rate constant ($K_{el}/K_{el,ss}$), apparent total clearance ($CL/F/CL/F_{,ss}$), and ratio 7-hydroxymitragynine/mitragynine concentrations were determined. During MD, minimum concentration ($C_{min,ss}$), average concentration ($C_{av,ss}$), trough concentration (C_{trough}), and fluctuation ($(C_{max,ss} - C_{min,ss})/C_{av,ss}$) during the dosing interval were also quantified. Accumulation of mitragynine and 7-hydroxymitragynine with MD was compared to SD AUC and C_{max} at each dose.

4.6. Statistical Analysis

Analysis sets included all participants with sufficient data for the calculation of at least one pharmacokinetic parameter after SD or MD (SAS version 9.4). A power model for C_{max} or $AUC_{0\text{-}T}$ after SD and $C_{max,ss}$ or $AUC_{0\text{-}tau}$ after MD investigated dose proportionality. The power model was pharmacokinetics parameter = α*dose$^{\beta}$. A regression analysis was conducted using the log form of this equation: ln (pharmacokinetics parameter) = $\ln\alpha + \beta * \ln(\text{dose}) + \varepsilon$. Estimates for α and the slope β were obtained, with a 90% confidence interval calculated for β. The criterion for dose proportionality was sufficiently met if this 90% confidence interval was entirely contained within the interval $(1 + [\ln(0.5)/\ln(r)], 1 + [\ln(2)/\ln(r)])$, where r is the ratio of highest dose divided by the lowest dose. The time required for mitragynine and 7-hydroxymitragynine to reach a steady state was estimated using a Helmert coding approach. The model included the day as a fixed effect, the subject as a random effect, and the logarithm of the concentration as a response variable. Pair-wise contrasts were then estimated, enabling an assessment of the time to reach steady state.

Author Contributions: Conceptualization, M.A.H. and R.A.; methodology, M.A.H. and R.A.; formal analysis, M.A.H., J.B., M.A.B., and R.A.; investigation, M.A.H. and R.A.; resources, R.A.; data curation, M.A.H., J.B., M.A.B., and R.A.; writing—original draft preparation, M.A.H.; writing—review and editing, M.A.H., J.B., M.A.B., and R.A.; visualization, M.A.H. and R.A.; supervision, R.A.; project administration, R.A. All authors have read and agreed to the published version of the manuscript.

Funding: Johnson Foods: LLC. paid for the clinical study on its dried kratom leaf powder product MitraLeaf to be performed by SGS Nutrasource, Guelph, Ontario, Canada. Huestis, Brett, Bothmer, and Atallah are paid consultants to Johnson Foods, LLC. Atallah is an employee of Della Terra Pharmaceuticals.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Advarra Institutional Review Board (Pro00048457). The Advarra

Institutional Review Board is organized and operates in compliance with the US and Canadian regulations and policies governing research with human subjects. Advarra's IRB is registered with the FDA and OHRP; IRB Organization (IORG) Number: 0000635, FWA Number: 00023875, IRB Registration Number: 00000971. Advarra is fully accredited by the Association for the Accreditation of Human Research Protection Programs (AAHRPP). The Advarra Institutional Review Board and Health Canada approved the study.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available upon request from Ramsey Atallah, Della Terra Pharmaceuticals.

Conflicts of Interest: Johnson Foods, LLC., paid for this clinical study on its dried kratom leaf powder product MitraLeaf to be performed by SGS Nutrasource, Guelph, Ontario, Canada. Huestis, Brett, Bothmer and Atallah are paid consultants to Johnson Foods, LLC. Atallah is an employee of Della Terra Pharmaceuticals. Huestis and Atallah are also consultants for the American Kratom Association, which had no role in this study or publication.

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Buprenorphine for the Management of 7-Hydroxymitragynine (7-OH) Use: A Retrospective Case Series

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Background: Kratom (*Mitragyna speciosa*) is increasingly used in the United States as a dietary supplement for pain, mood regulation, and opioid withdrawal. Its alkaloids, mitragynine and 7-hydroxymitragynine (7-OH), demonstrate μ-opioid receptor activity, with 7-OH exhibiting substantially greater opioid-like potency than mitragynine. There are concerns from the US Food and Drug Administration (FDA) regarding dependence and adverse effects. Purified 7-OH products may bypass metabolic pathways and deliver disproportionately strong opioid effects compared with natural kratom preparations, increasing the risk of dependence. Despite these concerns, 7-OH remains federally unscheduled, though listed as a "Drug of Concern" by the DEA. Although there is no standardized clinical guidance, buprenorphine has been used to manage kratom and 7-OH withdrawal with varying initiation strategies.

Objectives: To describe the clinical course and outcomes of buprenorphine initiation among patients with problematic 7-OH use, focusing on initiation timing, dosing strategies, and symptom monitoring.

Methods: Retrospective chart review of patients treated at a low-barrier telehealth addiction medicine clinic between April and October 2025.

Results: Nine patients with purified 7-OH product use (as opposed to kratom use) met the inclusion criteria. The mean age was 33.5 years; 7 patients were identified as male. Low-dose initiation was used in 6 cases and standard initiation in 3 cases. Successful initiation and stabilization occurred in 88.9%, with no precipitated withdrawal or adverse events. Symptom improvement was reported in 8 of the total cases at a median 6-week follow-up.

Conclusions: Buprenorphine initiation for problematic 7-OH use was feasible, well tolerated, and associated with symptom improvement, supporting development of pragmatic clinical approaches for this emerging substance use disorder.

Key Words: kratom, 7-hydroxymitragynine (7-OH), buprenorphine, psychoactive substance.

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Kratom (*Mitragyna speciosa*) is a plant-derived substance with increasing use in the United States, particularly for self-management of chronic pain, mood symptoms, and withdrawal from other substances. Its primary alkaloids, mitragynine and 7-hydroxymitragynine (7-OH), are partial agonists at the μ-opioid receptor, with 7-OH being a higher-potency partial agonist, though human clinical potency and risk remain incompletely characterized, prompting regulatory scrutiny in the face of risk of dependence. In July 2025, the FDA issued several warning letters to companies for illegally marketing 7-OH. Naturally occurring levels of 7-OH are found in minimal amounts (~0.01%) within kratom leaves and further undergo CYP-mediated conversion, leaving negligible remaining amounts.[1] Unlike naturally occurring 7-OH, purified and concentrated 7-OH products present a heightened risk of dependence and opioid-like adverse effects, which have been predominantly extrapolated from animal and mechanistic studies.[2] These purified preparations may bypass the slower metabolic conversion of mitragynine to 7-OH, thereby delivering disproportionately strong opioid-like activity, though the clinical impact of partial agonism on the μ-opioid receptor has not been well characterized.[1,3–8] 7-OH is ~4–7 times more potent than morphine in mouse antinociception models and has been shown to have minimal abuse liability, reduce opioid tolerance when administered before morphine, and have a lack of β-arrestin recruitment.[8–12] Such formulations raise safety concerns

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given their marketing as dietary supplements or "natural" remedies without standardized dosing or regulatory oversight. While kratom and 7-OH remain unscheduled at the federal level, their pharmacology raises significant concerns for both dependence and subsequent withdrawal. Clinical presentations suggest that individuals who consistently use 7-OH develop tolerance, escalating use, and withdrawal syndromes when attempting to stop, thereby meeting several standard DSM-5 criteria for a substance use disorder. Despite these risks, kratom and 7-OH continue to be sold as herbal supplements (often in low-barrier settings such as corner stores and gas stations under the guise of natural remedies for pain, mood, and energy), with limited regulation and clinical guidance for clinicians managing affected patients.

Emerging clinical reports describe problematic 7-OH use consistent with substance use disorder features as an underrecognized, yet growing phenomenon.[1,10,13] Unlike traditional opioid use disorders, 7-OH withdrawal may present with a somewhat unique symptom profile, including pronounced anxiety, irritability, insomnia, and restlessness, in addition to the classic autonomic and gastrointestinal features otherwise seen with kratom and opioid withdrawal.[14] Patients may use 7-OH or kratom specifically to avoid withdrawal from prescription opioids or fentanyl, yet ultimately develop physiological dependence on the substance itself. Increasingly, patients are presenting to primary care, emergency, and addiction medicine settings seeking help with discontinuation due to physical and financial dependence on the substance, though clinicians face uncertainty about effective and safe treatment strategies.[15]

Buprenorphine, a well-known partial μ-opioid receptor agonist, is the standard of care for opioid use disorder and has been used off-label (alongside methadone and naltrexone) in case reports and series to manage kratom dependence, though there is no dedicated research for the management of 7-OH use.[10,16] The absence of a structured, reproducible protocol limits generalizability and leaves uncertainty about how to approach buprenorphine initiation in patients using 7-OH. Furthermore, the natural history and outcomes of buprenorphine treatment for kratom use disorder, especially in outpatient settings, remain poorly described.

Given the rising prevalence of patients using 7-OH and the lack of clinically pragmatic approaches applicable in real-world addiction treatment settings amidst a growing clinical need, a structured examination of buprenorphine initiation in this population is of the utmost importance. This case series provides observational descriptions of patient presentations, initiation strategies, and short-term outcomes. By capturing both clinical course and patient-reported experiences, this case series aims to generate foundational knowledge that can guide safe and effective management of 7-OH withdrawal in clinical practice. These findings will serve as preliminary data to inform the development of standardized initiation protocols and ultimately support larger prospective studies.

METHODS

Study Design and Setting

This study was a retrospective, observational case series conducted within the Harm Reduction & BRidges to Care (HRBR) outpatient addiction medicine clinic. The case series was designed to descriptively characterize clinical presentations, buprenorphine initiation approaches, and stabilization trajectories among patients presenting with use of purified 7-OH. This case series was conducted and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for observational studies. This study was reviewed by the Oregon Health and Science University Institutional Review Board (Study #29198) and was determined not to constitute human subjects research and therefore was exempt from further IRB review. As the study involved retrospective review of existing clinical records with no direct patient contact, intervention, or collection of identifiable information, the requirement for informed patient consent was waived.

Case Identification

Cases were identified through review of the electronic health record (EHR) for patients receiving care at the HRBR clinic who reported current use of 7-OH and underwent buprenorphine initiation during routine clinical care. Identification was based on clinician documentation of 7-OH use in intake notes, follow-up encounters, or substance use histories. Inclusion and exclusion criteria are further detailed in Appendix A, Supplemental Digital Content 1, http://links.lww.com/JAM/A815

Data Sources and Abstraction

Data were extracted from the EHR and entered into a secure database. Each case was assigned a unique study identifier, and all data were de-identified before analysis.

Chart abstraction was performed using a structured abstraction form developed to capture demographic, clinical, and treatment-related variables. Extracted variables included patient demographics, reported 7-OH use patterns, buprenorphine initiation strategy, total daily buprenorphine dose during initiation as well as stabilization, adjunct medications, documented clinical outcomes, and incidence of concurrent or history of other substance use disorders.

A qualitative review of clinician notes was conducted to capture patient-reported symptoms, withdrawal experiences, tolerability of initiation, and functional impacts. These qualitative data were summarized narratively to illustrate representative clinical courses and contextualize quantitative findings.

Definitions

- Exposure (7-OH use): Defined as patient self-report of current use of purified 7-hydroxymitragynine products at the time of presentation. Reported daily use was documented either in milligrams or, when unavailable, as estimated daily expenditure.

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- Concurrent substance use disorders (SUDs): Defined by clinician-documented diagnoses in the medical record based on DSM-5 criteria or treatment history.
- Low-dose buprenorphine initiation: Defined as initiation of buprenorphine using incremental, subtherapeutic doses (eg, starting at ~0.5–1 mg total daily dose, with stepwise increases to 2–4 mg over the first 1–2 d) while a full opioid agonist is continued (over days 1–2), followed by gradual dose escalation over multiple days (typically reaching ≥8–16 mg/d) to achieve a therapeutic maintenance dose.
- Standard buprenorphine initiation: Defined as initiation of buprenorphine after the onset of moderate opioid withdrawal (COWS score ≥8–12 and often ≥24 hours following last opioid use), with administration of typical starting doses (eg, 2–4 mg initial dose, with additional 2–8 mg on day 1 as needed, up to ~8–12 mg total on day 1) and subsequent titration to a therapeutic maintenance dose (generally 8–24+ mg/d).
- High-dose buprenorphine initiation: Defined as a large initial dose (≥16 mg as the first dose) in patients with fentanyl exposure and severe withdrawal or >12 mg total buprenorphine on the first day.[17,18]
- Initiation success: Defined as completion of buprenorphine initiation (3 or more days on stable dose) without discontinuation due to intolerable symptoms or precipitated withdrawal.
- Stabilization: Defined as clinician-documented suppression of withdrawal symptoms and cravings with a stable total daily buprenorphine dose for three or more days.
- Symptom improvement: Defined as clinician-documented reduction in patient-reported withdrawal symptoms, cravings, or associated neuropsychiatric symptoms (eg, anxiety, insomnia, irritability) following buprenorphine initiation, as noted in follow-up clinical documentation.
- Precipitated withdrawal: Defined as a rapid and clinically significant worsening of opioid withdrawal symptoms within hours of buprenorphine administration, consistent with published operational criteria. Operationalized as an acute increase in withdrawal severity (eg, ≥6-point increase in COWS score or clear clinical deterioration) temporally associated with buprenorphine initiation and documented in the shared health record, without an alternative explanation.[19,20]
- Treatment engagement: Defined as receipt of MOUD during the 30 days following initiation.[21]

Urine drug screening was not uniformly obtained and, therefore, was not used as a required criterion for confirmation of exposure or concurrent substance use, particularly in the absence of routine laboratory testing for kratom and 7-OH. Ascertainment of precipitated withdrawal and adverse events relied on subjective history as well as review of outpatient clinic documentation and available emergency department or inpatient records within the same health system.

Outcomes

Clinical outcomes of interest included initiation approach, whether initiation was completed, achievement of stabilization, maximum buprenorphine dose required, presence or absence of precipitated withdrawal, symptom trajectories during initiation, and engagement in care at follow-up.

Data Analysis

Given the exploratory nature and small sample size of this retrospective case series, analyses were descriptive. Patient demographic and clinical characteristics were summarized using measures of central tendency and range. Continuous variables (eg, age, estimated daily 7-OH use, duration of use) were reported as a mean, median, or range, depending on the variable itself. Categorical variables (eg, sex, presence of concurrent SUDs, initiation strategy, stabilization status) were summarized using frequencies and proportions.

No inferential statistical analyses or hypothesis testing were performed due to the case series design, limited cohort size, and heterogeneity of presentations. Missing data were uncommon and occurred primarily when 7-OH exposure was documented in monetarily rather than milligram units. Analyses were conducted to descriptively characterize observed clinical practices and generate hypotheses for future studies.

RESULTS

A total of 9 clinical cases involving buprenorphine treatment for 7-OH use were identified and included in this case series (Table 1).

The mean age of patients was 33.5 years (range: 20–46). Most patients identified as male (n = 7, 77.7%), with one female patient and one who identified as nonbinary. The cohort was predominantly White (n = 7, 77.7%), with 2 patients reporting other racial or ethnic identities. Concurrent substance use disorders were present in 66.6% (n = 6) of cases.

Reported 7-OH use varied substantially. When reported in milligrams (n = 7), the median daily dose was 150 mg/d; when recorded as monetary expenditure (n = 2), the median expenditure was $33.14/d. The mean duration of 7-OH use before presentation was 11.4 months.

Buprenorphine initiation strategies included both standard and low-dose approaches, with 6 patients who underwent low-dose initiation and 3 patients who underwent standard initiation. Successful initiation was achieved in 88.9% of cases, and all 9 patients remained engaged in treatment with buprenorphine. Symptom improvement was reported in 8 of the total cases. No precipitated withdrawal was observed. The mean standard initial total daily dose (TDD) was 4 mg (median 2 mg), while the mean low-dose initial TDD was 4.25 mg (median 1.25 mg). Most patients (n = 8) required moderate to high stabilization doses (≥16 mg TDD), with several cases necessitating 24–32 mg/d to adequately suppress cravings and disrupt ongoing 7-OH use; 1 patient required a buprenorphine dose ≤16 mg TDD, which was predominantly driven by patient preference. Stabilization was achieved in 8 cases. One patient declined dose esca-

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TABLE 1. Clinical Cases of Buprenorphine for 7-OH Use

Patient Demographics	Daily 7-OH Dose (mg or $/d If Not Provided in mg)	Duration of 7-OH Use (mo)	Buprenorphine Stabilization Dose (TDD – mg) and Formulation	Initiation Method	Supportive Meds Used	Stabilization Achieved?	Important Case Considerations
Clinic patient #1 39 y/o M with h/o OUD, GAD, MDD, and ADHD	98–128	13	16, SL bupe-nal tablets	Low dose	Tizanidine	Yes	The initiation dose was too low and required 12 mg ~3 d following initial dosing. Stabilized at 16 mg
Clinic patient #2 28 y/o M with OCD, GAD	100–150	4	16, SL bupe-nal tablets	Standard	None	Yes	Was on 12 mg through HRBR clinic and eventually increased to 16 mg TDD through PCP follow-up
Clinic patient #3 25 y/o M with h/o OUD, PTSD, bipolar 2, ADHD, GAD	$30–40/d	1	24, SL bupe mono tablets	Low dose	Clonidine, tizanidine, ondansetron	Yes	Cravings on 16 mg TDD prompting increase to 24 mg
Clinic patient #4 46 y/o F with GAD and insomnia	300	12	16, SL bupe-nal tablets	Low dose	Trazodone, clonidine, ondansetron, hydroxyzine, tizanidine, loperamide, ibuprofen	Yes	Initially opted for detox from 7-OH, though later requested MAT
Clinic patient #5 24 y/o nonbinary patient with ADHD, GAD, chronic pain, gender dysphoria fibromyalgia, MDD, and unspecified psychosis	$100/d	5	24, SL bupe mono tablets	Standard	Ondansetron, clonidine, buspirone, ibuprofen	Yes	Initially had buprenorphine initiation while in active withdrawal before planned surgery, max dose 28 mg TDD though stabilized on 24 mg TDD
Clinic Patient #6 37 y/o M with GAD, MDD, OUD (in remission), and AUD (in remission)	400	48	28, SL bupe mono tablets	Low dose	Clonidine, tizanidine, gabapentin, hydroxyzine	Yes	Initial buprenorphine initiation was unsuccessful (3-d low dose). Reached 22 mg TDD but unable to disrupt use), went to medically managed withdrawal (detox) and was started on buprenorphine initial dose of 40 mg on day 1, then stabilized at 24–28 mg TDD. Later tapered with the intention to give LAI as taper off
Clinic patient #7 20 y/o M with GAD, MDD, insomnia, ADHD, and chronic pain	270	11	32, SL bupe mono tablets	Low dose	Trazodone, tizanidine	Yes	Difficulty decreasing 7-OH use despite high buprenorphine doses
Clinic patient #8 37 y/o M with chronic pain, OUD (in remission) and insomnia	150	3	4, SL bupe-nal tablets	Standard	Tizanidine, clonidine, hydroxyzine	No	Decreased 7-OH use on MAT, though did not want to increase higher than 4 mg TDD.
Clinic patient #9 46 y/o M with chronic pain, methamphetamine use disorder (in remission), insomnia, and MDD	300–400	6	28, SL bupe-nal tablets	Low-dose	Clonidine, tizanidine, ondansetron, hydroxyzine, loperamide, ibuprofen, trazodone, mirtazapine, bupropion	Yes	Significant anhedonia and insomnia with disruption of 7-OH requiring both mirtazapine temporarily and later bupropion

Bupe-nal, buprenorphine–naloxone; bupe mono, buprenorphine monoproduct; HRBR, Harm Reduction & BRidges to Care; LAI, long-acting injectable; MAT, medication-assisted treatment; SL, sublingual; TDD, total daily dose.

Clinical characteristics, 7-OH use patterns, and buprenorphine initiation and stabilization details among patients included in this observational case series.

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lation beyond 4 mg/d and did not achieve full stabilization despite reduced 7-OH use.

Adjunct medications, including clonidine, tizanidine, antiemetics, and antidepressants, were commonly used to manage withdrawal symptoms, insomnia, anxiety, or mood disturbance during initiation and stabilization. Several cases highlighted the need for dose adjustments, prolonged initiation periods, or higher-than-typical buprenorphine doses, reflecting the clinical complexity of treating 7-OH dependence.

Figure 1 illustrates the clinical course of buprenorphine initiation among patients with 7-hydroxymitragynine (7-OH) use (n = 9), demonstrating 2 distinct initiation pathways and their convergence on shared clinical outcomes while highlighting the absence of precipitated withdrawal in all cases. The majority of patients who underwent low-dose initiation achieved gradual symptom improvement and stabilization over the first 2 weeks of treatment. There was a modest positive relationship between self-reported 7-OH intake and the buprenorphine dose required for stabilization. Patients reporting higher estimated daily 7-OH consumption tended to require higher buprenorphine doses to achieve symptom control, though stabilization was achieved across all dosing ranges.

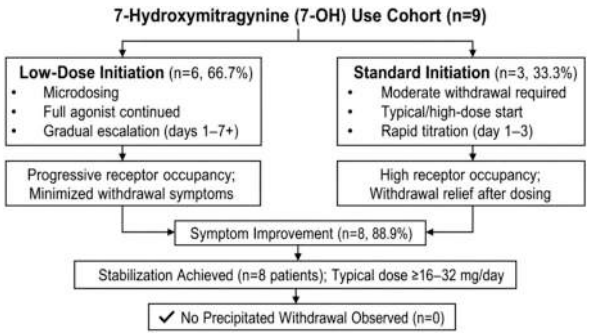


FIGURE 1. Flowchart illustrating initiation protocols, symptom improvement, stabilization, and absence of withdrawal symptoms in the 7-hydroxymitraginine cohort.

REPRESENTATIVE CASE NARRATIVES

Case 1

A 46-year-old male presented to Harm Reduction & BRidges to Care clinic (HRBR) using 300–400 mg/d of 7-OH in the preceding 6 months, ultimately spending $1500–2000/month. He made attempts to adjust his pain regimen when his chronic back pain worsened. He was given advice from the store owner that 7-OH would be more effective for pain control than the standard kratom he had been taking for the past 3 years. With provider guidance, the patient attempted low-dose buprenorphine initiation over a 7-day period, with an initial TDD of 0.5 mg and a max dose of 28 mg daily. The week following his disruption of 7-OH use was compounded by fatigue, anhedonia, and insomnia, making it challenging for him to return to work. HRBR providers trialed trazodone and clonidine for insomnia, which were ineffective. A short course (3 d) of mirtazapine 15 mg was then utilized with adjunct prescription bupropion 150 mg XL for new onset depression/fatigue. He later stabilized on a 28 mg total daily dose (TDD) with no return to 7-OH or kratom use. Following this, the patient returned to the corner store where he would typically buy 7-OH to advise store personnel to remove this dangerous substance from the shelves.

Case 2

A 37-year-old male presented to HRBR using ~400 mg/d 7-OH daily at the time of initial presentation. The patient attempted low-dose initiation with an initial TDD of 2 mg and eventually increased his buprenorphine to 22 mg TDD ~1 month following initial initiation, though he was unsuccessful at disrupting 7-OH use. He then presented to an inpatient medically managed withdrawal (detox) center, during which time he was started on buprenorphine via a high-dose approach. The patient reported that on the first day of buprenorphine initiation, he took 40 mg of buprenorphine TDD. On day 2, he took 32 mg TDD. Following this, he met with an HRBR provider at which time he noted he had reduced his dose to 24 mg TDD before discharge from the medically managed withdrawal center. He was stabilized at 24–28 mg buprenorphine TDD and remained engaged in treatment. He then successfully tapered his buprenorphine to 16 mg buprenorphine TDD over 3 months. His long-term goal is to taper off buprenorphine via a one-time dose of long-acting injectable (LAI) buprenorphine.

DISCUSSION

In this retrospective, observational case series, we report one of the first case series with clinical descriptions of off-label buprenorphine initiation for patients using 7-OH, an increasingly encountered kratom-derived alkaloid in clinical settings. As a convenience-sampled case series, these findings are descriptive and hypothesis-generating in nature and are not intended to demonstrate treatment efficacy, establish causality, or define standards of care. Clinicians in our case series utilized shared decision-making with patients to offer buprenorphine for off-label use as one component of an individualized treatment plan.

Across cases, patients exhibited features consistent with physiological opioid-like dependence and correspondingly, buprenorphine initiation appeared well tolerated in this cohort. However, this observation should be interpreted cautiously and reflects clinical experience rather than comparative effectiveness. Both low-dose and standard initiation approaches were used, potentially reflecting clinical uncertainty in the absence of established guidance for managing dependence on high-potency kratom-derived products.

Several patients required higher buprenorphine doses than anticipated (≥ 16 mg TDD) to achieve stabilization, which is higher than traditionally reported for lower-potency opioid exposures. Higher maintenance

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doses were often required compared with typical opioid use disorder treatment, suggesting that higher receptor occupancy may be necessary in this population. Following successful high-dose initiation, rapid down-titration (eg, from ≥ 32 mg to 24–28 mg total daily dose) appeared feasible in select cases without clinical destabilization once 7-OH use was fully disrupted, particularly when weighing the risks and benefits of continued buprenorphine treatment in an evidence-limited context. Many patients required doses like those used for fentanyl-associated opioid use disorder; these findings should not be interpreted as implying equivalence in pharmacologic potency or receptor dynamics but rather reflect observed clinical decision-making.

Importantly, disruption of high-dose 7-OH use was frequently associated with persistent neuropsychiatric symptoms (eg, anhedonia, fatigue, profound anxiety, and insomnia), which in some cases extended beyond the acute withdrawal period. Standard supportive medications, such as clonidine or trazodone, were not always sufficient to manage these symptoms, highlighting the need for anticipatory guidance and comprehensive symptom management strategies during early recovery. Despite these challenges, successful stabilization with buprenorphine was associated with sustained abstinence from 7-OH in many of our patients, including those with prolonged and high-dose use.

Table 2 is presented as a descriptive synthesis of observed dosing patterns, not as proposed treatment guidelines. Its purpose is to contextualize the range of buprenorphine doses used in this case series according to reported 7-OH exposure, facilitating clinical understanding and hypothesis generation. This approach parallels prior descriptive work in emerging substance use contexts, including the case series by Weiss and Douglas,[15] which similarly summarized clinical dosing experiences without asserting efficacy or prescriptive recommendations. Consistent with that precedent, Table 2 is intended to inform future systematic investigation rather than guide clinical practice. The case series by Weiss and Douglas included 9 patients using kratom products and found a linear relationship between kratom use and buprenorphine initiation dose, similar to the linear relationship that was evident in our cohort of patients using 7-OH. However, we must also acknowledge that the potency of 7-OH products varies significantly in a largely unregulated market and may contain mitragynine or mitragynine pseudoindoxyl, a more potent 7-OH metabolite. As such, this variability precludes specific buprenorphine dose recommendations based on this small sample.[22]

Our findings add to a small but growing literature suggesting that buprenorphine is a rational and effective treatment for kratom-derived alkaloid dependence. Given variability in pharmacokinetics between traditional kratom and purified 7-OH products, this focused analysis provides preliminary evidence that low-dose initiation may be particularly useful where potency and metabolic dynamics raise concerns for precipitated withdrawal.

TABLE 2. Observed Buprenorphine Dosing Patterns Based on Daily 7-OH Use

Estimated Daily 7-OH Use (mg/d)	Estimated Kratom Doses (per Weiss et al),[15] g kratom/d	Rationale	Proposed Buprenorphine Dose Range (mg/d)	Notes
< 50	< 20	Lower physiological tolerance, mild withdrawal risk	2–4	Consider standard initiation if already in moderate withdrawal
50–150		Moderate dependence, risk of dysphoria/anxiety with cessation	4–8	Low-dose initiation is optional if unable to tolerate abstinence/withdrawal symptoms
150–300	> 40	Significant opioid-like tolerance and withdrawal	8–16	Low-dose initiation preferred; consider split or incremental dosing
> 300		High-tolerance pattern similar to high-dose opioid use	12–24+	Low-dose initiation strongly recommended; higher doses may be required for stabilization

Observed buprenorphine total daily dose ranges associated with clinical stabilization in patients reporting use of purified 7-OH in this observational case series, with comparison to estimated kratom doses as demonstrated in the study by Weiss and colleagues.

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Limitations

This study has several important limitations. Small sample size, retrospective design, single-clinic setting, occasional incomplete dosing documentation, and reliance on patient self-report of 7-OH consumption not verified by toxicological testing limit generalizability. Selection bias is inherent in the harm reduction clinical context, as patients who engage in care may be different from those who do not, and follow-up duration varies across cases. Further, because the sample consisted primarily of White male patients, in part representative of the geographic area at large, these observations may not fully capture patterns of 7-OH use or treatment experiences among other diverse populations.

The variability of 7-OH products available and the risk of mislabeling limit the standardization of the exposure. While efforts were made to apply consistent operational definitions for key outcomes (eg, initiation success, stabilization, and symptom improvement), outcomes were based on clinician documentation rather than standardized assessment tools, which may limit comparability across cases. As such, conclusions regarding comparative initiation strategies, optimal dosing, or treatment effectiveness cannot be drawn.

Future Directions

Despite these limitations, this case series provides early descriptive insight into the clinical complexity of managing 7-OH use and highlights the range of buprenorphine-based approaches currently employed in practice. As kratom-derived alkaloids, including 7-OH, continue to increase in concentration and availability, systematic prospective studies are needed to better characterize withdrawal syndromes, initiation tolerability, and longer-term outcomes. Until such data are available, descriptive case series may help inform clinical awareness and identify priorities for future research.

CONCLUSIONS

This observational case series is one of the first that describes clinical experiences with buprenorphine initiation among patients presenting with use of purified 7-OH as feasible, well tolerated, and associated with high rates of stabilization and symptom improvement in a real-world outpatient setting. Both low-dose and standard initiation approaches were successfully utilized without precipitated withdrawal or adverse events.

As the availability and potency of 7-OH products continue to increase, further prospective studies are needed to better characterize withdrawal syndromes, optimize initiation strategies, and define longer-term outcomes to guide evidence-based care.

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A Case of 7-Hydroxymitragynine Use Requiring Inpatient Medically Managed Withdrawal

Rachel S. Wightman, MD and David Hu, MD

Objectives: There is an increased interest in kratom and kratom-related products such as 7-hydroxymitragynine (7-OH). Here we present the case of a patient seeking inpatient substance use treatment for daily 7-OH use and the clinical course of care including clinical opiate withdrawal scale (COWS) and medication treatment response.

Methods: This is a single case report from an inpatient medically managed withdrawal level of substance use treatment care.

Results: A 38-year-old man with a history of opioid use disorder reported escalation of use including kratom to 7-OH complicated by the development of use disorder, tolerance, and withdrawal. The patient experienced a withdrawal syndrome consistent with opioid withdrawal upon stopping use, with COWS peaking at 14. The patient was treated with a short course of buprenorphine sublingual (daily dosing 2–8 mg) based on signs and symptoms. He did not want to remain on buprenorphine after the medically managed withdrawal period and was successfully tapered off buprenorphine during his treatment stay and transitioned to residential level of care.

Conclusions: Given the widespread availability of kratom and kratom-related products it is important for health care providers to screen for use and development of use disorder and be prepared to provide counseling, support, and treatment for patients.

Key Words: buprenorphine, kratom, 7-hydroxymitragynine, kratom use disorder

(*J Addict Med* 2026;20:416–418)

Kratom (*Mitragyna speciosa*) is a tree native to Southeast Asia. Leaves are used to make tea, dried and turned into powder, and processed to create additional products including concentrates and semi-synthetics. Kratom alkaloids have shown activity at opioid, serotonergic, dopaminergic, and noradrenergic pathways, and although a clear dose-response relationship has not been established, kratom products are reported to have both stimulant and sedative effects depending on dose.[1]

The FDA has designated kratom as a new dietary ingredient based on inadequate information to suggest that kratom "does not pose a significant or unreasonable risk of illness or injury." As such, any product that contains kratom is considered adulterated under section 402(a)(2)(C)(i).[2] Despite this determination by the FDA, states and municipalities have taken different approaches to kratom use, sale, and regulation. Some states have scheduled mitragynine and 7-hydroxymitragine (7-OH) and some municipalities have banned sales of kratom and related products. Many states have adopted legislation promoted by industry to create a regulatory structure for kratom products which mirrors herbal and dietary supplement regulation in direct conflict with the FDA. Kratom products are available for sale in convenience stores, head shops, and over the internet in states with and without kratom bans.[3]

Recently, there has been increased interest in semi-synthetic kratom alkaloid products containing primarily 7-OH. In kratom leaf preparations, 7-OH presence is minimal, but 7-OH is formed in the body via metabolism of mitragynine, the most abundant alkaloid in kratom, and can also be formed from oxidation of dried plant material. Mitragynine and 7-OH both demonstrate partial agonism at the mu opioid receptor; however, 7-OH demonstrates higher efficacy and is more potent.[4] Previous case reports have documented physiologic dependence, withdrawal upon stopping use, and development of kratom use disorder related to chronic daily use.[5] Here we present, to our knowledge, the first published case of a patient seeking inpatient medically managed withdrawal for treatment of 7-OH use.

METHODS

This is a single case report. The case was identified prospectively and data were extracted manually from the electronic health record. Patient consent was obtained for case publication.

Case Details

CC: "I started using 7 hydroxy kratom a few months ago and it got a lot worse. I want to stop."

Case: A 38-year-old man with a history of opioid use disorder (OUD) presented to treatment seeking

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medically managed withdrawal from 7-OH. The patient reported a history of oxycodone use starting at age 16. He developed opioid use disorder in his late teens and later stabilized on prescription buprenorphine maintenance at age 28 for 2 years. The patient was able to remain abstinent from opioids for 1 year after stopping buprenorphine maintenance and then started using kratom at age 31. The patient reported starting to use kratom "to get high" and because it helped his mood, gave him energy, and helped with pain. The patient used kratom capsules 4–5 times daily, totaling ~30 g. Three months before presentation, the patient switched to 7-OH tablets to self-treat neck pain and anxiety. The patient reported improvement in his neck pain, but his use quickly escalated and he developed problems related to his use. The patient reported that 7-OH use caused memory and anxiety problems. He experienced difficulty concentrating at work as a store manager if he didn't take 7-OH throughout his day.

Patient reported 7-OH is readily accessible at gas stations and smoke shops near his workplace. His use caused interpersonal problems as he was hiding his use and spending less time with his family and his regular pastimes. Patient reported he was taking larger and more frequent amounts to avoid withdrawal symptoms that occurred on stopping use, which included anxiety, insomnia, rhinorrhea, abdominal discomfort, restlessness, diaphoresis, and chills. The patient reported spending up to $60 daily, using eight 30 mg tablets daily, and taking them every 1 to 2 hours. Patient reported attempts to cut down or quit on his own 4 or 5 times, but he had not remained abstinent for more than 24 hours.

On presentation, the patient reported cravings 8/10 and withdrawal symptoms of anxiety, restless legs, and abdominal discomfort. His last use of 7-OH was on the day of presentation at 2 PM. At intake breathalyzer was negative and point-of-care urine drug testing was negative for all substances other than cannabinoids. Reflex confirmatory testing was negative for delta-9-THC-COOH. Urine mitragynine testing was negative. The patient was admitted at ~10:30 PM to medically managed withdrawal (ASAM level of care 3.7) with a plan to monitor Clinical Opiate Withdrawal Scale (COWS) and Vital Signs every 4 hours while awake. The care plan included initiation of buprenorphine if COWS > 13.

The patient had a 5-day stay in a closed medically managed withdrawal unit. Please see Table 1 for a summary of patient vital signs, COWS scores, and medication administration during the inpatient stay. The COWS score peaked on day 2 and buprenorphine was initiated. The patient received a total of 4 mg of buprenorphine on day 2, 8 mg of buprenorphine on day 3, 6 mg of buprenorphine on day 4, 4 mg of buprenorphine on day 5, and 2 mg of buprenorphine on day 6. Throughout the inpatient stay, comfort medications including clonidine, methocarbamol, hydroxyzine, trazadone, melatonin, and ibuprofen were available. On day 6 the patient stopped buprenorphine and he was transitioned to the residential level of care.

TABLE 1. Clinical Course Including Vital Signs, Medications Administered, and Clinical Opiate Withdrawal Scales

Day and Time	Blood Pressure (mm Hg)	Heart Rate (beats/min)	COWS	Medications	Dose (mg)
Day 1					
22:20	148/81	96	2		
Day 2					
7:34	126/98	86	4	Clonidine	0.1
10:35			8		
10:41				Ibuprofen	600
10:42				Methocarbamol	500
15:30	130/83	82	14		
16:00				Bup/nal HCl	2–0.5
19:45	122/71	77	4		
20:00				Duloxetine	30
21:34				Methocarbamol	500
21:34				Ibuprofen	600
22:00				Bup/nal HCl	2–0.5
Day 3					
7:57	136/83	81	2		
8:00				Bup/nal HCl	2–0.5
12:00				Bup/nal HCl	2–0.5
12:35				Hydroxyzine	50
18:00				Bup/nal HCl	2–0.5
18:55				Hydroxyzine	50
20:00				Duloxetine	30
21:12	124/79	84	6		
21:14				Ibuprofen	600
21:14				Methocarbamol	500
21:57				Melatonin	5
22:00				Bup/nal HCl	2–0.5
22:30				Trazodone	50
Day 4					
7:30				Bup/nal HCl	2–0.5
8:07	112/74	82	3		
14:30				Bup/nal HCl	2–0.5
19:35	117/75	77	1		
20:00				Duloxetine	30
21:30				Bup/nal HCl	2–0.5
21:48				Methocarbamol	500
21:49				Hydroxyzine	50
22:44				Trazodone	50
Day 5					
8:00				Bup/nal HCl	2–0.5
8:36	103/71	78	2		
10:22				Clonidine	0.1
10:22				Hydroxyzine	50
10:24				Ibuprofen	600
10:24				Methocarbamol	500
19:20	99/68	81			
20:00				Duloxetine	30
20:00				Bup/nal HCl	2–0.5
22:12				Trazodone	50
Day 6					
6:48				Ibuprofen	600
6:52				Methocarbamol	500
6:53	105/69	79			
8:00				Bup/nal HCl	2–0.5
11:30			2		
12:34				Clonidine	0.1
19:09	112/76	80			

Bup/nal HCl indicates sublingual buprenorphine/naloxone HCl.

DISCUSSION

This case demonstrates escalation of use from kratom to 7-OH complicated by the development of use

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disorder, tolerance, and withdrawal. The patient transitioned from being fully abstinent from opioids, then started using kratom for psychoactive effects, and in time transitioned to 7-OH for anxiety and pain. He met criteria for substance use disorder using the DSM-V checklist adapted for 7-OH, including physiologic dependence (tolerance and withdrawal) and criteria within the domains of impaired control and social dysfunction with continued use despite interpersonal problems and problems at work.

Real-world detoxification cases including vital signs and COWS scores before and after pharmacotherapy, to our knowledge, have not been previously described for 7-OH. This patient tried unsuccessfully to quit use on his own on multiple occasions before seeking medically managed withdrawal (detox) for 7-OH. This individual was admitted to inpatient medically managed withdrawal treatment as the case was billed under an opioid use disorder diagnostic code. Other patients requesting inpatient medically managed withdrawal have been pushed to the residential level of care or turned away due to inability to obtain insurance prior authorization and reimbursement for admissions related to kratom or 7-OH. In addition, questions about kratom use and more recently available semi-synthesized kratom products such as 7-OH mitragynine and mitragynine pseudoindoxyl are not routinely asked as part of standard intake and are not tested for via standard urine drug screen panels.

In this case, confirmatory testing was only sent for mitragynine and not 7-OH. As the kratom alkaloidal product market expands to include semi-synthetic or other derived products, testing panels and libraries will need to be expanded. In addition, it has been reported that laboratory testing for 7-OH is more challenging than testing for mitragynine due to a variety of factors.[6] Knowledge of kratom alkaloidal testing limitations and patient product use history is critical to the correct interpretation of testing results for kratom-related product exposure. Further, in this case, the patient had a positive point-of-care test for cannabinoids, but confirmatory testing was negative for delta-9-THC metabolites; this could suggest use of alternative cannabinoids (eg, delta-8-THC) that may cross-react with the carboxy-THC immunoassay.[7]

During the treatment course the patient experienced withdrawal symptoms captured via standardized COWS assessments consistent with opioid withdrawal. Previous case reports and series have documented kratom use disorder and successful treatment with varied courses of buprenorphine.[8–11] While maintenance daily doses of buprenorphine have generally increased in the age of fentanyl, for this individual, smaller doses of buprenorphine over 5 days were effective with adjunctive medications to control withdrawal symptoms. It is important for providers to dose buprenorphine to clinical effect as guidelines do not exist for kratom or 7-OH products. Another option that could be considered would be a single dose of extended-release injectable buprenorphine. However, this approach may be complicated by cost and logistical challenges related to medication ordering and delivery, off-label use, and prior authorization requirements. The patient described did not want to remain on buprenorphine after the medically managed withdrawal period and was successfully tapered off buprenorphine during his treatment stay. The duration of buprenorphine treatment should be patient-centered and consider the patient's history of substance use disorder, symptoms, and goals and desires.

Researchers have previously highlighted concerns about 7-OH products and concentrated kratom formulations. Given the widespread availability of these products it is important for health care providers to understand the risks of kratom and kratom-derived products, to screen for use and development of use disorder, and be prepared to provide counseling, support, and treatment for patients.

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7-Hydroxymitragynine Is an Active Metabolite of Mitragynine and a Key Mediator of Its Analgesic Effects

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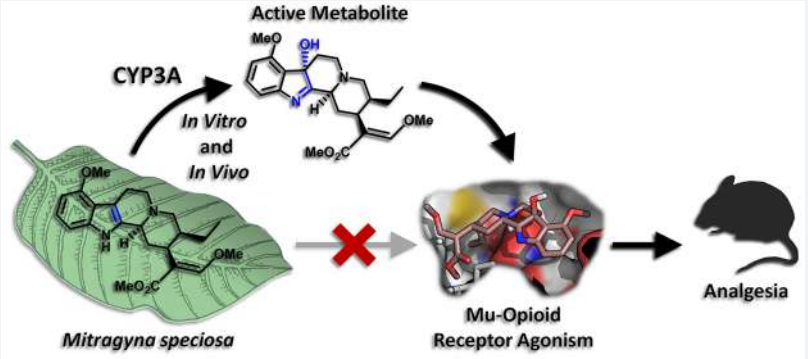
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Supporting Information



ABSTRACT: *Mitragyna speciosa*, more commonly known as kratom, is a plant native to Southeast Asia, the leaves of which have been used traditionally as a stimulant, analgesic, and treatment for opioid addiction. Recently, growing use of the plant in the United States and concerns that kratom represents an uncontrolled drug with potential abuse liability, have highlighted the need for more careful study of its pharmacological activity. The major active alkaloid found in kratom, mitragynine, has been reported to have opioid agonist and analgesic activity in vitro and in animal models, consistent with the purported effects of kratom leaf in humans. However, preliminary research has provided some evidence that mitragynine and related compounds may act as atypical opioid agonists, inducing therapeutic effects such as analgesia, while limiting the negative side effects typical of classical opioids. Here we report evidence that an active metabolite plays an important role in mediating the analgesic effects of mitragynine. We find that mitragynine is converted in vitro in both mouse and human liver preparations to the much more potent mu-opioid receptor agonist 7-hydroxymitragynine and that this conversion is mediated by cytochrome P450 3A isoforms. Further, we show that 7-hydroxymitragynine is formed from mitragynine in mice and that brain concentrations of this metabolite are sufficient to explain most or all of the opioid-receptor-mediated analgesic activity of mitragynine. At the same time, mitragynine is found in the brains of mice at very high concentrations relative to its opioid receptor binding affinity, suggesting that it does not directly activate opioid receptors. The results presented here provide a metabolism-dependent mechanism for the analgesic effects of mitragynine and clarify the importance of route of administration for determining the activity of this compound. Further, they raise important questions about the interpretation of existing data on mitragynine and highlight critical areas for further research in animals and humans.

INTRODUCTION

Mitragyna speciosa, often referred to by the common name kratom, is an evergreen tree native to the tropical jungles of

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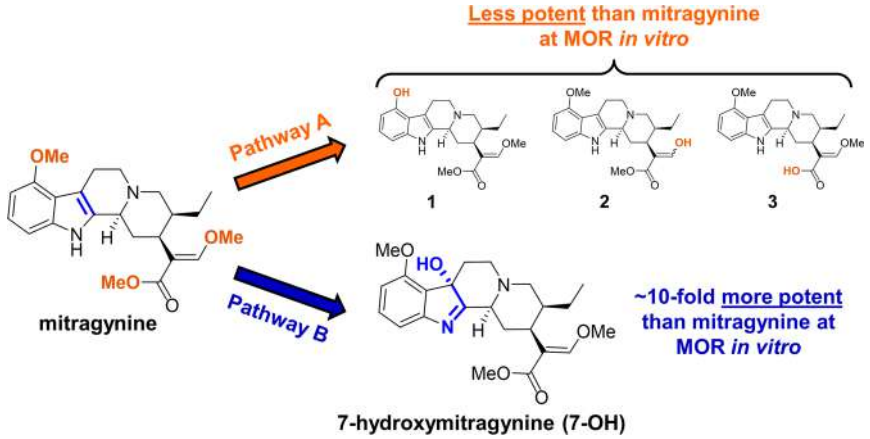


Figure 1. Known (Pathway A) and proposed (Pathway B) metabolic transformations of mitragynine.

Southeast Asia, where it has been used for at least the last century as a traditional medicine. In its native regions, leaves of the kratom tree are typically consumed as a tea or chewed directly, and are purported to induce stimulant and opioid-like analgesic effects in the user in a dose- and time-dependent manner. These effects have traditionally made kratom popular among agricultural workers, who use the plant to aid in long hours of hard labor. Kratom has also been reported to be useful in ameliorating withdrawal symptoms following cessation of opioid use, and it has achieved some popularity for this use in its native regions.[1−8]

Over the past decade, kratom use has become increasingly popular in the United States (US). Many thousands of US users have reported that consumption of kratom preparations (most often in the form of dried, powdered leaf material) is an efficacious treatment not only for physical pain, but also for mood and anxiety disorders, particularly in cases where other available treatments have either failed or elicited intolerable side effects.[9−13] A significant portion of users have also reported use of kratom as a tool to stop or reduce consumption of prescription or illicit opioids, a potential application that is presently of high interest given the ongoing opioid abuse epidemic in the US. Unfortunately, there have been no controlled clinical studies either examining these alleged therapeutic effects or quantifying any potential risks of kratom use. At the same time, kratom is primarily distributed through unregulated Internet sales and used outside medical supervision. Considering these factors, as well as the reported opioid-like effects and adverse health effects of the plant, the US Drug Enforcement Administration (DEA) and the US Food and Drug Administration (FDA) have both raised concerns that kratom represents a potential drug of abuse and public health threat.[14−16] In September 2016, the DEA announced its intent to place the active compounds found in kratom into Schedule I of the Controlled Substances Act but ultimately decided not to proceed with this action following a large public outcry from the kratom user community, including over 23 000 comments logged in the federal register docket for this issue.[13,17−19] Despite this turn of events, kratom at this time remains in regulatory limbo, with an uncertain future in the US. Accordingly, our laboratories have been working to develop a preclinical profile of kratom's biological effects, with the hope that such knowledge will prove useful for the development of future therapeutics based on kratom, as well as guide regulatory decision making on the plant itself.

Kratom contains a number of indole alkaloids that are believed to be the primary contributors to its psychoactive effects. Chief among these is mitragynine (Figure 1), which typically constitutes 1−2% of the dry leaf mass and up to approximately two-thirds of the total alkaloid content.[1,5] This compound is joined by 7-hydroxymitragynine (7-OH, Figure 1) in much lower concentrations, typically less than 0.05% of the dried leaf mass.[1,20] The other predominant alkaloids found in kratom leaf are speciogynine, speciociliatine, and paynantheine (each ∼0.2−0.5% of dry leaf mass), but at this time little is known about their pharmacology.[1,5]

We have reported that mitragynine and 7-OH are partial agonists of the human μ-opioid receptor (hMOR) in vitro, with 7-OH (EC_{50} = 34.5 nM; E_{max} = 47%) being ∼10-fold more potent than mitragynine (EC_{50} = 339 nM; E_{max} = 34%).[21] Further, both compounds are known to exhibit opioid-receptor-dependent analgesic effects in rodent models, with 7-OH again being significantly more potent.[6,22−25] At the same time, we have also found that both mitragynine and 7-OH are G protein-biased agonists of the mu-opioid receptor (MOR).[21,25] We have thus classified the kratom alkaloids as "atypical opioids" and have postulated that these compounds may provide a greater therapeutic window between analgesia and classical opioid side effects. For example, other MOR agonists that activate G protein signaling without recruiting the beta-arrestin pathway exhibit attenuated respiratory depression and reduced inhibition of gastrointestinal (GI) transit compared to classical opioids.[26−30] In fact, an early study on the pharmacology of mitragynine demonstrated its superiority compared to the classical opioid codeine in this regard, providing preliminary support for this hypothesis.[24] Similarly, mitragynine pseudoindoxyl, a chemical rearrangement product of 7-OH, has been found to be both G-protein biased and exhibit an improved therapeutic window in mice.[25]

Both methanolic and crude alkaloid extracts of kratom have also demonstrated analgesic properties in rodents.[1] However, it currently remains unsettled which of kratom's alkaloids are the most important contributors to the analgesic properties of such preparations or to those of the raw leaf. Despite its higher potency, we previously estimated on the basis of preclinical studies that the quantity of 7-OH contained in typical dry kratom leaf preparations is insufficient to induce opioid-like effects directly at typical doses consumed by human subjects.[1] At the same time, examination of the existing literature on the pharmacology of mitragynine revealed a number of unusual

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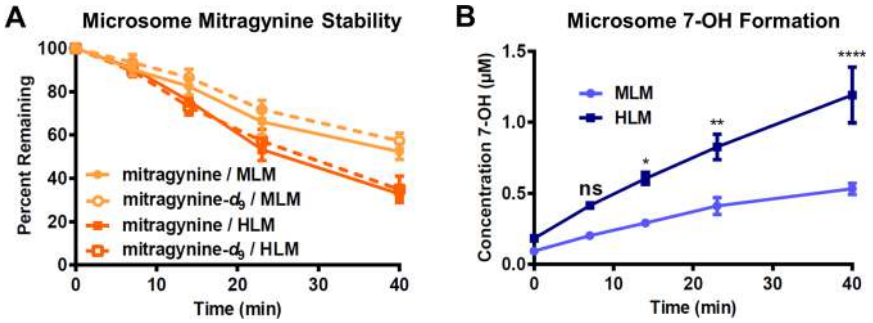


Figure 2. Decomposition of mitragynine and formation of 7-OH in liver microsomes was quantified by LC-MS/MS. (A) Mitragynine was unstable in both MLM and HLM, while deuteration of all three methyl groups, as in mitragynine-d_9, had little effect on microsomal stability. (B) Loss of mitragynine in microsomal preparations was accompanied by formation of 7-OH as a metabolite, and this conversion was more efficient in HLM. There was a strong main effect of species on 7-OH concentration. Two-way ANOVA: $F_{1,30} = 51.66$, $p < 0.0001$. ****$p < 0.0001$ relative to MLM, **$p < 0.01$ relative to MLM, *$p < 0.05$ relative to MLM, ns = $p > 0.05$ relative to MLM. All data points represent the means of two independent experiments with two incubations per experiment, with error bars representing ± SEM.

observations that complicated our understanding at the outset of the present investigations. First, despite low to moderate oral bioavailability (20−30% in rats), mitragynine has been found in prior investigations to be paradoxically *more* potent as an analgesic when administered by the oral (p.o.) and intraperitoneal (i.p.) routes compared to the subcutaneous (s.c.) route in rats and mice.[1,24,31] These findings suggested to us the involvement of an active metabolite produced via first-pass metabolism in mediating the analgesic activity of mitragynine.[1] At the same time, our own prior in vitro studies at the mouse mu-opioid receptor (mMOR) had shown that mitragynine acts as either a competitive *antagonist* or a partial agonist depending on the assay system used, calling into question the ability of this compound to have a direct agonist effect on MOR in rodents and consistent with a metabolite-driven analgesic effect.[21,25] However, mitragynine has also been found to induce analgesic effects when administered intracerebroventricularly (i.c.v), which argued conversely against the involvement of an active metabolite (unless metabolism occurs directly in the brain).[23] Therefore, we set out to further elucidate the pharmacological and metabolic mechanism(s) leading to mitragynine's analgesic effects.

RESULTS

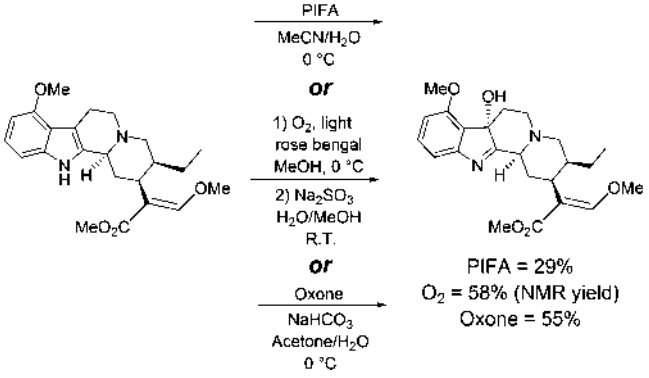
Known Metabolites Do Not Explain Analgesic Activity. Given our working hypothesis that an active metabolite might be involved in mediating the analgesic effects of mitragynine, we were interested in identifying metabolites that might exhibit agonist activity at MOR. The metabolites of mitragynine in rat and human urine had been previously studied, revealing a large number of metabolites modified primarily through demethylation at one or more of mitragynine's three O-methyl groups (Figure 1, Pathway A), followed by either glucuronidation or sulfation of the exposed nucleophile, or oxidative or reductive transformations of the acrylate moiety.[32] Considering previously observed structure−activity relationships (SAR) in the mitragynine scaffold, it was expected that the demethylated metabolites would exhibit similar or lower potency activity at MOR compared to mitragynine.[21] This was confirmed by synthesis of these compounds and testing using in vitro functional assays at hMOR (see Supporting Information, Table S1). Similarly, conjugated metabolites were also expected to be both inactive at MOR and have low penetration of the blood−brain barrier (BBB). Accordingly, when we began our investigation, known metabolites were insufficient to explain the opioid-mediated analgesic activity of mitragynine.

Deuteration of Mitragynine Has Little Effect on Metabolism in Vitro. Our above supposition was further strengthened by findings with a deuterated analogue. On the basis of the reported metabolic pathways of mitragynine, we synthesized an analogue of this compound fully deuterated at each of its three O-methyl groups, mitragynine-d_9 (see Supporting Information, Scheme S1), expecting that, through kinetic isotope effects, the metabolism of this derivative would be slowed. Surprisingly, when we examined the stability of mitragynine-d_9 head-to-head with its undeuterated counterpart in human liver microsomes (HLM) and mouse liver microsomes (MLM), stability was unchanged (Figure 2A). Therefore, we concluded that demethylation is only a minor pathway of hepatic metabolism (demethylated metabolites may accumulate slowly in the urine or be formed extrahepatically) and that the previously reported metabolites are unlikely to explain the hypothesized role of a hepatically formed metabolite in the analgesic activity of mitragynine.

Chemistry Suggests a New Site of Metabolism. With known metabolites seemingly unable to explain the observations pointing toward an active metabolite, we began exploring alternative metabolic pathways that might yield such a compound. For inspiration, we turned to known chemical transformations of indole alkaloids, including mitragynine, under oxidative conditions. For example, 2,3-disubstituted indoles are known to undergo functionalization under oxidation/halogenation conditions to afford the corresponding 3-substituted indolenines.[33] Specifically, mitragynine is known to be oxidized by [bis(trifluoroacetoxy)iodo]benzene (PIFA) to give 7-OH (Scheme 1).[6] During our ongoing synthetic explorations of the mitragynine scaffold, we also found that singlet oxygen[21] and potassium peroxymonosulfate (Oxone) were effective oxidants for the conversion of mitragynine into 7-OH (Scheme 1). Therefore, we postulated that the 2−3 indole double bond attacked by these chemical oxidants might also be a site for oxidation by cytochrome P450 enzymes (CYPs) to produce 7-OH as a metabolite (Figure 1, Pathway B). Given the much greater potency of 7-OH as an MOR agonist, we hypothesized that even minor conversion to this product might contribute significantly to mitragynine's analgesic activity and help to explain the apparent contradictions in the literature.

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Scheme 1. Conversion of Mitragynine to 7-OH by Chemical Oxidants



7-OH is a Mitragynine Metabolite in Vitro. To test our hypothesis, we monitored formation of 7-OH by liquid chromatography-tandem mass spectrometry (LC-MS/MS) during incubation of mitragynine with both HLM and MLM. In both microsome preparations, 7-OH was produced concomitant with disappearance of mitragynine (Figure 2B). Further, 7-OH appeared to be the major metabolite in each case, as the approximate decreases in molar mitragynine concentration from the starting level (2 μM) were accompanied by similar increases in 7-OH concentration. The metabolic conversion was more efficient in HLM, suggesting that an appreciation of interspecies differences is likely to be important for understanding the pharmacology of mitragynine. However, note that our findings conflict with an earlier report, which found that mitragynine was stable in microsomes.[34] The reason for this discrepancy remains unclear at this time, but it may relate to variable metabolic activity of microsome preparations from different sources. Further, we confirmed the low stability of mitragynine in HLM and concomitant formation of 7-OH in an independent laboratory (Biotranex, LLC, data not shown).

7-OH is Resistant to Phase I Metabolism. Having found that 7-OH was indeed formed from mitragynine in vitro, we next assessed whether this compound was itself stable in liver preparations. We found that 7-OH was stable in both HLM and MLM, with more than 90% remaining after 40 min of incubation (Figure S1). This finding that 7-OH was stable to oxidative Phase I metabolism was consistent with its accumulation during microsomal incubations of mitragynine.

Mitragynine and 7-OH are Stable in Plasma. Before proceeding further, we also tested the stability of mitragynine and 7-OH in blood plasma. Both mitragynine and 7-OH were highly stable in mouse plasma (Figure S2), indicating that plasma hydrolysis or other plasma metabolism does not contribute significantly to the biotransformation of these compounds.

Conversion of Mitragynine to 7-OH is Mediated by CYP3A Isoforms in Vitro. We next set out to determine which CYP isoform is responsible for conversion of mitragynine to 7-OH. Mitragynine was incubated in vitro with purified recombinant preparations of the five major human CYP isoforms (CYP3A4, 2C19, 2C9, 1A2, 2D6) alongside a reference substrate of each isoform as positive control. Decomposition of mitragynine was nearly complete in the presence of CYP3A4 (2% remaining at 60 min). In contrast, there was little or no decomposition in the incubations with CYP2C19, 2C9, 1A2, and 2D6 (77%, 99%, 96%, and 82% remaining at 60 min, respectively). The activity of each enzyme preparation was confirmed by nearly complete disappearance of the reference substrates (Figure 3A). During these incubations, the formation of 7-OH was also monitored by LC-MS/MS, revealing that formation of 7-OH was most robust in the presence of CYP3A4, whereas little conversion to 7-OH was observed in the incubations with other CYPs (Figure 3B).

Having identified CYP3A4 as the predominant metabolic pathway responsible for hepatic metabolism of mitragynine and concurrent formation of 7-OH using purified enzyme preparations, we next sought to confirm these results in liver microsomes, a more complex system with endogenous expression of relevant enzymes. To that end, mitragynine was incubated with HLM alone and in the presence of the CYP3A inhibitor ketoconazole (1 or 10 μM), the CYP2C19 inhibitor ticlopidine (20 μM), or a combination of the two. Ketoconazole robustly inhibited both decomposition of mitragynine (Figure 4A) and formation of 7-OH (Figure 4B), whereas ticlopidine had little effect. Analogous results were found in MLM (Figure S3). In sum, our findings

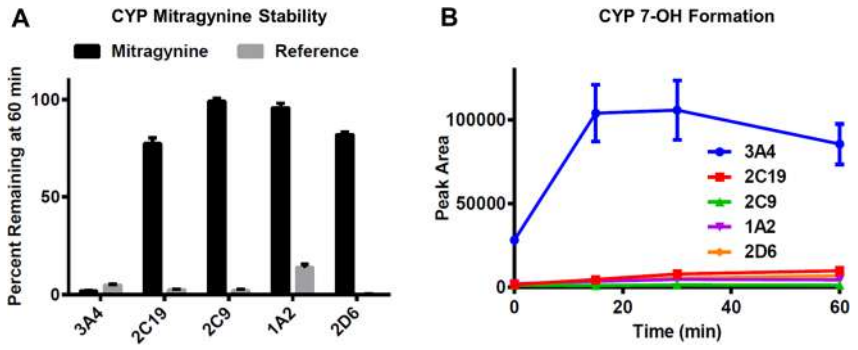


Figure 3. CYP3A4 mediates conversion of mitragynine to 7-OH. (A) Mitragynine was incubated in vitro with recombinant preparations of the five major human CYP isoforms alongside a reference substrate of each isoform as positive control. The relative percent remaining of mitragyine or reference substrate in each incubation was quantified by LC-MS/MS. Disappearance of mitragynine was most rapid in the presence of CYP3A4, whereas incubations with the other isoforms resulted in little or no decomposition. The activity of each preparation was confirmed by the nearly complete disappearance of the corresponding reference substrates. (B) Formation of 7-OH was monitored by LC-MS/MS during incubation of mitragynine with each of the recombinant CYP preparations. CYP3A4 resulted in the most robust conversion to 7-OH. All data points represent the means of two independent experiments with two incubations per experiment, with error bars representing ± SEM.

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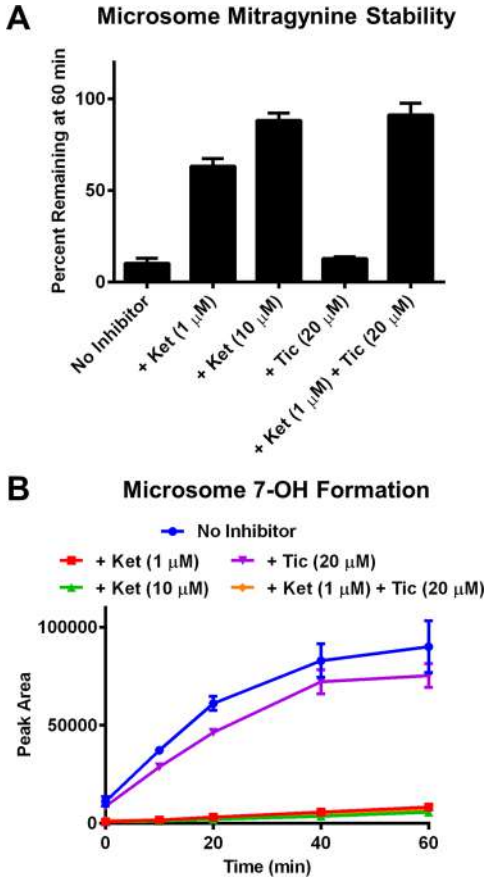


Figure 4. CYP3A-dependence of 7-OH formation was confirmed in HLM. Disappearance of mitragynine and formation of 7-OH was monitored by LC-MS/MS. (A) In HLM, the CYP3A isoform inhibitor ketoconazole (Ket) dose-dependently attenuated metabolic decomposition of mitragynine. In contrast, the CYP2C19 inhibitor ticlopidine (Tic) had little effect. (B) During incubation of mitragynine with HLM, ketoconazole (and to a small degree, ticlopidine) also inhibited formation of 7-OH. All data points represent the means of two independent experiments with two incubations per experiment, with error bars representing ± SEM.

demonstrate that mitragynine is converted to 7-OH in both mouse and human liver preparations and that this conversion is mediated by CYP3A isoforms (including CYP3A4 in humans).

Mitragynine is Analgesic in Mice Through an MOR-Dependent Mechanism. To build on the preceding results, we hoped to examine whether 7-OH is in fact formed as a metabolite of mitragynine in vivo. Further, we sought to test whether 7-OH, as a metabolite, could be a key mediator of the analgesic effects of mitragynine. However, before proceeding with these experiments, we first set out to confirm the analgesic activity of mitragynine in our own hands and to determine whether these effects are dependent on MOR.

For this purpose, we used the 129 mouse strain, which we found to be sensitive to mitragynine-induced analgesia, consistent with the documented high sensitivity of this strain to opioid agonists.[35,36] Further, genetic knockouts of opioid receptors are available on this genetic background. Dose–response curves were generated for mitragynine administered by both oral (p.o.) and subcutaneous (s.c.) routes using the tail-flick test, a classic rodent model for analgesic activity (Figure 5A). We found that mitragynine was much more potent when administered p.o. (ED_{50} = 2.1 mg/kg) than when administered s.c. (ED_{50} = 106 mg/kg), consistent with earlier literature reports. To establish the opioid receptor dependence of these effects, an efficacious analgesic dose of mitragynine (10 mg/kg, p.o.) was administered to MOR knockout (KO), κ-opioid receptor (KOR) KO, and δ-opioid receptor (DOR) KO mice, and to wild-type (WT) mice following pretreatment with the opioid receptor antagonist naloxone (1 mg/kg, s.c.). Both MOR KO and naloxone pretreatment significantly attenuated the analgesic effect of mitragynine, whereas KOR KO and DOR KO had no effect (Figure 5B). Therefore, the analgesic activity of mitragynine was found to be MOR-dependent, consistent with prior literature demonstrating pharmacological blockade of analgesia with the opioid receptor antagonist naloxone.[6,23]

7-OH is a Potent Analgesic in Mice Acting Through an MOR-Dependent Mechanism. Before further exploring our hypothesis that 7-OH is a key active metabolite of mitragynine, we also wanted to confirm the potent analgesic activity reported for this compound. A dose–response curve was generated for 7-OH in 129 mice using the tail-flick assay (Figure 5A). Consistent with previous reports,[22,25] it was found that 7-OH was a highly potent analgesic (ED_{50} = 0.6 mg/kg, s.c.), ~fivefold more potent than p.o. mitragynine. This high potency was consistent with our hypothesis that 7-OH formed as a metabolite, even in small quantities, could be responsible for mediating much of mitragynine's analgesic activity. Both MOR KO and naloxone pretreatment significantly attenuated the analgesic effects of 7-OH, whereas the analgesic effect was retained in KOR KO and DOR KO animals (Figure 5C). Therefore, the analgesic activity induced by 7-OH was also MOR dependent. We note that the reason for the slight enhancement of analgesia in KOR KO and DOR KO animals remains unclear, but some prior literature has suggested that blockade of DOR may enhance MOR-dependent analgesia.[37] Further, any potential antagonism of KOR or DOR by mitragynine or 7-OH, as has been suggested by in vitro studies,[21] would not be expected to contribute to analgesic effects directly, since antagonists of these receptors are not known to be analgesic on their own. Thus, our findings are consistent with agonism of MOR as the primary mechanism of analgesia for both compounds.

Conversion of Mitragynine to 7-OH also Occurs in Vivo. Having profiled the analgesic effects of mitragynine and 7-OH in mice, we next set out to confirm that the metabolic conversion that we had previously observed in vitro also occurred in living animals. Mice (129S1) were treated with mitragynine (10 mg/kg, s.c.), and plasma and brain samples were collected at 15 and 60 min and analyzed by LC-MS/MS for mitragynine and 7-OH. Both mitragynine (Figure 6A) and 7-OH (Figure 6B) were detected at both time points in both plasma and brain, confirming that 7-OH is indeed formed as a metabolite of mitragynine in vivo and that it enters the brain. These results also permitted a preliminary assessment of the BBB penetration of the two alkaloids by comparison of plasma to whole brain concentrations. By this metric, mitragynine brain penetration was very high (~1:1 brain/plasma), whereas 7-OH brain penetration was more modest (~1:5 brain/plasma). However, note that this estimation does not account for tissue binding and, thus, does not necessarily reflect free concentrations of drug.

It is interesting to note the differences between our findings in vitro in microsome preparations and those in vivo. In microsomes, we found that 7-OH was a major hepatic

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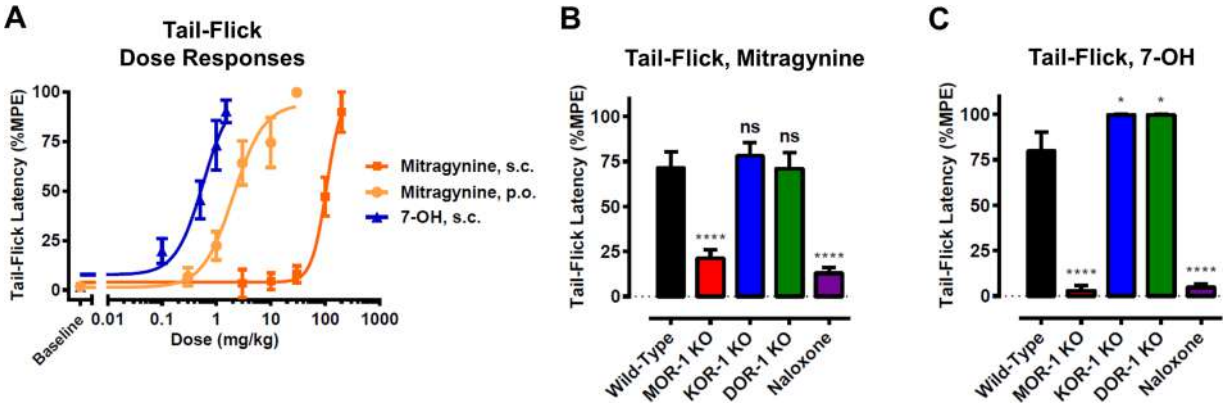


Figure 5. Analgesic activity of mitragynine in 129 mice. (A) Dose−responses of mitragynine and 7-OH in the tail-flick assay at time of maximal analgesic effect (15 min). Mice (n = 5−17 per dose, per treatment; 129S1 strain) were treated with ascending doses of mitragynine or 7-OH by the indicated route of administration in a cumulative dosing procedure, and tail-flick latency was recorded for each animal at 15 min post drug administration (peak analgesic effect). ED_{50}s (95% CI) were as follows: mitragynine, s.c. = 106 (57.4−195) mg/kg; mitragynine, p.o. = 2.05 (1.24−3.38); 7-OH, s.c. = 0.57 (0.19−1.7). The means of each point were calculated as percentage maximal possible effect (%MPE) [(observed latency − baseline latency)/(maximal latency − baseline latency)] × 100. Error bars represent ± SEM. (B) The analgesic effect of mitragynine (10 mg/kg, p.o.) in the tail-flick assay was evaluated 15 min post drug administration in WT, MOR-1 KO, KOR-1 KO, and DOR-1 KO mice, and WT mice following pretreatment with naloxone (1 mg/kg, s.c.), in two independent experiments (n = 5 per group per experiment, n = 10 total per group; 129S6 strain). The analgesic effect of mitragynine was attenuated by naloxone pretreatment and in MOR-1 KO, while the effect was found intact in KOR-1 KO and DOR-1 KO mice. One-way ANOVA: $F_{4,45}$ = 19.90, p < 0.0001. ****p < 0.0001 relative to WT, ns = p > 0.05 relative to WT. All values are expressed as the mean ± SEM. (C) The analgesic effect of 7-OH (1 mg/kg, s.c.) in the tail-flick assay was evaluated 15 min post drug administration in WT, MOR-1 KO, KOR-1 KO, and DOR-1 KO mice, and WT mice following pretreatment with naloxone (1 mg/kg, s.c.), in two independent experiments (n = 5 per group per experiment, n = 10 total per group; 129S6 strain). The analgesic effect of 7-OH was attenuated by naloxone pretreatment and in MOR-1 KO, while the effect was found intact in KOR-1 KO and DOR-1 KO mice. One-way ANOVA: $F_{4,45}$ = 101.7, p < 0.0001. ****p < 0.0001 relative to WT, *p < 0.05 relative to WT. All values are expressed as the mean ± SEM.

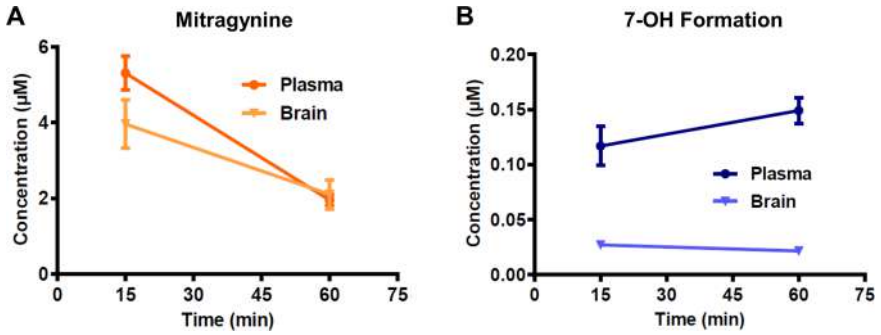


Figure 6. Conversion of mitragynine to 7-OH was confirmed in vivo in 129S1 mice using LC-MS/MS. (A) Mitragynine was detected in both the plasma and brains of mice treated with mitragynine (10 mg/kg, s.c.). n = 4 per time point for plasma; n = 8−9 per time point for brain. (B) At the same time, 7-OH was also detected in the plasma and brains of the same animals, but at lower concentrations.

metabolite. In contrast, in mice, 7-OH was found to be only a minor metabolite in terms of relative concentrations, with a mitragynine/7-OH ratio in plasma of ∼15:1 or more (dependent on time point). This suggests both inherent differences between the metabolic activity of in vitro and in vivo systems, as well as additional hepatic or extrahepatic routes of mitragynine and/or 7-OH metabolism not well accounted for by in vitro microsome preparations.

7-OH Contributes to the Analgesic Activity of Mitragynine as a Metabolite. Having shown that 7-OH was indeed a metabolite of mitragynine in mice, we last examined whether the concentration of this metabolite formed in vivo might be sufficient to contribute to mitragynine's opioid-mediated analgesic effects. To demonstrate this, we planned an experiment in which the brain concentration of 7-OH observed as a metabolite following administration of an analgesic dose of mitragynine would be compared to the brain concentration of 7-OH observed following direct administration of an equianalgesic dose of 7-OH. Under these conditions, we hypothesized that 7-OH concentrations would be similar if this metabolite was in fact playing a significant role in mediating the analgesic effects of mitragynine. In testing this hypothesis, we chose to administer both compounds by the s.c. route, as we found that it was difficult to obtain reliable dose−response curves for p.o. 7-OH, likely as a result of variable p.o. absorption of the compound. Thus, s.c. administration was necessary to allow the PK of mitragynine and 7-OH to be compared following the same route of administration, which was desirable from the standpoint of consistency.

Equianalgesic doses of mitragynine (140 mg/kg, s.c.) and 7-OH (0.7 mg/kg, s.c.) were selected based on the previously obtained dose−response curves (∼1.3-fold ED_{50}, Figure 5A). Mice (129S1) were treated with the selected doses, and analgesic activity was confirmed at 15 min in the tail-flick assay. As expected, there was no significant difference in tail-flick latency between the two groups (Figure 7A). Immediately after

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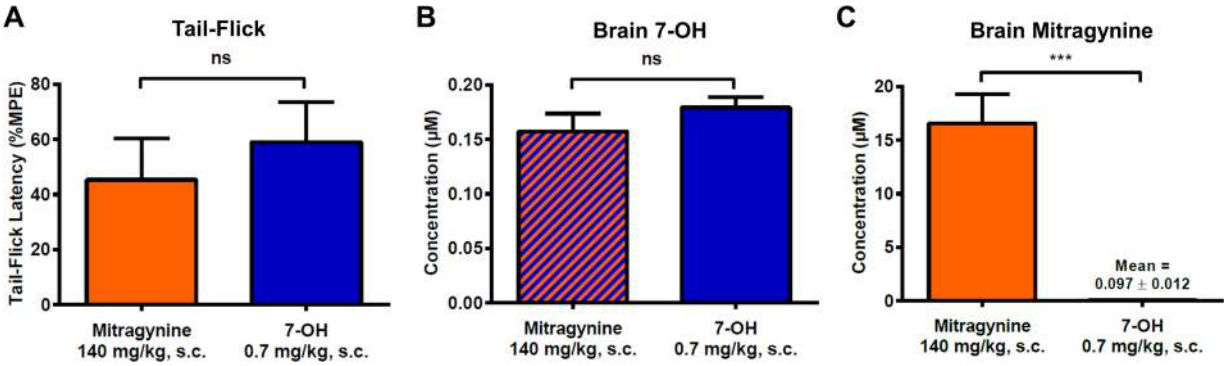


Figure 7. 7-OH formed as a metabolite is sufficient to explain mitragynine's analgesic effect in 129S1 mice. (A) Selected doses of mitragynine (140 mg/kg, s.c.) and 7-OH (0.7 mg/kg, s.c.) were equianalgesic at 15 min in the tail-flick assay and induced ∼50% MPE. n = 10 per treatment. Two-tailed t test, t(18) = 0.66, p = 0.52. ns = p > 0.05. (B) 7-OH was detected in the brains of the same animals at 15 min (sacrifice and sample collection immediately after tail-flick), and there was no difference in mean brain concentration between the two treatments. n = 9 for mitragynine, n = 10 for 7-OH. Two-tailed t test, t(17) = 1.19, p = 0.25. ns = p > 0.05. (C) Mitragynine was detected in the brains of the same animals at 15 min, and there was a significant difference between the two treatments, with a 170-fold higher brain concentration of mitragynine in the animals directly treated with this drug compared to those treated with 7-OH. n = 10 for mitragynine, n = 6 for 7-OH (four values below the lower limit of quantitation, 5 ng/g, were excluded from analysis). Two-tailed t test, t(13) = 4.88, p = 0.0003. ***p < 0.001. All bars represent mean ± SEM.

determination of tail-flick latency, mice were sacrificed, and brain samples were collected for analysis. There was no significant difference in the mean brain concentration of 7-OH found in the mitragynine group (formed as metabolite) compared to that found in the 7-OH group (from direct administration) (Figure 7B), consistent with 7-OH being the primary mediator of central analgesic activity in both cases. At the same time, very brain high concentrations (16.6 ± 2.7 μM; mean ± standard error of the mean (SEM)) of mitragynine were observed in the mitragynine-treated animals, while minimal concentrations (0.097 ± 0.012 μM; mean ± SEM) were observed in the 7-OH-treated animals (Figure 7C). This 170-fold difference in mitragynine brain concentration between the two groups had no significant effect on analgesia (Figure 7A), despite the fact that the mitragynine brain concentration in the mitragynine-treated group was ∼50-fold higher than the compound's binding affinity for mMOR (0.23 μM[21]). Accordingly, we conclude that 7-OH formed as a metabolite is sufficient to explain the opioid-mediated analgesic activity of mitragynine and that the parent compound does not make a significant contribution to its own analgesic activity in mice.

Although it does not appear to play an important role in mediating analgesia, the above observation of small quantities of mitragynine formed as a metabolite of 7-OH in vivo was surprising to us. We did not expect to observe this reductive metabolic process, which is effectively the reverse of the observed oxidative metabolism of mitragynine to 7-OH. However, prior literature had indeed reported the partial conversion of 7-OH to mitragynine in vitro in liver microsomes and simulated gastric fluid.[34] The exact mechanism of this transformation remains unclear at this time.

DISCUSSION

In the present report, we provide evidence that hepatic formation of 7-OH as a metabolite is important in mediating the analgesic activity (and presumably other MOR-mediated effects) of mitragynine, the major active alkaloid of the kratom plant. The analgesic effects of mitragynine and 7-OH each depended on activation of MORs. Further, the analgesia induced by mitragynine appears to depend largely on formation of 7-OH as a metabolite and not on the parent compound. Accordingly, the pharmacological profile of mitragynine may mirror that of 7-OH under certain circumstances, assuming appropriate dose corrections are made to account for the efficiency of conversion to this active metabolite in the system under study. It should be mentioned that during the preparation of this manuscript, other groups separately reported the formation of 7-OH as a metabolite of mitragynine in vitro and in rats, but the mechanistic significance of this finding was not established through pharmacological studies.[38,39] Here we confirmed these concurrent research findings and provided the essential in vivo link demonstrating the pharmacological relevance of 7-OH.

Our results highlight the critical need to consider metabolic pathways and interspecies differences when interpreting preclinical data on kratom or attempting translation of such to man. For example, greater or lesser conversion (relative to mice) of mitragynine to 7-OH in a given species would be expected to increase or decrease, respectively, the potency of mitragynine's opioid activity in that species. At the same time, if similar metabolic conversion is required for the expression of opioid agonist activity in humans, there exists the possibility that metabolic saturation at high doses might provide a built-in ceiling to the opioid effects of kratom, improving the inherent safety of this material. In fact, preliminary evidence of such metabolic saturation can be inferred from the present results, where the 7-OH brain concentration (0.157 μM) at 15 min after a high dose (140 mg/kg, s.c.) of mitragynine was only around sixfold higher than that (0.027 μM) formed from a 14-fold lower dose (10 mg/kg, s.c.). However, full PK profiles at multiple doses will be needed to confirm the existence of such a phenomenon. Likewise, metabolic differences between individuals will need to be considered, and some individuals may be more sensitive to the opioid agonist effects of kratom by virtue of more efficient CYP3A metabolism.

Indeed, here we demonstrated that 7-OH is formed from mitragynine in HLM and that this conversion appears more efficient compared to that in MLM. Accordingly, we expect that 7-OH will also be observed as a metabolite in humans. This hypothesis is supported by a post-mortem toxicological

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report from a kratom-associated fatality, where mitragynine and 7-OH were quantified in the blood and urine of the decedent.[40] Interestingly, the mitragynine/7-OH ratios found in blood and urine were ∼7:1 and ∼1.5:1, respectively. It is our opinion that such high relative concentrations of 7-OH compared to mitragynine would be impossible to achieve in the circulation through direct absorption of the minimal quantities of 7-OH found in kratom leaf preparations and commercial extracts, where the mitragynine/7-OH ratio is typically greater than 50:1. Instead, assuming no adulteration of the ingested material, the high concentrations of 7-OH observed in this case support the hypothesis that 7-OH is also formed as an important metabolite in humans. PK studies will ultimately be required to elucidate the importance of 7-OH as a mitragynine metabolite in man.

It is interesting to note the apparent parallel of mitragynine to codeine, a classical opioid that depends on metabolic conversion to the active metabolite morphine by CYP2D6 for inducement of opioid effects.[41] Accordingly, individuals with genetic mutations leading to abnormally high or low activity of CYP2D6 are more or less sensitive to codeine, respectively.[41] However, life-threatening respiratory depression with codeine is rare even in cases of intentional overdose, potentially due to metabolic saturation.[42] Considering that preclinical studies have found mitragynine to induce less respiratory depression than codeine,[24] which is itself relatively safe compared to other classical opioids, we hypothesize that severe respiratory depression or death induced through an opioid mechanism following oral mitragynine (or kratom) consumption is likely to be rare. Consistent with this hypothesis, mitragynine has been found to have low toxicity in mice when administered orally (LD_{50} > 400 mg/kg).[31,43] Further, the active metabolite 7-OH has itself been found to be a partial, G protein-biased agonist at MOR, signaling properties that are hypothesized to lead to improved respiratory safety among MOR agonists.[21,25] Overall, these considerations are consistent with the limited reports of kratom overdose deaths (44 in total for all time as of 2017, despite millions of worldwide users)[14] in comparison to all opioid overdose deaths (∼50 000 in 2017 alone).[44] However, such hypotheses must be confirmed in humans before definitive conclusions are reached regarding the respiratory safety of mitragynine or kratom.

Metabolic processes are also likely to be an important determinant of the abuse liability of kratom and its alkaloids. It has recently been found in two independent studies that mitragynine does not support self-administration (SA) in rats, while 7-OH is self-administered.[45,46] We suspect that this apparent contradiction may be explained, at least in part, by the metabolic effects reported here. Mitragynine has been shown to be more potent (in terms of analgesic effects) in rats by the p.o. route[24] versus the intravenous (i.v.) route used in the reported self-administration studies, thus complicating appropriate dose selection and challenging the relevance of i.v. SA studies for assessing the abuse liability of p.o. mitragynine (or kratom). Further, the likely necessity of metabolic conversion of mitragynine to 7-OH for the induction of opioid agonist effects might result in a temporal dissociation between drug administration and experience of rewarding effects, further complicating the interpretation of SA studies. Lastly, certain individuals with enhanced or deficient CYP3A4 activity may experience greater or lesser reinforcing effects following mitragynine (or kratom) consumption depending on the extent of conversion to 7-OH. Again, a comparison to codeine is warranted, as the abuse liability of this compound has been found to depend on formation of active metabolites and vary depending on interindividual differences in CYP metabolic efficiency.[47] There also remains the possibility that mitragynine exerts a buffering effect on the opioid agonist activity of its 7-OH metabolite, either through direct competitive antagonism at MOR or through other as-yet-unknown pharmacological mechanisms. Ultimately, only controlled clinical trials will be able to definitively resolve these issues for mitragynine.

In addition to interspecies metabolic differences, the in vitro pharmacology of mitragynine at the opioid receptors is also species-dependent and complicates the translation of the present results to humans. For example, we have previously shown in an in vitro assay that mitragynine acts as a competitive antagonist at the mMOR but as a weak partial agonist at the hMOR.[21] Accordingly, in humans, mitragynine itself might contribute to its own analgesic effects directly via the parent compound, in contrast to the present mouse study, where most of the analgesic effects appeared to be induced through the 7-OH metabolite. The interplay between mitragynine and 7-OH in mediating kratom's purported analgesic effects will thus require further exploration in humans.

Lastly, it is important to mention that extracts of the kratom plant have been found to both inhibit and induce various CYP enzymes.[48,49] Accordingly, it remains a possibility that the mixture of compounds contained in the plant or its extracts may present a distinct opioid pharmacology compared to isolated mitragynine, by virtue of interference of other chemical species with the hepatic conversion of mitragynine to 7-OH. Thus, additional study of metabolic drug−drug interactions among kratom compounds will also be necessary.

CONCLUSION

We have found that the opioid pharmacology of mitragynine is complicated by the important role of an active metabolite, 7-OH, in mediating its activity. This apparent requirement for metabolic activation is consistent with the higher analgesic potency of mitragynine administered orally versus parenterally in animals. Further, it suggests a possible explanation for the seemingly improved safety profile of mitragynine compared to classical opioid agonists. However, the critical involvement of hepatic metabolism also complicates our understanding of mitragynine's pharmacology and introduces the possibility of interindividual variability in the compound's potential therapeutic effects and side effects. We believe mitragynine and related compounds have great potential as future therapeutics, but metabolic processes must be carefully considered as the field continues to advance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscentsci.9b00141.

Opioid functional activity of desmethyl metabolites, additional microsome and plasma stability experiments, formation of 7-OH in mouse microsomes, synthetic scheme and procedures for preparation of deuterated analogues, biological procedures (PDF)

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Notes

The authors declare the following competing financial interest(s): A.C.K., D.S., J.A.J., S.M., and G.W.P. are named as inventors on patent applications related to mitragynine analogs, which may lead to royalties or other licensing revenues from future commercial products.

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Investigating Opioid Receptor Activity Through Biocatalytic Halogenation and Oxidation of Mitragynine

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Abstract

The development of safer opioid therapeutics remains an urgent challenge, given the limitations and adverse effects of current μ-opioid receptor (MOR) agonists. Natural products such as mitragynine, the predominant alkaloid from *Mitragyna speciosa* (kratom), provide structurally complex scaffolds for probing MOR signaling bias and structure–activity relationships. However, enzymatic strategies for mitragynine modification remain largely restricted to native plant or nonnative eukaryotic enzymes, which are often insoluble, low yielding, or incompatible with heterologous expression, limiting their use for late-stage derivatization. Here, we establish a modular biocatalytic platform that repurposes non-cognate oxidative enzymes for in vivo diversification of mitragynine. Using flavin-dependent monooxygenases (CtdE) and an orphan halogenase (D3) from *Saccharophagus degradans*, we achieved site-selective *N*-oxidation and C10 bromination, respectively, and demonstrate that these reactions can be combined in a single *E. coli* host to perform sequential halogenation and oxidation directly from the parent alkaloid. The resulting analogues, including 10-bromo-mitragynine and its *N*-oxide, exhibit distinct MOR binding and G protein signaling profiles while failing to recruit β-arrestin. Collectively, these findings introduce a generalizable whole-cell platform for programmable, multi-step alkaloid functionalization and highlight how biocatalysis can drive pharmacology to guide the discovery of novel bioactive scaffolds.

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BRIEFS

A platform integrating non-native enzyme catalysis, computational docking, and biosynthetic engineering enables selective derivatization of mitragynine and related alkaloids. This strategy expands the chemical space accessible for opioid receptor-targeted small molecules and offers a new approach to diversifying bioactive natural products.

Keywords

Mitragynine; Alkaloid biosynthesis; Biocatalysis; Flavin-dependent halogenase; Flavin-dependent monooxygenase; Opioid receptor ligands; Computational docking; Structurebased design; Biosynthetic pathway engineering; Chemodiversity; Natural Products

INTRODUCTION

The opioid epidemic remains a critical public health crisis, with addiction to both natural and synthetic opioids causing staggering mortality rates globally. In 2022, over 100,000 lives were lost to opioid overdoses in the United States alone, underscoring the urgent need for safer therapeutic options.[1] Beyond the immediate risk of overdose, chronic opioid use is fraught with severe dangers to human health, including tolerance development, diminished analgesic efficacy, and debilitating side effects such as nausea, constipation, and respiratory depression.[2,3] Although opioids are indispensable for managing acute and chronic pain, their significant risks necessitate alternative approaches that can provide effective analgesia without the associated liabilities.

Mitragyna speciosa (Korth.) Havil. Rubiaceae *(M. speciosa)*, commonly known as kratom, is a medicinal plant native to Southeast Asia that has been used for centuries to combat fatigue and pain. Traditionally, field workers in Thailand relied on kratom to enhance endurance during intensive labor. In recent years, kratom has gained popularity in the United States, with approximately 1.7 million Americans aged 12 and older reporting its use in 2021.[4] Despite its classification as a controlled substance in several regions of the world, growing preclinical and clinical evidence suggests that kratom's alkaloids possess promising therapeutic potential with a favorable safety profile.[5–20]

The primary alkaloid in kratom, mitragynine, exhibits unique pharmacological properties. Mitragynine is a partial agonist of the μ-opioid receptor (MOR), providing analgesic effects with a reduced risk of addiction and respiratory depression compared to conventional opioids.[7,9,11,19–22] It interacts with multiple opioid receptor subtypes, including μ-, κ-, and δ-opioid receptors (MOR, KOR, and DOR), resulting in a multifaceted pharmacological profile.[11,12,20,22] Activation of MOR and KOR receptors primarily mediates its analgesic effects, while interaction with DOR receptors has shown potential in alleviating morphine dependence and reducing alcohol intake.[7,8,20,21] These diverse effects underscore mitragynine's potential as a scaffold for developing safer pain management therapies and addressing substance use disorders.

Kratom contains over 50 structurally diverse indole alkaloids with varying functional groups, stereochemistry, and cyclization patterns.[9,18,19,23,24] These structural variations

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drive differences in receptor engagement and signaling, making kratom a promising reservoir for analgesic drug discovery.[6,7,11,19,21] Yet, efforts to expand the chemical space around mitragynine have largely relied on semi-synthesis or total synthesis, which also offer access to novel analogues with altered pharmacology, but can be labor intensive. Although several plant-derived biosynthetic enzymes have been used to access modified kratom alkaloids[25–27], these systems often suffer from poor solubility, limited expression in microbial hosts, or originate from orphan biosynthetic gene clusters making them less amenable to reconstitution and use for late-stage functionalization in heterologous systems. Currently, no studies to date have demonstrated the use of microbial nonnative enzymes—particularly flavin-dependent halogenases (FDHs) or monooxygenases (FMOs)—for direct modification of mitragynine or its analogues. This leaves a critical gap in the development of biocatalytic platforms for scaffold diversification.

In this study, we expand the chemical diversity of mitragynine using a microbial biocatalytic system that enables late-stage functionalization of mitragynine using non-cognate, readily expressed FDHs and FMOs. Through screening of candidate biocatalysts, we identified monooxygenases capable of catalyzing site-selective *N*-oxidation of mitragynine, as well as halogenases that install site-specific C-H bromination at non-activated positions on the mitragynine scaffold. Although the FMOs used in this study typically catalyze spirocyclization of native indole substrates, they instead selectively oxidized the mitragynine tertiary amine. To streamline our approach, we engineered *E. coli* to co-express select halogenases and monooxygenases, establishing a one-pot, multi-enzyme platform for derivatization directly from the parent scaffold. This work not only expands the chemical space around mitragynine but also demonstrates that readily available, heterologous enzymes from diverse microbial sources can be systematically deployed for scaffold modification, highlighting their potential as generalizable tools for diversifying complex alkaloids. For several analogues in this study, G protein-biased agonism at the MOR and revealed new structure activity relationships (SARs) at human opioid receptors, underscoring the utility of this platform for generating next-generation analgesic scaffolds.

RESULTS AND DISCUSSION

Establishment of a Panel of Native Mitragynine Analogues for SAR Benchmarking

To support structure–activity relationship (SAR) studies at the MOR, we extracted a panel of naturally occurring mitragynine analogues from green and red vein *Mitragyna speciosa* (kratom) cultivars. These included paynantheine (**2**), a partially saturated analogue; speciociliatine (**3**), a stereoisomer of mitragynine; speciofoline (**4**); and corynoxine A (**5**), which exhibit distinct ring rearrangements and stereoelectronic features (Figure 1, Supplementary Figures S1–S2). This panel provided a natural benchmark for assessing the pharmacological and structural consequences of enzymatic scaffold remodeling, serving as chemical reference points for later functional comparisons with biocatalytically generated derivatives.

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Biocatalytic Diversification of Mitragynine

To identify enzymes capable of late-stage and non-native diversification of the mitragynine (**1**) scaffold, we surveyed a panel of flavin-dependent monooxygenases (FMOs) and halogenases (FDHs) known to act on indole-containing alkaloids. This approach aimed to extend the catalytic repertoire of mitragynine biotransformation beyond plant-derived enzymes, which are often insoluble or poorly expressed in heterologous systems. Enzyme candidates were prioritized based on preliminary in silico docking analysis (Supplementary Figure S3).[28,29,30,31]

Nitrogen Oxidation of Mitragynine by Flavin-Dependent Monooxygenases (FMOs)

We initially targeted oxidative spirocyclization of mitragynine because its spirocyclic analogue, mitragynine pseudoindoxyl, exhibits full MOR agonism.[19] To probe whether a similar structural rearrangement could be accessed enzymatically, we screened four flavin-dependent monooxygenases (FMOs)—CtdE, PhqK, NotI, and NotI′—previously characterized for indole spirocyclization via 2,3-epoxidation in natural product biosynthesis.[28,29,32] Enzymatic assays revealed that CtdE, PhqK, and NotI each produced a single oxidized product (m/z 415.2), while NotI′ was inactive against mitragynine (Figure 2A&B, Supplementary Figure S4). Preparative scale reactions and full structural characterization identified this product as mitragynine-4(*N*)- oxide (**6**), formed by selective oxidation of the tertiary amine within the secologanin-derived substructure. Diagnostic HMBC correlations confirmed *N*-oxide formation (Supplementary Figures S23–S27).

In its native biosynthetic context, CtdE catalyzes the formation of a transient indole 2,3-epoxide intermediate that undergoes a semi-pinacol rearrangement, producing the characteristic spirocyclic scaffolds observed in multiple indole alkaloid pathways.[28,29,32] This mechanistic step—well established in prior work from our laboratory —defines the canonical trajectory of FMO-mediated spirocyclization.[28,29,32] However, when challenged with mitragynine, CtdE diverged sharply from this pathway, performing *N*-oxidation without detectable epoxidation or rearrangement.

This unexpected chemoselectivity likely reflects substrate-imposed steric and conformational constraints that disfavor productive alignment of the indole π-bond with the C4a- hydroperoxyflavin intermediate necessary for epoxide formation.[28,29,32,33,34] Instead, the more accessible tertiary amine is proposed to be positioned for nucleophilic attack on the flavin-peroxide species, redirecting the reaction toward amine oxidation (Supplementary Figure S8).

Mechanistically, this reactivity mirrors that of the fungal FMO OxaD, where subtle hydrogenbonding interactions (e.g., involving Asp63) dictate whether the flavin intermediate supports *N*- hydroxylation or indole epoxidation.[29,33,34] Such substrate-dependent remodeling underscores the inherent plasticity of FMO catalysis where small geometric or electronic perturbations can reroute oxidative trajectories away from canonical semi-pinacol rearrangement toward non-native *N*- oxidation.

Collectively, these results demonstrate that non-cognate FMOs can be co-opted to perform selective, non-native oxidation chemistry on alkaloid scaffolds, expanding their functional

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scope beyond native spirocyclization. This tunable balance between indole 2,3-epoxidation and tertiary- amine oxygenation highlights FMOs as versatile biocatalysts for late-stage derivatization of complex natural products, offering a platform for engineering oxidation specificity through substrate design or potentially active-site modification.

Biocatalytic Halogenation of Mitragynine

Initial attempts to halogenate mitragynine using the fungal FDH MalA[30] failed to generate a detectable product, underscoring its narrow substrate scope and selectivity for the bicyclo[2.2.2]diazaoctane alkaloid class. To overcome these limitations and identify more permissive halogenases, we turned to a family-wide FDH sequence similarity network (SSN) strategy, which had previously revealed subnetworks enriched in enzymes capable of acting on bulky substrates.[35] From two such subnetworks, we selected 21 genes, and expression of 12 in *E. coli* yielded soluble proteins. All 12 enzymes catalyzed bromination of at least one of six test substrates, though no chlorination activity was observed. (Supplementary Table 1).

Among these, an orphan FDH from *Saccharophagus degradans* (designated D3) exhibited the highest activity on bulky substrates and was selected for mitragynine screening. D3 catalyzed regioselective bromination at the indole C10 position, affording 10-bromo-mitragynine (**7**) in ~50% assay yield (Figure 2A&C). LC–MS and 2D NMR confirmed the site of halogenation, consistent with a canonical electrophilic aromatic substitution mechanism (Supplementary Figures S28–S32). D3 also brominated paynantheine (**2**) with ~65% conversion (Supplementary Figure S6), demonstrating its substrate tolerance and suitability as a general scaffold-modifying enzyme for kratom alkaloids.

This discovery underscores that naturally occurring, non-engineered bacterial halogenases can exhibit inherent flexibility toward complex plant alkaloids, thus expanding the synthetic biology toolbox for selective, late-stage halogenation.

Combinatorial Biocatalysis: N-oxidation of Halogenated Mitragynine Derivative

To evaluate the potential for sequential, orthogonal biocatalytic functionalization, we subjected 10-bromo-mitragynine (**7**) to oxidation by FMOs CtdE and NotI. Both enzymes converted **7** fully to a new product (**8**, m/z 493.1) corresponding to the *N*-oxide derivative (Figure 3, Supplementary Figures S33–S37). Remarkably, this transformation proceeded with higher efficiency than oxidation of unmodified mitragynine, revealing that the halogenated substrate (**7**) can enhance subsequent enzymatic turnover.

This synergy likely reflects halogen-induced substrate preorganization within the FMO active site: in silico modeling suggested that bromination improved binding affinity and oriented the tertiary amine directly toward the reactive flavin, facilitating oxygen transfer (Supplementary Figure S8). This cooperative reactivity between distinct oxidative enzymes demonstrates that noncognate FMOs and FDHs can be integrated into a modular cascade capable of multi-step scaffold diversification.

During extended handling in oxygenated aqueous solution, we observed that the isolated *N*- oxide intermediate (**8**) gradually converted to the corresponding 3-dehydro-10-bromo-

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mitragynine (**9**), as confirmed by LC–MS analysis (Supplementary Figures S9). Although freshly prepared material was supplied for all downstream activity assays, this spontaneous dehydration suggests that low-level formation of the dehydro species may still have occurred under assay conditions. This behavior aligns with the known lability of indole *N*-oxides and underscores their utility as transient yet chemically versatile intermediates for further enzymatic or synthetic diversification.[18]

Together, these findings establish a versatile biocatalytic framework in which halogenation and oxidation can be sequentially or concurrently leveraged to expand alkaloid chemical space—laying the foundation for programmable, multi-enzyme pathways that generate structurally diverse, pharmacologically tunable mitragynine analogues.

Engineering a Programmable Whole-Cell Platform for Alkaloid Diversification

To overcome the long-standing synthetic challenge of modifying complex alkaloids through multi-enzyme cascades in vitro[36–39], we established a programmable whole-cell biocatalytic platform in *E. coli* capable of late-stage mitragynine diversification in vivo. This system executes sequential C–H halogenation and *N*-oxidation of the mitragynine scaffold, eliminating the need for enzyme purification, intermediate isolation, or cofactor supplementation. Conceptually, this represents a shift from linear enzyme-centric biocatalysis to an integrated, modular framework for sustainable alkaloid tailoring.

We first validated this approach by co-expressing the orphan halogenase, D3, with the flavin reductase RebF[40] to regenerate reduced flavin cofactors in vivo. Upon mitragynine (**1**) feeding, engineered *E. coli* catalyzed site-selective bromination at C10 to generate 10-bromo-mitragynine (**7**) with ~50% conversion within two hours, demonstrating that oxidative halogenation can proceed efficiently in a living host (Figure 4).

Building on this biocatalytic platform, we integrated the D3-RebF system with the flavin-containing monooxygenase CtdE to construct a dual-enzyme cascade for consecutive C–H bromination and *N*-oxidation. Cultures expressing all three enzymes produced both 10-bromo- mitragynine (**7**) and its corresponding *N*-oxide analogue (**8**), with the latter reaching ~25% conversion at eight hours after mitragynine (**1**) feeding (Figure 5). Temporal analysis confirmed a directional sequence of halogenation followed by oxidation—reflecting the intrinsic substrate preferences of D3 and CtdE and validating their functional compatibility in vivo (Supplementary Figure S10).

This combinatorial enzymatic sequence—executed entirely within a living cell—demonstrates that evolutionarily unrelated oxidative enzymes can be co-expressed and functionally coupled to achieve cooperative late-stage alkaloid diversification. The resulting platform is programmable, sustainable, and can potentially be expanded to other natural product scaffolds, transforming how complex alkaloid analogues can be generated for structure–activity exploration. By coupling nonnative FDHs and FMOs within a microbial host, this work defines a generalizable framework for engineering multi-step oxidative pathways establishing a blueprint for in vivo modular biosynthetic tailoring.

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Biocatalytic Derivatization Uncovers Structure–Function Rules for Opioid Receptor Modulation

We next explored how site-selective enzymatic modifications reshape receptor engagement and signaling. Using competitive radioligand displacement assays, we quantified binding across human MOR, DOR, and KOR to delineate how chemical diversification impacts affinity and selectivity (Figure 6, Supplementary Table 3). Mitragynine (**1**) demonstrated moderate affinity at MOR (K_i = 323 ± 83 nM) and negligible binding at DOR and KOR, consistent with its reported MOR selectivity, providing a reference point for comparison to the other analogues.[18,19] Natural derivatives such as paynantheine (**2**) and speciociliatine **(3**) retained MOR affinity while modestly expanding KOR binding, underscoring the sensitivity of receptor recognition to subtle stereoelectronic differences. In contrast, speciofoline (**4**) failed to displace radioligand at MOR or KOR and showed weak DOR affinity (Ki = 853 ± 173 nM), reinforcing the importance of stereochemistry and substituent positioning in receptor recognition (Figure 6).

Among the natural series, corynoxine A (**5**), which is a tetracyclic analogue with a rigidified scaffold, exhibited the highest MOR affinity (K_i = 24 ± 2 nM) and showed cross-interactions at DOR and KOR, indicating pan-opioid potential[41] and identifying it as the most potent binder in the series. Among biocatalytically derived analogues, C10 bromination enhanced MOR and KOR binding (K_i = 68 ± 11 nM and 92 ± 15 nM, respectively), suggesting favorable lipophilic interactions at that position, whereas *N*-oxidation preserved moderate MOR affinity. However, combining *N*-oxidation with C10 bromination abolished binding at KOR and significantly decreased affinity for MOR, emphasizing the need for precise functional group placement (Figure 6).

This pattern reveals a finely tuned structure–activity relationship: single-atom substitutions can modulate affinity in opposite directions depending on their position and local electronic context. To evaluate how these binding differences translate to signaling outcomes, we assessed a series of natural and enzymatically modified mitragynine analogues using the [^{35}S]GTPγS binding assay in MOR-expressing membranes. Consistent with its well-characterized pharmacology, morphine exhibited full agonist activity (EC_{50} = 60 nM), while buprenorphine acted as a potent partial agonist (EC_{50} = 0.21 nM), reaching ~25% relative to the full agonist DAMGO (Figure 6, Supplementary Figure S11).

Among the mitragynine-derived compounds, only mitragynine-4(*N*)-oxide (**6**) and corynoxine A (**5**) elicited detectable MOR activation—both as low-efficacy partial agonists (~25–50% of DAMGO)—while all other analogues, including those with improved binding, failed to trigger G- protein signaling (Figure 6, Supplementary Figure S11). This result aligns with previous studies[42] using [^{35}S]GTPγS, though some reports in other assay systems (e.g., BRET) have suggested functional activity.[7,19,43,44] The absence of G protein signaling in our assay suggests that any opioid-like effects observed in vivo are more likely mediated by active metabolites or alternative receptor systems. Overall, this highlights that receptor occupancy alone does not predict signaling outcome; the oxidation state and three-dimensional conformation jointly dictate receptor activation dynamics.

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Importantly, none of the natural or engineered analogues recruited β-arrestin (Supplementary Figure S12), consistent with their low intrinsic efficacy. As expected, DAMGO and morphine robustly recruited β-arrestin, further underscoring that the mitragynine scaffold disfavors β- arrestin engagement due to its low-potency, low-efficacy nature, since ligands with much higher intrinsic efficacy are needed to effectively recruit β-arrestin.[45] This selective G-protein bias reflects the atypical signaling across the mitragynine scaffold and aligns with its non-morphinan pharmacology. Together, these data demonstrate that targeted biocatalytic diversification enables precise, atom-level interrogation of opioid receptor SAR, linking enzymatic oxidation or halogenation to discrete shifts in binding and signaling behavior.

Computational Docking Unifies Enzymatic Derivatization with Receptor Pharmacology

To rationalize the experimental trends in binding and efficacy, we performed molecular docking of mitragynine and key analogues into the MOR orthosteric site (PDB: 5C1M).[46] Across all natural derivatives, ligands adopted a distinct, noncanonical pose anchored by a salt-bridge to Asp147 and stabilized by hydrophobic contacts spanning TM3–TM7 (Figure 7).[47,48]

Corynoxine A exhibited the most favorable binding energy (–9.0 kcal/mol), consistent with its experimental potency, due to enhanced geometric complementarity and hydrogen bonding at the Q124–D147–Y148 triad.[47,48] In contrast, 10-bromo-mitragynine engaged a lipophilic pocket between TM3 and TM6, with the bromine substituent stabilizing an alternate orientation of the indole nitrogen— potentially accounting for its improved binding but absent efficacy. *N*-oxidation preserved the canonical salt-bridge while slightly altering hydrogen bond geometry, consistent with partial agonism. Conversely, dual C3 oxidation and C10 bromination disrupted this polar anchoring and displaced the ligand from the orthosteric site, possibly explaining the dramatic affinity loss (Figure 7).

These results mechanistically connect the enzymatic transformations achieved in vivo to their pharmacological consequences: modifications that maintain the Asp147 interaction preserve binding and partial efficacy, whereas those that distort this anchoring eliminate activity. Thus, the docking analysis in this context provides support in linking enzymatic site-selectivity, electronic perturbation, and receptor behavior—closing the loop from whole-cell biocatalysis to receptorlevel mechanism.

Conclusions

This work establishes a new paradigm in biosynthetic drug discovery: sustainable, programmable whole-cell system for alkaloid diversification that integrates biocatalysis, synthetic biology, and pharmacology. By demonstrating that orphan halogenases and previously characterized FMOs can be modularly combined in vivo to generate new-to-nature opioid analogues, we show that complex natural product scaffolds can be diversified with atomic precision directly in microbial culture. The resulting biocatalytic chassis not only bypasses synthetic bottlenecks but also enables systematic exploration of receptor structure–function relationships.

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At its core, this study reveals how synthetic biology can transform natural product modification from an empirical process into a programmable design problem—linking enzyme promiscuity, chemical reactivity, and pharmacological function within a unified mechanistic framework. Looking forward, this platform offers a starting point for high-throughput generation and screening of kratom derivatives with improved pharmacological profiles, providing a tractable path towards potentially safer opioid therapeutics.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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ABBREVIATIONS

HPLC	High-performance liquid chromatography
LC-MS	liquid chromatography mass spectrometry
NMR	Nuclear Magnetic Resonance
FDH	flavin-dependent halogenase
FMO	flavin-dependent monooxygenase
SAR	structure activity relationship
MOR	μ-opioid receptor
KOR	κ-opioid receptor
DOR	δ-opioid receptor
EFC	Enzyme Fragment Complementation
EC50	half maximal effective concentration
Emax	maximum effect
^{3}H-DPN	^{3}H-diprenorphine
RT	Retention time
EIC	Extracted ion chromatogram
m/z	Mass to charge ratio.

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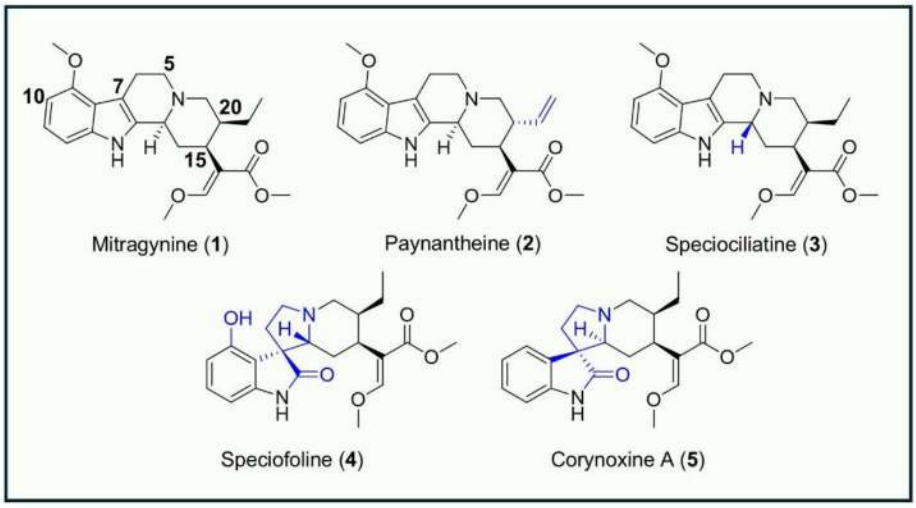


Figure 1. Alkaloids extracted and characterized from *Mitragyna speciosa* (kratom).

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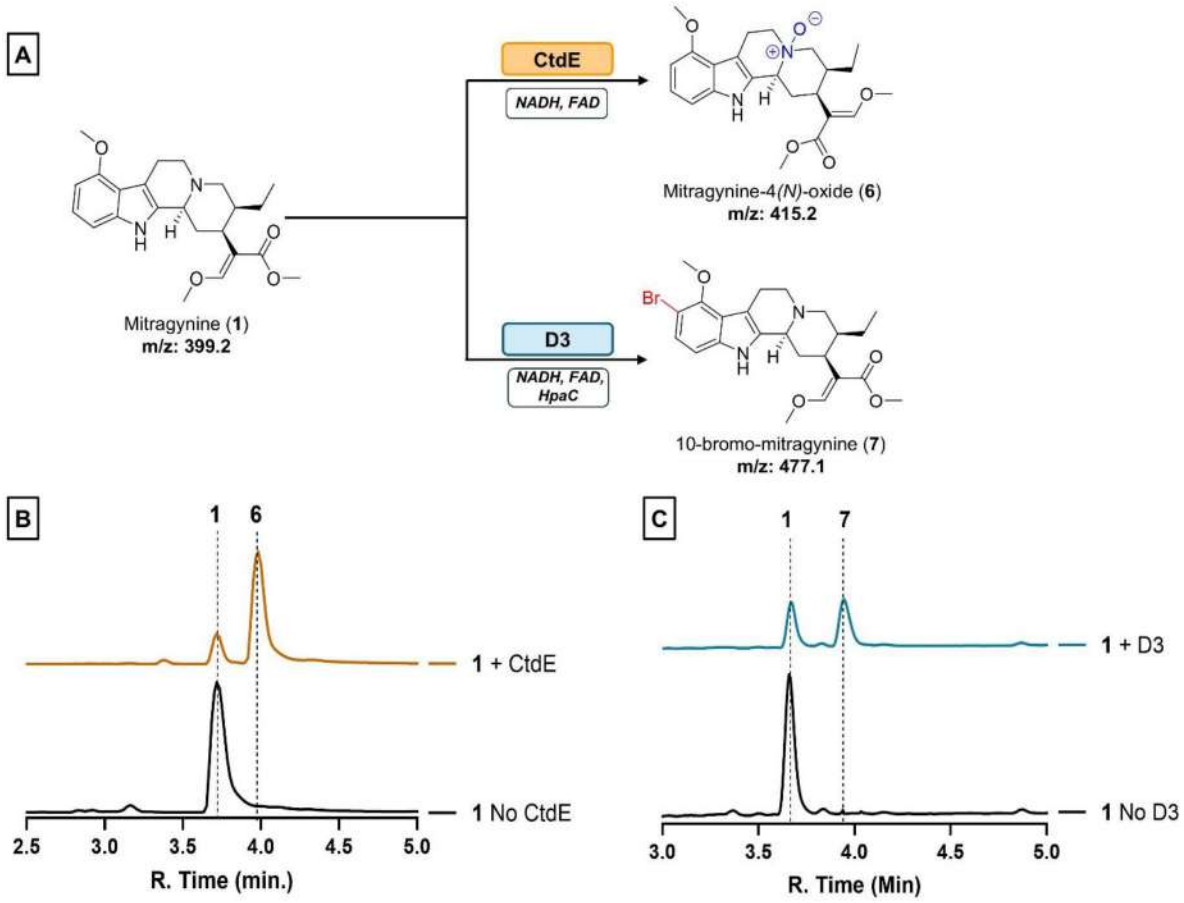


Figure 2. FMO- and FDH-catalyzed derivatization of mitragynine (1) generates distinct oxidative and halogenated products.

(**A**) Flavin-dependent monooxygenases (FMOs), initially hypothesized to induce spirocyclization of mitragynine (**1**), instead catalyze its conversion to mitragynine-4(*N*)-oxide (**6**), as validated by high-resolution MS and multidimensional NMR analyses. (**B**) Corresponding LC–MS extracted ion chromatograms (EICs) show depletion of the parent alkaloid (**1**) and formation of the oxidized product (**6**). (**A**) In parallel, biocatalytic halogenation of mitragynine (**1**) with the flavin-dependent halogenase D3 yields 10-bromo-mitragynine (**7**). (**C**) LC– MS EICs likewise demonstrate substrate consumption and formation of the brominated product (**7**), confirming selective C10 halogenation by D3.

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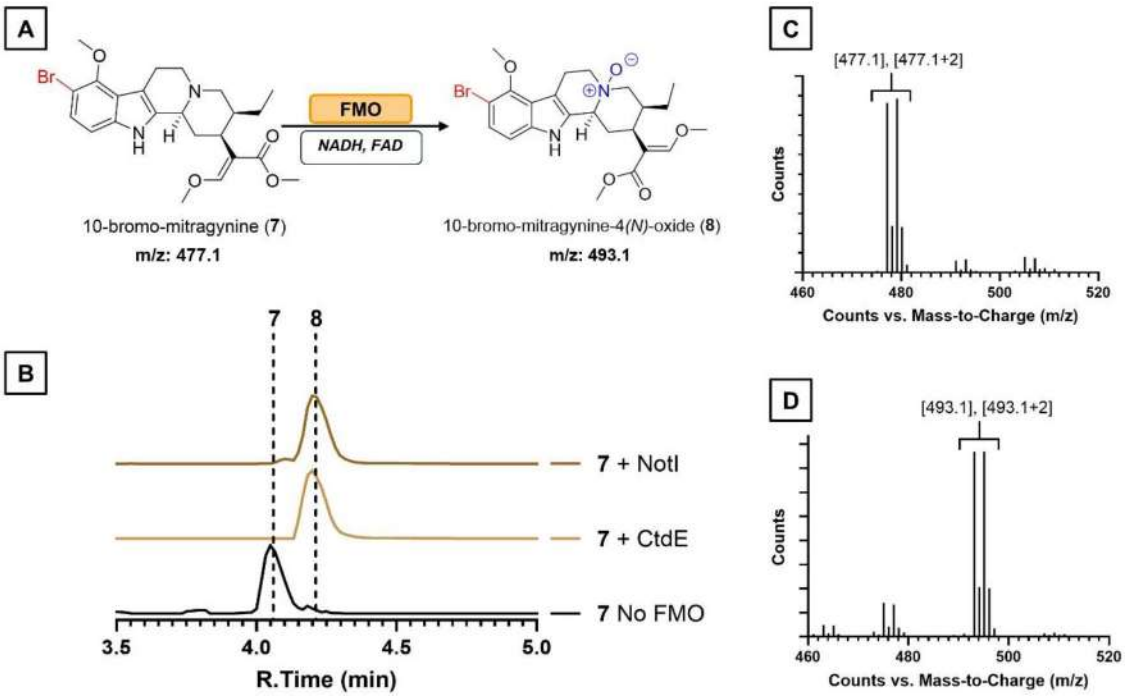


Figure 3. Oxidative functionalization of 10-bromo-mitragynine via flavin-dependent monooxygenases.

(**A**) Biorthogonal diversification of 10-bromo-mitragynine (**7**, *m/z* 477.1) using flavin-dependent monooxygenases (FMOs) CtdE and NotI yields the corresponding 4*(N)*-oxide (**8**, *m/z* 493.1). (**B**) LC-MS extracted ion chromatograms show efficient conversion of **7** to **8** upon incubation with either NotI or CtdE, with negligible activity in the absence of FMO. Notably, both enzymes achieve near-complete conversion, underscoring the compatibility of halogenated scaffolds with FMO-catalyzed oxidation. (**C**, **D**) Mass spectra corresponding to the eluted peaks from panel B confirm the expected mass-to-charge ratios for brominated species **7** (*m/z* 477.1) and **8** (*m/z* 493.1), consistent with *N*-oxide formation.

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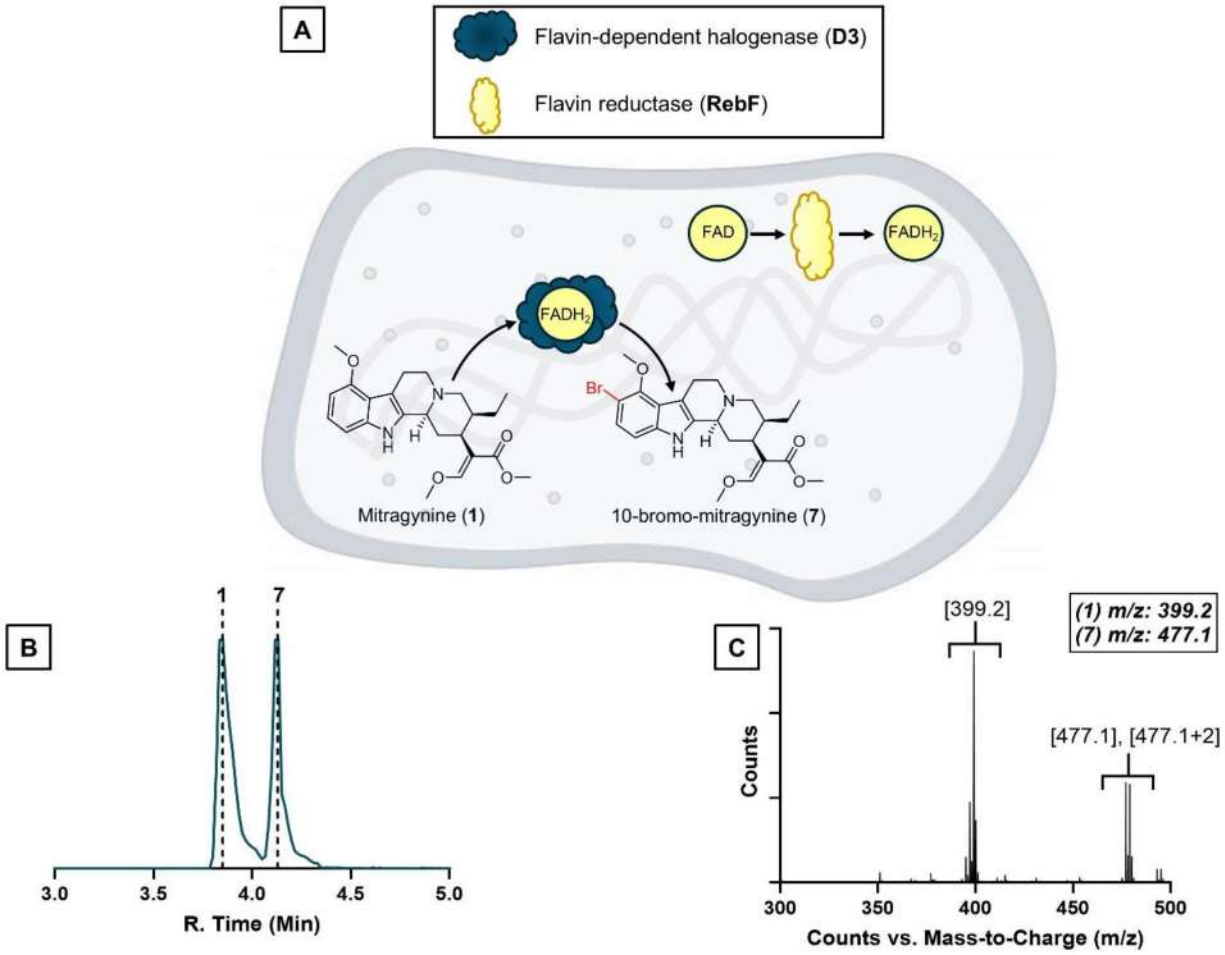


Figure 4. In vivo halogenation of mitragynine by engineered *E. coli* expressing D3 and RebF.

(**A**) Schematic overview of the engineered biosynthetic platform for whole-cell C–H bromination of mitragynine (**1**). *E. coli* BL21(DE3) was transformed with a plasmid co-expressing the flavin-dependent halogenase D3 and the flavin reductase RebF, which supplies reduced flavin (FADH2) to support halogenase activity. Upon feeding **1** to induced cultures, D3 catalyzes site-selective bromination at the C10 position in the indole ring to yield 10-bromo-mitragynine (**7**). (**B**) Representative LC-MS chromatogram depicting conversion of mitragynine (**1**, retention time = 3.9 min) to 10-bromo-mitragynine (**7**, retention time = 4.1 min) following a 2-hour incubation with D3/RebF-expressing E. coli. (**C**) Extracted ion chromatograms (EIC) showing the expected mass-to-charge ratios for mitragynine (**1**, m/z = 399.2) and 10-bromo-mitragynine (**7**, m/z = 477.1). The isotopic distribution of the brominated product further supports successful halogenation. Data are representative of three biological replicates.

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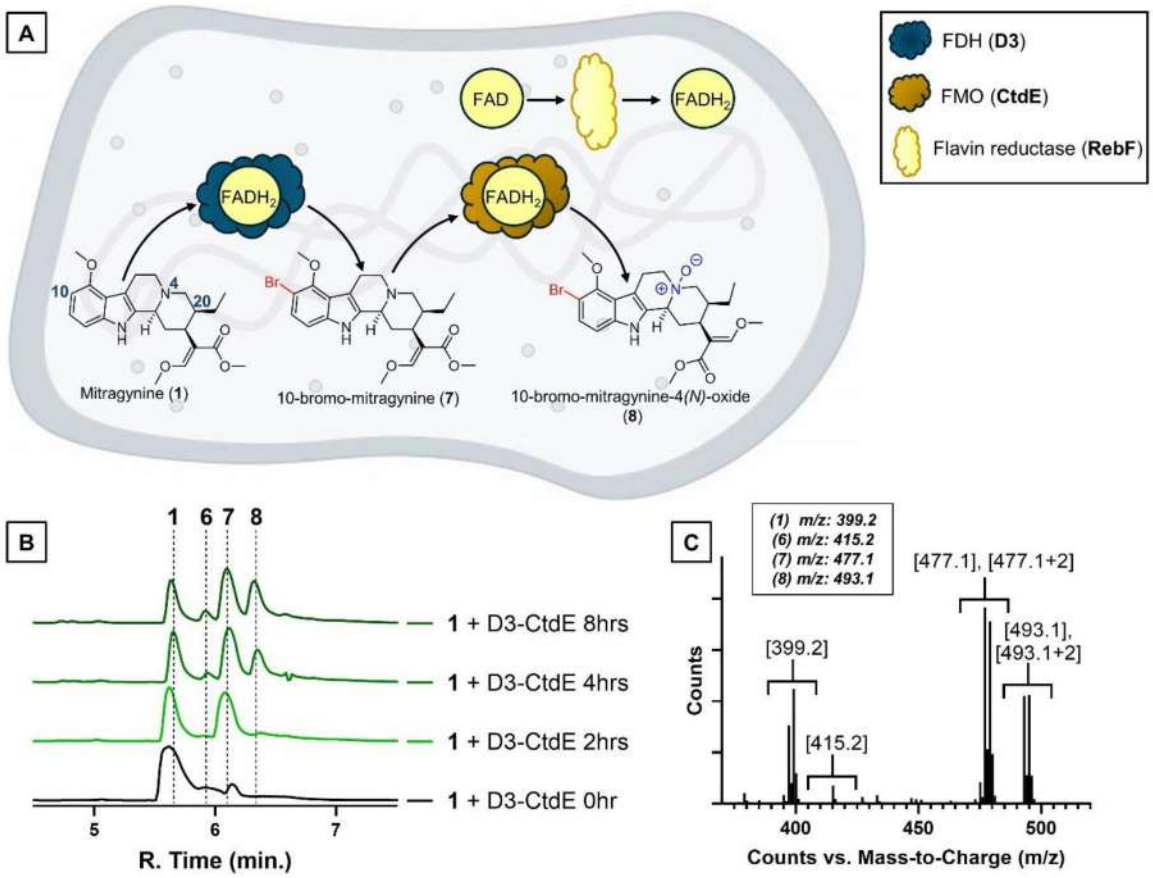


Figure 5. One-pot *in vivo* halogenation and oxidative derivatization of mitragynine by co-expression of D3 and CtdE in *E. coli*.

(**A**) Schematic representation of a dual-enzyme biocatalytic cascade for the site-selective halogenation and subsequent oxidative functionalization of mitragynine (**1**). Engineered *E. coli* co-expressing the flavin-dependent halogenase D3 (FDH, dark blue) and the flavin-containing monooxygenase CtdE (FMO, ochre) were supplemented with the flavin reductase RebF (yellow) to regenerate reduced flavin (FADH2). D3 catalyzes bromination at C10 of the indole moiety to yield 10-bromo-mitragynine (**7**), which is then oxidized by CtdE at the N4 position to generate the corresponding *N*-oxide, 10-bromo-mitragynine-4*(N)*-oxide (**8**). (**B**) Time-course LC-MS chromatogram showing substrate consumption and product formation. After 8 hours of incubation with induced *E. coli* expressing D3, CtdE, and RebF, mitragynine (**1**, RT = 3.1 min) is sequentially converted to 10-bromo-mitragynine (**7**, RT = 3.3 min) and 10-bromo-mitragynine-4*(N)*-oxide (**8**, RT = 3.6 min). (**C**) Extracted ion chromatograms confirm expected m/z values: mitragynine (**1**) at m/z = 399.2, 10-bromo-mitragynine (**7**) at m/z = 477.1, and 10-bromo-mitragynine-*4(N)*-oxide (**8**) at m/z = 493.1. The diagnostic $^{+}$16 Da shift between **7** and **8** supports monooxygenation at the nitrogen center. Data are representative of three independent biological replicates.

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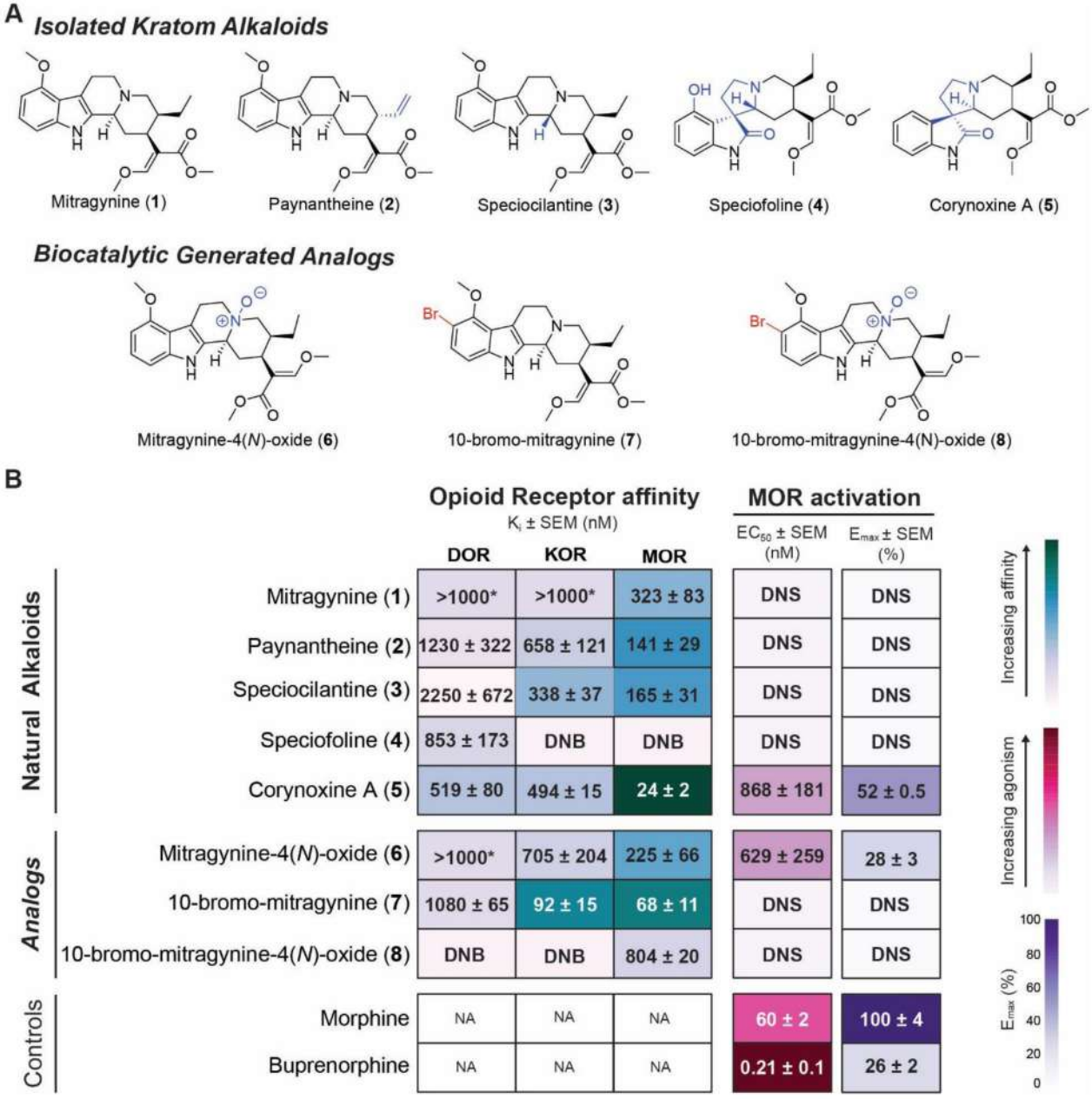


Figure 6. Biocatalytic diversification of kratom alkaloids and opioid receptor pharmacology.

(**A**) Structures of isolated kratom alkaloids (**1–5**) and biocatalytically generated analogues (**6–8**). Natural products include mitragynine (**1**), paynantheine (**2**), speciociliantine (**3**), speciofoline (**4**), and corynoxine A (**5**). Engineered whole-cell biocatalysis was used to generate mitragynine-4(*N*)-oxide (**6**), 10-bromo-mitragynine (**7**), and 10-bromo-mitragynine-4(*N*)-oxide (**8**). (**B**) Opioid receptor binding affinities and functional activities. Competition binding assays were performed against [^{3}H]-diprenorphine ([^{3}H]-DPN) at human δ- (DOR), κ- (KOR), and μ-opioid (MOR) receptors. Data are reported as Ki ± SEM (nM) from three independent experiments performed in duplicate. *For compounds that did not fully displace [^{3}H]-DPN at lower concentrations, the top of each curve was constrained

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to 100%, and percent displacement at 10 μM test ligand was determined. These values are reported as mean ± SEM: DOR: mitragynine (**1**, 49 ± 0.2%), mitragynine-4(*N*)-oxide (**6**, 66 ± 5%); KOR: mitragynine (**1**, 57 ± 5%). DNB indicates no detectable binding. Functional activity was assessed using [^{35}S]GTPγS binding to measure MOR- mediated G protein activation. Potency (EC50 ± SEM, nM) and efficacy (Emax ± SEM, %) are shown. DNS indicates no detectable stimulation. Morphine and buprenorphine were included as reference agonists. Heat map coloration reflects increasing receptor affinity and agonist efficacy.

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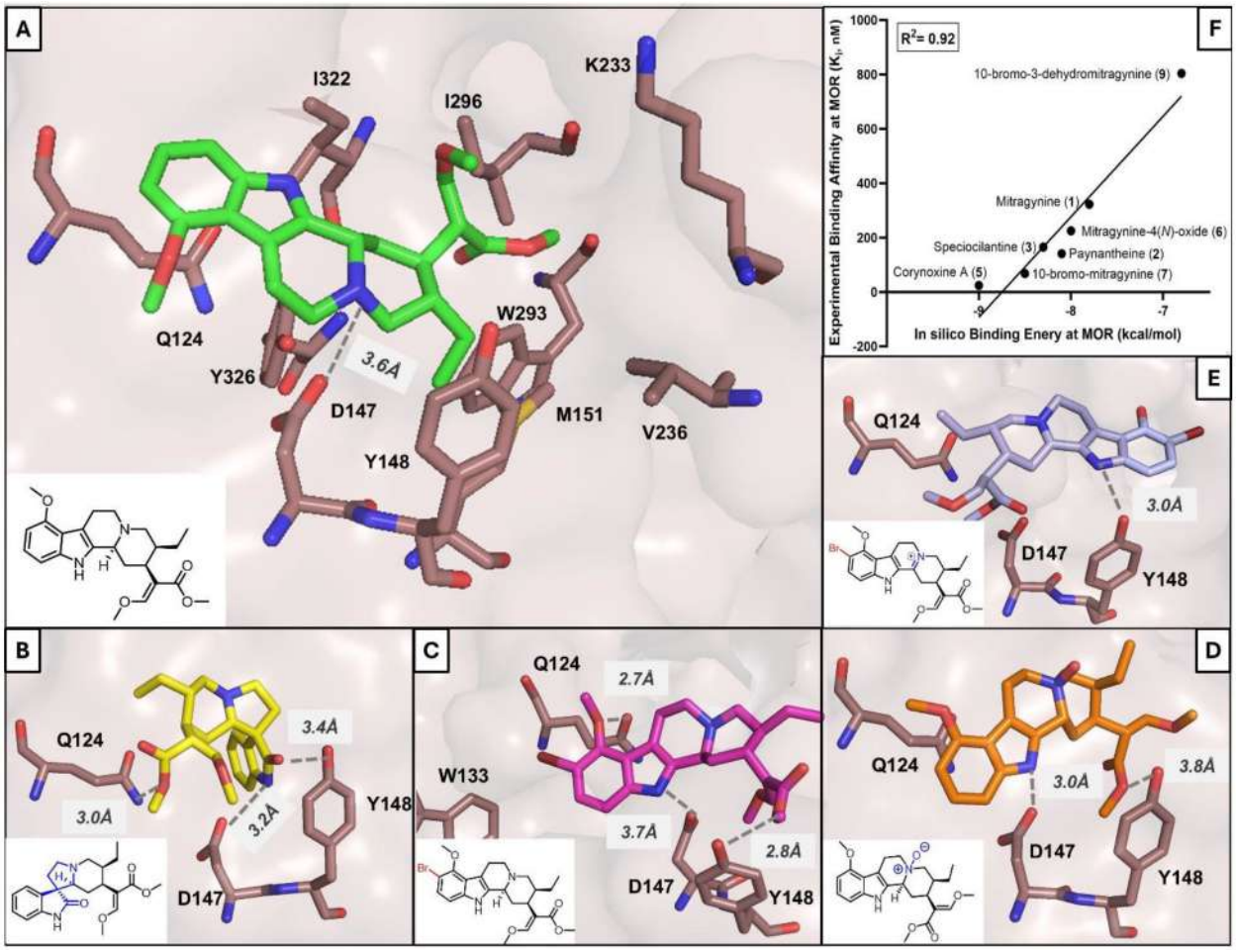


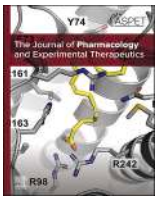
Figure 7. Comparative molecular docking of mitragynine and structural analogues at the μ-opioid receptor (MOR, PDB: 5C1M) reveals conserved and variant ligand–receptor interactions.

(**A**) Mitragynine (green) docked into the MOR orthosteric site, showing key polar and nonpolar interactions with residues identified from prior cryoEM and molecular dynamics studies.[21] The tertiary amine forms a salt-bridge with Asp147 (3.6 Å), while the methoxyacrylate moiety engages in hydrophobic contacts with Met151, Trp293, Ile296, Ile322, and Tyr326. (**B**) Corynoxine A (yellow) docked into the MOR orthosteric site, highlighting enhanced positioning at the Q124–D147– Y148 triad, suggesting improved electrostatic and hydrogen bonding interactions. (**C**) 10-bromo-mitragynine (pink) docked into the MOR orthosteric site, indicating bromine substitution does not disrupt canonical interactions but may enhance lipophilic contacts within the TM3–TM6 cleft. (**D**) Mitragynine-4(*N*)-oxide (orange) docked into the MOR orthosteric site showing retention of binding pose and salt-bridge geometry, consistent with conserved receptor affinity. (**E**) 10-bromo-3-dehydromitragynine (light blue) docked into the MOR orthosteric site, showing displacement from the optimal pose, likely due to steric or electronic effects from dual substitution. (**F**) Scatter plot displaying the correlation between in silico binding energy and experimental binding affinity at the MOR. Dashed lines indicate atom–atom distances in angstroms.

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ARTICLE

Mitragynine and 7-hydroxymitragynine: Bidirectional effects on breathing in rats

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ABSTRACT

The use of kratom as an alternative to conventional opioids has surged, driven largely by anecdotal reports of its efficacy for pain relief and opioid withdrawal management. The growing prevalence of kratom products enriched with 7-hydroxymitragynine (7-HMG), an active metabolite of mitragynine (MG), necessitates evaluating the respiratory effects of these alkaloids and determining whether naloxone reverses their potential respiratory depressant effects. Respiratory parameters were measured in awake, freely moving female and male Sprague-Dawley rats using whole body plethysmography. To minimize handling-induced artifacts and ensure precise respiratory recordings, drugs were administered intravenously. Morphine and 7-HMG induced significant respiratory depression, evidenced by reductions in breathing frequency, tidal volume, and minute volume. The potency of 7-HMG to decrease minute volume by 50% was 4.5-fold greater than that of morphine. In contrast, MG administration unexpectedly increased respiratory frequency. Naloxone fully reversed the respiratory depression induced by both morphine and 7-HMG but did not alter the respiratory stimulant effects produced by MG. These findings demonstrate that 7-HMG exhibits significant respiratory depressant properties similar to classical opioids, and importantly, such depressant effects are effectively antagonized by naloxone. Conversely, MG exerts respiratory stimulant effects through mechanisms independent of opioid receptor pathways. Collectively, these data highlight crucial pharmacological distinctions between kratom alkaloids, underscoring the risk associated with high 7-HMG–containing kratom products and suggesting that the predominant alkaloid MG may offer a safer respiratory profile.

Significance Statement: The prevalence of kratom products containing 7-hydroxymitragynine (7-HMG), a μ-opioid receptor agonist, underscores the need to evaluate respiratory effects of kratom-related alkaloids and their reversal by naloxone. 7-HMG induced significant respiratory depression comparable with morphine, which was reversed by naloxone. Conversely, mitragynine, kratom's most abundant alkaloid, unexpectedly increased respiratory frequency unaffected by naloxone. These findings highlight critical pharmacological differences between kratom-related alkaloids, emphasizing potential risks associated with products containing high concentrations of 7-HMG.

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1. Introduction

Opioids have long been recognized for their analgesic properties, yet their clinical utility is hindered by their potentially life-threatening side effects, like respiratory depression.[1–3] Opioid-induced respiratory depression (OIRD) represents a significant risk in both clinical settings and abuse-related overdoses.[4,5] Preclinical

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studies have elucidated the mechanisms underlying OIRD, implicating μ-opioid receptor (MOR) activation in brainstem respiratory centers.[6,7] Clinical research has further highlighted the risk of OIRD in vulnerable populations such as in older adults and those with sleep-disordered breathing.[8,9]

These safety concerns have driven the search for alternative analgesics with improved respiratory profiles. Kratom (*Mitragyna speciosa*) has garnered increasing attention for its opioid-like effects and potential therapeutic applications. In recent years, the use of kratom as an alternative to conventional opioids within the public domain has surged, driven by anecdotal reports of efficacy in managing pain and mitigating opioid withdrawal symptoms.[10–12] Beyond these anecdotal reports, scientific interest in kratom continues to grow, with several studies focusing on its primary alkaloid, mitragynine (MG), and its hydroxylated metabolite, 7-hydroxymitragynine (7-HMG).

Although both MG and 7-HMG function as biased agonists at the MOR,[13–16] such biased agonism does not necessarily confer improved safety over conventional opioids. The concept of biased agonism, which involves selective activation of specific signaling pathways over others, was initially expected to provide a path to safer opioid medications by preferentially activating analgesic pathways while avoiding those associated with adverse effects. However, compounds developed as biased MOR agonists have failed to demonstrate improved safety profiles.[17–20] This disconnect highlights the complexity of MOR signaling and suggests that biased agonism alone may not be sufficient to overcome the fundamental challenges of opioid pharmacology.

Pharmacological characterization of the abuse liability of MG and 7-HMG revealed striking differences. MG did not substitute for morphine in the drug discrimination assay and was not self-administered, whereas 7-HMG fully substituted for morphine and demonstrated reinforcing effects.[21–24] Additionally, 7-HMG in vivo has been demonstrated to be 3- to 13-fold more potent than morphine.[24–26] These findings suggest that 7-HMG, but not MG, has greater liability risk and abuse potential similar to other opioid agonists despite biased agonism.

The differential abuse potential between these alkaloids is particularly concerning given the current market trends. A significant concern is the increasing availability of kratom products that contain high concentrations of 7-HMG.[27,28] Although analyses indicate that 7-HMG is either not detectable or present in extremely small amounts (<0.1%) in the living plant,[29,30] likely formed as a postharvest artifact,[31] some manufacturers are marketing products containing 14–25 mg of 7-HMG per recommended serving, with labeled purities as high as 98%.[28] This poses a risk, particularly for kratom-naïve individuals who may unknowingly purchase these products under the assumption that they are typical kratom preparations. These concerns underscore the need to evaluate the safety of both 7-HMG and MG, particularly regarding their effects on respiration.

The current state of knowledge regarding the respiratory effects of kratom and its related alkaloids remains limited and somewhat conflicting. One study found that oral MG did not yield significant changes in blood gases in rats, even at high doses[32] though this study measured blood gases rather than direct respiratory parameters. Another study in mice reported respiratory depression but observed a ceiling effect in MG-induced respiratory depression, whereas 7-HMG showed dose-dependent effects.[33] However, it is important to note that mice exhibit hyperlocomotion following MOR agonist administration,[34–38] potentially making rats the more appropriate species for studying OIRD. Additionally, the naloxone-reversibility of these respiratory effects remains to be determined. This is crucial as it would inform clinicians whether naloxone is effective in reversing 7-HMG induced respiratory depression.

In this study, we investigated the respiratory effects of the prototypical opioid, morphine, and kratom-related alkaloids (MG and 7-HMG) using whole body plethysmography in rats in 2 distinct experimental phases. In the first phase, we evaluated the direct respiratory effects of morphine, MG, and 7-HMG. In the second phase, we examined the potential for naloxone as a rescue intervention. This investigation of the respiratory effects and naloxone-reversibility of these compounds provides new insights into the role of opioid receptor mechanisms in kratom-induced respiratory depression.

2. Material and methods

2.1. Subjects

A total of 32 Sprague-Dawley rats (Charles River; $n = 16$ females, $n = 16$ males), aged 12 weeks, were used in this study. The rats were purchased with preimplanted indwelling jugular catheters. For the first experimental phase, 16 animals were divided into 2 groups ($n = 8$, morphine group; $n = 8$, MG/7-HMG group). All testing followed a repeated-measures (RM), within-subjects design where each animal received multiple doses of their respective drug(s). One male rat from the kratom-related alkaloids group expired before the completion of the experiment owing to postsurgical complications, resulting in a final sample size of $n = 7$ for this group. The remaining 16 animals were used in the second experimental phase, with 8 animals assigned to the morphine–naloxone group and 8 animals assigned to the kratom-related alkaloids–naloxone group. Animals were housed individually within a temperature- and humidity-controlled vivarium and kept on a 12:12 light-dark cycle (lights on at 07:00 AM). Food and water were provided ad libitum. This study was conducted in accordance with the Guide for the Care and Use of Laboratory Animals[39] and was approved by the Institutional Animal Care Use Committee at Texas Tech University Health Sciences Center.

2.2. Drugs

Morphine sulfate and naloxone hydrochloride (Fagron) were diluted in normal saline. Mitragynine (MG) HCl and 7-hydroxymitragynine (7-HMG) were isolated as previously described[40] and diluted in a vehicle solution consisting of 2.5% Tween 20, 2.5% Tween 80, 25% polyethylene glycol, 25% propylene glycol, and 45% normal saline. Heparin sodium salt from porcine intestinal mucosa (MilliporeSigma), gentamicin (Covetrus), ketamine (Covetrus), and midazolam (Zoetis) were all diluted in normal saline.

2.3. Apparatus

A whole body plethysmography system (Data Sciences International) was used to measure and record the respiratory parameters in awake, freely moving rats. The system consisted of 8 cylindrical acrylic chambers measuring 18.4 cm in diameter and 13.1 cm in height. The top interior portion of the chamber houses an infusion tether that is connected magnetically to the implanted vascular access button on the rats. The exterior portion of the chamber is connected to a pinport that facilitates aseptic intravenous drug delivery (PNP3M; Instech laboratories). The intravenous route was chosen to eliminate handling-induced respiratory artifacts during drug delivery, ensuring undisturbed respiratory recordings while enabling maximal effects with lower drug doses. Similarly, white noise was continuously played in the testing room

to mask external sounds and minimize respiratory disturbances that could be caused by environmental noises.

The whole body plethysmography system measures box flow (milliliters per second), which refers to the airflow in and out of the chamber caused by the animal's respiration. Changes in air temperature and humidity during the breathing cycle primarily drive pressure/box flow changes recorded using whole body plethysmography.[41] When an animal inhales, the air is warmed and humidified. When it exhales the warmed air, the air temperature and humidity in the chamber increase, and pressure subsequently increases. Therefore, exhalations are recorded as upward deflections in pressure/box flow, and inhalations are recorded as downward deflections in pressure/box flow. These pressure/box flow changes are detected by a pressure transducer and processed by the whole body plethysmography software Finepointe v.3.1 (DSI) to calculate various respiratory parameters.

Respiratory rate (breaths per minute) is directly measured from fluctuations in the positive and negative box flow. Tidal volume (milliliters per breath) and minute volume (milliliters per minute) are estimated from the box flow signal. Tidal volume is equal to the product of the volume measured from the box flow and a compensation factor calculated using the formula by Drorbaugh and Fenn,[42] which accounts for temperature, humidity, and barometric pressure. Minute volume is equal to the product of respiratory rate and tidal volume (rate $\times$ box flow volume $\times$ compensation factor). In addition to these parameters, the system also calculates apneas. An apnea is defined as an unusually long breath (at least 1 second long and 100% greater than the breaths 5 minutes before).

2.4. Study design

The study was conducted in 2 distinct experimental phases. In the first phase, we evaluated the direct respiratory effects by administering either saline or morphine (10 and 32 mg/kg, i.v.) in 1 group ($n = 8$). A separate group ($n = 7$) received either vehicle or the kratom-related alkaloids MG (5.6 and 10 mg/kg, i.v.) and 7-HMG (1, 3.2, and 10 mg/kg, i.v.). The order of drug administration was counterbalanced between subjects in both groups. For MG, we observed stimulatory effects but no respiratory depression at doses up to 10 mg/kg. On the basis of these findings, all animals in the kratom alkaloid group received a higher 17.8 mg/kg (i.v.) MG dose to assess potential respiratory depression at higher doses. This highest dose still produced no significant respiratory depression and resulted in seizure-like activity in 2 of 3 male subjects, with 1 animal sustaining physical injury to the digits of its hind paw. This adverse reaction established 10 mg/kg as our upper safety limit for subsequent testing.

In the second phase, using entirely different animals, we examined the extent to which the opioid antagonist naloxone could reverse respiratory effects. The doses used in this phase (32 mg/kg for morphine, 10 mg/kg for MG, and 10 mg/kg for 7-HMG) were selected on the basis of their significant effects on respiratory parameters during the first phase. One cohort ($n = 8$) underwent 4 treatment conditions in a fully counterbalanced Latin square design: saline followed by saline, saline followed by naloxone (1 mg/kg, i.v.), morphine (32 mg/kg, i.v.) followed by saline, and morphine (32 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.). Another cohort ($n = 8$) underwent 6 treatment conditions using Williams design: vehicle followed by saline, vehicle followed by naloxone (1 mg/kg, i.v.), MG (10 mg/kg, i.v.) followed by saline, MG (10 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.), 7-HMG (10 mg/kg, i.v.) followed by saline, and 7-HMG (10 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.). In all second-phase treatments, naloxone or saline was administered 10 minutes after the initial compound injection.

Dose selection for this study was determined through preliminary testing in our laboratory. For morphine, doses of 10 and 32 mg/kg (i.v.) were selected on the basis of pilot studies demonstrating that higher doses were necessary to produce reliable respiratory depression under normal air conditions without CO_2 challenge, which typically reveals respiratory depression at lower doses. For the kratom-related alkaloids, we implemented a systematic dose-escalation approach due to the absence of published intravenous administration data. For the naloxone antagonism studies, we selected a dose of 1 mg/kg (i.v.) to provide effective blockade of opioid receptor-mediated effects given the relatively high doses of the test compounds used in our paradigm.

Animals were given a 1-week habituation to new housing conditions before experiments commenced. After habituation, rats were transported to an experimental room mimicking conditions of the housing room and were placed individually in a whole body plethysmography chamber for a 3-hour acclimation period. During this time, rats received 2 vehicle injections to familiarize them with the injection procedure before returning to the housing room. For experimental sessions, each weekly testing began with a 60-minute acclimation period followed by a 20-minute baseline recording. After delivery of test compounds, respiratory parameters were recorded continuously for 60 minutes in both phases. Injections were manually administered over 15–30 seconds, and infusion lines were flushed with saline to ensure complete drug delivery.

To maintain catheter patency and prevent infections or blood clots, catheters were flushed with a solution containing gentamicin (4 mg/mL) and heparin (30 IU/mL) following catheter manipulation and at least every 3 days.[43] On the day after each test session (Fridays), catheter patency was verified by infusing 0.2 mL of a solution containing ketamine (15 mg/mL) and midazolam (0.5 mg/mL) into the catheter.[44] Loss of righting reflex within 5 seconds indicated a patent catheter.

2.5. Statistical analysis

The study timeline consisted of 60 minutes of acclimation to whole body plethysmography chambers, followed by a 20-minute baseline recording period. The acclimation period was neither plotted nor included in any analyses. After baseline recording, test compounds were administered intravenously, and postdrug measurements continued for 60 minutes, with data collected in 5-minute bins. To account for individual differences and associated baseline measures, all respiratory parameters (frequency, tidal volume, and minute volume) were normalized by expressing each postdrug value as a percentage of the corresponding predrug baseline value. This normalization is particularly important for tidal volume, which is directly influenced by mass across species.[45–48] For each experimental session, baseline values were calculated as the average of the 20-minute recording period before drug infusion. Raw trace data for significant doses of each compound are shown in Supplemental Figs. 1–4, and the average raw frequency data are shown in Supplemental Table 1. For quantitative analysis, the time course data were transformed to area under the curve (AUC) values using the trapezoidal method, calculated over 2 specific time intervals: 0–30 minutes and 0–60 minutes postadministration.

For single-drug experiments with morphine, MG, and 7-HMG, AUC values were analyzed using 1-way RM ANOVAs separately for each respiratory parameter at the 30- and 60-minute intervals. Greenhouse–Geisser correction was applied to adjust degrees of freedom when the assumption of sphericity was violated.

Following significant main effects, Dunnett's multiple comparisons test was used to compare drug treatment groups against the vehicle control. ED_{50} values were calculated using simple linear regression if a drug decreased minute volume below 50% of baseline.

In the naloxone antagonism experiments, morphine, MG, or 7-HMG administration was followed by naloxone or saline 10 minutes later. AUC values were calculated as described earlier and analyzed using 2-way RM ANOVAs with naloxone treatment and drug treatment as factors. The analysis examined the main effects of naloxone treatment and drug treatment, and their interaction at both time intervals. Tukey multiple comparisons test was used for multiple comparisons between treatment conditions. Sex differences, ie, differences between male and female rats, were analyzed using 2-way RM ANOVA (Supplemental Table 2), and no significant differences were observed. Statistical significance was established at $P < .05$ for all analyses. All statistical procedures were performed using GraphPad Prism (version 10.3.1) software.

3. Results

3.1. Respiratory effects of morphine, MG, and 7-HMG alone

Morphine produced dose-dependent respiratory depression, as illustrated by time courses for respiratory frequency, tidal volume, and minute volume (Fig. 1A–C). Representative raw trace data for saline control and morphine are shown in Supplemental Figs. 1 and 2, respectively. Morphine had no significant effect on respiratory frequency at either 30 minutes (Fig. 1D) ($F[1.57,11.0] = 3.65$; $P = .069$) or 60 minutes (Fig. 1E) ($F[1.58,11.1] = 3.06$; $P = .096$). Morphine significantly affected tidal volume at both 30 minutes (Fig. 1F) ($F[1.25,8.76] = 10.9$; $P = .007$) and 60 minutes (Fig. 1G) ($F[1.35,9.42] = 16.9$; $P = .001$). Post hoc analysis revealed that only the higher dose of morphine (32 mg/kg) significantly reduced tidal volume at both time points ($P < .001$). Similarly, morphine significantly decreased minute volume at both 30 minutes (Fig. 1H) ($F[1.31,9.15] = 8.79$; $P = .012$) and 60

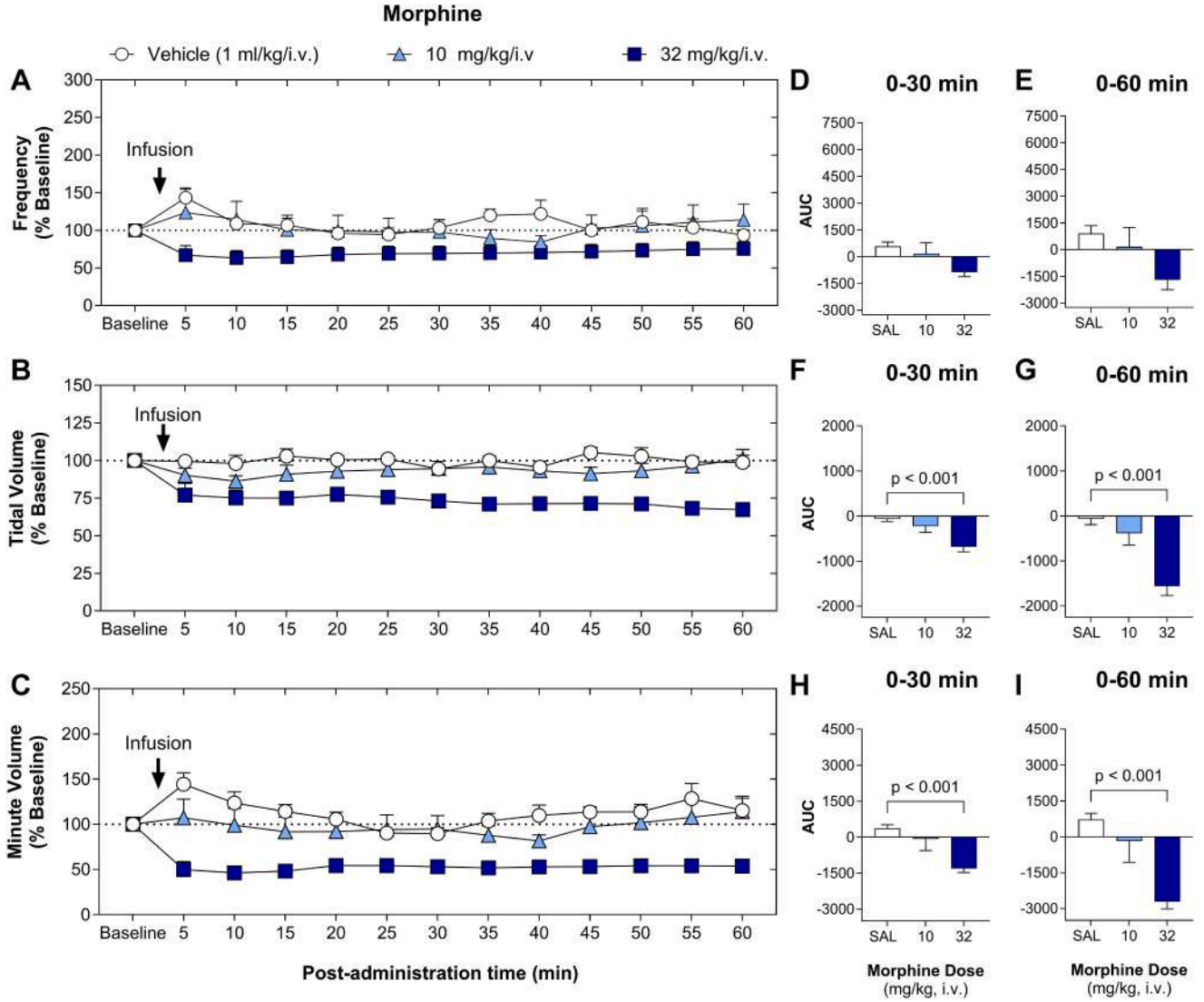


Fig. 1. Time-course and AUC of morphine effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels), following intravenous administration of morphine in rats. Morphine was administered at doses of 10 and 32 mg/kg, with saline (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between saline and different doses are indicated (P values). The ED_{50} for morphine to decrease minute volume by 50% was 31.5 mg/kg (24.0–47.8 mg/kg).

minutes (Fig. 1I) (F[1.21,8.44] = 9.73; $P = .011$). Post hoc analysis showed that only the higher dose of morphine (32 mg/kg) significantly reduced minute volume at both time points ($P < .001$), whereas the lower dose (10 mg/kg) did not produce significant respiratory depression. The ED_{50} for morphine to decrease minute volume by 50% was 31.5 mg/kg (24.0–47.8 mg/kg). Quantification of apneas during the first 15 minutes showed a nonsignificant trend for the highest dose of morphine (Supplemental Fig. 5).

MG administration produced respiratory effects distinct from morphine, as shown in the time courses for frequency, tidal volume, and minute volume (Fig. 2A–C). Representative raw trace data for significant MG doses are shown in Supplemental Fig. 3. MG significantly increased respiratory frequency at both 30 minutes (Fig. 2D) (F[1.99,11.9] = 4.00; $P = .047$) and 60 minutes (Fig. 2E) (F[1.78,10.7] = 4.77; $P = .036$). Post hoc analysis revealed that the 10-mg/kg dose significantly elevated respiratory frequency compared with vehicle control at both 30 minutes ($P = .035$) and 60 minutes ($P = .046$). MG did not significantly affect tidal volume at either 30 minutes (Fig. 2F) (F[1.53,9.20] = 3.40; $P = .086$) or 60 minutes (Fig. 2G) (F[1.35,8.09] = 3.93; $P = .075$). Similarly, minute volume was not significantly altered by MG administration at either 30 minutes (Fig. 2H) (F[1.53,9.17] = 4.07; $P = .062$) or 60 minutes (Fig. 2I) (F[1.59,9.53] = 2.65; $P = .128$).

Administration of 7-HMG produced robust, dose-dependent respiratory depression. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 3A–C. Representative raw trace data for significant 7-HMG doses are shown in Supplemental Fig. 4. Respiratory frequency was significantly decreased by 7-HMG at both 30 minutes (Fig. 3D) (F[1.91,11.5] = 12.7; $P = .001$) and 60 minutes (Fig. 3E) (F[2.26,13.6] = 7.72; $P = .005$). Post hoc analysis revealed that both 3.2- and 10-mg/kg doses significantly reduced respiratory frequency at 30 minutes ($P = .04$ and $P = .03$, respectively), but these effects were not maintained at 60 minutes. For tidal volume, 7-HMG approached but did not reach significance at both 30 minutes (Fig. 3F) (F[1.55,9.31] = 4.19; $P = .058$) and 60 minutes (Fig. 3G) (F[1.57,9.40] = 4.05; $P = .061$). Minute volume was significantly

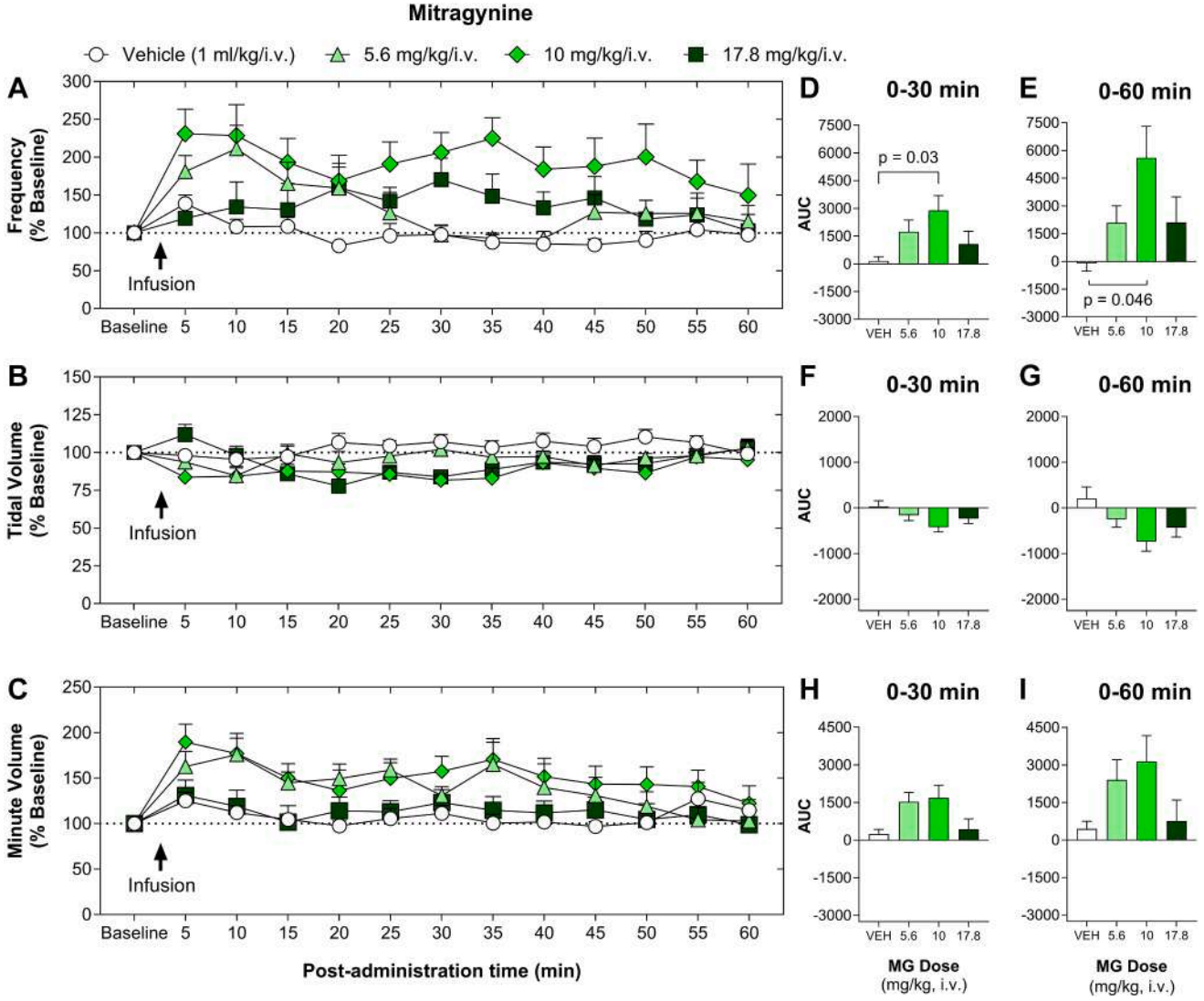


Fig. 2. Time-course and AUC of MG effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of MG in rats. MG was administered at doses of 5.6, 10, and 17.8 mg/kg, with vehicle (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM (n = 7), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM (n = 7) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between vehicle and different doses are indicated (P values).

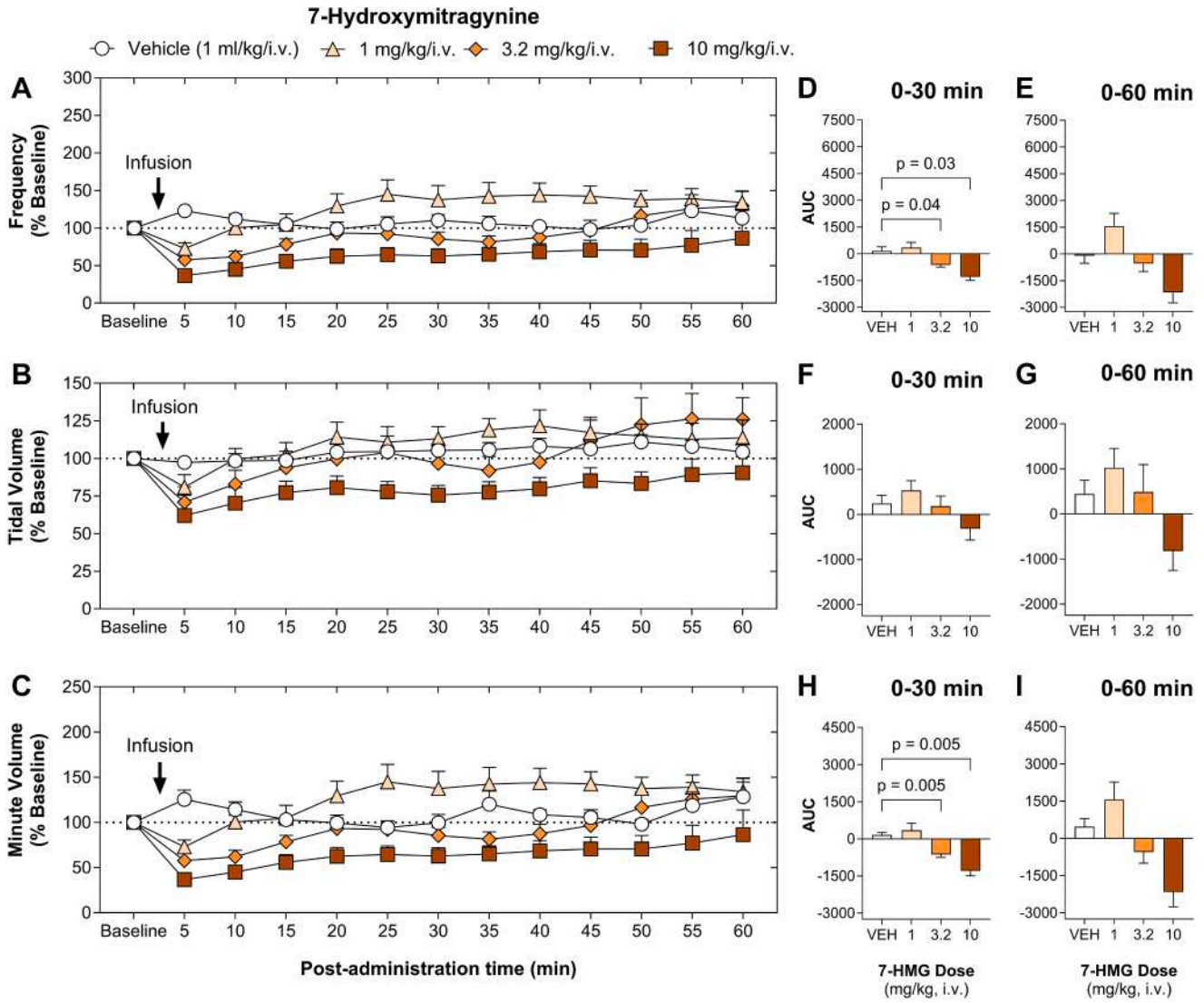


Fig. 3. Time course and AUC of 7-HMG effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of 7-HMG in rats. 7-HMG was administered at doses of 1, 3.2, and 10 mg/kg, with vehicle (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM ($n = 7$), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between vehicle and different doses are indicated (P values). The ED_{50} for 7-HMG to decrease minute volume by 50% was 6.98 mg/kg (5.20–10.23 mg/kg).

reduced by 7-HMG at both 30 minutes (Fig. 3H) ($F[2.17,13.0] = 17.9$; $P < .001$) and 60 minutes (Fig. 3I) ($F[2.35,14.1] = 8.52$; $P = .003$). Post hoc analysis revealed that the highest doses (3.2 and 10 mg/kg/i.v.) significantly decreased minute volume at 30 minutes ($P = .005$ for both doses), but these effects were not statistically significant at 60 minutes. The ED_{50} for 7-HMG to decrease minute volume by 50% was 6.98 mg/kg (5.20–10.23 mg/kg). Quantification of apneas during the first 15 minutes revealed a significant effect for the highest dose of 7-HMG (Supplemental Fig. 5).

3.2. Naloxone antagonism studies

3.2.1. Morphine + naloxone

Naloxone (1 mg/kg) treatment effectively reversed the respiratory depression induced by morphine (32 mg/kg). Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 4A–C. For respiratory frequency, 2-way RM ANOVA at 30 minutes revealed nonsignificant main effects of naloxone ($F[1,7] = 3.74$; $P = .09$) or morphine ($F[1,7] = 4.71$; $P = .07$), but a significant naloxone × morphine interaction ($F[1,7] = 15.5$; $P = .009$) (Fig. 4D). Multiple comparisons tests revealed that neither morphine + saline nor morphine + naloxone administration significantly impacted respiratory frequency compared with control, consistent with the results of the morphine dose-response experiment. At 60 minutes, these effects accumulate with significant main effects of naloxone ($F[1,7] = 7.09$; $P = .03$), morphine ($F[1,7] = 7.58$; $P = .03$), and naloxone × morphine interaction ($F[1,7] = 18.1$; $P = .004$) (Fig. 4E). At this time point, morphine + naloxone was significantly different from morphine + saline ($P = .04$), demonstrating an overshoot effect.

For tidal volume at 30 minutes, a significant effect of morphine ($F[1,7] = 34.2$; $P < .001$) was observed, whereas naloxone ($F[1,7] = 1.99$; $P = .20$) and naloxone × morphine interaction ($F[1,7] = 0.185$; $P = .68$) did not reach significance (Fig. 4F). At 60 minutes, both naloxone ($F[1,7] = 6.52$; $P = .04$) and morphine ($F[1,7] = 34.5$; $P < .001$) showed significant effects but not the naloxone × morphine

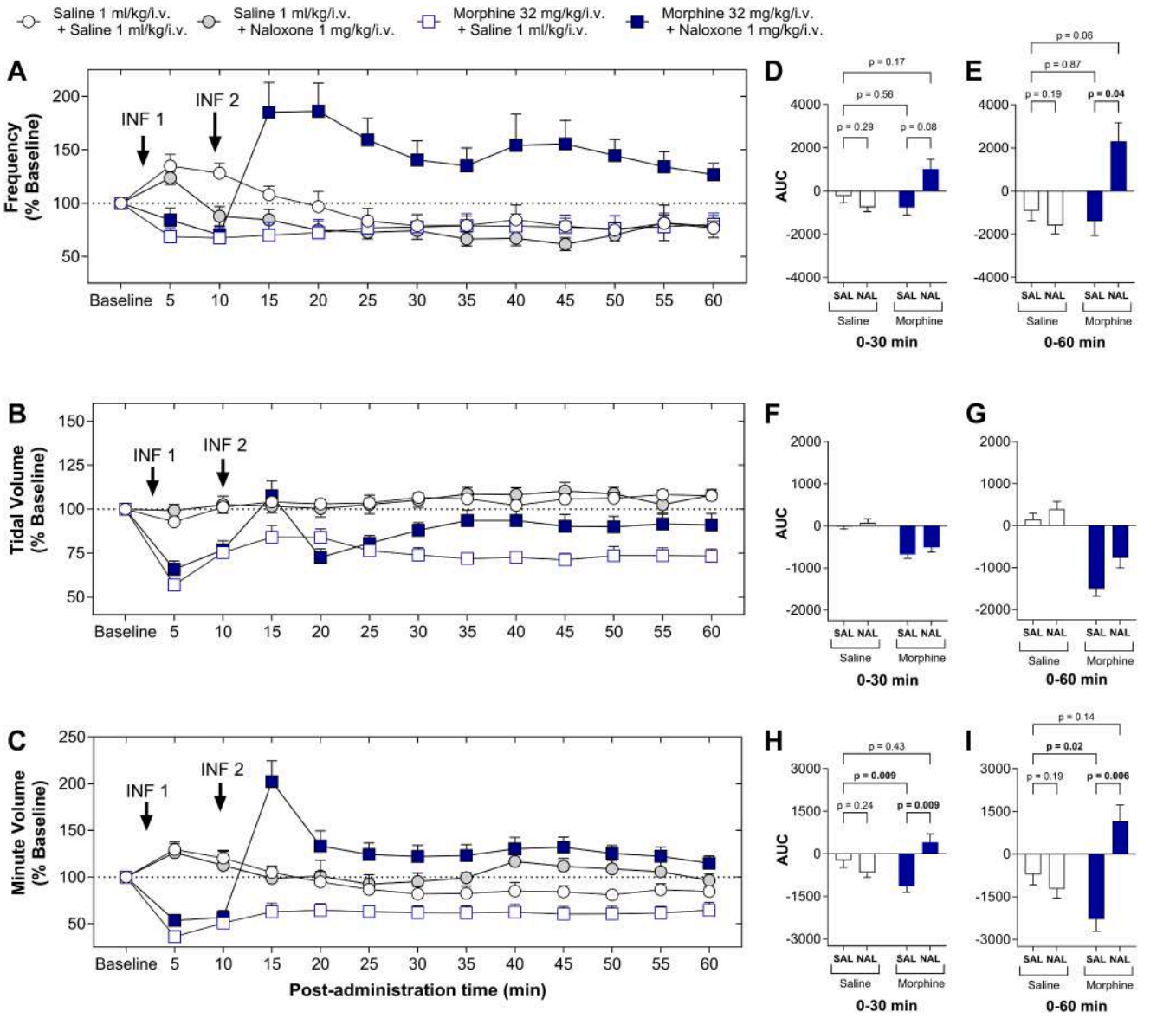


Fig. 4. Time course and AUC of morphine and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of morphine with naloxone rescue treatment in rats. Morphine was administered (32 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 7$), recorded in 5-minute bins continuously for up to 60 minutes postmorphine infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postmorphine infusion. Statistically significant differences between treatments are indicated (P values).

interaction ($F[1,7] = 2.07$; $P = .19$) (Fig. 4G). No post hoc analyses were conducted for tidal volume owing to the absence of significant interactions at either time point.

For minute volume at 30 minutes, 2-way RM ANOVA showed significant effects of naloxone ($F[1,7] = 6.46$; $P = .04$) and naloxone × morphine interaction ($F[1,7] = 36.5$; $P < .001$) (Fig. 4H). Post hoc analysis revealed significant depression with morphine alone compared with that with control ($P = .009$), which was not significant when naloxone was coadministered. Additionally, morphine + saline and morphine + naloxone treatments were significantly different from each other ($P = .009$). At 60 minutes, significant effects of naloxone ($F[1,7] = 13.1$; $P = .008$) and naloxone × morphine interaction ($F[1,7] = 40.2$; $P < .001$) persisted (Fig. 4I). The same pattern of post hoc differences was observed at this time point, with morphine alone causing significant depression compared with control ($P = .02$), whereas morphine + naloxone was not significantly different from control but was significantly different from the morphine + saline group ($P = .006$).

3.2.2. MG + naloxone

Administration of naloxone failed to antagonize MG's respiratory effects. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 5A–C, respectively. Two-way RM ANOVA for respiratory frequency showed that MG had a significant effect at both 30 (Fig. 5D) ($F[1,7] = 14.4$; $P = .007$) and 60 minutes (Fig. 5E) ($F[1,7] = 8.99$; $P = .020$), whereas naloxone effects and interactions were not significant at either time point. Despite the lack of naloxone effects, the significant main effect of MG indicates that both MG + saline and MG + naloxone treatments resulted in elevated respiratory frequencies compared with controls.

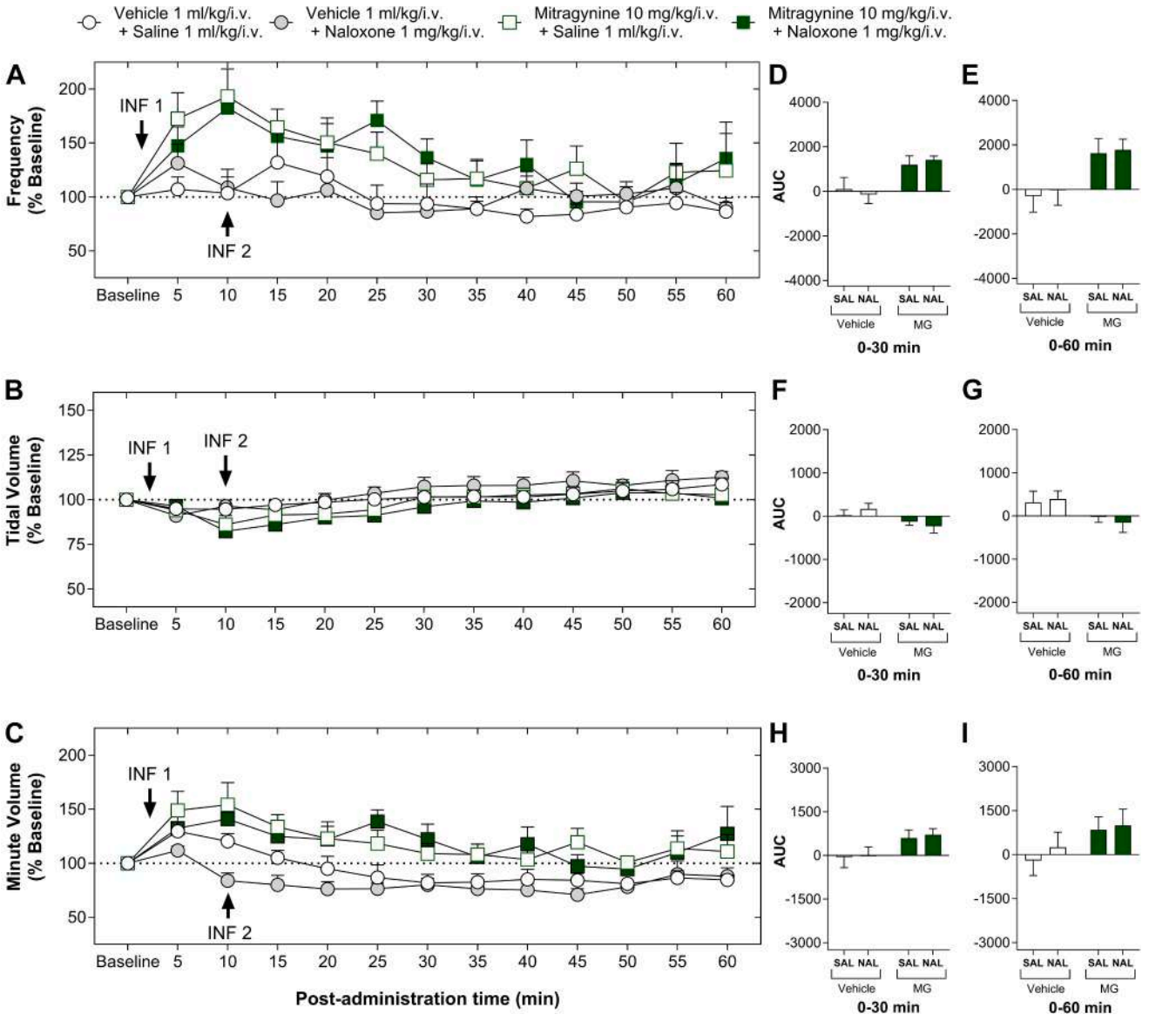


Fig. 5. Time course and AUC of MG and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of MG with naloxone rescue treatment in rats. MG was administered (10 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes post-MG infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes post-MG infusion. Statistically significant differences between treatments are indicated (P values).

For tidal volume, MG showed a near-significant effect at 30 minutes (Fig. 5F) ($F[1,7] = 5.26$; $P = .055$) and a significant effect at 60 minutes (Fig. 5G) ($F[1,7] = 7.18$; $P = .032$), with no significant naloxone effects or interactions. For minute volume, 2-way RM ANOVA showed no significant effects of MG, naloxone, or their interaction at either 30 (Fig. 5H) or 60 minutes (Fig. 5I). These results indicate that naloxone did not effectively antagonize the respiratory effects of MG, suggesting a mechanism distinct from typical opioid activity.

3.2.3. 7-HMG + naloxone

Naloxone effectively reversed the respiratory depression induced by 7-HMG. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 6A–C, respectively. Two-way RM ANOVA for respiratory frequency at 30 minutes showed that while main effects of naloxone ($F[1,7] = 0.460$; $P = .519$) and 7-HMG ($F[1,7] = 0.0922$; $P = .770$) were not significant, a significant naloxone × 7-HMG interaction ($F[1,7] = 12.5$; $P = .010$) was observed (Fig. 6D). Further analysis of the interaction using multiple comparisons demonstrated that neither 7-HMG + saline nor 7-HMG + naloxone administration significantly impacted respiratory frequency compared with control. The direct comparison between 7-HMG + saline and 7-HMG + naloxone further revealed a significant difference ($P = .03$). At 60 minutes, no significant main effects or interactions were detected (Fig. 6E).

For tidal volume, 2-way RM ANOVA demonstrated significant effects of naloxone ($F[1,7] = 18.0$; $P = .004$), 7-HMG ($F[1,7] = 10.4$; $P = .014$), and naloxone × 7-HMG interaction ($F[1,7] = 11.7$; $P = .011$) at 30 minutes (Fig. 6F). Despite a significant main effect of naloxone, there was no significant difference between vehicle + naloxone and saline + saline control. Post hoc analysis also revealed significant depression with 7-HMG + saline compared

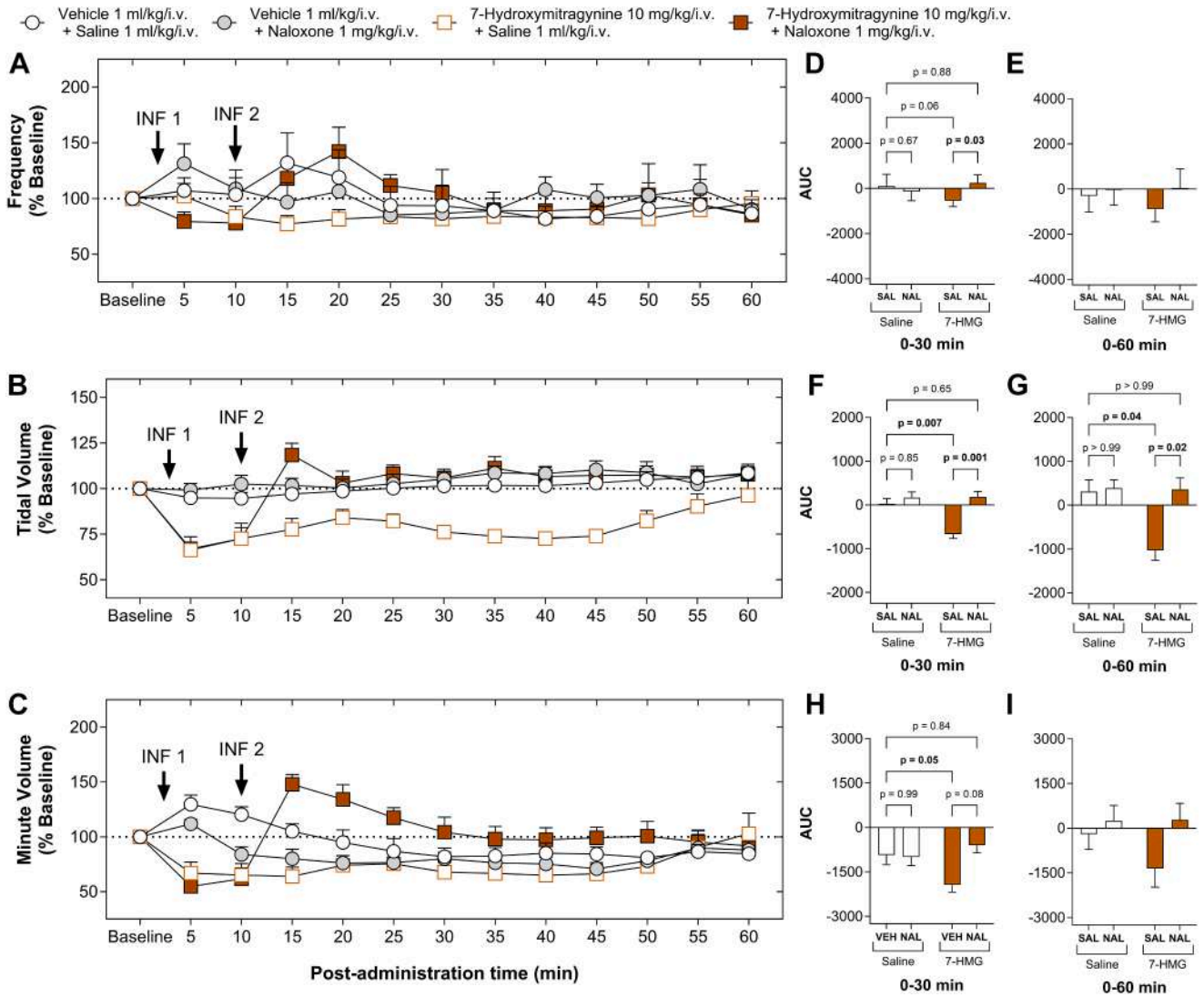


Fig. 6. Time course and AUC of 7-HMG and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of 7-HMG with naloxone rescue treatment in rats. 7-HMG was administered (10 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes post-7-HMG infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes post–7-HMG infusion. Statistically significant differences between treatments are indicated (P values).

with that with control ($P = .007$), which was not observed when naloxone was coadministered ($P = .65$). Additionally, the 7-HMG + saline and 7-HMG + naloxone treatments were significantly different from each other ($P = .001$). These effects on tidal volume persisted at 60 minutes for naloxone ($F[1,7] = 18.0$; $P = .004$), 7-HMG ($F[1,7] = 10.4$; $P = .014$), and naloxone × 7-HMG interaction ($F[1,7] = 11.7$; $P = .011$) (Fig. 6G). At 60 minutes, 7-HMG + saline showed significant depression compared with control ($P = .04$), whereas the 7-HMG + naloxone group was not significantly different from control. The difference between 7-HMG + saline and 7-HMG + naloxone treatments remained significant ($P = .02$).

For minute volume, despite nonsignificant main effects of naloxone ($F[1,7] = 3.85$; $P = .090$) and 7-HMG ($F[1,7] = 1.58$; $P = .249$) at 30 minutes, a significant naloxone × 7-HMG interaction ($F[1,7] = 26.7$; $P = .001$) was observed (Fig. 6H). Post hoc analysis at 30 minutes revealed a significant minute volume depression in the 7-HMG + saline group compared with that in the control group ($P = .05$), which was nonsignificant when naloxone was also administered. At 60 minutes, the interaction approached but did not reach significance (Fig. 6I) ($F[1,7] = 5.07$; $P = .059$).

4. Discussion

In the present study, both morphine and 7-HMG produced significant respiratory depression as indicated by decreases in breathing frequency, tidal volume, and minute volume, whereas MG unexpectedly increased respiratory frequency. Naloxone significantly reversed (ie, antagonized) the respiratory depressant effects of both morphine and 7-HMG, whereas the respiratory stimulant effects of MG were not antagonized or otherwise modified by naloxone administration. These results provide strong evidence that 7-HMG shares respiratory depressant effects with classical opioids, whereas MG increases respiration through mechanisms not involving opioid receptors.

Morphine, our reference compound, exhibited clear respiratory depression evidenced by decreases in tidal volume and minute

volume. At the highest dose tested (32 mg/kg), morphine significantly reduced tidal volume and minute volume, whereas no significant effects were observed at the lower dose (10 mg/kg), indicating dose-dependent respiratory depression under ambient air conditions. These effects were completely reversed by naloxone with an overshoot in respiration, confirming the well-established role of opioid receptor mechanisms in morphine-induced respiratory depression.[49–51] Rebound hyperventilation, or an overshoot in respiration, can occur following administration of naloxone after morphine-induced respiratory depression. Administration of naloxone rapidly displaces morphine from MORs in the brainstem, abruptly restoring excitatory input to respiratory centers and together with compensatory physiological mechanisms, triggering the overshoot observed.[2,3,52,53] Our findings are consistent with previously documented morphine-induced decreases in respiration across multiple species using various routes of administration.[1,54–56]

In the current study, MG did not produce respiratory effects typical of most opioid receptor agonists, that is, did not decrease respiration up to 17.8 mg/kg. Doses of MG larger than 17.8 mg/kg produced overt signs of toxicity and were not fully tested. Instead, MG produced significant increases in respiratory frequency. In previously published studies, MG was reported to decrease some respiratory parameters.[33] Lyophilized kratom tea also decreased respiration rates.[57] However, the experimental parameters of the previous studies (eg, oral administration in mice) differed from the experimental parameters of the current study (eg, intravenous administration in rats). With respect to route of administration, cytochrome 450-mediated conversion of MG into its active metabolites is expected to be greater following oral as compared with intravenous administration.[58–61] Although metabolism to sufficiently large amounts of 7-HMG could potentially contribute to the effects produced by MG administration, our group has demonstrated MG-induced antinociception in mice is evident when the amounts of 7-HMG measured in the brain are far less than the amounts required for antinociceptive effects.[62] Therefore, experimental support for a MG prodrug hypothesis is currently lacking, and any differences in 7-HMG formation across routes of administration do not appear to explain variable outcomes in studies of antinociception. Alternatively, the effects of MG appear to differ consistently between mice[33,57] and rats.[32] For example, under experimental conditions in rats that are highly sensitive to the antinociceptive effects of 7-HMG and other opioid agonists, our group has demonstrated that MG administration is devoid of antinociceptive effects.[24] The present study extends these findings to include respiratory effects, which in rats are not merely lacking but rather opposite in direction to the respiratory depressant effects of opioids. Intravenous administration of MG produced no significant change in temperature up to 10 mg/kg,[63] indicating that the MG-induced increase in frequency may not be due to an increase in the metabolic rate. The MG-induced increase in frequency might be due to its in vitro α2-adrenergic antagonistic activity,[64] which would prevent norepinephrine negative feedback,[65,66] resulting in increased alertness, arousal, and overall activity.

Results for 7-HMG demonstrated significant dose-dependent respiratory depression, particularly at the 10-mg/kg dose. Importantly, we observed distinct time-dependent effects in the respiratory parameters. At 30 minutes postadministration, 7-HMG significantly decreased respiratory frequency and minute volume at both 3.2- and 10-mg/kg doses, but these effects were not maintained at 60 minutes. This temporal profile indicates that 7-HMG's respiratory depressant effects peak earlier and begin to diminish over time, contrasting with morphine, which maintained effects across both 30 and 60 minutes. The reductions in respiratory frequency, tidal volume, and minute volume by 7-HMG observed in this study parallel the characteristic respiratory profile of classical μ-opioid agonists such as morphine, which are consistent with prior findings where 7-HMG showed respiratory depressant effects in mice.[33] Notably, the reversal of 7-HMG-induced respiratory depression by naloxone provides pharmacological evidence for opioid receptor-mediated effects. The efficacy of naloxone antagonism, particularly evident in tidal volume measurements, supports substantial MOR involvement in 7-HMG's respiratory effects. The ED_{50} of 7-HMG (6.98 mg/kg) to decrease minute volume was 4.5-fold more potent than morphine (31.5 mg/kg). These findings align with previous in vivo characterization, demonstrating that 7-HMG has 3- to 13-fold higher potency than morphine.[24–26] Collectively, these results suggest that 7-HMG functions primarily as an MOR agonist.

The opposing respiratory effects of MG and 7-HMG suggest that MG—the most abundant kratom alkaloid, constituting up to 66% of the total alkaloidal fraction[67]—may counteract the respiratory depressive effects of 7-HMG. In the study by Hill et al,[33] MG was shown to reduce minute volume; however, its respiratory depressant effects exhibited a ceiling effect, such that respiratory depression did not continue to worsen at higher doses of MG. In contrast, 7-HMG did not display a ceiling effect.[33] Furthermore, Wilson et al[57] demonstrated that although lyophilized kratom tea (containing all alkaloids in the plant) decreased respiratory rate, the reduction in respiration caused by morphine was significantly greater than that produced by the kratom tea. These findings suggest that MG could mitigate the respiratory depressant effects of 7-HMG, given that the magnitude of 7-HMG-induced respiratory depression is similar to that of morphine. MG is metabolized to 7-HMG in liver microsomes, with this conversion occurring more efficiently in human liver microsomes than in mouse liver microsomes.[68,69] However, the extent to which 7-HMG contributes to the respiratory depressant effects of MG may be limited because MG produces a ceiling effect on respiration.[33] Furthermore, MG-induced antinociception in mice is observed even when brain concentrations of 7-HMG are significantly lower than those required to elicit antinociceptive effects on their own.[62] In rats, ketoconazole was shown to inhibit the metabolism of MG to 7-HMG, indicating that rat liver microsomes also mediate this metabolic pathway.[59] In humans, co-administration of itraconazole with kratom tea reduced 7-HMG exposure by 56%, suggesting that 7-HMG exposure following kratom consumption may be higher in humans than in rodents.[70] Taken together, these results support the idea that MG contributes to the improved safety profile of kratom. However, caution is warranted with higher doses of MG because administration of a high dose of MG (17.8 mg/kg) induced seizure-like activity without causing respiratory depression. This observation aligns with poison control center data (2011–2017), documenting 113 seizures among 1174 kratom exposures.[71]

Adding to these concerns is the rapid proliferation of kratom-associated products specifically marketed for their high 7-HMG content.[27,28] An online market analysis conducted in February 2025 identified 304 semisynthetic kratom products. The majority (82.2%) contained only 7-HMG and were formulated as chewable or sublingual tablets, shots, or gummies. Products containing 7-HMG in combination with another semisynthetic compound, mitragynine pseudoindoxyl, accounted for 14.5% and were most commonly marketed as kratom.[72] Such formulations may be mistakenly perceived as typical kratom products, increasing the risk of unintentional exposure to pharmacologically significant doses of 7-HMG. The ability of 7-HMG to induce respiratory depression similar to morphine, highlights the urgent need for the regulation of these 7-HMG–containing products.

Although the findings of this study contribute significantly to our understanding of the respiratory effects of MG and 7-HMG and their interaction with naloxone, several limitations should be considered. First, we examined respiration after acute administration only; repeated administration might yield different respiratory profiles. Second, although we focused on respiratory parameters, a comprehensive safety evaluation would require assessment of additional physiological systems, such as cardiovascular and carcinogenic toxicity. Third, kratom contains numerous alkaloids; additional research is needed to determine the respiratory effects of other constituent alkaloids and their combinations. Future studies are also needed to investigate the molecular mechanisms underlying the distinct respiratory profile of MG, which was not affected by naloxone.

5. Conclusions

These respiratory depression findings have direct implications for evaluating kratom's safety profile and informing regulatory decisions. The observation that MG does not produce respiratory depression, and may even stimulate respiration, is particularly intriguing. However, the potent respiratory depressant effects of 7-HMG raise significant safety concerns, especially considering the increasing presence of novel semisynthetic products that may contain up to 98% 7-HMG.[28,72] Importantly, the respiratory depression produced by 7-HMG is sensitive to naloxone, which may have lifesaving implications in emergency medicine.

Abbreviations

AUC, area under the curve; 7-HMG, 7-hydroxymitragynine; MG, mitragynine; MOR, μ-opioid receptor; OIRD, opioid-induced respiratory depression; RM, repeated-measures.

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Conflict of interest

The authors declare no conflicts of interest.

Data availability

The data that support the findings of this study are available within the article. Any additional information is available on request from the corresponding authors.

CRediT authorship contribution statement

Julio Zuarth Gonzalez: Conceptualization, Formal analysis, Investigation, Writing – Original Draft, Writing – Review and Editing. **Alexandria Ragsdale**: Investigation, Writing – Review and Editing. **Sushobhan Mukhopadhyay**: Resources, Writing – Review and Editing. **Christopher McCurdy**: Conceptualization, Funding acquisition, Resources, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing. **Lance McMahon**: Conceptualization, Funding acquisition, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing. **Samuel Obeng**: Formal analysis, Writing – Original Draft, Writing – Review and Editing. **Jenny Wilkerson**: Conceptualization, Formal analysis, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing.

Supplemental material

This article has supplemental material available at jpet.aspetjournals.org.

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Site selective C–H functionalization of *Mitragyna* alkaloids reveals a molecular switch for tuning opioid receptor signaling efficacy

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Mitragynine (MG) is the most abundant alkaloid component of the psychoactive plant material "kratom", which according to numerous anecdotal reports shows efficacy in self-medication for pain syndromes, depression, anxiety, and substance use disorders. We have developed a synthetic method for selective functionalization of the unexplored C11 position of the MG scaffold (C6 position in indole numbering) via the use of an indole-ethylene glycol adduct and subsequent iridium-catalyzed borylation. Through this work we discover that C11 represents a key locant for fine-tuning opioid receptor signaling efficacy. 7-Hydroxymitragynine (7OH), the parent compound with low efficacy on par with buprenorphine, is transformed to an even lower efficacy agonist by introducing a fluorine substituent in this position (11-F-7OH), as demonstrated in vitro at both mouse and human mu opioid receptors (mMOR/hMOR) and in vivo in mouse analgesia tests. Low efficacy opioid agonists are of high interest as candidates for generating safer opioid medications with mitigated adverse effects.

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In search of molecules with robust clinical effects in the area of central nervous system (CNS) disorders and ability to repair synaptic function in the brain, we have been led to atypical modulators of endogenous opioid signaling[1–4]. In this context, we became interested in the psychoactive plant *Mitragyna speciosa* that has been used for centuries in Southeast Asia for treatment of pain, fatigue, opium dependence, and a number of other ailments. In the US, the use of the dry leaf material, known as "kratom", has been on the rise in the last decade, along with the number of anecdotal reports that point to the efficacy of kratom in a range of disorders with limited therapeutic options, including opioid dependence, treatment-resistant depression, anxiety, and pain syndromes[5–12]. A number of alkaloids in this plant, including mitragynine (MG) and its oxidation product 7-hydroxymitragynine (7OH, Fig. 1), have been found to bind to opioid receptors and represent molecular scaffolds for the development of opioid receptor modulators[13,14]. MG, as an indole alkaloid, shows no structural resemblance to traditional morphine-type compounds and represents an atypical opioid ligand with distinct signaling properties and physiological effects compared to clinically used opioid analgesics. For example, we have shown that MG is a partial MOR agonist (mu-opioid receptor agonist) with a potential bias for G protein signaling (showing no β-arrestin-2 recruitment) in cell-based assays[15,16]. Further, we have found that 7OH is a more potent and efficacious partial MOR agonist compared to MG, and that it acts as a potent analgesic in mice[15,16]. The relative activities of MG and 7OH are highly relevant, as we recently reported that 7OH is formed in vivo from MG and mediates MG's analgesic effects in mice[17]. In additional preclinical studies, others have found that MG exhibits antinociceptive effects in dogs comparable to those of codeine but with less respiratory depression[18], and that the compound is not self-administered by rats, but instead, inhibits self-administration of morphine and heroin[19,20].

The study and development of safer opioids is a long-standing scientific and societal goal, and a part of the scientific strategy proposed by the National Institute of Health (NIH) initiative to address the opioid crisis[21,22]. Further, there is a growing interest in the use of opioid receptor modulators as medicaments for depression and anxiety disorders (and other psychiatric diseases)[1,23]. Therefore, safer opioid modulators represent promising treatments for disorders spanning a wide spectrum of physical and emotional pain. Accordingly, further understanding of kratom and its alkaloids has major implications for public health.

Our previous report provided a basic structure–activity relationship (SAR) map of MG, in terms of in vitro opioid receptor pharmacology, with respect to the substituents on the saturated rings of MG. This work was enabled by the total enantioselective synthesis of MG developed in our laboratories[15]. Several other research teams have also reported total syntheses of MG and related compounds[24–28]. These de novo synthetic approaches offer nearly unlimited exploration of complex MG-related molecules, but they are labor intensive due to the high number of synthetic steps. Thus, the need to access many different derivatives at multiple positions demands more efficient synthetic strategies. MG can be extracted from kratom leaf matter in multigram quantities (~1% of dry kratom mass[14,15]), and therefore, there is a strong incentive to develop synthetic methods for direct functionalization of MG, for example, via late-stage C–H functionalization, rather than laborious total synthesis.

With respect to derivatization of the indole nucleus, the methoxy group at the C9 position is synthetically accessible by selective demethylation and subsequent functionalization of the free phenol[16]. For the adjacent C10 position, a small series of compounds has been prepared[14,29], while the C11 position remained unexplored due to the lack of functionalization methods (see below). Our preliminary docking studies suggested that relatively small substituents (e.g., F, Cl, and Me) would be tolerated at C11 (Fig. 1)[15]. However, it is presently difficult to predict the effect of such substituents on opioid receptor activation potency, efficacy, or signaling bias, demonstrating the need for C–H bond functionalization approaches compatible with the structural complexity presented by *Mitragyna* alkaloids.

More than a decade ago, we formulated the general concept of "C–H bonds as ubiquitous functionality" and demonstrated the strategic impact of C–H functionalization in both the construction of molecular frameworks and modification of existing complex cores[30–33]. For the latter, termed "complex core

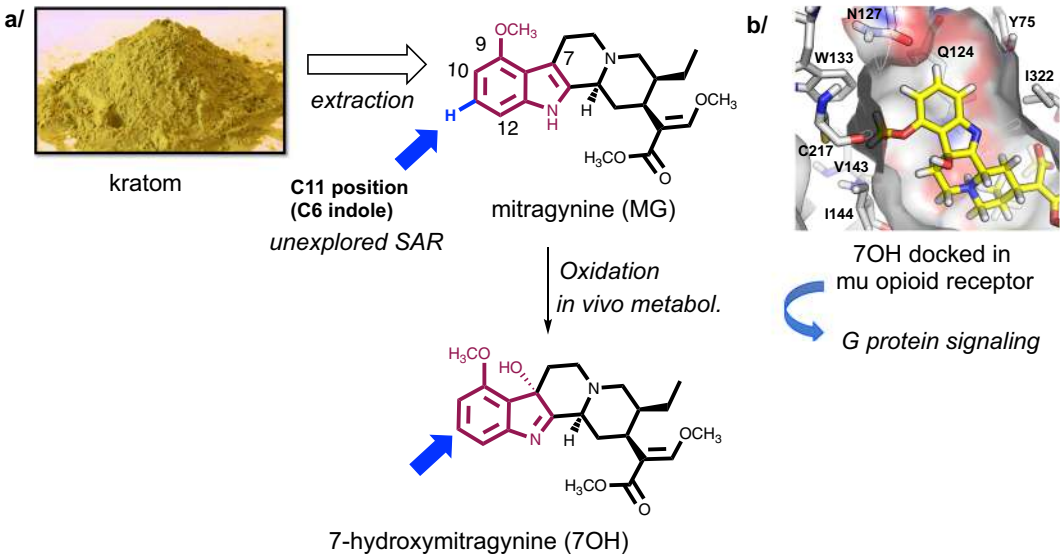


Fig. 1 The rationale for selective C–H bond functionalization of mitragynine (MG) and 7-hydroxymitragynine (7OH). a MG is readily obtained in multigram quantities from kratom powder by extraction, hence there is an incentive to develop selective functionalization of MG and related scaffolds via late-stage C–H bond functionalization. Specifically, the C11 position has not been explored in terms of mu-opioid receptor (MOR) signaling and other biological effects, due to the lack of chemistries for functionalization of this position. **b** The top-right panel shows a docking pose of 7OH in human MOR highlighting the aromatic ring and the proposed binding pocket in the receptor.

diversification" or "late-stage functionalization", the synthetic power of this concept is readily apparent as the positional (as well as stereoisomeric) analogs of complex starting materials are accessed with high efficiency when compared to lengthy de novo approaches[30–36]. Late-stage functionalization approaches have since been widely adopted and become a common part of chemists' armamentarium[37–41]. These concepts are being extended beyond C–H bonds to include the possibilities of skeleton modification via C–C and other bond activation[42–45].

In this paper, we describe application of these concepts to *Mitragyna* alkaloids in the context of mapping their neuropharmacology. We introduce a strategic temporary modification of the MG alkaloid skeleton ("complex core restructuring"), to create a distinct chemotype disposed toward the desired C–H functionalization chemistry, namely C11 functionalization of MG's indole nucleus. In this manner, we offer a solution that provides for rapid and selective functionalization of the aromatic ring of MG at the C11 and C12 positions (C6 and C7 positions in indole numbering). We subsequently found that these analogs enable fine-tuning of opioid receptor signaling efficacy, which represents one of the currently most promising strategies for creating safer opioid therapeutics.

Results

C–H borylation of MG yields C12-substituted analogs. MG is a complex natural product of a corynanthidine alkaloid type decorated with a number of functional groups, including enol ether, ester, tertiary amine, indole nitrogen, and aromatic methyl ether, arranged in a specific constitutional and geometrical configuration that underlies its opioid activity. It therefore poses an exciting challenge for late-stage functionalization. In this study, we focused on functionalization of the indole arene ring (the rationale is discussed in "Introduction"). Iridium-catalyzed arene C–H borylation was selected due to its wide substrate scope, including basic heterocyclic compounds[46,47].

After exploration and optimization of reaction conditions (Supplementary Table 1), we found that borylation is compatible with the MG chemotype. Using bis(pinacolato)diboron (B_2Pin_2) under catalytic $[Ir(COD)OMe]_2$ and 4,4′-di-tert-butyl-2,2′-dipyridyl (dtbpy) as the ligand in absolute heptane at 80 °C for ≥17 h, C12-boronate ester **1** was obtained as the single isomer (Fig. 2a). This compound proved to be unstable and decomposed during silica gel column chromatography, and we thus confirmed its formation in the crude material via nuclear magnetic resonance spectroscopy (NMR), thin-layer chromatography (TLC), and mass spectroscopy (MS). The crude ester **1** was transformed to several derivatives without further purification in good yields. Namely, 12-Cl- (**2**) and 12-Br-MG (**3**) were prepared in two steps in 70% and 65% yield, respectively (Fig. 2a)[48]. The observed regioselectivity is consistent with the known directing effect of the indole nitrogen and C7-borylation of 2,3-disubstituted indoles[49,50].

Known C6 indole derivatization methods do not produce C11 analogs of MG. After achieving C12 functionalization we turned our attention to the C11 position in MG (C6 position in indole numbering), which remained unexplored due to the limited reaction repertoire available for C6 indole functionalization[51]. We considered several known approaches for C6 indole functionalization to gain access to these derivatives. Specifically, we examined the possibility of blocking the indole nitrogen's directing effect with a protecting group, along the path demonstrated in the context of tryptophan substrates, where protection with a bulky triisopropyl silane (TIPS) group led to C6-borylation under optimized C–H borylation conditions (Fig. 2b)[52]. However, this approach was not applicable to MG as the TIPS-protected MG decomposed under the catalytic conditions. On the other hand, *p*-methoxybenzyl (PMB)-MG was either unreactive or formed undesired products under the reaction conditions. Tert-butoxycarbonyl (Boc)-protected MG provided the C11-boronate ester, but the subsequent functionalization reaction was inefficient (Supplementary Figs. 3 and 4). The latter approach was not further optimized as an entirely different protection method was found to be successful (see below).

Another avenue of enquiry was inspired by an intriguing thermal rearrangement of 3-bromoindolenines to 6-bromoindoles[53], which had been successfully applied in complex substrates en route to stephacidine A alkaloids[42]. However, this approach also failed in the MG scaffold under similar reaction conditions (Fig. 2c); namely, MG was unreactive in the presence of *N*-bromosuccinimide (NBS) under neutral conditions, while under acidic conditions, 12-Br-MG (**3**) was the major product (Fig. 2c). Apparently, the C9 methoxy group exerts an activating and directing effect in the benzene ring of the indole nucleus once the bromination reagent is sufficiently activated[54].

Finally, the methods relying on remote directing effect of groups attached to the indole nitrogen, such as the copper-catalyzed C6 functionalization of indoles using the phosphinimide directing group[55], were not examined, as (1) the deprotection step involves harsh reduction conditions (e.g., $LiAlH_4$) incompatible with MG's functionalities or (2) the C6 functionalization is too restrictive in terms of the functionalization chemistry or substrate requirements[56,57].

Complex core restructuring: C–H borylation of mitragynine-ethylene glycol adduct (MG-EG). Direct functionalization of MG gave C12-halo or -boronate ester analogs, while re-routing this regioselectivity was unsuccessful via either catalyst optimization or indole nitrogen protection. The known methods for C6 indole functionalization—that possess the required generality and scope to pursue systematic SAR studies—failed to provide an efficient synthetic route to C11-substituted MG analogs.

We therefore resorted to changing the reactivity of the MG core by "removing" the indole double bond (2,3-π bond in indole numbering) by either reduction or oxidation, or oxidation followed by rearrangement (Fig. 3). 7OH was examined as an indolenine substrate in the Ir-catalyzed borylation reaction, but there was no conversion to a borylated product. 7OH can be readily rearranged to mitragynine pseudoindoxyl with zinc triflate as reported by us previously[16], however this pseudoindoxyl system afforded C12 borylation (Fig. 3), while the corresponding 12-boronate ester (not shown) was unreactive in the subsequent halogenation reaction. Next, we investigated the possibility of using a reduced indole substrate, namely dihydromitragynine[58]. With this compound, we observed a partial conversion of starting material to the corresponding 11-boronate ester (50% conversion as determined by ^1H NMR). However, the following halogenation reaction gave complex mixtures of MG and 2,3-dihydromitragynine as the major products, and 11-Br-MG and 11-Br-2,3-dihydromitragynine as the minor products (Fig. 3).

Lastly, we focused on oxidative attachment of ethylene glycol to the indole nucleus, rendering a distinct chemotype: mitragynine-ethylene glycol (MG-EG, Fig. 4a). This unusual compound was first introduced by Takayama and co-authors to provide access to the C10-substituted analogs via electrophilic aromatic substitution, and this scaffold showed potent activity at MOR[14,59]. Aside from these studies, this indole-ethylene glycol adduct has not been explored in the context of indole chemistry. We became interested in this compound as the ethylene glycol group not only masks the indole, which is particularly relevant in this study, but

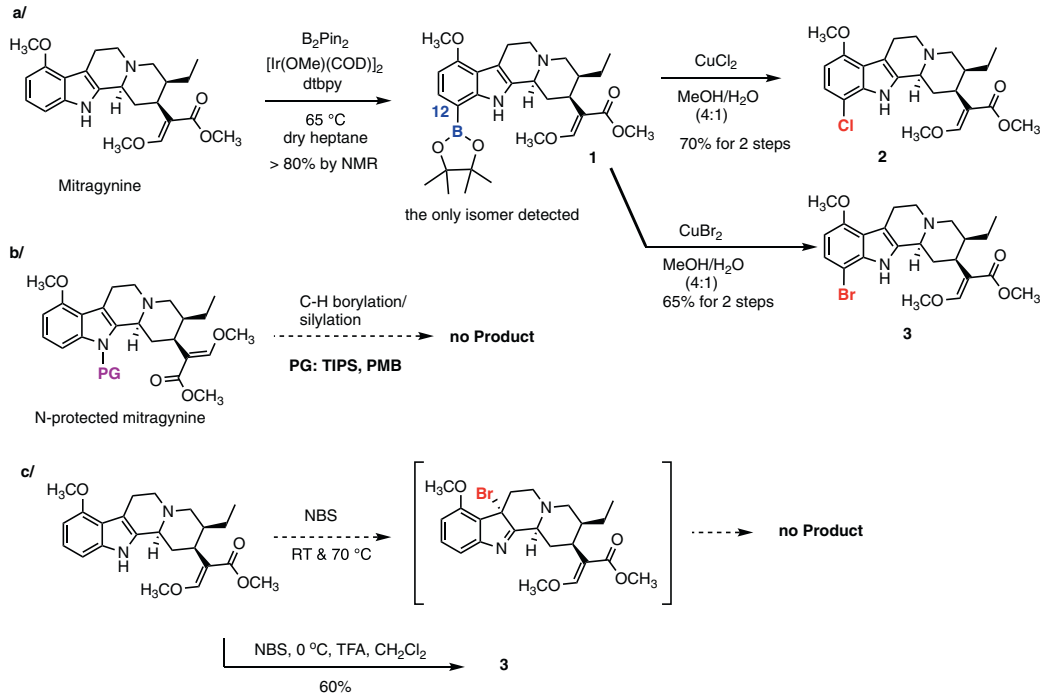


Fig. 2 Examination of direct functionalization of mitragynine indole nucleus. a C–H borylation of MG provides exclusively C12-derivatives consistent with indole directing effects (C7 in indole numbering). The yields are a general range based on >10 repetitions by two experimenters. **b** The positional selectivity cannot be re-routed to C11 functionalization via previously reported TIPS protection of the indole nitrogen. **c** The C3-indole halogenation–rearrangement process[53] is not applicable to MG. Bromination under acidic conditions gave 12-Br-MG as the major product.

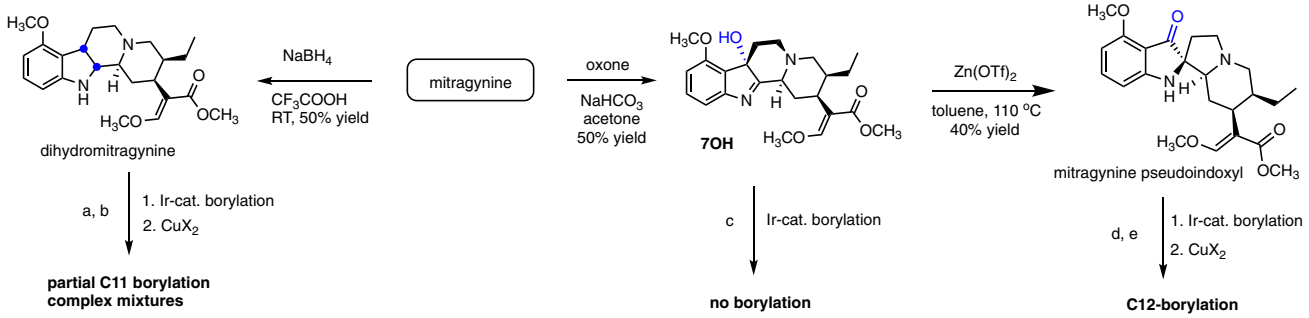


Fig. 3 Catalytic borylation of MG analogs with reduced, oxidized, or rearranged indole nucleus. a B_2Pin_2, dtbpy/Me_4-phen, $[Ir(COD)(OMe)]_2$, 65 °C, dry heptane; 50% conversion to 11-boronate ester by ^1H NMR. **b** $CuBr_2$, MeOH/H_2O (4:1), 80 °C; using the crude borylation mixture, provided mixtures of mitragynine, 2,3-dihydromitragynine (major products, ~60% by ^1H NMR) and 11-Br-dihydromitragynine and 11-Br-MG (minor products, ~30% by ^1H NMR). **c** B_2Pin_2, dtbpy/Me_4-phen, $[Ir(COD)(OMe)]_2$, 65 °C, dry heptane; no borylation product was detected. **d** B_2Pin_2, dtbpy/Me_4-phen, $[Ir(COD)(OMe)]_2$, 65 °C, dry heptane; complete conversion to C12-boronate by ^1H NMR). **e** $CuBr_2$, MeOH/H_2O (4:1), 80 °C; no reaction.

also dramatically alters the shape of the entire alkaloid scaffold. We crystallized MG-EG from methanol and confirmed its 3D structure (Fig. 4b, CCDC 1905559), which revealed the chair-like conformation of the dioxane ring and the propeller-like arrangement of the three rings converging on the C–C bond of the former indole ring.

When we applied the catalytic borylation protocol (with the dtbpy ligand) to MG-EG, a 1:1 ratio of C11 (**4a**) and C12 (**4b**) products was observed (Fig. 4c), clearly confirming the different reactivity property of this chemotype, and unlocking the possibility of optimizing the reaction conditions to favor C11 functionalization.[52,60,61] Indeed, ligand screening (Supplementary Table 3) showed that 3,4,7,8-tetramethyl-1,10-phenanthroline (Me_4-phen) as the ligand, together with B_2Pin_2 and $[Ir(COD)OMe]_2$ in heptane at 65 °C, gave the 11-borylated product (**4a**) as the major product with ratio >16:1 of C11 (**4a**): C12 (**4b**) products (Fig. 4c, confirmed by the subsequent bromination).

We reasoned that by converting the indole nucleus to an unusual aniline derivative, the directing effects of the nitrogen would be diminished while its steric effects would gain importance, resulting in functionalization of the unhindered C11 position (after catalyst–ligand optimization). Boronates **4a** and **4b** were converted to the bromo derivatives **5** and **6**, respectively[48], using copper(II) bromide. Thus, this sequence provided two 11-substituted MG-EG intermediates—boronate **4a** and bromide **5**—with versatile synthetic potential.

Preparation of C11-substituted analogs of mitragynine and related scaffolds. The 11-boronate ester (**4a**) was converted to the compounds 11-Cl-MG-EG (**13**) and 11-OH-MG-EG (**14**) using the appropriate substitution methods (Fig. 5a)[48,62]. For the preparation of additional derivatives, the 11-Br-MG-EG (**5**) served as the key intermediate enabling preparation of compounds **8**–**12** in one synthetic step (Fig. 5a), where X = I (**8**)[63], Me (**9**)[64], Ph

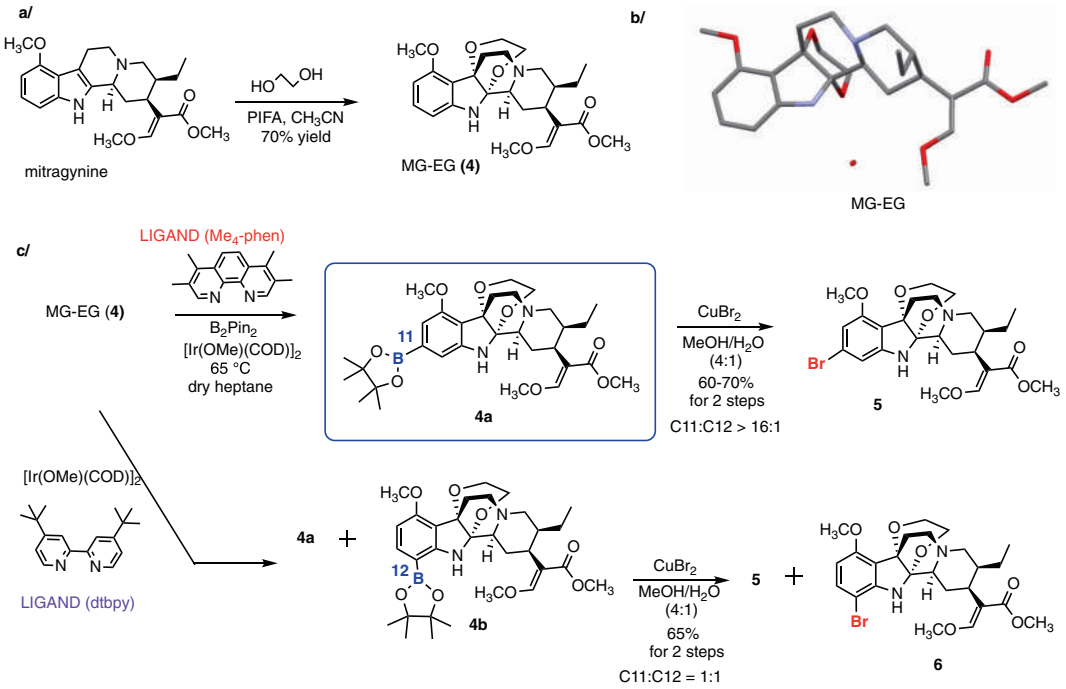


Fig. 4 C11-selective C–H borylation via conversion of MG to mitragynine-ethylene glycol adduct (MG-EG). a One-step synthesis of MG-EG, a stable derivative of MG. **b** The 3D structure of MG-EG was confirmed by X-ray crystallography. ORTEP representation of MG-EG structure is shown without hydrogen atoms for clarity. **c** C–H borylation of MG-EG gives a mixture of C11- and C12-boronates and the subsequent functionalization products. The high C11 selectivity was achieved by ligand optimization. The ratio was determined by ^{1}H NMR of crude reaction mixtures. The yields, given as a general range, are based on >20 repetitions by 3 experimenters, on a scale up to 100 mg of substrate.

(**10**)[65], $CONH_2$ (**11**)[66], and CN (**12**)[67]. Considering the complexity of these compounds, the yields were satisfactory and more than sufficient to produce practical amounts of the compounds for preliminary pharmacological evaluation. However, the synthesis of 11-F-MG-EG (**7**) proved more difficult compared to its other halogen relatives. Relevant reported procedures were not effective in this context; for example, direct fluorination of 11-boronate ester **4a** using copper-mediated fluorination with $(^{t}BuCN)_2CuOTf$ or $Cu(OTf)_2py_4$ failed[68,69], so did an indirect sequence of stannylation–fluorination[70]. Deoxyfluorination of the corresponding phenol **14** using Phenofluor™ Mix was also unsuccessful[71]. Eventually, 11-F-MG-EG (**7**) was synthesized from bromide **5** via the sequence of stannylation and fluorination (Fig. 5b)[72]. The triflate **14a** represents an alternative intermediate (to the corresponding bromide **5**) for further functionalization; for example, (1) 11-F-MG-EG (**7**) was also synthesized from triflate through a series of stannylation and fluorination (Fig. 5c) and (2) carboxamide **11** was prepared in 57% yield (versus 35% via the bromide, Supplementary information).

The ethylene glycol moiety can be readily removed under mild reductive condition in one step to yield 11-substituted MG analogs (**15**–**21**, Fig. 5d) in moderate-to-good yields. Our strategy enables functionalization of the C11 position with a wide range of substituents starting directly from the MG natural product, thus permitting systematic exploration of SAR at this position.

To convert the MG analogs to the corresponding 7OH series, we optimized a set of conditions using OXONE® in acetone (Fig. 5d)[73,74]. This procedure was successfully applied to the 11-substituted MG analogs, as demonstrated by preparation of the 11-halo derivatives' **22**–**24** analogs in yields around 50%. This oxidation protocol is superior to the previously reported approaches, namely PIFA gives complex mixtures of products and a lower yield of 7OH, while lead tetraacetate involves a toxic heavy metal and requires a second hydrolysis step[75].

Table 1 Binding affinities of 7OH analogs at the mouse opioid receptors.

Compound	K_i ± SEM (nM)[a]		
	mMOR	mKOR	mDOR
7OH	21.5 ± 0.8	119.0 ± 2.1	88.5 ± 9.9
11-F-7OH (**22**)	13.7 ± 1.0	21.0 ± 3.2	35.8 ± 2.0
11-Cl-7OH (**23**)	27.1 ± 1.1	30.7 ± 11.4	47.2 ± 2.4
11-Br-7OH (**24**)	32.4 ± 1.4	81.1 ± 7.1	57.7 ± 6.7

[a]All data points represent mean ± SEM (nM) of $n = 3$. [^{125}I]BNtxA was used as the standard radiolabeled ligand[98].

In summary, we developed access to 11-substituted analogs in three MG-related molecular series, compounds previously inaccessible via electrophilic substitution or other approaches, providing the means for a systematic SAR exploration of this position in multiple MG-type scaffolds.

Neuropharmacology: in vitro modulation of opioid receptors by the 11-substituted 7OH analogs. With efficient access to the C11 analogs, we examined the effect of C11 substitution on the opioid receptor pharmacology of MG analogs. We focused the initial set of biological assays on the 7OH series, as 7OH is an active metabolite of MG and exhibits an order of magnitude greater potency compared to MG as an MOR agonist[17]. First, we examined the 11-halo analogs of 7OH in radioligand binding studies to assess the affinity for the opioid receptors (Table 1). We found that the C11 halogen modulated affinity across all three opioid receptors: the 11-F compound **22** exhibited greater affinity compared to the parent 7OH, but this gain in binding was

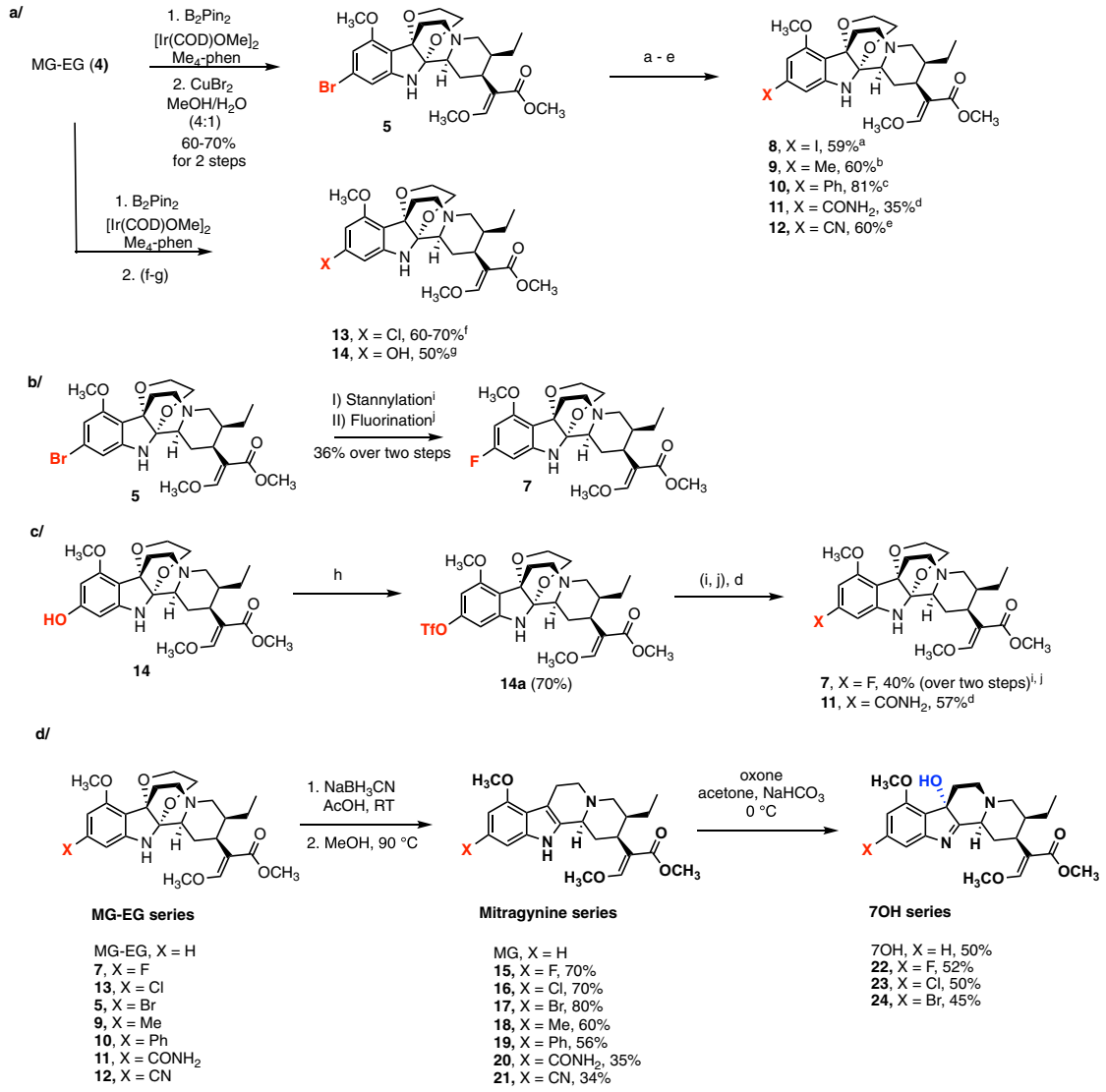
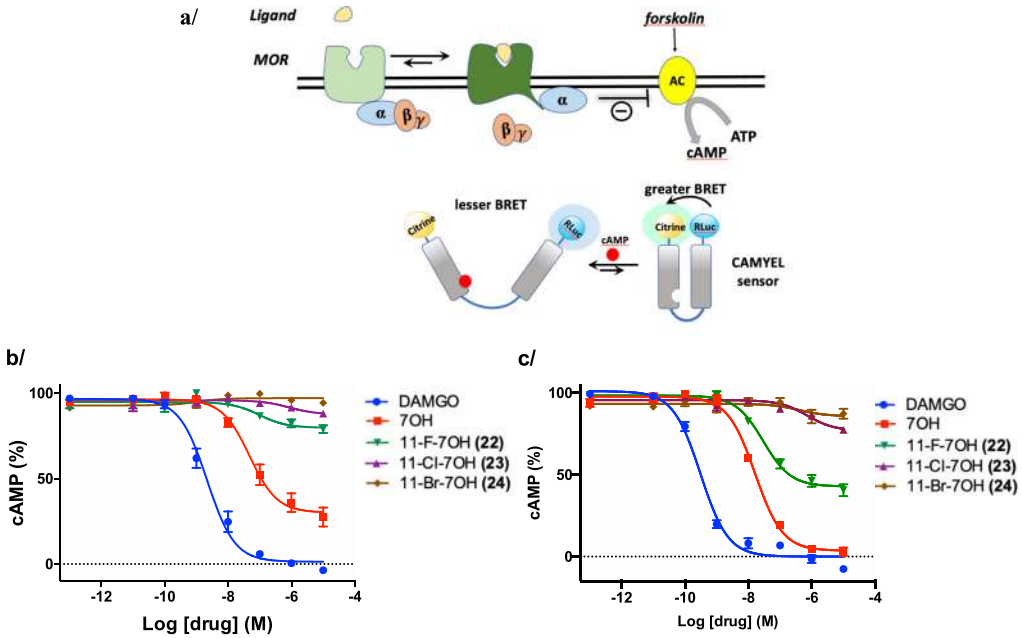


Fig. 5 The MG-EG scaffold is compatible with a wide array of functionalization chemistries. a, **b**, **c** Synthesis of C11-substituted analogs of MG-EG. (a) NaI, CuI, *N,N'*-dimethylethylenediamine, 110 °C, 46 h. (b) $Pd_2(dba)_3$, XPhos, DABAL-Me_3, THF, 60 °C, 2 h. (c) Pd(dppf)Cl_2·CH_2Cl_2, $PhB(OH)_2$, CsOAc, THF, 70 °C, 7 h. (d) $Pd(OAc)_2$, dppf, $Co_2(CO)_8$, DIPEA, NH_4Cl, imidazole, dioxane, 90 °C, 15 h. (e) $Pd_2(dba)_3$, Zn dust, $Zn(CN)_2$, $[HPtBu_3]BF_4$, DMF, RT, 3 h. (f) $CuCl_2$·$2H_2O$, MeOH: H_2O (4:1), 80 °C, 12 h. (g) H_2O_2, THF, 0 °C to RT, 30 min. (h) $PhNTf_2$, DIPEA, DMF, 50 °C, overnight. (i) $(^nBu_3Sn)_2$, $Pd(PPh_3)_4$, LiCl, dioxane, 100 °C, overnight. (j) AgOTf, F-TEDA-PF_6, RT, 20 min. The yields are based on >2 repetitions. **d** Reductive removal of ethylene glycol. This step renders 11-substituted MG analogs, which in turn can be oxidized to the corresponding 7OH derivatives. The yields are average based on at least 3 repetitions.

progressively lost as the halogen became larger (compounds **23** and **24**, Table 1). This effect was most pronounced at kappa-opioid receptor (KOR), where the affinity of **22** was six times greater than that of the parent compound 7OH. The compound **22** also had more than 2-fold greater affinity for delta-opioid receptor (DOR) compared to 7OH. The effect of the C11 halogen on MOR binding was subtle across the halogen series. Since the parent compound 7OH acts as a potent agonist, we examined the functional modulation of mouse MOR (mMOR) activation in living cells. Specifically, we examined the signaling consequences of G protein activation via detection of the downstream signaling molecule cyclic adenosine monophosphate (cAMP). cAMP is formed from adenosine triphosphate (ATP) by adenylyl cyclase (AC), which is inhibited by activation of the MOR and its associated G proteins. Changes in the level of cAMP were measured using a bioluminescence resonance energy transfer (BRET) functional assay. The BRET signal increases in response to decreasing amounts of cAMP as a result of conformational changes in the sensor. cAMP levels were raised prior to the assay by addition of forskolin as described previously (Fig. 6a, illustration created by the authors)[1,76].

In the cAMP assay, 7OH was a potent agonist with relatively high efficacy ($E_{max} = 70\%$, compared to DAMGO ([D-Ala^2, N-Me-Phe^4, Gly^5-ol]-enkephalin), $EC_{50} = 48$ nM, Fig. 6b, d). Introduction of the 11-fluoro substitution in compound **22** dramatically reduced the efficacy, rendering a low-efficacy agonist ($E_{max} \sim 21\%$) (Fig. 6b, d). For the **23** and **24** derivatives, the agonist activity at mMOR was nearly abolished ($E_{max} = 13\%$ for **23** and $E_{max} < 10\%$ for **24**). Accordingly, 11-halo substituents in the 7OH scaffold do not interfere with binding of these compounds to mMOR, but profoundly modulate the signaling efficacy at this receptor.

To determine if the efficacy modulation trend holds at the human receptor, we examined the 11-X-7OH series in



d/

Compound	E_{max} [a]	pEC_{50} [b] ± SE (EC_{50} nM)
DAMGO	100	8.6 ± 0.1 (2.2)
7OH	70	7.3 ± 0.1 (48.0)
11-**F**-7OH (**22**)	21	7.0 ± 0.3 (97.0)
11-**Cl**-7OH (**23**)	13	6.1 ± 1.2 (841.0)
11-**Br**-7OH (**24**)	<10	Not determined

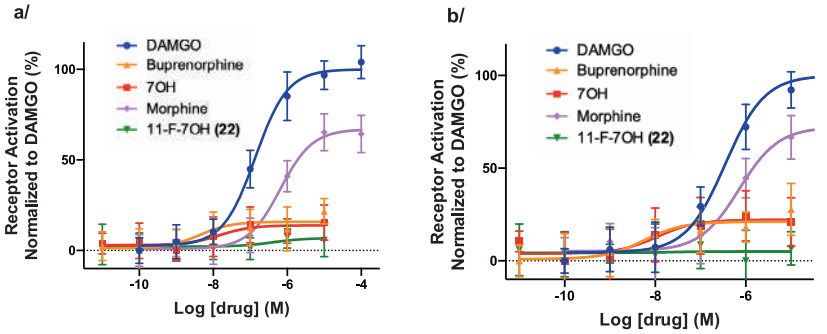
e/

Compound	E_{max} [a]	pEC_{50} [b] ± SE (EC_{50} nM)
DAMGO	100	9.5 ± 0.10 (0.3)
7OH	96	7.8 ± 0.10 (16.4)
11-**F**-7OH (**22**)	57	7.5 ± 0.10 (29.4)
11-**Cl**-7OH (**23**)	23	6.1 ± 0.6 (716.7)
11-**Br**-7OH (**24**)	14	6.4 ± 1.8 (401.0)

Fig. 6 Activity of 7-hydroxymitragynine (7OH) analogs at the mu-opioid receptor (MOR). a Conceptual representation of the MOR CAMYEL BRET assay. To measure G protein activation, MOR (light green) was coexpressed with G protein subunits $G\alpha_{oB}$, β_1, γ_2, and the BRET CAMYEL sensor. Forskolin activates adenylyl cyclase (AC), which converts ATP to cyclic adenosine monophosphate (cAMP). On activation of mMOR by ligand, α subunit inhibits AC, resulting in a decrease of cAMP accumulation. The CAMYEL sensor is comprised of citrine (mustard yellow) and *Renilla* luciferase (light blue) with human Epac1 (blue linker) interspaced between them. **b** Agonist activity of 7OH at mMOR, presented as a percentage of cAMP inhibition produced by the positive control DAMGO ([D-Ala^2, N-Me-Phe^4, Gly^5-ol]-enkephalin); **c** Agonist activity of 7OH analogs at hMOR; curves represent the average of $n = 3$, independent experiments with error bars representing ±SEM. **d** Functional activity of 7OH analogs at mMOR as determined in CAMYEL BRET assays. **e** Functional activity of 7OH analogs at hMOR as determined in CAMYEL BRET assays [[a]Agonist activity indicated by EC_{50} values, maximal efficacy is expressed as maximal inhibitory effect on cAMP levels relative to DAMGO ($E_{max} = 100\%$ - cAMP(min)%). [b]All data points represent $-\log EC_{50}$ ± SE (EC_{50} nM) for $n = 3$ repetitions].

hMOR-CAMYEL assay (Fig. 6c) using the same experimental setup as illustrated above (Fig. 6a). The parent compound 7OH was potent and highly efficacious ($E_{max} = 96\%$, compared to the reference ligand DAMGO, $EC_{50} = 16.4$ nM, Fig. 6c, e), whereas 11-F-7OH (**22**) was a partial agonist with markedly reduced efficacy ($E_{max} = 57\%$, $EC_{50} = 29.4$ nM, Fig. 6c, e). The **23** and **24** derivatives showed very low efficacy ($E_{max} = 23\%$ for **23** and $E_{max} = 14\%$ for **24**). Hence, we can conclude that the trend in efficacy signaling modulation in the halogen 7OH series is comparable between human and mouse MOR.

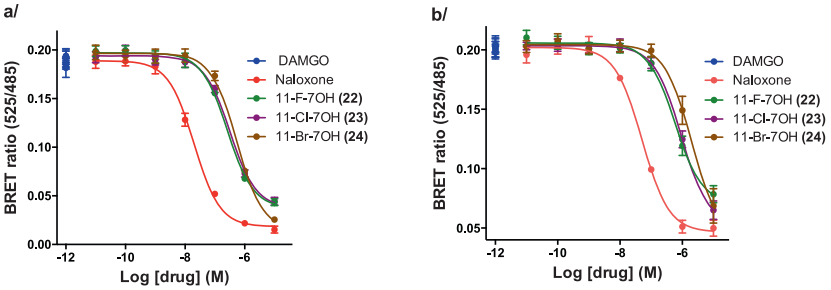
The measured efficacy of receptor agonists depends greatly on the receptor reserve pools and the extent of downstream signaling amplification of each functional readout. Under the conditions typically used in in vitro cell-based assays, such as the cAMP BRET assay discussed above, the efficacy of partial agonists appear exaggerated owing to large receptor pools produced by receptor overexpression and downstream signaling amplification[77]. To examine this effect for 7OH and its analogs, we adopted an assay that detects an active conformation of MOR, using the Nb33 nanobody sensor, and thus limits the signaling-related amplification (Nb33 BRET assay using a luciferase-tagged MOR and Venus-tagged Nb33, see Supplementary information)[78]. This assay was initially described by Stoeber et al.[79] and has recently been rigorously validated by Gillis et al.[78] via comparison to a number of independent signaling assays, and calibrated by direct comparison of several opioid drugs including well-established partial agonists (buprenorphine and morphine) and experimental opioids. Using mMOR and hMOR Nb33-BRET assays, we found that morphine and buprenorphine act as partial agonists, in comparison to DAMGO, with markedly different signaling efficacies (Fig. 7). While morphine appears as a partial agonist with relatively high efficacy ($E_{max} \sim 67\%$ (mMOR) and 72% (hMOR), Fig. 7),



c/

Compound	E_{max}[a] mMOR	E_{max}[a] hMOR	pEC_{50}[b] ± SE (EC_{50} μM) mMOR	pEC_{50}[b] ± SE (EC_{50} μM) hMOR
DAMGO	100	100	6.9 ± 0.10 (0.14)	6.5 ± 0.10 (0.35)
Morphine	67	72	6.2 ± 0.20 (0.67)	6.1 ± 0.20 (0.72)
Buprenorphine	16	21	8.4 ± 0.6 (0.004)	8.2 ± 0.50 (0.006)
7OH	14	22	7.8 ± 1.0 (0.015)	7.9 ± 0.90 (0.01)
11-**F**-7OH **(22)**	<10	<10	Not determined	Not determined

Fig. 7 Receptor activation by 7OH series in comparison to known opioids in BRET-based Nb33 recruitment assay at mMOR and hMOR. Intrinsic receptor activation efficacy was determined by the extent of recruitment of the conformationally selective nanobody Nb33 to hMOR or mMOR fused as its C terminus to NanoLuc, as measured by BRET. Positive controls used are: DAMGO, buprenorphine, morphine, and parent 7OH. **a** Agonist activity of 11-F-7OH (**22**) as compared to positive controls at mMOR. **b** Agonist activity of 11-F-7OH (**22**) as compared to positive controls at hMOR. Curves represent the average of $n = 3$, independent experiments with error bars representing ±SEM. **c** Functional activity of 11-F-7OH (**22**) and positive controls at mMOR and hMOR in Nb33 BRET Assay [[a]Agonist activity indicated by EC_{50} values, maximal efficacy is expressed relative to DAMGO (E_{max}). [b]All data points represent $-\log EC_{50}$ ± SE (EC_{50} μM) for $n = 3$ repetitions].



c/

Compound	pIC_{50}[c] ± SE (IC_{50} μM)
Naloxone	7.7 ± 0.1 (0.02)
11-**F**-7OH **(22)**	6.6 ± 0.1 (0.26)
11-**Cl**-7OH **(23)**	6.5 ± 0.1 (0.32)
11-**Br**-7OH **(24)**	6.3 ± 0.1 (0.55)

d/

Compound	pIC_{50}[c] ± SE (IC_{50} μM)
Naloxone	7.3± 0.1 (0.050)
11-**F**-7OH **(22)**	6.2 ± 0.1 (0.59)
11-**Cl**-7OH **(23)**	6.0 ± 0.1 (0.92)
11-**Br**-7OH **(24)**	5.7 ± 0.1 (1.97)

Fig. 8 Antagonist activity of 7OH analogs at mMOR and hMOR. To assay G protein activation, mMOR and hMOR was coexpressed with G protein subunits $G\alpha_{oB}$, β_1, γ_2, and the BRET CAMYEL sensor. Antagonism was measured by the inhibition of DAMGO's effect on cAMP. **a** Competitive antagonist activity of 7OH analogs at mMOR; positive control = naloxone. **b** Competitive antagonist activity of 7OH analogs at hMOR; positive control = naloxone. Curves represent the average of $n = 3$, independent experiments with error bars representing ±SEM. **c** Functional antagonist activity of 7OH analogs at mMOR as determined in CAMYEL BRET assays. **d** Functional antagonist activity of 7OH analogs at hMOR as determined in CAMYEL BRET assays [[c]Antagonist activity indicated by IC_{50} values for inhibition of a reference agonist (DAMGO), all data points represent $-\log IC_{50}$ ± SE (IC_{50} μM) for $n = 3$ repetitions].

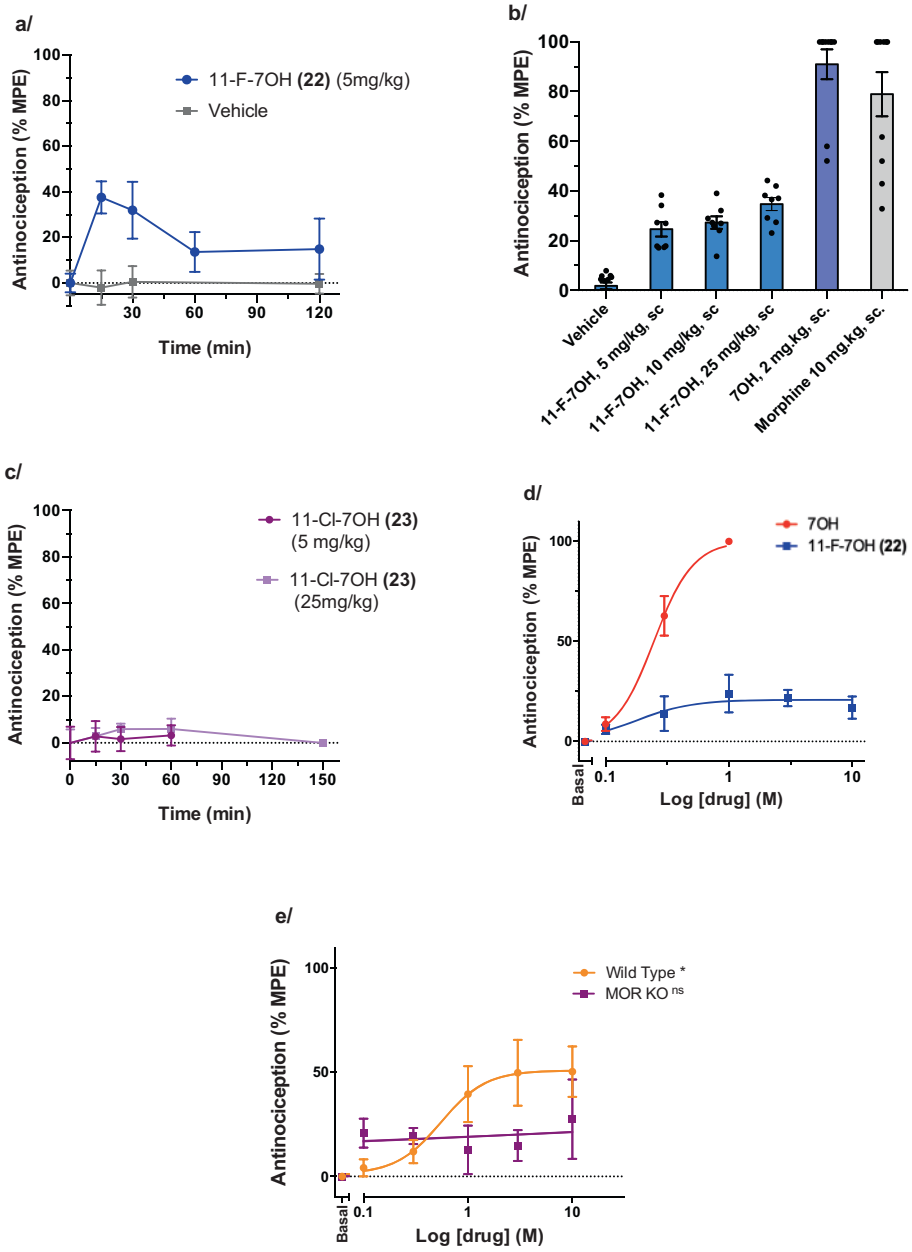


Fig. 9 Antinociceptive effects of 7OH analogs in tail-flick assay in mice. a Time course of the antinociceptive effect of **22** (5 mg/kg. s.c., $n = 11$ CD1 mice) and vehicle ($n = 6$ CD1 mice) in the tail-flick assay. Each point represents mean ± SEM. % MPE, a percentage of the maximum possible effect as set in this assay (see Supplementary information). **b** Tail-flick dose-response for the high-dose range (5, 10, and 25 mg/kg, s.c.; $n = 8$ CD1 mice each group) of **22**, in direct comparison to controls 7OH and morphine. CD1 mice ($n = 10$), at peak analgesic time point. (Vehicle (−5.4, 7.8, 4.0), 11-F-7OH 5 mg/kg (17.3, 22.4, 38.3), 11-F-7OH 10 mg/kg (13.7, 39.0, 27.1), 11-F-7OH 25 mg/kg (23.1, 44.2, 36.8), 7OH 2 mg/kg (52.1, 100.0, 100.0), Morphine 10 mg/kg (32.8, 100, 100)) **c** Time course of **23** at 2 doses (5 and 25 mg/kg, s.c., $n = 8$ CD1 mice each). Each point represents mean ± SEM. **d** Dose-response curves in mouse tail-flick assay for the parent 7OH (s.c.) and **22** (s.c.), 15 min time point, C57BL/6 mice ($n = 10$). Each point represents mean ± SEM. **e** Antinociceptive effect of 11-F-7OH (**22**) is MOR dependent. Dose-response curves of **22** (s.c.), in the tail-flick assay was evaluated 15 min post drug administration in WT, C57 mice ($n = 8$) and MOR KO mice ($n = 5$). Each point represents mean ± SEM. One-way RM ANOVA shows statistically significant result for WT* mice with $^*p = 0.006$ and statistically non-significant result (MOR KO^{ns}) with $p = 0.832$ for MOR KO mice.

buprenorphine shows much lower efficacy (E_{max} ~16% (mMOR) and 21% (hMOR)), results consistent with the previous report[78]. Remarkably, we found that 7OH exhibits low "intrinsic" efficacy (E_{max} ~14% (mMOR) and 22% (hMOR), Fig. 7) comparable to that of buprenorphine. In direct comparison, the efficacy of **22** is further reduced to near the detection limit of the assay using both mMOR and hMOR (Fig. 7).

The two assays employed provide complementary windows into MOR signaling efficacy measurements. The Nb33 BRET assay, with limited signaling amplification, positioned 7OH close to buprenorphine in terms of the broad range of signaling efficacies of different opioids. Despite the narrow dynamic range of this assay for low-efficacy compounds, it showed that fluorination of the 11-position further impaired the signaling

efficacy of 7OH to near the limit of detection. In the amplified cAMP CAMYEL BRET assay, with a greater dynamic range in the low end of efficacy readout, **22** exhibits greatly diminished but readily detectable agonist signaling as compared to 7OH, whereas the efficacy was essentially lost for the much less efficacious compounds **23** and **24**.

To correlate the very low agonist efficacy with antagonism, we next determined the antagonist activity of the 11-X-7OH series in a CAMYEL antagonism assay at mMOR (Fig. 8a, c) and hMOR (Fig. 8b, d) compared to the positive control naloxone. As expected, the compounds were able to inhibit the response elicited by the standard full agonist DAMGO at hMOR and mMOR down to the levels predicted based on their agonist efficacy as determined above.

To complete the examination of this series at the opioid receptors, we performed the analogous cAMP BRET assays with mDOR and hDOR, and rat KOR (rKOR) and hKOR. The parent 7OH was found to act as a partial agonist at both mDOR (E_{max} = 72%, EC_{50} = 71.4 nM) and hDOR (E_{max} = 45%, EC_{50} = 99.9 nM, Supplementary Fig. 7)[80]. Again, **22** showed considerable reduction in signaling efficacy at the receptors of both species, when compared to 7OH, without a major shift in potency (E_{max} = 41%, EC_{50} = 102.1 nM at mDOR; E_{max} = 26%, EC_{50} = 150.8 nM at hDOR,). The **23** and **24** compounds had very low efficacy at mDOR and hDOR (Supplementary Fig. 7). Thus, the C11 halogen trend described for MORs was also observed at DORs. In contrast, this compound series showed no agonist activity at the KOR receptors (Supplementary Fig. 8). Finally, we selected one representative compound of this series, 11-Br-7OH **(24)**, and tested it for antagonism at the KOR and DOR (Supplementary Figs. 9 and 10) revealing full inhibition of reference agonists DPDPE ([D-Pen^{2,5}]-enkephalin) for DOR and U50,488 for KOR.

In summary, these pharmacological studies indicate an important role for the C11 substitution in selective and profound modulation of the signaling efficacy at MORs and DORs of the MG-related scaffolds and provide a strong rationale for examining this effect in vivo.

11-Fluoro-7-hydroxymitragynine (22) is a low-efficacy partial agonist in vivo in mouse analgesia tests. We set out to examine how the gradual signaling efficacy modulation found in vitro, within the 7OH series, would translate to in vivo effects in living rodents. There is a strong rationale for pursuing such studies, as partial MOR agonists may provide a path to safer opioid therapeutics (see "Discussion" below for more details). We recently reported that 7OH was a potent and efficacious analgesic in mice using a tail-flick assay (thermal nociception assay)[16,17], consistent with a previous report by others[81,82]. On the basis of the previously determined ED_{50} value for 7OH in mouse tail-flick assay (ED_{50} = 0.57 (0.19–1.7) mg/kg, 95% CI)[17], and similar in vitro binding and signaling potencies of 7OH and **22** at MOR, we selected a relatively high dose of **22** (5 mg/kg, subcutaneous (s.c.) administration) for the initial time course profiling of the analgesic effect in CD1 mice (Fig. 9a). The analgesic effect peaked at 15 min post injection, indicating rapid bioavailability in vivo comparable to the parent 7OH, followed by a decay to a residual analgesic effect that lasted for at least 120 min. Noteworthy is the partial analgesic efficacy of <40% MPE (maximum possible effect) in the tail-flick assay. To confirm that the maximum effect was reached, we examined even higher doses at a 15 min time point, namely 10 and 25 mg/kg (s.c.), which showed that indeed the maximal analgesic effect was reached already at the 5 mg/kg dose (Fig. 9b). The control compounds, 7OH and morphine, elicited a greater analgesic effect in the same mouse strain. In contrast, **23** showed no analgesic effect at 5 or 25 mg/kg dose (Fig. 9c).

Thus, the in vitro MOR pharmacological profile of the 7OH analogs translates well to the in vivo tail-flick test, an established physiological readout for MOR agonists that measures antinociceptive effects of drugs to painful thermal stimuli. We next set out to rigorously confirm these results by: (1) using a different and widely used mouse strain, C57BL/6J mice; (2) performing the experiments in a different location and research group; and (3) determining a full dose-curve for both the control (7OH) and the compound **22** (Fig. 9d). 7OH showed a potent and high-efficacy analgesic effect (ED_{50} = 0.25 (0.20–0.31) mg/kg, 95% CI, analgesic efficacy normalized to 100%, comparable to morphine), consistent with previous results[17], while **22** also elicited a potent analgesic effect (ED_{50} = 0.18 (0.04–0.78) mg/kg, 95% CI), but with dramatically attenuated analgesic efficacy (E_{max} = 21%, Fig. 9d). Thus, **22** is equipotent to 7OH but exhibits low efficacy in antinociceptive effects, whereas 7OH is comparable to morphine in efficacy.

To confirm that the low-efficacy analgesia is driven by MOR, and no other targets or mechanisms (e.g., other opioid receptors or non-opioid off-targets), we performed the mouse tail-flick assay in wild-type (WT), and MOR knockout (MOR KO) mice with the same genetic background (Fig. 9e). Despite the low range of the efficacy readout afforded by **22**, we were able to show that the dose-dependent effect of **22** was statistically significant in WT mice, but not in MOR KO animals.

These results support an explanatory model where the analgesic effect of 11-F-7OH is largely driven by MOR, and the reduced analgesic efficacy is a consequence of low-efficacy MOR activation/signaling induced by this compound. Thus, 7OH and 11-F-7OH provide an excellent probe pair for examining the effects of relative signaling efficacy on therapeutic (desired) and adverse effects, and the resulting preclinical measure of a therapeutic window.

Discussion

Most recent estimates of kratom use, based on the tonnage of imported kratom and survey reports of dosing ranges, suggest 10–16 million users in the US alone[83]. Numerous anecdotal reports, including those collected by the United States Drug Enforcement Administration (>23,000 publicly available reports)[8], point to remarkable medicinal effects of kratom in self-medication for chronic pain, depression, anxiety, substance use disorders, and other ailments. Inspired by these clinical indicators, our laboratories have been studying the chemistry and biology of kratom alkaloids with the long-term goal of developing drug leads based on these compounds. In this context, the aromatic ring SAR of MG has not been mapped systematically, as only limited series of analogs have been synthesized and studied (e.g., C10 analogs)[14,16], and the C11 position has not been examined at all. To address this task, and to complement our total synthesis efforts[15], we set out to examine the late stage functionalization of MG, in this study focusing on the C11 position of the indole nucleus.

Years ago, we formulated the ideas of C–H bond functionalization and late-stage functionalization as general concepts with transformational potential for the thought and practice of chemical synthesis of carbon-based substances[30–36]. These concepts have since been widely adopted in both academia and industry and the C–H functionalization mindset, as well as corresponding methods, are indispensable components of today's synthetic repertoire[37–40]. Our team has been developing methods for systematic and programmable functionalization of heteroarenes, which are essential building blocks in medicinal chemistry, including indoles, pyrroles, pyrazoles, imidazoles, pyridines, and others[30–36]. In the present study, we focused on

the benzene ring of highly complex indoles, in the context of specific neuropharmacological goals.

Our approach provided a practical route to C11-analogs of MG, 7OH, and MG-EG, from the natural product, and systematic SAR exploration of the C11 position. The results indicate that substitution of this position plays an important role with respect to MOR signaling, specifically in modulating the efficacy of receptor-triggered signaling events. Designing the receptor signaling parameters of GPCR ligands a priori remains an unfulfilled potential of computational methods, and thus, a systematic examination of functional characteristics of receptor ligands and probes still relies largely on chemical synthesis. The chemical approach described here provides a means to advance our examination and understanding of how molecular interactions of MG-type scaffolds with the opioid receptors underlie functional outcomes such as G protein signaling and maximal signaling efficacy.

There has been considerable excitement and hope for generating MOR agonists that bias its signaling toward G protein-initiated pathways as a means for rational design of improved safety and therapeutic index of opioid analgesics ("G protein bias hypothesis" of opioids)[84]. It was suggested that some of the typical adverse effects of opioids such as respiratory depression or reinforcing effects are linked to arrestin signaling. According to this approach, biasing the MOR-triggered signaling toward the G protein and away from arrestin would result in attenuation of the adverse effects, relative to the desired analgesic activity[84]. However, this hypothesis has been seriously challenged using both genetic and pharmacological tools[78,85–90]. An emerging alternative hypothesis is that partial agonism of MOR can lead to reduced side effects while preserving therapeutic effects ("signaling efficacy hypothesis")[78,91]. An example is the superior safety profile of buprenorphine (a partial agonist in vitro) compared to the standard prescription opioids[92]. Buprenorphine is as efficacious as full agonists in treating pain but has substantially reduced ability to produce respiratory depression and thus reduced risk of opioid overdose death[92]. Such low-efficacy compounds may have been incorrectly identified as G protein biased due both to receptor reserve and to the much greater amplification of the G protein signaling arm as compared to the much less sensitive arrestin pathway assays[78]. Lower efficacy opioids are also of great interest for the treatment of opioid use disorder (OUD) where such compounds may provide efficacious opioid maintenance while having a reduced abuse potential. Therefore, close structural derivatives with a varying degree of G protein signaling efficacy are needed for direct and rigorous probing of this hypothesis.

In this context the 11-halo series of 7OH provide valuable pharmacological tools. We demonstrated that 7OH is a potent low-efficacy agonist at mMOR and hMOR in the efficacy range comparable to buprenorphine (E_{max} ~ 20% versus DAMGO), in a cell-based assay without signaling amplification (Nb33 BRET assay). Comparatively, 11-F-7OH showed only minimal receptor activation in this assay. In the amplified system (cAMP BRET assay) with a greater dynamic range on the low end of efficacy spectrum, the enhancement in apparent efficacy of 7OH was much smaller for **22**. Thus, using two assays with complementary dynamic ranges of efficacy readouts, we showed that fluorination in the 11-position markedly reduces MOR signaling efficacy of an already low-efficacy agonist, the parent 7OH.

Interestingly, the correlation between the cAMP signaling efficacy obtained in vitro in cultured cells and the in vivo analgesic efficacy determined in living animals, for the 11-X-7OH series, suggests that the in vitro amplified system may mimic the receptor reserve and/or receptor-effector situation in the neurons of relevant pain circuitry, and thus provides a useful model system. Considering the recent surveys that document much improved safety of kratom compared to the prescription or illicit opioids[83,93,94], a further decrease of MOR's G protein signaling efficacy of the key mediator of kratom's opioid-like effects, 7OH[17], may lead to even safer compounds, and illustrates the potential importance of the present work. Further, a recent report showed that 7OH attenuated alcohol intake in drinking mice and that this effect was mediated by DOR activation, while 7OH exhibited rewarding effects on its own[80]. Thus, the attenuated MOR signaling efficacy of **22**, while maintaining sufficient DOR signaling, may provide a therapeutic lead for OUD and other substance use disorders with no or much diminished abuse potential. That we see only partial analgesia for **22** and effective analgesia with 7OH suggests that the ideal efficacy may be somewhere between that of these two analogs, which can guide subsequent synthetic efforts in search of the ideal balance between therapeutic effects and side effects.

In summary, we describe a systematic examination of late-stage functionalization of kratom alkaloids, which provided efficient access to MG analogs and identified 11-F-7OH (**22**) as an important lead compound for further investigations.

Methods

General procedure for synthesis of 11-boronate ester MG-EG (4a). Starting material MG-EG (**4**) (50 mg, 0.11 mmol), $[Ir(COD)OMe]_2$ (3.6 mg, 5.5 µmol, 5 mol %), 3,4,7,8-tetramethyl-1,10-phenanthroline (3.9 mg, 16 µmol, 15 mol%) and B_2Pin_2 (111 mg, 0.44 mmol, 4 equiv.) were balanced into an oven dried vial. The vial was purged with argon, dry heptane (2.5 mL) was added under argon, and the vial was sealed with a Teflon-lined screw cap and heated to 65 °C. The RM became a dark red-brown solution after 5–15 min of heating. After 17–24 h, when LR-MS indicated complete consumption of SM, the RM was concentrated to give the crude boronate ester. This intermediate was immediately used to prepare the **5**, **13**, and **14** derivatives without further purification. Compound **7** was synthesized either from **14** through a sequence of triflation, stannylation and fluorination or from **5** through a sequence of stannylation and fluorination.

General procedure for synthesis of 11-X-MG derivatives (15–17). Starting material (**5**, **7**, or **13**, 0.11 mmol) was dissolved in AcOH (2.0 mL) under argon and $NaBH_3CN$ (13.7 mg, 0.22 mmol, 2 equiv.) was added to the solution. After stirring at RT for 15 min, another portion of $NaBH_3CN$ (13.7 mg, 0.22 mmol, 2 equiv.) was added and stirring was continued for 1 h. After this time, MeOH (81 μL) was added and the RM was heated to 90 °C for 1 h (for **15**) and 14 h (for **16** and **17**). The reaction mixture was added into a cold concentrated NH_4OH solution and extracted with DCM. After drying over Na_2SO_4, the DCM extract was evaporated. Product was purified by preparative TLC using an appropriate solvent mixture.

General procedure for synthesis of 11-X-7OH derivatives (22–24). Starting material (**15**, **16**, or **17**; 73 µmol) was dissolved in acetone (2.2 mL), sat. aq. $NaHCO_3$ (1.5 mL) was added, and the stirred suspension was cooled in an ice bath (0 °C). OXONE® (1.4–1.5 equiv.) in H_2O (0.7 mL) was added dropwise over 20 min with vigorous stirring (care should be taken that the RM does not form lumps and should be stirred thoroughly). The reaction was monitored during the addition of OXONE® by TLC. After 25 min from the first addition, the reaction mixture was diluted with H_2O (10 mL) and extracted with EtOAc (3 × 10 mL). The combined extracts were washed with brine (10 mL), dried over Na_2SO_4, and concentrated. Product was purified by preparative TLC using an appropriate solvent mixture to synthesize compounds **22**, **23**, and **24**.

Tail-flick mice assay

Mice. For analgesic dose–response experiments, male CD1 mice (20–32 g), 6–8 weeks were obtained from Charles River Laboratories and male C57BL/6 mice (22–30 g), 8–15 weeks were obtained from Jackson Lab (Bar Harbor, ME) and housed 5 mice per cage in a vivarium following an IACUC-approved protocol. For male C57BL/6 mice temperature was kept constant at 22 ± 2 °C, and relative humidity was maintained at 50 ± 5%. For male CD1 mice (20–32 g) the temperature was in the range of 20–26 °C and relative humidity maintained within the range of 30–70%. Mice were given access to food and tap water ad libitum. All mice used throughout the manuscript were opioid naïve. All mice were maintained on a 12 h light/dark cycle with Purina rodent chow and water available ad libitum and housed in groups of five until testing.

For analgesic testing in KO animals, wild-type, male C57BL/6 mice (22–33 g), 10–12 weeks were purchased from the Jackson Lab (Bar Harbor, ME). These mice were kept at a constant temperature of 22 ± 2 °C, and relative humidity was maintained at 40–50%. Exon-1/Exon-11 MOR-1 KO mice on a C57 background

were bred in the Pintar laboratory at Rutgers University. All mice were maintained on a 12-hour light/dark cycle with food and water available ad libitum, and housed in groups of five until testing. All testing was done in the light cycle.

All animal studies were preapproved by the Institutional Animal Care and Use Committees of Washington University School of Medicine and Columbia University, in accordance with the 2002 National Institutes of Health Guide for the Care and Use of Laboratory Animals.

Tail flick (dose–response). Tail-flick antinociception was determined using the radiant heat tail-flick technique using an Ugo Basile model 37360 instrument as previously described[95,96]. The intensity was set to achieve a baseline between 2 and 3 s. Baseline latencies were determined before experimental treatments for all mice. Tail-flick antinociception was assessed as an increase in baseline latency, with a maximal 15 s latency to minimize damage to the tail. Data were analyzed as percent maximal effect, % MPE, and was calculated according to the formula: % MPE = [(observed latency – baseline latency)/(maximal latency – baseline latency)] × 100. Compounds were injected s.c. and antinociception was assessed at the peak effect. Mice were tested for analgesia with cumulative subcutaneous doses of the drug until the mouse can withstand the maximal latency. Once the mouse reached the maximal latency, the mouse was no longer given higher doses. The analgesia experiments were performed by blinding the experimenter to the identity of 7OH versus 11-F-7OH. In vivo experiments were evaluated using GraphPad Prism 8, San Diego, CA as described above.

Tail flick (MOR KO animals). Analgesia was tested in wild-type and MOR KO animals by the radiant heat tail-flick technique using an IITC Model 33 Tail Flick Analgesia Meter as previously described[97]. The intensity was set to achieve a baseline between 2 and 3 s. Tail-flick antinociception was assessed as an increase in baseline latency, with a maximal 10 s latency to minimize damage to the tail. Data were analyzed as percent maximal effect, % MPE, which was calculated according to the formula: % MPE [(observed latency − baseline latency)/(maximal latency − baseline latency)] × 100. Compounds were administered s.c. as indicated in the figures, and analgesia was assessed at the peak effect (15 min). Mice were tested for analgesia with cumulative subcutaneous doses of the drug until the mouse can withstand the maximal latency. Once the mouse reached the maximal latency, the mouse was no longer given higher doses. In vivo experiments were evaluated using GraphPad Prism 8, San Diego, CA as described above.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The authors declare that all the data supporting the findings of this study are available within the article and Supplementary Information files which contains synthetic procedures and NMR spectra for the featured compounds, additional functional data at rodent and human receptors, biological protocols for receptor binding, activity and antinociception assays. The X-ray crystallographic coordinates for structure **4** reported in this study have been deposited at the Cambridge Crystallographic Data Centre (CCDC), under deposition numbers 1905559. This data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. A preprint version of this work has been deposited in the public depository ChemRxiv platform https://doi.org/10.26434/chemrxiv.12799787. Source data are provided with this paper.

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Author contributions

D.S. conceptualized and supervised the work. S.B. and J.G. carried out majority of the synthetic work including reaction development and optimization, and compound characterization. V.H. contributed to the synthesis and characterization of compounds, T.F. contributed to reaction optimization and compound characterization. M.N. performed the BRET functional assays, A.C.K. contributed to the design of compounds and supervised the interpretation of the pharmacological data, S.M. supervised the binding assays, tail-flick mice assay and contributed to global planning of the project, A.H. performed the binding assays, A.F. and B.B. performed the tail-flick mice assay, M.A. performed the tail-flick mice assay comparing WT and MOR/KO mice, J.E.P. supervised the tail-flick mice assay with MOR/KO mice, J.A.J. supervised the BRET functional assays and interpretation of the corresponding results. S.B. and D.S. wrote the manuscript with significant help from J.G., V.H., J.A.J., and A.C.K. All authors contributed to editing of the manuscript.

Competing interests

S.B., J.G., V.H., A.C.K., J.A.J., S.M., and D.S., are named as inventors on patent applications related to mitragynine analogs. A.C.K., J.A.J., and D.S. are co-founders of Kures, Inc. S.M. is a co-founder of Sparian biosciences. The authors M.N., A.F., B.B., M.A., T.F., A.H., and J.E.P. declare no competing interests.

Additional information

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RESEARCH ARTICLE

The respiratory depressant effects of mitragynine are limited by its conversion to 7-OH mitragynine

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Background and Purpose: Mitragynine, the major alkaloid in *Mitragyna speciosa* (kratom), is a partial agonist at the μ opioid receptor. CYP3A-dependent oxidation of mitragynine yields the metabolite 7-OH mitragynine, a more efficacious μ receptor agonist. While both mitragynine and 7-OH mitragynine can induce anti-nociception in mice, recent evidence suggests that 7-OH mitragynine formed as a metabolite is sufficient to explain the anti-nociceptive effects of mitragynine. However, the ability of 7-OH mitragynine to induce μ receptor-dependent respiratory depression has not yet been studied.

Experimental Approach: Respiration was measured in awake, freely moving, male CD-1 mice, using whole body plethysmography. Anti-nociception was measured using the hot plate assay. Morphine, mitragynine, 7-OH mitragynine and the CYP3A inhibitor ketoconazole were administered orally.

Key Results: The respiratory depressant effects of mitragynine showed a ceiling effect, whereby doses higher than 10 $mg \cdot kg^{-1}$ produced the same level of effect. In contrast, 7-OH mitragynine induced a dose-dependent effect on mouse respiration. At equi-depressant doses, both mitragynine and 7-OH mitragynine induced prolonged anti-nociception. Inhibition of CYP3A reduced mitragynine-induced respiratory depression and anti-nociception without affecting the effects of 7-OH mitragynine.

Conclusions and Implications: Both the anti-nociceptive effects and the respiratory depressant effects of mitragynine are partly due to its metabolic conversion to 7-OH mitragynine. The limiting rate of conversion of mitragynine into its active metabolite results in a built-in ceiling effect of the mitragynine-induced respiratory depression. These data suggest that such 'metabolic saturation' at high doses may underlie the improved safety profile of mitragynine as an opioid analgesic.

Abbreviations: MPE, maximum possible effect; MV, minute volume; OST, opioid substitution therapy.

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1 | INTRODUCTION

Mitragyna speciosa, more commonly known as kratom, is a plant popular in labour-intensive agricultural communities and native to Southeast Asia (Adkins et al., 2011), where it is mostly prepared as a recreational drink similar to tea. Kratom tea is consumed both for work-time stimulation, much like caffeinated drinks, and as a relaxation aid (Singh et al., 2016). Over 40 unique alkaloids have been identified in the kratom plant, with mitragynine accounting for over 60% by mass of the alkaloid extracts (Basiliere et al., 2020; Kruegel et al., 2016; Prozialeck et al., 2012; Suhaimi et al., 2016; Takayama, 2004). Mitragynine and its oxidised metabolite 7-OH mitragynine have recently been shown to bind and activate opioid receptors, displaying higher affinity for the **μ opioid receptor** subtype (Chakraborty, Uprety, Slocum, et al., 2021; Kruegel et al., 2016; Matsumoto et al., 2004; Shamima et al., 2012; Todd et al., 2020). The anti-nociceptive effects of both mitragynine and 7-OH mitragynine were abolished in μ receptor knockout mice, but not in mice lacking either the **δ** or the **κ opioid receptors** (Kruegel et al., 2019). The analgesic effects of kratom, together with the low rates of acute toxicity or overdose following its ingestion (Veltri & Grundmann, 2019), have attracted considerable attention in the search for novel, safer opioid-based analgesics amid the current opioid crisis (Bhowmik et al., 2021; Chakraborty, Diberto, Faouzi, et al., 2021; Chakraborty, Uprety, Daibani, et al., 2021; Gutridge et al., 2020).

Kratom also has been utilised as an aid for recovering opioid users that have previously maintained themselves on abusable opioids, such as **heroin**, **oxycodone** and **fentanyl**, or have been on opioid substitution therapy (OST) such as **methadone** or suboxone (Grundmann, 2017; Singh et al., 2016; Swogger et al., 2015). The use of kratom as a recovery aid is not limited to Southeast Asia but is increasingly being sought and used in Europe and North America as a preferred recovery aid to existing prescription OST (Grundmann, 2017; Swogger et al., 2015). Importantly, mitragynine has been shown to produce less respiratory depression than **codeine** and, to our knowledge, no fatal kratom-induced respiratory depression has been reported, supporting the recent interest in this compound as a potential route to design safer opioids (Kruegel et al., 2016, 2019; Matsumoto et al., 1996; Smith et al., 2019; Suhaimi et al., 2016; Swogger et al., 2015; Varadi et al., 2016; Veltri & Grundmann, 2019).

The observations that mitragynine elicited anti-nociception upon oral or intraperitoneal administration, but not when administered subcutaneously, suggested the existence of an active metabolite generated through first-pass metabolism that would be mediating such effects. Recently, 7-OH mitragynine has been identified as the product of mitragynine conversion by cytochrome P450 3A4 (**CYP3A**) (Basiliere & Kerrigan, 2020; Chakraborty, Uprety, Slocum, et al., 2021; Kruegel et al., 2019). 7-OH mitragynine binds to μ receptors, is significantly more efficacious than mitragynine in vitro and has been detected in brain at concentrations sufficient to explain the analgesic effects of its precursor compound (Chakraborty, Uprety, Slocum, et al., 2021; Kruegel et al., 2019). It has also been suggested that, similar to codeine and **morphine**, the opioidergic actions of mitragynine would depend on its metabolic conversion to 7-OH mitragynine. This metabolically limited production of the more efficacious compound may offer an avenue to limit the side effect profile of these drugs (Kruegel et al., 2019). However, while there is evidence showing that 7-OH mitragynine formed as a metabolite is sufficient to explain the anti-nociceptive effects of mitragynine (Chakraborty, Uprety, Slocum, et al., 2021; Kruegel et al., 2019), the respiratory depressant effects of 7-OH mitragynine, relative to mitragynine, have not yet been studied.

Here, we have characterised the respiratory depressant and anti-nociceptive effects of mitragynine and 7-OH mitragynine in mice. Both drugs were administered orally, as this route demonstrated better bioavailability (Kruegel et al., 2019) and is also the route of consumption in humans. We show that mitragynine-induced respiratory depression has a ceiling effect, whereby doses above 10 $mg \cdot kg^{-1}$ produce similar responses. In contrast, 7-OH mitragynine induces respiratory depression in a dose-dependent manner. Inhibition of CYP3A reduces the respiratory depressant and anti-nociceptive effects of mitragynine but not those of 7-OH mitragynine, supporting the hypothesis that the metabolic conversion of mitragynine underlies the built-in ceiling respiratory effect of this drug.

What is already known

- Mitragynine oxidation results in the generation of 7-OH mitragynine, a more efficacious μ receptor agonist.
- The anti-nociceptive effects of mitragynine are partly mediated by 7-OH mitragynine.

What does this study add

- Mitragynine and 7-OH mitragynine induced respiratory depression, but only mitragynine showed a ceiling effect.
- The ceiling effect of mitragynine-induced respiratory depression is due to metabolic saturation of CYP3A enzyme

What is the clinical significance

- Metabolic regulation of opioidergic effects offers potential to improve safety profiles of opioid agonists.

2 | METHODS

2.1 | Animals

All animal care and experimental procedures were performed in accordance with the UK Animals (Scientific Procedures) Act 1986 and the European Communities Council Directive (2010/63/EU) and were approved by ethical review board at the University of Nottingham (project licence P3C7EE0BA). Animal studies are reported in compliance with the ARRIVE guidelines (Percie du Sert et al., 2020) and with the recommendations made by the British Journal of Pharmacology (Lilley et al., 2020).

Male CD-1 mice (Charles River, UK) weighing approximately 31 ± 1 g were group housed, 3 per cage, in an environment maintained at 22°C, on a reversed 12 h/12 h dark–light cycle with food and water available ad libitum. Experiments were performed in the dark (active) phase. The experimenter was blinded to all drug treatments, both during the experiment and data analysis. Cages of mice were randomly designated to drug treatments, such that all mice in one cage received the same drug treatment in order to minimise social conflict between different drug effects. A total of 250 mice were used in the study.

2.2 | Mouse respiration

Respiration was measured in freely moving mice using plethysmography chambers (EMKA Technologies, Paris, France) supplied with room air as described previously (Hill et al., 2016, 2018). Mice were habituated to respiratory chambers for 30 min, on the day prior to experimentation. Respiratory parameters were recorded (IOX software - EMKA Technologies, Paris, France) and averaged over 5 min periods.

Data are presented both as minute volume (MV) and as percentage change from the pre-drug MV baseline, calculated from data for each individual mouse before collating and plotting as a mean. Mice within cohorts may vary in size and this varies their individual MVs, accordingly; data are presented as a percentage change from each mouse's pre-drug baseline controls for these inherent variations.

2.3 | Calculation of equi-effective doses

To determine equi-effective doses for each opioid, linear and polynomial regressions were fitted to the peak dose response for each drug. Linear regressions were found to produce the best fit for each agonist. Slope equations were used with a set *Y*-value of 40% (the degree of respiratory depression) to determine the *X*-axis dose value predicted to induce this level of effect.

2.4 | Measurement of anti-nociception

Prior to the day of experimentation, mice were handled onto and off the inactive hot plate a minimum of 5 times; more aggressive or anxious mice were habituated up to a maximum of 10 times. On the day of experimentation, mice were placed on a hot plate at 52.5°C and the latency to exhibit a pain-like response (defined as paw withdrawal, jumping, paw licking and fluttering of the hind limbs) was measured. A maximum cut-off time of 20 s was used to prevent tissue damage. For the same reason, measurements were taken no more frequently than every 15 min. Anti-nociception was quantified and is presented as both latency to response in seconds and calculated as the percentage of maximum possible effect (%MPE), which was calculated as $\%MPE = [(\text{test latency} - \text{control latency}) / (20 - \text{control latency})] \times 100$.

2.5 | Experimental design and data analysis

Data from previous experiments (Hill et al., 2016, 2018) where respiratory depression or anti-nociception was measured following acute opioid administration in naïve mice were used to guide a priori power analyses using G*Power (Version 3.1.9). These calculations indicated that $n = 6$ (respiration experiments) or $n = 10$ (anti-nociception experiments) for each individual group would produce a significant result if an actual effect occurred. GraphPad Prism 8 was used for all statistical analyses. All data are displayed as mean ± SEM. All statistical tests are shown in the figure legends and Table S1. The data and statistical analyses comply with the recommendations of the *British Journal of Pharmacology* on experimental design and analysis in pharmacology (Curtis et al., 2015).

2.6 | Materials

Morphine hydrochloride (Macfarlan Smith, Edinburgh, UK— a kind gift from Graeme Henderson, University of Bristol, UK) - was dissolved in distilled water for oral administration. Mitragynine was extracted from the dried leaves of M. speciosa and 7-OH mitragynine was chemically synthesised as previously described (Kruegel et al., 2016). Both drugs were supplied as free base and dissolved by the addition of 3 molar equivalents of acetic acid followed by dilution in de-ionised water before oral administration. **Ketoconazole** (Tocris, UK) was dissolved in a 4:16:80 solution of DMSO:Tween80:distilled water for oral administration. All drugs were administered in 0.1 ml volumes and made up freshly on the day of experiment.

2.7 | Nomenclature of targets and ligands

Key protein targets and ligands in this article are hyperlinked to corresponding entries in http://www.guidetopharmacology.org and are permanently archived in the Concise Guide to PHARMACOLOGY 2021/22 (Alexander, Christopoulos et al., 2021; Alexander, Fabbro et al., 2021).

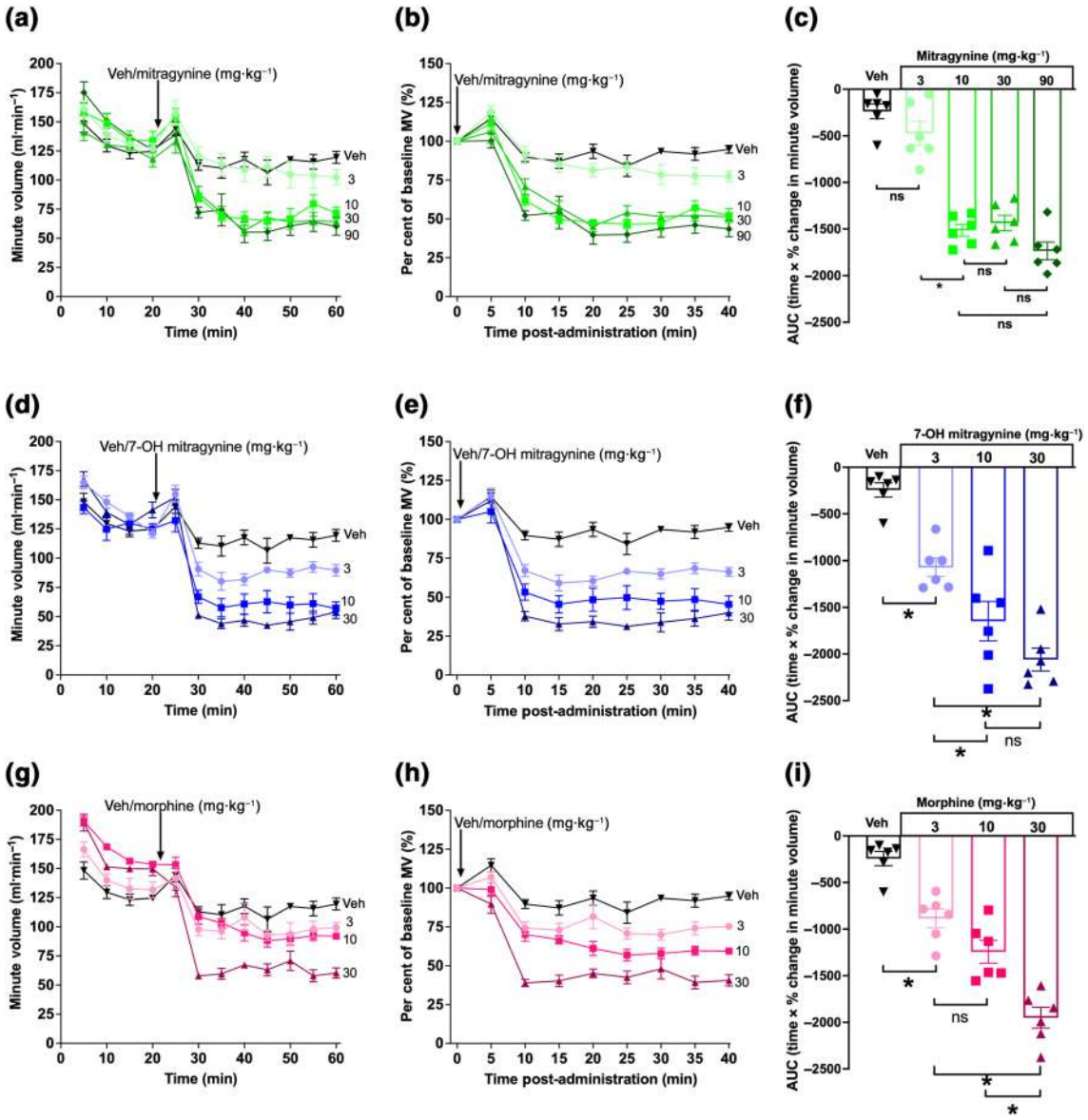


FIGURE 1 Mitragynine and 7-OH mitragynine depress respiration in awake mice. (a–c) Mitragynine (10–90 mg·kg^{-1}) but not 3 mg·kg^{-1} significantly depressed minute volume (MV) when compared with effects of vehicle alone. There was no significant difference in the degree of depression induced by mitragynine over a similar dose range (10–90 mg·kg^{-1}). (d–f) 7-OH mitragynine significantly depressed mouse MV at all doses in a dose-dependent manner. (g–i) Morphine significantly depressed mouse MV at all doses in a dose-dependent manner similar to 7-OH mitragynine. $N = 6$ for all groups. All drugs administered by oral gavage dosing. In (c), (f) and (i), * $P < 0.05$, significantly different; ns, not significantly different, as indicated; one-way ANOVA (Kruskal–Wallis) with Dunn's multiple comparison. See Table S1 for full test statistics

3 | RESULTS

3.1 | Respiratory depressant and anti-nociceptive effects of mitragynine and 7-OH mitragynine

The respiratory depressant activity of mitragynine, 7-OH mitragynine and morphine was assessed using whole body plethysmography in mice breathing air. All compounds were administered orally at increasing doses. Both morphine (3–30 mg·kg^{-1}) and 7-OH mitragynine (3–30 mg·kg^{-1}) dose-dependently depressed mouse MV for the measured 40 min post-administration (Figure 1 and Table S1). While a low dose of mitragynine (3 mg·kg^{-1}) did not produce significant changes in respiration, a 10 mg·kg^{-1} dose induced significant and prolonged respiratory depression that was not further increased at higher doses (30 and 90 mg·kg^{-1}) (Figure 1 and Table S1). These data suggest that the ability of mitragynine to induce respiratory depression is limited by a ceiling effect. Given that 7-OH mitragynine is a more potent active metabolite of mitragynine that has already been suggested as the primary mediator of kratom-induced effects (Kruegel et al., 2019), this ceiling effect may be due to a metabolic bottleneck (metabolic saturation) limiting mitragynine oxidation and accumulation of 7-OH mitragynine.

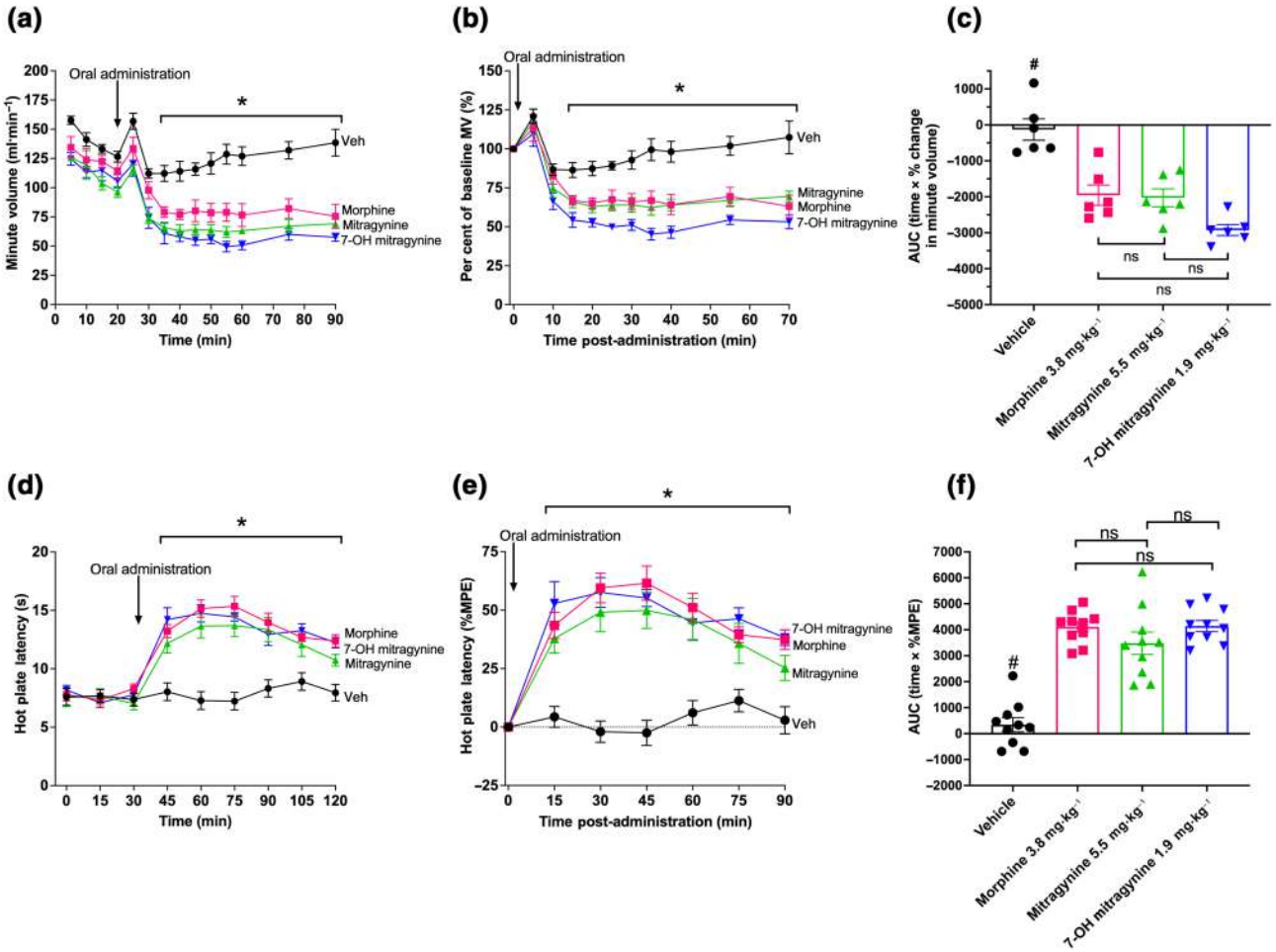


FIGURE 2 Equi-respiratory depressant doses induce similar levels of anti-nociception: (a,b) Orally administered doses of morphine (3.8 mg·kg^{-1}), mitragynine (5.5 mg·kg^{-1}) and 7-OH mitragynine (1.9 mg·kg^{-1}) all significantly depressed minute volume (MV) when compared with effects of vehicle alone. (c) There was no significant difference in the degree of respiratory depression induced by morphine, mitragynine and 7-OH mitragynine. (d, e) The same doses of morphine, mitragynine and 7-OH mitragynine induced significant increases in hot plate latency when compared with vehicle administered mice. (f) When comparing the overall degree of anti-nociception induced by each drug, measured as AUC (time × %MPE), all drugs induced a significant and similar overall increase. $N = 6$ for (a)–(c) and $N = 10$ for (d)–(f). All drugs administered by oral gavage dosing. In (a), (b), (d) and (e), * $P < 0.05$, significantly different; ns, not significantly different, from vehicle; two-way ANOVA with Dunnett's multiple comparisons. In (c) and (f), # $P < 0.05$, significantly different, ns, not significantly different, from opioid groups; one-way ANOVA (Kruskal–Wallis) with Dunn's multiple comparison. See Table S1 for full test statistics

The maximum decrease in respiration (peak effect) induced by each drug and dose was used to estimate a dose for each opioid such that they would produce approximately the same degree of respiratory depression (Figure S1; see Section 2). Oral administration of calculated equi-effective doses for mitragynine (5.5 mg·kg^{-1}), 7-OH mitragynine (1.9 mg·kg^{-1}) and morphine (3.8 mg·kg^{-1}) induced significant respiratory depression, and this was maintained for the entire period of observation (Figure 2a, b and Table S1). Administration of vehicle resulted in a transient decrease in mouse respiration (10–35 min post-dose), but it was not significantly different from baseline following this period. Although 7-OH mitragynine produced significantly more respiratory depression than morphine or mitragynine at some individual time points, AUC analysis of equi-effective doses did not highlight any significant difference in the overall degree of respiratory depression induced over the entire observation period (Figure 2c and Table S1).

The equi-effective doses of mitragynine, 7-OH mitragynine and morphine in respiratory depression were then tested in hot plate anti-nociception assays. All three agonists induced significant increases in mouse hot plate latency at all time points after opioid administration (90 min) (Figure 2d,e and Table S1). There were no significant differences at any time points between the three agonists, suggesting similar degrees of anti-nociception and duration of action. AUC analysis of these data shows that the overall degree of anti-nociception induced by each agonist was not significantly different from each other, but was significantly different from the effects of the vehicle (Figure 2f and Table S1).

3.2 | Inhibition of CYP3A reduces mitragynine-induced respiratory depression

The data above showing the ceiling effect of mitragynine in respiratory depression together with previous reports suggesting the need of metabolic conversion of mitragynine to 7-OH mitragynine by CYP3A to induce anti-nociception (Kruegel et al., 2019) led us to hypothesise that such metabolic conversion is also required to mediate the respiratory effects of mitragynine and underlies the built-in ceiling for the effects of this compound. To address this hypothesis, we treated mice with ketoconazole (50 mg·kg^{-1}, p.o.), a known inhibitor of CYP3A (Vermeer et al., 2016). Previous research has shown that a 30 min pre-treatment with ketoconazole (35 mg·kg^{-1}, p.o.) significantly inhibited the metabolic transformation of atorvastatin via CYP3A (Chang et al., 2014). Other work has demonstrated that a dose of 50 mg·kg^{-1} ketoconazole significantly inhibited loperamide metabolism while leaving **P-glycoprotein** function unaffected (Seneca et al., 2010). Thus, a 30 min pretreatment with a dose of 50 mg·kg^{-1} p.o. was selected for these studies.

While ketoconazole pretreatment seemed to produce a decrease in baseline respiration, this was not significantly different to the vehicle/control group (Figure S2). Pretreatment with ketoconazole significantly reduced mitragynine (5.5 mg·kg^{-1})-induced respiratory depression (Figure 3a,b and Table S1). AUC analysis of these data shows that in mice pretreated with vehicle, mitragynine induced a significant degree of overall respiratory depression compared with the acute vehicle that was significantly reduced by ketoconazole pretreatment (Figures 3e and S3 and Table S1). This supports the hypothesis that the effects of mitragynine are likely to be mediated by its active metabolite.

However, as ketoconazole is known to affect other metabolic pathways other than CYP3A in rodents (Eagling et al., 1998), its effect on 7-OH mitragynine was also investigated as a control. Mice administered 7-OH mitragynine (1.9 mg·kg^{-1}) showed significant respiratory depression at all time points following administration in both vehicle and ketoconazole-pretreated mice (Figure 3c,d and Table S1), though when these data were analysed as percentage of baseline respiration, neither group produced significant respiratory depression at the final time point (40 min) compared with vehicle control (Figure 3d and Table S1). Further analysis by AUC demonstrated that not only did 7-OH mitragynine depress respiration in both vehicle and ketoconazole pretreated mice but also that the degree of respiratory depression induced by 7-OH mitragynine was not significantly different between pretreatments (Figures 3f and S3B and Table S1).

We obtained similar results when we assessed the effects of ketoconazole pretreatment on mitragynine and 7-OH mitragynine-induced anti-nociception. Both mitragynine and 7-OH mitragynine induced significant increases in hot plate latency compared with vehicle (at all time points) for mice that received vehicle pretreatment (Figure 4a,b and Table S1). In ketoconazole-pretreated mice, 7-OH mitragynine, but not mitragynine, induced a significant and prolonged increase in hot plate latency at any time point over the observation period (Figure 4b and Table S1). AUC analysis of these data further demonstrated that in mice pretreated with vehicle or ketoconazole, 7-OH mitragynine induced a significant degree of anti-nociception compared with the acute vehicle (Figure 4d and Table S1), whereas in ketoconazole pretreated mice, mitragynine failed to induce significant anti-nociception compared with acute vehicle (Figure 4c and Table S1).

4 | DISCUSSION

In this study, we report the respiratory depressant effects of the kratom alkaloid mitragynine and its oxidised derivative 7-OH mitragynine. We also showed that the respiratory depressant effects of mitragynine were partly due to its metabolic conversion to 7-OH mitragynine by CYP3A. These results are in line with previous reports suggesting that mitragynine-induced anti-nociception can be mediated by its active metabolite (Kruegel et al., 2019).

Available evidence suggests that mitragynine, when administered orally, has an improved side-effect profile, compared with typical opioids such as morphine or codeine, with reports showing reduced respiratory depression (Macko et al., 1972) and GI transit inhibition, although some signs of withdrawal have been reported (Hassan et al., 2021; Yusoff et al., 2016). Altogether, these data suggest that mitragynine may exhibit a greater safety margin for overdose compared with other opioids. Mitragynine has been associated with both low toxicity in mice (LD_{50} > 400 mg·kg^{-1}) (Sabetghadam et al., 2013; Smith et al., 2019) and an extremely low toxicity compared with other opioids in humans (44 total vs. ~50,000 all opioid deaths 2017) (NIDA, 2018; Veltri & Grundmann, 2019). The prevalence of kratom as an increasingly sought-after alternative to traditional opioid antagonist/agonist combinations such as suboxone or the replacement opioid methadone has suggested a perceived benefit in terms of intoxication, overdose protection or withdrawal management, either individually or in combination (Grundmann, 2017; Veltri & Grundmann, 2019).

The mechanism underlying the improved safety profile for mitragynine has been suggested to be 'metabolic saturation', namely, when the generation of the active metabolite of the drug is limited by the rate of conversion of the enzyme(s) responsible for its metabolism. As such, the extent of induction of opioidergic effects by mitragynine will be determined by its conversion to 7-OH mitragynine by CYP3A (Kruegel et al., 2019). This would explain the built-in ceiling effect observed in this study, where doses of mitragynine above 10 mg·kg^{-1} produce the same respiratory depressant effects. This hypothesis is also supported by the experiments using the CYP3A inhibitor ketoconazole at a dose that has been shown to modulate CYP3A opioid metabolism (Chakraborty, Uprety, Slocum, et al., 2021; Seneca et al., 2010), which inhibited the effects of mitragynine but not those of 7-OH mitragynine. Importantly, the effects observed in respiratory depression measurements are mirrored in anti-nociceptive measurements. However, the possibility that ketoconazole affects differently the absorption rates of mitragynine and 7-OH mitragynine and the pharmacodynamic profile of ketoconazole as it relates to opioid receptors remains to be further investigated.

In the context of metabolic conversion, mitragynine could be compared with codeine whose conversion into morphine (and

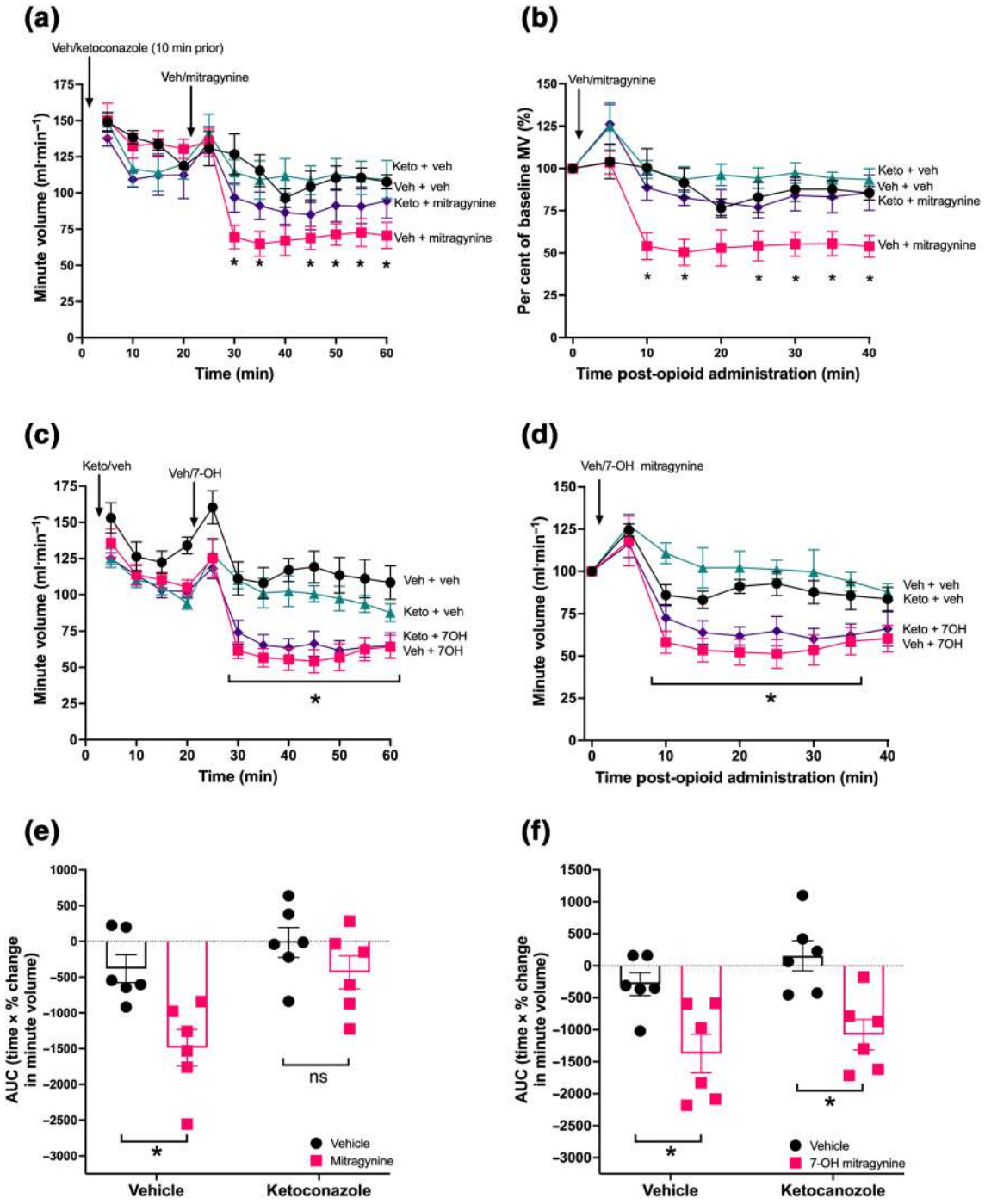


FIGURE 3 Effect of CYP3A inhibition on mitragynine and 7-OH mitragynine induced respiratory depression: (a,b) Mitragynine (5.5 mg·kg^{-1}) induced significant respiratory depression in mice pretreated with vehicle (veh) but not in mice pretreated with ketoconazole (keto) (50 mg·kg^{-1}, 30 min), (c, d) 7-OH mitragynine (1.9 mg·kg^{-1}) induced significant respiratory depression in vehicle and ketoconazole (50 mg·kg^{-1}) pretreated mice. Ketoconazole pretreatment significantly reduced overall respiratory depression induced by mitragynine (e) but not that induced by 7-OH mitragynine (f). $N = 6$ for all groups. All drugs administered by oral gavage dosing. In (a) - (d), * $P < 0.05$, significantly different from vehicle control or as indicated; two-way ANOVA with Dunnett's multiple comparisons. In (e) and (f), * $P < 0.05$, significantly different; ns, not significantly different, as indicated; 2 × 2 factorial by two-way ANOVA with Bonferroni's multiple comparisons. See Table S1 for full test statistics

induction of opioidergic effects) depends on the enzyme **CYP2D6**. It could be anticipated that, as with codeine, individuals with genetic mutations leading to alterations in the levels of activity of its metabolising enzyme would have different sensitivities to the effects of kratom. From a drug discovery standpoint, however, this metabolic regulation of opioidergic effects offers significant potential to improve safety profiles of opioid agonists.

The interaction between opioids and their metabolic enzymes is also likely to be altered in polydrug exposure either with prescription or illicit drugs (Darke, 2003). In particular, the combination of

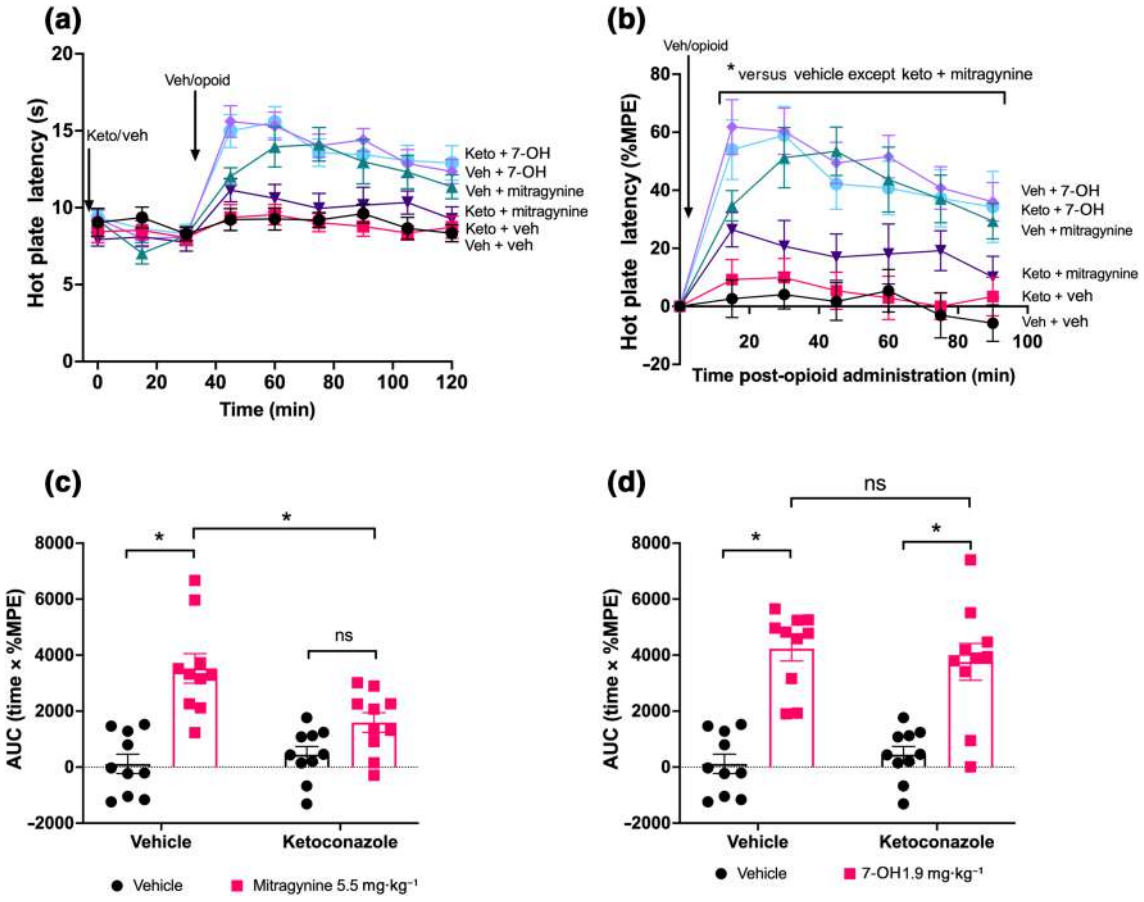


FIGURE 4 Inhibition of CYP3A by ketoconazole pretreatment prevents mitragynine but not 7-OH mitragynine-induced anti-nociception: (a, b) 7-OH mitragynine (1.9 mg·kg^{-1}) and mitragynine (5.5 mg·kg^{-1}) induced significant anti-nociception in vehicle-pretreated mice; however, only 7-OH induced significant anti-nociception in ketoconazole-pretreated mice. (c, d) 7-OH mitragynine induced the same degree of anti-nociception in both vehicle- and ketoconazole-pretreated mice whereas mitragynine was significantly reduced by ketoconazole treatment as well as not being significant from vehicle. $N = 10$ for all groups. All drugs administered by oral gavage dosing. In (b), * $P < 0.05$, significantly different as indicated; two-way ANOVA with Dunnett's multiple comparisons. In (c) and (d), * $P < 0.05$, significantly different, ns not significantly different, as indicated; 2 × 2 factorial by two-way ANOVA with Bonferroni's multiple comparisons. See Table S1 for full test statistics

benzodiazepines and opioids is known to be extremely dangerous, not only through the combined respiratory depressant effects of both drugs but also because benzodiazepines can act as substrates for cytochrome P450 enzymes (Gudin, 2012; Smith, 2009). This has been postulated to be responsible for altering the pharmacodynamics of oxycodone and enhancing its potential lethality (Ji et al., 2021). Similarly, barbiturates (e.g., **phenobarbital** and pentabarbital) and antibiotics, such as **rifampicin**, are strong inducers of CYP3A enzymes and thus, when used in combination, would be expected to increase the effects of mitragynine. Therefore, as with other opioids, the safety profile of mitragynine and its metabolites must also be considered in a polydrug context in guiding dosage adjustment in potential cases of poly therapy.

Taking into consideration that mitragynine itself has been shown to bind and activate μ receptors (Kruegel et al., 2016), albeit with lower efficacy than 7-OH mitragynine, it is possible that some of its actions might be directly mediated by mitragynine. It would then follow that the overall effect of mitragynine is likely to result from the activation of μ receptors in the presence of two ligands with different efficacies, whereby binding of a lower efficacy ligand such as mitragynine will decrease the response of a higher efficacy agonist such as 7-OH mitragynine. This could function alongside metabolic saturation as a mechanism that limits the effects of mitragynine.

A recent study has also demonstrated that 7-OH mitragynine can be metabolised further to mitragynine pseudoindoxyl in a manner that is specific to human plasma when compared with primates, rodents and dogs (Kamble et al., 2020). Although the exact metabolic pathway responsible for this conversion is yet to be elucidated, mitragynine pseudoindoxyl has been reported to have greater efficacy at μ receptors than 7-OH mitragynine, as well as being a δ receptor antagonist (Varadi et al., 2016). Thus, it is likely that following human consumption of mitragynine (kratom), the overall physiological response will be induced by multiple mitragynine alkaloids with the metabolic rate conversion playing a key role in the observed response (Chakraborty, Uprety, Slocum, et al., 2021). Further studies should aim to establish the relative intrinsic efficacy of these atypical opioids to assess whether their effects, in vivo, reflect such efficacies.

Two interesting comparisons can be made between the actions of mitragynine and codeine and oxycodone. Codeine is a relatively safe opioid in the management of moderate pain with reduced respiratory liability, though it still presents a significant risk in terms of dependence and addiction (Nielsen et al., 2018). In contrast, management of moderate to severe pain using oxycodone has been associated with significant numbers of overdose deaths over the last 10–15 years (Kibaly et al., 2021). Interestingly, both agonists are metabolised by cytochrome P450 enzymes with codeine primarily metabolised by **CYP2D6** (Crews et al., 2021) and oxycodone metabolised by both CYP2D6 and CYP3A (Umukoro et al., 2021). Codeine may become less safe following the induction of CYP2D6 activity, because of the formation of morphine as an active metabolite (Crews et al., 2021; Haertter, 2013). Conversely, inhibition of CYP3A metabolism may cause a delay in the clearance of oxycodone, exacerbating its opioidergic side effects (Rytkönen et al., 2020; Umukoro et al., 2021). In the case of mitragynine, the results presented here suggest that induction of the CYP3A metabolic pathway may increase the chance of toxicity being achieved with kratom, due to enhanced formation of 7-OH mitragynine.

Of note, the experiments presented here assessed respiratory depression in mice breathing air to facilitate comparisons with previous studies of novel opioid agonists. While opioids do directly inhibit neurons responsible for respiratory rhythmogenesis (Saunders & Levitt, 2020), they are also responsible for inhibiting the hypercapnic reflex to stimulate breathing (Palkovic et al., 2020) and thus further exacerbate respiratory distress and increase the likelihood of a fatal overdose. Previous research has shown that the potency of an opioid to depress respiration is enhanced under hypercapnic conditions (Czapla & Zadina, 2005), suggesting that a progressively worsening feedback loop may occur during overdose. The effects of hypercapnic challenges on mitragynine-induced respiratory depression will be an important future step in defining its full safety profile, particularly with regard to escalating concentrations of CO_2 that aim to mimic the progressively deteriorating respiratory pathology of an overdose.

In summary, our results suggest that the respiratory depressant effects of mitragynine are likely to be mediated by the generation of the active metabolite 7-OH mitragynine. The metabolic saturation of the enzymatic oxidation of mitragynine provides a built-in ceiling for respiratory depression that, together with the very weak partial agonist activity displayed by mitragynine when bound to μ receptors (acting to reduce the effects of 7-OH mitragynine), may underlie its improved safety profile as an analgesic.

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AUTHOR CONTRIBUTIONS

RH, JRL and MC designed experiments. RH performed experiments and analysed the data. RH, JRL and MC wrote the manuscript. ACK and JAJ provided reagents. All authors edited and reviewed the manuscript.

CONFLICT OF INTEREST

ACK is a co-founder and shareholder of Kures, Inc. (a subsidiary of ATAI Life Sciences) and Gilgamesh Pharmaceuticals, Inc. Kures is currently developing a mitragynine analogue as a potential pharmaceutical. ACK is named as an inventor on mitragynine-related patent applications (WO 2020/160280 A1, WO 2020/037136 A1, US Patent 10,961,244 B2) owned by Columbia University and Kures. JAJ is a co-founder of Kures and named as an inventor on mitragynine-related patents (WO 2020/160280 A1, WO 2020/037136 A1, US Patent 10,961,244 B2) owned by Columbia University. No funding was provided by Kures for this work.

DECLARATION OF TRANSPARENCY AND SCIENTIFIC RIGOUR

This Declaration acknowledges that this paper adheres to the principles for transparent reporting and scientific rigour of preclinical research as stated in the *BJP* guidelines for Design and Analysis, and Animal Experimentation, and as recommended by funding agencies, publishers and other organisations engaged with supporting research.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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Oxidative Metabolism as a Modulator of Kratom's Biological Actions

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S.C. and R.U. contributed equally. S.M., J.M., D.S., and A.C.K. conceived the experiments. SM coordinated and directed the multilab efforts. S.C., R.U., S.S., T.I., V.L.R., X.L., M.A., A.V., S.E., A.H., A.A., S.K., and J.X. performed the experiments and analyzed the data. Y.X.P./G.W.P. supervised behavior experiments along with S.M. J.P. supervised M.A., M.C. supervised X.L., B.K. supervised B.S. and A.A., J.P.M. supervised L.L.W., and S.O.E. and S.S. carried out PRESTO-Tango assays. The manuscript was written by S.M., S.C., R.U., J.P.M., and A.C.K. with input from everybody.

Supporting Information

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Time course of antinociception of oral and systemic kratom, mitragynine and metabolites, kratom, mitragynine, and its metabolites show MOR-dependent antinociception, oral mitragynine is more potent as an antinociceptive agent over systemically administered mitragynine in mice, mitragynine is the principal component found in kratom for producing antinociception in mice, metabolism of mitragynine to 7OH and MP, MP kappa/delta antagonism, ambulation effect in C57BL/6J mice, binding affinity (Ki ± SEM nM/pKi ± SEM) of kratom alkaloids and mitragynine metabolites at select CNS receptors, tables of binding assay conditions, concentration of parent compounds and metabolites detected from mitragynine-dosed mice (μM), ^{1}H–^{13}C NMR spectra, and UPLC chromato-grams of new compounds (PDF)

Molecular formula strings for compounds (CSV)

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jmedchem.1c01111

The authors declare the following competing financial interest(s): SM, GWP, YXP are co-founders of Sparian biosciences. DS and ACK are co-founders of Kures, Inc. GWP, ACK, DS, AK, AV and SM have filed a provisional patents on mitragynine and related molecules. The other authors declare no other competing interests.

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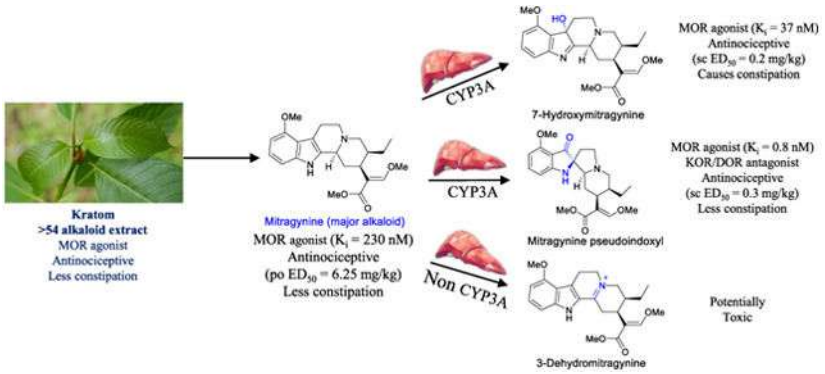
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Abstract

The leaves of *Mitragyna speciosa* (kratom), a plant native to Southeast Asia, are increasingly used as a pain reliever and for attenuation of opioid withdrawal symptoms. Using the tools of natural products chemistry, chemical synthesis, and pharmacology, we provide a detailed *in vitro* and *in vivo* pharmacological characterization of the alkaloids in kratom. We report that metabolism of kratom's major alkaloid, mitragynine, in mice leads to formation of (a) a potent mu opioid receptor agonist antinociceptive agent, 7-hydroxymitragynine, through a CYP3A-mediated pathway, which exhibits reinforcing properties, inhibition of gastrointestinal (GI) transit and reduced hyperlocomotion, (b) a multifunctional mu agonist/delta-kappa antagonist, mitragynine pseudoindoxyl, through a CYP3A-mediated skeletal rearrangement, displaying reduced hyperlocomotion, inhibition of GI transit and reinforcing properties, and (c) a potentially toxic metabolite, 3-dehydromitragynine, through a non-CYP oxidation pathway. Our results indicate that the oxidative metabolism of the mitragynine template beyond 7-hydroxymitragynine may have implications in its overall pharmacology *in vivo*.

Graphical Abstract

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INTRODUCTION

Effective pain management remains one of the challenges of modern medicine. A recent study found that over 20 million adults in the United States (U.S.) are affected by chronic pain.[1] Opioid agonists targeting the mu opioid receptor (MOR), such as morphine, are effective analgesics, but have high addiction potential and potentially lethal side effects in overdose attributed to profound respiratory suppression.[2] Another lingering effect of MOR agonist use is opioid-induced constipation.[3,4] As effective antinociceptives are essential to minimize pain and suffering, the identification of safer antinociceptive agents with diminished side effects and abuse potential is critical for breaking the vicious cycle fueling the opioid epidemic.

In the search for safer opioid agonists, we have focused on the alkaloids of *Mitragyna speciosa* (kratom),[5] specifically mitragynine,[6,7] 7-hydroxymitragynine (7OH),[6,7] and mitragynine pseudoindoxyl (MP),[7,8] showing that these alkaloids are MOR partial agonists[9] with reduced *β*arrestin-2 recruitment *in vitro* and mitigate lesser side effects compared to morphine and other clinical MOR agonists.[7,10,11] The lower intrinsic efficacy[12] of these alkaloids and negligible arrestin recruitment[13] has been independently replicated and supported by a mechanistic conformational model for the activation of MOR by these alkaloids.[14] Recent studies across numerous chemical templates suggest the possibility of designing safer opioids using the synthesis of low efficacy partial agonists.[9,15–18]

Kratom, a plant native to Southeast Asia, has been used for traditional treatment for various medical disorders, including pain, diarrhea, cough, alcoholism, and depression. Many aspects of the medicinal, pharmacological, and neurobiological potential of kratom-based natural products have been reviewed.[19–23] Among the nearly 54 alkaloids found in kratom, mitragynine is the major alkaloid (approximately 1–2% mass of dry leaf matter, representing up to 66% of the total alkaloid content) thought to be largely responsible for the plant's biological actions.

In 2016, the US Drug Enforcement Administration (DEA) released a notice of intent to place mitragynine and its oxidized analogue, 7OH, into Schedule I of the Controlled Substances Act (CSA). The notice was withdrawn two months later after outcry from the national media and public.[24] However, kratom[25] remains under regulatory scrutiny at

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the state and federal level. For example, the U.S. Food and Drug Administration (FDA) issued a statement on kratom in February 2018, describing its potential for abuse, addiction, and serious health consequences including death, concluding that kratom consumption has no health benefits.[26] In contrast, an independent analysis that applied the FDA's 7-point procedure to evaluate the risks and safety of drugs concluded that kratom presents a vastly safer profile compared to prescription and illicit opioids.[27] Regardless, kratom-based products remain widely used in the U.S., with reported sales on an estimated 10,000 retail outlets and a market value of $207 million.[27] At the same time, many aspects of kratom's pharmacology remain understudied, although new findings are emerging. For example, recent reports have shown that mitragynine is metabolized into 7OH.[28–30] The contribution of other metabolites to the various pharmacological effects of mitragynine remains unclear, as does a complete understanding how other component alkaloids might contribute to the effects of kratom as a whole. As part of the present study, we have examined the antinociceptive activity of the major individual alkaloids in kratom and a large variety of possible mitragynine metabolites in mice in order to better understand the mechanism of kratom's biological action. We had previously used the 129S1 strain for antinociception and PK studies.[29] In the present report, we used 129Sv6 mice for antinociception and peak effect PK assays to facilitate a direct comparison to our previous studies. In addition, we also used the inbred C57BL/6J strain which are well validated for pharmacokinetic investigations[31] and the studies of opioid-induced antinociception[32] and adverse effects such as reinforcement.[33] We find that the antinociceptive actions of kratom are mediated by mitragynine through its first-pass metabolic product, namely, 7OH and MP. We also identify another previously unknown toxic metabolite, 3-dehydromitragynine (3DM), which seemed to be responsible for the lethality of mitragynine in mice. Further, we assess the locomotor effect, constipation, and conditioned-place preference induced by kratom, mitragynine, and its opioid-active metabolites, in order to better clarify the interaction of these compounds in mediating the side effect profile of the kratom plant.

RESULTS

Chemistry.

Extraction of Natural Products and Synthesis of Metabolites.—The natural products mitragynine, paynantheine, speciogynine, and speciociliatine (Figure 1A) were extracted from kratom leaves as described before by Varadi *et al.*[7] The alkaloid derivatives, namely, 7-hydroxymitragynine (7OH),[34] 9-hydroxy corynantheidine (9OH), glucuronyl 9-hydroxycorynantheidine (9G), 3-dehydromitragynine (3DM),[35] and mitragynine pseudoindoxyl (MP),[7] were found as the major metabolites of mitragynine (Figure 1B, discussed in a later section) in mice dosed *per os* (po). The syntheses of 7-hydroxymitragynine, 9-hydroxycorynantheidine (9OH), and MP were accomplished by following our previously reported procedure.[7] Briefly, the synthesis of 7OH was accomplished by the oxidation of mitragynine using oxone.[7] The synthesis of 9G, 3DM, and mitragynine *N*-oxide (MNO) was completed by starting with mitragynine (Scheme 1). 3-dehydromitragynine could not be isolated in the pure form when mitragynine was heated with lead tetraacetate in acetic acid by following a previous report.[35] Thus, the 3-dehydromitragynine (3DM) was prepared by an acid-catalyzed dehydration of the

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7OH mitragynine using trifluoroacetic acid (TFA) in dichloromethane (DCM), isolated in quantitative yields as a TFA salt. The synthesis of glucuronyl 9-hydroxycorynantheidine (9G) was accomplished by the coupling of 9-hydroxycorynantheidine (9OH) with 2,3,4-tri-*O*-acetyl-*α*-D-glucuronic acid methyl ester and trichloroacetimidate in the presence of a mild Lewis acid $BF_3.OEt_2$.[36] The crude glucuronide ester intermediate product was further subjected to hydrolysis directly since the ester intermediate was found not to be stable in column chromatography using silica or basic alumina. Hydrolysis of both the acetyl groups and methyl ester was performed in a one-pot reaction with a slight excess of NaOMe/MeOH in DCM for 10 min followed by alkaline hydrolysis of the methyl ester. Finally, the desired product 9G was purified by using the anion exchange resin (Amberlite(R) IRA-410 chloride, 20–25 mesh). Mitragynine *N*-oxide (MNO)[37] has recently been reported to be a natural product present in kratom which was synthesized directly from mitragynine. The synthesis of MNO was achieved in three steps starting from mitragynine. Direct *N*-oxide formation[38] in mitragynine using *m*CPBA resulted in various byproducts in our hands resulting in a complication during the purification of MNO. However, a clean reaction of Boc-mitragynine *N*-oxide formation was found using Boc-mitragynine with *m*CPBA in 79% yield. Finally, MNO was obtained from the Boc deprotection of Boc-mitragynine *N*-oxide using TFA/DCM.

Pharmacology of Kratom Alkaloid Extract.—To investigate the molecular actions of kratom, we utilized a robust extraction protocol developed to isolate kratom alkaloids from dry kratom leaf powder commercially available in the US.[7] As it is well established that distribution of alkaloids in kratom is dependent on plant strain, growing climate, and when the leaves are harvested,[37,39] our studies used the well-validated “Red Indonesian Micro Powder” from moon kratom, Austin, TX, USA. To begin, we studied the *in vivo* pharmacology of a kratom alkaloid extract (KAE) containing the major alkaloids, mitragynine (48.6%), paynantheine (5.8%), speciogynine (3.5%), speciocilliatine (13.7%), and MNO (not present in our kratom batches) (Figure 1A).

Since kratom is commonly consumed orally, we examined the effects of KAE in mice after oral (po) administration as our starting point to understand the pharmacology of kratom. Antinociception of the alkaloid fraction in rats following po administration has been reported previously using the hot plate assay at a single dose.[40] In our hands, po KAE further showed potent, dose-dependent antinociception in a tail-flick assay in two strains of mice, C57BL/6J and 129Sv6. The peak antinociceptive effect of KAE was seen at 20 min in C57BL/6J mice (Figure 2A) and 15 min (Figure S2A) in 129Sv6 mice. The oral antinociceptive ED_{50} (and 95% C.I.) values of KAE were 17.1 (11.87–23.92) mg/kg in C57BL/6J mice (Figure 2A) and 35 (25.3–74.3) mg/kg in 129Sv6 (Figure S2A). The antinociception of KAE tested at an ED_{80} oral dose (45 mg/kg in C57BL/6J and 100 mg/kg in 129Sv6 mice) and was lost in MOR KO mice, while antinociception remained intact in delta opioid receptor (DOR) and kappa opioid receptor (KOR) KO mice in both C57BL/6J and 129Sv6 strains (Figure 2E and S2E, respectively). The antinociception was also attenuated by the opioid antagonist naloxone (Figure S2E), further confirming opioid-receptor mediation of KAE antinociception.

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In addition to examining the antinociception induced by KAE and its molecular mechanism, we also compared this substance's profile to that of established opioids in terms of three serious side effects: locomotor activity, inhibition of gastrointestinal (GI) transit, and addictive liability in C57BL/6J mice. Oral KAE (45 mg/kg) displayed significantly less inhibition of GI transit (Figure 2I) and no hyperlocomotion (Figure S7A) compared to morphine. Oral KAE did however show conditioned place preference (CPP) similar to morphine in C57BL/6J mice at doses 6-fold (100 mg/kg, po) higher than the antinociceptive ED_{50} (Figure 2M). Thus, in mice, KAE administered orally is antinociceptive (Figures 2A and S2A), and the antinociception is MOR-dependent (Figures 2E and S2E), yet associated with reduced constipation as well as the locomotor effect (Figures 2I and S7A). However, reward behavior as assessed by the CPP assay persists (Figure 2M), suggestive of addictive potential.

Pharmacology of Mitragynine.—We next evaluated *in vitro* binding and functional activity of the major kratom alkaloids in opioid-receptor-transfected CHO cells and antinociception in 129Sv6 mice after either subcutaneous (sc) or po administration, in order to identify the alkaloid responsible for the oral antinociceptive actions of KAE (Table 1). Briefly, among the natural products, mitragynine showed moderate binding affinity at mMOR (K_i = 230 nM), while speciocilliatine showed the highest affinity at mMOR (K_i = 79 nM). Both displayed partial agonism at mMOR (E_{max} = 63% and 45% for mitragynine and speciocilliatine, respectively) with potencies in line with the observed binding affinity. The other natural products had lower affinity as well as potency at opioid receptors. Among the six major natural individual alkaloids tested (Figure 1A) in 129Sv6 mice, only mitragynine showed antinociceptive activity and was more potent when given po (ED_{50} = 4 (2.21–7.16) mg/kg, Figure S3B) than when given sc (ED_{50} = 57.3 (44.6–73.5) mg/kg, Figure S3B). The po and parenteral antinociceptive actions of mitragynine in the hot plate assay in rats[41,42] have been described by others, and our finding of higher potency by the oral route is consistent with previous work.[43,44] Paynantheine, speciogynine, and speciocilliatine were inactive in the tail-flick assay at doses as high as 100 mg/kg (Table 1), suggesting that mitragynine is the major contributor to kratom's antinociceptive effects. To further confirm that only mitragynine mediates antinociception, an alkaloid extract fraction without mitragynine (alkaloid 2) was administered (po and sc) to 129Sv6 mice. No antinociceptive activity was seen at doses as high as 100 mg/kg. The alkaloid extract fraction without mitragynine also showed no antagonism of morphine and/or mitragynine po antinociception, suggesting no functional opioid antagonist activity (Figure S4). The oral antinociceptive potency of mitragynine was similar to that of po morphine (ED_{50} = 3.3 (2.075–5.5) mg/kg, Figure S2B). However, unlike mitragynine, morphine's antinociceptive potency was higher when administered sc (ED_{50} = 0.7 (0.47–0.99) mg/kg, Figure S3D). Similar to the 129Sv6 strain, in C57BL/6J mice, mitragynine was a more potent antinociceptive when given po (ED_{50} = 6.25 (4.49–8.68) mg/kg, Figures 2B–S3A) than when given sc (ED_{50} = 25.95 (15.43, 43.4) mg/kg, Figure S3A). In both strains and by both po and sc administration, the antinociceptive effect of mitragynine was rapid, reaching a peak in 15–20 min (Figure S1A–E). As with KAE, mitragynine po antinociception was mediated by MOR (but not DOR and KOR) in both C57BL/6J (Figure 2F) and 129Sv6 strains (Figure S2F), and could be reversed by naloxone (Figure S2F).

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The enhanced po *versus* sc potency of mitragynine (in two mice strains, Figures S3A,B) suggests the formation of a more potent active metabolite. The prototypic mu opioid agonist morphine also undergoes extensive first pass metabolism when given orally to yield both less active entities, like morphine 3-glucuronide and normorphine, and more active entities, like morphine 6-sulphate[45] and morphine 6-gluronide.[46,47] However, in contrast to mitragynine, morphine produces greater antinociception following parenteral *versus* oral administration; in humans, morphine given intravenously is ~3-fold more potent than when given orally.[48] A similar trend was seen in mice. In the 129Sv6 strain, po morphine [ED_{50} = 3.32 (2.08–5.32) mg/kg] was less potent than sc morphine [ED_{50} = 0.68 (0.47–0.99) mg/kg] (Figure S3D), while in the C57BL/6J strain, morphine showed ED_{50} = 4.73 (3.24–6.85) mg/kg, po and 2.48 (1.57–3.87) mg/kg, sc (Figure S3C).

We also examined the locomotor, inhibition of GI transit, and addictive side effect profile of mitragynine in C57BL/6J mice. At doses 5-fold higher than its antinociceptive ED_{50}, mitragynine (20 mg/kg, po in C57BL/6J mice) displayed attenuated inhibition of GI transit and locomotor compared to equi-antinociceptive doses of morphine (Figures 2J and S7B). However, po mitragynine produced reward behavior in C57BL/6J mice similar to morphine at a dose 5-fold higher than its antinociceptive ED_{50} dose (30 mg/kg, po) (Figure 2N). Mitragynine-conditioned place preference has been reported when administered parentally,[49–51] but to the best of our knowledge, it has not been examined following po administration, despite the common oral consumption of kratom.

In light of these data, we conclude that mitragynine is responsible for kratom's antinociceptive effects. When administered orally, it mediates MOR-dependent antinociception with reduced hyperlocomotion and constipation compared to morphine but displays rewarding behavior similar to morphine. The other tested alkaloids from kratom did not demonstrate antinociception nor did they antagonize mitragynine- or morphine-mediated antinociception.

Metabolic Mapping of Mitragynine Metabolites: PK Assays Show Formation of 7OH, MP, and 3DM as Important Entities.—Next, we sought to explore the potential contribution of active metabolites to the antinociceptive activity of mitragynine. To identify mitragynine metabolites, C57BL/6J and 129Sv6 mice were treated with mitragynine at doses of 30 mg/kg (po) and 10 mg/kg (po), respectively, at doses approximating the antinociceptive ED_{80} value. Plasma and brain samples were collected from C57BL/6J and samples analyzed at 20 min, 40 min, 60 min, 90 min, 2 h, 4 h, 8 h, and finally 24 h. In the case of 129Sv6 mice, samples were collected at peak antinociceptive effects of 15 min only. Liquid chromatography–tandem mass spectrometry (LC–MS/MS) was used to analyze and quantify parent drug and metabolites (See Table S3 and Figure 3). The identified mitragynine metabolites (shown in Figure 1B) were synthesized (see Scheme 1) and validated by LC–MS/MS using these authentic standards. Confirmed metabolites, namely 7OH,[52] MP,[7] 3-dehydromitragynine (3DM), glucuronyl 9-hydroxycorynantheidine (9G),[53] and 9-hydroxycorynantheidine (9OH),[52] were evaluated for *in vitro* binding/functional activity in opioid-transfected cell lines and for antinociception in mice (after the sc route). Consistent with previous findings,[6,7] *in vitro*, MP was a 200-fold more potent binder and 120-fold more potent agonist than mitragynine at MOR, while 7OH was a 6-fold more

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potent binder and 4-fold more potent agonist than mitragynine (Table 1). Among the other metabolites, only 3DM exhibited notable binding affinity for MOR (see the next section). 7OH and MP also showed potent antinociceptive activity in 129Sv6 mice [7OH ED_{50} (and 95% C.I.) value = 0.2 (0.07–0.6) mg/kg, sc; MP ED_{50} (and 95% C.I.) value = 0.3 (0.2–0.4) mg/kg, sc] (Figure S2C,D, respectively). Both metabolites displayed >100-fold more antinociceptive potency than mitragynine (sc),[7] suggesting that these two metabolites might play a key role in mediating the oral antinociceptive effects of mitragynine.

The highest concentrations in both brain and plasma of mitragynine (administered po in C57BL/6J mice) 7OH and MP (metabolites formed from mitragynine po) matched the antinociceptive peak effect in C57BL/6J mice at 20 min generally (See Figure 3 for concentrations observed and Figure S1A,C for antinociception time course). The levels of parent drug mitragynine and metabolites 7OH and MP decreased with time with no quantifiable 7OH or MP found at 8 h. Mitragynine and 3DM levels reached below detection limits at the 24 h point. Moreover, at the time of peak antinociception, 129Sv6 mice possessed the same set of metabolites as seen in the C57BL/6J mice.

We next closely examined 7OH and MP administered alone in PK assays on plasma and brains collected from C57BL/6J and 129Sv6 mice at the time of peak antinociception. Plasma concentrations of 343 ± 80 and 830 ± 50 nM of 7OH and 20 ± 4 and 13 ± 0 nM of MP were detected, respectively, in C57BL/6J and 129Sv6 mice (Figure S5A,C). Similarly, 7OH at concentrations of 96 ± 24 and 1070 ± 12 nM and MP at concentrations of 40 ± 16 and 180 ± 0 nM were detected in the brains of C57BL/6J and 129Sv6 mice, respectively (Figure S5B,C). Next, we sought to compare the levels of 7OH and MP reached when formed as metabolites following mitragynine administration *versus* those achieved when administered directly at equi-antinociceptive dosing, in order to demonstrate that the metabolites were responsible for antinociception. 7OH and MP were administered alone in C57BL/6J mice at the ~ED_{80} antinociceptive dose sc. Brain and plasma samples were analyzed at the time point of the peak antinociceptive effect to determine concentrations of the intact drug. Concentrations of 270 ± 13 and 450 ± 30 nM were found for 7OH in plasma and brain, respectively, consistent with the mitragynine results. MP administered directly sc to C57BL/6J mice at its ED_{80} antinociceptive dose resulted in concentrations of 410 ± 120 and 70 ± 10 nM in plasma and brain, respectively (Figure S5A,B, Table S3).

These data suggest that po administration of mitragynine at antinociceptive doses yields concentrations of 7OH and MP as a metabolite that are 7–10-fold higher than the agonist potencies (EC_{50} = 53 ± 4 nM for 7OH and 1.7 ± 0.1 nM for MP) of these drugs at MOR, suggesting that these metabolites are present to activate MOR at relevant doses.

We also sought to explore the role that intact mitragynine plays in its antinociceptive activity. When mitragynine was administered po to mice at an antinociceptive dose equivalent to its ED_{80} value, 6.63 ± 0.67 μM and 12.22 ± 3.4 μM of intact mitragynine were found in the brains of C57BL/6J and 129Sv6 mice at concentrations 30-to-60-fold greater than the demonstrated MOR agonist potency. When mitragynine was administered subcutaneously in C57BL/6J or 129Sv6 mice at 30 and 10 mg/kg doses, respectively, brain concentrations of 9.74 ± 3.05 μM (in C57BL/6J mice) and 30.06 ± 8.12 μM (in 129Sv6

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mice) were subsequently detected (Table S3). These mitragynine levels were approximately 45-to-150-fold greater than the demonstrated MOR agonist potency (EC_{50} = 203 nM) concentration in brain. Thus, these results suggest the parent mitragynine plays relatively a little role in the MOR-mediated antinociception associated with oral kratom or oral mitragynine.

Formation of Mitragynine Active Metabolites Is Dependent on CYP3A.—To determine if the oxidation of mitragynine to 7OH and MP is carried out by the liver, mitragynine was incubated with mouse liver microsomes from C57BL/6J or 129Sv6 mice or from human liver microsomes (HLM) and S9 fraction (Figure 4). Mitragynine was found to be labile in these assays (Figure 4A). The use of CYP3A inhibitor ketoconazole at 1 μM (Figure 4B) and liver microsomes from Cyp3a KO mice in the C57BL/6J strain (Figure 4C) decreased the decomposition of mitragynine. The half-life of mitragynine in microsomes was also increased by ~5 fold in the presence of ketoconazole and ~2 fold in microsomes from Cyp3a KO mice. The formation of 7OH and MP from mitragynine was also observed in mouse (129Sv6 and C57BL/6J) microsomes and HLM (Figure 4D,E). Lastly, the selective CYP3A inhibitor ketoconazole attenuated the formation of 7OH and MP from mitragynine in liver microsomes (Figures 4G,H).

To evaluate NADPH dependency, we reevaluated metabolism in HLM and S9 fractions in presence and absence of NADPH (Figure 5). Depletion of mitragynine and formation of 7OH and MP was seen in HLM (Figure 5A) and S9 (Figure 5C) in the presence of NADPH, with 7OH being the major metabolite in both cases. As expected from a CYP-dependent mechanism, the formation of 7OH as well as MP was minimal in the absence of NADPH in both HLM (Figure 5B) and S9 incubations (Figure 5D). Together, these findings suggest that mitragynine is oxidized to 7OH and MP in the liver through a CYP3A-dependent mechanism.

In summary, among mitragynine's known metabolites, only 7OH and MP are MOR agonists and antinociceptive. Further, formation of 7OH (major) and MP (minor) from mitragynine in mice occurs in the liver through a CYP3A-dependent mechanism, and these metabolites in turn play an important role in mediating mitragynine's antinociceptive effects.

Antinociception of 7OH and MP is Mediated through MOR, but Associated with Reduced Locomotor Effects.—Opioids are associated with a myriad of physiological effects including (though not limited to) antinociception, hyperlocomotion (in mice), constipation, and reward. We quantified the potency of the active metabolites of mitragynine in multiple mouse behavioral assays in order to evaluate their activity and side effect profiles relative to morphine. Subcutaneous 7OH produced potent antinociception, with ED_{50} (and 95% C.I.) values of 0.3 (0.19–0.48) mg/kg, sc in C57BL/6J (Figure 2C) and 0.2 (0.07–0.6) mg/kg, sc in 129Sv6 mice (Figure S2C), approximately 3-to-10-fold more potent than morphine in these strains. The antinociceptive actions were mediated through MOR, independent of DOR or KOR, and antinociception was naloxone-sensitive (Figures 2G and S2G). Mitragynine pseudoindoxyl was 2–10-fold more potent than morphine [ED_{50} (and 95% C.I.) values of 0.26 (0.15–0.43) mg/kg, sc in C57BL6J, Figure 2D and 0.3 (0.2–0.4) mg/kg, sc in 129Sv6 mice (Figure S2D)]. The antinociception of both metabolites was

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mediated through MOR, independent of DOR/KOR, and was naloxone-sensitive (Figures 2G,H and S2G,H). To further evaluate MP-mediated antagonism of the KOR and DOR free of MOR agonist effects, we utilized MOR KO mice. In MOR KO mice, MP (30 mg/kg, sc) antagonized *icv* U50, 488H antinociception (100 nmol), as well as *icv* SNC 80 antinociception (100 nmol) (Figure S6). These results are in line with our *in vitro* [^{35}S]-GTPγS assays, where MP was an antagonist at these receptors. It also further corroborates that MP is centrally active when administered systemically.

To determine whether other targets might contribute to the antinociceptive actions of these compounds, we also screened the major alkaloids and mitragynine metabolites for binding affinity at ~50 targets (Table S1) and βarrestin-2 recruitment (7OH and 3DM only, ~318 targets, Figure 6A,B) in a broad panel of CNS receptors. Mitragynine, 7OH, and MP selectively bound opioid receptors, consistent with the sensitivity of antinociception to naloxone and genetic disruption of MOR, and consistent with recent reports on the minimal activity of these alkaloids at non-opioid receptors.[42,64] Above, we showed that po administration of either KAE or mitragynine induced less hyperlocomotion or constipation than did morphine when administered at doses 5-fold higher than their respective antinociceptive ED_{50} values. Similarly, 7OH (1 mg/kg, sc in C57BL/6J mice, Figure S7C) and MP (1.5 mg/kg, sc in C57BL/6J, Figure S7D) induced less hyperlocomotion than morphine at equi-antinociceptive doses. However, 7OH did inhibit GI transit (Figure 2K), while MP in the same assay was reported previously to have a ceiling effect in GI transit assays[7] (Figure 2L).

Similar to po mitragynine, testing of the metabolite 7OH at 1 mg/kg, sc produced conditioned place preference (Figure 2O), while CPP was not reported in previous testing with MP[7] in the C57BL/6J strain at doses 10-fold higher than its antinociceptive ED_{50} (3.2 mg/kg, sc in the C57BL/6J strain; Figure 2P). Notably, the finding of CPP following systemic administration of 7OH is consistent with previous reports examining this metabolite.[56]

Identification of 3-Dehydromitragynine as a Toxic Metabolite of Mitragynine.—We next closely evaluated the pharmacology of 3DM. The formation of 3DM[35] from mitragynine was observed *in vivo* in mice of both strains (Figure 7D). Administration of the ED_{80} po antinociceptive dose of mitragynine resulted in detection of 0.094 ± 0.03 and 0.035 ± 0.01 μM concentrations of 3DM in the brains of C57BL/6J and 129Sv6 mice, respectively, coinciding with concentrations of 0.037 ± 0.01 and 0.028 ± 0.01 μM in plasma of C57BL/6J and 129Sv6 mice, respectively (Figure 7D). To further characterize the formation of 3DM, we looked at liver microsomes (Figure 4F), finding the formation of 3DM unlike 7OH or MP was not catalyzed by CYP3A (Figure 4I) and did not occur through 7OH as an intermediate, since incubation of 7OH with microsomes showed no formation of 3DM. The formation of 3DM from mitragynine was NADPH independent. 3DM formation was seen in both the presence and absence of NADPH in both HLM as well as S9 and in heat-inactivated HLM (Figure 5). Taken together, 3DM formation as a minor metabolite of mitragynine was found to be CYP3A independent. *In vitro*, 3DM had similar MOR affinity as 7OH in binding assays (K_i = 34.1 ± 5.9 nM), was a partial agonist at MOR (EC_{50} = 67.5 ± 8.3 nM; E_{max} = 44 ± 1%), and showed no other notable off–target activity beyond opioid receptors (Table 1

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and Figure 6B). Poor (μM) activity at BB3, CXCR7, and CCR3 receptors (Figure 6C–E) in the PRESTO-Tango assay[54] rule them out as possible targets for this molecule. Surprisingly, 3DM was not antinociceptive when administered directly to mice.

While not itself antinociceptive, 3DM was found to be toxic with 100% lethality after a 50 mg/kg, sc dose in C57BL/6J mice. Confirming this, 3DM demonstrated LD_{50} (and 95% C.I.) values of 48.4 (71.27–342.3) mg/kg sc in 129Sv6 and 74 (48.08–120.5) mg/kg, sc in CD1 mice (Figure 7A–C). However, the toxicity of this molecule seemed independent of mu, kappa, and delta opioid receptors, as either MOR KO mice or C57BL/6J mice treated with the kappa and delta opioid receptor antagonists, norbinaltorphimine (nor-BNI) and naltrindole (NTI), still demonstrated 3DM-mediated toxicity (Figure 7C). Supraspinal administration (10 μg) of 3DM to either CD1 or 129Sv6 mouse strains also led to fatality. At this time, the concentrations and half-life of 3DM achieved in plasma and brain following direct administration remain unknown, so interpretation of the toxicological relevance of its formation as a metabolite is admittedly difficult. Regardless, the present data demonstrates that mitragynine metabolizes to 3DM through a non-CYP mediated liver microsomal pathway and is lethal in multiple strains of mice.

DISCUSSION

The leaves of *M. speciosa*, also known as kratom, have been consumed by Southeast Asians for centuries. The pain-relieving properties of kratom have previously been attributed to the major indole alkaloid mitragynine. However, it has remained unclear how mitragynine acts as such a potent opioid antinociceptive despite having 50-fold weaker MOR binding affinity (230 nM) than morphine (4.6 nM) and only modest *in vitro* agonist potency (EC_{50} = 203 nM) and low efficacy at MOR (Table 1). Previous reports had shown that mitragynine administered by the oral route displayed more antinociceptive potency than when administered by parenteral routes,[29,43] suggesting the potential involvement of active metabolites. Further, the opioid pharmacology of the other major alkaloids in kratom and the role they might play in the antinociceptive effects of the plant have remained less well understood. To better understand kratom's pharmacological actions, we rigorously characterized *in vitro* and *in vivo* the opioid pharmacology of KAE, all the major natural kratom alkaloids (mitragynine, paynantheine, speciogynine, and speciociliatine) and minor (MNO) in pure form, and important metabolites of mitragynine. We show that mitragynine is converted to three biologically important metabolites (7OH, MP, and 3DM) in mice through two independent mechanisms. Formation of 7OH, as we have shown previously,[29] was CYP3A-dependent and produced antinociception. A CYP3A-independent conversion of mitragynine led to 3DM, a newly identified and toxic metabolite. In this context, we would also like to state that mitragynine is very stable in pH 7.4 buffer. We recorded the ^{1}H NMR spectroscopy (at 1, 5, 10, and 24 h time intervals) of mitragynine in D_2O to confirm that it is stable in D_2O and does not convert to 3DM at room temperature. We also confirmed this using LC/MS of mitragynine (at 1, 5, 10 and 24 h time intervals) incubated in pH 7.4 buffer. Again there was no formation of 3DM from mitragynine. We only see one peak that corresponds to mitragynine (^{1}H NMR and LC/MS traces at different time intervals are provided in the SI).

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The indole alkaloid mitragynine has at least five metabolically labile sites, including the C-9 methoxy group, C-7 unsaturated moiety (indole 2–3 double bond), C-3 axial hydrogen, and methyl (C-22), methoxy (C-17) acrylate functionalities (Figure 1B). Previous work suggested metabolism at these sites, and in particular, at the acrylate moiety.[29,53] However, to our surprise, less than 1% of total plasma metabolites in mice involved the acrylate moiety. Further, previous literature has shown that hydrolysis of the acrylate ester to a carboxylic acid leads to a complete loss of opioid activity,[19,21,29] suggesting that such metabolites do not play a major role in the pharmacological effects of mitragynine. In our PK studies in mice, we also observed demethylation of the 9-methoxy group to form 9OH, followed by glucuronidation to give 9G. Although 9OH showed an affinity similar to mitragynine at MOR (K_i = 216.2 ± 45.6 nM; Table 1), it did not produce antinociception when given subcutaneously at doses of up to 100 mg/kg and thus is not likely to contribute to mitragynine's antinociception. Likewise, 9G was a weak MOR agonist (EC_{50} = 5.7 ± 1.36 μM) *in vitro* but not significantly antinociceptive due to lower potency; (Table 1).

The formation of 3DM[35] from mitragynine, a natural product with unknown biosynthetic origins, was observed for the first time in both liver microsomes, S9 fractions, and *in vivo* in mice. 3DM has also been proposed to be a precursor which may lead to inhibition of CYP3A4 by Paine and co-workers.[57] Though not antinociceptive in mice, 3DM was identified as a metabolite of mitragynine with lethal effects. These results with 3DM may mediate potential lethality of mitragynine and kratom, given the reported LD_{50} of po mitragynine in mice is 477 and 591 mg/kg for po kratom alkaloid fractions.[58,59] In our hands, the LD_{50} of po mitragynine in the 129Sv6 strain was 750 mg/kg, while following administration of a 1 g/kg, po dose to C57BL/6J mice, 100% lethality was observed (data not shown). Importantly, administration of po mitragynine (1 g/kg) to C57BL/6J or MOR KO mice led to the death of all animals, confirming that the toxicity of mitragynine, similar to that of 3DM, does not seem to depend on MOR. While further study is warranted, this mechanistic similarity further suggests that 3DM may mediate the toxicity of mitragynine in rodents. Notably, the dose discrepancy between antinociception and fatality has precedence. In our previous work, the LD_{50} of morphine in CD1 mice was 750 mg/kg, sc while the antinociceptive ED_{50} value was 5 mg/kg.[60] The therapeutic indices, calculated as the ratio of antinociceptive and lethal ED_{50} doses, are ~150 for both morphine sc and mitragynine po. Given the difference in metabolism between humans and rodents, it is premature to determine whether 3DM has toxic effects in humans. Further, when corrected by standard allometric scaling, the oral LD_{50} value of mitragynine observed in rodents corresponds to >2.5 g of mitragynine in an adult human. At typical concentrations of mitragynine found in dry leaf powder available in the US (~1%), such an exposure would require consumption of >250 g of plant matter, a situation that seems unlikely given that most human users report consumption of <10 g per dose. Nevertheless, considering a number of reported deaths associated with kratom use[26,61] mediated by unknown causes, caution in the human usage of kratom is advisable until further studies of 3DM are available.

Our mouse PK studies also identified 7OH (major) and MP (minor) as metabolites of mitragynine. Inconsistent with its affinity and functional activity at MOR, po mitragynine in both mouse strains produced potent antinociception, with an ED_{50} value comparable to that of morphine. However, the higher potency of 7OH and MP in functional assays and

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antinociception in mice suggest that they are likely contributors to the analgesic effects of mitragynine. The brain/plasma ratio of 7OH was found to be 1.5:1, while MP appeared less brain permeable in our studies with ratios of 1:6. The higher brain permeability of 7OH might explain why despite its moderate binding (37 nM) and MOR agonistic potency (EC_{50} = 53 nM), 7OH displays a 10-fold greater analgesic potency over morphine and comparable potency to MP in the C57BL/6J strain.

Detailed PK studies with multiple time points and doses are required to conclusively prove this hypothesis, which is beyond the scope of this paper. The present results corroborate previous findings identifying 7OH as the key element in the antinociceptive actions of mitragynine.[29]

KAE, mitragynine, 7OH, and MP showed reduced hyperlocomotion compared to morphine while KAE, mitragynine, and 7OH all induced CPP in mice, suggesting that these substances are likely to be habit forming (Figure 2M–O). Recent reports have shown that mitragynine is not self-administered in rats given by the intravenous (iv) route.[62,63] Our results contrast with these studies, perhaps because iv administration of mitragynine enables less first pass metabolism of the drug, and in turn, attenuates formation of the 7OH concomitant to MOR-mediated reward. It is important to note that this discrepancy may also be due to differences in species (rats instead of mice), assay performed (self-administration *vs* CPP), or low doses tested in the self-administration assay (0.5 mg/kg/injection in Hemby *et al.* and 3 mg/kg/injection in Yue *et al.*), doses at which the drug does not show antinociception in mice in our hands.

Differences between the KAE, mitragynine and 7OH were seen in their ability to cause constipation. KAE and mitragynine showed reduced constipation while 7OH was similar to morphine in this assay. MP in the same assay behaved like a partial agonist. It is possible that the non-opioid actions[42,64,65] of mitragynine may be responsible for reduced constipation over 7OH which exclusively labels MOR. More studies are required to test this observation.

As an indole alkaloid, the metabolism of mitragynine may provide broader insight into the metabolism of other indole-based drugs, which is important given that over 41% of all drugs approved by the FDA are natural products or derived analogues.[66,67] The oxidative lability of the C-3 position of an indole ring (C-7 position in the case of mitragynine) has been previously reported[62,68] and conversion of mitragynine to mitragynine pseudoindoxyl *via* the transformation of an indole ring to a spirocyclic core through microbial fermentation is known.[69] However, we believe that the importance of metabolic oxidation at this position is still underappreciated. Here, we report for the first time a CYP-mediated transformation of indole to a spirocyclic pseudoindoxyl core. Incubation of 7OH with liver microsomes showed no conversion to MP (though a recent report shows formation in human plasma),[70] suggestive of a distinct CYP3A-mediated conversion of an indole (mitragynine) to a spirocyclic pseudoindoxyl (MP), which does not involve oxidation of the C-3 position (7OH) as a precursor. In addition, we present the first insight into a potential biosynthetic route to 3DM *via* C-3 oxidation of mitragynine.

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CONCLUSIONS

To summarize, we present a comprehensive evaluation of kratom's pharmacological actions *in vitro* and *in vivo*. We show that KAE, mitragynine, and the mitragynine metabolites 7OH and MP are potent MOR antinociceptive agents. Oral administration of mitragynine displays antinociceptive activity similar to sc morphine. The antinociception of po mitragynine results from its bioactivation to potent 7OH (major) and MP (minor) metabolites, while the parent compound mitragynine is not itself antinociceptive. We also report for the first time that metabolism of mitragynine leads to a highly toxic metabolite, 3DM. Further study is undoubtedly required to elucidate the relevance, if any, of this toxic metabolite to reported kratom toxicities and fatalities in man, but caution among kratom consumers in the interim is advised. Kratom, mitragynine, and the metabolite 7OH may be safer antinociceptive agents in view of their attenuated locomotor activity and reduced ability to cause GI transit inhibition for mitragynine and kratom. However, they still retain at least some degree of abuse potential, as evidenced by our findings in the CPP paradigm.

EXPERIMENTAL SECTION

Chemistry.

General Methods.—Reagents purchased from Sigma-Aldrich Chemicals, Fisher scientific, Alfa Aesar, were used without further purification. Reaction mixtures were purified by silica gel flash chromatography on E. Merck 230–400 mesh silica gel 60 using a Teledyne ISCO CombiFlash R_f instrument with UV detection at 280 and 254 nm. RediSep R_f silica gel normal phase columns were used with a gradient range of 0–10% MeOH in DCM. The yields reported are isolated yields (UPLC purity ≥95%). NMR spectra were recorded on Bruker Avance III 500, Avance III 600 with DCH CryoProbe instruments. Chemical shifts are reported in parts per million (ppm) relative to residual solvent peaks at the nearest 0.01 for proton and 0.1 for carbon ($CDCl_3$ ^1H: 7.26, ^{13}C: 77.1; and CD_3OD ^1H: 3.31, ^{13}C: 49.0). Peak multiplicity is reported as the NMR spectra were processed with MestreNova software, namely, s—singlet, d—doublet, t—triplet, q—quartet, and m—multiplet for examples. Coupling constant (*J*) values are expressed in Hz. Mass spectra were obtained at the MSKCC Analytical Core Facility using The Waters Acuity SQD LC MS by electrospray ionization (ESI). High-resolution mass spectra were obtained on a Waters Acuity Premiere XE TOF LC–MS by ESI, and the accurate masses are reported for the molecular ion $[M + H]^+$. All compounds are >95% pure by UPLC analysis. Purity of the products (≥95%) was confirmed by Waters Acquity UPLC: equipped with a binary solvent manager system, Waters XBridge C18 column (1.7 μm × 2.1 × 100 mm), PDA, ELS, and QDa mass detectors; mobile phase: solvent A: water with 0.1% TFA; solvent B: acetonitrile with 0.1% TFA. The flow rate was 1 mL/min in a reversed phase; total run time 8 min.

Isolation of Mitragynine from M. speciosa (Kratom).—Mitragynine was extracted from the powdered leaves by following our previously reported methods,[7] which is a modified method with an extra step of petroleum ether wash of the acidic aqueous phase reported by Ponglux *et al.*[71] Kratom powder (500 g) was heated to reflux in MeOH 700 mL for 40 min. The suspension was filtered, and the methanolic extraction process was repeated

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(3 × 500 mL). The solvent of combined methanolic extract was removed under reduced pressure and the content was dried using high vacuum. The dry residue was resuspended in 20% acetic acid solution (1 L) and washed with petroleum ether (4 × 500 mL). The aqueous layer was then cooled on ice bath and basified (pH ~ 9) with aqueous NaOH solution (3.5 M. ~ 1 L) slowly. Alkaloids were extracted in DCM (4 × 400 mL) from the aqueous layer. The combined DCM layer was washed with brine 300 mL and dried over anhydrous Na_2SO_4 and filtered. The solvent was removed under reduced pressure, and the residue was dried under high vacuum to obtain kratom alkaloid extract (11 g). Then, 200 mg of the kratom extract was subjected to silica gel column chromatography, using 0–40% EtOAc in hexanes to isolate mitragynine (97.2 mg); paynantheine (11.6 mg), speciogynine (7 mg), and 0–15% MeOH in DCM to isolate speciociliatine (27.4 mg), which correspond to approximately 48.6, 5.8, 3.5, and 13.7% of each alkaloids in the kratom extract. After removing the mitragynine fraction out, the total alkaloid content in the enriched **alkaloid extract 2** (**alkaloid 2**; devoid of mitragynine), when isolated as a combined fraction using the above protocol, approximately contains paynen-theine (24%), speciogynine (14%), and speciociliatine (57%) as the major alkaloids.

Synthesis and Characterization of New Compounds.

(2S,3S)-2-((E)-1,3-Dimethoxy-3-oxoprop-1-en-2-yl)-3-ethyl-8-methoxy-2,3,4,6,7,12-hexahydro-1H-indolo[2,3-a]quinolizin-5-ium TFA Salt (3DM TFA).—An ice-cold solution of trifluoroacetic acid (150 μL, 4.2 mmol) dissolved in DCM (1.0 mL) was added to the stirred solution of 7 hydroxymitragynine (330 mg, 0.79 mmol) in DCM (6 mL) at 0 °C under inert atmosphere. LC/MS indicated that the dehydration reaction was clean and completed in 2 h at 0 °C. Solvent and the excess of reagent were removed under reduced pressure to yield 3-dehyromitragynine TFA salt. The salt was redissolved in a mixture of acetonitrile in water (1:3, 15 mL) and was lyophilized to get **3DM TFA** salt as a yellow solid, (405 mg, quantitatively). ^{1}H NMR (600 MHz, chloroform-*d*): δ 12.24 (s, 1H), 7.49 (s, 1H), 7.28−7.25 (m, 1H), 7.14 (d, *J* = 8.4 Hz, 1H), 6.41 (d, *J* = 7.7 Hz, 1H), 3.90 (s, 3H), 3.88−3.80 (m, 2H), 3.75 (s, 3H), 3.61 (s, 3H), 3.59−3.51 (m, 3H), 3.50−3.40 (m, 3H), 3.28 (t, *J* = 12.9 Hz, 1H), 2.04 (ddt, *J* = 10.3, 7.7, 5.3 Hz, 1H), 1.46 (ddd, *J* = 13.7, 7.5, 5.9 Hz, 1H), 1.19−1.13 (m, 1H), 0.97 (t, *J* = 7.4 Hz, 3H). ^{13}C NMR (150 MHz, $CDCl_3$): δ 168.3, 166.9, 162.4, 155.9, 143.3, 130.1, 125.2, 122.4, 107.7, 107.0, 100.0, 62.1, 55.3, 54.3, 52.9, 51.5, 38.5, 31.5, 28.0, 23.0, 21.3, 11.6. HRMS calcd for $C_{23}H_{29}N_2O_4$ $[M]^+$, 397.2127; found, 397.2114.

(2S,3S,5R,6S)-6-(((2S,3S)-2-((E)-1,3-Dimethoxy-3-oxoprop-1-en-2-yl)-3-ethyl-1,2,3,4,6,7,12,12b-octahydroindolo[2,3-a]quinolizin-8-yl)oxy)-3,4,5-trihydroxytetrahydro-2H-pyran-2-carboxylic Acid (9G).—The **9G** was prepared by following a previously reported protocol with the modification.[72] 2,3,4-Tri-*O*-acetyl-α-D-glucuronic acid methyl ester and trichloroacetimidate (120.4 mg, 0.10 mmol) were added to the solution of 9 hydroxy corynantheidine (50 mg, 0.31 mmol) stirred in DCM (6 mL), over a molecular sieve, at rt under argon. The mixture was cooled to −78 °C bath for 5 min, $BF_3{\cdot}OEt_2$ (42.1 μL, 0.15 mmol) was added, and the reaction was continued for 10 min at the temperature. The cold bath was removed, and the reaction was continued over ice bath (at 0 °C) for 5 h. The reaction mixture was quenched

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with MeOH (30 μL) at the temperature, and the volume was concentrated down under reduced pressure. The concentrate was redissolved in DCM (15 mL), and the organic layer was washed with saturated aqueous $NaHCO_3$ solution (2 × 15 mL), brine (15 mL), dried over anhydrous Na_2SO_4, filtered, concentrated, and dried under reduced pressure. The crude material was redissolved in DCM (6 mL) and cooled to 0 °C. A solution of NaOMe (62 μL, 0.31 mmol; 0.5 M) was added at 0 °C, and the reaction was continued for 10 min; MS indicated that deacetylation was completed. Water (30 μL) was added, and the reaction was continued for 10 min to complete the hydrolysis of methyl ester. The reaction mixture was diluted with water (3 mL), acidified to pH 6 using acetic acid (20 μL), and the volatilities were removed under reduced pressure. The pH of the aqueous layer (pH ~ 6) was adjusted to slightly basic (pH ~ 8) using saturated aqueous $NaHCO_3$ solution, and the aqueous layer was washed with diethyl ether (3 × 5 mL) to ensure the removal of unreacted 9-hydroxy corynantheidine and the small amount of 9-acetyl corynantheidine byproduct; MS indicated the **9G** as the major peak. Then, the aqueous layer was loaded into the anion exchange resin, (Amberlite IRA-410 chloride, 20–25 mesh, 5 g pre-incubated in MeOH for 30 min and washed) and incubated in 10 min. The flow through the aqueous layer was collected with argon positive pressure and reloaded to the resin; and the process was repeated 5–6 times to optimize loading, MS of the final flow through indicated only a trace amount the product. The resin was washed with acetonitrile, EtOAc, and water (5 mL of each) to remove trace amount of impurities. Finally, the product was eluted with aqueous acetic acid (0.5%, 14 mL), and the aqueous layer was lyophilized to obtained the desired product **9G** as a white solid; (yield: 9.2 mg, 16%).^{1}H NMR (600 MHz, methanol-d_4): δ = 7.63 (d, J = 3.4 Hz, 1H), 7.19−7.00 (m, 2H), 6.77 (dd, J = 6.9, 1.7 Hz, 1H), 5.14 (d, J = 7.5 Hz, 1H), 4.23 (d, J = 12.3 Hz, 1H), 3.99 (d, J = 9.7 Hz, 1H), 3.86 (s, 3H), 3.72 (s, 5H), 3.68−3.61 (m, 2H), 3.60−3.47 (m, 3H), 3.46−3.32 (m, 2H), 2.80−2.60 (m, 1H), 2.40 (d, J = 14.5 Hz, 1H), 1.99 (s, 1H), 1.65 (td, J = 13.5, 12.8, 7.0 Hz, 1H), 1.51−1.42 (m, 1H), 0.96 (t, J = 7.4 Hz, 3H). ^{13}C NMR (150 MHz, MeOD): δ 172.2, 169.8, 163.2, 153.0, 140.0, 128.3, 124.3, 118.86, 110.0, 107.6, 106.6, 105.7, 102.9, 77.7, 76.6, 74.9, 73.1, 64.2, 62.6, 56.9, 55.0, 52.0, 40.3, 38.6, 28.4, 21.8, 19.3, 12.6. HRMS calcd for $C_{28}H_{35}N_2O_{10}$ $[M + H]^+$, 561.2448; found, 561.2443.

(2S,3S)-12-(tert-Butoxycarbonyl)-2-(€-1,3-dimethoxy-3-oxoprop-1-en-2-yl)-3-ethyl-8-methoxy-1,2,3,4,6,7,12,12b-octahydro-5H-indolo[2,3-a]quinolizine 5-Oxide (Boc-MNO).—Di-*tert*-butyl dicar-bonate (41.0 mg, 0.18 mmol) and DMAP (5 mg, 0.04 mmol) were added to the stirred solution of mitragynine (50 mg, 0.12 mmol) dissolved in THF (5.0 mL) at rt. under argon. The reaction mixture was heated to reflux at 80 °C for 4 h. The reaction mixture was concentrated under reduced pressure, the residue was redissolved in EtOAc (20 mL), and the organic layer was washed with saturated aqueous $NaHCO_3$ (20 mL), brine (20 mL), dried over anhydrous Na_2SO_4, filtered, concentrated, and the crude product was purified by SiO_2 column chromatography (ISCO, 4 g) using 1–20% MeOH in DCM to get **Boc-mitragynine** as a white solid, (yield: 58 mg, 93%); ^{1}H NMR (500 MHz, chloroform-d): δ 7.72 (d, J = 8.4 Hz, 1H), 7.39 (s, 1H), 7.13 (t, J = 8.2 Hz, 1H), 6.62 (d, J = 8.0 Hz, 1H), 3.86 (s, 3H), 3.69 (s, 3H), 3.68 (s, 3H), 3.60−3.54 (m, 1H), 3.10 (dt, J = 13.2, 3.6 Hz, 1H), 3.07−3.01 (m, 2H), 2.91−2.83 (m, 2H), 2.72−2.61 (m, 2H), 2.23 (td, J = 13.0, 10.4 Hz, 1H), 1.95 (d, J = 12.8 Hz, 1H), 1.72 (ddd, J = 13.8, 11.0, 7.1 Hz, 1H), 1.59 (s, 9H), 1.56 (d, J = 16.1 Hz, 1H), 1.21 (qd, J = 7.4, 2.9 Hz, 1H), 0.86 (t, J =

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7.4 Hz, 3H). ESI-MS m/*z*: 499.2 $[M + H]^+$.Then, the *meta*-Chloroperoxybenzoic acid (41.8 mg, 0.24 mmol) was added to the stirred solution of **Boc-mitragynine** (110 mg, 0.22 mmol) dissolved in DCM (3.0 mL) at 0 °C under argon and the reaction was continued overnight. The solvent was removed under reduced pressure, and the crude product was purified by alumina basic column chromatography (ISCO, 8 g) using MeOH (3–5%) in DCM to get Boc-mitragynine *N*-oxide (**Boc-MNO**) as a white solid, (yield: 90 mg, 79%); ^{1}H NMR (600 MHz, chloroform-*d*): δ 7.70 (d, J = 8.4 Hz, 1H), 7.42 (s, 1H), 7.15 (t, J = 8.2 Hz, 1H), 6.63 (d, J = 8.0 Hz, 1H), 4.93−4.87 (m, 1H), 4.05−3.99 (m, 1H), 3.87 (s, 3H), 3.82 (ddd, J = 15.5, 9.8, 4.6 Hz, 2H), 3.78 (s, 3H), 3.68 (s, 3H), 3.65 (dd, J = 11.4, 5.7 Hz, 1H), 3.51 (dt, J = 12.8, 5.0 Hz, 1H), 3.34 (dd, J = 11.6, 6.0 Hz, 1H), 3.17 (dd, J = 16.8, 4.5 Hz, 1H), 2.40−2.35 (m, 1H), 2.29 (dt, J = 15.2, 3.4 Hz, 1H), 2.22 (s, 1H), 1.63 (s, 9H), 1.46 (ddq, J = 21.5, 14.3, 7.2 Hz, 2H), 0.98 (t, J = 7.4 Hz, 3H). ^{13}C NMR (150 MHz, $CDCl_3$): δ 168.4, 161.1, 154.2, 149.7, 137.7, 130.3, 125.0, 118.3, 112.1, 109.6, 109.0, 103.64, 84.2, 74.3, 73.8, 61.9, 57.9, 55.4, 53.5, 51.5, 32.8, 28.2, 21.2, 20.4, 12.5. ESI-MS m/*z*: 515.2 $[M + H]^+$.

(2S,3S)-2-((E)-1,3-Dimethoxy-3-oxoprop-1-en-2-yl)-3-ethyl-8-methoxy-1,2,3,4,6,7,12,12b-octahydro-5H-indolo[2,3-a]quinolizine 5-Oxide (MNO).—An ice-cold solution of trifluoroacetic acid in DCM (1:1, 1.0 mL) was added to the stirred solution of **Boc-MNO** (15 mg, 0.03 mmol) dissolved in DCM (0.5 mL) at 0 °C under an argon. After 10 min, the ice bath was removed, and the reaction was continued for 60 min. Volatilities were removed under reduced pressure, and the crude product was purified by HPLC using acetonitrile/water with 0.1% TFA to get MNO as a white solid, (yield: 5.8 mg, 48%); ^{1}H NMR (600 MHz, methylene chloride-d_2): δ 8.92 (s, 1H), 7.28 (s, 1H), 7.06 (t, J = 8.0 Hz, 1H), 6.92 (d, J = 8.2 Hz, 1H), 6.45 (d, J = 7.8 Hz, 1H), 5.09 (dd, J = 9.3, 3.6 Hz, 1H), 4.44 (s, 1H), 3.98 (dt, J = 12.2, 5.3 Hz, 1H), 3.87 (ddd, J = 12.2, 7.5, 4.2 Hz, 1H), 3.82 (s, 3H), 3.75−3.69 (m, 1H), 3.66−3.53 (m, 3H), 3.50 (s, 3H), 3.45 (t, J = 7.4 Hz, 1H), 3.31−3.17 (m, 2H), 2.55−2.35 (m, 3H), 1.29 (dq, J = 14.5, 7.2 Hz, 1H), 1.21 (dq, J = 14.4, 7.2 Hz, 1H), 0.88 (t, J = 7.4 Hz, 3H). ^{13}C NMR (150 MHz, CD_2Cl_2): δ 168.6, 161.5, 154.7, 138.3, 125.6, 123.9, 117.4, 116.2, 115.5, 108.4, 105.2, 104.9, 100.1, 68.9, 62.6, 62.0, 55.4, 51.6, 34.6, 30.9, 22.0, 20.3, 12.0. HRMS calcd for $C_{23}H_{30}N_2O_5$ $[M + H]^+$, 415.2233; found, 415.2228.

Biological Methods.

Drugs and Materials.—Opiates were provided by the Research Technology Branch of the National Institute on Drug Abuse (Rockville, MD). IBNtxA and [^{125}I]BNtxA were synthesized in our laboratory as previously described.[73,74] Na^{125}I and [^{35}S]GTPγS were purchased from PerkinElmer (Waltham, MA). Selective opioid antagonists were purchased from Tocris Bioscience or Sigma-Aldrich. Miscellaneous chemicals and buffers were purchased from Sigma-Aldrich. Moon Kratom “Red Indonesian Micro Powder was purchased from Moon Kratom (Austin, TX).

Mice.—Mice including CD1, 129Sv6, and C57BL/6J (24–38 g) were purchased from Jackson Laboratories (Bar Harbor, ME). Exon-1/Exon-11 MOR KO mice were bred in the Pasternak/Pan laboratory at Sloan Kettering, while DOR and KOR KO mice in 129Sv6 strain were bred in the Pintar laboratory at Rutgers University. Exon 2 MOR KO mice

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on the C57BL/6J background were bred in the McLaughlin laboratory at University of Florida, from progenitors obtained from commercial sources [Jackson Laboratories (Bar Harbor, ME)]. All mice used throughout the manuscript were opioid naïve. All mice were maintained on a 12-h light/dark cycle with Purina rodent chow and water available ad libitum and housed in groups of five until testing. All animal studies were preapproved by the Institutional Animal Care and Use Committees of the Memorial Sloan Kettering Cancer Center or University of Florida or Rutgers University, in accordance with the 2002 National Institutes of Health Guide for the Care and Use of Laboratory Animals. The formulation used for all mice experiments was 1:1:8 of DMSO/solutol/saline. The volume of injection varied with the weight of mice, with 250 μL for a 25 g mouse typically used.

Power analysis to justify minimal numbers of animals was used. Mice are a valuable resource, prompting the application of power analysis to determine the minimal significant subject number. On the basis of analysis aiming for 95% power with an alpha significance of $p = 0.05$ and an ANOVA analysis with repeated measures within-between interaction, G*Power 3.19.2 was used to calculate a minimum number mice per antinociceptive of place conditioning testing condition. For the tail-withdrawal ("tail-flick"), a moderate effect size ($f = 0.7$) and 10 measurements assumed over time, resulting in an estimated sample size of eight mice, with a predicted critical F-value of 1.80 and actual power = 0.996. Assuming a large effect size ($f = 2.6$) and four treatment groups, eight mice were estimated needed for the GI transit testing, with a predicted critical F-value of 6.59 and actual power = 0.960. Assuming a small effect size ($f = 0.5$) and two measures, G*Power estimated 24 mice per condition for conditioned place preference/aversion were needed to achieve statistical significance in this work, defined with a critical F-value of 3.098, with actual power = 0.973.

Radioligand Competition Binding Assays.—[^{125}I]IBNtxA binding was carried out in membranes prepared from Chinese Hamster Ovary (CHO) cells stably expressing murine clones of MOR, DOR, and KOR, as previously described.[73,74] Binding in MOR/CHO was carried out in 50 mM potassium phosphate buffer with 5 mM $MgSO_4$ and 20 μg/mL protein, while binding in KOR/CHO and DOR/CHO was carried out in 50 mM potassium phosphate pH = 7.0 buffer and 40 μg/mL protein at 25 °C for 90 min. After the incubation, the reaction was filtered through glass-fiber filters (Whatman Schleicher & Schuell, Keene, NH) and washed (3 × 3 mL of ice-cold 50 mM Tris–HCl, pH 7.4) on a semiautomatic cell harvester. Nonspecific binding was determined by the addition of levallorphan (8 μM) to matching samples and was subtracted from total binding to yield specific binding. K_i values were calculated by nonlinear regression analysis (GraphPad Prism, San Diego, CA). Protein concentrations were determined using the Lowry method[75] with BSA as the standard.

[^{35}S]GTPγS Functional Assay.—[^{35}S]GTPγS binding was performed on membranes prepared from transfected cells stably expressing opioid receptors in the presence and absence of the indicated compound for 60 min at 30 °C in the assay buffer (50 mM Tris–HCl, pH 7.4, 3 mM $MgCl_2$, 0.2 mM EGTA, and 10 mM NaCl) containing 0.05 nM [^{35}S]GTPγS; 2 μg/mL each leupeptin, pepstatin, aprotinin, and bestatin; and 30 μM GDP, as previously described.[7,76,77] After the incubation, the reactions were filtered through glass fiber filters (Whatman Schleicher & Schuell, Keene, NH) and washed (3 × 3 mL of

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ice-cold buffer, 50 mM Tris–HCl, pH 7.4) on a semi-automatic cell harvester. Filters were transferred into vials with 3 mL of Liquiscint (National Diagnostics, Atlanta, GA), and the radioactivity in vials was determined by scintillation spectroscopy in a Tri-Carb 2900TR counter (PerkinElmer Life and Analytical Sciences). Basal binding was determined in the presence of GDP and the absence of drug. Data were normalized to 100 nM DAMGO, DPDPE, and U50, 488H for MOR, DOR, and KOR binding, respectively. EC_{50}, IC_{50}, and $\%E_{max}$ values were calculated by nonlinear regression analysis (GraphPad Prism, San Diego, CA).

Antinociception. Pasternak/Pan Labs.—Tail flick antinociception in 129Sv6 mice was determined using the radiant heat tail flick technique using an Ugo Basile model 37360 instrument as previously described.[7,76,78] The intensity was set to achieve a baseline between 2 and 3 s. Baseline latencies were determined before experimental treatments for all mice. Tail flick antinociception was assessed quantally as a doubling or greater of the baseline latency, with a maximal 10 s latency to minimize damage to the tail. Data were analyzed as the percent maximal effect, %MPE, and was calculated according to the formula: % MPE [(observed latency − baseline latency)/(maximal latency − baseline latency)] × 100. Compounds were administered either orally (po) or subcutaneously (sc) or intracerebroventricularly (icv), and antinociception was assessed at the peak effect. Intracerebroventricular dosing (icv) was carried out as described previously. In brief, mice were anesthetized using isoflurane. A small incision was made, and the drug (2 μL/mouse) was injected (using a 10 μL Hamilton syringe fitted to a 27-gauge needle) into the right lateral ventricle at the following coordinates: 2 mm caudal to bregma, 2 mm lateral to sagittal suture, and 2 mm in depth. Mice were tested for antinociception at the peak effect post injection. For the antagonism studies, naloxone (1 mg/kg, sc) was administered 15 min prior to the drug administration.

McLaughlin Lab.—The 55 °C warm-water tail-withdrawal assay was conducted in C57BL/6J mice as a measure of acute thermal antinociception as described previously.[79] Briefly, each mouse was tested for baseline tail-withdrawal latency prior to drug administration. Following drug administration, the latency for each mouse to withdraw the tail was measured every 10 min until latency returned to the baseline value. A maximum response time of 15 s was utilized to prevent tissue damage. If the mouse failed to display a tail-withdrawal response within 15 s, the tail was removed from the water and the animal was assigned a maximal antinociceptive score of 100%. Data are reported as percent antinociception, calculated by the equation: % antinociception = 100 × [(test latency − baseline latency)/(15 − baseline latency)]. This was utilized to account for innate variability between mice. Compounds were administered subcutaneously (sc), and the antinociceptive action of compounds was assessed at the peak effect.

Conditioned Place Preference and Aversion.—Mice were conditioned with a counterbalanced place conditioning paradigm using similar timing as detailed previously.[7,10] Groups of mice (n = 18–24) were habituated to freely explore both sides of a three-compartment apparatus for 3 h each for 2 days prior to testing. The amount of time subjects spent in each of three compartments was measured over a 20 min testing period. Prior to

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place conditioning, the animals did not demonstrate significant differences in their time spent exploring the left *versus* right compartments. During each of the next 2 days, mice were administered vehicle (0.9% saline) and consistently confined in a randomly assigned outer compartment for 20 min, half of each group in the right chamber, half in the left chamber. Four hours later, mice were administered drugs morphine (10 mg/kg/d, sc), U50, 488H (30 mg/kg/d, sc), **Kratom alkaloid extract** (100 mg/kg/d, po), **mitragynine** (30 mg/kg/d, po), or **7-OH** (1 mg/kg/d; 3.2 mg/kg/d, sc) and were confined to the opposite compartment for 20 min. Conditioned place preference data are presented as the difference in time spent in drug- and vehicle-associated chambers and were analyzed *via* repeated measures two-way ANOVA with the difference in time spent on the treatment-*versus* vehicle-associated side as the dependent measure and conditioning status as the between-groups factor. Where appropriate, Tukey's HSD or Sidak's multiple comparison post hoc tests were used to assess group differences. Effects were considered significant when $p < 0.05$. All effects are expressed as mean ± SEM.

Assessment of GI Transit.—C57BL/6J mice (8 per drug treatment) were administered saline (0.9%, s.c.), morphine (10 mg/kg, s.c.), KAE (85.5 mg/kg, p.o.), mitragynine (25 mg/kg, p.o.), 7OH (1.5 mg/kg, s.c.), or mitragynine pseudoindoxyl (0.75–4 mg/kg, s.c.) 20 min prior to oral gavage with 0.3 mL of a 5% aqueous solution of charcoal meal. After 3 h, mice were euthanized, and the intestines were removed. The progression of charcoal through the intestines was measured as distance traveled from the jejunum to the cecum as utilized elsewhere.[80,81]

Pharmacokinetic Study.—Mitragynine was administered to mice either orally or subcutaneously at appropriate doses in C57BL/6J and 129Sv6 mice.[82] At peak effects, after 15 or 20 min post administration of the drug, mice were anesthetized under isoflurane, blood was removed, and animals were sacrificed for brain removal. Blood was separated to plasma in K2EDTA tubes by centrifugation and then plasma was extracted with 50% volume of methanol, vortexed, pelleted, and supernatants dried down. Brains were quickly rinsed off with PBS, blot-dried, homogenized 50% methanol in water, pelleted, and supernatants dried down to analyze metabolites. Each brain sample reflects material recovered from half a brain. The dried-down metabolites content was resuspended in the acetonitrile/water mixture (100 μL) prior to the LC–MS/MS analysis. In the case of direct plasma analysis, 5 μL of plasma was mixed with 100 μL of 80/20 acetonitrile/water (containing 5 ng/mL carbamazepine, an internal standard), centrifuged, and passed through a filter plate (Millipore Multiscreen Solvinter 0.45 μm, low binding PTFE hydrophilic filter plate). Samples were analyzed by LC–MS/MS.

Liver Microsomal Assays.—Liver microsomal assays were performed by following the previously reported protocols.[82] Briefly, microsome stability of the kratom alkaloid was evaluated by incubating mitragynine(1 μM) with 1 mg/mL hepatic microsomes (human, rat, or mouse) in a 100 mM potassium phosphate buffer, pH 7.4. The reactions were held at 37 °C with continuous shaking. The reaction was initiated by adding NADPH, 1 mM final concentration. The final incubation volume was 300, and 40 μL aliquots were removed at 0, 5, 10, 20, 40, and 60 min. The removed aliquot was added to 160 μL acetonitrile

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to stop the reaction and precipitate the protein. NADPH dependence of the reaction was evaluated in parallel incubations without NADPH. Mitragynine metabolism by CYP3A4 was verified using a validated 3A4 inhibitor assay using 1 μM mitragynine in the presence of 1 μM ketoconazole for selective 3A4 blockade. At the end of the assay, the samples are centrifuged through a 0.45-μm filter plate (Millipore Solventer low binding hydrophilic plates, cat# MSRLN0450) and analyzed by LC–MS/MS. The data were log-transformed, and the results are reported as half-life. Note: Protein concentration for both liver micrsomes ans well as S9 was 1 mg/mL. For S9 fraction experiments, we simply swapped microsomes with S9. S9 was bought from XenoTech. Varapamil was used as control for S9 experiemnts. The half-life in min for verapamil was 24.14 min (data not shown).

Assessment of Off–target Activity of Kratom alkaloids and Metabolite Using PRESTO-Tango GPCR-ome.—To identify potential off–target activity of 7OH and 3DM, we used the National Institutes of Mental Health Psychoactive Drug Screen Program. 3DM and 7OH was first tested for activity against 330 non-olfactory GPCRs using the PRESTO-Tango GPCRome screening β-arrestin2 recruitment assay at 10 μM. The activity at each receptor was measured in quadruplicate. Screening of compounds was accomplished using previously described methods with several modifications (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4424118/).[54] First, HTLA cells were plated in DMEM with 2% dialyzed FBS and 10 U/mL penicillin–streptomycin. Next, the cells were transfected using an in-plate PEI method (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4012321/).[55] PRESTO-Tango receptor DNAs were resuspended in OptiMEM and hybridized with PEI prior to dilution and distribution into 384-well plates and subsequent addition to cells. After overnight incubation, drugs diluted in DMEM with 1% dialyzed FBS were added to cells without replacement of the medium. The remaining steps of the PRESTO-Tango protocol were followed as previously described.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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ABBREVIATIONS

C.I.	confidence interval
CLAMS	comprehensive lab animal monitoring system
DCM	dichloromethane

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3-DM	3-dehydromitragynine
DOR	delta opioid receptor
9G	9-glucuronyl corynantheidine
HLM	human liver microsomes
KAE	kratom alkaloid extract
KOR	kappa opioid receptor
MNO	mitragynine *N*-oxide
MOR	mu opioid receptor
MP	mitragynine pseudoindoxyl
MPE	maximum possible effect
7OH	7-hydroxymitragynine
po	*per os*
Rt	room temperature
sc	subcutaneous

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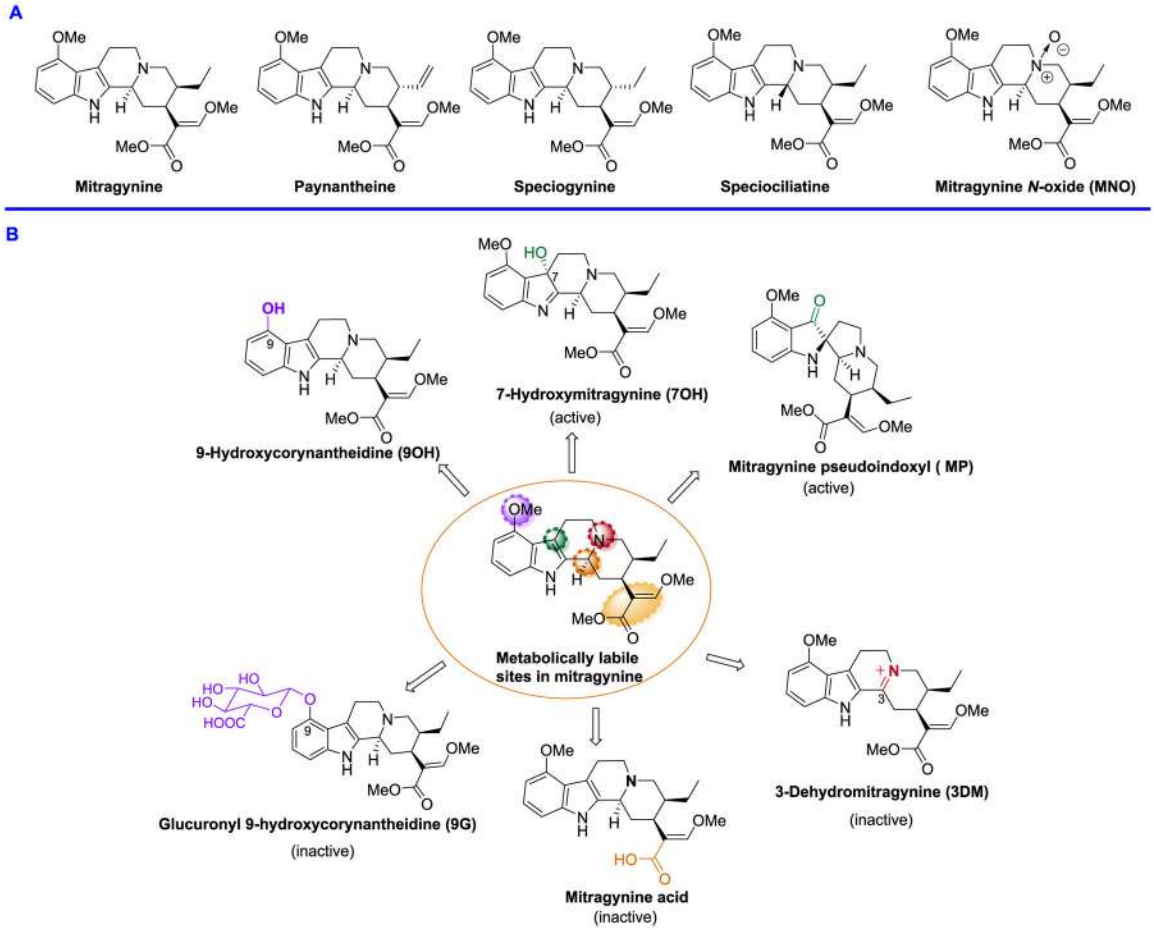


Figure 1.

(A) Kratom alkaloids; major alkaloid natural product (B) metabolites of mitragynine in mice.

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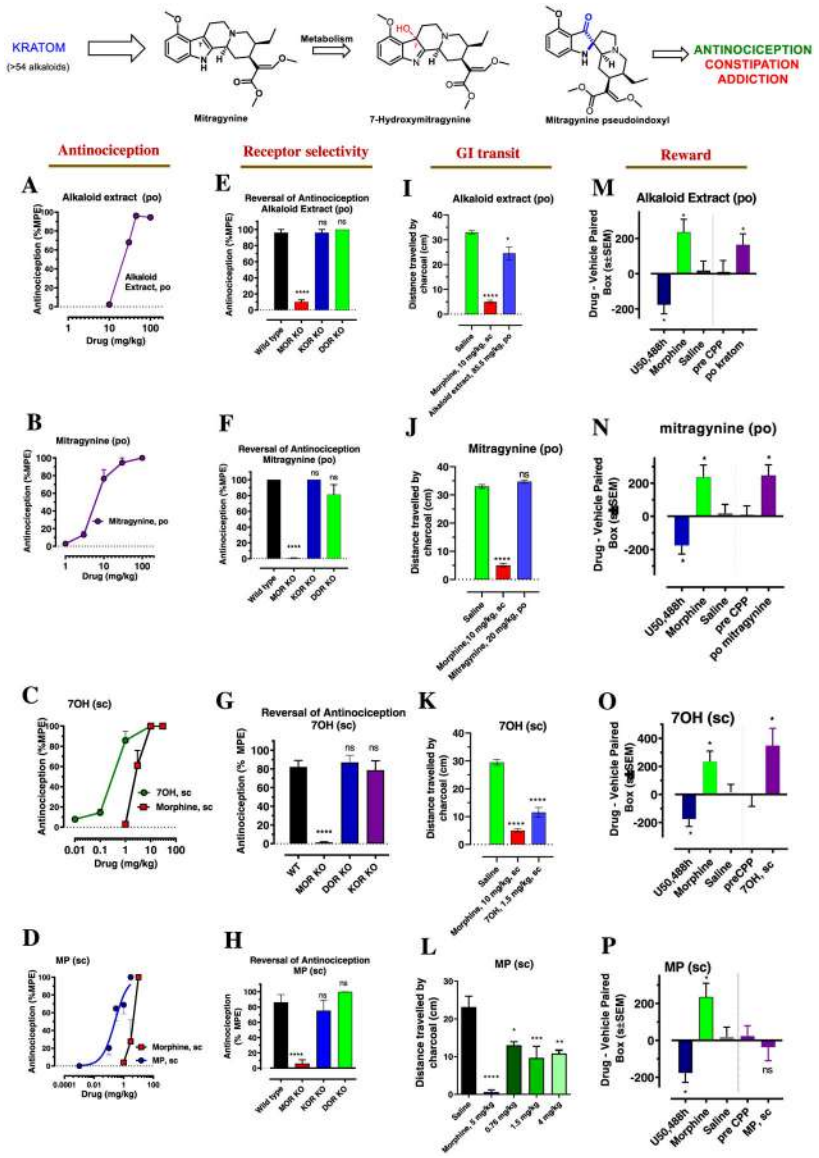


Figure 2.

Kratom, mitragynine, and its metabolites show MOR-dependent antinociception with reduced constipation except for 7OH but retain reinforcing behavior except for MP in C57BL/6J mice. Oral tail flick dose–response curves (A,B): groups of mice ($n = 8$) per dose were given different doses of po kratom (A) or po mitragynine (B) by gavage and tested for the antinociceptive response at 20 min. ED_{50} (and 95% CI) was 17.1 (11.87–23.92) mg/kg for po kratom and 6.25 (4.49–8.68) mg/kg for po mitragynine. Subcutaneous tail flick dose–response curves (C,D): Groups of mice ($n = 8$) per dose were given different doses of sc 7OH and sc morphine (**C**) and sc MP and sc morphine (D) and tested for antinociceptive response at 20 min for 7OH and MP and 30 min for morphine. ED_{50} (and 95% CI) was 0.3 (0.19–0.48) mg/kg for sc 7OH, 0.26 (0.15–0.43) mg/kg for sc MP, and 2.48(1.57–3.87) mg/kg for sc morphine. The means of each point in each determination were calculated as percentage maximal possible effect (%MPE) [(observed latency −

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baseline latency)/(maximal latency − baseline latency)] × 100. Kratom alkaloid fraction, mitragynine showed antinociception given orally while the putative metabolites 7OH and MP showed antinociception given sc and about 10 fold more potent over sc morphine. Antinociception in KO mice (E–H): An ~ED_{80} antinociceptive dose of alkaloid extract (45 mg/kg, po), mitragynine (45 mg/kg, po), 7OH (1 mg/kg, sc), and MP (3 mg/kg, sc) was evaluated in group of ($n = 8$) of wildtype (WT), MOR KO, KOR KO, and DOR KO mice. Antinociceptive effects of all these drugs were found attenuated in MOR KO, whereas the effect was found intact in KOR KO and DOR KO mice (****$p < 0.0001$, one-way ANOVA followed by Dunnett's multiple comparison comparisons test). Effects on GI transit in C57BL/6J mice (I–L): a group of C57BL/CJ mice ($n = 8$) were administered with either saline, morphine (10 mg/kg, sc), alkaloid extract (85.5 mg/kg, po), mitragynine (20 mg/kg, po) 7OH (1.5 mg/kg, sc), or MP (0.76, 1.5, 4 mg/kg, sc) and then were fed a charcoal meal. After 3 h, morphine significantly reduced the distance traveled by the charcoal through the intestines, consistent with the action of a MOR agonist (5.07 ± 0.57 cm), compared to 33 ± 0.68 cm for saline-treated mice (****$p < 0.0001$, one-way ANOVA followed by Dunnett's multiple comparison test). In contrast, the alkaloid extract (24.6 ± 2.54 cm) showed constipation compared to saline (*$p < 0.0019$, one-way ANOVA followed by Dunnett's multiple comparison test) but less than morphine, while mitragynine (34.7 ± 0.53 cm) showed no significant effect (I,J). 7OH (1.5 mg/kg, sc) showed significant reduction in the distance traveled by the charcoal (11.6 ± 1.8 cm) through the intestines (****$p < 0.0001$, one-way ANOVA followed by Dunnett's multiple comparison test) (K). MP showed constipation at all doses (*$p = 0.016$, ***$p = 0.001$, **$p = 0.005$, one way ANOVA followed by Dunnett's multiple comparison test) compared to saline but less than morphine. Conditioned place preference or aversion in C57BL/CJ mice (CPP/CPA) (M–P): A group of C57BL/CJ mice ($n = 18$–24) were habituated to freely explore both sides of a two-compartment apparatus for 3 h each for 2 days prior testing. The baseline preferences of 20 min were determined as the pre-condition test prior to the administration of a drug on the conditioning day. Mice were conditioned for 20 min on each session with either saline, morphine (10 mg/kg/d, sc), U50, 488H (30 mg/kg/d, sc), alkaloid extract (100 mg/kg/d), mitragynine (30 mg/kg/d) 7OH (1 mg/kg/d, sc), and MP (3.2 mg/kg/d, sc) after being habituated for 1 h. On the testing day, animals were placed in the side paired with saline and are allowed to freely explore both compartments for 20 min. The difference score was calculated to determine CPP/CPA. MP did not display significant CPP or CPA, whereas kratom alkaloid fraction, mitragynine, and 7OH displayed CPP (M–P); ($p < 0.05$) as determined by ANOVA followed by Tukey's multiple-comparison test. Note: Figure 2F,G; data reprinted (adapted or reprinted in part) with permission from [Kruegel, A. C.; Uprety, R.; Grinnell, S. G.; Langreck, C.; Pekarskaya, E. A.; Le Rouzic, V.; Ansonoff, M.; Gassaway, M. M.; Pintar, J. E.; Pasternak, G. W.; Javitch, J. A.; Majumdar, S.; Sames, D. 7-Hydroxymitragynine Is an Active Metabolite of Mitragynine and a Key Mediator of Its Analgesic Effects. *ACS Cent. Sci.* **2019**, *5* (6), 992–1001.] Copyright [2019/American Chemical Society]. Figure 2L,P data reprinted (adapted or reprinted in part) with permission from [Váradi, A.; Marrone, G. F.; Palmer, T. C.; Narayan, A.; Szabó, M. R.; Le Rouzic, V.; Grinnell, S. G.; Subrath, J. J.; Warner, E.; Kalra, S.; Hunkele, A.; Pagirsky, J.; Eans, S. O.; Medina, J. M.; Xu, J.; Pan, Y. X.; Borics, A.; Pasternak, G. W.; McLaughlin, J. P.; Majumdar, S. Mitragynine/Corynantheidine Pseudoindoxyls As Opioid Analgesics with Mu

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Agonism and Delta Antagonism, Which Do Not Recruit β-Arrestin-2. *J. Med. Chem.* **2016**, *59* (18), 8381–8397.] Copyright [2016/American Chemical Society].

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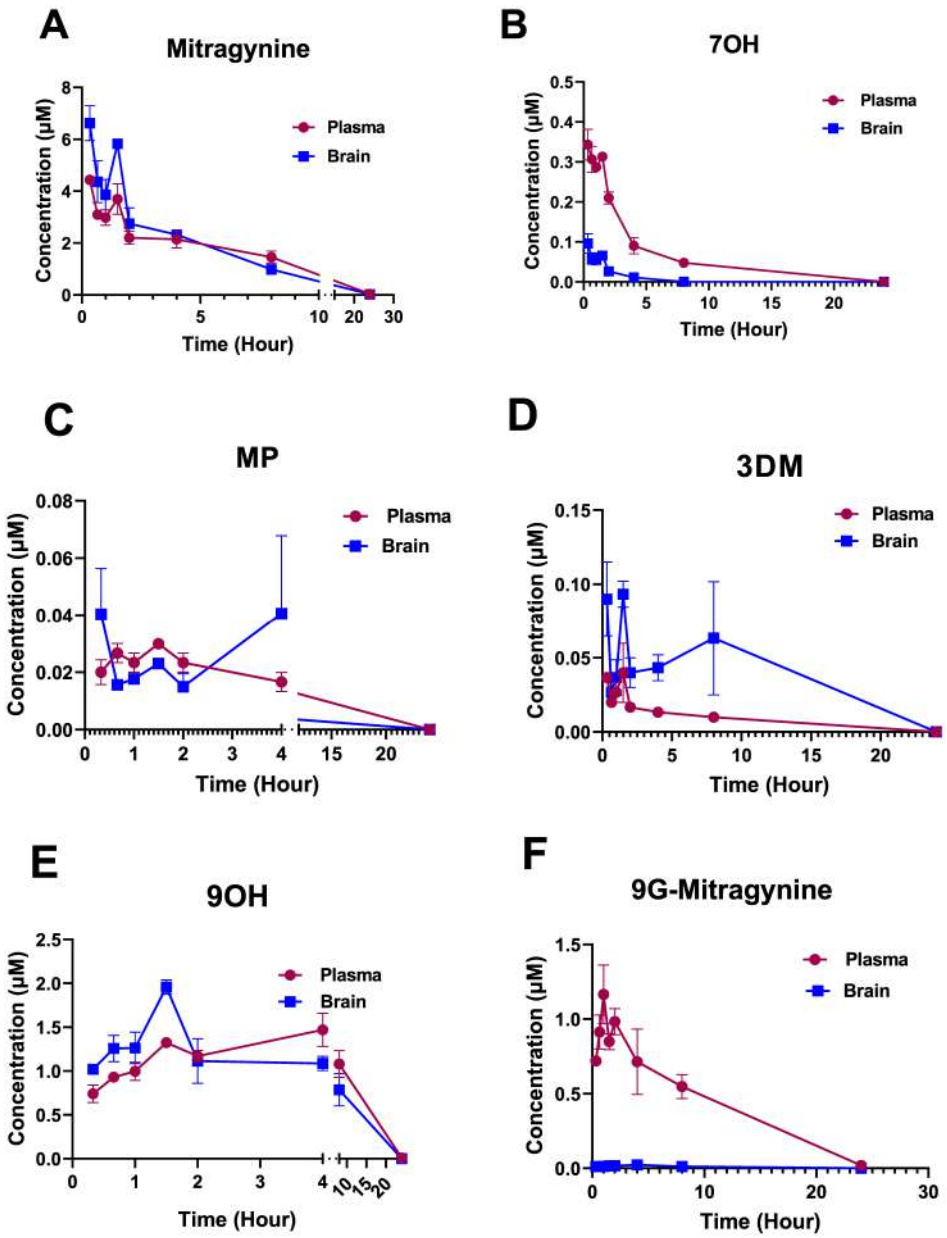


Figure 3.

Concentration time profile for mitragynine to its metabolites was confirmed *in vivo* in C57BL/6 mice using LC–MS/MS. (A) Mitragynine was detected in both the plasma and brains of mice treated with mitragynine (30 mg/kg, po). $n = 3$ per time point for plasma; $n = 3$ per time point for brain. At the same time gradient, metabolites formed from mitragynine, namely, 7-OH (B), MP (C), 3DM (D), 9OH (E), and 9G-mitragynine (F) were also detected in the plasma and brains of the same animals, but at lower concentrations. 7OH was the major metabolite followed by MP and 3DM as other important metabolites.

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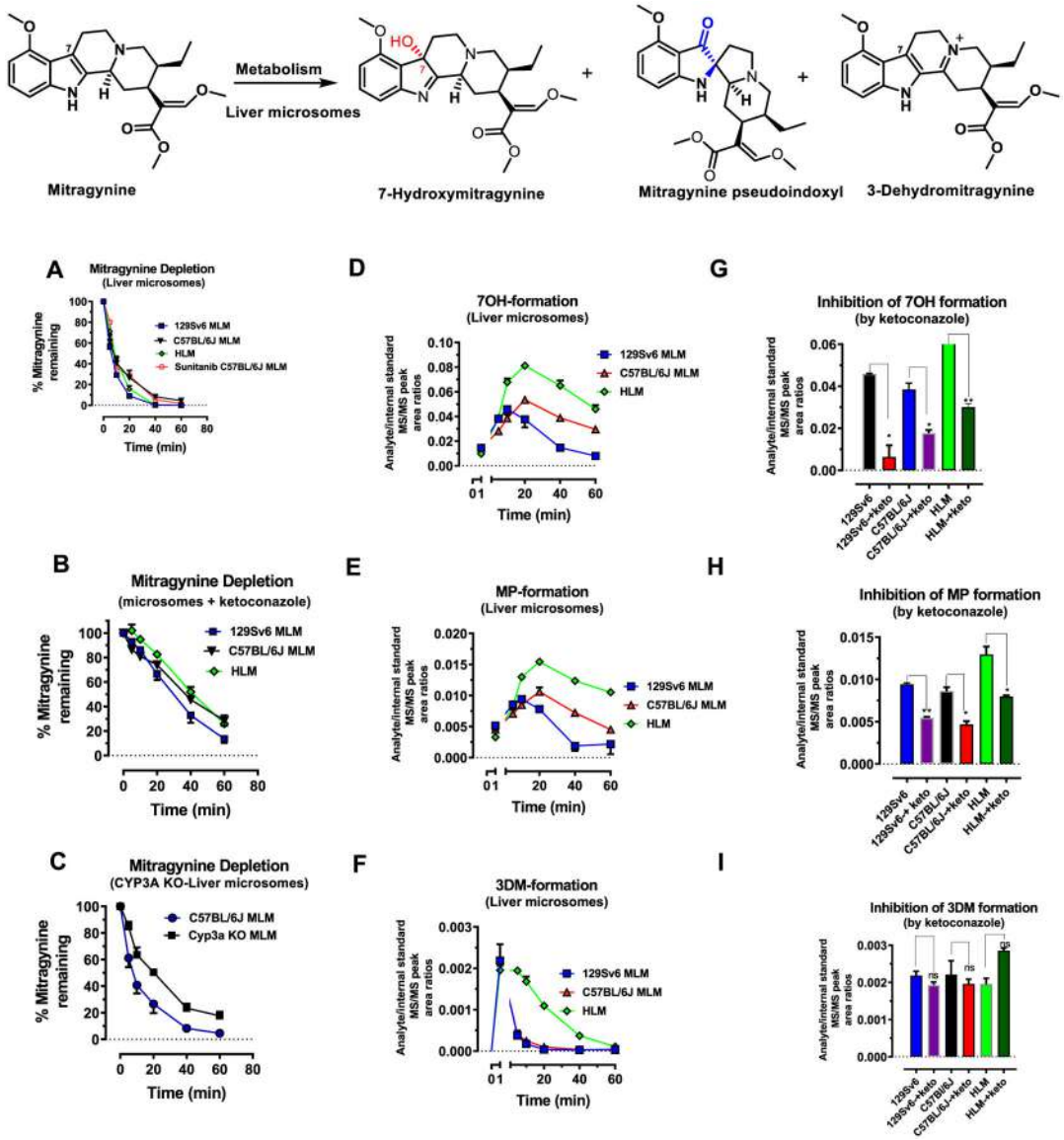


Figure 4.

Metabolism of mitragynine to **7OH**, **MP**, and **3DM** in liver microsomes: mitragynine (1 μM) was incubated in the potassium phosphate buffer (0.1 M; pH 7.4) at 37 °C in the presence of liver microsomes of mice (129Sv6 and C57BL/6J) or human (HLM) for 60 min. Each point in the curve represents three independent determinations. Level of intact mitragynine decreases over time due to microsomal disintegration. The half-lives of mitragynine were 7.5 ± 0.82, 5.9 ± 0.3, and 13.5 ± 1.1 min in HLM, 129Sv6, and C57BL/6J mice liver microsomes. (**A**). Usage of a CYP3A blocker ketoconazole (1 μM). The half-lives of mitragynine in the presence of ketoconazole were 37.4 ± 7.5, 24.4 ± 4.9, and 35.8 ± 4 min in HLM, 129Sv6, and C57BL/6J mice liver microsomes (B). Liver microsomes from a Cyp3a KO male mice also inhibit the depletion of mitragynine. The half-life of mitragynine was 25.2 ± 2.05 min in Cyp3a KO mice microsomes (C). These liver microsomal assays indicated the formation of three major metabolites **7OH**, **MP**, and **3DM** (D–F). A decrease

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in the formation metabolites from mitragynine including **7OH** and **MP** was observed in the presence of ketoconazole at the peak effect compared to control, where ketoconazole was not included in microsomal incubations ($p < 0.05$ as determined by unpaired *t*-test) (G–H). The formation of 3DM was found to be independent of CYP3A (I).

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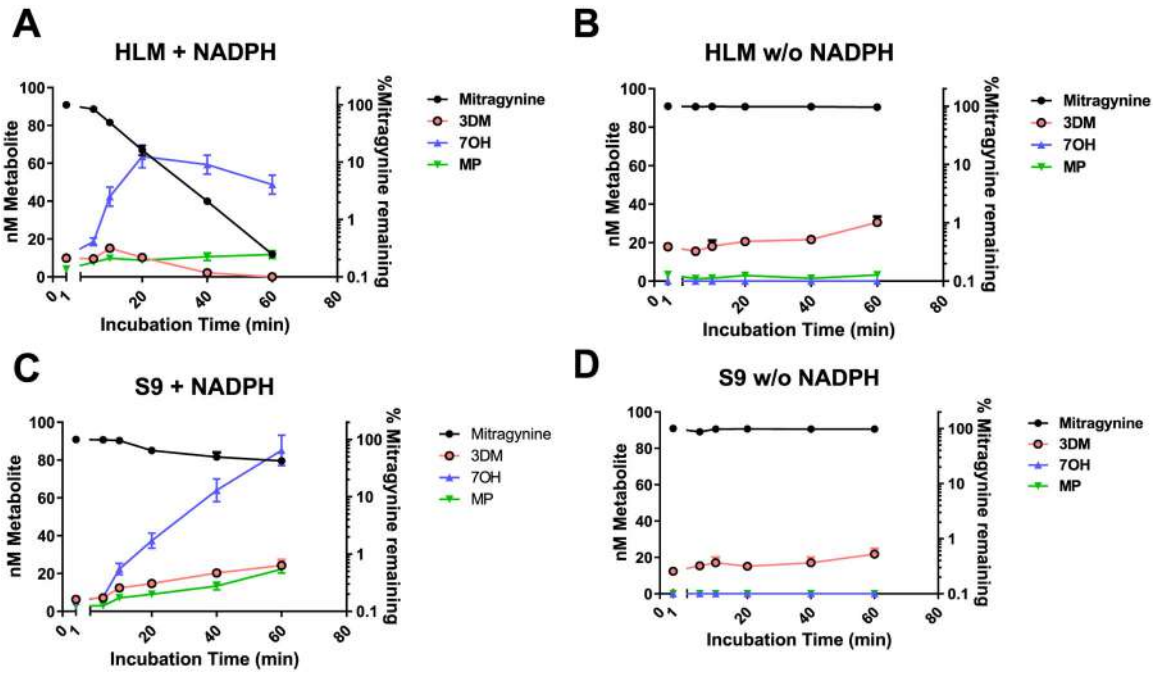


Figure 5.

Metabolism of mitragynine to **7OH**, **MP**, and **3DM** in **HLM** and **S9** fractions in the presence and absence of NADPH: mitragynine (1 μM) was incubated in the potassium phosphate buffer (0.1 M; pH 7.4) at 37 °C in the presence of HLM for 60 min. Level of intact mitragynine decreases over time due to microsomal disintegration. (A). In the presence of NADPH in HLM, concentration of mitragynine decreases with formation of 7OH, MP; 7OH being the major metabolite. (B) In the absence of NADPH in HLM, the depletion of mitragynine was inhibited with minimal formation of 7OH and MP, while formation of 3DM is less effected. (C) In the presence of NADPH in S9, the fraction concentration of mitragynine decreases with formation of 7OH, MP, and 3DM; 7OH being the major metabolite. (D) A decrease in the formation metabolites from mitragynine including **7OH** and **MP** was observed in S9 fraction in the absence of NADPH, while 3DM levels were less effected.

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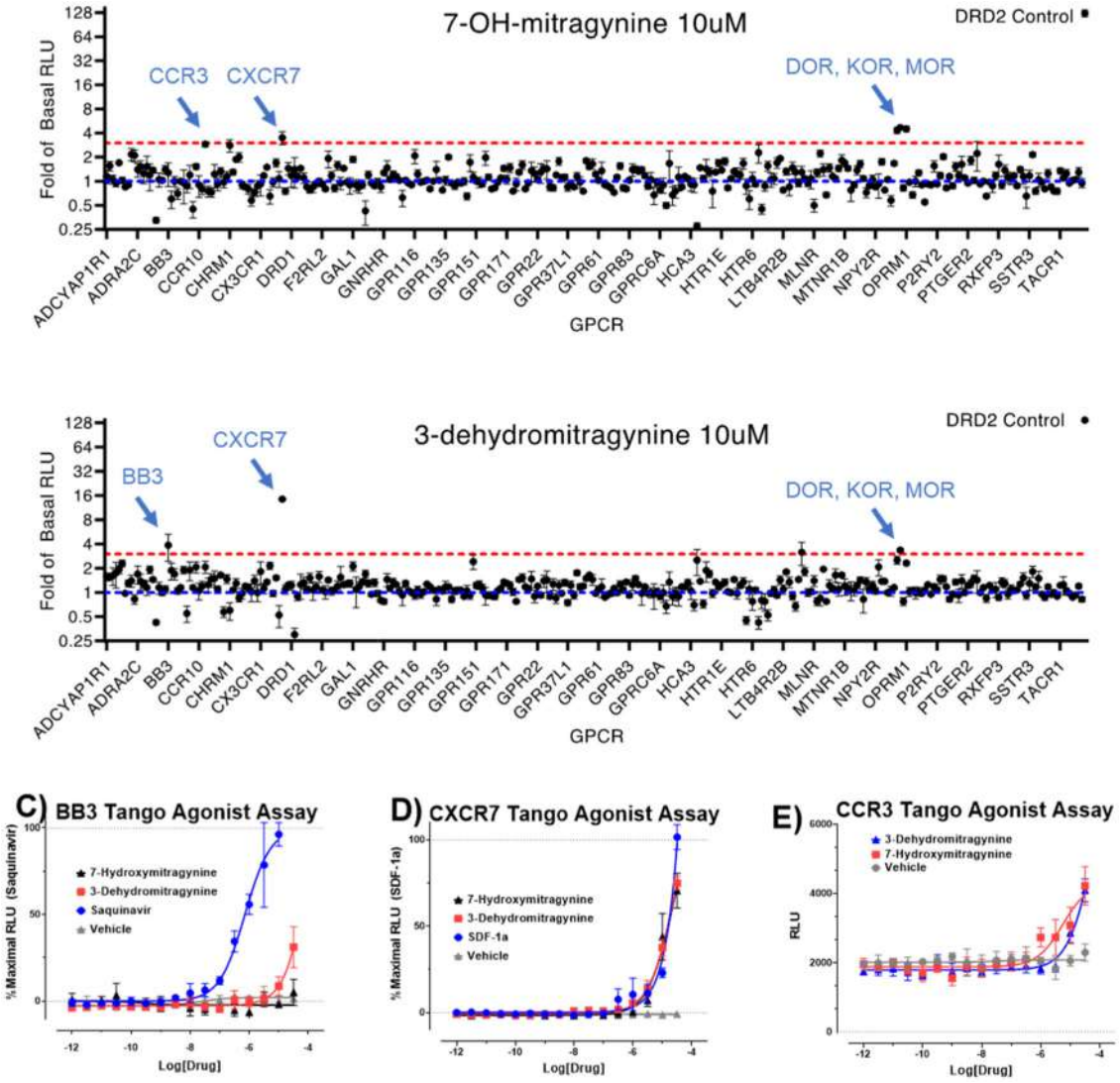


Figure 6.

GPRome screening of **7OH** and **3DM**: 7OH and 3DM were screened against 318 non-olfactory GPCRs for agonism in the β-arrestin2 recruitment assay PRESTO-Tango. Each point shows change in the luminescence signal as fold-of-basal at a given GPCR with (A) 10 μM 7OH and (B) 10 μM 3DM. For clarity, GPCRs are listed alphabetically along the X-axis, and every 10th receptor is labeled. A full list of receptors is reported previously (Kroeze *et al.* PRESTO-Tango Nat. Struc. Mol. Bio. 2015[54]). Both 7OH and 3DM induce increases in signal >3-fold of basal at opioid receptors. Non-opioid receptor screening hits (C) BB3, (D) CXCR7, and (E) CCR3 were tested against a concentration–response of 3DM, 7OH, and reference (if available) using PRESTO-Tango. Experiments were performed in triplicate and independently replicated three times.

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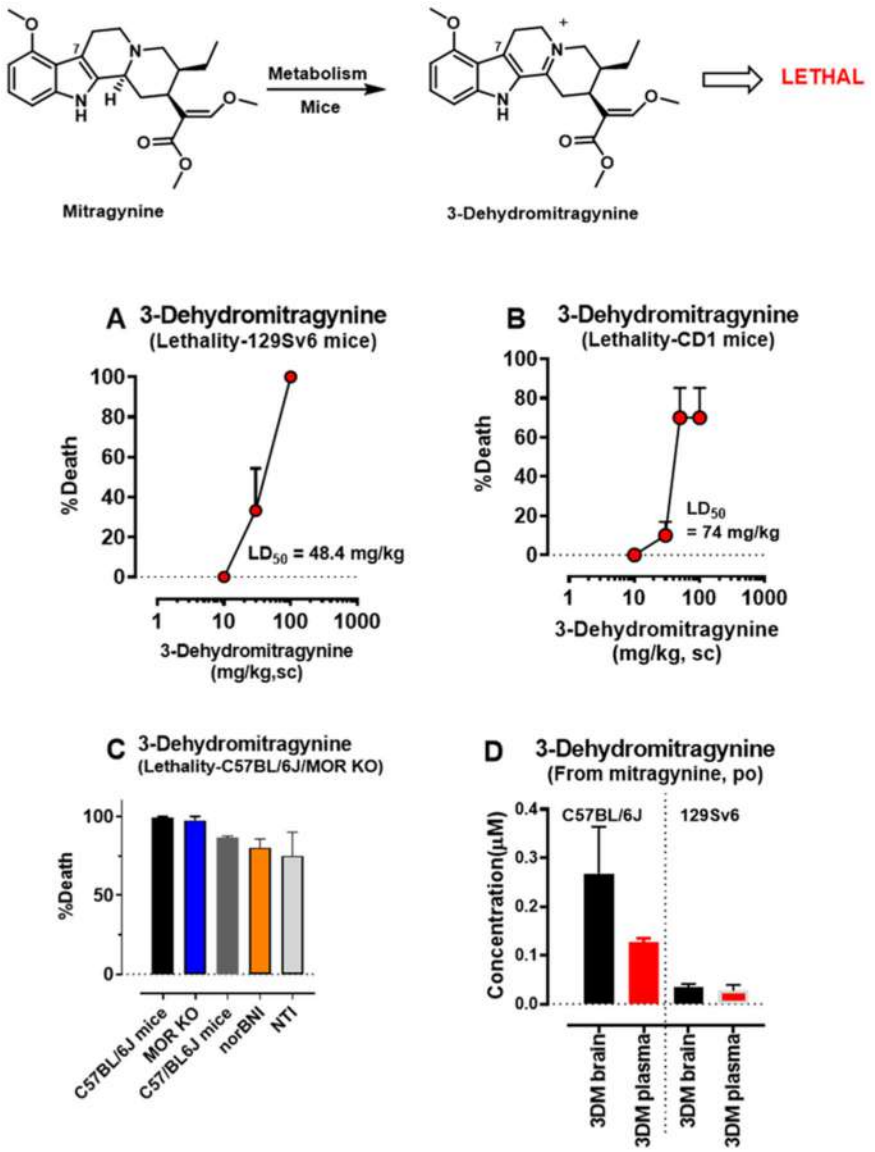
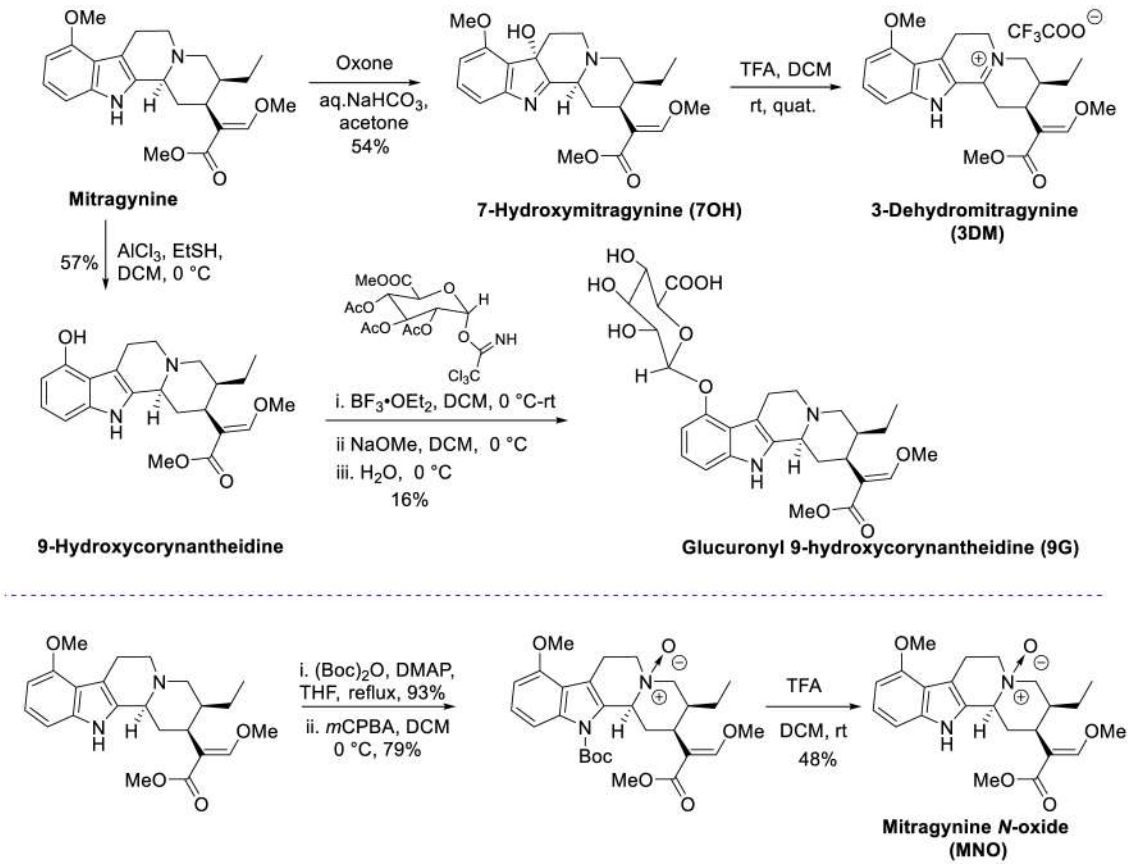


Figure 7.

Detection of metabolic product 3-dehydromitragynine in brain and plasma in mice and its toxicity. Groups of mice (n = 6 per dose) were given different doses of sc 3DM to 129Sv6 mice (**A**) or CD1 mice (B) and tested for lethality. Similarly, C57BL/6J mice (n = 4) at 50 mg/kg, sc and MOR KO mice (n = 4) (C) at 50 mg/kg, sc and C57BL/6J mice pretreated with norBNI (10 mg/kg, sc; n = 15) for 24 h and NTI (20 mg/kg, sc; n = 12) for 15 min were tested for lethality. While 3-dehydromitragynine was not an antinociceptive agent, it was found to be toxic with a LD_{50} of 48.4 (20.85–129.8) mg/kg in 129Sv6 (A) and 74 (48.08–120.5) mg/kg in CD1 mice (B). It caused 100% lethality at 50 mg/kg, sc in C57BL/6J mice (C) and at 50 mg/kg, sc in MOR KO mice, suggesting that the toxicity is not MOR-mediated. Similarly, in the presence of KOR antagonist (nor-BNI) and DOR antagonist (NTI), the lethality was still observed. Results suggest lethality is opioid receptor-independent. The metabolite 3DM was detected at 0.035 ± 0.01 and 0.09 ± 0.025 μM in

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the brain of 129Sv6 and C57BL/6J mice when mice were dosed at ED_{80} po antinociceptive doses of mitragynine (D). Similarly, about 0.0275 ± 0.01 and 0.037 ± 0.01 μM were seen in plasma of 129Sv6 and C57BL/6J mice (D).



Scheme 1.

Synthesis of Mitragynine Metabolites and Mitragynine *N*-Oxide

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Table 1.

Receptor Affinities, [^{35}S]GTPγS Functional Assays in Transfected Cell Lines, and Subcutaneous Antinociception in Mice

	affinity (K_i nM)[a]			[^{35}S]GTPγS functional assays[b]							analgesia[c]
	mMOR	mKOR	mDOR	mMOR			mKOR			mDOR	
compound				EC_{50}	$\%E_{max}$	IC_{50}	EC_{50}	$\%E_{max}$	IC_{50}	IC_{50}	sc, mg/Kg, (CI)
mitragynine	230.4 ± 47.2	231.1 ± 21.2	1011.2 ± 48.9	203.3 ± 13.19	63 ± 5		no activity		>4 μM	>4 μM	57.3 (44.6–73.5)
paynanthine	665.7 ± 82.9	883.33 ± 293.72	3395 ± 962	9384 ± 801	54 ± 1		3586.3 ± 2621.4	57 ± 2		>10 μM	>100
speciogynine	578.23 ± 63.46	2198 ± 764	7947 ± 823	6225.7 ± 3288	43 ± 0			no activity			>100
speciociliatine	78.6 ± 10.9	649 ± 168.8	1154.3 ± 200.1	72.7 ± 15	45 ± 1	10851 ± 2758	1355 ± 95.1	58 ± 2		12562.7 ± 1637	>100
7-hydroxymitragynine	36.62 ± 4.12	131.67 ± 6.97	90.64 ± 8.49	52.99 ± 4.02	77 ± 5		no activity		2524 ± 551	691 ± 434	0.2 (0.07–0.6)
mitragynine pseudoindoxyl	0.75 ± 0.18	24.23 ± 0.94	3.03 ± 1.32	1.66 ± 0.10	84 ± 5		no activity		30.9 ± 3.07	61.23 ± 6.34	0.3 (0.2—0.4)
9-hydroxy corynanthedine	216.2 ± 45.6	1177.2 ± 282.2	1882.4 ± 76	422.4 ± 134.5	62 ± 2		857.8 ± 33.5	87 ± 3		no activity	>100
9-glucuronoyl mitragynine	1674.7 ± 42.1	1274.3 ± 379.9	3239.3 ± 345.1	5778 ± 1357	102 ± 15		1143.5 ± 287.4	98 ± 2		no activity	>30
3-dehydromitragynine	34.1 ± 5.9	124.7 ± 14.9	49.1 ± 2.3	67.5 ± 8.3			166.1 ± 25.8	64 ± 9		1096 ± 64.7	>100
mitragynine *N*-oxide	6902 ± 493	7180 ± 822	4143 ± 232								
DAMGO[d]	3.34 ± 0.43			1.39 ± 0.67							
U50, 488H[d]		0.73 ± 0.32					9.5 ± 1.8				
DPDPE[d]			1.39 ± 0.67								
NTI[d]			0.46 ± 0.32							0.72 ±0.11	
morphine[d]	4.6 ± 1.8										0.68 (0.47, 0.99)

[a]Competition binding studies were performed for indicated compounds against ^{125}IBNtxA (0.1 nM) in membranes from CHO cells stably expressing cloned mouse opioid receptors. K_i values were calculated from the IC_{50} values and represent the means ± SEM of at least three independent replications.

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[b] Efficacy data were determined using an agonist-induced stimulation of [^{35}S]GTPγS binding assay. Efficacy is presented in EC_{50} (nM) and percent maximal stimulation (E_{max}) relative to standard agonists DAMGO (mMOR), DPDPE (mDOR), or U50, 488H (mKOR) at 1000 nM. To determine the antagonist properties of a compound, membranes were incubated with 100 nM of the appropriate agonist by varying its concentrations. Results are presented as nM ± SEM from three independent experiments performed as triplicate.

[c] Cumulative dose–response curves were carried out on **129Sv6** mice (n = 10) using radiant heat tail flick assays with drugs at the indicated doses (sc), and antinociception was measured at 15 min (at peak effect). Results from two independent experiments are shown as mean (95% Cl).

[d] Literature binding values.[7] Binding and [^{35}S]GTPγS data for mitragynine, 7OH, and MP are from Váradi *et al.*[7]

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Research

Biosynthesis of kratom opioids

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Summary

- Mitragynine, an analgesic alkaloid from the plant *Mitragyna speciosa* (kratom), offers a safer alternative to clinical opioids such as morphine, owing to its more favorable side effect profile. Although kratom has been traditionally used for stimulation and pain management in Southeast Asia, the mitragynine biosynthesis pathway has remained elusive.
- We embarked on a search for mitragynine biosynthetic genes from the transcriptomes of kratom and other members of the Rubiaceae family. We studied their functions *in vitro* and *in vivo*.
- Our investigations led to the identification of several reductases and an enol methyltransferase that forms a new clade within the SABATH methyltransferase family. Furthermore, we discovered a methyltransferase from *Hamelia patens* (firebush), which catalyzes the final step. With the tryptamine 4-hydroxylase from the psychedelic mushroom *Psilocybe cubensis*, we accomplished the four-step biosynthesis for mitragynine and its stereoisomer, speciogynine in both yeast and *Escherichia coli* when supplied with tryptamine and secologanin.
- Although we have yet to pinpoint the authentic hydroxylase and methyltransferase in kratom, our discovery completes the mitragynine biosynthesis. Through these breakthroughs, we achieved the microbial biosynthesis of kratom opioids for the first time. The remarkable enzyme promiscuity suggests the possibility of generating derivatives and analogs of kratom opioids in heterologous systems.

Introduction

Mitragynine from the plant *Mitragyna speciosa* (kratom) and its derivatives and semisynthetic analogs including 7-hydroxymitragynine and mitragynine pseudoindoxyl are new opioids with superior side effect profiles (Kruegel *et al.*, 2016, 2019). Comparing to the widely used opioids such as morphine, these monoterpenoid indole alkaloid (MIA)-based opioids have distinct opioid receptor agonism and signal transduction pathways, which do not recruit β-arrestin-2 that is linked with respiration depression (Takayama, 2004; Kruegel *et al.*, 2016, 2019; Váradi *et al.*, 2016; Bhowmik *et al.*, 2021). In animal models, kratom opioids did not promote self-administration (Yue *et al.*, 2018), suggesting their use to be less addictive compared with opioids responsible for the devastating overdose epidemic around the world. Recent structure activity relationship (SAR) studies have shown that the scaffold of kratom opioids is amenable to modifications and substitutions, which could further improve the safety of these drugs (Kruegel *et al.*, 2019; Bhowmik *et al.*, 2021). While kratom has been historically used as an ethnobotanical remedy for hundreds of years in Southeast Asia and the total synthesis has been reported (Kruegel *et al.*, 2016), the biosynthesis of kratom opioids from the central precursor strictosidine, from which over 3000 MIAs derive, has not been studied.

It is well known that strictosidine deglycosylation leads to numerous unstable strictosidine aglycone structures that exist in spontaneous equilibrium (Fig. 1). The aglycones are further reduced by many iminium reductases to more stable forms responsible for the early diversification of MIA skeletons (Stavrinides *et al.*, 2016; Qu *et al.*, 2017, 2018a). The mitragynine biosynthetic pathway (Fig. 1) from strictosidine likely involves four steps (red arrows in Fig. 1): (a) reduction in strictosidine aglycones to demethylcorynantheidine (**2**); (b) 17-enol methylation of **2** to form corynantheidine (**9**); (c) indole 9-hydroxylation; and (d) 9-*O*-methylation to form mitragynine (**18**).

We reasoned that the strictosidine reduction step is likely catalyzed by an iminium reductase homologous to other characterized reductases such as geissoschizine synthase (GS) and heteroyohimbine synthases (HYS) that also reduce the same substrates (Figs 1, 2a; Stavrinides *et al.*, 2016; Qu *et al.*, 2017, 2018a). The following 17-enol methylation is unique since, to our knowledge, none of the characterized *O*-methyltransferases (OMTs) catalyze nonaromatic enol methylation, suggesting the involvement of a different OMT. While it is common for MIAs to be hydroxylated at C10 or C11 on the indole moiety, the indole C9 hydroxylation is rare and has only been reported in few MIA producing species such as kratom and *Uncaria gambier* (Merlini *et al.*, 1967; Martins &

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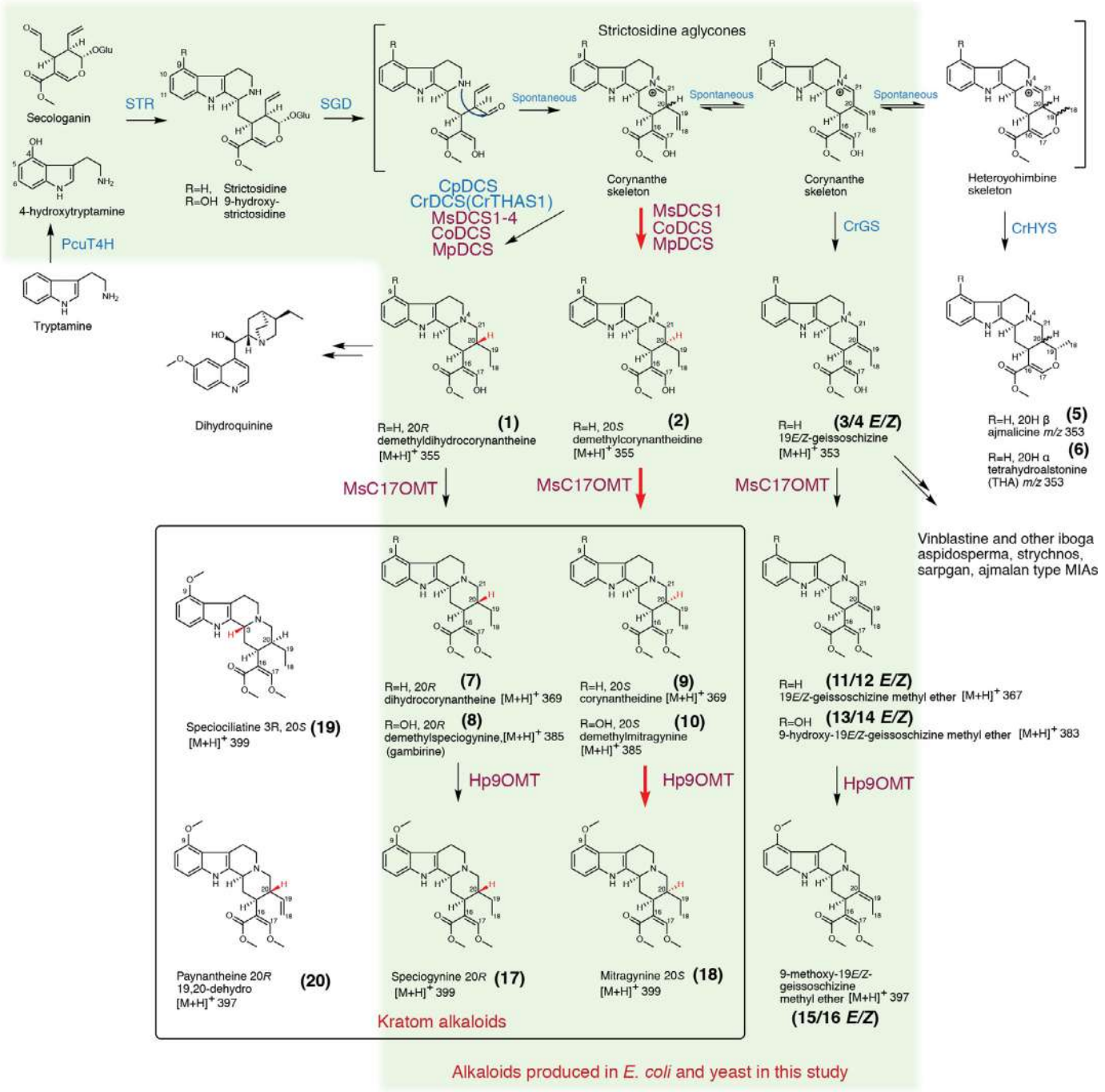


Fig. 1 Biosynthetic pathway for kratom alkaloids and other related monoterpenoid indole alkaloids. The enzymes in dark red are new enzymes identified and characterized in this study. The enzymes in blue are previously characterized enzymes, while new demethyldihydrocorynantheine synthase activity was reported for the *Catharanthus roseus* tetrahydroalstonine synthase 1 (CrTHAS1). The red arrows indicate the biosynthesis of mitragynine (20*S*). The black box shows the naturally accumulating alkaloids in kratom leaves. The alkaloids in green shade are biosynthesized in *Escherichia coli*, yeast, and *in vitro* in this study.

Nunez, 2015). Like MIA C10/C11 hydroxylations, these hydroxyls are often methylated by OMTs, which are commonly found in the cation-independent OMT group that includes many aromatic hydroxyl OMTs such as the MIA 11-hydroxytabersonine OMT (Cr11OMT or Cr16OMT, Fig. 2b; Lam *et al.*, 2007; Murata *et al.*, 2008; Lashley *et al.*, 2022).

In this study, we report the discovery and characterization of nine reductases, a C17-enol OMT, and a 9-OMT that are responsible for the biosynthesis of mitragynine and related corynanthe alkaloids in kratom and other MIA-producing plant species. Among these new enzymes, two of them were also reported in a recent study (Schotte *et al.*, 2023). While completion of the

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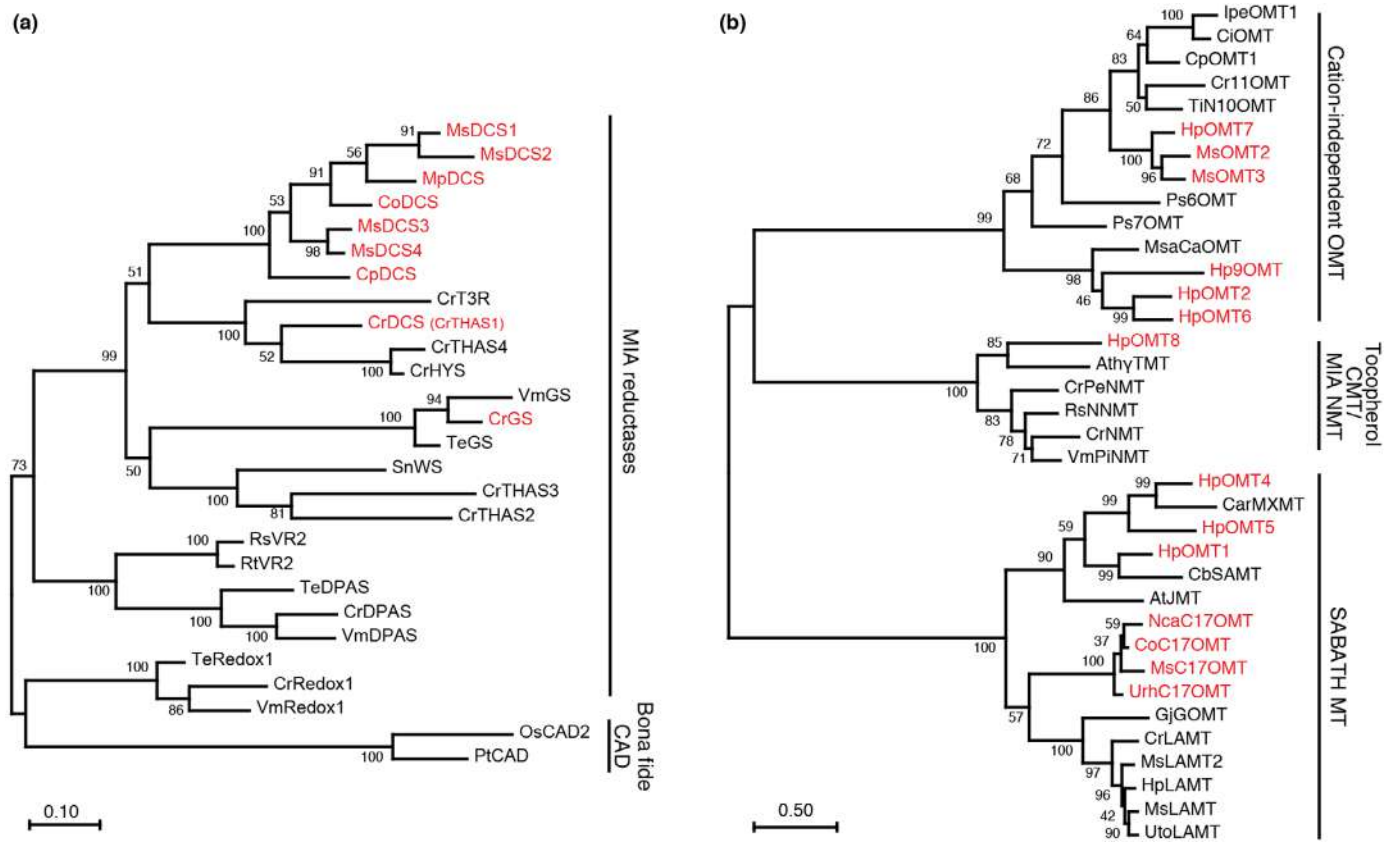


Fig. 2 Phylogenetic analysis of monoterpenoid indole alkaloid (MIA)-producing reductases (a) and methyltransferases (b) in this study. The plant species, amino acid sequences, and Genbank accession numbers are included in Supporting Information Dataset S1. The enzymes labeled in red were investigated in this study. Evolutionary analyses were conducted in MEGA11. The evolutionary history was inferred by using the Maximum Likelihood method and JTT matrix-based model. The percentage of trees in which the associated taxa clustered together is shown next to the branches (100 bootstrap replicates). Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the JTT model, and then selecting the topology with superior log likelihood value. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site (scale bar). CAD, cinnamyl alcohol dehydrogenase; DCS, demethyldihydrocorynantheine/demethylcorynantheidine synthase; DPAS, dihydroprecondylocarpine acetate synthase; GS, geissoschizine synthase; HYS, heteroyohimbine synthase; MT, methyltransferase; THAS, tetrahydroalstonine synthase; VR2, vomilenine reductase 2; WS, Wieland-Gumlich aldehyde synthase.

mitragynine pathway still requires the discovery of the corynantheidine 9-hydroxylase, this study documents the microbial biosynthesis of mitragynine and its epimer speciogynine in *Escherichia coli* using kratom enzymes when supplying the unnatural substrate 4-hydroxytryptamine, which bypasses the need of a hydroxylase. We also show the biosynthesis of mitragynine and speciogynine in *Saccharomyces cerevisiae* (baker's yeast) from tryptamine substrate by including a tryptamine 4-hydroxylase (PcuT4H) from the psychedelic mushroom *Psilocybe cubensis* (Fricke *et al.*, 2017). These results document the first microbial biosynthesis of mitragynine pharmacophore-based analgesics.

Materials and Methods

Chemicals standards

The standards for mitragynine, speciogynine, speciociliatine, paynantheine, corynantheidine, 4-hydroxytryptamine, and 5-hydroxytryptamine were purchased from Cayman Chemicals (Ann Arbor, MI, USA). The standard for 6-hydroxytryptamine was purchased from Toronto Research Chemicals (Toronto, ON, Canada), and the standard for 19*E*-geissoschizine methyl ether was purchased from AvaChem Scientific (San Antonio, TX, USA). The substrate 19*E*-geissoschizine and 19*Z*-geissoschizine was prepared and purified as described previously (Qu *et al.*, 2017). The substrate demethyldihydrocorynantheine was prepared from *in vitro* reaction (10 ml) containing 20 mM Tris–HCl pH 7.5, 1 mM NADPH, 100 μg trptamine, 100 μg secologanin, and purified recombinant proteins: CrSTR (200 μg), CrSGD (20 μg), and MsDCS4 (200 μg). The reaction took place at 30°C for 1 h, and then it was extracted with 10 ml of ethyl acetate. The evaporated extract was reconstituted in methanol and separated by thin-layer chromatography using TLC-silica gel 60 f254 (Sigma-Aldrich) with solvent ethyl acetate: methanol (9 : 1, v/v), which afforded 20 μg demethyldihydrocorynantheine.

Plant materials, crude protein isolation, and RNA/cDNA synthesis

The plants *Mitragyna speciosa* Korth., *Mitragyna parvifolia* (Roxb.) Korth, *Cephalanthus occidentalis* L., *Cinchona pubescens* Vahl, *Catharanthus roseus* (L.) G. Don, and *Hamelia patens* Jacq.

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were grown in a glasshouse at 28°C with 16 h : 8 h, light : dark photoperiod. Leaf tissues (3 g) and 0.2 g polyvinylpolypyrrolidone were ground in liquid nitrogen with mortar and pestle, which were extracted with ice-cold sample buffer (20 mM Tris–HCl pH 7.5, 100 mM NaCl, 10% (v/v) glycerol). The extracts were centrifuged at 15 000 ***g*** for 30 min, and desalted into the same sample buffer with a PD10 desalting column (Cytiva, Marlborough, MA, USA) according to the manufacture's protocol. The total proteins were desalted one more time, and the final samples were stored at −80°C. Leaf tissues (100 mg) were collected for RNA extraction using standard TRIzol RNA isolation reagent according to the manufacture's protocol (ThermoFisher Scientific, Waltham, MA, USA). The resulting RNA was used to generate cDNA using the LunaScript®RT SuperMix Kit according to the manufacture's protocol (New England Biolabs, Ipswich, MA, USA).

Cloning

For His-tagged protein purifications, MsDCS1, MsDCS2, and MpDCS were amplified using the same primer set (1/2) from respective leaf cDNA. MsDCS3 and MsDCS4 were amplified using the same primer set (3/4). CpDCS was amplified using the primer set (5/6). CoDCS was amplified using primer set (7/8). MsC17OMT (MsOMT1), MsOMT2, and MsOMT3 were amplified using primer set (9–14), respectively. HpOMT1-8 were amplified using the primer sets (15–30), respectively. The primers are listed in Supporting Information Table S1. MsOMsDCS1-4, MpDCS, CoDCS, MsC17OMT, and MsOMT2 were cloned in pET30b+ vector within BamHI/SalI sites. MsOMT3 was cloned in pET30b+ vector within SalI/NotI sites. HpOMT4 was cloned in pRSF-duet vector within SacI/NotI site. All remaining HpOMTs were cloned in pRSF-duet vector within SalI/NotI site. All these vectors were mobilized to *E. coli* BL21DE3 for expression. CpDCS were gateway-cloned into pDEST17 vector by Gateway™ BP and LR clonase™ II Enzyme mix according to the manufacturer's protocol (ThermoFisher Scientific), and mobilized to *E. coli* BL21A1 for expression. The Genbank nos. for MsDCS1-4, Ms/Co/Nca/UrhC17OMT, and Hp9OMT are OQ129427–129436, and OQ515471–515478.

For pathway assembly in *E. coli*, N-terminal truncated CrSTR was subcloned into both pACE and pDC vectors (MultiColi™ system; Geneva Biotech, Pregny-Chambésy, Switzerland) by releasing the gene with XbaI/XhoI from pET30b + dCrSTR previously described (Qu *et al.*, 2017) and ligating it within the same sites. CrSGD was subcloned into pDS vector by releasing the gene with XbaI/XhoI from pET30b + CrSGD (Qu *et al.*, 2017) and ligating it within the same sites. CrGS was subcloned into pACE vector by releasing the gene with BamHI/XhoI from pET30b + CrGS vector (Qu *et al.*, 2017) and ligating it within the same sites. The triple expression vector containing CrdSTR, CrSGD, and one of the CrGS, MsDCS1, MsDCS4 was created by Cre-recombinase (New England Biolabs) according to the manufacturer's protocols (Geneva Biotech). MsDCS1 and MsDCS4 were subcloned from respective pET30b+ vectors using BamHI/XhoI into pDC vectors. MsC17OMT was subcloned to pRSF-duet vector by releasing it with NdeI/SalI from the respective pET30b+ vector and ligating it within NdeI/XhoI sites. Into the same pRSF-duet vector, Hp9OMT was subcloned by SalI/NotI. The biosynthetic pathway was constructed by cotransforming the above vectors into *E. coli* BL21A1 cells.

For pathway assembly in yeast, codon-optimized dCrSTR (Shahsavarani *et al.*, 2022) was subcloned from pUG19 vector into pESC-His vector within NotI/ClaI site. MsDCS1 was subcloned from the respective pET30b+ vector into pESC-His-dCrSTR within the BamHI/SalI sites. The expression cassette of codon-optimized CrSGD was subcloned from pESC-His-CrSGD vector (Gao *et al.*, 2022) by the primer set (31/32), and ligated into pESC-His-dCrSTR-MsDCS1 vector within the DraIII site. The PcuT4H open reading frame was synthesized (Twist Biosciences, South San Francisco, CA, USA) and subcloned into pESC-Leu-CrCPR vector (Shahsavarani *et al.*, 2022) within the BamHI/SalI sites. MsOMT1 was subcloned from the respective pET30b+ vector into pESC-Ura vector within BamHI/SalI sites. Hp9OMT was amplified by the primer set (33/34) and cloned into pESC-Ura-MsOMT1 vector within NotI/SpeI sites. The vectors were mobilized into yeast strain BY4747 (MATα his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 YPL154c::kanMX4; Qu *et al.*, 2018a).

Recombinant protein expression and purifications

An overnight culture (2 ml) of *E. coli* BL21DE3 strains containing MsDCS1-4, MpDCS, MsC17OMT, MsOMT2, and MsOMT3 in pET30b+ vectors were used to inoculate 200 ml of LB media, which were cultured at 200 rpm and 37°C until OD600 reached 0.6–0.7. The cultures were induced with 0.1 mM IPTG at 15°C, 200 rpm overnight. For CpDCS and CoDCS, an overnight culture (2 ml) of *E. coli* BL21A1 was used to inoculate 200 ml of LB media, which were cultured at 200 rpm and 37°C until OD600 reached 0.3. The cultures were induced with 0.1% (w/v) galactose at 15°C, 200 rpm overnight. The induced cultures were sonicated in ice-cold sample buffer (20 mM Tris–HCl pH 7.5, 100 mM NaCl, 10% (v/v) glycerol) and purified using standard Ni-NTA affinity chromatography. After eluting with 250 mM imidazole in sample buffer, the purified recombinant proteins were desalted using a PD-10 desalting column (Cytiva) according to the manufacturer's protocol into the same sample buffer and stored at −80°C.

In vitro assays and kinetics

A standard *in vitro* reaction (50 µl) included 20 mM Tris–HCl pH 7.5, 1 mM NADPH, and some or all these components: 60 µM SAM, 2 µg CrSTR, 0.2 µg CrSGD, 2 µg MsDCS1-4, CoDCS, MpDCS, CrGS, or CrDCS(CrTHAS1), 2 µg MsC17OMT, and 2 µg Hp9OMT. The substrates included 1 µg secologanine and 1 µg tryptamine or its hydroxylated derivatives. The reaction was incubated at 30°C for 1 h and terminated by adding 150 µl methanol. The kinetics triplicated assays (50 µl) included 20 mM Tris–HCl pH 7.5, 100 µM SAM, 1 µg MsC17OMT, and substrate 19*E*-, or 19*Z*-geissoschizine, or demethyldihydrocorynantheine at 1.5, 2.5, 4, 12, 24, 36, 60, and

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100 μM concentrations. The kinetics assays were performed at 30°C for 2 min before they were terminated by adding 150 μl methanol to the reactions. The products were quantified using a standard curve, to generate the enzyme velocity. The kinetics parameters and saturation curves were approximated using the software PRISM 9.5.0 (GraphPad Software, LLC., Boston, MA, USA).

In vivo biotransformation

Escherichia coli strain BL21A1 containing various biosynthetic pathway genes were inoculated in 1 ml of LB media with appropriate antibiotics and 2% glucose (w/v) overnight at 37°C in a shaking incubator. The co-expression of CrSTR and CrSGD significantly impacted the *E. coli* growth and 2% glucose (w/v) sufficiently inhibited leaky protein expression in uninduced BL21A1 cells. The overnight cultures were used to inoculate 10 ml of fresh LB media (1 in 100 dilution) with appropriate antibiotics and 2% glucose (w/v), which were further grown at 37°C in a shaking incubator until OD_{600} reached 1.0. The cells were then collected by centrifugation, washed once with water, and resuspended in 10 ml of fresh LB media with appropriate antibiotics and 0.1% arabinose (w/v) to induce protein expression. The cultures were incubated in a shaking incubator at 15°C overnight. The induced cells were collected and resuspended in 2 ml Tris HCl pH 7.5 supplemented with 10% (v/v) LB broth. The substrates secologanin (10 μg), tryptamine (10 μg), and/or 4-hydroxytryptamine (10 μg) were added to the biotranformation mixture, which was incubated in a shaking incubator at 30°C for MsDCS1/4 and 15°C for CrGS for 24 h. The cultures were mixed with equal volume of methanol and used for liquid chromatography tandem mass spectrometry (LC–MS/MS) analyses.

Yeast strain BY4747 (MATα his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 YPL154c::kanMX4; Qu *et al.*, 2018a) containing various biosynthetic genes were inoculated in 1 ml of standard synthetic complete (SC) media with 2% glucose (w/v) overnight at 30°C in a shaking incubator. The cells were collected by centrifugation, washed once with water, and resuspended in 1 ml of SC media with 2% galactose (w/v) for 24 h at 30°C in a shaking incubator. The induced cells were collected and resuspended in 1 ml Tris HCl pH 7.5 with the substrates secologanin (10 μg), tryptamine (10 μg), and/or 4-hydroxytryptamine (10 μg). The biotransformation mixtures were incubated in a shaking incubator at 30°C overnight, and then equal volume of methanol was added for LC–MS/MS analyses.

LC–MS/MS

Liquid chromatography tandem mass spectrometry was performed on an Agilent Ultivo Triple Quadrupole LC–MS equipped with an Avantor® ACE® UltraCore™ SuperC18™ column (2.5 μm, 50 × 3 mm), which included the solvent systems: solvent A, methanol : acetonitrile : ammonium acetate 1 M : water at 29 : 71 : 2 : 398; solvent B, methanol : acetonitrile : ammonium acetate 1 M : water at 130 : 320 : 0.25 : 49.7. The following linear gradient (8 min, 0.6 ml min^{-1}) were used: 0 min 80% A, 20% B; 0.5 min, 80% A, 20%B; 5.5 min 1% A, 99% B; 5.8 min 1% A, 99% B; 6.5 min 80% A, 20% B; 8 min 80% A, 20% B. The photodiode array detector records from 200 to 500 nm. The MS/MS was operated with gas temperature at 300°C, gas flow of 10 l min^{-1}, capillary voltage 4 kV, fragmentor 135 V, collision energy 30 V with positive polarity. The Qualitative Analysis 10.0 software by Agilent was used for all LC analyses.

Results

Discoveries of four demethyldihydrocoryanthine/demethylcorynantheidine synthases in kratom

Previously, our group and others characterized the key reductase geissoschizine synthase (CrGS) from the plant *Catharanthus roseus* (Madagascar's periwinkle; Qu *et al.*, 2017, 2018a; Tatsis *et al.*, 2017), which reduces strictosidine aglycones to corynanthe type MIAs 19*E*- and 19*Z*-geissoschizine ($[M+H]^+$ *m/z* 353, 19,20-dehydro, **3**, **4**). The 19*E* intermediate is the precursor to major strychnos, sarpagan, iboga, and aspidosperma MIA skeletons (Fig. 1). In addition to geissoschizine, the heteroyohimbine MIAs such as ajmalicine (**5**), tetrahydroalstonine (THA, **6**), and mayumbine are also derived from reducing strictosidine aglycones by homologous reductases such as heteroyohimbine synthase (CrHYS) and tetrahydroalstonine synthase (CrTHAS1-4; Stavrinides *et al.*, 2016). Studies with *Cinchona pubescens* showed that another reductase, demethyldihydrocorynantheine synthase (CpDCS; Trenti *et al.*, 2021), reduced strictosidine aglycones to demethyldihydrocorynantheine (*m/z* 355, corynanthe type, 20*R*, **1**), the precursor to the antimalarial drug quinine (Fig. 1). By contrast, the biosynthesis of mitragynine (**18**) involves the formation of demethylcorynantheidine (**2**), the 20*S*-epimer of demethyldihydrocorynantheine (**1**). In addition to mitragynine and its intermediates, kratom and other related Rubiaceae species also make several stereo- and geometric isomers differing at C3 and C18-19-20 (Fig. 1; Takayama, 2004; Martins & Nunez, 2015). These diverse structures, such as specioginine (20*R*, **17**), speciociliatine (3*R*, **19**), and paynantheine (20*R*, 18,19-dehydro, **20**), suggest that different reductases are likely involved in their biosynthesis.

Using the sequences of CrGS, CrTHAS1, and CpDCS for comparison led to the identification and cloning of four homologous reductases from the *M. speciosa* leaf transcriptomes (http://mpgr.uga.edu) named demethyldihydrocorynantheine/demethylcorynantheidine synthase 1–4 (MsDCS1-4). While MsDCS1 was cloned by Trenti *et al.* (2021; Genbank MW456555), the gene product and its biochemical function were not characterized. The four MsDCS shared 80–95% amino acid sequence identity and were clustered together with other characterized MIA iminium reductases and short/medium chain alcohol dehydrogenases in a phylogenetic analysis (Fig. 2a).

We expressed and purified *N*-terminal His-tagged MsDCS1-4 in *E. coli* (Fig. S1) and tested their biochemical activities *in vitro*. The strictosidine aglycones (*m/z* 351) were produced by the *C. roseus* strictosidine synthase (CrSTR) and strictosidine β-glucosidase (CrSGD) using the substrate tryptamine and

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secologanin (Fig. 1), which were coupled to MsDCS *in vitro* reactions. We also compared these reactions to those with the characterized enzymes CrGS, CrTHAS1, and CpDCS. As expected from previous studies, CrGS reduced strictosidine aglycones (*m/z* 351) to produce both 19*E*- and 19*Z*-geissoschizine (*m/z* 353, **3, 4**) with THA (*m/z* 353, **6**) by-product (Fig. 3a). As expected from previous studies, CpDCS produced a major product of demethyldihydrocorynantheine (*m/z* 355, 20*R*, **1**), which is intermediate to mitragynine 20*R*-epimer speciogynine (**17**) (Fig. 3a). Several isomeric MIAs (*m/z* 353 and *m/z* 355) were also produced as minor products in these reactions. We also noticed that CrTHAS1 produced not only THA but also significant amounts of **1**, which has not been reported (Fig. 3a). Like CpDCS, MsDCS2-4 produced **1** as a major product together with THA (**6**) (*m/z* 353) and other minor (*m/z* 355) products (Fig. 3). By contrast, a major MsDCS1 product was a new *m/z* 355 MIA (Fig. 3a,b) with an elution time different from that of **1**. In the same reaction, MsDCS1 also produced 19*Z*-geissoschizine (*m/z* 353, **4**), THA (**6**), yohimbine (*m/z* 355), and several other minor *m/z* 353 MIA products when compared to authentic standards (Fig. 3b). The Electrospray Ionization tandem Mass Spectrometry (ESI-MS/MS) fragment patterns of this new *m/z* 355 MIA was nearly identical to that of **1** (Fig. 3c). We later identified this MIA as demethylcorynantheidine (20*S*, **2**), because it was enzymatically converted to corynantheidine (**9**) and mitragynine (**18**) by the two methyltransferases discovered in this study. The C17-enol methylation on DCS and GS products dramatically improved their poor chromatograms (wide peaks), which allowed us to properly identify the methylated derivatives with authentic standards (Fig. 3a, panel 2–4). The ESI-MS/MS fragment patterns of standards, known MIA products from enzymatic reactions, and MIAs produced in this study are also included in Fig. S2.

Discovery of the corynanthe C17-enol OMT and two additional demethyldihydrocoryanthine/demethylcorynantheidine synthases

With these findings, we continued to discover the methyltransferase (MT) responsible for 17-enol methylation. To our knowledge, no MTs discovered to date methylate a nonaromatic enol oxygen. The well-studied plant MTs include the SABATH (Salicylic acid carboxyl methyltransferase, Benzoic acid carboxyl MT, and Theobromine synthase) MTs that methylate carboxylic acids such as salicylic acid and jasmonic acid (Zubieta *et al.*, 2003). The loganic acid MT (LAMT) in *C. roseus* involved in MIA biosynthesis is also a member of this family (Murata *et al.*, 2008). Other characterized MTs include the aromatic *O*-methyltransferases (OMT) that methylate the aromatic hydroxyls such as flavonoids and benzylisoquinoline alkaloids, steroid *C*-methyltransferases (CMT) that methylate many triterpenoids, and the tocopherol CMTs, some of which have evolved to perform MIA *N*-methylation (Lam *et al.*, 2007; Levac *et al.*, 2008; Liscombe *et al.*, 2010; Cázares-Flores *et al.*, 2016; Fig. 2b).

Using the sequences of *C. roseus* CrLAMT (SABATH type) and 11-hydroxytabersonine OMT (Cr11OMT, also known as Cr16OMT, aromatic OMT) in MIA biosynthesis, we identified three candidate MTs from the kratom transcriptome, which included a SABATH-type MsOMT1 and two homologous (70% amino acid sequence identity) aromatic OMT type: MsOMT2 and 3. We further identified other MsOMT1 homologs from the transcriptomes of *Uncaria rhynchophylla* (cat's claw), *Nauclea cadamba* (burflower tree), and *Cephalanthus occidentalis* (buttonbush), which all produce MIAs and reside in the Naucleeae tribe of the Rubiaceae family. No MsOMT1 homologs of > 55% identity could be identified outside this tribe when searching in National Center for Biotechnology Information (NCBI) total nonredundant protein database, nor could such homologs be identified in another Rubiaceae species *Cinchona pubescens* in the tribe Cinchoneae or in MIA producing species *C. roseus*, *Vinca minor*, and *Tabernaemontana elegans* in the Apocynaceae family. The phylogenetic analysis (Fig. 2b) showed that MsOMT1 and its Naucleeae homologs form a distinct clade within the SABATH family, and they are clearly distinguished from other OMTs where MsOMT2 and 3 locate.

We expressed and purified *N*-terminal His-tagged MsOMT1-3 in *E. coli*, and tested their activities using purified 19*E*-, 19*Z*-geissoschizine (**3**, **4**), and demethyldihydrocorynantheine (20*R*, **1**). Clearly MsOMT1 was the expected C17-enol OMT, as it methylated (gain of 14 amu) all three substrates to respective geissoschizine methyl ethers (**11**, **12**) and dihydrocorynantheine (**7**). The identity of 19*E*-geissoschizine methyl ether was also supported by comparing it to a commercial standard. MsOMT1 showed typical Michaelis–Menten enzyme kinetics for all three substrates (K_M 5.3–28.7 μM, Fig. S3). MsOMT1 did not show activity with other tested MIAs including reserpic acid, yohimbic acid, yohimbine, corynanthine, 3-hydroxy-2,3-dihydrotabersonine, 11-hydroxytabersonine, vincamine and ajmaline, or simple phenolic acids including gallic acid, syringic acid, salicylic acid, benzoic acid, caffeic acid, *p*-coumaric acid, ferulic acid and trans-cinnamic acid. The results suggested that MsOMT1 likely only accepts corynanthe MIA with a free 17-enol. We therefore renamed MsOMT1 as corynanthe 17-OMT (MsC17OMT). In comparison, MsOMT2 and 3 did not accept any tested MIAs as substrates.

With MsC17OMT, we revisited the activities of MsDCS1-4 in coupled reactions involving these enzymes. With MsC17OMT, the methylated dihydrocorynantheine (**7**) formed at the expense of demethyldihydrocorynantheine (20*R*, **1**) in MsDCS2-4 reactions (Figs 3a, S4). With MsC17OMT, the reaction involving MsDCS1 clearly showed three major products: corynantheidine (20*S*, **9**) that is intermediate to mitragynine (**18**), dihydrocorynantheine (**7**), and 19*Z*-geissoschizine methyl ether (**12**) (Fig. 3a). Owing to the 17-*O*-methylation, the methylated products could no longer exist in enol-keto resonance, and they elute as sharp peaks during LC separation. The improved liquid chromatograms and availability of authentic corynantheidine, 19*E*-geissoschizine methyl ether standards allowed confident compound identification (Fig. 3a,c). The identification of corynantheidine (**9**) by MsDCS1 and MsC17OMT confirmed our previous identification of demethylcorynantheidine (**2**) when MsC17OMT was not involved. With this finding, we cloned two more homologous

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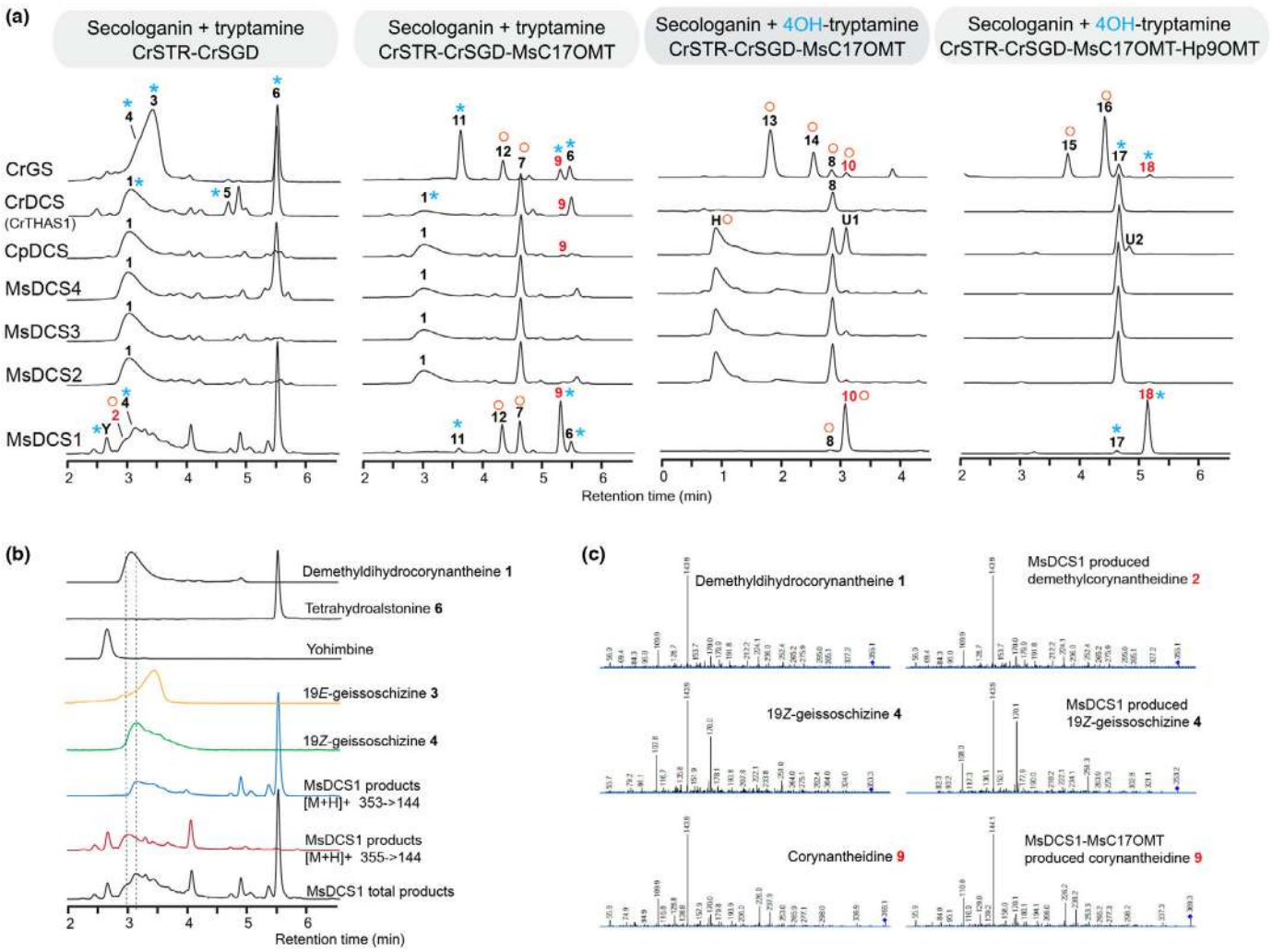


Fig. 3 Biosynthesis of mitragnine, speciogynine, 9-methoxygeissoschizine methyl ether epimers by recombinant enzymes. (a) Liquid chromatography tandem mass spectrometry (LC–MS/MS) chromatograms of coupled enzyme activities from CrSTR, CrSGD, CrGS, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3) with secologanin and tryptamine/4-hydroxytryptamine substrates. The structures of peaks with numeric numbers can be found in Fig. 1. The peaks labeled with red numbers indicate that they are intermediate to mitragynine. The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either enzymatically converted to authentic standards or enzymatically converted from authentic standards. Unless noted, all LC–MS/MS chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry $[M+H]^+$ *m/z* 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. Peak Y, yohimbine; Peak H, 9-hydroxy-demethyldihydrocorynantheine; Peak U1, unknown monoterpenoid indole alkaloids (MIAs) with $[M+H]^+$ 383; Peak U2, $[M+H]^+$ 397 (383 + 14, methylation). (b) Comparing the LC–MS/MS chromatograms of MsDCS1 products to authentic standards. (c) Comparing the MS/MS product ion fingerprints of MsDCS1 and MsC17OMT products to authentic standards.

reductases CoDCS and MpDCS from the plants *Cephalanthus occidentalis* and *Mitragyna parvifolia*. When assayed in the same coupled reactions with MsC17OMT, both CoDCS and MpDCS generated two products, dihydrocorynantheine (**7**) and corynantheidine (**9**) (Fig. S4). It is also worth noting that the previously characterized reductases CrGS, CrTHAS1, and CpDCS also produced various amounts of corynantheidine (20*S*, **9**) in these MsC17OMT-coupled assays (Fig. 3a), which could not be easily identified without 17-*O*-methylation due to the overlapping LC chromatograms of the nonmethylated MIAs. These results indicated the plasticity of strictosidine aglycone reduction by these homologous reductases, which is key to the earlier diversification of MIA skeletons.

Using 4-hydroxytrypamine substrate allowed the discovery of the 9-hydroxyl OMT from firebush

The remaining steps of mitragynine biosynthesis include 9-hydroxylation and 9-*O*-methylation. We reasoned that the 9-hydroxyl group could be artificially introduced earlier in the biosynthesis by using 4-hydroxytryptamine as a substrate based on the promiscuity of these biosynthetic enzymes. Despite the reduced activities, purified recombinant CrSTR, CrSGD, MsDCS1, and MsC17OMT converted secologanin and 4-hydroxytryptamine to demethylmitragynine (*m/z* 385, 20*S*, **10**) and demethylspeciogynine (also known as gambirine, *m/z* 385, 20*R*, **8**) epimers (Fig. 3a). We obtained similar results when we

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replaced MsDCS1 with MpDCS or CoDCS, while the assays containing MsDCS2-4, CrTHAS, and CpDCS only made demethylspeciogynine (*m/z* 385, 20*R*, **8**) as a major product (Figs 3a, S4). Interestingly, CpDCS also produced a less reduced MIA (*m/z* 383, peak U1 in Fig. 3a) that we have not identified. When we swapped MsDCS1 with GS, the reaction led to the formations of 9-hydroxygeissoschizine 19-*E*/*Z* epimers (**13**, **14**) that have not been reported in nature, as well as small amounts of both demethylmitragynine and demethylspeciogynine (Fig. 3a). When we swapped the 4-hydroxytryptamine with 5, or 6-hydroxylated tryptamine in the same reactions, we detected the formation of 10, or 11-hydroxylated isomers of demethylmitragynine and demethylspeciogynine (Fig. S5), suggesting that both the reductases and MsC17OMT tolerate indole hydroxylation at these positions.

To confirm the identification of demethylmitragynine (**10**) and demethylspeciogynine (**8**) from our *in vitro* reactions, we subjected these *in vitro* produced MIAs to the total leaf proteins isolated from kratom, buttonbush, *C. roseus*, and *Hamelia patens* (firebush), which were also supplemented with the methyl donor *S*-Adenosyl methionine (SAM). While the leaf proteins of kratom, buttonbush, or *C. roseus* did not catalyze the final 9-*O*-methylation, firebush leaf proteins converted demethylmitragynine and demethylspeciogynine to the final products mitragynine (**18**) and speciogynine (**17**) that accumulate naturally in kratom leaves, when compared to commercial standards (Fig. 4). The results confirmed the identification of other MIA intermediates and indicated that the final methylation is SAM-dependent.

After searching the firebush leaf transcriptome with sequences of CrLAMT and Cr11OMT, we identified and cloned eight OMT candidates that were named as HpOMT1-8 with decreasing expression levels (Table S2). By feeding the demethylmitragynine (**10**) and demethylspeciogynine (**8**) substrate mixtures to *E. coli* expressing each HpOMT, we discovered that only HpOMT3 methylated both substrates (Fig. S6), which was therefore renamed as Hp9OMT. We purified *N*-terminal His-tagged Hp9OMT (Fig. S1), and included it in the *in vitro* reactions containing CrSTR, CrSGD, various DCS/GS, MsC17OMT, and the substrates 4-hydroxytryptamine and secologanin. As hoped the addition of Hp9OMT to the coupled reaction containing MsDCS1 generated mitragynine and speciogynine rather than demethylmitragynine and demethylspeciogynine (Fig. 3a). Similarly, the coupled reactions generated speciogynine as the major products for MsDCS2-4, CpDCS, and CrTHAS with Hp9OMT, while both mitragynine and speciogynine were formed with CoDCS and CpDCS (Figs 3a, S4). The reaction containing CrGS and Hp9OMT also formed small amounts of mitragynine and speciogynine, but the 9-methoxygeissoschizine methyl ether 19-epimers were the dominant products (Fig. 3a).

In addition to its corynanthe MIA 9-hydroxylation activity, Hp9OMT also accepted 10-hydroxylated corynanthe MIAs as substrates. When we replaced 4-hydroxytryptamine with 5-hydroxytryptamine in the same coupled reactions, all corresponding MIAs were methylated by Hp9OMT (Fig. S5). By contrast, Hp9OMT did not methylate the 11-hydroxylated corynanthe MIAs when we used 6-hydroxytryptamine substrate. Therefore, Hp9OMT activity is restricted to C9 and C10 positions.

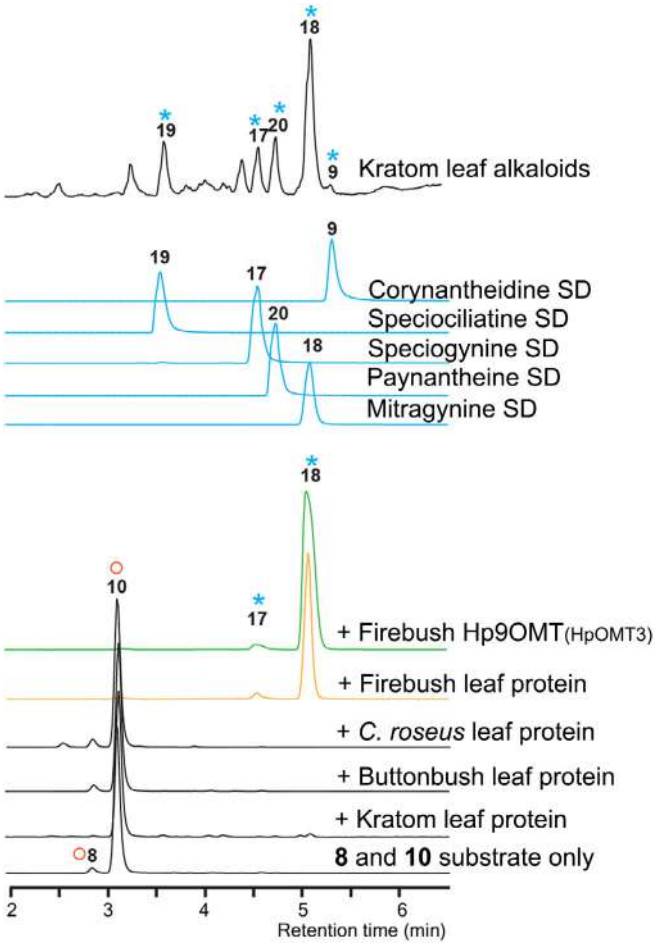


Fig. 4 Demethylmitragynine (**10**) and demethylspeciogynine (**8**) were methylated to form mitragynine (**18**) and speciogynine (**17**) by Hp9OMT (HpOMT3) and the total leaf proteins of *Hamelia patens* (firebush). The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either converted to authentic standards or enzymatically converted from authentic standards. Except for kratom leaf alkaloids that were recorded with a total ion chromatogram, all remaining liquid chromatography tandem mass spectrometry (LC–MS/MS) chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry $[M+H]^+$ *m/z* 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. Five alkaloid standards were used to identify *in vitro* enzyme products and the total kratom leaf alkaloids in these experiments. The structures can be found in Fig. 1.

In vivo biosynthesis of mitragynine in *E. coli* and yeast

The discoveries and characterizations of DCSs, C17OMT, and 9OMT enzymes allowed us to assemble the almost complete mitragynine biosynthetic pathway in both *E. coli* and *Saccharomyces cerevisiae* (baker's yeast). We co-expressed five genes CrSTR, CrSGD, MsDCS1, MsC17OMT, and HpOMT9 on plasmids in *E. coli* and fed the culture with 4-hydroxytryptamine

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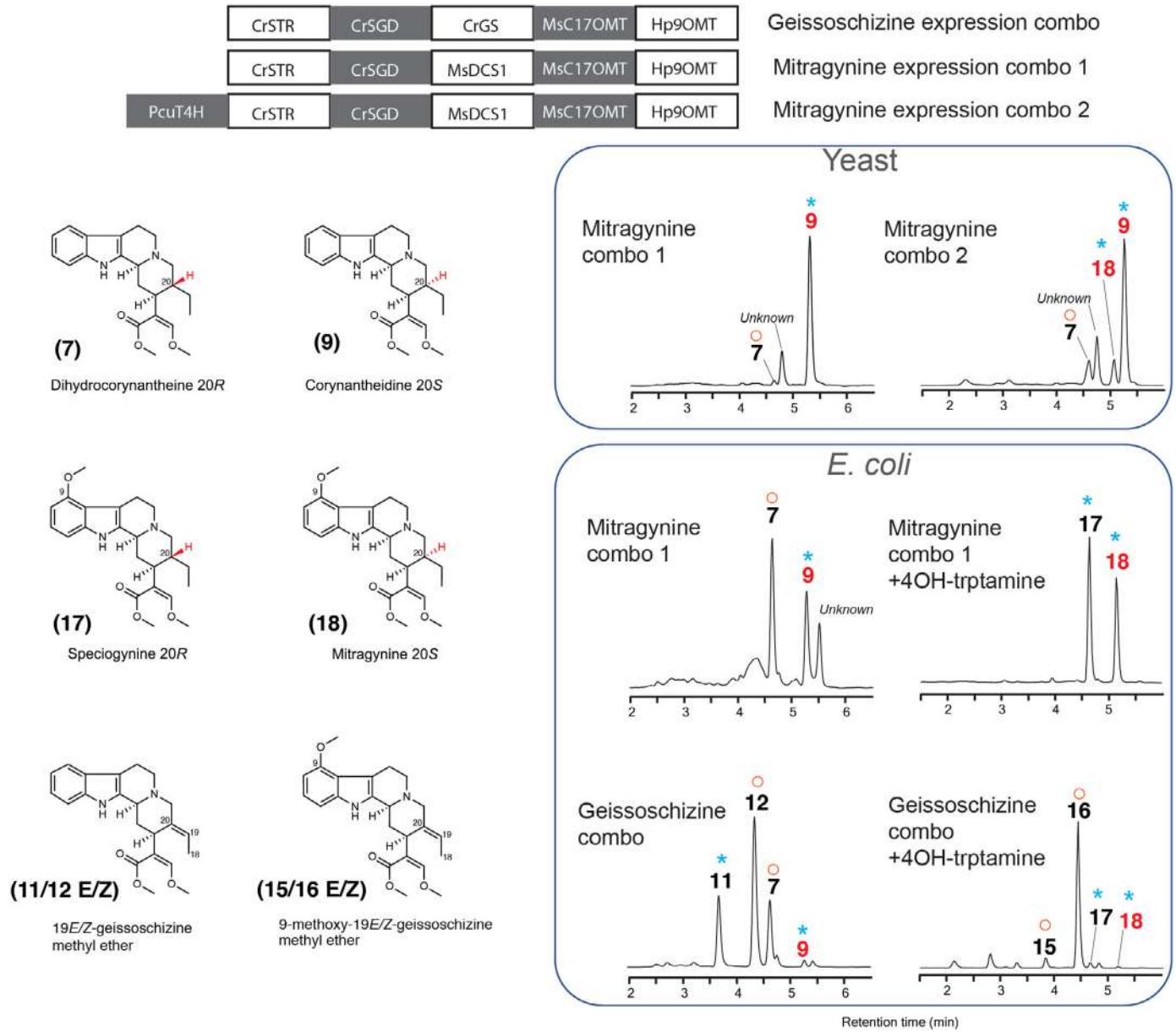


Fig. 5 Biosynthesis of kratom alkaloids mitragynine, speciogynine, and unnatural alkaloid 9-methoxylgeissoschizine methyl ether epimers in *Escherichia coli* and *Saccharomyces cerevisiae* (baker's yeast). Five enzymes CrSTR, CrSGD, MsDCS1 (or CrGS), MsC17OMT, and Hp9OMT were co-expressed in *E. coli*, while the same five enzymes with an additional PcuT4H were co-expressed in yeast. The chromatograms show the alkaloid biosynthesis in culture media after feeding secologanin and either tryptamine or 4-hydroxytryptamine to the cultures. Red numbers indicate that the monoterpenoid indole alkaloids (MIAs) are intermediate to mitragynine. The blue stars indicate that the alkaloids were confirmed with authentic standards, while the orange circles indicate that the alkaloids were either converted to authentic standards or enzymatically converted from authentic standards. All liquid chromatography tandem mass spectrometry (LC-MS/MS) chromatograms are shown by combining these ion transitions together: electrospray ionization tandem mass spectrometry *m/z* $[M+H]^+$ 353➔144, 355➔144, 367➔144, 369➔144, 383➔160, 385➔160, 397➔174, and 399➔174. The unknown peaks all have *m/z* 367.

and secologanin substrates. After overnight incubation, the culture successfully converted the substrates to both mitragynine (18.7 ng ml^{-1}) and speciogynine (24.7 ng ml^{-1}) with minimal by-products (Figs 5, S7). The result again confirmed our identifications of all these enzymes, the intermediates, and their stereochemistry. Replacing MsDCS1 with MsDCS4 led to the sole accumulation of speciogynine (32.9 ng ml^{-1}, Fig. S7), similar to the results from the *in vitro* reactions (Fig. 3a). Replacing MsDCS1 with CrGS mostly resulted in the biosynthesis of 9-methoxy-19*Z*-geissoschizine methyl ether (**16**, 16.7 ng ml^{-1}), and minimal amounts of other MIAs (Figs 5, S7). The opioid activities of 9-methoxygeissoschizine methyl ethers are not known since they are neither natural products nor synthesized for pharmacology studies. Removing Hp9OMT expression in *E. coli* instead led to the biosynthesis of the nonmethylated, 9-hydroxyl versions of respective MIAs for all three reductases (Fig. S7), and feeding tryptamine also led to the formations of the non-hydroxylated MIAs (Figs 5, S7).

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While the corynantheidine 9-hydroxylase remains to be discovered, we attempted the biosynthesis of mitragynine in yeast by using a recently characterized tryptamine 4-hydroxylase (PcuT4H, *pcuH*, genbank MF000993) from the psychedelic mushroom *Psilocybe cubensis* (Fricke *et al.*, 2017), which would supply 4-hydroxytryptamine. Yeast co-expressing the six enzymes CrSTR, CrSGD, MsDCS1, MsC17OMT, HpOMT9, and PcuT4H successfully produced mitragynine (145 ng ml^{-1}) from tryptamine and secologanin substrates, although corynantheidine accumulated as the major product (1190 ng ml^{-1}, 1 : 8 ratio, Fig. 5). Removing PcuT4H from yeast co-expression led to the biosynthesis of corynantheidine as the major product, since no tryptamine hydroxylation occurred (Fig. 5). In both cases, a minor unknown MIA (*m/z* 367) also accumulated, which might be a less reduced version of corynantheidine (*m/z* 369) or its isomer. It is also worth noting that the *in vivo* stereo-specificity of MsDCS1 when expressed in *E. coli* and yeast were different. The MsDCS1-bacteria generated equal levels of 20*S* and 20*R* MIAs, while the 20*S* MIAs (corynantheidine, mitragynine) were dominant in yeast (Fig. 5). In comparison, MsDCS1 produced mostly 20*S* stereochemistry *in vitro* (Fig. 3a,b).

Discussion

Kratom alkaloids such as mitragynine, 7-hydroxymitragynine, and derivatives appear to cause significantly fewer side effects and may have superior safety profiles (Takayama, 2004; Kruegel *et al.*, 2016, 2019; Váradi *et al.*, 2016; Bhowmik *et al.*, 2021). It is interesting that a pharmacophore based on corynanthe MIA skeleton can induce agonism with several opioid receptors, and such a pharmacophore is amenable to further modifications that may lead to even safer opioids (Bhowmik *et al.*, 2021). Structure activity relationship studies showed that the 9-hydroxyl, 17-methoxyl, and the stereochemistry of C15 and C20 are all required for opioid receptor activation. For example, switching the C20 stereochemistry from *S* (mitragynine) to *R* (speciogynine) causes significant loss of opioid activity (Kruegel *et al.*, 2016), indicating the importance of this position with respect to overall receptor interaction. Therefore, it is critical to identify biosynthetic steps responsible for these structural features. The biosynthesis of kratom opioids described here complements the recent invention of their total organic synthesis and reveals an alternative method for acquiring these promising pharmaceuticals that may be developed commercially to mitigate the opioid overdose epidemics around the world.

In this study, we identified six homologous reductases that reduce strictosidine aglycones to the corynantheidine (20*S*) and dihydrocorynantheine (20*R*) skeletons and compared their biochemical properties with three already characterized MIA reductases. The results show their instrumental roles in forming the signature chemical structures required for kratom's opioid activities. The identifications of the demethyldihydrocorynantheine/demethylcorynantheidine synthases (DCSs) were based on previous characterizations of these short/medium chain dehydrogenase type reductases from *C. roseus* and *Cinchona pubescens* (Fig. 2a; Qu *et al.*, 2015, 2017, 2018a,b; Stavrinides *et al.*, 2016; Tatsis *et al.*, 2017; Eng *et al.*, 2022). The recombinant enzymes clearly showed great plasticity in their reduction product spectra since every enzyme produced multiple products, comparable to the well-known structural diversity of the strictosidine aglycones that exist in equilibrium (Fig. 1). Such product diversity was not easily discerned due to the low abundance of some products and their overlapping liquid chromatography behaviors. Remarkably, the C-17 enol methylation led to clear distinctions of these isomeric products since the structures could no longer exist in enol-keto resonance. This allowed us to properly identify and characterize these reduced strictosidine aglycones and to identify demethylcorynantheidine (**2**) and demethyldihydrocorynantheine (**1**) end-products also generated by *C. roseus* enzymes CrGS and CrTHAS1 (Fig. 3a), which have not been described in previous studies. Based on the high production of demethyldihydrocorynantheine (**2**), we also suggest renaming CrTHAS1 as CrDCS.

In particular, the SABATH type methyltransferase MsC17OMT was able to methylate the C17-enol on a corynanthe skeleton, which was demonstrated by using the geissoschizine 19-epimers, and demethylcorynantheidine 20-epimers (Figs 3a, S4, S5). To our knowledge, this is the first example of enzymatic methylation of a nonaromatic enol in plant natural products. Other members of the SABATH methyltransferases included the founding members that methylate the carboxyl groups of plant hormones such as salicylic acid and jasmonic acid, as well as theobromine synthase that *N*-methylates 7-methylxanthine in coffee plants. The loganic acid methyltransferases (LAMTs) that performs carboxyl methylation of the MIA precursor loganic acid also belongs to this family (Fig. 2b). The identification and characterization of MsC17OMT therefore demonstrated a new and notable member in the SABATH methyltransferase family, which expands our understanding of natural product methylations.

We could only find C17OMT homologs in the plants from the Naucleeae tribe of the Rubiaceae family, such as *Uncaria rhynchophylla* (cat's claw), *Nauclea cadamba* (burflower tree), and *Cephalanthus occidentalis* (buttonbush). These plants naturally accumulate C17-methylated corynanthe MIAs and their simple derivatives such as the corresponding oxindoles and are not known to rearrange the corynanthe skeletons to complex MIA skeletons, such as strychnos, aspidosperma, or iboga types that are commonly found in MIA-producing plants in the Apocynaceae plants (Phillipson *et al.*, 1974; Martins & Nunez, 2015; Flores-Bocanegra *et al.*, 2020). *Catharanthus roseus* and *Rauvolfia serpentina*, both from the Apocynaceae family, can rearrange geissoschizine to form a strychnos (stemmadenine) or sarpagan (polyneuridine aldehyde) skeletons owing to the reactivity of the C17-enol and the cytochrome P450 monooxygenases that catalyze the intramolecular cyclization (Tatsis *et al.*, 2017; Qu *et al.*, 2018a). It is likely that the C17OMT evolved in these Naucleeae species to mask the reactive C17-enol, which is otherwise prone to nonspecific electrophilic addition to other cellular components, to reduce their cytotoxicity. However, such hypotheses require further experimental confirmation.

In a recent work (Schotte *et al.*, 2023), MsDCS1 and MsC17OMT were also discovered independently. This earlier

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study provided insights into the key residues involved in determining the corynanthe C20 stereochemistry between MsDCS1 (20*S* dominant) and another 20*R* dominant MsDCS, which has 64–68% amino acid identity to the MsDCS1-4 in this study. MsDCS1 and MsC17OMT function was also supported by their transient expression in *Nicotiana benthamiana* (tobacco) and co-infiltration of 4-methoxytryptamine and secologanin, leading to the formation of mitragynine.

While the 9-hydroxylase that completes mitragynine opioid biosynthesis awaits discovery, we further identified an alternative 9OMT from *Hamelia patens* (firebush). We demonstrated the Hp9OMT activity both *in vitro* and *in vivo* (Figs 3a, 5, S4–S7). While Hp9OMT has been useful to produce mitragynine in synthetic biology experiments, it is unlikely that Hp9OMT is involved in MIA 9-hydroxylation in firebush plant, since firebush naturally accumulates 10-hydroxylated heteroyohimbine type MIAs such as aricine and palmirine (Paniagua-Vega *et al.*, 2012) and is not known for producing either 9-hydroxylated MIAs or corynanthe type MIAs. Hp9OMT also accepted 10-hydroxycorynantheidine, its 20*R* isomer, and 10-hydroxygeissoschizine methyl ether 19-epimers, but not 11-hydroxylated versions of these MIAs as substrates (Fig. S5). Therefore, its activity is restricted to C9 and C10 on corynanthe MIAs. We hope to provide further updates on substrate specificity of Hp9OMT on heteroyohimbine MIA skeletons and the activity of other HpOMTs in future studies.

Interestingly, we could not identify a close homolog (> 50% amino acid identity) of Hp9OMT in the kratom transcriptomes (http://mpgr.uga.edu), which likely indicated that the Ms9OMT differs significantly from Hp9OMT in its primary sequence. In addition, 9OMT activity was not detected in assays using total kratom leaf protein extracts (Fig. 4). It is possible that the protein extraction method did not preserve 9OMT activity since we were also unable to detect the MsC17OMT activity from the same protein extract. It is also possible that the 9-hydroxylation and 9-*O*-methylation in kratom require different biosynthetic mechanism. Nonetheless, a bona fide Ms9OMT still needs to be discovered.

Finally, we successfully reconstituted the biosynthesis of kratom alkaloids mitragynine and speciogynine in both *E. coli* and yeast supplied with secologanin, tryptamine, and 4-hydroxytryptamine substrates. Both *in vitro* and *in vivo* studies confirmed our identifications of reaction products, which were either converted from known standards or converted enzymatically into known standards. Remarkably, the promiscuity of pathway enzymes allowed incorporation 4-hydroxytryptamine, which was generated by a fungal cytochrome P450 monooxygenase PcuT4H in yeast, into final alkaloid products. With enzyme promiscuity, we also showed the biosynthesis of unnatural MIAs such as 9-methoxygeissoschizine methyl ether epimers and 10-methoxycorynantheidine in *E. coli* (Figs S4, 5), which may also bear exciting opioid activities. It will be interesting to evaluate the opioid activity of 9-methoxygeissoschizine methyl ether epimers, since they are very similar to mitragynine except that the C20-19-18 vinyl side chain takes on a different confirmation. Based on the enzyme promiscuity demonstrated in our study, it is likely the remaining enzymes may tolerate or could be engineered to accept other indole substitutions for enzymatic production of kratom opioid derivatives for their pharmaceutical development. The identifications of the 9-hydroxylase and Ms9OMT will complete the mitragynine biosynthesis, which will also lead to their complete biosynthesis without the need for enzymes from other plant sources or organic synthesis.

Acknowledgements

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Competing interests

None declared.

Author contributions

KK, MS, VDL and YQ conceived and designed the research. KK, MS, JJOG-G, JEC, JG and YQ performed experiments and analyzed data. VDL and YQ wrote the manuscript. KK and MS contributed equally to this work.

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Data availability

The authors confirm that the data substantiating the findings of this study can be accessed within the article and its Supporting Information.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Dataset S1 Plant species, amino acid sequences, and Genbank numbers for the enzymes used in the phylogenetic study.

Fig. S1 SDS-PAGE image of His-tagged, purified recombinant proteins in this study.

Fig. S2 Electrospray Ionization tandem Mass Spectrometry fragment patterns of alkaloids involved in this study.

Fig. S3 Michaelis–Menten saturation kinetics of MsC17OMT.

Fig. S4 Liquid chromatography tandem mass spectrometry chromatograms of coupled enzyme activities from CrSTR, CrSGD, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MpDCS, CoDCS, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3), with secologanin and tryptamine/4-hydroxytryptamine substrates.

Fig. S5 Liquid chromatography tandem mass spectrometry chromatograms of the coupled enzyme activities from CrSTR, CrSGD, CrGS, CrDCS (CrTHAS1), CpDCS, MsDCS1-4, MpDCS, CoDCS, MsC17OMT (MsOMT1), and Hp9OMT (HpOMT3), with secologanin and 5-hydroxytryptamine/6-hydroxytryptamine substrates.

Fig. S6 Screening 9OMT activities from eight *Hamelia patens* HpOMT candidates *in vivo* in *Escherichia coli* with demethylmitragynine (10) and demethylspeciogynine (9) substrates.

Fig. S7 Mitragynine biosynthesis in *Escherichia coli* by co-expressing combinations of CrSTR, CrSGD, CrGS, MsDCS1, MsDCS4, MsC17OMT, and Hp9OMT.

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Table S1 Primers used in this study.

Table S2 Top expressing OMT candidates in *Hamelia patens* (firebush) leaf transcriptome.

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Synthetic and Receptor Signaling Explorations of the *Mitragyna* Alkaloids: Mitragynine as an Atypical Molecular Framework for Opioid Receptor Modulators

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Abstract

Mu-opioid receptor agonists represent mainstays of pain management. However, the therapeutic use of these agents is associated with serious side effects, including potentially lethal respiratory depression. Accordingly, there is a longstanding interest in the development of new opioid analgesics with improved therapeutic profiles. The alkaloids of the Southeast Asian plant *Mitragyna speciosa*, represented by the prototypical member mitragynine, are an unusual class of opioid receptor modulators with distinct pharmacological properties. Here we describe the first receptor-level functional characterization of mitragynine and related natural alkaloids at the mu-, kappa-, and delta-opioid receptors. These results show that mitragynine and the oxidized analog 7-hydroxymitragynine, are partial agonists of the human mu-opioid receptor and competitive antagonists at the kappa- and delta-opioid receptors. We also show that mitragynine and 7-hydroxymitragynine are G-protein-biased agonists of the mu-opioid receptor, which do not recruit β-arrestin following receptor activation. Therefore, the *Mitragyna* alkaloid scaffold represents a novel framework for the development of functionally biased opioid modulators, which may exhibit improved therapeutic profiles. Also presented is an enantioselective total synthesis of both (-)-

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Supporting Information

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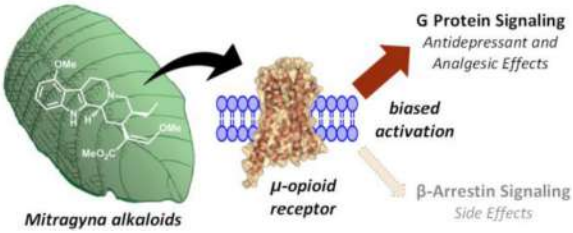
Synthetic procedures and spectral data for all compounds, copies of NMR spectra, additional binding data at rodent receptors, additional functional data at rodent and human receptors

The authors declare no competing financial interest.

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mitragynine and its unnatural enantiomer, (+)-mitragynine, employing a proline-catalyzed Mannich-Michael reaction sequence as the key transformation. Pharmacological evaluation of (+)-mitragynine revealed its much weaker opioid activity. Likewise, the intermediates and chemical transformations developed in the total synthesis allowed the elucidation of previously unexplored structure-activity relationships (SAR) within the *Mitragyna* scaffold. Molecular docking studies, in combination with the observed chemical SAR, suggest that *Mitragyna* alkaloids adopt a binding pose at the mu-opioid receptor that is distinct from that of classical opioids.

Graphical Abstract



Introduction

The opioid receptors, and in particular, the mu-opioid receptor (MOR), are among the longest and most intensely studied molecular signaling systems in the central nervous system.[1] Likewise, the prototypical small molecule agonist of these receptors, morphine, has been used by humans as an important analgesic and recreational euphoriant since ancient times. Indeed, MOR agonists, including not only morphine itself, but also a vast number of synthetic and semi-synthetic opioids, remain the gold standard of pain therapy. Unfortunately, acute MOR activation is also associated with serious side effects, including respiratory depression, constipation, sedation, nausea, and itching.[1,2] At sufficiently high doses, the evoked respiratory depression can be severe enough to cause death. Further, the pronounced euphoria produced by MOR agonists makes them major drugs of abuse. These properties have made overdose from prescription opioid analgesics a leading cause of accidental death in the United States, killing more than 18,000 people in 2014.[3,4] Another shortcoming of MOR agonists is the rapid development of tolerance to their analgesic effects. Thus, continuing escalation of dose is required to maintain an equivalent level of pain management. Similarly, when they are abused, tolerance to the euphoric effects of opioids is also rapidly developed. Thus in either case, chronic use often results in severe physical dependence on MOR agonists due to cellular- and circuit-level adaptations to continuous receptor stimulation. Accordingly, much effort has been dedicated to the development of new MOR agonists retaining potent analgesic effects, while mitigating or eliminating the deleterious side effects of the agents currently in use.[1–8]

Historically, MOR agonists have also been applied in the treatment of mood disorders, notably including major depressive disorder (MDD). Indeed, until the mid-20th century, low doses of opium itself were used to treat depression, and the so called “opium cure” was purportedly quite effective.[9] With the advent of tricyclic antidepressants (TCAs) in the 1950s however, the psychiatric use of opioids rapidly fell out of favor and has been largely

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dormant since, likely due to negative medical and societal perceptions stemming from their abuse potential. However, there have been scattered clinical reports (both case studies and small controlled trials) since the 1970s indicating the effectiveness of MOR agonists in treating depression. The endogenous opioid peptide β-endorphin, as well as a number of small molecules, have all been reported to rapidly and robustly improve the symptoms of MDD and/or anxiety disorders in the clinical setting, even in treatment resistant patients.[10–17] These results have been recapitulated in rodent models, where a variety of MOR agonists show antidepressant effects.[18–21] Most recently, we have found that the atypical antidepressant tianeptine, which has been used clinically for several decades and extensively studied in rodents and other mammalian species, is an MOR agonist, suggesting that this agent exerts its antidepressant effects via direct MOR activation.[22] When taken together, this body of work establishes a clear precedent for the effectiveness of MOR agonists in the treatment of depression and anxiety. Unfortunately, the treatment of mood disorders with conventional MOR agonists is expected to suffer from the same liabilities as their use in the context of pain management. Accordingly, our laboratories have been concerned with the exploration of structurally and pharmacologically distinct classes of MOR agonists, with the aim of developing new opioid-based treatments for mood disorders and pain, which lack the classical side effects of these agents.

It was in this context that we became interested in the psychoactive plant *Mitragyna speciosa* (Figure 1), known as “kratom” in Thailand, or “biak biak” in Malaysia, a substance which has been used by humans in Southeast Asia for centuries to treat a variety of ailments. The plant material is typically consumed as a tea or chewed directly. At low doses, kratom is primarily used for its stimulating effects. At higher doses, opioid-like effects predominate, and the plant has been used as a general analgesic and as a substitute for opium or to treat opium withdrawal symptoms. Other medicinal applications are also known, including use as a treatment for fever, cough, diarrhea, and depression. There is also a precedent for recreational use of the plant, a fact that has contributed to legal control of *Mitragyna speciosa* in both Thailand and Malaysia, but the plant remains uncontrolled outside its endemic regions.[23–27]

In light of its well documented medicinal properties, the molecular constituents of *Mitragyna speciosa* responsible for its psychoactive effects have been studied, with more than 20 unique indole alkaloids having been identified in the plant.[23–25,28] The indole alkaloid mitragynine (Figure 1) has been universally cited as the primary alkaloid constituent of *Mitragyna speciosa*, accounting for up to 66% by mass of crude alkaloid extracts.[23] The other major alkaloids in the plant have been found to include paynantheine, speciogynine, and speciociliatine (Figure 1).[23] The quantities of these major alkaloids, along with a wide variety of minor alkaloids, are considerably varied amongst different regional varieties of the plant and also depend on plant age, facts that considerably complicate the interpretation of reported psychoactive effects from the raw plant material.[23–25,28] Amongst the minor alkaloids, the oxidized derivative 7-hydroxymitragynine (7-OH)[29] (Figure 1) is of particular interest, as it has been reported to induce analgesic effects mediated through agonist activity at the mu-opioid receptor (MOR), exceeding in potency those of the prototypical opioid agonist morphine.[27,30]

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The mechanism of action of *Mitragyna* alkaloids has also been studied both *in vitro* and *in vivo*. In particular, Takayama and coworkers have accumulated a large body of evidence implicating the opioid receptor system as the primary mediator of the psychoactive effects of these alkaloids. Specifically, both mitragynine and 7-OH exhibit nanomolar binding affinities for the MOR and possess functional activity in tissue assays.[27,30] Likewise, the antinociceptive effects of mitragynine and 7-OH in several rodent models are also inhibited by naloxone.[27,30,31] Despite this evidence for the involvement of opioid receptor systems, and specifically MOR, in mediating the analgesic effects of *Mitragyna* alkaloids, there are conflicting reports in the literature. Most notably, an early study with mitragynine indicated that the behavioral effects in cats, and analgesic effects in rats, were not reversed by treatment with nalorphine, an opioid antagonist, while at the same time, mitragynine was found to produce markedly less respiratory depression than codeine.[32] Mitragynine has been shown to bind in some degree to several non-opioid CNS receptors, including alpha-2 adrenergic receptor (α2R), adenosine A_{2a}, dopamine D_2, and the serotonin receptors 5-HT_{2C} and 5-HT_7, but the strength of these affinities has not been reported.[33] Mitragynine analgesia has also been shown to be inhibited by α2R antagonist idazoxan and by the non-specific serotonin antagonist cyproheptadine.[34]

Considering the many possible confounding factors present at the tissue or system level, such *in vivo* and *ex vivo* tissue assays are not ideal for positive confirmation of a functional effect at a particular receptor. Although receptor-level functional activity studies ([^{35}S]GTPγS binding) using cloned opioid receptors have recently been reported for several synthetic oxidized analogs[35], no similar functional studies have been reported for mitragynine itself, or for other naturally occurring alkaloids in *Mitragyna speciosa*.[36] Given the unique and still unclear molecular pharmacology surrounding *Mitragyna* alkaloids, we undertook a thorough examination of mitragynine and a number of natural and novel analogs at the opioid receptors, measuring receptor activation and intracellular signaling.

Results and Discussion

Mitragynine is a Partial Agonist of Human MOR

We isolated the four major alkaloids (Figure 1) from the Thai strain of *Mitragyna speciosa* and prepared 7-OH by photochemical oxidation of mitragynine (Scheme S1). Interestingly, only trace quantities of 7-OH were observed (by mass spectrometry) in our extractions of the raw plant material, and it was not possible to isolate any measurable quantity of this derivative. Therefore, it is doubtful that this alkaloid is a universal constituent of all *Mitragyna speciosa* preparations, and is unlikely to generally account for the psychoactive properties of this plant.

We then set out to evaluate the activity of these compounds in HEK cells expressing the MOR, delta-opioid receptor (DOR), or kappa-opioid receptor (KOR), using bioluminescence resonance energy transfer (BRET) functional assays. In these G protein-dependent assays, the G_α subunit is fused with a luciferase (RLuc8) donor, and the G_γ subunit is fused with a fluorescent protein (mVenus) acceptor. Thus, on receptor activation, the G protein subunits move apart and the observed BRET signal decreases (Figure 2A).

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At the human opioid receptors (hMOR, hKOR, and hDOR), mitragynine acted as a partial agonist at hMOR (Table 1 and Figure 2B, EC_{50} = 339 ± 178 nM; maximal efficacy (E_{max}) = 34%). This result represents the first demonstration of functional opioid agonist activity at human receptors with a natural *Mitragyna* alkaloid, all prior functional studies having been conducted *in vivo* in rodents or *ex vivo* in rodent tissue. In contrast, at hKOR, mitragynine was a competitive antagonist (Table 1, IC_{50} = 8.5 ± 7.6 μM; pA_2 = 1.4 ± 0.40 μM, and Figure S2 and S4), fully inhibiting the activity of the reference agonist U-50,488. Similarly, mitragynine acted as an antagonist at hDOR, but with very low potency (Figure S3). The other major natural alkaloids paynantheine, speciogynine, and speciociliatine, showed no measurable agonist activity at any of the human opioid receptors at concentrations up to 100 μM, and only weak antagonist effects were observed (Table 1 and Figures S1, S2, and S3). The oxidized analog, 7-OH, was also characterized and found to be a potent, partial agonist at hMOR (Table 1 and Figure 2B, EC_{50} = 34.5 ± 4.5 nM; E_{max} = 47%). Further, it acted as a competitive antagonist at both hKOR (Table 1, IC_{50} = 7.9 ± 3.7 μM; pA_2 = 490 ± 131 nM, and Figures S2 and S5) and hDOR (Table 1, IC_{50} = 15.6 ± 9.1 μM, and Figure S3). The partial agonist activity of mitragynine and 7-OH at the human receptors was further confirmed in antagonist experiments, as both compounds were able to partially inhibit the response elicited by the full agonist [D-Ala^2, *N*-Me-Phe^4, Gly-ol^5]-enkephalin (DAMGO) (Figure S1).

In order to further confirm the opioid activity of the *Mitragyna* alkaloids, we performed radioligand binding studies to assess the affinity of the natural alkaloids and 7-OH for the opioid receptors (Table 2). As expected from their activity in our functional assays, both mitragynine and 7-OH exhibited significant binding affinities for hMOR. Similarly, the other natural alkaloids also bound to hMOR, consistent with their antagonist activity in the functional assays. Binding was also observed for mitragynine and 7-OH at hKOR and hDOR, again in agreement with the results of the functional assays. Further, the K_i values determined in the binding experiments were in accord with the EC_{50} and IC_{50} (after correction by Cheng-Prusoff equation) values determined in the functional experiments, or with the pA_2 values (an estimate of K_d) determined by Schild analysis. Similar binding results were also obtained at the mouse opioid receptors (Table S2). Further, our binding data with 7-OH was consistent with previous literature reports (using guinea pig brain homogenates), but the affinities determined for mitragynine at MOR and DOR were much weaker than previously described.[27,30] Given the better agreement between our *in vitro* data and the relative potencies of mitragynine and 7-OH *in vivo*, this new data may be more reliable. Taken together, the functional and binding results provide a rigorous and internally consistent assessment of the *in vitro* activity of *the Mitragyna* alkaloids at all three human opioid receptors.

Surprisingly, analogous assays using rodent receptors (mouse or rat) revealed that mitragynine exhibited no agonist activity, and it was instead found to act as a competitive antagonist at mouse MOR (mMOR) (Figure S6 and Table S1, IC_{50} = 1.1 μM; pA_2 = 807 ± 573 nM), fully inhibiting the activation induced by the reference agonist DAMGO. The other isolated natural products, paynantheine, speciogynine, and speciociliatine, showed no notable agonist or antagonist activity at any of the rodent opioid receptors. In contrast, the activity of 7-OH was similar at the rodent receptors (Figure S6 and Table S1). These

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observations complicate the interpretation of prior reports of analgesic effects elicited by mitragynine in rodent models. Further, this interspecies variation may present significant challenges in the development of compounds in this class as novel therapeutics, as activity in rodent models may not be easily translatable to man. In this regard, it is fortunate that the more potent derivative, 7-OH, retains its agonist activity at mMOR, as the oxidized scaffold represents a more promising starting point for the development of analgesics in this class. With respect to mitragynine, it is also conceivable that the in vitro assays used in our study do not exactly replicate the receptor activation and signaling in native murine brain tissue. For example, a low efficacy partial agonist in the brain may appear as an antagonist (i.e. agonist with 0% efficacy) in the cell-based assays. Thus, consideration of both interspecies and interassay variability is advised for the future development of these compounds.

Mitragyna Alkaloids Bias Intracellular Signaling Toward G Proteins

Functionally selective (biased) agonists of the opioid receptors, which preferentially activate only certain downstream signaling pathways, hold promise as analgesics or antidepressants with reduced side effects. Given this potential to separate the well established therapeutic benefits of MOR activation from negative side effects, the study of functionally selective MOR ligands is a very active area of research at present.[41–44] In particular, some evidence indicates that MOR agonists biased toward G protein signaling over β-arrestin signaling display less respiratory depression, tolerance development, and constipation, while remaining potent analgesics.[45–48] Therefore, we were interested in exploring the potential for mitragynine's unique molecular scaffold to serve as a starting point for the development of biased MOR agonists. To assess the level of β-arrestin recruitment induced by *Mitragyna* alkaloids, we employed a recently described BRET assay in transfected HEK cells, which was adapted for use with the opioid receptors.[49] In this assay, the unlabeled receptor (in this case hMOR) is transfected alongside β-arrestin-2 (arrestin-3), fused with luciferase (RLuc8) and an SH3 binding domain (Sp1), and the membrane protein GAP43, fused with a fluorescent protein (citrine) and an SH3 domain. Accordingly, when an agonist induces β-arrestin recruitment to MOR, the Sp1 domain on β-arrestin is brought into proximity with the membrane-localized SH3 domain, and binding occurs. This results in concomitant association of the luciferase donor and fluorescent protein acceptor, and a BRET signal is observed (Figure 3A). An advantage of this format is that native receptor can be used, avoiding potential confounds related to C-terminal tagging on expression or function.

As measured in this cell signaling assay, DAMGO induced robust recruitment of β-arrestin. Interestingly, both mitragynine and 7-OH were found to elicit no measurable β-arrestin binding, even in the presence of G protein-coupled receptor kinase 2 (GRK2), which typically enhances coupling to β-arrestin (Figure 3B). Due to this extremely weak signal, we were not able to quantitate the magnitude of functional selectivity for mitragynine and 7-OH. However, since both compounds exhibit significant efficacy for activation in the G protein dissociation assay (Figure 2), there is a strong qualitative bias in favor of G protein signaling in these cells. Such selectivity for G protein-dependent signaling may explain the reduced respiratory depression previously reported for mitragynine compared to codeine[32], although any such connection between upstream receptor signaling and physiological properties must be considered speculative. In any case, both mitragynine and 7-OH trigger

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downstream signaling significantly different from that induced by DAMGO. These results provide a good rationale for future experiments aimed at examining cellular signaling in relevant neuronal cells and mapping the effects of structural modifications on the level of functional selectivity.

SAR Exploration: Strategy

Having confirmed the MOR agonist activity of both mitragynine and 7-OH, we chose to explore several key structure-activity relationships (SAR) within this scaffold. In particular, we were interested in the effects of modifications at three positions: the methoxy group at position 9, the ethyl group on ring D, and the β-methoxyacrylate moiety (Figure 4). The aryl methoxy group at position 9 was of particular interest, as in the classical morphinan and benzomorphan opioid scaffolds, the change from aryl methoxy to phenol at key position 3 (e.g. codeine to morphine), results in a dramatic increase in potency. Likewise, unsubstituted phenyl analogs in these scaffolds also exhibit weak activity.[51] The importance of a phenolic moiety in many known MOR ligands is proposed to be due to formation of a water-mediated hydrogen bonding network with H297$^{6.52}$ (superscript numbers refer to Ballesteros and Weinstein's generic numbering scheme[52]), as shown in two recent X-ray crystal structures of MOR.[37,53] Accordingly, it was hypothesized that if mitragynine binds in a similar pose, then the corresponding desmethyl analog **1** (Figure 4) would be expected to exhibit significantly increased potency. We also envisioned simplification of the D ring substituents by selective deletion of the enol ether (**2**) or pendant ethyl group (**3**) (Figure 4), analogs which would probe previously unexplored SAR in this region.

Enantioselective Total Synthesis of Mitragynine and Analogs

Although 7-OH and the desmethyl derivative **1** were prepared from the isolated natural product (Scheme S1), we also desired a total synthesis of mitragynine that would allow for the construction of more extensively modified analogs. Accordingly, we envisioned a synthesis starting from 4-methoxyindole, where ring C could be installed via Bischler-Napieralski reaction to give a 3,4-dihydro-β-carboline. This intermediate would then be subjected to an enantioselective, proline-catalyzed Mannich-Michael-type cyclization with the appropriate enone to install ring D and set the stereocenter at position 3. This transformation was developed by Itoh and coworkers, and has been successfully employed in the total synthesis of several structurally similar indole alkaloids.[54–56] The resulting tetracyclic ketone would serve as a versatile intermediate for the synthesis of a variety of analogs, including mitragynine itself (Figure 5).

The required 3,4-dihydro-β-carboline (**4**) was successfully prepared in a six step sequence starting from commercially available 4-methoxyindole (21% overall yield, Scheme S2A). The required enone (**5**) was synthesized from methyl 2-ethyl-3-oxobutanoate according to previously described procedures (Scheme S2B).[57]

With the necessary building blocks in hand, we attempted the key proline-mediated step for formation of ring D. Initial experiments with crude β-carboline **4** treated with an excess of enone **5** in the presence of proline, afforded the desired ketones in moderate yield as a mixture of epimers **6** and **7**. Importantly, we found that the major diastereomer **6** was

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obtained in high enantiomeric excess (*ee*), while the minor diastereomer **7** was obtained in much lower *ee*. After optimization of the reaction conditions (it proved crucial to purify the β-carboline by chromatography), the desired ketone isomer **6*S*** could be obtained in good yield and excellent enantiomeric excess (Scheme 1A). To rationalize the stereoselectivity of this transformation, a two step mechanism (Scheme 1B, D-proline shown) can be imagined, in which a Mannich-like reaction between the proline enamine of enone **5** and the β-carboline **4**, first sets the stereocenter at position 3. This is followed by intramolecular Michael addition to close the ring. In the last step, proline controls the face of protonation during hydrolysis, thereby also influencing the stereocenter at position 19. Proline is believed to be involved in control of the stereochemistry at the 19-position, as the diastereomeric ratio obtained in the reaction varies, and is different from that observed when either ketone product (**6** or **7**) is epimerized under basic conditions. Thus, the orientation of this stereocenter is not due simply to thermodynamic equilibration of the ketone product.

With key intermediate ketone **6*S*** in hand, the next challenge of the synthesis was to effect epimerization of the pendant ethyl group into the appropriate β stereochemical configuration (as in mitragynine). This could not be accomplished by thermodynamic equilibration of the ketone with base or acid, as the desired stereochemistry positions the ethyl group in the less favorable axial configuration. As an alternative strategy, we envisioned that pseudo-allylic strain in the ene-ester product of a Horner-Wadsworth-Emmons (HWE) reaction, or in the corresponding transition state, might be used to perturb the equilibrium away from the equatorial alignment.[58] Therefore, ketone **6*S*** was treated with the carbanion formed from methyl diethylphosphonoacetate, and partial epimerization was indeed observed (Scheme 2A). The ene-ester with the desired axial ethyl group was obtained as a mixture of *E* and *Z* isomers, **8*E*** and **8*Z*** (55 mol% of total product by NMR), while the ene-ester with retention of configuration at the ethyl group, **9**, was obtained exclusively as the *E* isomer (45 mol% of total product by NMR). Unfortunately, the low reactivity of ketone **6*S*** necessitated the use of large excesses of phosphonoaceate, a requirement that led to partial transesterification of the methyl ester (products **8** and **9** contained ~25 mol% of the corresponding ethyl ester).

With the ethyl group successfully epimerized to the appropriate position, the mixed ene-esters **8*E*/*Z*** were converted to (-)-mitragynine by a sequence of several key steps, including the simultaneous reduction of the ene-ester and detosylation with magnesium, transesterification, Claisen condensation (formylation), and *O*-methylation[59] of the enol-ester intermediate **10** (Scheme 2A). The final *O*-methylation step provided both (-)-mitragynine and the isomeric analog (*Z*)-mitragynine (**11**) in modest yields. Starting from ketone **6*R*** and employing the same synthetic sequence, the unnatural enantiomer, (+)-mitragynine, was also obtained (Scheme 2B). Comparison of this material to the natural product by chiral HPLC confirmed their opposite stereochemistry, and thus, the absolute stereochemistry of the ketone products **6** and **7** obtained in the proline-catalyzed cyclization.

Synthesis of Desethylmitragynine

Considering that speciogynine, which is epimeric with mitragynine at the pendant ethyl group on ring D, was inactive as an opioid agonist *in vitro*, we reasoned that the nature and orientation of the substituent at this position was important for activity, and thus decided to

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further define the tolerance of the scaffold in this region via synthesis of the desethylmitragynine analog **3** (Figure 4). This dramatically modified analog was prepared by a synthetic sequence analogous to that applied for the synthesis of mitragynine, starting with the treatment of β-carboline **4** with methyl vinyl ketone in the presence of D-proline (Scheme S3).

Opioid Activity of Synthetic Analogs

With a variety of semi- and fully synthetic mitragynine derivatives in hand, key SAR of this molecular scaffold at the opioid receptors could be further explored. In particular, the synthesized analogs allowed investigation of the key moieties highlighted above, namely the aryl methoxy group at position 9, the β-methoxyacrylate moiety, and the ethyl group on ring D (see above, **SAR Exploration: Strategy**), as well as the importance of the absolute stereochemistry. As expected, the unnatural enantiomer (+)-mitragynine was found be a much weaker agonist at hMOR, in terms of both potency and maximal efficacy (Table 3). Surprisingly however, in contrast to naturally occurring (-)-mitragynine, it was found to be a low potency partial agonist at hKOR (EC_{50} = 9.1 ± 4.3 μM; E_{max} = 31%). Apparently, inversion of the stereochemistry in this scaffold is sufficient to switch from antagonist to agonist activity at hKOR.

In examining the specific functional groups identified above (Figure 4), we observed several important trends. For one, the switch from a methoxy group to a hydroxy group at position 9 had very little effect on activity at hMOR, with the phenolic analog **1** being of similar potency and efficacy to mitragynine itself (Table 3). This result suggests that the mitragynine scaffold is likely to adopt a unique binding pose compared to classical morphinan- and benzomorphan-derived opioids, which display a dramatic relationship between their MOR activity and an exposed phenol group. Moving to the β-methoxyacrylate moiety, removal of the enol ether, as in the total synthesis intermediate **2**, completely eliminated activity at hMOR (both agonist and antagonist). In contrast, the very close analog (*Z*)-mitragynine (**11**), was found to possess very similar activity to the natural product, indicating that stereochemical inversion of the β-methoxyacrylate moiety is tolerated (Table 3). Lastly, we turned our attention to the ethyl group on ring D. Results with speciogynine (Table 1) had already demonstrated that epimerization of this group was sufficient to switch from agonist to antagonist activity at hMOR, while also reducing binding affinity (see above, Tables 1 and 2, mitragynine vs. speciogynine). Examining the desethylmitragynine analog **3**, we found that this derivative retained agonist activity at hMOR, but with much lower potency (Table 3). Thus, the substituent at position 19 is important for controlling both efficacy (agonist vs. antagonist) and potency. The synthetic derivatives described here ((+)-mitragynine, **1, 2, 3**, and **11**) were also found to be without activity at hKOR and hDOR (agonist and antagonist mode, EC_{50} and IC_{50} both >30 μM).

Molecular Docking Reveals a Unique Binding Pose

In order to rationalize our empirical SAR observations in the mitragynine scaffold and inform future rational design, we conducted preliminary molecular docking studies. Specifically, AutoDock[60] was used to dock mitragynine and several of the analogs described

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above in the binding pocket of the agonist-bound X-ray crystal structure of mMOR (PDBID = 5C1M)[37] (Figure 6).

The top-scored binding pose of (-)-mitragynine in the orthosteric site of the receptor partially overlapped with the binding pose of the co-crystallized morphinan-derived agonist, BU72, but exhibited some dramatic differences (compare Figure 6A to Figure S7). Although (-)-mitragynine and BU72 share a polar interaction between their protonated amines and D147$^{3.32}$, which is well known to be critical for the binding of classical opioid agonists and antagonists[37,53], other important ligand-receptor contacts are significantly different. In BU72 (Figure S7), the phenol occupies a hydrophobic pocket formed by TM5 and TM6 and forms a water-mediated hydrogen bonding network with H297$^{6.52}$, an interaction that is common to opioid ligands in the morphinan class (see above for discussion, **SAR Exploration: Strategy**).[36,51,53] In contrast, our docking studies suggest that the methoxyindole domain of (-)-mitragynine is preferentially directed toward an alternative hydrophobic pocket formed by residues of TM2 and TM3 (Figure 6A), which appears to lack a means for forming an analogous hydrogen bonding network. This result is in agreement with the minimal activity difference observed *in vitro* between (-)-mitragynine and phenolic derivative **1** (Table 3). Instead, the enol ether domain (part of the β-methoxyacrylate) of mitragynine appears to be directed into the same region as the phenol of BU72 (toward TM5 and TM6). Thus, it can be speculated that this functional group participates in an analogous hydrogen bonding network with H297$^{6.52}$, consistent with the complete loss of activity when this group was removed (see compound **2**, Table 3).

The importance of the ethyl group on ring D is also explained by the predicted lowest-energy binding pose of (-)-mitragynine. In comparing the orientation of this ligand with that of BU72 (Figure 6A and Figure S7, respectively), it can be seen that the ethyl group occupies a nearly identical region as the *N*-methyl group of BU72, being in the vicinity of several hydrophobic residues (M151$^{3.36}$, Y148$^{3.33}$, and Y326$^{7.43}$). Accordingly, loss of this group, as in compound **3**, would be expected to diminish hydrophobic ligand-receptor interactions and reduce activity, as was observed (Table 3).

Docking of other mitragynine analogs described above also produced results that were in agreement with the *in vitro* activity data. For example, in accord with its similar *in vitro* activity to (-)-mitragynine, (*Z*)-mitragynine (**11**) adopted a nearly identical binding pose (Figure 6B). Likewise, the predicted low-energy binding pose of 7-OH was also similar, with all key functionalities (the acrylate, ethyl group, tertiary amine, and indole) occupying very similar positions (Figure 6C). Interestingly, the 7-position hydroxy group does not appear to make a close contact with the receptor surface, and thus, potential hydrogen bonding interactions do not seem to be involved in the significant increase in potency afforded by oxidation at position 7. Instead, the slight bend introduced to the core structure by this modification appears to be more important. Notably, while (-)-mitragynine interacts with W293$^{6.48}$ through its acrylate group, 7-OH interacts with this residue through both ethyl and acrylate groups, according to the chosen cutoff for apolar interactions (4.5 Å between heavy atoms; see Table S3). In contrast, docking of the unnatural enantiomer, (+)-mitragynine, revealed a dramatically different binding pose, reversed in its overall orientation in the

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pocket (i.e. with the acrylate projecting toward TM2 and TM3, see Figure S8A and compare with Figure 6A), in agreement with its weak activity in the functional assays.

We also docked paynantheine, speciogynine, and speciociliatine, compounds which bind to hMOR, but exhibit no agonist activity, instead serving as competitive antagonists. Considering paynantheine and speciogynine, the inverted stereochemistry at position 19 (vinyl or ethyl group) induces a significant shift of the indole moiety toward TM1 and away from TM3 (Figure 6D), which results in the formation of additional contacts with TM1 ($Y75^{1.39}$) and TM6 ($H319^{7.36}$) that are not seen in the docking of the active compounds. Moreover, the hydrophobic interactions between the indole methoxy group and $V143^{3.28}$ and $I144^{3.29}$ observed with (-)mitragynine are lost in these analogs, potentially resulting in reduced affinity for the active conformation of the receptor and corresponding antagonist activity. Consistent with the inverted stereochemistry of the core structure at position 3, docking of speciociliatine revealed a completely different binding pose with its indole group extending towards TM3 and forming a polar interaction with $Y148^{3.33}$ (Figure S8B and Table S3). Notably, unlike all other compounds reported here, the acrylate moiety in speciociliatine is in axial rather than equatorial position, pointing towards TM5 and TM6. All in all, a number of different stereochemical configurations in the *Mitragyna* alkaloid family appear to be sufficient to retain binding at hMOR, while the absolute stereochemistry found in mitragynine is required for activation of the receptor.

Conclusion

In this work, we have presented the first thorough, *in vitro* characterization of the opioid receptor pharmacology and signaling of the natural *Mitragyna* alkaloids in receptor-level functional assays. The partial agonist activity of (-)-mitragynine at MOR is likely to account for many of the effects of *Mitragyna speciosa* extracts in humans. However, the other major alkaloids isolated from the plant, paynanthiene, speciogynine, and speciociliatine, all exhibited competitive antagonist activity at this receptor. Considering that cumulatively, these secondary alkaloids accounted for an approximately equal percentage of the total alkaloid content compared to mitragynine in our extracts, the gross psychoactive effects of crude plant material are likely to represent a complex interplay of competing agonist and antagonist effects at the opioid receptors. It was also found that the potent oxidized analog 7-OH is not necessarily present in all extracts of plant material (as previously suggested), so the potential contribution of this alkaloid to the effects of *Mitragyna speciosa* is not general, unless 7-OH is formed as a metabolite.

Exploration with several synthetic mitragynine analogs allowed us to define basic SAR at each of the existing substituents, including in regions (the D ring) that were only accessible to derivatization via total synthesis. This work revealed that opioid activity in the mitragynine scaffold is quite sensitive to structural modification and suggested that the *Mitragya* alkaloids adopt a unique binding pose at MOR, distinct from that of classical opioids. This possibility was further supported by molecular docking studies at MOR. The first total synthesis of the unnatural enantiomer, (+)-mitragynine, likewise allowed us to confirm that the naturally occurring isomer is the more active opioid agonist.

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We also found that mitragynine and 7-OH display biased signaling at MOR, being selective for the G protein pathway. Thus, it is hoped that this scaffold may represent a starting point for the development of novel opioid analgesics with reduced side effects. Relatedly, the finding that, in addition to their activity at MOR, both mitragynine and 7-OH are KOR antagonists, suggests that *Mitragyna* alkaloids or their synthetic derivatives may also serve as leads for novel dual-action antidepressants. In this regard, mitragynine itself is of particular interest, as it acts as both a low efficacy MOR partial agonist, and a KOR antagonist. Both of these mechanisms have been shown to elicit antidepressant effects in animals, and in the case of MOR, humans as well (see above). However, modulators of the KOR have received considerably more attention in the psychiatry community in recent years, likely due to their apparent lack of abuse potential. Antagonists of this receptor are antidepressant in animal models and oppose a variety of stress-related responses that are believed to be mediated through the endogenous KOR agonists, dynorphins.[61–63] Further, several agents in this class have entered clinical trials for MDD, but none have yet reached the market.[64] Given the ability of mitragynine to perturb both of these opioid signaling systems in the direction consistent with mood enhancement, it is unsurprising that it has been shown to exhibit antidepressant activity in the forced swim test (a rodent model of depression).[65] Therefore, *Mitragyna* alkaloids also hold promise in the treatment of mood disorders.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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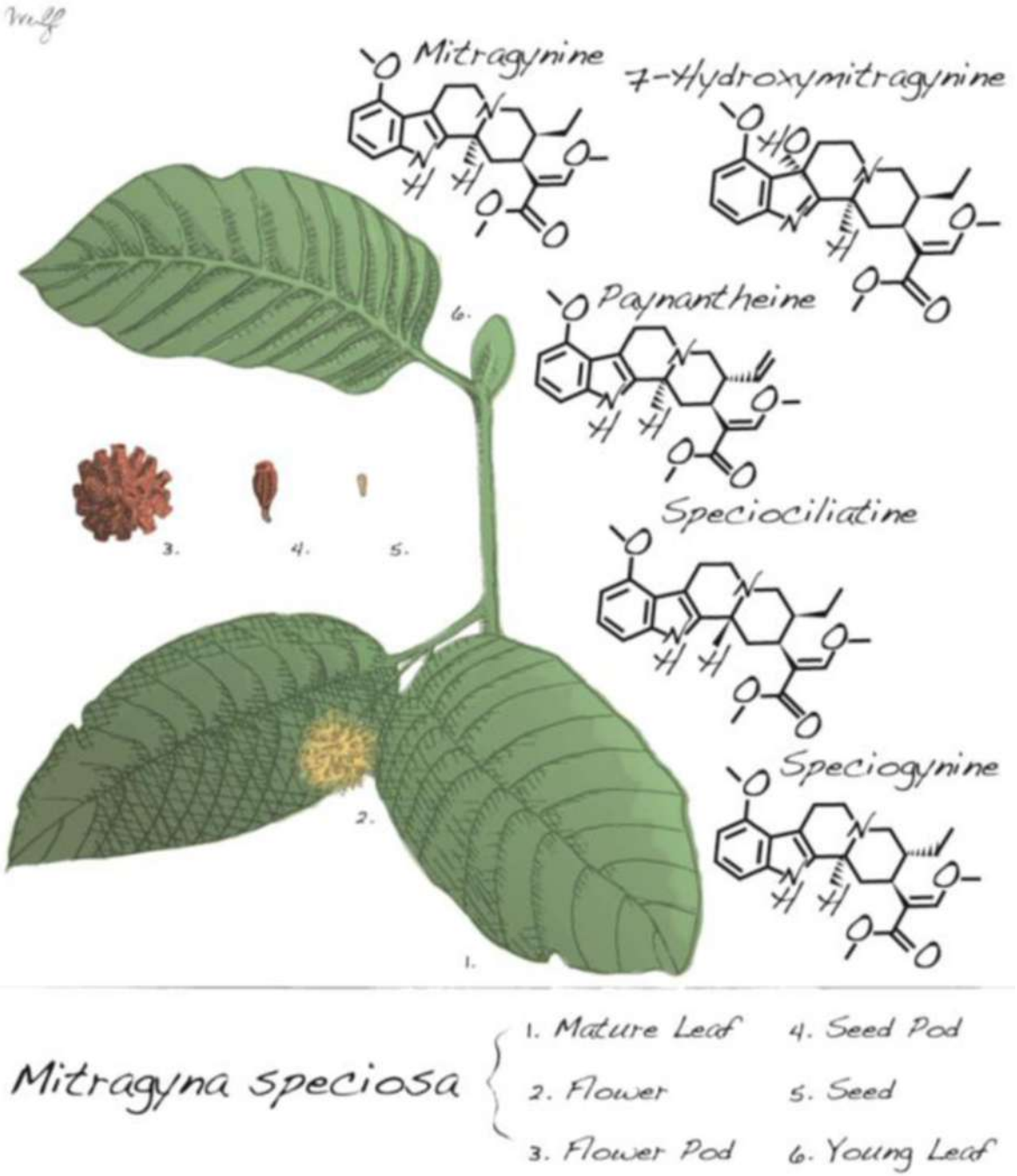


Figure 1.
Leaves and fruiting bodies of *Mitragyna speciosa* and chemical structures of notable alkaloids.

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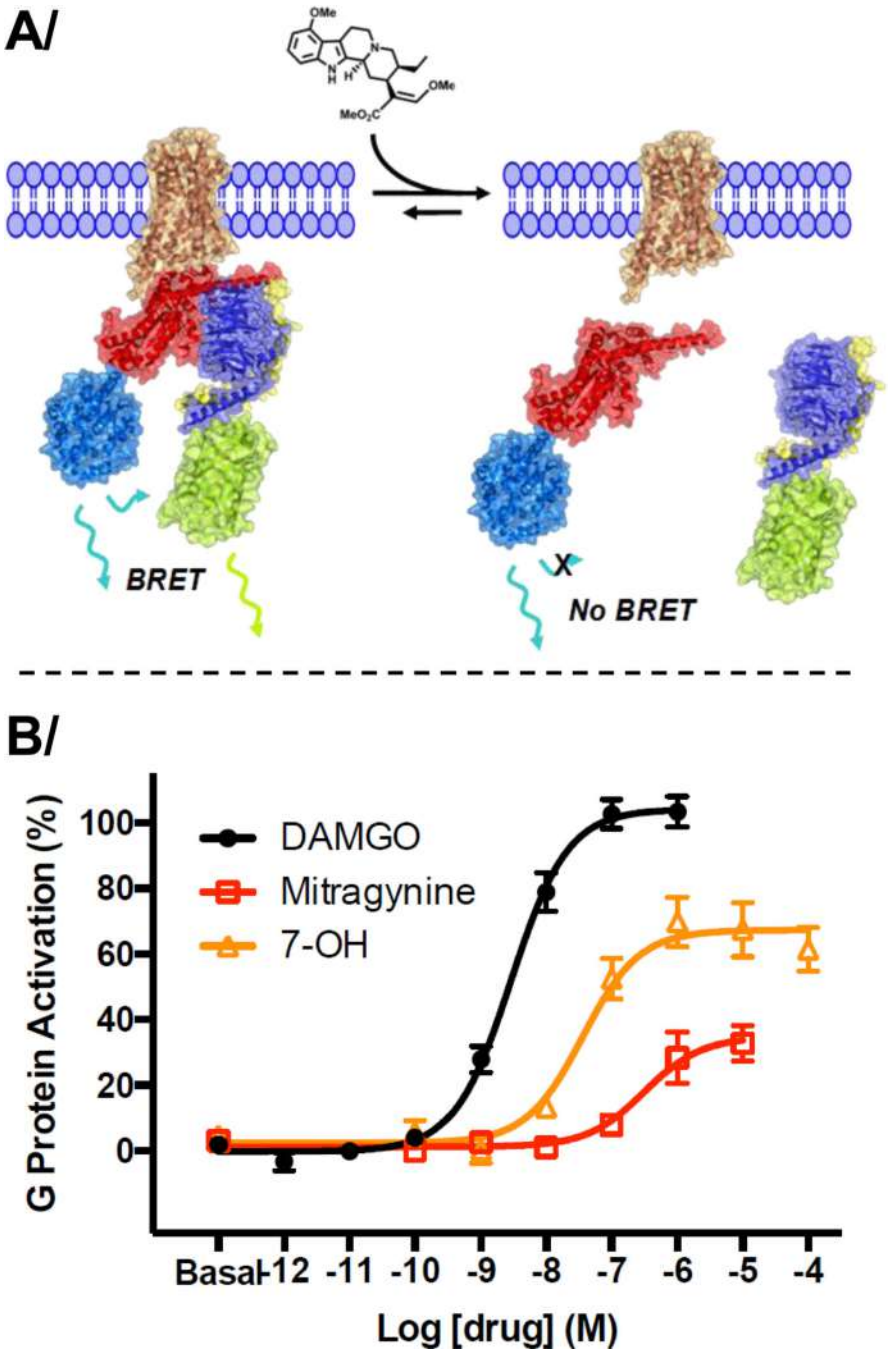


Figure 2.

Activity of mitragynine and 7-hydroxymitragyine (7-OH) at the human mu-opioid receptor (hMOR). (A) Conceptual representation of the G protein BRET assay employing atomistic cartoons derived from available x-ray crystal structures. To measure G protein activation, hMOR[37] (brown) was co-expressed with $G\alpha o_B$[38] (red) fused to RLuc8[39] (teal), $\beta 1$[38] (blue), and mVenus[40] (yellow-green) fused to $\gamma 2$[38] (yellow). (B) Agonist activity at hMOR; positive control = [D-Ala^2*N*-Me-Phe^4, Gly-ol^5]-enkephalin (DAMGO); curves represent the average of n≥3, with error bars representing ±SEM.

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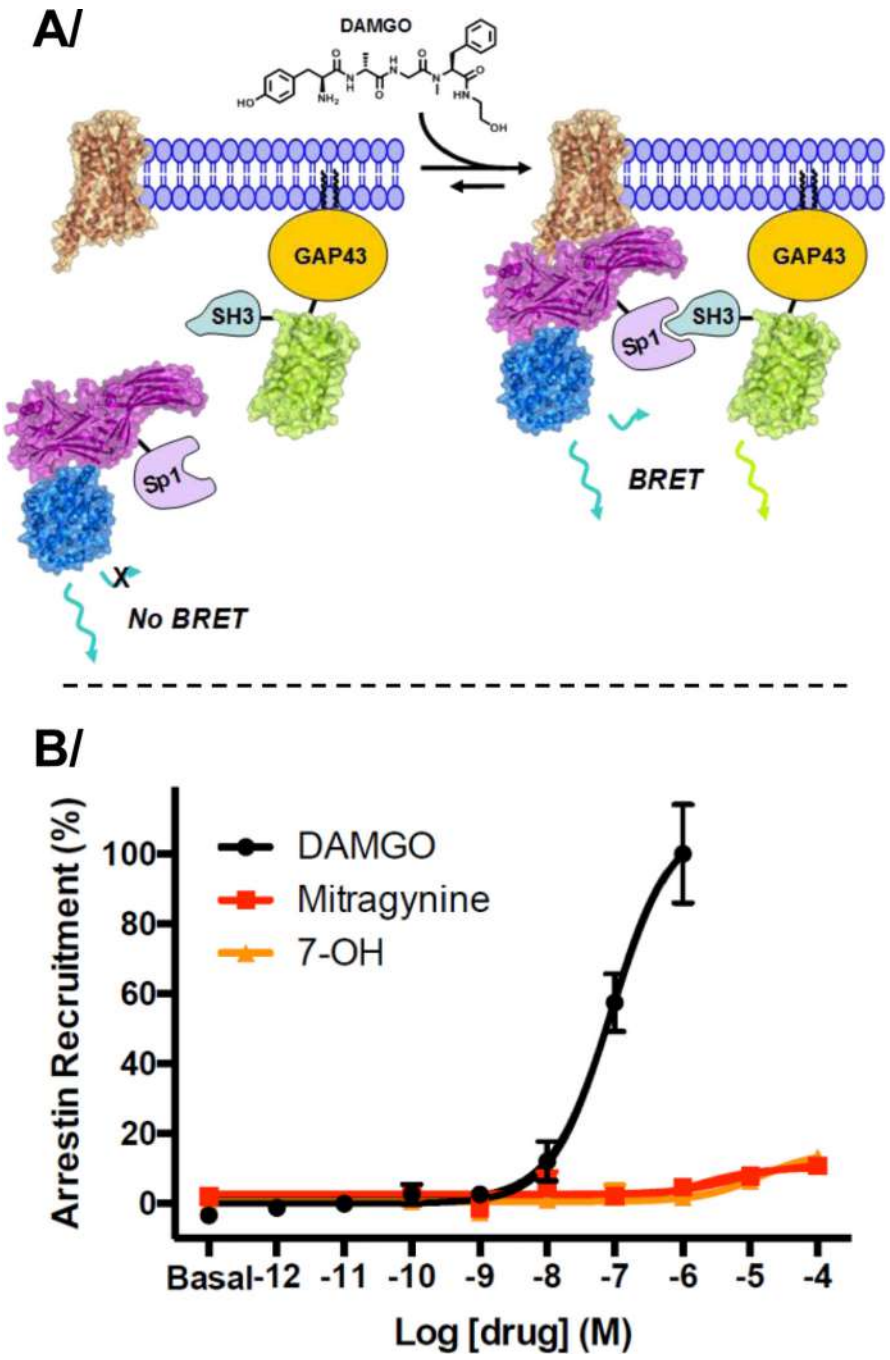


Figure 3.

Mitragynine and 7-OH do not recruit β-arrestin. (A) Conceptual representation of the β-arrestin BRET assay employing atomistic cartoons derived from available x-ray crystal structures. To measure β-arrestin recruitment, hMOR[37] (brown) was co-expressed with β-arrestin-2[50] (purple) fused to RLuc8[39] (teal) and Sp1, GAP43 fused to citrine[40] (yellow-green) and SH3, and G protein-coupled receptor kinase 2 (GRK2). On β-arrestin recruitment, association of the Sp1 and SH3 domains results in an increase in the BRET

signal between RLuc8 and citrine. (B) β-arrestin recruitment at hMOR, positive control = DAMGO; curves represent the average of n≥3, with error bars representing ±SEM.

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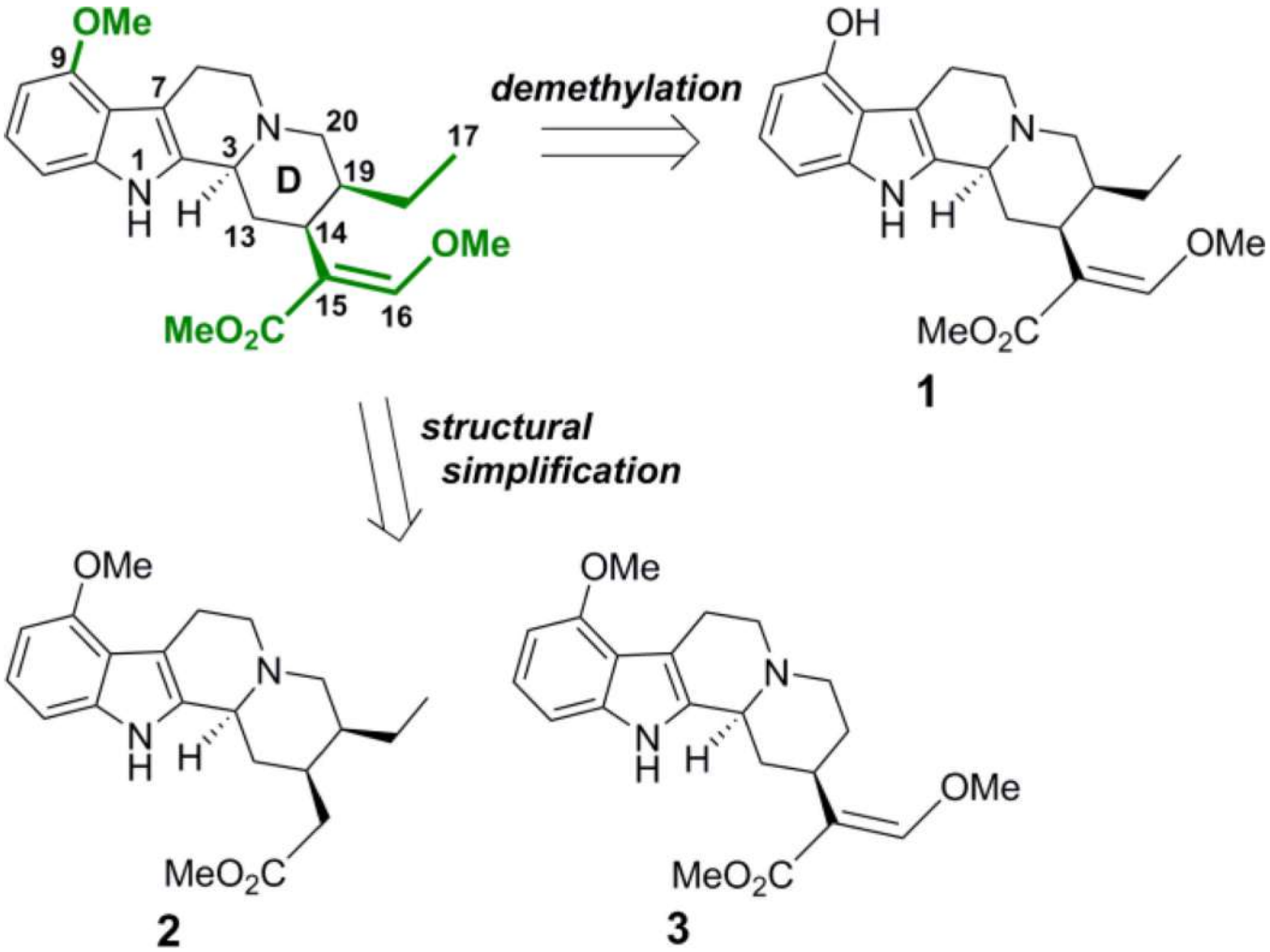


Figure 4.

Key SAR locants of *Mitragyna* alkaloids (highlighted in green) and select analogs designed to interrogate each position.

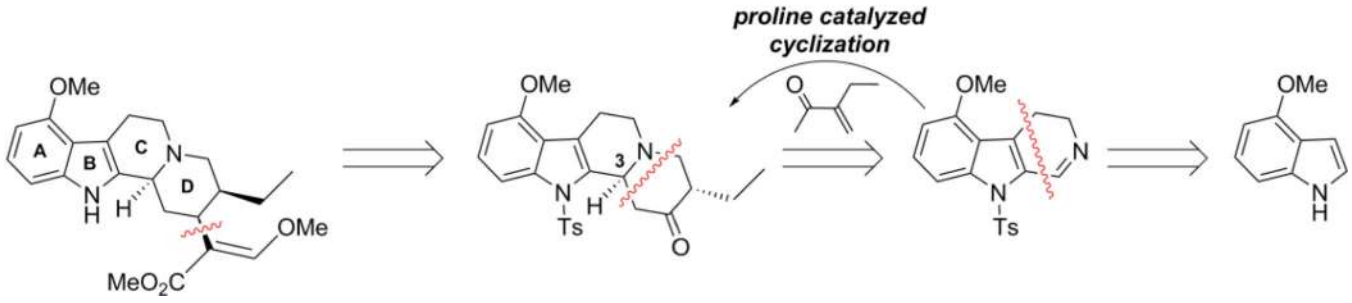


Figure 5.
Retrosynthesis of mitragynine.

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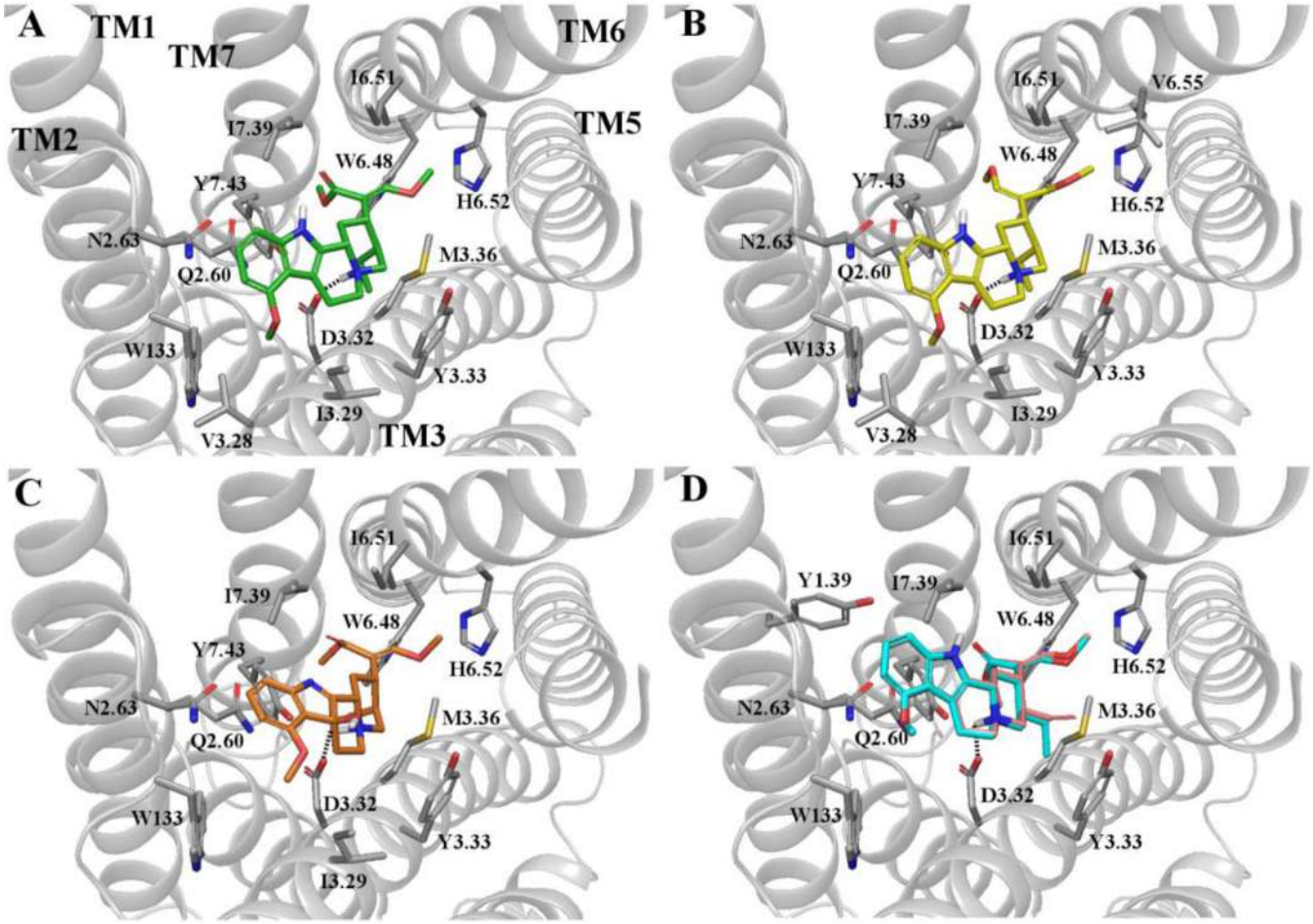
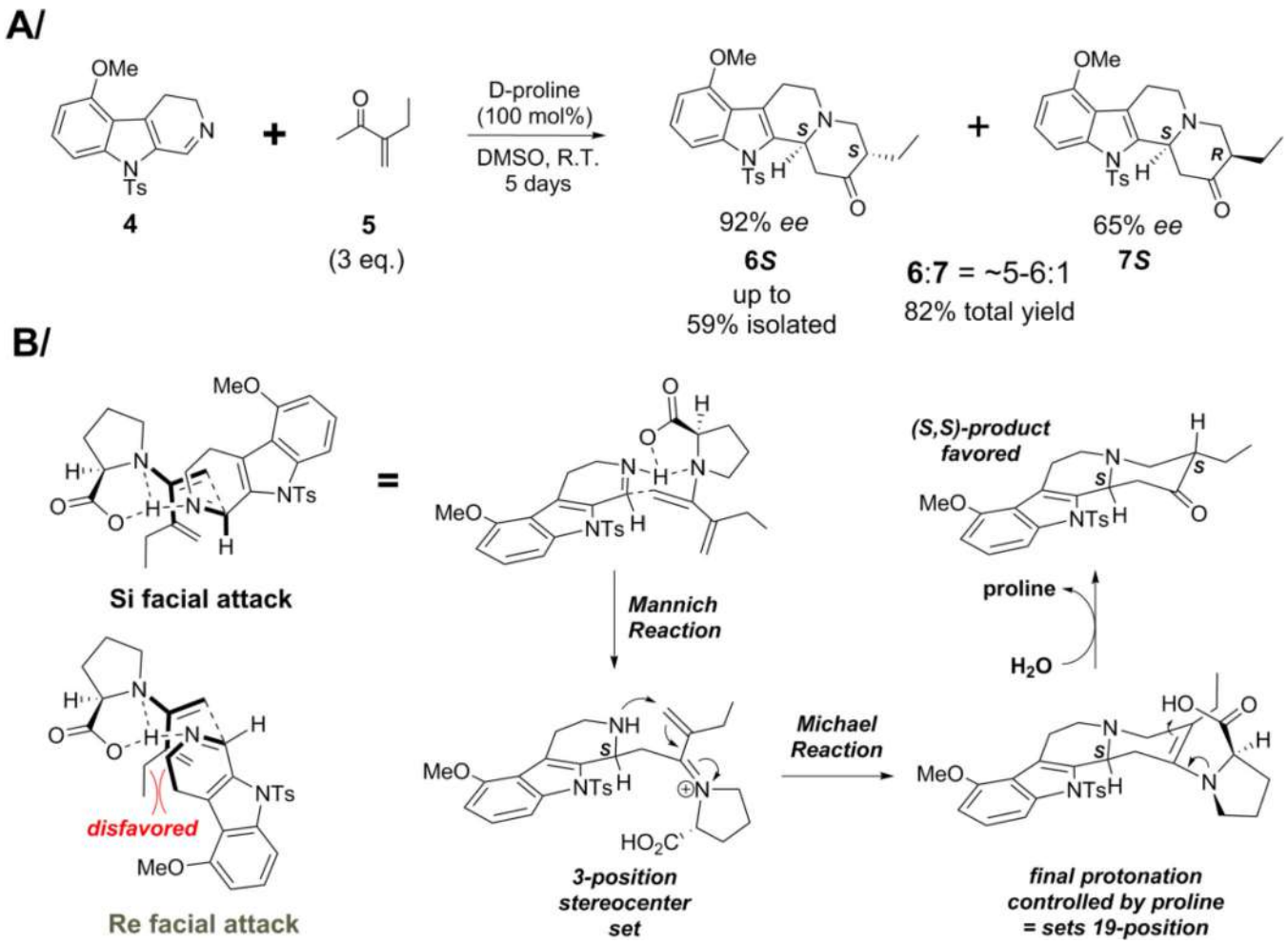


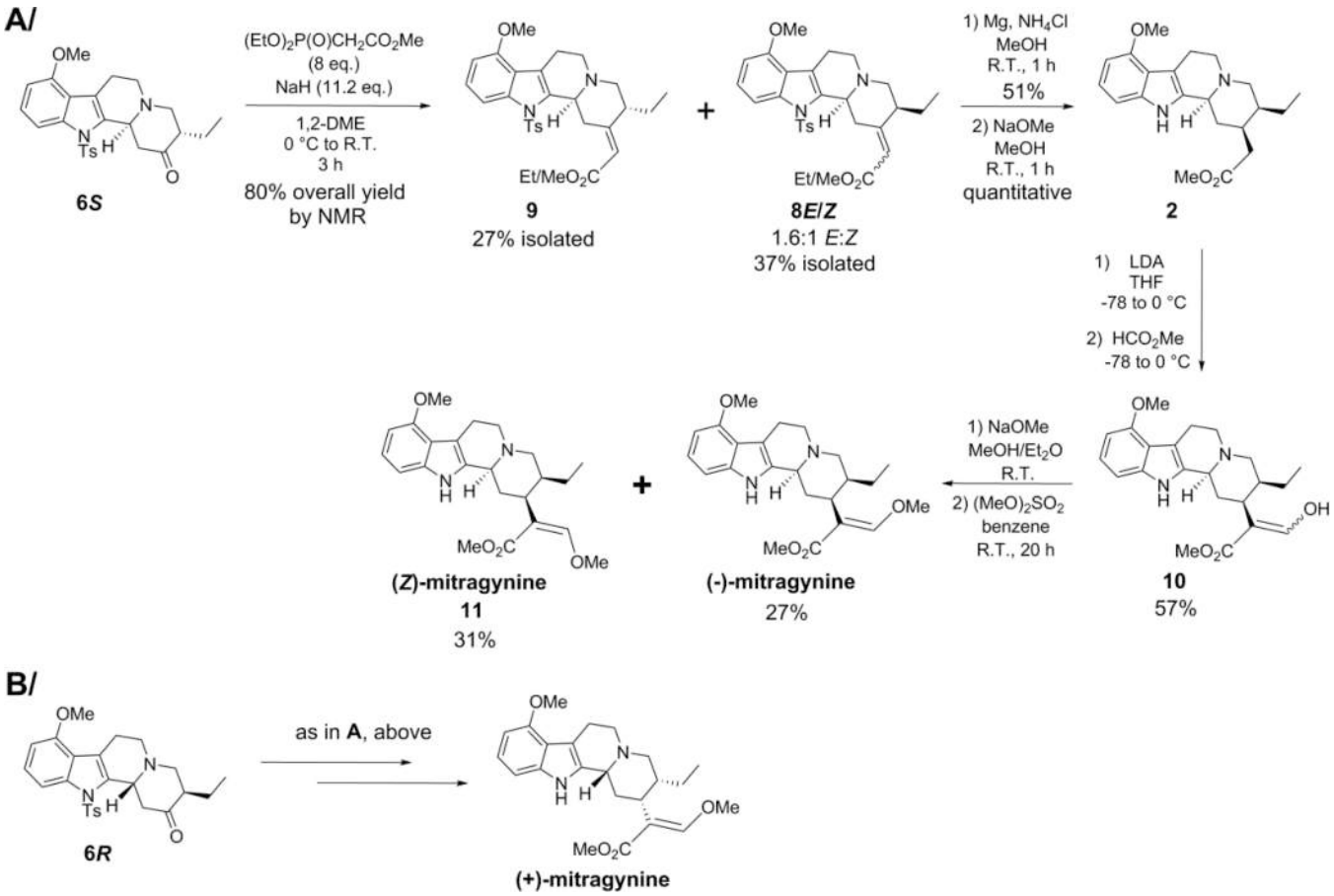
Figure 6.

Docking of (-)-mitragynine and other analogs to the active μ-opioid receptor crystal structure. Top-scoring binding poses of (A) (-)-mitragynine, (B) (Z)-mitragynine, (C) 7-hydroxmitragynine, and (D) antagonists paynantheine (pink) and speciogynine (cyan). Only residue sidechains within 4 Å of the ligand are reported. Polar interactions are shown as dotted lines. TM helices are shown in cartoon representation (in gray). ECL2 and part of TM5 have been omitted for clarity. Residues are labeled using one-letter amino acid code and Ballesteros and Weinstein's generic numbering scheme.

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Scheme 1.
Stereoselective Formation of D-ring



Scheme 2.

Total Synthesis of Mitragynine in Both Enantiomeric Forms

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Table 1

Functional Activity of *Mitragyna* Alkaloids at Human Opioid Receptors as Determined in G Protein BRET Assays.

	EC_{50} ± SEM (E_{max})[a] or [IC_{50} ± SEM (pA_2)][b] (μM)		
Compound	***hMOR***	***hKOR***	***hDOR***
mitragynine	**0.339 ± 0.178 (34%)**	***[8.5 ± 7.6 (1.4)]***	***[>10]***
paynantheine	***[2.2 ± 10]***	***[>10]***	***[>10]***
speciogynine	***[5.7 ± 2.8]***	***[>10]***	***[>10]***
speciociliatine	***[4.2 ± 1.6]***	***[>10]***	***[>10]***
7-OH	**0.0345 ± 0.0045 (47%)**	***[7.9 ± 3.7 (0.49)]***	***[>10]***

[a] Agonist activity indicated by EC_{50} values, maximal efficacy (E_{max}) relative to DAMGO in parentheses;

[b] Antagonist activity indicated by IC_{50} values for inhibition of a reference agonist, pA_2 determined from Schild analysis in parentheses; All data points represent mean ± SEM (μM) of $n \geq 3$.

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Table 2

Binding Affinities of *Mitragyna* Alkaloids at Human Opioid Receptors

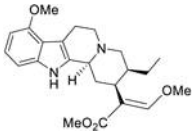
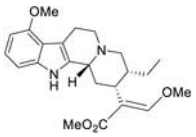
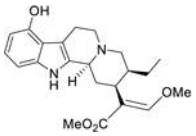
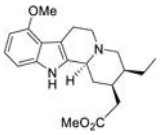
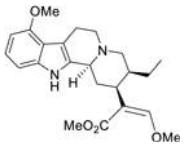
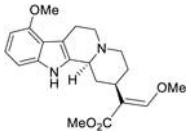
	K_i ± SEM (μM)[a]		
Compound	***hMOR***	***hKOR***	***hDOR***
mitagynine	**0.233 ± 0.048**	**0.772 ± 0.207**	**> 10**
paynantheine	**0.410 ± 0.152**	**2.56 ± 0.37**	**> 10**
speciogynine	**0.728 ± 0.061**	**3.20 ± 0.36**	**> 10**
speciociliatine	**0.560 ± 0.168**	**0.329 ± 0.112**	**> 10**
7-OH	**0.047 ± 0.018**	**0.188 ± 0.038**	**0.219 ± 0.041**

[a]All data points represent mean ± SEM (μM) of n ≥ 3.

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Table 3

Agonist Activity of Synthetic Analogs at hMOR as Determined in G Protein BRET Assays

Compound	*Structure*	*EC_{50} ± SEM* (% efficacy) (μM) [a]
(-)-mitragynine		**0.339 ± 0.178 (34%)**
(+)-mitragynine		**3.34 ± 1.1 (18%)**
1		**0.681 ± 0.379 (29%)**
2		**>50**
11		**0.219 ± 0.071 (38%)**
3		**12 ± 7.6 (59%)**

[a] Agonist activity at hMOR indicated by EC_{50} values, maximal efficacy (E_{max}) relative to DAMGO in parentheses; All data points represent mean ± SEM (μM) of n ≥ 3.

Addiction Biology

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ORIGINAL ARTICLE

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Abuse liability and therapeutic potential of the *Mitragyna speciosa* (kratom) alkaloids mitragynine and 7-hydroxymitragynine

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ABSTRACT

Kratom, derived from the plant *Mitragyna speciosa,* is receiving increased attention as an alternative to traditional opiates and as a replacement therapy for opiate dependence. Mitragynine (MG) and 7-hydroxymitragynine (7-HMG) are major psychoactive constituents of kratom. While MG and 7-HMG share behavioral and analgesic properties with morphine, their reinforcing effects have not been examined to date. 7-HMG, but not MG, substituted for morphine self-administration in a dose-dependent manner in the rat self-administration paradigm. Following the substitution procedure, re-assessment of morphine self-administration revealed a significant increase following 7-HMG and a significant decrease following MG substitution. In a separate cohort, 7-HMG, but not MG, engendered and maintained intravenous self-administration in a dose-dependent manner. The effects of pretreatment with nalxonaxine (NLXZ), a μ1 opiate receptor antagonist, and naltrindole (NTI), a δ opiate receptor antagonist, on 7-HMG and morphine self-administration were also examined. Both NLXZ and NTI reduced 7-HMG self-administration, whereas only NLXZ decreased morphine intake. The present results are the first to demonstrate that 7-HMG is readily self-administered, and the reinforcing effects of 7-HMG are mediated in part by μ and δ opiate receptors. In addition, prior exposure to 7-HMG increased subsequent morphine intake whereas prior exposure to MG decreased morphine intake. The present findings indicate that MG does not have abuse potential and reduces morphine intake, desired characteristics of candidate pharmacotherapies for opiate addiction and withdrawal, whereas 7-HMG should be considered a kratom constituent with high abuse potential that may also increase the intake of other opiates.

Keywords 7-hydroxymitragynine, addiction, mitragynine, opiate, reinforcement, self-administration.

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INTRODUCTION

The increasing use of kratom (*Mitragyna speciosa korth*; aka *thang, kakuam, thom, ketum, biak biak*), a plant indigenous to Southeast Asia, has emerged as a public health concern in the US. Kratom has been used traditionally to combat fatigue and increase work productivity amongst farm populations in Southeast Asia. Kratom leaves are chewed or made into an extract and brewed (Hassan *et al.* 2013; Warner, Kaufman, & Grundmann 2016), and consumption is reported to produce stimulation (at low doses) and opiate-like effects (at higher doses) including analgesia, antitussive, antidiarrheal, and anti-inflammatory effects. Historically, kratom has also been used to reduce the intensity and duration of opiate withdrawal symptoms (Boyer *et al.* 2008; Ward *et al.* 2011; Cinosi *et al.* 2015; Warner *et al.* 2016); however, studies assessing the clinical efficacy of kratom are limited. In Southeast Asia, regular kratom use has been associated with physical dependence and withdrawal symptoms (Suwanlert 1975; Saingam *et al.* 2013; Singh, Muller, & Vicknasingam 2014), effects attributed in large part to mitragynine (MG). In the United States, there is growing concern regarding the safe use of kratom based on reports of addiction (Sheleg & Collins 2011; Galbis-Reig 2016) and toxicity and fatalities associated with use

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(McWhirter & Morris 2010; Neerman, Frost, & Deking 2013; Singh, Narayanan, & Vicknasingam 2016; Drago *et al.* 2017; Fluyau & Revadigar 2017).

Various strains of kratom are widely available over the internet as well as in various locations throughout the country. Forecasting models indicate kratom use will continue to increase in the United States (Stogner 2015). Currently, kratom consumption remains legal in the majority of states in the US. FDA's associate commissioner for regulatory affairs has stated that the FDA has 'identified Kratom as a botanical substance that poses a risk to public health and has the potential for abuse' (Food and Drug Administration, 2016). Concerns about Kratom have led the FDA to issue an import alert and public health advisory and the DEA to include kratom on the list of Drugs and Chemicals of Concern.

Kratom leaves contain more than 25 identified alkaloids (Hassan *et al.* 2013). Mitragynine (MG) and 7-hydroxymitrgynine (7-HMG), the main psychoactive alkaloids of kratom, constitute approximately 60 and 2 percent of the plant's alkaloids, respectively (Prozialeck, Jivan, & Andurkar 2012). MG and 7-HMG are partial agonists at the μ opiate receptor and weak antagonists at δ and κ opiate receptors (Kruegel *et al.* 2016; Varadi *et al.* 2016), with 7-HMG exhibiting approximately 5-fold greater affinity at the μ opiate receptor compared to MG. Assays of opiate receptor-mediated G-protein function reveal similar potencies for MG and 7-HMG (Kruegel *et al.* 2016; Varadi *et al.* 2016). While both compounds exhibit naloxone-sensitive antinociceptive activity, 7-HMG exhibits 40-fold greater potency than MG and 10-fold greater potency than morphine in these assays (Takayama *et al.* 2002; Matsumoto *et al.* 2004). Repeated administration of 7-HMG produces tolerance to the compound's analgesic effects as well as cross-tolerance to morphine's antinociceptive action (Matsumoto *et al.* 2005). Chronic consumption of kratom as well as repeated administration of 7-HMG induces physical dependence as determined by naloxone-precipitated withdrawal (Matsumoto *et al.* 2005).

Few studies have examined the abuse/addiction potential of MG and 7-HMG. The absence of controlled studies in humans creates space for basic science studies to provide critical information to help guide use and policy decisions. The abuse liability of compounds is generally assessed in animal models using drug discrimination, place conditioning and/or self-administration paradigms —all of which address different aspects of abuse. Studies in animal models have provided considerable insight into the behavioral effects of MG and 7-HMG as well as the potential neurobiological mechanisms underlying those effects. Acute administration of MG increases locomotor activity, induces anxiolytic effects and induces conditioned place preference (Yusoff *et al.* 2016), while repeated MG administration induces locomotor sensitization but impairs performance on a variety of cognitive tasks (Yusoff *et al.* 2016; Ismail *et al.* 2017). Moreover, MG and 7-HMG fully generalize to the discriminative stimulus effects of morphine, suggesting the potential for abuse (Harun et al., 2015).

The reinforcing effects of drugs are an important indicator of abuse liability which is typically evaluated using drug self-administration procedures. Inherent in the operational definition of reinforcement is the contingency between behavior and drug administration, an important differentiation between self-administration and place conditioning and drug discrimination. Drug self-administration is widely accepted as the gold standard of measurements for abuse liability (Katz 1989; Hemby 1999; Lynch & Hemby 2011). In spite of the concern of the potential abuse liability of MG and 7-HMG in humans, no published studies to date have examined the ability of these compounds to maintain self-administration in experimental subjects. To that end, we assessed that ability of MG and 7- HMG to substitute for morphine self-administration and to engender and maintain self-administration in drug naïve animals. Additionally, the contribution of μ and δ opiate receptors on the reinforcing effects of 7-HMG was examined. Results of these studies provide an objective assessment of the abuse potential of MG and 7-HMG in a rodent model that recapitulates key features of human drug taking.

MATERIALS AND METHODS

Subjects

Male Fischer 344 rats (100–130 days; Charles River, Wilmington, MA) were housed in a temperature-controlled vivarium on a 12-hour reversed light/dark cycle (lights on at 6:00 PM). Rats were group-housed two per cage with water available *ad libitum*. Food was restricted such that rats were maintained at 90 percent of their free feeding weight throughout the experiment. Experimental sessions were conducted during the dark phase of the light/dark cycle. All procedures were performed in accordance with the High Point University Institutional Animal Care and Use Committee and the National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publication No. 80-23) revised in 1996.

Chemicals

Mitragynine was isolated according to published procedures (Ponglux *et al.* 1994). 7-HMG was synthesized from mitragynine by McCurdy's research group, Department of BioMolecular Sciences, University of Mississippi, as previously published (Ponglux *et al.* 1994; Takayama *et al.* 2002). Mitragynine and 7-HMG were analyzed by ^{1}H NMR, ^{13}C NMR, elemental analysis, HPLC and HR-MS,

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and were found to be >99 percent pure. Morphine SO_4 was purchased from Gallipot, Inc. (St. Paul, MN), penicillin G procaine from Butler Company (Columbus, OH), propofol, ketamine HCl (Ketaset), xylazine (Xylamed) from Patterson Veterinary Supply, Inc. (Greely, CO), naltrindole and naloxonazine from Tocris Bioscience/Biotechne (Minneapolis, MN). Drugs were dissolved in heparinized saline. Morphine, MG and 7-HMG were infused in a volume of 200 µl.

Behavioral apparatus and training

Operant apparatus

Experiments were conducted in operant conditioning chambers (ENV-008CT; Med Associates, St. Albans, VT) enclosed in sound-attenuating cubicles (ENV-018; Med Associates). The front panel of the operant chambers contained a response lever (4 cm above the floor and 3 cm from the side wall), a cue light (3 cm above the lever) and a food chute centered on the front wall (2 cm above the floor) that was connected to a food pellet dispenser (ENV-023; Med Associates) located behind the front wall and a tone generator to mask extraneous noise. A syringe pump (PHM-100; Med Associates) holding a 20-ml syringe delivered infusions. A counterbalanced arm containing the single channel liquid swivel was located 8–8.5 cm above the chamber and attached to the outside of the front panel. An IBM compatible computer was used for session programming and data collection (Med Associates Inc., East Fairfield, VT).

Lever training

Subjects were transferred to the operant chambers for daily experimental sessions, and responding was engendered and maintained by delivery of food pellets (45-mg pellets; Noyes, Lancaster, NH) under an FR 1 schedule of reinforcement that was gradually increased to FR3 (every third response produced a food pellet). The lever light was illuminated when the schedule was in effect. Completion of the response requirement extinguished lights, delivered food and was followed by a 20-second timeout (TO) period during which all lights were extinguished, and responses had no scheduled consequences. After the TO, the lights were illuminated, and the FR schedule was again in effect. Sessions lasted 20 minutes or until 30 food pellets were delivered. Responding was considered stable when there was less than 10 percent variation in the number of reinforcers for three consecutive sessions.

Intravenous jugular surgery

After operant responding was acquired and maintained by food, subjects surgically implanted with an intravenous jugular catheter. Venous catheters were inserted into the right jugular vein following administration of ketamine (90 mg/kg; IP) and xylazine (5 mg/kg; IP) for anesthesia as described previously (Pattison *et al.* 2012; Pattison *et al.* 2014; McIntosh *et al.* 2015). Catheters were anchored to muscle near the point of entry into the vein. The distal end of the catheter was guided subcutaneously to exit above the scapulae through a Teflon shoulder harness. The harness provided a point of attachment for a spring leash connected to a single-channel fluid swivel at the opposing end. The catheter was threaded through the leash and attached to the swivel. The other end of the swivel was connected to a syringe (for saline and drug delivery) mounted on a syringe pump. Rats were administered penicillin G procaine (75 000 units in 0.25 ml, i.m.) and allowed a minimum of 5 days to recover before self-administration studies were initiated. Hourly infusions of heparinized saline were administered through the catheter to maintain functional catheters. The health of the rats was monitored daily by the experimenters and weekly by institutional veterinarians per the guidelines issued by the Institutional Animal Care and Use Committee and the National Institutes of Health. Infusions of propofol (6 mg/kg; i.v.) were manually administered as needed to assess catheter patency.

Rats were transferred to the operant chambers for daily two-hour self-administration sessions. Before each session, the swivel and catheter were flushed with 500 µl of heparinized saline before connecting the catheter to the syringe via a 20 ga luer hub and 28 ga male connector. The start of each session was indicated by the illumination of the house light, stimulus light above the lever and the extension of the lever. Completion of the response requirement was followed by a 20-second time out (FR3:TO 20 seconds) during which time the subject received a 200-µl intravenous infusion over the first 6 seconds, retraction of the lever, extinguishing of lever light, generation of a tone and illumination of the house light. At the end of the TO, the lever was extended, lever light illuminated, tone silenced and the house light extinguished (Hemby, Smith, & Dworkin 1996; Hemby *et al.* 1999; McIntosh *et al.* 2015).

Experiment 1: Substitution for morphine self-administration

Self-administration was engendered using 100 µg/inf of morphine sulfate. When responding was stable (two consecutive sessions in which the number of reinforcers did not vary by more than 20 percent), the dose was changed to 50 µg/inf. Following stable responding at this dose, saline was substituted for morphine until responding stabilized. Following the conclusion of extinction testing, rats were assigned to one of two groups to

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receive MG (n = 9; 25, 50, 100 and 150 μg/inf) or 7-HMG (n = 8; 2.5, 5, 10 and 20 μg/inf)—doses were randomized. The dose range for MG self-administration was based on the finding that equivalent doses of morphine and MG induced place conditioning in rats (Yusoff *et al.* 2016). The dose range for 7-HMG was based on the finding that 7-HMG substitutes for morphine in the drug discrimination procedure at a dose five-fold lower than the training dose of morphine. Once responding stabilized for a particular dose, the next dose was made available the following session until all doses mentioned above were assessed. After all doses had been assessed for a subject within a group, the rat was allowed to self-administer morphine (50 and 100 μg/inf) to determine the effect of prior drug history on subsequent morphine intake.

Experiment 2: Acquisition of self-administration

Whereas substitution procedures may be more sensitive for determining the reinforcing effects of a compound, acquisition procedures assess only the reinforcing effects of a single compound, without the potential confound of prior drug associations. The ability of MG and 7-HMG to engender and maintain responding without prior drug history was determined in a separate cohort of rats. Rats were assigned to one of three groups to self-administer MG (n = 8; 100 μg/infusion), 7-HMG (n = 8; 10 μg/infusion) or morphine (n = 6; 100 μg/infusion). Following stable responding at the initial dose, rats in the MG group were given access to 25 and 50 μg/infusion MG, 7-HMG group was allowed to self-administer 5.0 and 2.5 μg/infusion 7-HMG, while rats in the morphine group were allowed to self-administer 50 μg/infusion morphine. The presentation of doses for MG and 7-HMG was randomized following the initial dose.

Experiment 3: Selective opiate receptor antagonism of morphine and 7-HMG self-administration

To determine the contribution of μ and δ opiate receptors on the reinforcing effects of 7-HMG, rats that had previously acquired morphine and 7-MHG self-administration were administered naloxonazine (NLXZ), a selective μ1 receptor antagonist or the δ receptor antagonist naltrindole (NTI). NLXZ (5 and 15 mg/kg, i.p.) and NTI (0.5 and 5 mg/kg, i.p.) were administered 30 minutes before the session. The effects of saline pretreatment and saline extinction on responding were also assessed.

Data analysis

Experiment 1

Morphine self-administration as well as MG and 7-HMG substitution was analyzed using a one-factor ANOVA (Dose). Morphine self-administration before and after MG and 7-HMG substitution was analyzed using a two-way ANOVA (Dose × Pre/Post). Analysis of three days following MG and 7-HMG substitution was conducted using a two-way repeated measures ANOVA (Dose × Pre/Post) with Sessions as the repeated measure. **Experiment 2.** Acquisition of self-administration was analyzed using one-factor ANOVA (Dose) ANOVA and number of infusions as the dependent measure. **Experiment 3.** Comparisons between baseline intake and intake following saline pretreatment were conducted using two-tailed paired t-test for both the morphine and 7-HMG groups. The effects of NLXZ and NTI on morphine and 7-HMG were analyzed independently using a one-factor ANOVA (Antagonist Dose). Analysis of the session in which the antagonists or saline were administered, and the following session was conducted using a two-way repeated measures ANOVA (Dose × Day) with Sessions as the repeated measure. The number of infusions was the dependent variable for analyses. Where appropriate, post hoc analyses were conducted using Tukey's test.

RESULTS

Substitution and reintroduction of morphine

Responding was engendered and maintained by morphine at the doses tested [$F(2,17) = 24.4$, $P < 0.0001$]. Morphine self-administration was dose-dependent with both morphine doses significantly greater than saline and 50 μg greater than 100 μg/infusion ($P < 0.05$) (Fig. 1a). MG was not reliably self-administered when substituted for morphine [$F(4,38) = 0.51$, $P = 0.73$]. MG intake was not significantly different from vehicle, suggesting that MG does not function as a reinforcer at the doses tested (Fig. 1b). In contrast, 7-HMG substituted for morphine with intake dependent on the dose available [$F(4,32) = 6.3$, $P = 0.0009$]. The number of infusions obtained for 5 and 10 μg/inf were significantly greater than vehicle ($P < 0.05$), confirming that these doses of 7-HMG functioned as reinforcing stimuli. 7-HMG resulted in an inverted 'U'-shaped dose-effect function with maximal intake observed at 5 and 10 μg/inf (Fig. 1c).

Following completion of the substitution protocol, morphine self-administration was re-assessed. Morphine intake was significantly altered by both MG [$F(1,32) = 5.5$, $P = 0.025$] and 7-HMG [$F(1,24) = 12.92$, $P = 0.0015$], albeit in opposite directions. MG exposure significantly reduced morphine self-administration of 50 μg ($P < 0.01$), but not 100 μg [Fig. 1d (top panel)]. The decrease in morphine self-administration (50 μg) was not observed on the first day following MG exposure but was significantly decreased on days two and three Fig. 1c (middle panel)].

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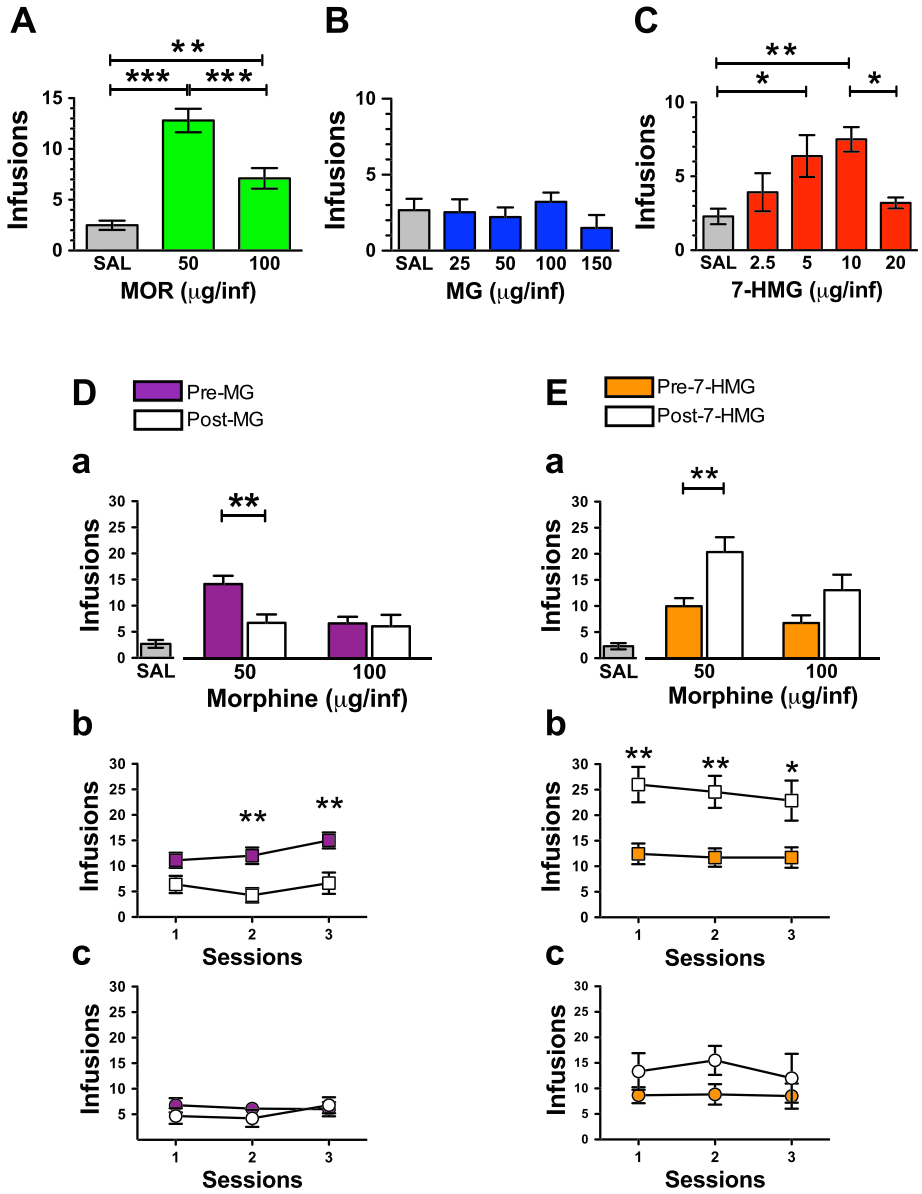


Figure 1 Substitution of MG and 7-HMG following morphine self-administration and the impact on subsequent morphine self-administration. (a) Morphine was reliably self-administered in a dose-dependent manner (***$P < 0.001$ saline versus 50 μg/inf and 50 versus 100 μg/inf, **$P < 0.01$ saline versus 100 μg/inf). (b) MG did not substitute for morphine (100 μg/inf) at any of the doses tested. (c) 7-HMG substituted for morphine (100 μg/inf) in a dose-dependent manner (*$P < 0.05$ saline versus 5 μg/inf, 10 versus 20 μg/inf, $P < 0.01$ saline versus 10 μg/inf). (d, e) Comparisons were made between morphine self-administration pre-exposure and post-exposure to MG ($n = 9$), and 7-HMG ($n = 7$) substitution. (d) Top panel: Comparison of morphine intake pre-MG exposure versus post-MG exposure revealed a significant difference in self-administration (**$P < 0.01$, pre-MG exposure versus post-MG exposure for 50 μg/inf). (d) Middle panel: Assessment of morphine self-administration for 3 days following MG exposure revealed a decrease in intake of 50 μg/inf morphine compared to pre-MG exposure levels (**$P < 0.01$ sessions 2 and 3 pre-exposure versus post-exposure to MG). (d) Bottom panel but had no change in morphine self-administration of 100 μg/inf across the 3 days. (e) Top panel, Comparison of morphine intake pre-7-HMG exposure versus post-7-HMG exposure revealed a significant difference in self-administration (**$P < 0.01$, pre-7-HMG exposure versus post-7-HMG exposure for 50 μg/inf morphine). (e) Middle panel: Substitution of 7-HMG resulted in a significant increase in morphine self-administration (50 μg/inf) for the 3 days following 7-HMG exposure (**$P < 0.01$ sessions 1 and 2, *$P < 0.05$ session 3 pre-exposure versus post-exposure to 7-HMG). (e) Bottom panel whereas self-administration of 100 μg/inf of morphine was not altered by 7-HMG exposure. Data are expressed as mean ± s.e.m.

Intake of 100-μg morphine did not differ from saline before or following exposure to MG [Fig. 1d (bottom panel)].

In contrast, 7-HMG administration significantly increased self-administration of 50-μg morphine ($P < 0.01$), and there was a trend towards increased intake of 100-μg morphine ($P = 0.054$) [Fig. 1e (top panel)]. Morphine self-administration (50 μg) was significantly elevated for the three days following 7-HMG exposure by 109, 110 and 95 percent, respectively

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[Fig. 1e (middle panel)]. Self-administration of 100-μg morphine was also increased for the 3 days following 7-HMG exposure by 53, 75 and 41 percent, respectively [Fig. 1e (bottom panel)]. We also determined whether morphine self-administration was altered by forced abstinence for an equivalent number of days to the substitution procedure. Morphine intake was not altered by the forced abstinence period [$F(1,20) = 8$, $P = 0.1523$], suggesting that the amount of time between morphine self-administration availability did not influence intake (*data not shown*).

Acquisition of self-administration

Following the assessment of the ability of MG and 7-HMG to substitute for morphine self-administration, we assessed the ability of MG and 7-HMG to engender and maintain responding in a separate cohort of rats. Morphine was readily self-administered at the doses tested [$F(2,17) = 24$, $P < 0.0001$]. Self-administration of 50 and 100 μg was greater than saline and the number of infusions obtained for the 50 μg/inf was significantly greater than 100 μg/inf (Fig. 2a). In contrast, access to MG did not reliably engender or maintain self-administration at any of the doses tested [$F(3,31) = 2.255$, $P = 0.1038$] as intake was not significantly different from vehicle, indicating that MG does not function as a reinforcer in this procedure at the doses tested (Fig. 2b). Similar to morphine, rats self-administered 2.5, 5 and 10 μg/infusion of 7-HMG [$F(3,31) = 6.2$, $P = 0.0024$]—the same doses that substituted for morphine self-administration (Fig. 2c). The number of infusions obtained 5 and 10 μg/inf was significantly greater than vehicle ($P < 0.05$), demonstrating that these doses served as reinforcing stimuli. Intake for 2.5 and 20 μg/inf 7-HMG did not significantly differ from vehicle. Comparison of saline intake for all groups revealed there was no statistically significant difference between the groups [$F(2,22) = 2.99$, $P = 0.0731$]. Tukey's post hoc test revealed there was no statistically significant difference between any of the group pairs.

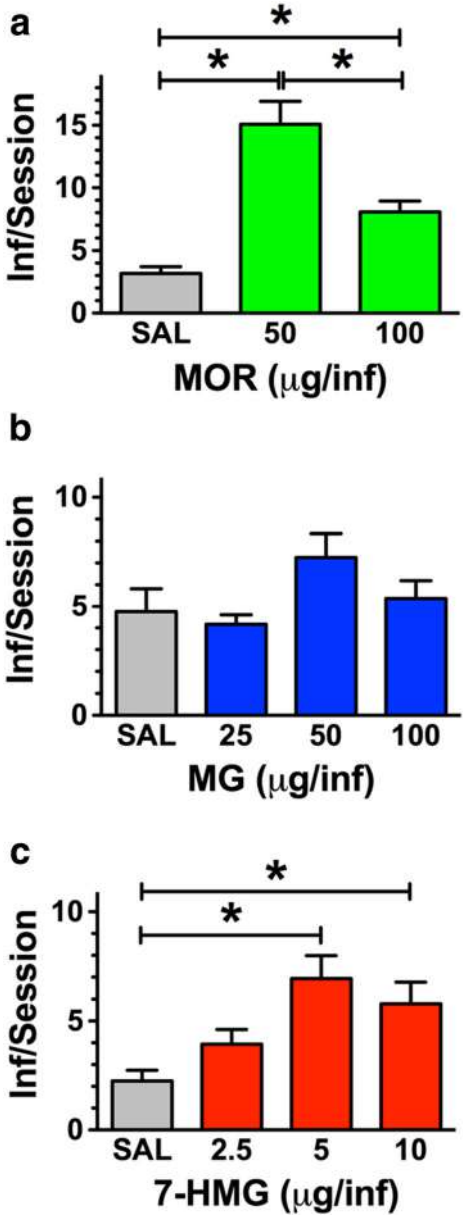


Figure 2 Acquisition of intravenous self-administration with 7-HMG but not MG. (a) Rats acquired morphine self-administration, and intake was dose dependent. (b) Rats did not acquire self-administration of MG at any of the doses tested. (c) However, a separate cohort of rats acquired self-administration of 7-HMG. $n = 8$ rats. Data are expressed as mean ± s.e.m. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{\#}P < 0.0001$

Antagonist effects on 7-HMG and morphine self-administration

The effects of the selective μ opiate receptor antagonist NLXZ and the δ receptor antagonist NTI on morphine (50 μg/inf) and 7-HMG (5 μg/inf) self-administration were examined. The selection was based on doses that maintained the highest intake for morphine and 7-HMG self-administration, respectively. NLXZ pretreatment attenuated morphine self-administration [$F(2,15) = 76$, $P < 0.0001$] [Fig. 3a (left panel)]. Both 5 and 15 mg/kg NLXZ produced a significant decrease in the number of infusions for morphine ($P < 0.05$); however, there was no significant difference in the number of infusions between the doses. Morphine intake was significantly decreased the day of NLXZ pretreatment for both doses ($P < 0.001$) and returned to control levels the following session [Fig. 3a (right panel)]. Pretreatment with NLXZ also significantly reduced 7-HMG self-administration [$F(2,22) = 39$, $P < 0.0001$] [Fig. 3b (left panel)]. Both NLXZ doses significantly reduced 7-HMG

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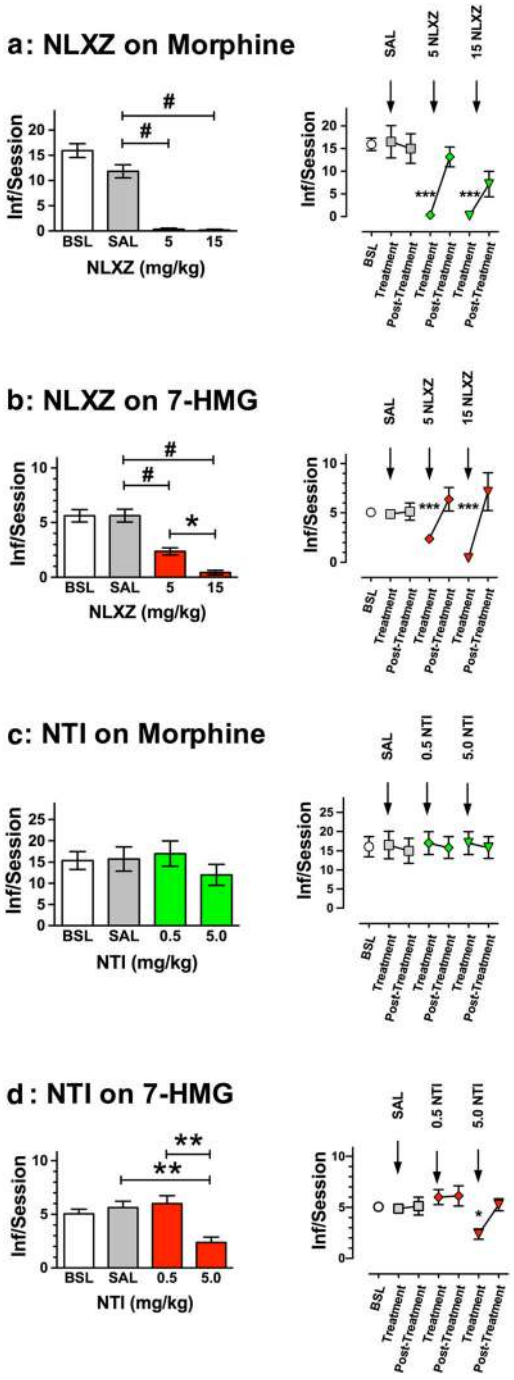


Figure 3 Selective μ and δ opiate receptor antagonist pretreatment alters 7-HMG self-administration. Effects of NLXZ on morphine (50 μg/inf) and 7-HMG (5 μg/inf) self-administration. Baseline drug intake was not significantly different than intake following saline pretreatment for morphine analysis or 7HMG analysis. (a) Left panel NLXZ significantly reduced the number of morphine infusions compared to saline pretreatment ($^{\#}P < 0.0001$, saline versus 5 and 15 mg/kg NLXZ). (a) Right panel There was a significant interaction between NLXZ doses and sessions with infusions on the day of pretreatment significantly less than following saline pretreatment for the 15 mg/kg NLXZ dose ($^{***}P < 0.001$). (b) Left panel NLXZ decreased 7-HMG self-administration, the effect was dose dependent ($^{\#}P < 0.0001$, $^{*}P < 0.05$) and (b) right panel There was a significant interaction between NLXZ doses and sessions with infusions on day of pretreatment significantly less than following saline pretreatment for both NLXZ doses. We also examined the effects of NTI on morphine (50 μg/inf) and 7-HMG (5 μg/inf) self-administration. Baseline drug intake was not significantly different than intake following saline pretreatment for morphine analysis and 7HMG analysis. (c) Left panel NTI did not significantly alter morphine self-administration at either dose tested. (c) Right panel No significant interaction of NTI dose and session was observed. (d) Left panel Pretreatment with NTI decreased 7-HMG self-administration in a dose-dependent manner ($^{**}P < 0.01$, saline versus 5.0 mg/kg NTI, 0.5 versus 5.0 mg/kg NTI) and (d) right panel There was a significant interaction between NTI doses and sessions although Bonferroni's post hoc analysis did not reveal any significant differences between saline and NTI pretreatment for either day. Data are expressed as mean ± s.e.m.

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intake compared to saline ($P < 0.0001$), although intake following 15 mg/kg was significantly less than following 5 mg/kg ($P < 0.05$). For both doses, 7-HMG intake was significantly reduced the day of NLXZ pretreatment ($P < 0.001$) and returned to control levels the following session [Fig. 3b (right panel)]. Comparison of baseline intake was not significantly different than intake following saline pretreatment for the morphine [$t = 2.5$, df = 5, $P = 0.057$] or 7-HMG groups [$t = 0.03$, df = 7, $P = 0.98$]; therefore, the number of infusions following saline was used for comparison with NLXZ.

NTI pretreatment produced differential effects on morphine and 7-HMG self-administration. Neither 0.5 nor 5.0 mg/kg NTI significantly altered morphine self-administration [$F(2,17) = 0.88$, $P = 0.434$] [Fig. 3c (left panel)]. Morphine intake following either dose of NTI did not differ significantly from saline pretreatment the day of or the day following pretreatment [Fig. 3c (right panel)]. In contrast, 5.0 mg/kg NTI pretreatment significantly altered 7-HMG self-administration [$F(2,23) = 11$, $P < 0.0007$] [Fig. 3d (left panel)]. The effect of NTI on 7-HMG intake was dose dependent as intake following the 5 mg/kg dose was significantly lower than saline ($P < 0.05$) as well as the 0.5 g/kg dose ($P < 0.05$). 7-HMG intake was significantly reduced the day of 5.0 mg/kg NTI pretreatment ($P < 0.05$) and returned to control levels the following session [Fig. 3 (right panel)]. For naltrindole (NTI), comparison of baseline intake was not significantly different than intake following saline pretreatment for the morphine [$t = 0.13$, df = 5, $P = 0.90$] or 7-HMG groups [$t = 0.94$, df = 7, $P = 0.38$].

DISCUSSION

The present study provides the first characterization of the reinforcing effects of MG and 7-HMG in an animal model of human drug consumption. Using the intravenous self-administration procedure, the study demonstrates 7-HMG, but not MG, substitute for morphine self-administration and engender and maintain self-administration in drug-naïve rats. Under both the substitution and acquisition self-administration procedures, 7-HMG self-administration exhibited an inverted 'U' dose–effect curve, typical of other drugs of abuse under similar experimental conditions, wherein low to moderate doses increase responding and higher doses decrease responding (Wilson, Hitomi, & Schuster 1971; Pickens 1978). In both self-administration procedures, 7-HMG maintained intake above levels observed for saline, indicating the compound serves as a reinforcing stimulus within the range of the doses examined. Previous studies have shown MG induces a conditioned place preference (Sufka *et al.* 2014; Yusoff *et al.* 2016; Yusoff *et al.* 2017) and both MG and 7-HMG generalize to the discriminative stimulus effects of morphine (Harun *et al.* 2015). Both procedures provide valuable information about the abuse liability of compounds such as the motivational effects of contextual cues associated with a drug (place conditioning) and whether a compound shares subjective effects with a known drug of abuse (drug discrimination). Self-administration is used to examine the reinforcing effects of drugs, a key feature influencing the potential risk for abuse. With regard to opioids, findings from studies using rat self-administration models are highly consistent with clinical measures of abuse liability and have proven to be a reliable predictor of abuse liability in humans (O'Connor *et al.* 2011). While MG shares a similar mechanism of action with morphine, the present results demonstrate a difference in the reinforcing effects and thus the abuse potential between the two compounds. The discrepancy between the present finding and those of aforementioned place conditioning and drug discrimination studies are intriguing and warrant further assessment of routes of administration as well as non-human primate and human abuse potential studies. The self-administration of 7-HMG along with the shared subjective effects of morphine (Harun *et al.* 2015) indicate that 7-HMG has a significant potential to be abused. Additional studies are needed to examine the ability of 7-HMG self-administration to induce physical dependence and withdrawal, and the effects of 7-HMG on relapse to morphine, other opiates and other drug classes. The present findings along with the results of additional studies will provide critical information as to whether the compounds warrant control under the Controlled Substances Act.

In addition to abuse liability assessments, we also explored the impact of exposure to MG and 7-HMG to subsequent morphine intake following completion of the substitution procedure. Exposure to less than 2 mg of MG over a 2-week period (when examining the ability of MG to substitute for morphine) significantly reduced subsequent morphine self-administration up to 3 days following the last administration of MG. These findings raise the possibility that MG may contribute to the reported reduction in opiate intake in dependent individuals. Buprenorphine, a μ receptor partial agonist like MG, also reduced morphine (Harrigan & Downs 1981) as well as heroin self-administration (Mello, Bree, & Mendelson 1983) when administered continuously or intravenously twice daily but the effects on intake were acute. The effects of MG on morphine intake in the present study are intriguing but should be interpreted with caution for several reasons including the lack of morphine dose-dependent effects, the absence of a MG dose-effect determination on morphine and other commonly abused opiates, and the need to assess the effect of MG on animal

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models of opiate relapse and withdrawal. The finding that MG, in contrast to buprenorphine which has abuse liability (Jones et al., 2017), does not appear to have abuse liability and reduces morphine intake is intriguing and warrants further investigation to determine the efficacy of MG as a potential pharmacotherapy for opiate addiction. Comparison of MG with buprenorphine as well as methadone, on the self-administration of morphine as well as other opiates, is needed to determine the therapeutic potential and potency of MG. The effects of 7-HMG on morphine intake are opposite to those observed with MG. An average total intake of approximately 700 μg of 7-HMG over a 2 to 3-week period (when examining substitution for morphine) induced a significant elevation in subsequent morphine self-administration over 3 days following the last exposure to the compound, indicative of tolerance to morphine. Previous reports indicate that kratom users develop tolerance, increasing intake over time (Suwanlert 1975; Hassan *et al.* 2013). While tolerance to the analgesic effects of 7-HMG has been reported (Matsumoto *et al.* 2005; Matsumoto *et al.* 2008), no published studies to date report tolerance or cross-tolerance to morphine (or other opiates) intake. The increase in morphine intake in the present study could reflect a compensatory response to withdrawal from 7-HMG; however, this is unlikely as no overt signs of withdrawal were observed. Additional studies are warranted to determine if the observed increase in morphine intake following 7-HMG self-administration is related to a change in the reinforcing effects of morphine, a decrease in the adverse effects or a combination of the two. The demonstration that 7-HMG has significant abuse potential and increases the intake of morphine and possibly other opiates has significant potential clinical relevance.

The present study utilized two variations of the intravenous self-administration procedure to determine abuse liability of MG and 7-HMG: substitution and acquisition of self-administration. The substitution procedure assessed the ability of MG and 7-HMG to substitute for morphine in rats with a history of morphine self-administration and provided a clinically relevant evaluation given the potential for cross-generalization of opiates, the use of kratom to reduce intake of and withdrawal from other opiates, and the use of kratom as an affordable substitute for heroin (Boyer, Babu, & Macalino 2007; Boyer *et al.* 2008; Vicknasingam *et al.* 2010). The ability of 7-HMG to substitute for morphine in the present study suggests that the reinforcing effects may be mediated by similar neural mechanisms such as μ and δ opiate receptors. The effects of NLXZ on 7-HMG and morphine self-administration may reflect nonspecific effects on responding; however, we do not consider this a viable interpretation inasmuch as NLXZ doses within this range do not affect responding maintained by saccharin or food (Liu & Jernigan 2011; Peana *et al.* 2011). The present results confirm and extend previous studies of the effects of NLXZ on opiates including decreasing heroin self-administration (Negus *et al.* 1993), decreasing morphine-induced place conditioning in rats (Piepponen *et al.* 1997) and attenuating the discriminative stimulus effects of morphine (Suzuki *et al.* 1995). These results indicate that the reinforcing effects of morphine 7-HMG are mediated in part by μ opiate receptors.

NTI, the selective δ opiate receptor antagonist, attenuated 7-HMG in a dose-dependent manner. The lack of effect of NTI on morphine self-administration is supported by previous studies, which indicates that NTI does not alter morphine place conditioning (Suzuki *et al.* 1994; Piepponen *et al.* 1997) or the discriminative stimulus effects of morphine (Stevenson *et al.* 2000). In contrast, NTI and naltrindole-5′-isothiocyanate have been shown to attenuate the reinforcing effects of heroin (Negus *et al.* 1993; Martin *et al.* 2000); however, δ opiate receptors are not considered to directly mediate heroin reinforcement. The reinforcing effects of heroin are mediated via μ opiate receptors which bind the major heroin metabolites, morphine and 6-monoacetylmorphine, neither of which bind to the δ receptor with appreciable affinity. The difference in the effects of NTI between the Negus *et al.* study is likely attributable to the finding that only the 10 and 17 mg/kg doses of NTI were effective in attenuating heroin self-administration—compared with 0.5 and 5 mg/kg in the present study. Given that the 5 mg/kg NTI dose slightly decreased morphine self-administration, higher doses of NTI may have significantly attenuated intake. The significant attenuation of the reinforcing effects of 7-HMG by NTI reflects involvement of δ opiate receptors in the reinforcing effects of this compound. Given that 7-HMG is a weak δ opiate receptor antagonist, the finding that antagonism of the δ receptor partially attenuates the reinforcing effects of this compound seems counterintuitive. However, two potential mechanisms may account in part for the finding. Martin *et al.* suggested that morphine-induced activation of μ opiate receptors in the pallidum stimulates release of met-enkephalin that then binds to δ opiate receptors to exert reinforcing effects (Martin *et al.* 2000). Because 7-HMG is a partial μ opiate receptor agonist, the compound may exert effects similar to morphine via this mechanism. An alternative hypothesis suggests that chronic μ opiate receptor stimulation results in δ opiate receptor recruitment and μ opiate receptor desensitization. However, the dose-dependent effect of NLXZ on 7-HMG indicates that μ receptors were not de-sensitized. The manner in which δ opiate receptors are involved in 7-HMG reinforcement remains to be determined. Nonetheless, results from the present study indicate that both μ and δ opiate receptors contribute to the reinforcing

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effects of 7-HMG. Current pharmacotherapies for opiate addiction including methadone, buprenorphine and naloxone do not bind appreciably to δ opiate receptors and therefore may be less effective in treating kratom abuse.

The current results are the first demonstration of self-administration of MG and 7-HMG in the preclinical literature; however, several issues related to study design and methodology deserve to be addressed. First, food was restricted for all subjects in the present study in order to enhance acquisition and maintenance of self-administration in both the substitution and acquisition procedures. Previous studies have demonstrated that food restriction facilitates the acquisition of opiate self-administration and increases intake during the maintenance phase as well as enhances drug seeking (Piazza & Le Moal 1998). The absence of MG acquisition or substitution under food restriction conditions provides confidence that administration of this alkaloid is not reinforcing even under conditions that enhance the probability of intake. Nonetheless, negative results in self-administration procedures are difficult to interpret and may be due to multiple experimental variables that are not optimal for the drug being investigated including, but not limited to, the schedule of reinforcement, response contingencies, selected doses, rate of infusion, route of administration and drug availability. For example, the lack of MG self-administration may be attributed in part to the route of administration. Kratom is administered orally by humans whereas intravenous administration was used in the present study—routes which would yield different rates of onset of drug effects, drug levels, duration of effect and metabolism. The concern is mitigated to some extent by the finding that 7-HMG, a structurally similar compound from the same plant, is readily self-administered intravenously using the same procedure. Thus, the relevance of the current findings using the intravenous self-administration is not diminished, although future studies need to address the abuse liability of MG and 7-HMG using voluntary oral consumption methods (food and liquid). Another caveat to the present experimental design is the use of one versus two levers in the self-administration procedure. The determination of reinforcement is based on the statistical comparison of intake maintained by doses of a drug versus a negative control, which is vehicle in the present study. Alternatively, comparisons can be made between the rates of responding on an 'active' versus the 'inactive' lever. In either case, drugs that engender and/or maintain responding at levels that exceed responding maintained by the negative control are considered to be self-administered (function as reinforcing stimuli) (O'Connor *et al.* 2011).

In summary, the present findings from the rodent self-administration model indicate that MG, the main kratom alkaloid, does not have abuse or addiction potential and reduces morphine intake—desired characteristics of candidate pharmacotherapies for opiate addiction and withdrawal. In contrast, 7-HMG should be considered a kratom constituent with high abuse potential that may also increase the intake of other opiates. Although 7-HMG constitutes only 2 percent of the alkaloid content of kratom, purified extracts of 7-HMG are widely available on the internet and are consumed for their euphoric effects. Additional danger is posed by adulteration or the presence of high concentrations of 7-HMG in commercially available kratom products (Lydecker *et al.* 2016) which may increase the abuse liability.

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Conflict of Interest

The authors have no competing financial interests.

Authors Contribution

S.E.H., S.J.C. and C.R.M. jointly conceived the study; S.E.H. designed the experiments; J.F., S.J.C. and C.R.M. isolated, purified and prepared salts of mitragynine and 7-hydroxymitragynine; S.M. and S.E.H. performed the experiments, collected the data and analyzed the data; S.E.H. wrote and revised the manuscript and S.M., S.J.C. and C.R.M. edited the manuscript.

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